

# A/C-HEATER SYSTEM - MANUAL

1993 Toyota Celica

1993 Manual A/C-Heater Systems

Celica

## A/C SYSTEM SPECIFICATIONS

### SPECIFICATIONS TABLE

| Application                       | Specification                             |
|-----------------------------------|-------------------------------------------|
| Compressor Type                   |                                           |
| 4A-FE Engine .....                | Nippondenso 10PA15C 10-Cyl.               |
| 3S-GTE & 5S-FE Engine ...         | Nippondenso 10PA17C/VC 10-Cyl.            |
| Compressor Belt Tension (1)       |                                           |
| 1.6L Engine                       |                                           |
| New .....                         | 135-185 lbs. (61.2-83.9 kg)               |
| Used .....                        | 80-120 lbs. (36.2-54.4 kg)                |
| 2.0L Engine                       |                                           |
| New .....                         | 155-175 lbs. (70.3-79.4 kg)               |
| Used .....                        | 69-99 lbs. (31.3-44.9 kg)                 |
| 2.2L Engine                       |                                           |
| New .....                         | 155-175 lbs. (70.3-79.4 kg)               |
| Used .....                        | 100-120 lbs. (45.4-54.4 kg)               |
| System Oil Capacity .....         | 3.4-4.7 ozs.                              |
| Refrigerant (R-12) Capacity ..... | 24-27 ozs.                                |
| System Operating Pressures        |                                           |
| High Side .....                   | 206-213 psi (14.5-15 kg/cm <sup>2</sup> ) |
| Low Side .....                    | 21-28 psi (1.5-2.0 kg/cm <sup>2</sup> )   |

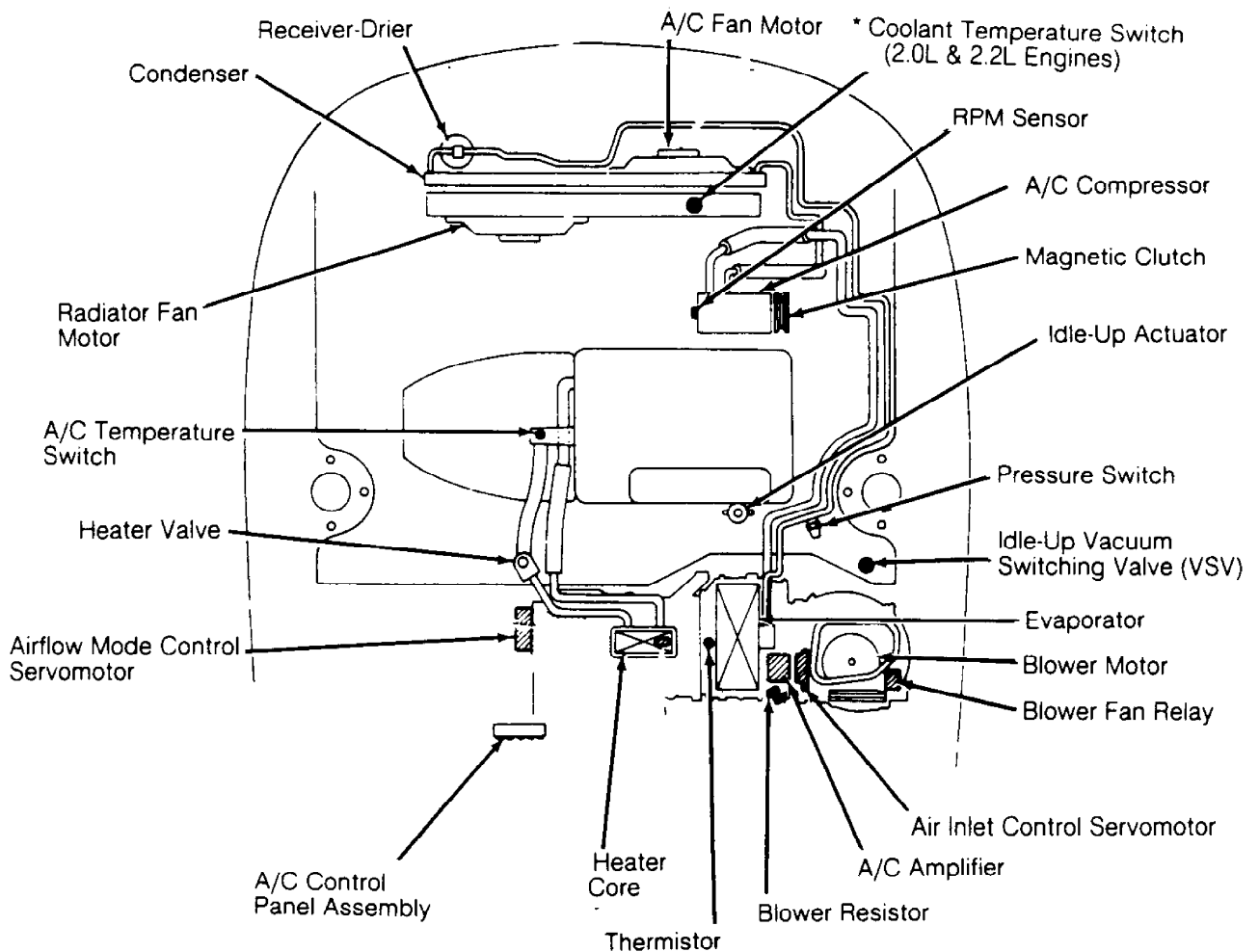
(1) - Using belt tension gauge, measure at longest run of belt.

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

## DESCRIPTION

The 2.0L and 2.2L A/T equipped models use a Nippondenso 10-cylinder variable displacement compressor. All other models use a Nippondenso 10-cylinder fixed displacement compressor. An electric condenser fan operates at 2 speeds, depending on coolant temperature and A/C switch position.

System components include A/C amplifier, evaporator, thermistor, triple-pressure switch, engine coolant temperature switch, A/C compressor, condenser, receiver-drier and pipes and hoses. See Fig. 1. Air door operation is controlled through cables or servomotors. A/C compressor operation and A/C modes are electrically controlled.



**NOTE:** Coolant Temperature Switch (for cooling fan control) on 1.6L engine is located on thermostat housing.

**93119610**

Fig. 1: Identifying Manual A/C-Heater System Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## OPERATION

### SYSTEM CONTROLS

A/C modes are controlled by push buttons and an A/C on-off switch. A dial type switch is used to control fan speed functions. See Fig. 2. A/C controls operate air supply selection (fresh or recirculating air), mode, temperature selection and blower speeds.

Temperature control knob operates the blend-air door in the A/C-heater unit. This mixes cooled and heated air so the selected air temperature can be obtained. The system will provide cooled air when A/C switch is on and blower motor is in any position other than OFF. The temperature control knob should be rotated counterclockwise (maximum cooling) when maximum A/C operation is desired.

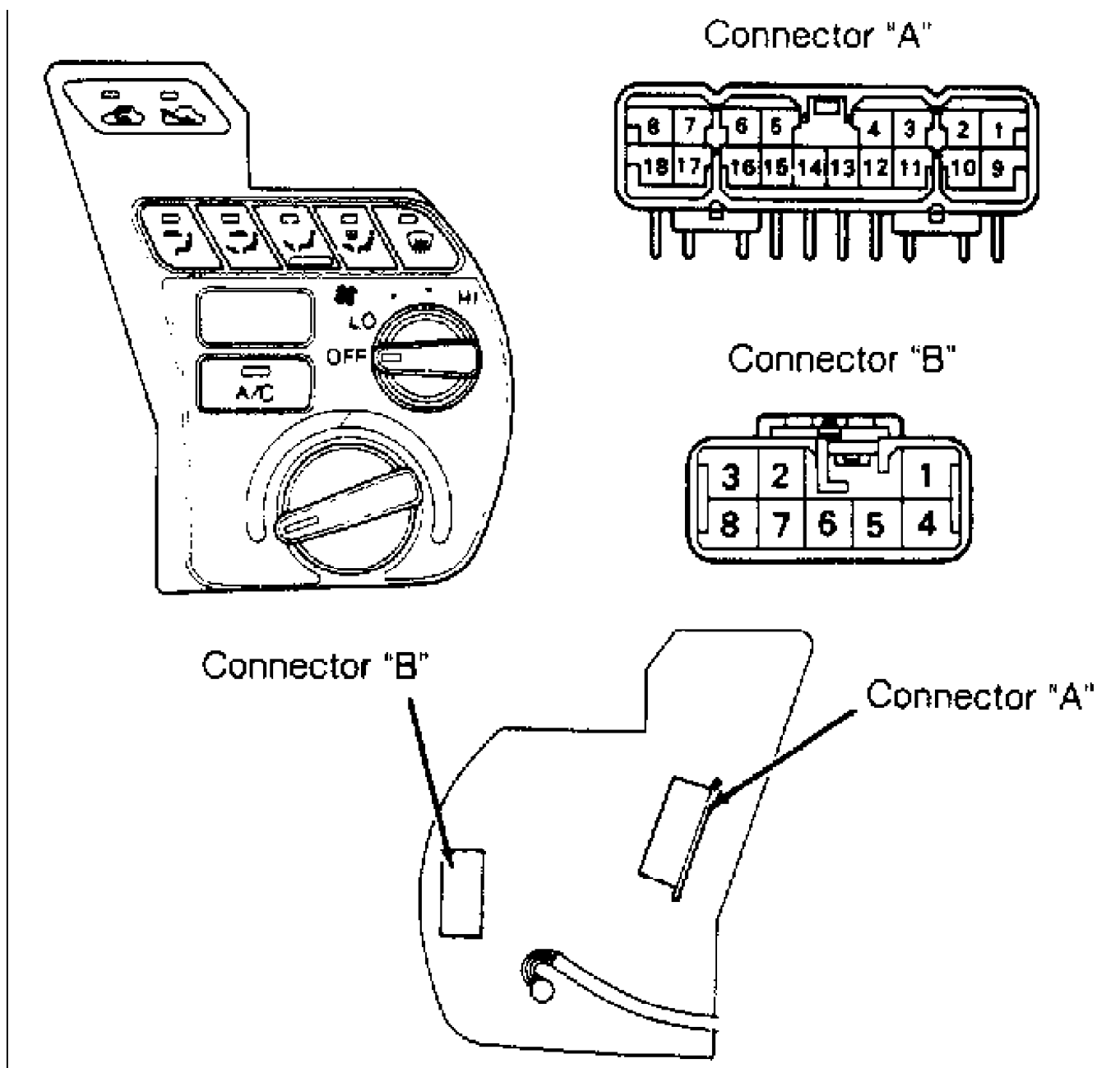


Fig. 2: Identifying A/C-Heater Control Panel Terminals  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## SYSTEM COMPONENTS

### A/C Switch

When A/C switch is pushed, A/C will operate if the blower motor control is in any position other than off. When activated, the A/C switch allows the compressor (magnetic) clutch to engage and operate the compressor. When activated, a light will illuminate on the A/C push button.

### Triple-Pressure Switch

The triple-pressure switch is located on the liquid line, and

is wired in series with the compressor (magnetic) clutch and the electric fan motor. The compressor clutch control portion of the switch cuts off electrical power to the clutch when refrigerant pressures have gone above or below the control point of the switch. When pressures have returned to normal operating ranges, compressor clutch receives power to resume operation.

The fan control portion of the switch cuts off electrical power to the condenser fan motor when refrigerant pressures have gone above the control point of the switch. When pressures have returned to normal operating ranges, condenser fan receives power to resume operation.

#### Thermistor

The thermistor is a thermocouple, mounted in front of the evaporator (air outlet side) to sense airflow temperature. The thermistor is electronically wired in series with the compressor clutch. The evaporator thermistor is used to prevent the evaporator from freezing up. The amplifier uses value received from thermistor to send appropriate electrical signal to compressor clutch for proper on-off cycling.

#### Vacuum Switching Valve (VSV)

A solenoid valve is used to assist in smooth engine operation during compressor on cycle. The VSV holds the throttle at slightly above idle (spring loaded to this position) when A/C system is operating. When system is off, vacuum is directed to VSV diaphragm, allowing throttle to return to normal idle position.

## ADJUSTMENTS

### AIR MIX DAMPER

With control cable of air mix damper removed, set air mix damper and temperature control switch to cool position. See Fig. 3. Install control cable and clamp.

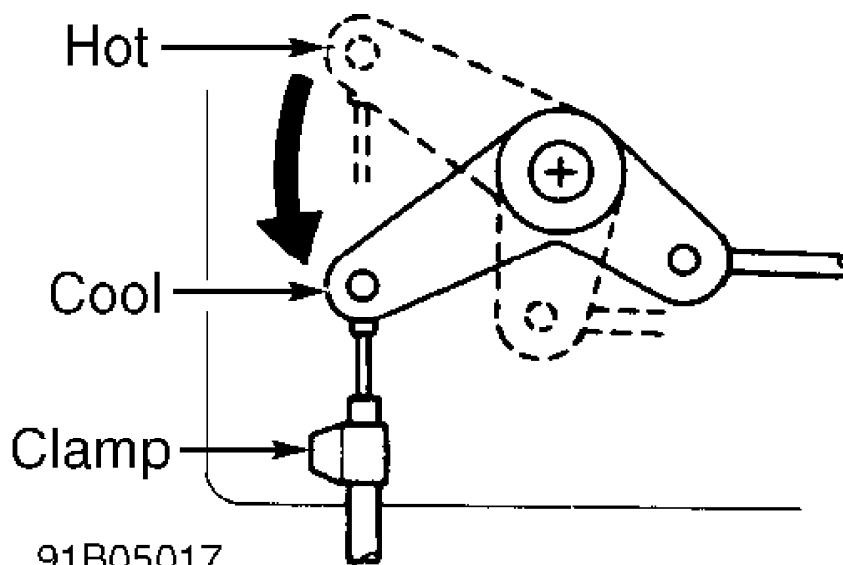
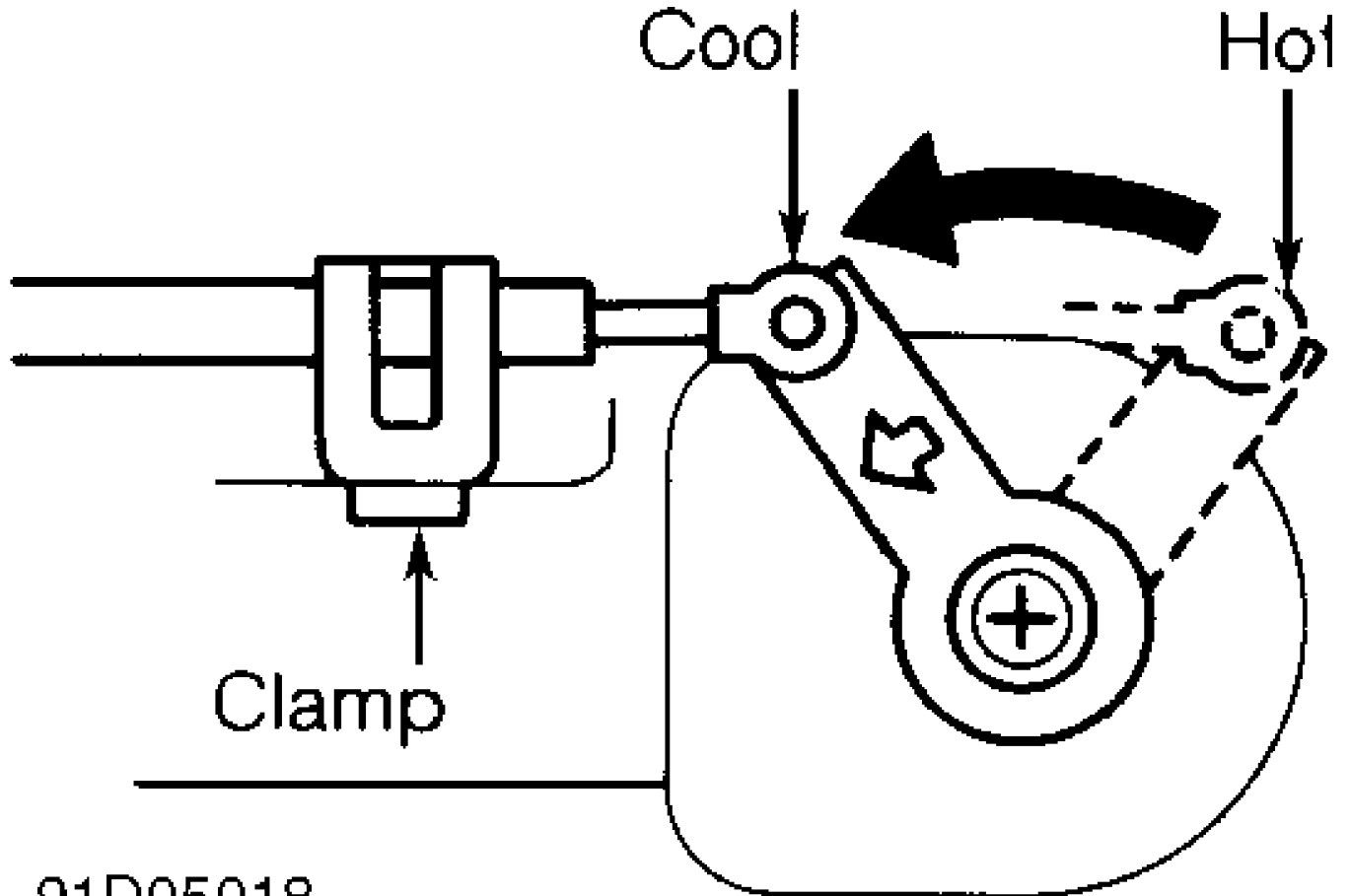


Fig. 3: Adjusting Air Mix Damper  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### HEATER VALVE



On servomotor type temperature control switch, turn ignition on, set temperature control switch to cool position, and turn blower motor on. Set heater valve and temperature control switch to cool position. See Fig. 4. Install control cable and clamp.



91D05018

Fig. 4: Adjusting Heater Valve Control Cable  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## **TROUBLE SHOOTING**

### **NO BLOWER OPERATION**

Problem may be due to blown fuses, faulty heater relay, or faulty blower motor or resistor. Also check for faulty blower fan relay or blower switch, faulty A/C-heater control panel, faulty wiring or bad ground.

### **NO COOL AIR**

Problem may be due to blown fuses, incorrect refrigerant charge, incorrect A/C compressor belt tension, faulty compressor (magnetic) clutch relay, or faulty pressure switch or A/C compressor. Check for plugged receiver-drier, condenser, expansion valve or evaporator. Also check for faulty A/C-heater control panel, faulty thermistor or A/C amplifier, faulty wiring or bad ground.

### **INTERMITTENT COOL AIR**

Problem may be caused by incorrect A/C compressor belt tension, incorrect refrigerant charge or faulty compressor clutch. Check for plugged expansion valve.

### **COOL AIR ONLY AT HIGH SPEED**

Problem may be caused by incorrect A/C compressor belt tension, incorrect refrigerant charge or faulty A/C compressor. Check for plugged condenser, faulty A/C fan motor or faulty fan relay.

### **INSUFFICIENT COOLING**

Problem may be caused by incorrect A/C compressor belt tension, incorrect refrigerant charge or A/C control cable out of adjustment. Check for clogged condenser, faulty A/C compressor, or faulty A/C fan or relay. Also check for faulty expansion valve, faulty A/C-heater control panel, faulty air mix servomotor or faulty wiring.

### **INSUFFICIENT COOL AIR VELOCITY**

Problem may be caused by faulty blower motor, blocked air inlet, clogged/frosted evaporator, or air leakage from evaporator case or air duct.

## **TESTING**

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

### **A/C SYSTEM PERFORMANCE**

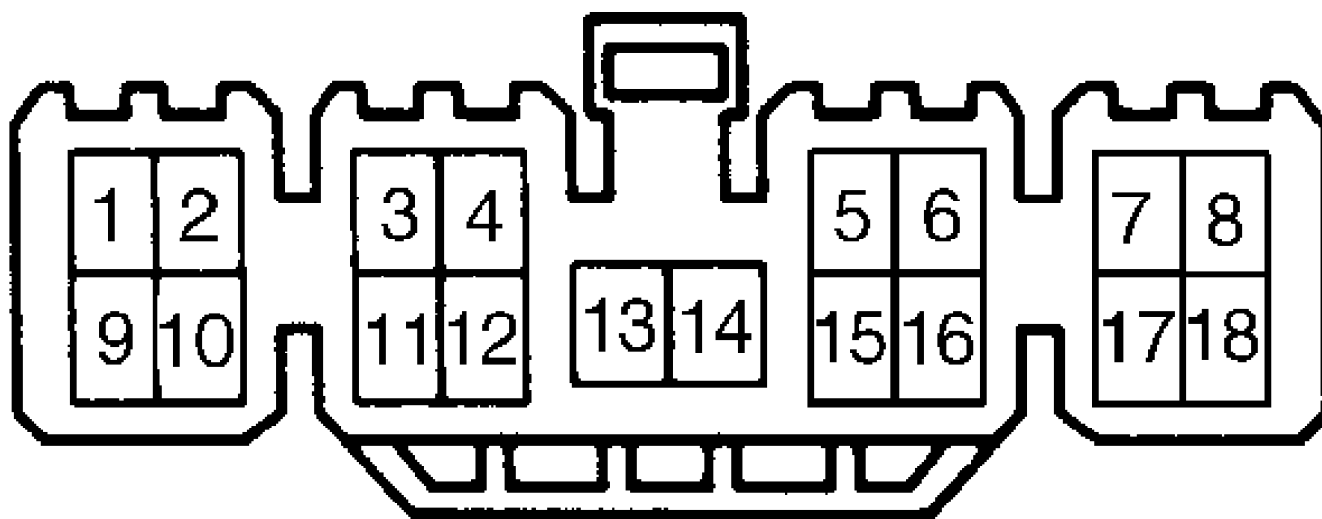
Connect manifold gauge set. Operate engine at 2000 RPM. Place blower fan on high speed. Place temperature control switch on maximum cooling. Set A/C-heater control panel in recirculated air mode. Ensure air inlet temperature is between 86-95°F (30-35°C). Ensure system operating pressures are within specification. See SPECIFICATIONS table at beginning of article.

### **A/C SWITCH**

Disconnect negative battery cable. Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Check continuity between terminals No. 14 and 16. There should be continuity when switch is on and no continuity when switch is off. If continuity is not as specified, replace A/C-heater control panel.

### **A/C AMPLIFIER**

Disconnect A/C amplifier connector. See Fig. 1. Test wiring harness side of connector. See Fig. 5. Ensure circuit tests as specified in A/C AMPLIFIER CIRCUIT TEST table. If circuit is not as specified, repair as necessary.



**91F05019**

Fig. 5: Identifying A/C Amplifier Harness Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### A/C AMPLIFIER CIRCUIT TEST TABLE

| Terminal No. & Test Condition (1)            | Specification       |
|----------------------------------------------|---------------------|
| 2 & Ground (2)                               |                     |
| Coolant Temp. Less Than 203°F (95°C) .....   | Continuity          |
| Coolant Temp. Greater Than 212°F (100°C) ... | No Continuity       |
| 15 & Ground .....                            | Continuity          |
| 10 & Ground                                  |                     |
| 3S-GTE (2.0L) .....                          | No Continuity       |
| 5S-FE (2.2L) .....                           | Continuity          |
| 5 & Ground (2) .....                         | Approx. 12 Ohms     |
| 9 & 14 .....                                 | Approx. 115 Ohms    |
| 16 & 14                                      |                     |
| Air Temp. @ 77°F (25°C) .....                | Approx. 15,000 Ohms |
| 1 & Ground                                   |                     |
| A/C Switch On .....                          | Voltage             |
| A/C Switch Off .....                         | No Voltage          |
| 3 & Ground                                   |                     |
| A/C Switch On .....                          | Battery Voltage     |
| A/C Switch Off .....                         | No Voltage          |
| 6 & Ground .....                             | Battery Voltage     |
| 8 & Ground .....                             | Battery Voltage     |
| 13 & Ground .....                            | Battery Voltage     |
| 18 & Ground                                  |                     |
| Start Engine .....                           | Approx. 10-14 Volts |
| Stop Engine .....                            | No Voltage          |

- (1) - Basic test condition: ignition on, temperature control in max. cool position and blower switch on HI position.  
(2) - Models with variable displacement compressor only.

#### BLOWER MOTOR

Disconnect wiring harness connector. Apply battery voltage to motor connector. See Fig. 6. Motor should operate smoothly. If motor does not operate smoothly, replace motor.

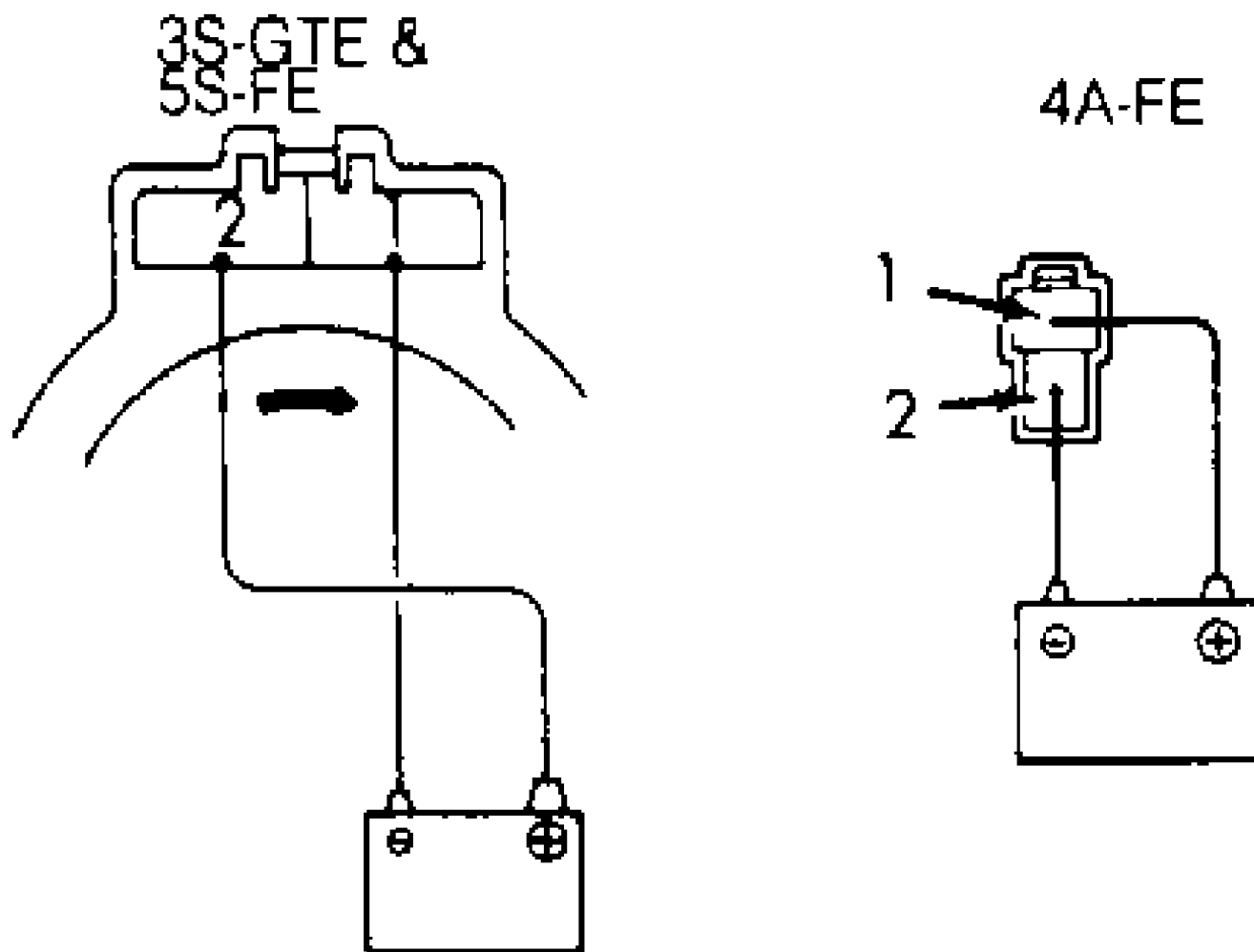


Fig. 6: Identifying Blower Motor Connector Terminals  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## COOLANT TEMPERATURE SWITCH

When coolant temperature reaches 212°F (100°C), switch opens and breaks continuity. When coolant temperature drops to approximately 203°F (95°C), switch closes and restores continuity. If switch does not function as specified, replace switch.

## CONDENSER FAN MOTOR

Disconnect fan motor 2-pin connector. Apply battery voltage to fan motor connector. Fan motor should rotate smoothly and current draw should be 6.0-7.4 amps. If operation is not as specified, replace condenser fan motor.

## INDICATOR & ILLUMINATION LIGHTS

A/C Indicator

Disconnect harness connector "A" of A/C-heater control panel. See Fig. 2. Apply battery voltage to terminal No. 1, and ground

terminal No. 15. Depress A/C switch and ensure indicator light comes on. If indicator light does not come on, replace A/C-heater control panel.

#### Air Inlet Indicator

Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Apply battery voltage to terminal No. 1, and ground terminal No. 2. Ensure recirculated and fresh air indicator lights come on each time air inlet control switch is pressed. If operation is not as specified, replace A/C-heater control panel.

#### Illumination Light Operation

Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Apply battery voltage to terminal No. 18, and ground terminal No. 17. Illumination light should come on. If illumination light does not come on, remove and check bulb.

#### Indicator Light Dimming Operation

Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Apply battery voltage to terminal No. 1, and ground terminal No. 2. Apply battery voltage to terminal No. 3. Indicator lights should dim. If operation is not as specified, replace A/C-heater control panel.

#### Mode Indicator

Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Apply battery voltage to terminal No. 1, and ground terminal No. 2. Press each mode button and ensure proper indicator light comes on. If any light fails to come on, replace A/C-heater control panel.

## MAGNETIC CLUTCH

1) Inspect pressure plate and rotor for oil contamination. Check clutch bearing for noisy operation and grease leakage. Using an ohmmeter, measure resistance of stator coil between compressor connector terminal No. 1 and ground. See Fig. 7.

2) Resistance should be 3.4–3.8 ohms at 68°F (20°C). If reading is not as specified, replace stator coil. Apply 12 volts to coil side of A/C compressor connector. If clutch does not energize, replace stator coil.

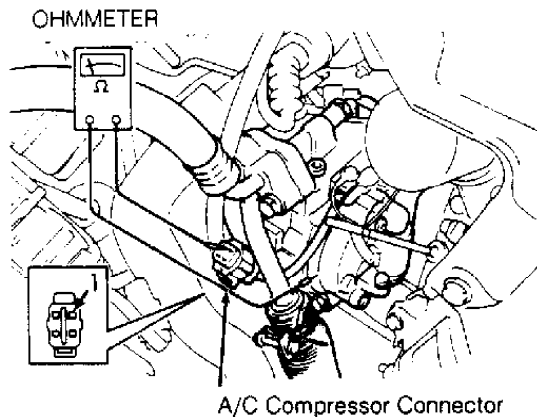


Fig. 7: Identifying A/C Compressor Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## RELAYS

#### Magnetic Clutch Relay

Remove relay. Using an ohmmeter, ensure continuity exists between terminals No. 1 and 2. See Fig. 8. Apply battery voltage to terminals No. 1 and 2. Continuity should exist between terminals No. 3 and 4. If continuity is not as specified, replace relay.

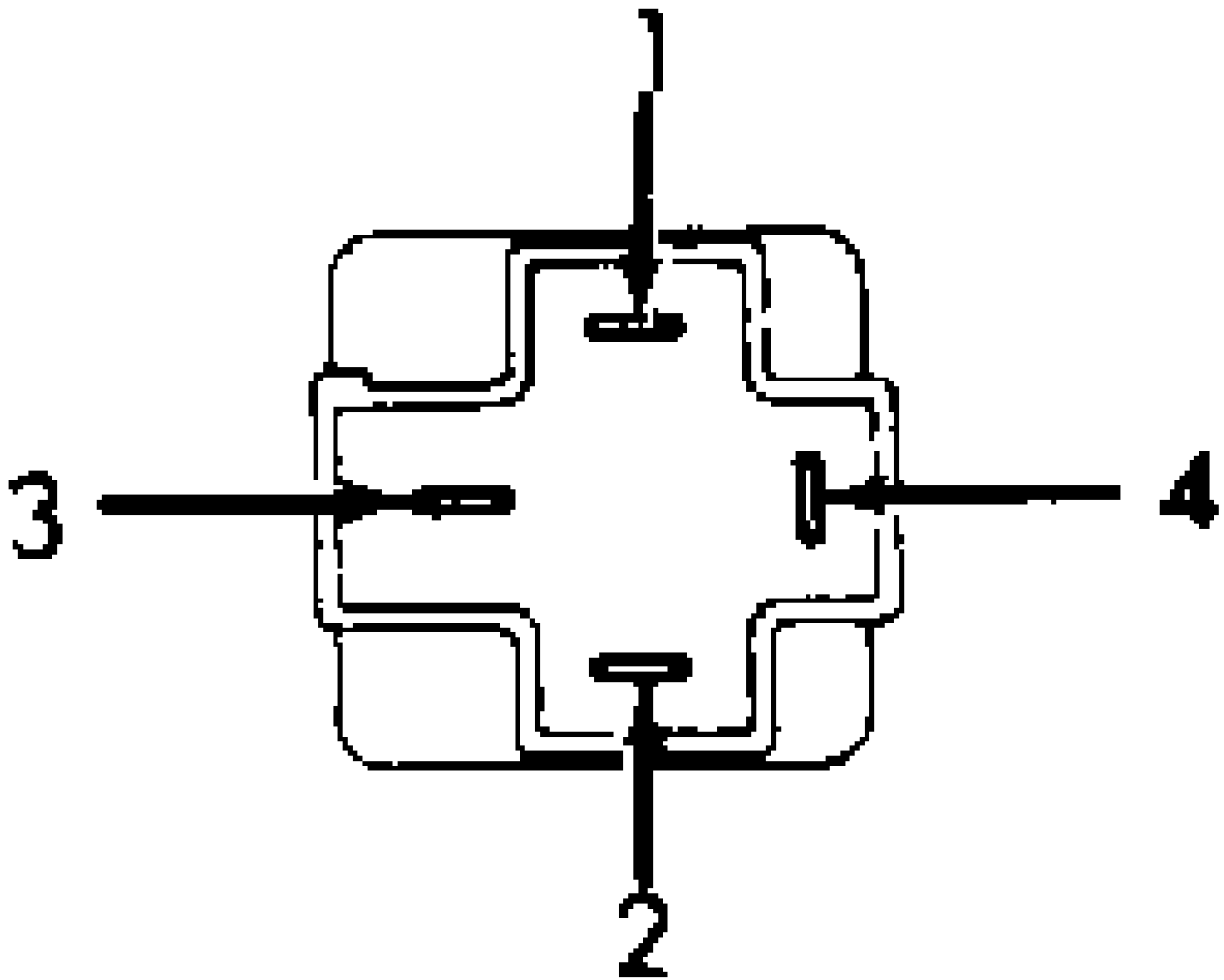


Fig. 8: Identifying Magnetic Clutch Relay Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Fan Relay No. 1

Remove relay. Using ohmmeter, ensure continuity exists between terminals No. 1 and 2 and terminals No. 3 and 4. See Fig. 9. Apply battery voltage between terminals No. 1 and 2. Continuity should not exist between terminals No. 3 and 4. If continuity is not as

specified, replace relay.

Fan Relay No. 2

Remove relay. Using ohmmeter, check continuity between terminals No. 1 and 2 and terminals No. 3 and 4. See Fig. 9. If continuity does not exist, replace relay. Apply battery voltage between terminals No. 1 and 2. Check continuity between terminals No. 3 and 5. If continuity does not exist, replace relay.

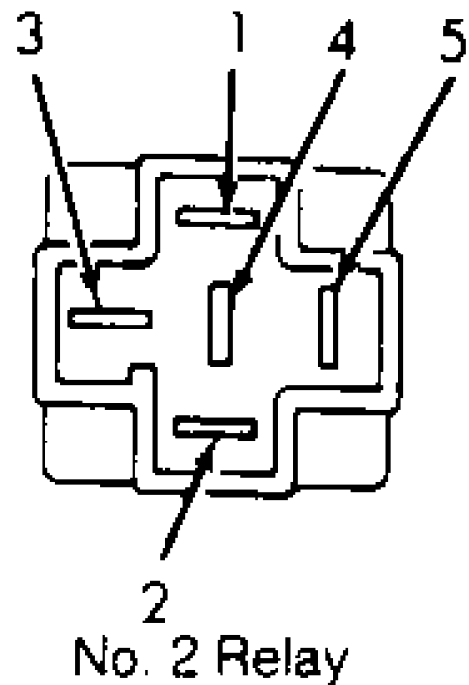
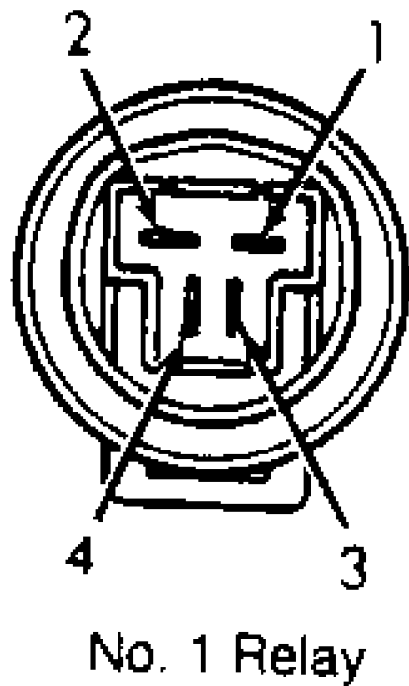


Fig. 9: Identifying A/C Fan Relay Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

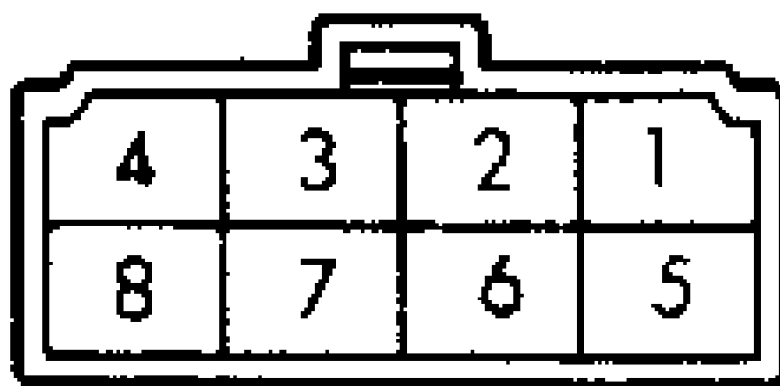
Blower Fan Relay

Disconnect blower fan relay wiring harness connector. Test relay as specified in TESTING BLOWER FAN RELAY table. See Fig. 10. If relay fails any test, replace relay.

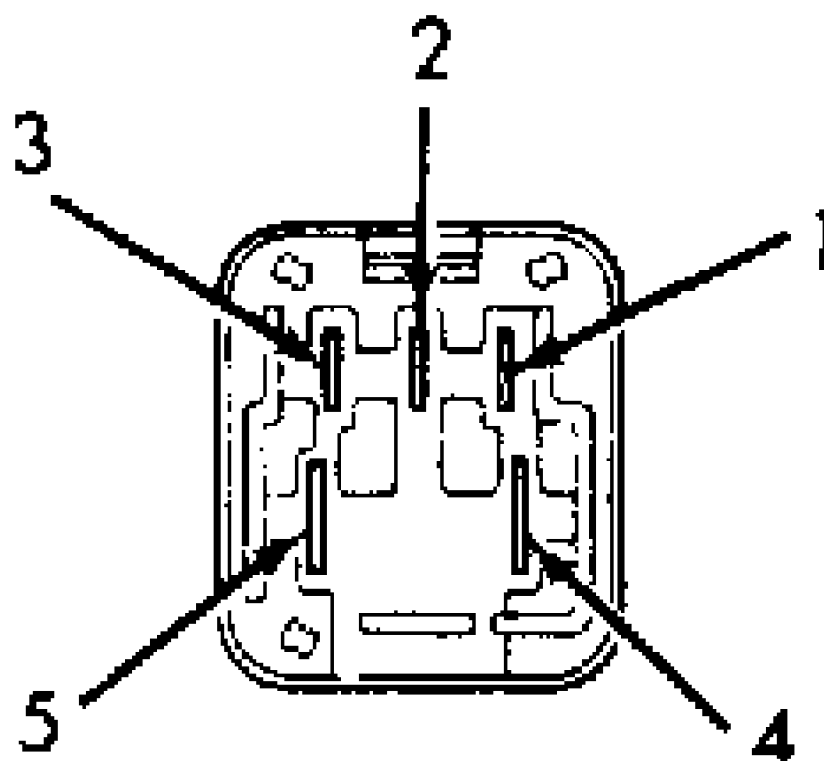
TESTING BLOWER FAN RELAY TABLE

| Terminal No. & Test Condition | Specification                         |
|-------------------------------|---------------------------------------|
| 5 & 6, 5 & 7, 5 & 8 .....     | Constant Continuity                   |
| 5 & 6 (1) .....               | Continuity Between Terminal No. 1 & 3 |
| 5 & 7 (1) .....               | Continuity Between Terminal No. 3 & 4 |
| 5 & 8 (1) .....               | Continuity Between Terminal No. 2 & 3 |

(1) - Apply battery voltage between terminals specified.



## BLOWER FAN RELAY CONNECTOR



## HEATER MAIN RELAY CONNECTOR

Fig. 10: A/C Blower Fan & Heater Main Relay Connector Terminal ID  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

Heater Main Relay  
 Disconnect heater main relay wiring harness connector. Test



relay as specified in TESTING HEATER MAIN RELAY table. See Fig. 10. If relay fails any test, replace relay.

#### TESTING HEATER MAIN RELAY TABLE

| Terminal No. & Test Condition | Specification                         |
|-------------------------------|---------------------------------------|
| 1 & 3, 2 & 4 .....            | Constant Continuity                   |
| 1 & 3 (1) .....               | Continuity Between Terminal No. 4 & 5 |

(1) - Apply battery voltage between terminals specified.

#### RPM SENSOR

Disconnect wiring harness connector from A/C compressor. Using an ohmmeter, measure resistance between RPM sensor terminals No. 2 and 3. See Fig. 11. Resistance should be 100-130 ohms at 68°F (20°C). If resistance is not within specification, replace RPM sensor.

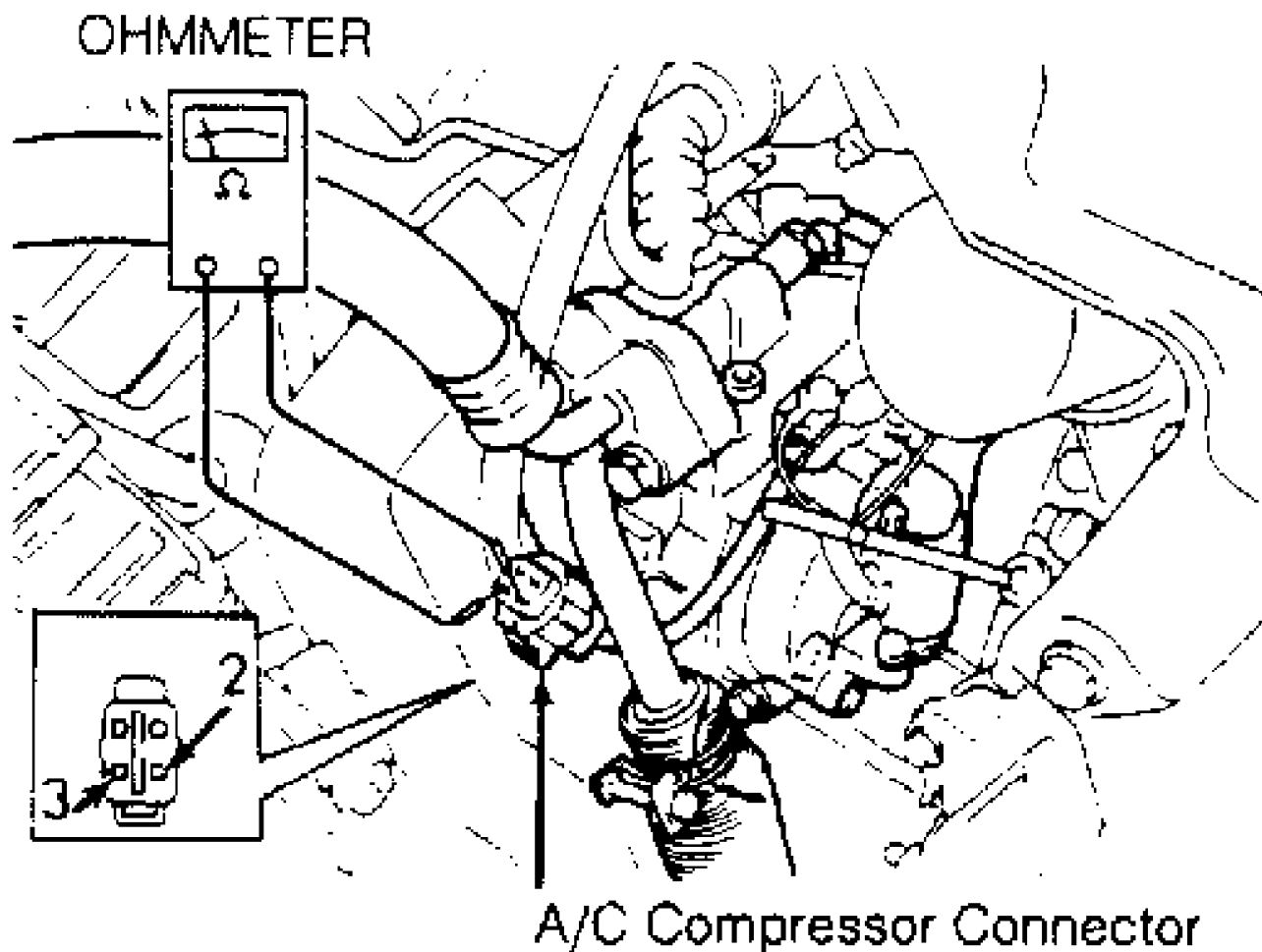


Fig. 11: Identifying RPM Sensor Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### SERVOMOTORS

Air Inlet Control Servomotor

1) Disconnect servomotor connector. See Fig. 1. Apply battery voltage to terminal No. 1, and ground terminal No. 3. See Fig. 12. Ensure arm rotates smoothly to fresh air side.

2) Apply battery voltage to terminal No. 1, and ground terminal No. 2. See Fig. 12. Ensure arm rotates smoothly to recirculated air side. If operation is not as specified, replace servomotor.

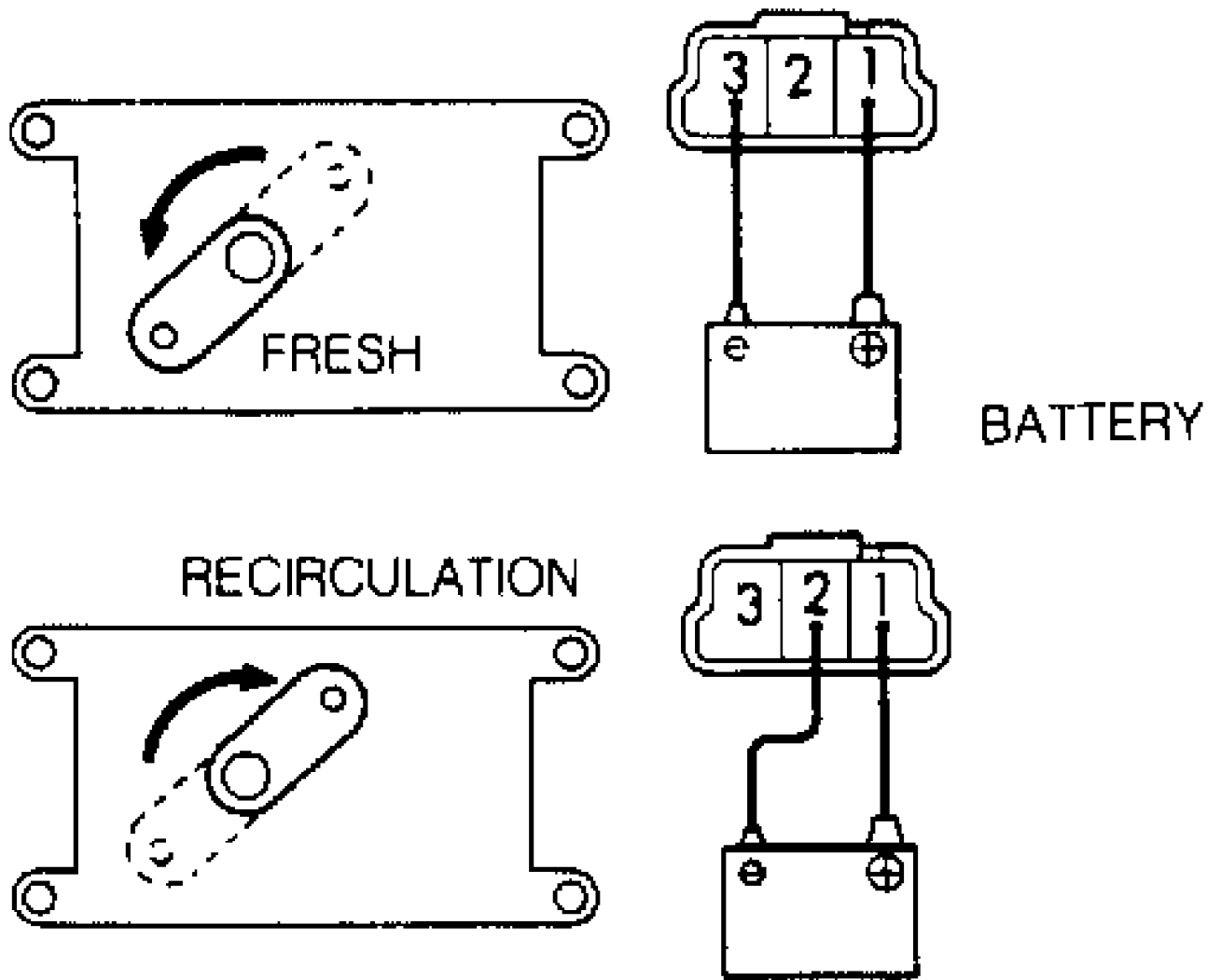


Fig. 12: Testing Air Inlet Control Servomotor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Airflow Mode Control Servomotor

Disconnect servomotor connector. See Fig. 1. Apply battery voltage to terminal No. 5, and ground terminal No. 6. See Fig. 13. Ground each specified terminal and ensure arm rotates smoothly to correct position. See TESTING AIRFLOW CONTROL SERVOMOTOR table.

#### TESTING AIRFLOW CONTROL SERVOMOTOR TABLE

| Ground Terminal No. | Arm Position |
|---------------------|--------------|
| 1                   | Vent         |
| 2                   | Bi-Level     |

|   |       |              |
|---|-------|--------------|
| 3 | ..... | Foot 2       |
| 4 | ..... | Foot/Defrost |
| 7 | ..... | Defrost      |
| 8 | ..... | Foot 1       |

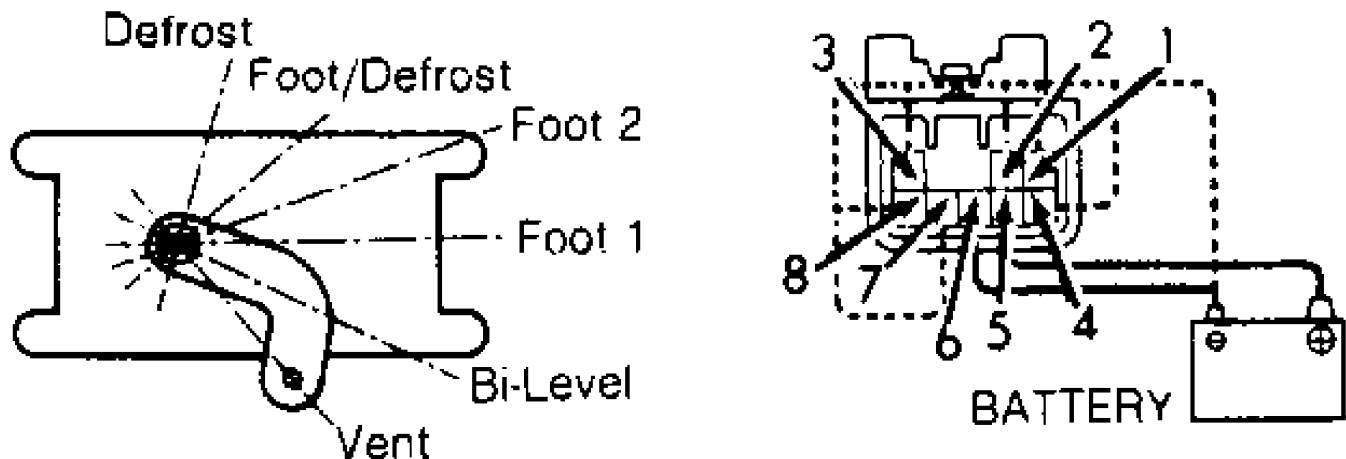


Fig. 13: Testing Airflow Mode Control Servomotor  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## SWITCHES

### Air Inlet Control Switch

- 1) Disconnect wire harness connector "A" of A/C-heater control panel. See Fig. 2. With recirculated air button depressed, continuity should exist between terminals No. 2 and 7.
- 2) With fresh air button depressed, continuity should exist between terminals No. 2 and 8. Switch contains diodes. Check continuity in both directions before assuming switch is faulty. Continuity should exist in one direction only. If continuity does not exist or exists in both directions, replace A/C-heater control panel.

### Blower Speed Control Switch

Disconnect wire harness connector "B" of A/C-heater control panel. See Fig. 2. Check continuity at specified terminals. See TESTING BLOWER SPEED CONTROL SWITCH table. If continuity is not as specified, replace A/C-heater control panel.

### TESTING BLOWER SPEED CONTROL SWITCH TABLE

| Switch Position | Continuity Between Terminals |
|-----------------|------------------------------|
| OFF .....       | No Continuity                |
| LO .....        | 1, 3 & 7                     |
| (f) (1) .....   | 2, 3 & 6                     |
| (f) (2) .....   | 2, 3 & 4                     |
| HI .....        | 2, 3 & 5                     |

- (1) - Square (f) closest to LO switch position.  
 (2) - Square (f) closest to HI switch position.

### Mode Control Switch

Disconnect wiring harness connector "A" of A/C-heater control panel. See Fig. 2. Check continuity at specified terminals. See

TESTING MODE CONTROL SWITCH table. If continuity is not as specified, replace A/C-heater control panel.

TESTING MODE CONTROL SWITCH TABLE

| Switch Position    | Continuity Between Terminals |
|--------------------|------------------------------|
| Face .....         | 2 & 9                        |
| Bi-Level .....     | 2 & 10                       |
| Foot .....         | 2 & 11                       |
| Foot/Defrost ..... | 2 & 12                       |
| Defrost .....      | 2 & 13                       |

## THERMISTOR

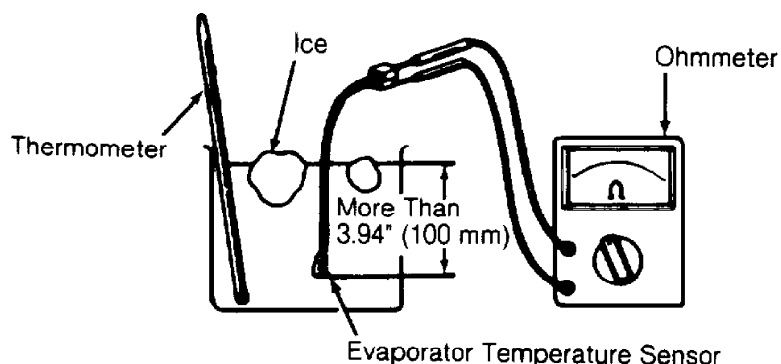
1) Disconnect negative battery cable. Remove lower trim panel and glove box. Check thermistor installed operation. Using an ohmmeter, measure resistance of thermistor. Resistance should be 1500 ohms at 77°F (25°C). If resistance is not as specified, go to next step.

2) Remove evaporator case. See EVAPORATOR ASSEMBLY under REMOVAL & INSTALLATION. Disassemble evaporator case, and remove thermistor. Check thermistor operation. Place thermistor at least 3.94" (100 mm) deep in cold water.

3) Measure resistance at connector while measuring temperature of water using thermometer. See Fig. 14. Compare readings with THERMISTOR RESISTANCE VALUES table. Use ice or hot water to change temperature of water. If readings are not within specification, replace thermistor.

THERMISTOR RESISTANCE VALUES TABLE

| Temperature °F (°C) | Ohms      |
|---------------------|-----------|
| 41 (5) .....        | 3500-4100 |
| 39 (4) .....        | 3800-4300 |
| 37 (3) .....        | 3900-4500 |
| 36 (2) .....        | 4100-4800 |
| 34 (1) .....        | 4300-4900 |
| 32 (0) .....        | 4500-5200 |
| 30 (-1) .....       | 4700-5400 |



36420  
Fig. 14: Testing Thermistor Resistance  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TRIPLE-PRESSURE SWITCH

#### Magnetic Clutch Control

1) Install A/C manifold gauges. Start engine, turn A/C on and observe gauge readings. If low side pressure is less than 30 psi (2.1 kg/cm<sup>2</sup>), system pressure is too low and pressure switch (terminals No. 3 and 4) should be open and clutch disengaged.

2) If low side pressure is greater than 33 psi (2.4 kg/cm<sup>2</sup>), switch should be closed and clutch engaged. If pressure drops below 33 psi (2.4 kg/cm<sup>2</sup>), switch will open and disengage clutch.

3) During system operation, switch opens and disengages clutch if high side pressure becomes greater than 384 psi (27 kg/cm<sup>2</sup>). Switch remains open until high side pressure drops to less than 299 psi (21 kg/cm<sup>2</sup>). If switch does not operate as specified, replace switch. If switch is okay, proceed to ELECTRIC FAN CONTROL.

#### Electric Fan Control

If high side pressure becomes greater than 220 psi (15.5 kg/cm<sup>2</sup>), switch closes and condenser fan operates at high speed. When pressure drops to approximately 178 psi (12.5 kg/cm<sup>2</sup>), switch opens (terminals No. 1 and 2) and condenser fan returns to low speed operation. If switch does not function as specified, replace switch.

### VACUUM SWITCHING VALVE (VSV)

1) Remove vacuum connections from fittings "A" and "B" on VSV. See Fig. 15. Connect VSV terminal connector to battery. Blow air through fitting "A". Air should pass from fitting "A" out through fitting "B". Air should not be felt at "C" (atmospheric port).

2) Disconnect battery from VSV terminal connector. Blow air through fitting "A". Air should pass from fitting "A" out through "C" (atmospheric port). Air should not come out of fitting "B".

3) Use ohmmeter to check for short between each terminal and VSV body. Also check resistance between terminals. Reading should be 38-43 ohms at 68°F (20°C).

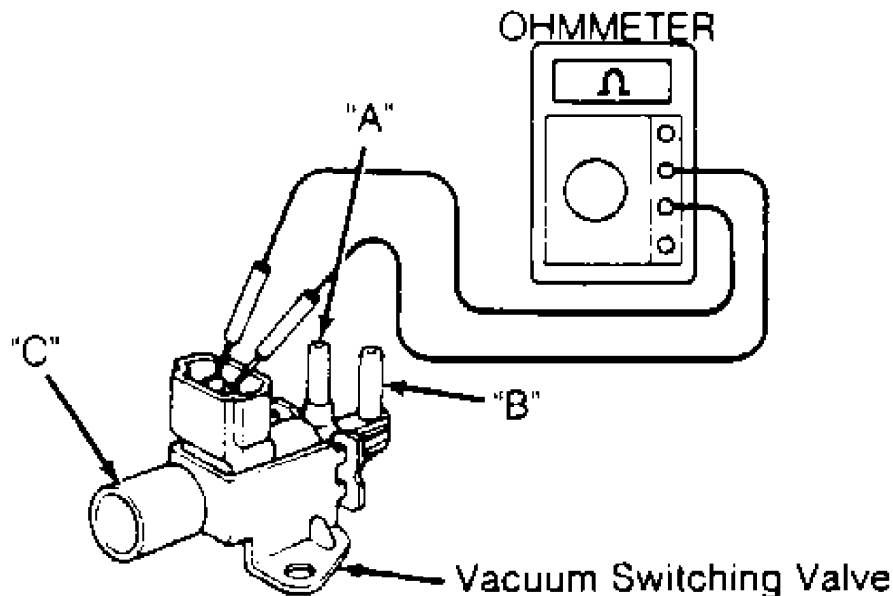


Fig. 15: Testing Vacuum Switching Valve  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### REMOVAL & INSTALLATION

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

## COMPRESSOR

### Removal

1) If possible, run system longer than 10 minutes before starting removal procedure. Disconnect battery cables and remove battery. Disconnect A/C wiring harness connectors. Discharge A/C system using approved refrigerant recovery/recycling equipment.

2) Disconnect hoses from service valves. Plug all openings. Loosen and remove compressor belt from pulley. Remove bolts and compressor.

### Installation

To install, reverse removal procedure. Evacuate, recharge and leak test system.

## EVAPORATOR ASSEMBLY

### Removal

Disconnect negative battery cable. Discharge A/C system using approved refrigerant recovery/recycling equipment. Disconnect inlet and outlet lines and grommets from evaporator. Plug openings. Disconnect electrical leads from evaporator. Remove glove box and reinforcement. Remove nuts, bolts and evaporator assembly.

### Disassembly

Release spring clips holding covers together. See Fig. 16. Remove any screws at case joints. Separate upper and lower cases from evaporator core. Remove thermistor with holder. Remove heat insulator from outlet tube. Remove high side (inlet) line from expansion valve, and remove expansion valve.

### Reassembly & Installation

To reassemble and install evaporator assembly, reverse disassembly and removal procedures. If installing new evaporator core, add 1.4-1.7 ounces of refrigerant oil to core before installing. Evacuate, recharge and leak test system.

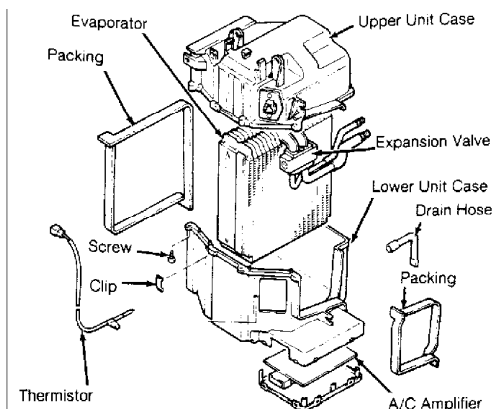


Fig. 16: Exploded View Of Evaporator Assembly  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## CONDENSER

### Removal

Discharge A/C system using approved refrigerant recovery/recycling equipment. Remove lower engine cover. Remove grille and hood lock brace. Remove horns. Detach lines from condenser. Plug all openings. Remove bolts and condenser.

### Installation

To install, reverse removal procedure. If installing new condenser, add 1.4-1.7 ounces of refrigeration oil before installing. Evacuate, recharge and leak test system.

## RECEIVER-DRIER

### Removal

Discharge A/C system using approved refrigerant recovery/recycling equipment. Remove lines from receiver-drier. Plug all openings. Remove bolts and receiver-drier.

### Installation

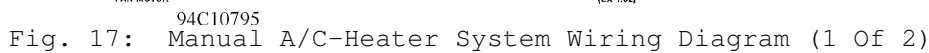
To install, reverse removal procedure. Add 0.7 ounce of refrigerant oil. Evacuate, recharge and leak test system.

## TORQUE SPECIFICATIONS

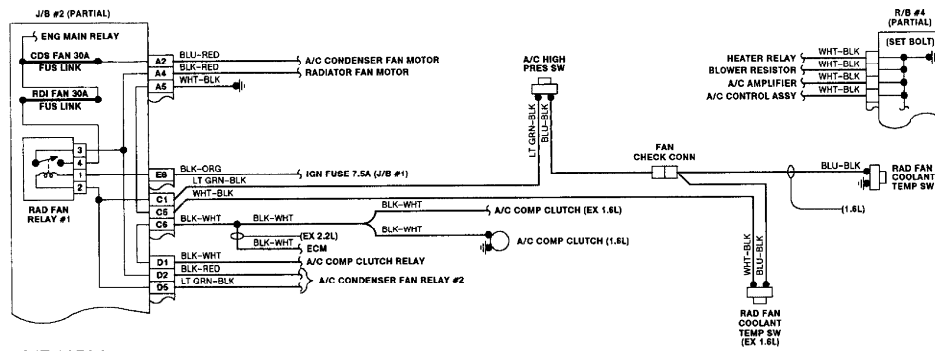
### TORQUE SPECIFICATIONS TABLE

| Application                         | Ft. Lbs. (N.m)  |
|-------------------------------------|-----------------|
| A/C Compressor .....                | 18 (25)         |
| A/C Compressor Bracket (1.6L) ..... | 35 (47)         |
| Condenser                           |                 |
| Discharge Hose .....                | 17 (23)         |
| Liquid Tube .....                   | 10 (14)         |
| Evaporator Suction Tube .....       | 24 (32)         |
|                                     | INCH Lbs. (N.m) |
| Evaporator Liquid Tube .....        | 115 (13)        |
| Expansion Valve Allen Bolt .....    | 48 (5.4)        |
| Receiver-Drier .....                | 115 (13)        |

## WIRING DIAGRAMS







94D10796  
Fig. 18: Manual A/C-Heater System Wiring Diagram (2 Of 2)

# A/C-HEATER SYSTEM - AUTOMATIC

1993 Toyota Celica

1993 Automatic A/C-Heater Systems

Celica

## SPECIFICATIONS

### SPECIFICATIONS TABLE

| Application                                                                                        | Specification                               |
|----------------------------------------------------------------------------------------------------|---------------------------------------------|
| Compressor Type                                                                                    |                                             |
| 1.6L .....                                                                                         | Nippondenso 10PA15C 10-Cyl.                 |
| 2.0L & 2.2L .....                                                                                  | Nippondenso 10PA17C/VC 10-Cyl.              |
| Compressor Belt Tension (1)                                                                        |                                             |
| 1.6L                                                                                               |                                             |
| New .....                                                                                          | 135-185 Lbs. (61.2-83.9 kg)                 |
| Used .....                                                                                         | 80-120 Lbs. (36.3-54.4 kg)                  |
| 2.0L                                                                                               |                                             |
| New .....                                                                                          | 155-175 Lbs. (70.3-79.4 kg)                 |
| Used .....                                                                                         | 69-99 Lbs. (31.3-44.9 kg)                   |
| 2.2L                                                                                               |                                             |
| New .....                                                                                          | 155-175 Lbs. (70.3-79.4 kg)                 |
| Used .....                                                                                         | 100-120 Lbs. (45.4-54.4 kg)                 |
| System Oil Capacity .....                                                                          | 3.4-4.1 ozs.                                |
| Refrigerant (R-12) Capacity .....                                                                  | 24-27 ozs.                                  |
| System Operating Pressures (2)                                                                     |                                             |
| High Side .....                                                                                    | 206-213 psi (14.5-15.0 kg/cm <sup>2</sup> ) |
| Low Side .....                                                                                     | 21-28 psi (1.5-2.0 kg/cm <sup>2</sup> )     |
| (1) - Using a belt tension gauge, measure at longest run of belt.                                  |                                             |
| (2) - Specification is with ambient temperature at 86-95°F (30-35°C) and engine speed at 2000 RPM. |                                             |

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES/SAFETY EQUIPMENT section.

**CAUTION:** When battery is disconnected, radio will go into anti-theft protection mode. Obtain radio anti-theft protection code from owner prior to servicing vehicle.

## DESCRIPTION

Automatic temperature control system is a cycling clutch type with an expansion valve. See Fig. 1.

## OPERATION

### AMPLIFIERS & SENSORS

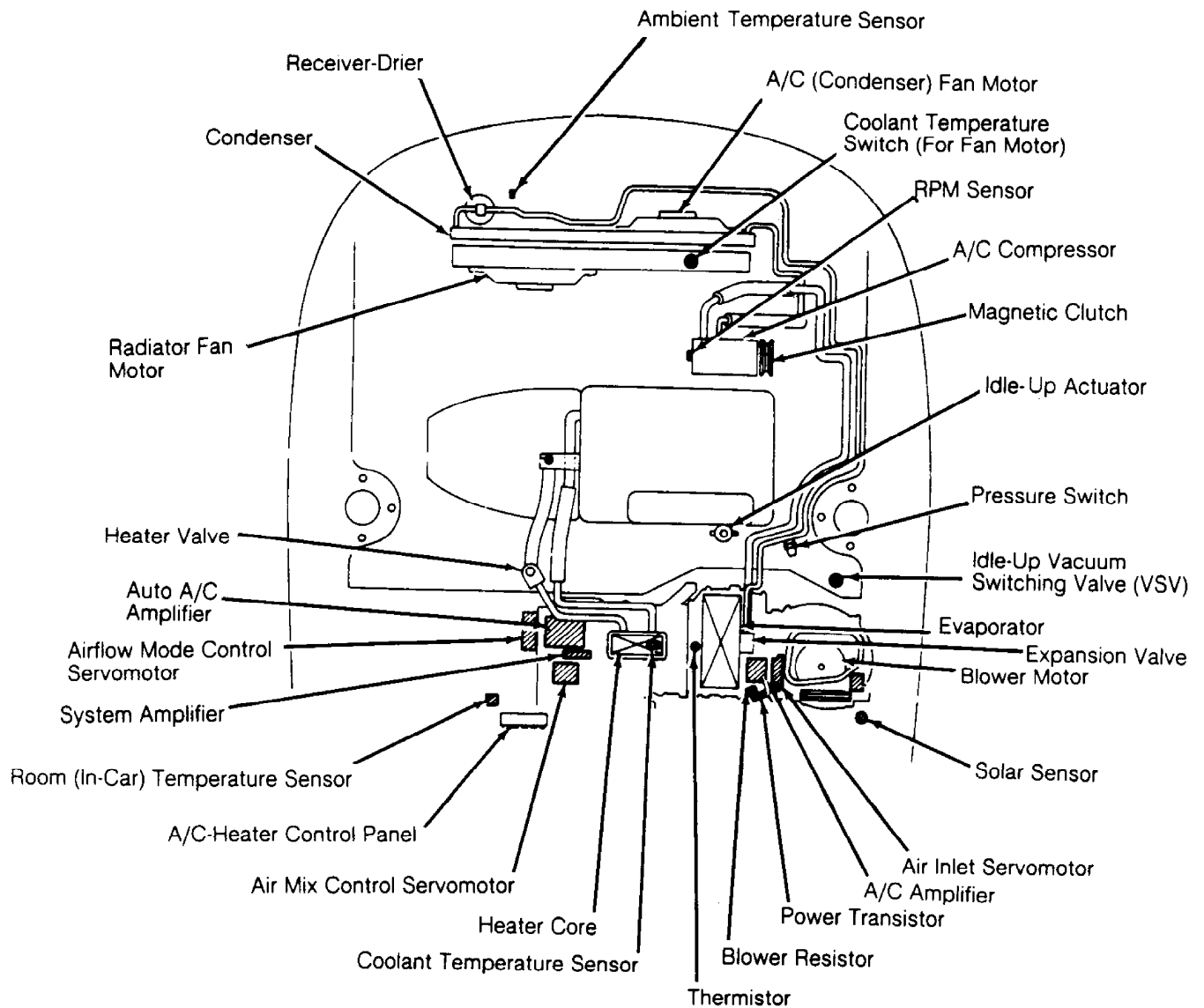
Amplifiers monitor system conditions through sensors. Based on signals from sensors, amplifiers control operation of compressor clutch and air control servomotors.

Ambient temperature sensor monitors outside air temperature.

Coolant temperature sensor monitors engine coolant temperature. Room (in-car) temperature sensor monitors passenger compartment air temperature. Solar (sunload) sensor monitors sunlight load. Thermistor monitors evaporator temperature. RPM sensor monitors compressor speed.

## ENGINE IDLE-UP CONTROL

When A/C compressor clutch is engaged, Vacuum Switching Valve (VSV) solenoid is energized, allowing vacuum to idle-up actuator. See Fig. 1. Idle-up actuator rod opens throttle slightly, increasing engine speed.



91E04873

Fig. 1: Locating Automatic A/C-Heater System Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TESTING

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES/SAFETY EQUIPMENT section.

**NOTE:** For testing of components not listed in this article, see A/C-HEATER SYSTEM - MANUAL article in the AIR CONDITIONING & HEAT section.

## A/C SYSTEM PERFORMANCE

Start engine and allow it to idle at 2000 RPM. Turn A/C on. Select recirculated air. Ensure temperature of inlet air is 86-95°F (30-35°C). Turn blower to highest speed. Select maximum cool temperature setting. Allow several minutes for system to stabilize. Ensure high side and low side pressures are within specification. See SPECIFICATIONS table at beginning of article.

## A/C COMPRESSOR CLUTCH TEST

1) Ensure compressor clutch pressure plate, rotor and bearings are okay, and air gap is .002-.006" (.05-.15 mm). Disconnect compressor connector. Check resistance between Black/White wire terminal of compressor connector and compressor body (ground).

2) If resistance is not 3.4-3.8 ohms at 68°F (20°C), replace clutch coil. If resistance is as specified, connect positive battery lead to Black/White wire terminal, and negative lead to compressor body (ground). Clutch is okay if it engages. Replace clutch if it does not engage.

## A/C-HEATER CONTROL PANEL ILLUMINATION

Disconnect negative battery cable. Disconnect A/C-heater control panel connector "A". See Fig. 2. Connect positive battery lead to terminal No. 18, and negative lead to terminal No. 17 of connector "A". If illumination lights DO NOT come on, check bulbs. If bulbs are okay, replace A/C-heater control panel.

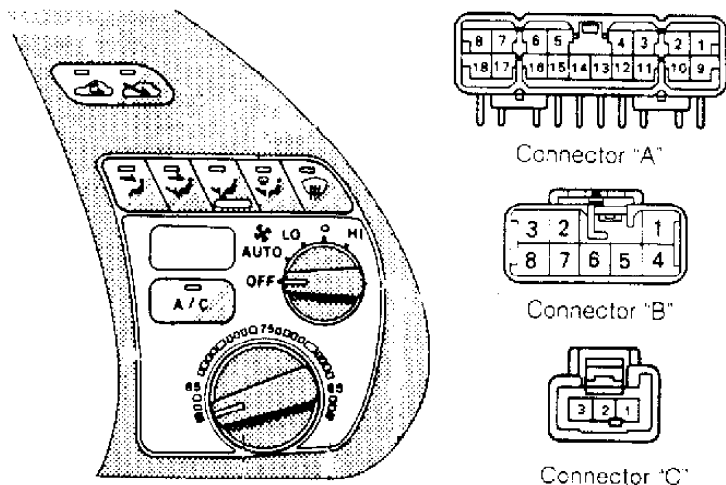


Fig. 2: Identifying A/C-Heater Control Panel Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## A/C-HEATER CONTROL PANEL INDICATORS

#### Air Inlet & Airflow Mode Indicators Test

Disconnect negative battery cable. Disconnect A/C-heater control panel connector "A". See Fig. 2. Connect positive battery lead to terminal No. 1, and negative lead to terminal No. 2 of connector "A". If indicator lights on air inlet (fresh and recirculated) buttons and airflow mode buttons DO NOT come on when respective button is pressed, replace A/C-heater control panel.

#### A/C Switch Indicator Test

Disconnect negative battery cable. Disconnect A/C-heater control panel connector "A". See Fig. 2. Connect positive battery lead to terminal No. 1, and negative lead to terminal No. 15 of connector "A". Turn A/C switch on. If A/C switch indicator light does not come on, replace A/C-heater control panel.

#### Indicator Light Dimming Test

Disconnect negative battery cable. Disconnect connector "A" from A/C-heater control panel. See Fig. 2. Connect positive battery lead to terminal No. 1, and negative lead to terminal No. 2 of connector "A". Connect another positive battery lead to terminal No. 3. If indicator lights DO NOT dim, replace A/C-heater control panel.

### A/C SWITCH TEST

Disconnect negative battery cable. Disconnect connector "A" from A/C-heater control panel. See Fig. 2. Check continuity between terminals No. 14 and 16 of connector "A". With A/C switch turned off, there should be no continuity. With A/C switch turned on, there should be continuity. If continuity is not as specified, replace A/C-heater control panel.

### AIR INLET SWITCHES TEST

NOTE: Air inlet switches contain diodes. Check continuity in both directions (polarity) before assuming switch is faulty.

Disconnect connector "A" from A/C-heater control panel. See Fig. 2. With recirculated air button pressed, continuity should exist between terminals No. 2 and 7 of connector "A". With fresh air button pressed, continuity should exist between terminals No. 2 and 8. If continuity is not as specified, replace A/C-heater control panel.

### AIRFLOW MODE CONTROL SWITCHES TEST

Disconnect connector "A" from A/C-heater control panel. See Fig. 2. Check continuity between specified terminals of connector "A". See AIRFLOW MODE CONTROL SWITCHES CONTINUITY TEST table. If continuity is not as specified, replace A/C-heater control panel.

#### AIRFLOW MODE CONTROL SWITCHES CONTINUITY TEST TABLE

| Switch Position    | Continuity Between Terminals |
|--------------------|------------------------------|
| Face .....         | 2 & 9                        |
| Bi-Level .....     | 2 & 10                       |
| Foot .....         | 2 & 11                       |
| Foot/Defrost ..... | 2 & 12                       |
| Defrost .....      | 2 & 13                       |

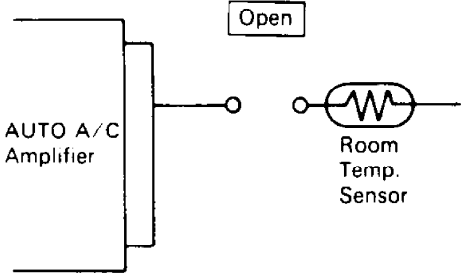
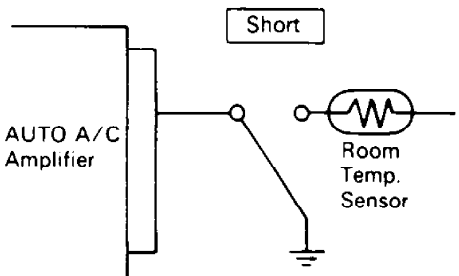
### A/C AMPLIFIER CIRCUIT TEST

Disconnect A/C amplifier connector. Turn ignition on. Turn temperature control dial to maximum cool position. Turn blower switch to HI position. At harness side of A/C amplifier connector, check continuity, resistance and voltage at specified terminals. See A/C AMPLIFIER CIRCUIT TEST table. See Fig. 4. If continuity, resistance or voltage is not as specified, repair appropriate circuit.

A/C AMPLIFIER CIRCUIT TEST TABLE

| Terminals & Test Condition                 | Specification       |
|--------------------------------------------|---------------------|
| 2 & Ground (1)                             |                     |
| Coolant Temp. Less Than 203°F (95°C) ..... | Continuity          |
| Coolant Temp. Greater Than 212°F (100°C) . | No Continuity       |
| 15 & Ground .....                          | Continuity          |
| 10 & Ground                                |                     |
| 1.6L .....                                 | No Continuity       |
| 2.2L .....                                 | Continuity          |
| 5 & Ground (1) .....                       | Approx. 12 Ohms     |
| 9 & 14 .....                               | Approx. 115 Ohms    |
| 16 & 14                                    |                     |
| Air Temp. @ 77°F (25°C) .....              | Approx. 15,000 Ohms |
| 1 & Ground                                 |                     |
| A/C Switch On .....                        | Battery Voltage     |
| A/C Switch Off .....                       | No Voltage          |
| 3 & Ground                                 |                     |
| A/C Switch On .....                        | Battery Voltage     |
| A/C Switch Off .....                       | No Voltage          |
| 6 & Ground                                 |                     |
| A/C Switch On .....                        | No Voltage          |
| A/C Switch Off .....                       | Battery Voltage     |
| 8 & Ground .....                           | Battery Voltage     |
| 13 & Ground                                |                     |
| A/C Switch On .....                        | Battery Voltage     |
| A/C Switch Off .....                       | No Voltage          |
| 18 & Ground                                |                     |
| Engine Running .....                       | Approx. 10-14 Volts |
| Engine Off .....                           | No Voltage          |

(1) - Vehicles with variable displacement compressor only.

| False Signal | A                                                                                                                               | B                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Condition    | <p>Interior room temperature is very low.</p>  | <p>Interior room temperature is very high.</p>  |
| Your Work    | Disconnect room temperature sensor connector.                                                                                   | Disconnect room temperature sensor connector. Ground appropriate terminal of room temperature sensor female connector.            |

| System Main Parts               | False Signal | Motion                                                         |          |       |       |
|---------------------------------|--------------|----------------------------------------------------------------|----------|-------|-------|
| Air Mix Control Servomotor      | A            | Air mix control servo motor shaft moves towards max-hot side.  |          |       |       |
|                                 | B            | Air mix control servo motor shaft moves towards max-cool side. |          |       |       |
| Airflow Mode Control Servomotor |              | Airflow Mode Door Position                                     |          |       |       |
|                                 |              | VENT                                                           | BI-LEVEL | HEAT  | DEF   |
|                                 | A            | Close                                                          | Close    | Open  | Close |
|                                 | B            | Open                                                           | Close    | Close | Close |

| System Main Parts            | False Signal  | Motion                              |
|------------------------------|---------------|-------------------------------------|
| Blower Motor                 | A             | Blower motor rotates at high speed. |
|                              | B             |                                     |
| Heater Water Valve           | A             | OPEN                                |
|                              | B             | CLOSE                               |
|                              |               |                                     |
| Air Inlet Control Servomotor | FRE Switch ON | Fresh air is ventilated.            |
|                              | REC Switch ON | Recirculated air is ventilated.     |

93D19771

Fig. 3: Testing Auto A/C Amplifier  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

# Wiring Harness Connector

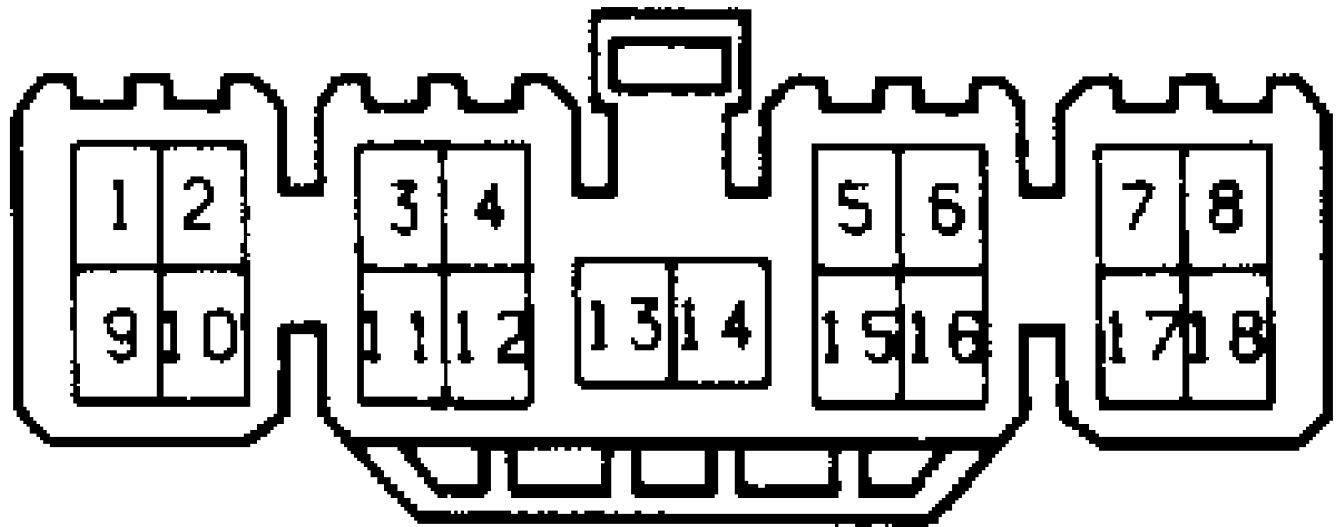


Fig. 4: Identifying A/C Amplifier Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## AIR MIX CONTROL SERVOMOTOR

### Servomotor

1) Disconnect air mix control servomotor wiring harness connector. Apply battery voltage to terminal No. 2, and ground terminal No. 6. See Fig. 5. Ensure arm rotates smoothly from hot to cool position.

2) Apply battery voltage to terminal No. 6, and ground terminal No. 2. Ensure arm rotates smoothly from cool to hot position. If operation is not as specified, replace servomotor.

### Position Sensor

Measure resistance between terminals No. 1 and 3. See Fig. 5. Reading should be about 6000 ohms. Set arm to cool position. While rotating arm to hot position, measure resistance between terminals No. 1 and 4. Resistance should decrease from about 4800 ohms to 1200 ohms. If readings are not as specified, replace air mix control servomotor.

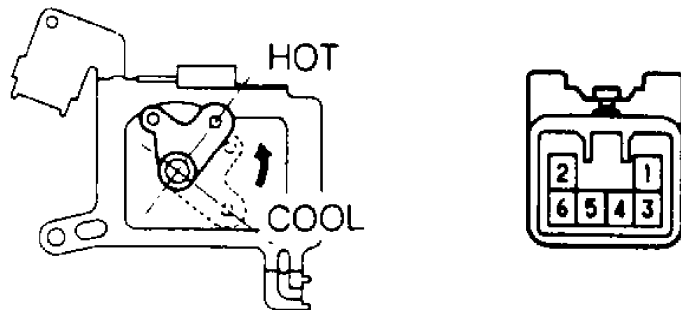


Fig. 5: Testing Air Mix Control Servomotor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## AUTO A/C AMPLIFIER TEST



1) Set temperature control dial to 77°F (25°C). Disconnect room temperature sensor connector. This simulates very low room temperature (FALSE SIGNAL "A"). See Fig. 3. System should operate at maximum heating, and components indicated in chart should operate as specified.

2) Ground appropriate terminal of room temperature sensor connector. This simulates very high room temperature (FALSE SIGNAL "B"). System should operate at maximum cooling, and components indicated in chart should operate as specified.

3) If system and components operate as specified, replace room temperature sensor. If system and components DO NOT operate as specified, replace auto A/C amplifier.

## BLOWER SPEED CONTROL SWITCH TEST

Disconnect connector "B" from A/C-heater control panel. See Fig. 2. Check continuity between specified terminals of connector "B". See BLOWER SPEED CONTROL SWITCH CONTINUITY TEST table. If continuity is not as specified, replace A/C-heater control panel.

BLOWER SPEED CONTROL SWITCH CONTINUITY TEST TABLE

| Switch Position | Continuity Between Terminals |
|-----------------|------------------------------|
| OFF .....       | None                         |
| AUTO .....      | 1, 3 & 7                     |
| LO .....        | 2, 3 & 6                     |
| f .....         | 2, 3 & 4                     |
| HI .....        | 2, 3 & 5                     |

## TEMPERATURE CONTROL DIAL TEST

1) Disconnect connector "C" from A/C-heater control panel. See Fig. 2. Measure resistance between terminals No. 2 and 3 of connector "C". Resistance should be approximately 3000 ohms.

2) Measure resistance between terminals No. 1 and 3 while rotating temperature control dial from cool position to hot position. If resistance does not vary from zero to about 3000 ohms, replace A/C-heater control panel.

## AMBIENT TEMPERATURE SENSOR TEST

Disconnect electrical connector from ambient temperature sensor, located at front grille, near horns. Check resistance across ambient temperature sensor terminals. If resistance is not as specified, replace sensor. See AMBIENT TEMPERATURE SENSOR RESISTANCE TEST table.

AMBIENT TEMPERATURE SENSOR RESISTANCE TEST TABLE

| Temperature °F (°C) | Ohms |
|---------------------|------|
| 77 (25) .....       | 1700 |
| 122 (50) .....      | 620  |

## COOLANT TEMPERATURE SENSOR TEST

Remove coolant temperature sensor. Submerge sensing portion of sensor in container of water. Monitor water temperature with thermometer, and measure resistance between sensor terminals. See

COOLANT TEMPERATURE SENSOR RESISTANCE TEST table. Use ice or hot water to change temperature of water. Replace sensor if resistance is not as specified.

| COOLANT TEMPERATURE SENSOR RESISTANCE TEST TABLE |       |               |
|--------------------------------------------------|-------|---------------|
| Temperature °F (°C)                              |       | Ohms          |
| 34 (1)                                           | ..... | 15,000-19,000 |
| 104 (40)                                         | ..... | 2500-2700     |
| 158 (70)                                         | ..... | 800-1000      |

### ROOM (IN-CAR) TEMPERATURE SENSOR TEST

NOTE: If room temperature sensor circuit is open, system will operate at maximum heating. If circuit is shorted, system will operate at maximum cooling.

Disconnect electrical connector from room temperature sensor, located under driver knee bolster. Measure resistance across sensor terminals. If resistance is not as specified, replace the sensor. See ROOM TEMPERATURE SENSOR RESISTANCE TEST table.

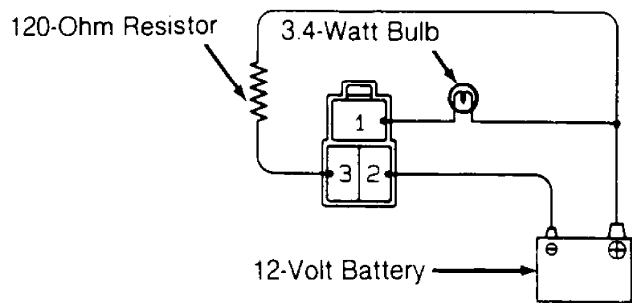
| ROOM TEMPERATURE SENSOR RESISTANCE TEST TABLE |       |      |
|-----------------------------------------------|-------|------|
| Temperature °F (°C)                           |       | Ohms |
| 77 (25)                                       | ..... | 1700 |
| 122 (50)                                      | ..... | 620  |

### RPM SENSOR TEST

Disconnect compressor connector. Measure resistance between White/Blue wire and White/Red wire terminals of compressor connector. If resistance is not 100-130 ohms at 68°F (20°C), replace RPM sensor.

### POWER TRANSISTOR TEST

Disconnect power transistor connector. Connect battery, 120-ohm resistor and 3.4-watt light bulb to terminals as shown. See Fig. 5. If light bulb does not come on, replace power transistor.



91B04876

Fig. 6: Testing Power Transistor  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

### SOLAR (SUNLOAD) SENSOR TEST

Disconnect solar sensor connector. Check continuity across solar sensor terminals. If there is no continuity, replace solar sensor.

## SYSTEM AMPLIFIER TEST

1) Disconnect system amplifier connector. Turn ignition on. Turn temperature control dial to maximum cool position. At system amplifier connector, check continuity and voltage at specified terminals. See SYSTEM AMPLIFIER TEST (PART 1 OF 2) table. See Fig. 7.

SYSTEM AMPLIFIER TEST (PART 1 OF 2) TABLE

| Terminal No.     | Specified Value |
|------------------|-----------------|
| 1 & 6 .....      | Continuity      |
| 9 & Ground ..... | Continuity      |
| 7 & Ground ..... | Battery Voltage |

2) If continuity and voltage are not as specified, repair appropriate circuit. If continuity and voltage are as specified, turn off ignition. Reconnect system amplifier connector. Start engine. Turn A/C on. Turn blower switch to HI position.

3) Set temperature dial in appropriate position, and check voltage at specified terminals of system amplifier connector (backprobe terminals). See SYSTEM AMPLIFIER TEST (PART 2 OF 2) table. If voltage is as specified, system amplifier is okay. If voltage is not as specified, replace system amplifier.

SYSTEM AMPLIFIER TEST (PART 2 OF 2) TABLE

| Terminals & Test Condition | Specification             |
|----------------------------|---------------------------|
| 3 & Ground                 |                           |
| Maximum Hot .....          | Approximately Zero Volts  |
| Maximum Cool .....         | (1) Approximately 5 Volts |
| 4 & Ground                 |                           |
| Maximum Hot .....          | (1) Approximately 5 Volts |
| Maximum Cool .....         | Approximately Zero Volts  |

(1) - When temperature control dial is turned, reading may drop to zero volts and then slowly return to 5 volts.

## WIRING HARNESS CONNECTOR

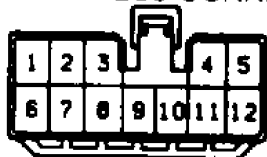


Fig. 7: Identifying System Amplifier Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## REMOVAL & INSTALLATION

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES/SAFETY EQUIPMENT section.

**NOTE:** For testing of components not listed in this article, see A/C-HEATER SYSTEM - MANUAL article in the AIR CONDITIONING & HEAT section.

## COMPRESSOR

### Removal

Run engine with A/C on for at least 10 minutes (if possible). Turn engine off. Remove battery. Disconnect A/C wire harness connector. Discharge A/C system using approved refrigerant recovery/recycling equipment. Disconnect A/C hoses from service valves. Plug all openings. Remove compressor drive belt. Remove compressor bolts and compressor.

### Installation

To install, reverse removal procedure. Evacuate and charge A/C system.

## EVAPORATOR, EXPANSION VALVE & THERMISTOR

### Removal

1) Disconnect negative battery cable. Discharge A/C system using approved refrigerant recovery/recycling equipment. Disconnect inlet and outlet lines and grommets from evaporator at engine compartment firewall. Plug openings.

2) Disconnect electrical connectors from evaporator case as necessary. Remove glove box and reinforcement. In passenger compartment, remove 3 nuts and 4 bolts securing evaporator case to firewall. Remove evaporator case.

### Disassembly

Remove evaporator case clips and screws. See Fig. 8. Separate evaporator case halves. Remove thermistor, evaporator and expansion valve.

### Reassembly & Installation

To reassemble and install evaporator assembly, reverse disassembly and removal procedures. If installing new evaporator core, add 1.4-1.7 ounces of refrigerant oil to system. Evacuate and charge A/C system.

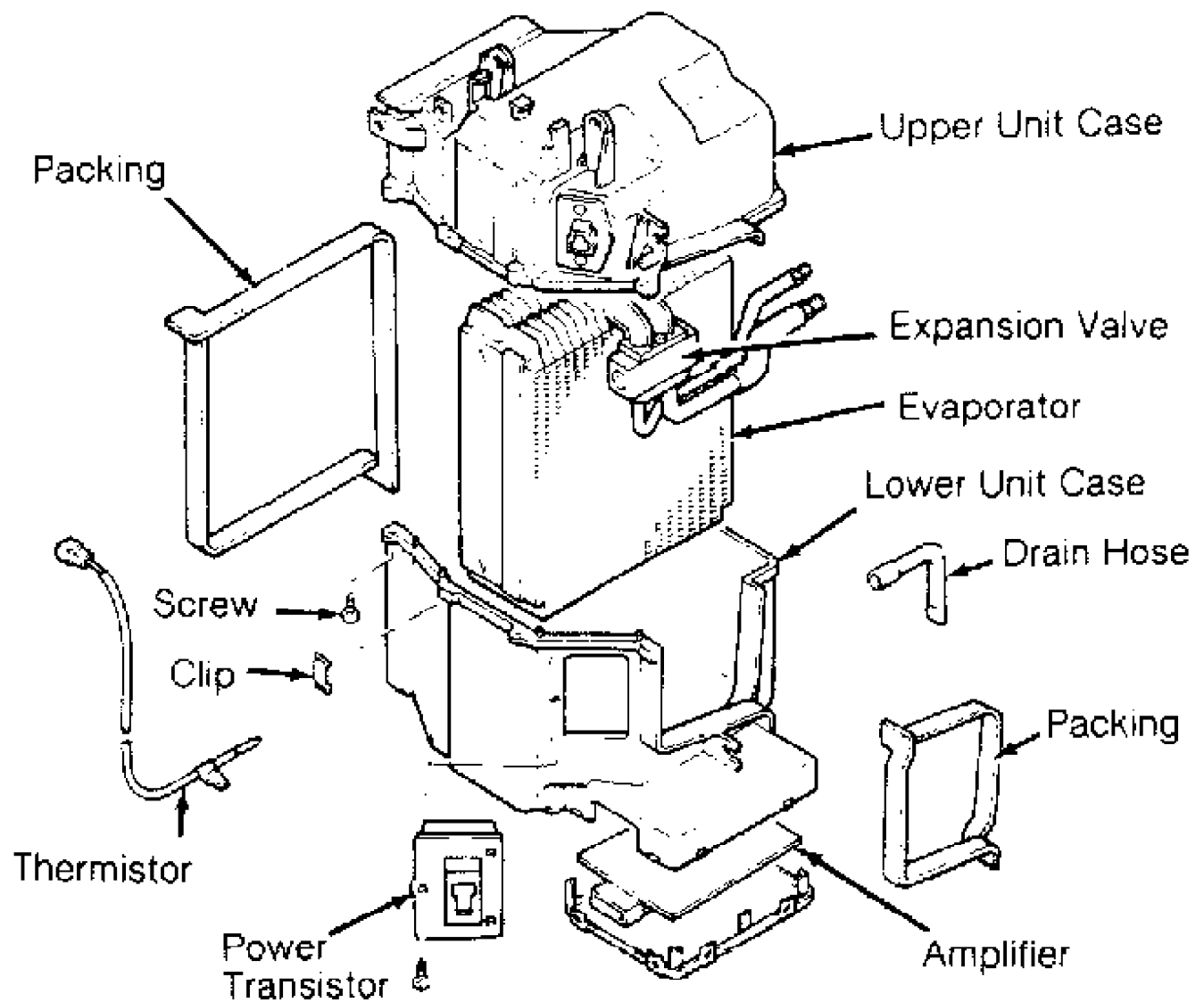


Fig. 8: Exploded View Of Evaporator Case  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## CONDENSER

### Removal

Discharge A/C system using approved refrigerant recovery/recycling equipment. Remove lower engine cover. Remove grille and hood lock brace. Remove horns. Disconnect A/C lines from condenser. Plug all openings. Remove mounting bolts and condenser.

### Installation

To install, reverse removal procedure. If installing new condenser, add 1.4-1.7 ounces of refrigeration oil to system. Evacuate and charge A/C system.

## RECEIVER-DRIER

### Removal

Discharge A/C system using approved refrigerant

recovery/recycling equipment. Disconnect A/C hoses from receiver-drier. Plug all openings. Remove receiver-drier bolts and receiver-drier.

Installation

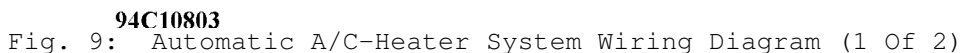
To install, reverse removal procedure. Add 0.7 ounce of refrigerant oil to system. Evacuate and charge A/C system.

TORQUE SPECIFICATIONS

TORQUE SPECIFICATIONS TABLE

| Application              | Ft. Lbs. (N.m) |
|--------------------------|----------------|
| Compressor Bolts         |                |
| 1.6L .....               | 18 (24)        |
| 2.0L & 2.2L .....        | 25 (34)        |
| Refrigerant Hose Fitting |                |
| At Compressor .....      | 18 (24)        |
| At Condenser             |                |
| Liquid Line .....        | 10 (14)        |
| Discharge Line .....     | 17 (23)        |
| At Evaporator            |                |
| Liquid Line .....        | 10 (14)        |
| Suction Line .....       | 24 (32)        |
| At Receiver-Drier .....  | 10 (14)        |

WIRING DIAGRAMS



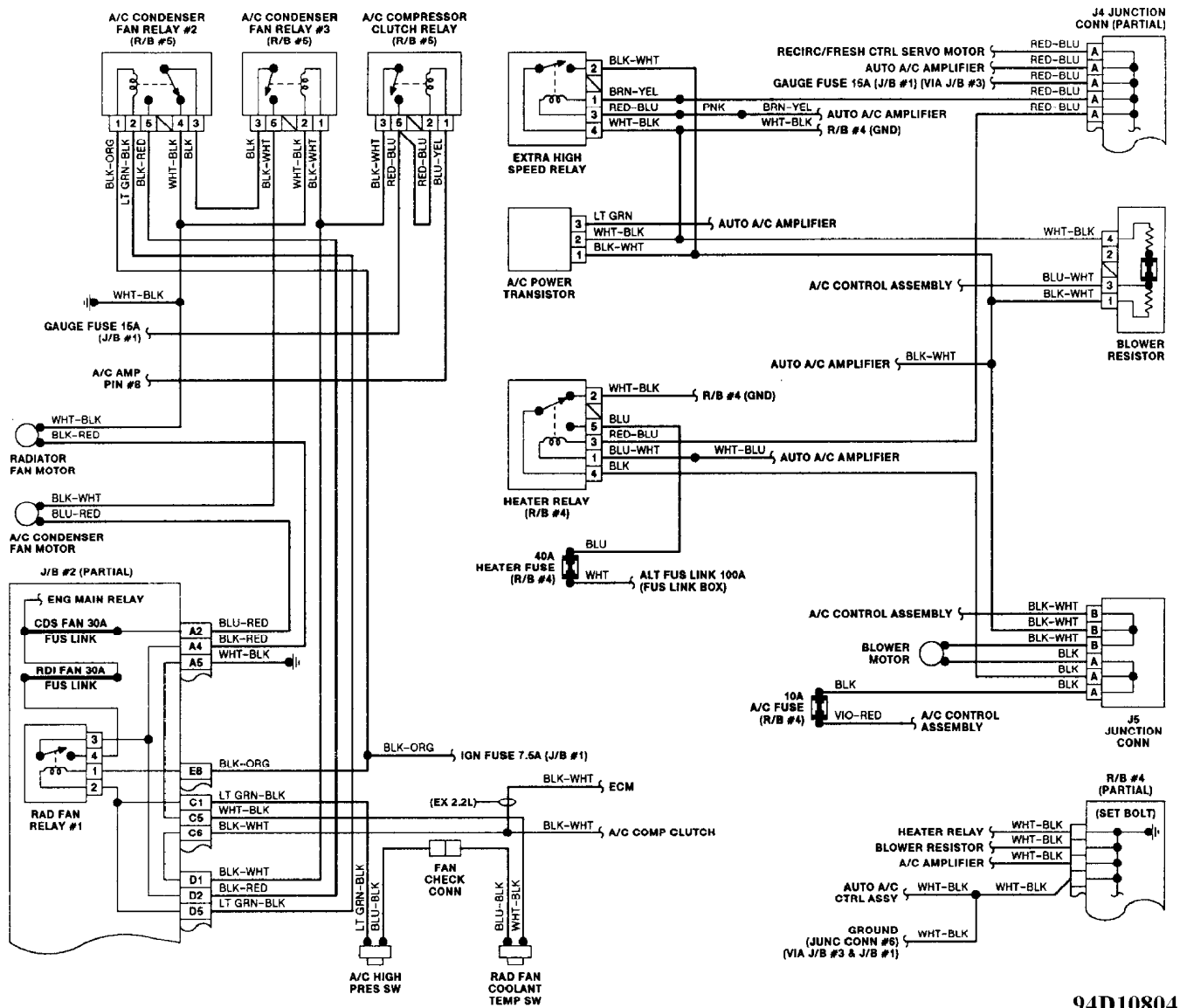


Fig. 10: Automatic A/C-Heater System Wiring Diagram (2 Of 2)

94D10804



# AIR BAG RESTRAINT SYSTEM

1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Air Bags

Celica

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all WARNINGS and SERVICE PRECAUTIONS.

## DESCRIPTION & OPERATION

The Supplement Restraint System (SRS) consists of an AIR BAG/SRS warning light in the instrument cluster, left and right front impact sensors, steering wheel pad, spiral cable and center air bag sensor. Steering wheel pad contains inflator and bag assembly. Center air bag sensor assembly contains the back-up power source circuit, safety circuit, safing sensor, memory circuit, diagnostic circuit, ignition control and drive circuits. See Fig. 1.

The SRS is designed to deploy when the front-to-rear shock is greater than a specified value. The ignition control and drive circuits calculate signals from the center air bag sensor, deploying air bag.

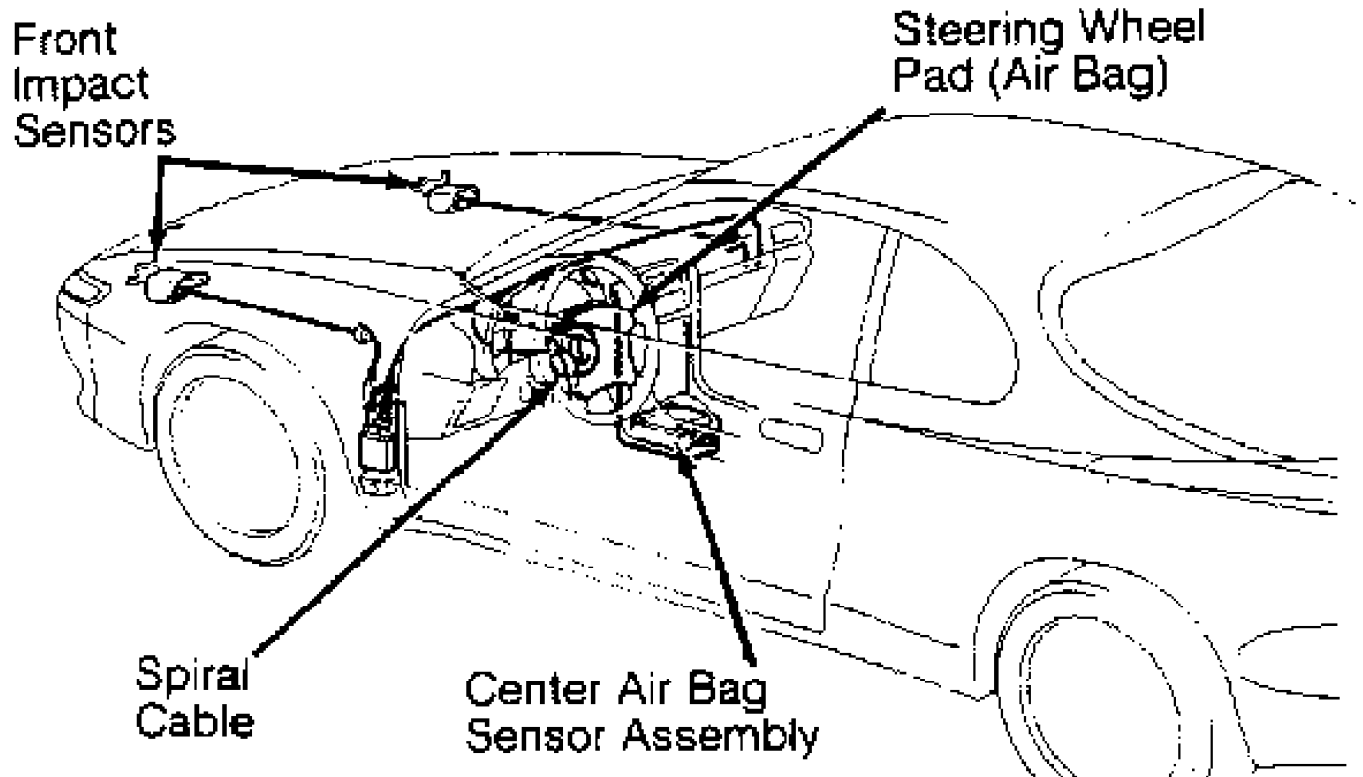


Fig. 1: SRS Component Location  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## SYSTEM OPERATION CHECK

Turn ignition switch to ACC or ON position. AIR BAG/SRS warning light in instrument cluster should glow and go out after about 6 seconds. If AIR BAG/SRS warning light glows for more than 6 seconds with ignition switch in ACC or ON position, SRS system is malfunctioning and needs repair. If AIR BAG/SRS warning light glows with ignition off, a short circuit is likely in AIR BAG/SRS warning light circuit. See DIAGNOSIS & TESTING.

## SERVICE PRECAUTIONS

Observe following precautions when working with air bag systems:

- \* Disable SRS before servicing any SRS or steering column component. Failure to do this could result in accidental air bag deployment and possible personal injury. See DISABLING & ACTIVATING AIR BAG SYSTEM.
- \* When trouble shooting SRS, always check for diagnostic codes before disconnecting battery.
- \* After turning ignition switch to LOCK position and disconnecting negative battery cable, wait at least 90 seconds before working on SRS. SRS is equipped with a back-up power source that may allow air bag to deploy up to 90 seconds after negative battery cable is disconnected.
- \* In a minor collision in which air bag does not deploy, front air bag sensors and steering wheel pad should be inspected.
- \* NEVER use air bag parts from another vehicle. Replace air bag parts using new parts.
- \* Remove air bag sensors if shocks are likely to be applied to sensors during repairs.
- \* Center air bag sensor assembly contains mercury. After replacement, DO NOT destroy old part. When scrapping vehicle or replacing center air bag sensor assembly, remove center air bag sensor assembly and dispose of as toxic waste.
- \* Never disassemble and repair front air bag sensors, center air bag sensor assembly or steering wheel pad.
- \* If front air bag sensors, center air bag sensor assembly or steering wheel pad is dropped or if cracks, dents or other defects exist in case, bracket or connector, replace parts using new ones.
- \* DO NOT expose front air bag sensors, center air bag sensor assembly or steering wheel pad directly to hot air or flame.
- \* Use a volt-ohmmeter with high impedance (10 k/ohm minimum) for trouble shooting electrical circuit.
- \* Information labels are attached to air bag components. Follow all notices on labels.
- \* After work on SRS is completed, check AIR BAG/SRS warning light to ensure system is functioning properly. See SYSTEM OPERATION CHECK.
- \* Always wear safety glasses when servicing or handling an air bag.
- \* When placing a live air bag on a bench or other surface, always face air bag and trim cover up, away from surface. This will reduce motion of module if it is accidentally deployed.
- \* After deployment, air bag surface may contain deposits of sodium hydroxide, which irritates skin, from gas generant combustion. Always wear safety glasses, rubber gloves and long-sleeved shirt during clean-up, and wash hands using mild soap and water.
- \* When carrying a live air bag module, trim cover should be pointed away from your body to minimize injury in case of

accidental deployment.

- \* If SRS is not fully functional for any reason, vehicle should not be driven until system is repaired and again becomes operational. DO NOT remove bulbs, modules, sensors or other components or in any way disable system from operating normally. If SRS is not functional, park vehicle until it is repaired and functions properly.

## **DISABLING & ACTIVATING AIR BAG SYSTEM**

**WARNING:** Back-up power supply maintains SRS voltage for about 90 seconds after battery is disconnected. After disabling SRS, wait at least 90 seconds before servicing SRS to prevent accidental air bag deployment and possible personal injury.

To disable SRS, turn ignition switch to LOCK position and disconnect negative battery cable. Wait at least 90 seconds before servicing SRS. To activate SRS, reconnect negative battery cable. Perform SYSTEM OPERATION CHECK.

## **DISPOSAL PROCEDURES**

### **DEPLOYED AIR BAG**

Deployed air bag modules can be disposed of as would any other part. Handle air bag module wearing gloves and safety glasses.

### **SCRAPPED VEHICLE**

**NOTE:** Some vehicles to be scrapped may have an undeployed air bag. When scrapping vehicles equipped with SRS, deploy air bag module.

1) Before proceeding, follow service precautions. See SERVICE PRECAUTIONS. Disable SRS. See DISABLING & ACTIVATING AIR BAG SYSTEM. Ensure steering wheel and steering wheel pad are not loose. If steering wheel and steering wheel pad are loose, air bag cannot be deployed using this procedure. Follow procedure listed under UNDEPLOYED AIR BAG.

2) Remove instrument panel lower finish panel. Disconnect spiral cable air bag connector, located on lower steering column. Connect Deployment Tool (09082-00700) connector to spiral cable air bag connector. Position deployment tool at least 33 feet from front of vehicle.

3) Close all doors and windows of vehicle. Connect deployment tool Red clip to positive battery terminal and Black clip to negative battery terminal. Ensuring no one is inside or within 33 feet of vehicle, press activation switch to deploy air bag. Because of heat, DO NOT touch air bag for at least 30 minutes after deployment.

### **UNDEPLOYED AIR BAG**

1) Never dispose of a steering wheel pad with an undeployed air bag. Never deploy an air bag inside a vehicle, unless vehicle is to be scrapped.

2) Before proceeding, see SERVICE PRECAUTIONS. Disable SRS. See DISABLING & ACTIVATING AIR BAG SYSTEM. Remove steering wheel pad from vehicle. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

3) To deploy a loose steering wheel pad, manufacturer recommends installing pad to a scrap vehicle wheel rim and tire

assembly. To do so, install 4 bolts in holes provided in rear of steering wheel pad. Tighten bolts by hand until they become difficult to turn. DO NOT overtighten bolts.

4) Wrap strong wire at least twice around bolts on left and right sides of steering wheel pad. See Fig. 2. Ensure no slack is present in wire. If slack is present, or wire is not strong enough, steering wheel pad may become loose due to shock when air bag is deployed.

5) Position steering wheel pad on wheel rim and tire assembly with pad side facing upward. Ensuring wire is tight, separately tie left and right sides of steering wheel pad to wheel rim through lug nut holes. See Fig. 3.

6) Connect Deployment Tool (09082-00700) to steering wheel pad connector. Position deployment tool at least 33 feet from steering wheel pad. Place a large cardboard box (weighted at sides) or 3 scrap tires on top of steering wheel pad. Ensure no one is within 33 feet of steering wheel pad. Press activation switch to deploy air bag.

7) Because of heat, wait 30 minutes before handling air bag. Use Gloves and safety glasses when handling a steering wheel pad with deployed air bag.

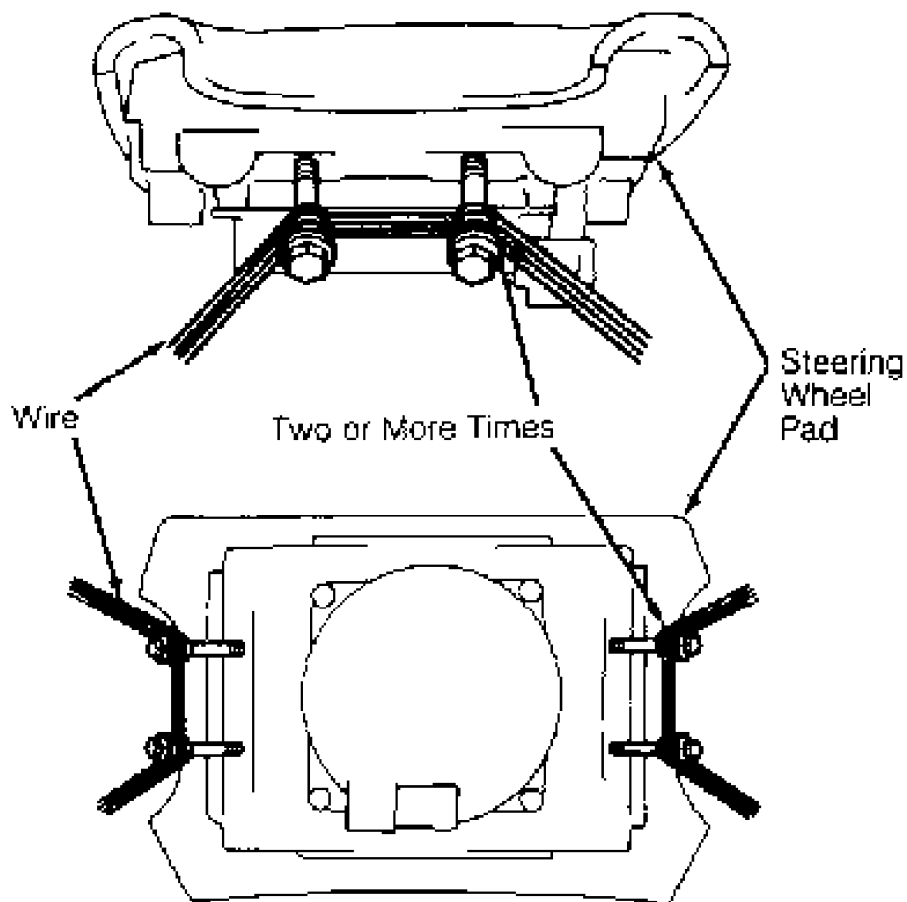


Fig. 2: Installing Wire On Steering Wheel Pad  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

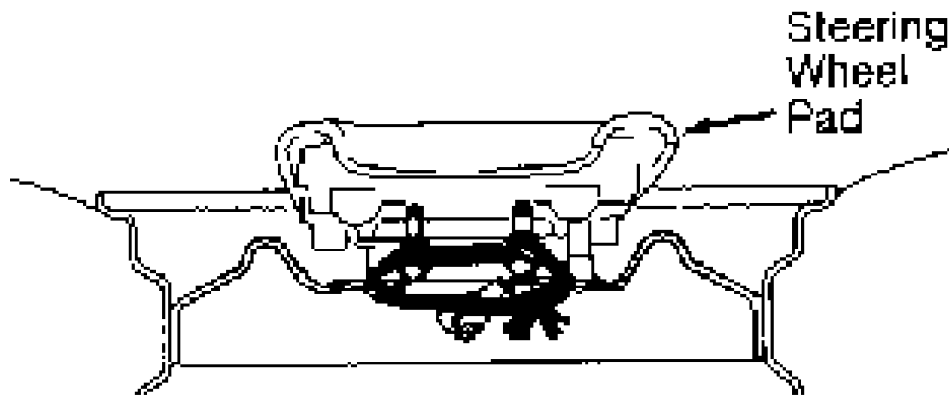


Fig. 3: Installing Steering Wheel Pad On Wheel Assembly  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## POST-COLLISION INSPECTION

Check diagnostic system. See SELF-DIAGNOSTIC SYSTEM under DIAGNOSIS & TESTING. Replace center air bag sensor if air bag deployed. Remove steering wheel pad and air bag assembly. Check for following conditions and replace components as necessary:

- \* Cut, cracked or markedly discolored steering wheel pad top surface and steering wheel pad grooved portion.
- \* Cut, cracked or chipped connectors or wire harnesses.
- \* Deformity of horn button contact plate and front air bag sensor bracket.
- \* Peeling off of label or damage to series number on front air bag sensor.

**WARNING:** If horn button contact plate is deformed, never repair it. Instead, replace entire steering wheel assembly. Ensure steering wheel pad does not contact steering wheel. Clearance must be uniform all the way around steering wheel pad.

## REMOVAL & INSTALLATION

**WARNING:** Failure to follow air bag service precautions may result in air bag deployment and personal injury. See SERVICE PRECAUTIONS. After component replacement, perform a system operational check to ensure proper system operation. See SYSTEM OPERATION CHECK.

## CENTER AIR BAG SENSOR

Removal & Installation

1) Before proceeding, follow air bag service precautions. See SERVICE PRECAUTIONS. Disable SRS. See DISABLING & ACTIVATING AIR BAG

SYSTEM.

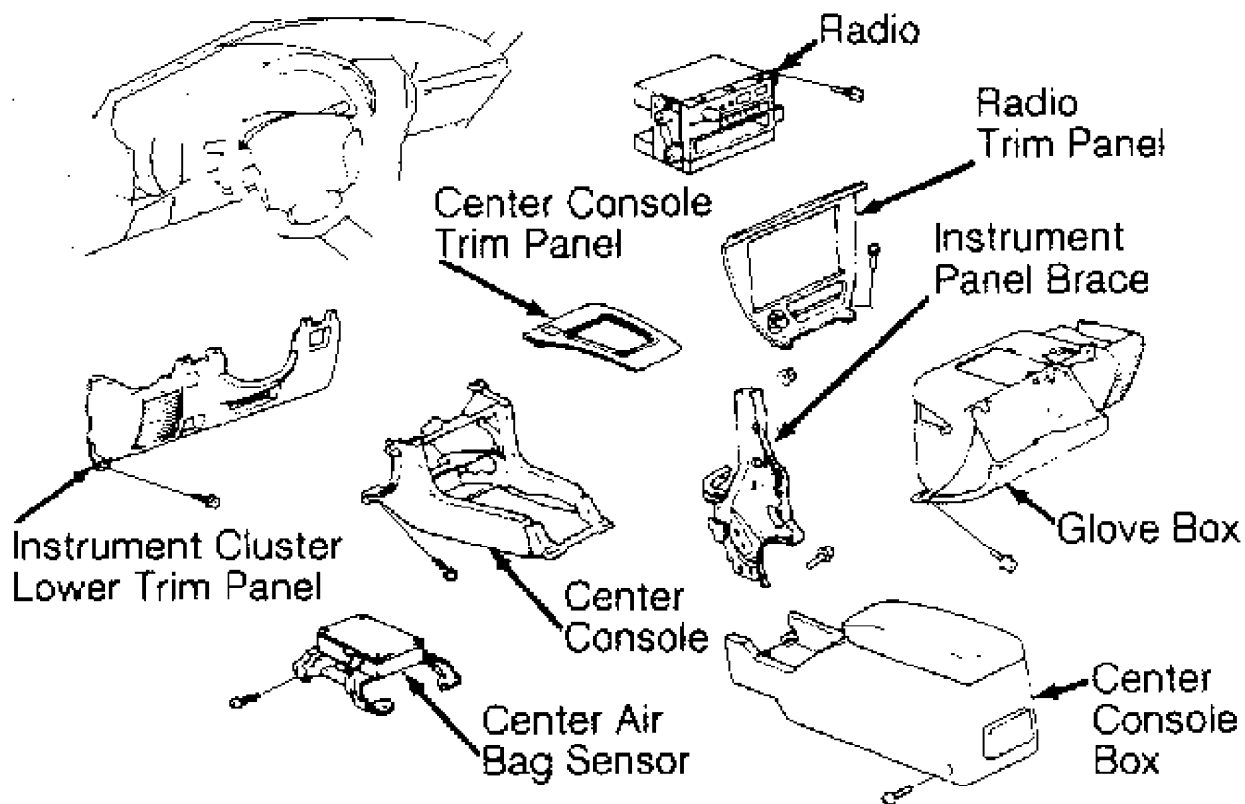
2) Center air bag sensor is located underneath console. Remove center console trim panel. See Fig. 4. Remove 4 screw covers from sides of center console box. Remove 4 center console screws and 2 bolts. Remove center console box.

3) Remove scuff plates. Remove 4 screws, glove box and passenger-side lower trim panel. Remove engine hood release lever. Remove screw covers, 6 screws and instrument cluster lower trim panel.

NOTE: Disconnect center air bag sensor electrical connector before removing sensor attaching screws.

4) Remove 2 screws and radio trim panel. Remove 4 screws and radio. Remove screw covers and 4 screws from center console. Remove instrument panel brace. See Fig. 4. Disconnect center air bag sensor assembly electrical connector. Remove 4 screws and center air bag sensor assembly.

5) To install, reverse removal procedure. Tighten center air bag sensor screws to specification. See TORQUE SPECIFICATIONS. Reactivate SRS. Ensure system is functioning properly. See SYSTEM OPERATION CHECK.



91G02870

Fig. 4: Removing Center Air Bag Sensor Assembly  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## FRONT AIR BAG SENSORS

#### Removal & Installation

1) Before proceeding, follow air bag service precautions. See SERVICE PRECAUTIONS. Disable SRS. See DISABLING & ACTIVATING AIR BAG SYSTEM.

2) Front air bag sensors are located in left and right fender areas. Remove screws and clips attaching inner fender shield to vehicle.

3) Remove hood lock protector plate or inner fender shield. Disconnect front air bag sensor electrical connector. Remove 2 bolts attaching sensor to fender. Remove front air bag sensor.

4) To install, reverse removal procedure. Ensure arrow marks on sensors face front of vehicle. Tighten front air bag sensor bolts to specification. See TORQUE SPECIFICATIONS. Reactivate SRS. Check AIR BAG warning light to ensure system is functioning properly. See SYSTEM OPERATION CHECK.

### STEERING WHEEL PAD & SPIRAL CABLE

#### Removal & Installation

1) Ensure front wheels are in straight-ahead position. Before proceeding, see SERVICE PRECAUTIONS. Disable SRS. See DISABLING & ACTIVATING AIR BAG SYSTEM. Remove driver-side scuff plate.

2) Remove engine hood release lever. Remove screw covers, 6 screws and lower instrument cluster trim panel(s). Loosen 4 steering wheel pad Torx screws until groove along screw circumference catches on screw case. See Figs. 5 and 6.

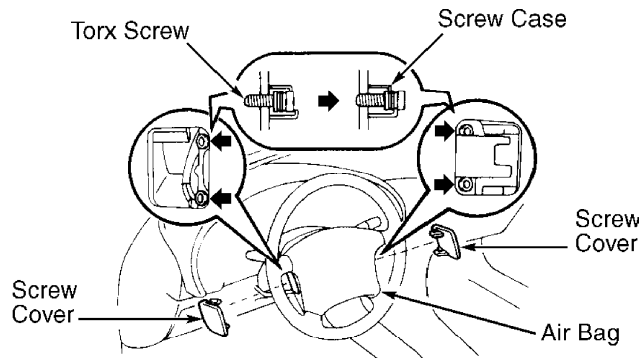
3) Pull steering wheel pad from steering wheel and disconnect steering wheel pad (squib) connector. Remove steering wheel pad assembly. Place steering wheel pad assembly on a flat surface with pad cover facing up.

4) Place a mark on steering wheel and main shaft for installation reference. Using steering wheel puller, remove steering wheel. Remove 4 screws from upper and lower steering column covers.

5) Remove screws attaching spiral cable to combination (headlight/turn signal/wiper) switch. Disconnect spiral cable and remove from vehicle.

6) To install, reverse removal procedure. Before installing spiral cable, ensure spiral cable is properly aligned. See SPIRAL CABLE under ADJUSTMENTS. Tighten steering wheel nut and steering wheel pad screws to specification. See TORQUE SPECIFICATIONS.

7) After spiral cable and steering wheel pad are installed, reactivate SRS. Ensure proper SRS operation. See SYSTEM OPERATION CHECK.



91A02872

Fig. 5: Removing Steering Wheel Pad  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

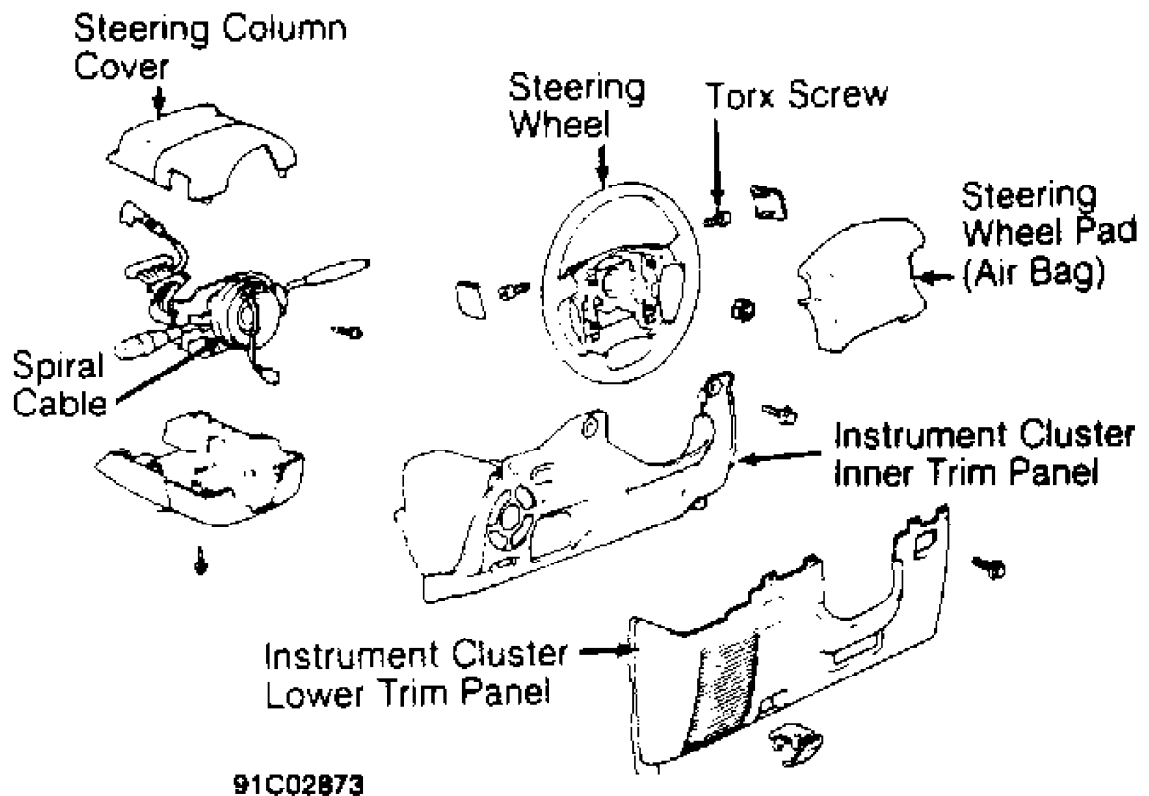


Fig. 6: Exploded View Of Steering Wheel Pad  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## ADJUSTMENTS

### SPIRAL CABLE

Ensure front wheels are in straight-ahead position. Turn spiral cable counterclockwise until it stops. Turn spiral cable clockwise 2.5 turns. Mating marks should align with Red mark. See Fig. 7. Ensure mating marks are aligned and install steering wheel.

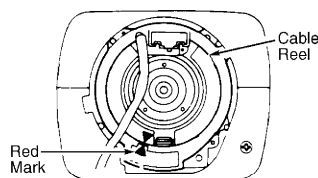


Fig. 7: Aligning Spiral Cable  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



## WIRE REPAIR

**CAUTION:** If air bag wiring harness is damaged, replace complete wiring harness assembly. Following WIRE REPAIR procedure is to be used only to repair connector to front air bag sensors.

1) Repair wire using 2 Pressure-Contact Sleeves (82988-50010); sleeves are available for repairing air bag connector problems. To repair, uncover front air bag sensor connector to be repaired. Leaving wires as long as possible, cut wiring harness behind connector. Carefully strip .31-.43" insulation from each wire to be repaired. DO NOT damage wire during this operation. If any damage exists, perform stripping operation again.

2) Overlap 2 stripped wire ends inside pressure-contact sleeve. Using Crimper (169060-2), place sleeve in correct color-keyed section of tool. With center of sleeve between crimping jaws, squeeze tool until either end comes in contact at section of tool marked CLOSE HERE.

3) Pull joined wires on both ends to ensure a secure crimp. Crimp both ends of sleeve using crimper at INS position. See Fig. 8. Thoroughly wrap silicon tape around joint for protection.

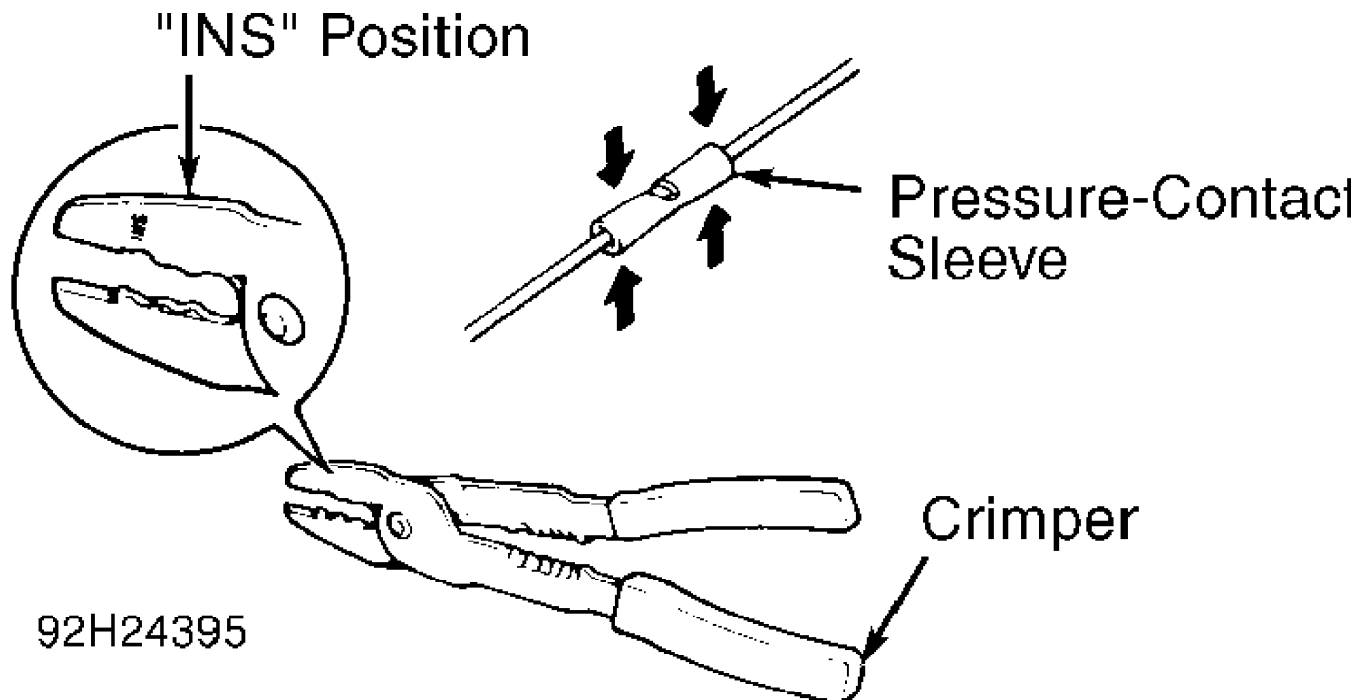


Fig. 8: Crimping Pressure-Contact Sleeve  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TORQUE SPECIFICATIONS

### TORQUE SPECIFICATIONS

| Application                               | Ft. Lbs. (N.m)  |
|-------------------------------------------|-----------------|
| Front Air Bag (Impact) Sensor Bolts ..... | 19 (26)         |
| Steering Wheel Nut .....                  | 26 (35)         |
|                                           | INCH Lbs. (N.m) |

|                                      |          |
|--------------------------------------|----------|
| Center Air Bag Sensor Screws .....   | 108 (12) |
| Steering Wheel Pad Torx Screws ..... | 65 (7)   |

---

## DIAGNOSIS & TESTING

**WARNING:** Failure to follow air bag service precautions may result in air bag deployment and personal injury. See SERVICE PRECAUTIONS. After component replacement, ensure proper system operation. See SYSTEM OPERATION CHECK.

### SELF-DIAGNOSTIC SYSTEM

#### AIR BAG/SRS Warning Light Check

1) Turn ignition switch to ACC or ON position. Ensure AIR BAG/SRS warning light glows, and then goes out after approximately 6 seconds.

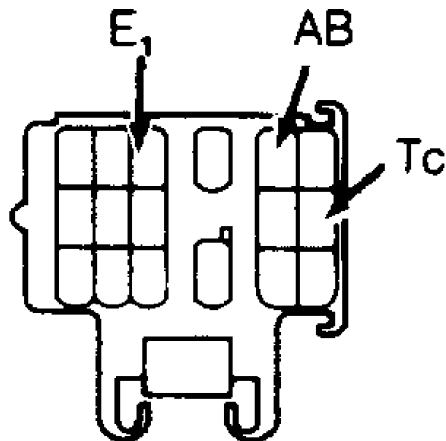
2) When ignition switch is in ACC or ON position and AIR BAG/SRS warning light remains on, center air bag sensor assembly has detected a malfunction code. To identify malfunction code, see DIAGNOSTIC CODE CHECK.

3) If after approximately 6 seconds, air bag warning light sometimes glows or glows even when ignition switch is in OFF position, a short in AIR BAG/SRS warning light circuit is present. See TROUBLE SHOOTING.

**NOTE:** Celica is equipped with one Data Link Connector (DLC), located in engine compartment.

#### Diagnostic Code Check

1) Jump DLC terminals Tc and E1 using Diagnosis Check Wiring (09843-18020) jumper harness. See Fig. 9.



# 93B75508

Fig. 9: Data Link Connector (DLC) Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

2) Read diagnostic codes by noting number of AIR BAG/SRS warning light flashes. See Fig. 10. Normal code indication will flash

light twice per second. If a malfunction code is present, first number of code number will equal first digit of a 2-digit diagnostic code. After a 1.5-second pause, second number of code number will equal second digit.

3) If 2 or more codes are present, a 2.5-second pause will occur between codes. After all codes are displayed, a 4-second pause will occur and codes will be repeated. After code has been identified, see appropriate diagnostic code and perform tests as specified.

4) If more than one malfunction code is present, codes will flash from smallest numbered code to largest. If malfunction code is not displayed or is continuously displayed, see TROUBLE SHOOTING. If AIR BAG/SRS warning light remains illuminated and diagnostic code is normal code, a source voltage drop is present. This malfunction is not stored in memory. If power source voltage returns to normal after approximately 10 seconds, AIR BAG/SRS warning light will automatically go out. See DIAGNOSTIC CODE NORMAL, SOURCE VOLTAGE DROP under TROUBLE SHOOTING.

5) Code 22 is recorded when a malfunction is present in air bag warning light system. If an open circuit is present in air bag warning light circuit, AIR BAG/SRS warning light will not glow. Diagnostic codes (including Code 22) cannot be confirmed until warning light circuit is repaired.

6) When a malfunction is present in SRS, malfunction Codes 11 through 31 are displayed. Codes 11 through 31 are cleared from memory after repair of malfunction, but Code 41 will remain until cleared.

| Code No. | Blink Pattern                                                                       | Diagnosis                                                                                      | Trouble Area                                                                                                                 |
|----------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| (Normal) |    | • System normal                                                                                | —                                                                                                                            |
|          |                                                                                     | • Source voltage drop                                                                          | • Battery<br>• Center airbag sensor assembly                                                                                 |
| 11       |    | • Short in squib circuit or front airbag sensor circuit (to ground)                            | • Steering wheel pad (squib)<br>• Front airbag sensor<br>• Spiral cable<br>• Center airbag sensor assembly<br>• Wire harness |
| 12       |    | • Short in squib circuit or front airbag sensor circuit (to +B)                                | • Steering wheel pad (squib)<br>• Front airbag sensor<br>• Spiral cable<br>• Center airbag sensor assembly<br>• Wire harness |
| 13       |   | • Short in squib circuit (between D <sup>+</sup> wire harness and D <sup>-</sup> wire harness) | • Steering wheel pad (squib)<br>• Spiral cable<br>• Center airbag sensor assembly<br>• Wire harness                          |
| 14       |  | • Open in squib circuit                                                                        | • Steering wheel pad (squib)<br>• Spiral cable<br>• Center airbag sensor assembly<br>• Wire harness                          |
| 15       |  | • Open in front airbag sensor circuit                                                          | • Front airbag sensor<br>• Center airbag sensor assembly<br>• Wire harness                                                   |
| 22       |  | • Airbag warning light system malfunction                                                      | • Airbag warning light<br>• Center airbag sensor assembly<br>• Wire harness                                                  |
| 31       |  | • Center airbag sensor assembly malfunction                                                    | • Center airbag sensor assembly                                                                                              |
| * 41     |  | • Malfunction stored in memory                                                                 | • (Center airbag sensor assembly)                                                                                            |

\* Not used on Corolla, Paseo, Previa or Tercel.

## 93F75510

Fig. 10: Diagnostic Code ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### Clearing Malfunction Codes

1) Connect wire probes to DLC terminals Tc and AB. See

Fig. 9. Turn ignition switch to ACC or ON position, and wait about 6 seconds. Starting with Tc terminal, apply body ground alternately to terminal Tc and terminal AB twice each, in cycles of .5-1.5 seconds.

2) When alternating probes between body ground, simultaneously release one probe from body ground while applying ground to other terminal. If time interval is too long, code will not clear. After several seconds, when AIR BAG/SRS warning light starts to blink a regular cycle, code cancellation is complete.

GJ

## **TROUBLE SHOOTING**

### **AIR BAG/SRS WARNING LIGHT REMAINS ILLUMINATED**

Turn ignition switch to LOCK position. Disconnect negative battery cable. Disconnect center air bag sensor assembly connector. Reconnect negative battery cable. If warning light remains illuminated, replace center air bag sensor assembly. If warning light goes out, check warning light circuit or Data Link Connector (DLC) terminal AB for an open circuit. See Fig. 9.

### **AIR BAG/SRS WARNING LIGHT INTERMITTENT**

Intermittent malfunctions can make AIR BAG/SRS warning light glow erratically. To diagnose intermittent problems, apply vibration, heat (as with a hair dryer), or humidity (to entire vehicle, not directly to electrical components) to check whether malfunction reoccurs.

### **DIAGNOSTIC CODE NOT DISPLAYED**

1) If code is not displayed after performing diagnostic code check, turn ignition switch to ACC or ON position. Measure voltage between Data Link Connector (DLC) terminals Tc and E1. If battery voltage is not present, go to next step. If battery voltage is present, go to step 3).

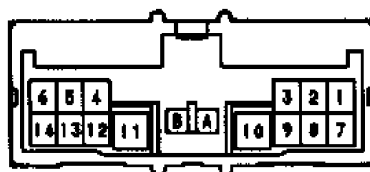
2) Measure voltage between DLC terminal Tc and ground. If battery voltage is present, check harness between terminal E1 of DLC and body ground. If battery voltage is not present, go to next step.

3) Using diagnosis check wiring, jump center air bag sensor assembly connector terminal Tc to ground. If warning light glows, check harness between center air bag sensor assembly and DLC. If warning light does not glow, replace center air bag sensor assembly.

### **DIAGNOSTIC CODE CONTINUOUSLY DISPLAYED**

1) If code is displayed continuously after performing diagnostic code check, turn ignition switch to LOCK position. Disconnect center air bag sensor assembly connector.

2) Check resistance between center air bag sensor assembly connector terminal Tc and ground. See Fig. 11. If resistance is infinite ohms, replace center air bag sensor assembly. If resistance is not infinite ohms, repair or replace harness or connector.



| No. | Symbol          | Terminal Name                         | No. | Symbol         | Terminal Name                         |
|-----|-----------------|---------------------------------------|-----|----------------|---------------------------------------|
| 1   | IG <sub>1</sub> | Power Source (ECU-IG Fuse)            | 8   | E <sub>2</sub> | Ground                                |
| 2   | -SR             | RH Front Airbag Sensor ⊖              | 9   | LA             | Airbag Warning Light                  |
| 3   | +SR             | RH Front Airbag Sensor ⊕              | 10  | D <sup>-</sup> | Squib ⊖                               |
| 4   | +SL             | LH Front Airbag Sensor ⊕              | 11  | D <sup>+</sup> | Squib ⊕                               |
| 5   | -SL             | LH Front Airbag Sensor ⊖              | 12  | Tc             | Diagnosis                             |
| 6   | +B              | Battery (ECU-B Fuse)                  | 13  | E <sub>1</sub> | Ground                                |
| 7   | IG <sub>2</sub> | Power Source (IGN Fuse)               | 14  | Acc            | Power Source (CIG Fuse)               |
| A   | -               | Electrical Connection Check Mechanism | B   | -              | Electrical Connection Check Mechanism |

Fig. 11: Center Air Bag Assembly Connector Terminal ID  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## DIAGNOSTIC CODE NORMAL, SOURCE VOLTAGE DROP

### Circuit Description

1) SRS is equipped with a voltage increase circuit in center air bag sensor assembly. This circuit increases voltage of SRS to normal voltage if source voltage drops.

2) Diagnostic system malfunction display for this circuit is different than other circuits. When AIR BAG/SRS warning light remains lit and diagnostic code is a normal code, source voltage drop is indicated. A malfunction in this circuit is not recorded in center air bag sensor assembly. About 10 seconds after source voltage returns to normal, air bag warning light goes off.

### Diagnosis

1) Turn ignition switch to LOCK position. Disconnect center air bag sensor connector. Turn ignition switch to ON position (engine off). Measure voltage between ground and terminals IG<sub>1</sub> and ACC on connector side of center air bag sensor assembly. See Fig. 11. Operate electrical components (defogger, wiper, headlight, heater blower, etc.).

2) Approximately 6.0-11.5 volts should be present at terminals. Turn all electrical components off. Turn ignition switch to LOCK position. Remove voltmeter, and reconnect connector.

3) Turn ignition switch to ON position. Turn on electrical accessories. If air bag warning light goes out after approximately 10 seconds, check battery and charging system. If light does not go out, check diagnostic code. If a malfunction code is present, perform appropriate trouble shooting. If a normal code is present, replace center air bag sensor assembly.

## CODE 11, SHORT IN SQUIB CKT, FRONT AIR BAG SENS CKT (TO

GROUND)

#### Circuit Description

Squib circuit consists of center air bag assembly, spiral cable and steering wheel pad (squib). Squib causes air bag to deploy when deployment conditions have been met. Front air bag sensors detect deceleration force in frontal collision and are located in front left and right sides of vehicle. Code 11 is recorded when a ground short is detected in squib circuit or front air bag sensor circuit.

#### Diagnosis

1) Disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

2) Disconnect the center air bag sensor assembly connector. Measure resistance between terminals +SR and -SR and terminals +SL and -SL of harness side connector of center air bag sensor assembly. See Fig. 11. If resistance is not 755-885 ohms, go to step 9).

3) If resistance is 755-885 ohms, measure resistance between terminals +SR and +SL of harness side connector of center air bag sensor assembly and body ground. If infinite resistance is present, go to next step. If infinite resistance is not present, repair or replace harness or connector between center air bag sensor assembly and front air bag sensor.

4) Measure resistance between terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad and body ground. See Fig. 12. If resistance is infinite, go to next step. If resistance is not infinite, go to step 10).

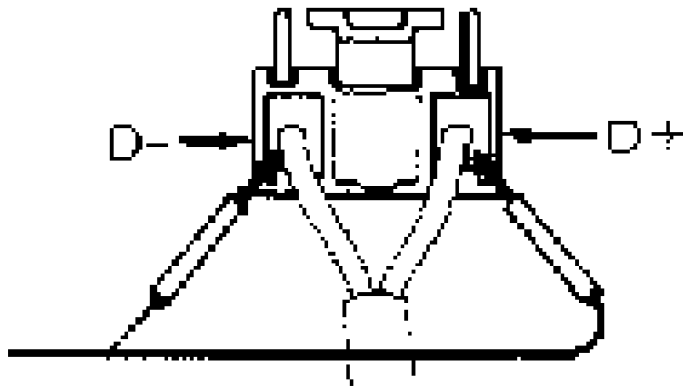


Fig. 12: Spiral Cable-To-Steering Wheel Pad Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

5) Reconnect center air bag sensor assembly. Using a jumper wire, jump terminals D+ and D- on spiral cable side of connector

between spiral cable and steering wheel pad. Connect negative battery cable to battery, and wait at least 2 seconds.

6) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using Diagnosis Check Wiring (09843-18020), jump DLC terminals Tc and E1. See Fig. 9. Check diagnostic code. If Code 11 is not displayed, go to next step. If Code 11 is displayed, replace center air bag sensor assembly.

7) Turn ignition switch to LOCK position. Disconnect negative battery cable and wait at least 90 seconds. Reconnect steering wheel pad (squib) connector. Reconnect negative battery cable and wait 2 seconds.

8) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 11 is not displayed, go to next step. If Code 11 is displayed, replace steering wheel pad assembly.

9) Disconnect front air bag sensor connector. Measure resistance between terminals +S and +A (755-885 ohms), +S and -S (infinite ohms), and -S and -A (less than one ohm). See Fig. 13. If resistance is not to specification, replace front air bag sensor. If resistance is okay, repair or replace harness or connector between center air bag sensor assembly and front air bag sensor.

10) Disconnect connector between center air bag sensor assembly and spiral cable. Measure resistance between terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad and body ground. See Fig. 12. If resistance is infinite, repair or replace harness or connector between center air bag sensor assembly and spiral cable. If resistance is not infinite, repair or replace spiral cable.

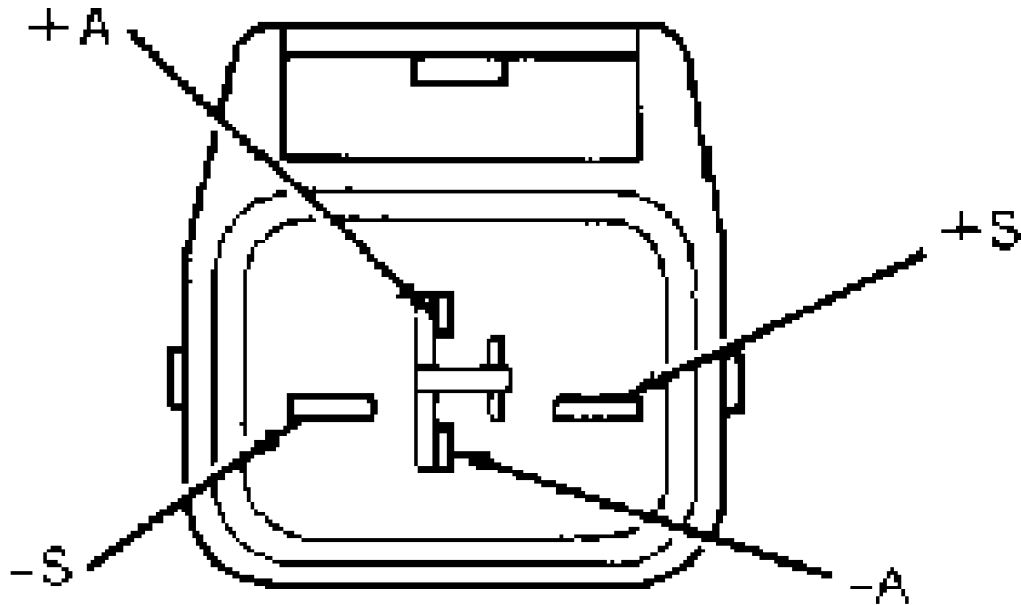


Fig. 13: Front Air Bag Sensor Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

CODE 12, SHORT IN SQUIB CKT, FRONT AIR BAG SENS CKT (TO  
VOLTAGE)



#### Circuit Description

Squib circuit consists of center air bag sensor assembly, spiral cable and steering wheel pad (squib). Squib causes air bag to deploy when deployment conditions have been satisfied. Front air bag sensors (located on left and right sides of vehicle) detect deceleration force in a frontal collision.

#### Diagnosis

1) Disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

2) Disconnect center air bag sensor assembly connector. Measure resistance between center air bag sensor harness connector terminals +SR and -SR and terminals +SL and -SL. See Fig. 11. If resistance is not 755-885 ohms, go to DIAGNOSTIC CODE 15, OPEN IN FRONT AIR BAG SENSOR CIRCUIT.

3) If resistance is 755-885 ohms, reconnect negative battery cable. Turn ignition switch to ON position. Measure voltage between ground and terminal +SR or +SL of harness side connector of center air bag sensor assembly. If any voltage is present, repair or replace harness or connector between center air bag sensor assembly and front air bag sensor.

4) If no voltage is present, measure voltage at terminal D+ on spiral cable side of connector between spiral cable and steering wheel pad. See Fig. 12. If no voltage is present, go to step 5). If voltage is present, turn ignition switch to LOCK position. Disconnect connector between center air bag sensor assembly and spiral cable. Turn ignition switch to ON position. Measure voltage at terminal D+ on spiral cable side of connector between spiral cable and steering wheel pad. If no voltage is present, repair or replace harness or connector between center air bag sensor assembly and spiral cable. If voltage is present, repair or replace spiral cable.

5) Turn ignition switch to LOCK position. Disconnect negative battery cable. Reconnect center air bag sensor connector. Using Diagnosis Check Wiring (09843-18020), jump terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad. See Fig. 12. Reconnect negative battery cable.

6) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 12 is displayed, replace center air bag sensor assembly.

7) If Code 12 is not displayed, turn ignition switch to LOCK position. Disconnect negative battery cable, and wait at least 90 seconds. Reconnect steering wheel pad (squib) connector. Reconnect negative battery cable and wait at least 2 seconds.

8) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 12 is displayed, replace steering wheel pad. If Code 12 is not displayed, system is okay.

### CODE 13, SHORT IN SQUIB CKT (BETWEEN D+ HARNESS & D- HARNESS)

#### Circuit Description

Squib circuit consists of center air bag sensor assembly, spiral cable and steering wheel pad (squib). Squib causes air bag to deploy when air bag deployment conditions are satisfied. Code 13 is recorded when a short is detected in D+ wire harness and D- wire harness of squib circuit. See Fig. 12.

#### Diagnosis

1) Disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL

CABLE under REMOVAL & INSTALLATION.

2) Measure resistance between D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad. See Fig. 12. If resistance is 1000 ohms or more, go to next step. If resistance is not 1000 ohms or more, go to step 6).

3) Reconnect negative battery cable. Clear diagnostic code stored in memory. See CLEARING MALFUNCTION CODES under SELF-DIAGNOSTIC SYSTEM under DIAGNOSIS & TESTING. Turn ignition switch to LOCK position, and wait at least 2 seconds. Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using Diagnosis Check Wiring (09843-18020), jump DLC terminals Tc and E1. See Fig. 9. If Code 13 is present, replace center air bag sensor assembly. If Code 13 is not present, go to next step.

4) Turn ignition switch to LOCK position. Disconnect negative battery cable and wait at least 90 seconds. Reconnect steering wheel pad (squib) connector. Reconnect negative battery cable. Clear diagnostic code.

5) Turn ignition switch to LOCK position and wait at least 2 seconds. Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 13 is present, replace steering wheel pad. If Code 13 is not present, system is okay.

6) Disconnect connector between center air bag sensor assembly and spiral cable. Release air bag activation prevention mechanism on center air bag sensor assembly side of spiral cable connector. Measure resistance between terminals D+ and D- on spiral cable side of connector, between spiral cable and steering wheel pad. See Fig. 12. If infinite resistance is present, go to next step. If infinite resistance is not present, repair or replace spiral cable.

7) Disconnect center air bag sensor assembly connector. Release air bag activation prevention mechanism on center air bag sensor assembly connector. Measure resistance between terminals D+ and D- on center air bag sensor assembly side of connector between center air bag sensor assembly and spiral cable. See Fig. 12. If resistance is infinite, go to next step. If resistance is not infinite, repair or replace harness or connector between center air bag sensor assembly and spiral cable.

8) Reconnect center air bag sensor assembly connector. Measure resistance between terminals D+ and D- on center air bag sensor assembly side of connector between center air bag sensor assembly and spiral cable. See Fig. 12. If resistance is 1000 ohms or more, system is okay. If resistance is not 1000 ohms or more, replace center air bag sensor assembly.

## DIAGNOSTIC CODE 14, OPEN IN SQUIB CIRCUIT

### Circuit Description

Squib circuit consists of center air bag sensor assembly, spiral cable and steering wheel pad (squib). Squib causes air bag to deploy when deployment conditions are satisfied. Code 14 is recorded when an open is detected in squib circuit.

### Diagnosis

1) Disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

2) Disconnect center air bag sensor assembly connector. Measure resistance between terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad. See Fig. 12. If resistance is more than one ohm, go to next step. If resistance is less than one ohm, go to step 5).

3) Disconnect connector between center air bag sensor assembly and spiral cable. Measure resistance between terminals D+ and

D- on spiral cable side of connector between spiral cable and steering wheel pad. If resistance is less than one ohm, go to next step. If resistance is more than one ohm, repair or replace spiral cable.

4) Measure resistance between terminals D+ and D- on center air bag sensor assembly side of connector between center air bag sensor assembly and spiral cable. If resistance is less than one ohm, go to next step. If resistance is more than one ohm, repair or replace harness or connector between center air bag sensor assembly and spiral cable.

5) Reconnect connector to center air bag sensor assembly. Connect connector between center air bag sensor assembly and spiral cable. Using Diagnosis Check Wiring (09843-18020), jump terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad. Connect negative battery cable and wait at least 2 seconds.

6) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 14 is displayed, replace center air bag sensor assembly. If Code 14 is not displayed, go to next step.

7) Turn ignition switch to LOCK position. Disconnect negative battery cable, and wait at least 90 seconds. Reconnect steering wheel pad (squib) connector. Connect negative battery cable, and wait at least 2 seconds.

8) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 14 is displayed, replace steering wheel pad. If Code 14 is not displayed, system is okay.

## DIAGNOSTIC CODE 15, OPEN IN FRONT AIR BAG SENSOR CIRCUIT

### Circuit Description

Front air bag sensors (located on left and right side of vehicle) detect deceleration force in a frontal collision. Code 15 is recorded when an open circuit is detected in front air bag sensor circuit.

### Diagnosis

1) Disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

2) Disconnect center air bag sensor assembly connector. Measure resistance between harness side of center air bag sensor assembly connector terminals +SR and -SR and terminals +SL and -SL. See Fig. 11. If resistance is 755-885 ohms, go to next step. If resistance is not 755-885 ohms, go to step 5).

3) Reconnect center air bag sensor connector. Using Diagnosis Check Wiring (09843-18020), jump terminals D+ and D- on spiral cable side of connector between spiral cable and steering wheel pad. See Fig. 12. Connect negative battery cable, and wait at least 2 seconds.

4) Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using diagnosis check wiring, jump DLC terminals Tc and E1. See Fig. 9. If Code 15 is not present, go to next step. If code is present, replace center air bag sensor assembly.

5) Disconnect front air bag sensor connector. Measure resistance between front air bag sensor terminals +S and +A (755-885 ohms), terminals +S and -S (infinite ohms) and terminals -S and -A (less than one ohm). See Fig. 13. If resistances are not to specification, replace front air bag sensor. If resistances are okay, go to next step.

6) Disconnect center air bag sensor assembly connector. Using diagnosis check wiring, jump center air bag sensor assembly connector terminals +SR and -SR and terminals +SL and -SL. See Fig. 11. Measure resistance between terminals +SR and -SR and terminals +SL and -SL of

harness side connector of front air bag sensor. If resistance is less than one ohm, replace front air bag sensor connector. If resistance is more than one ohm, repair or replace harness or connector between center air bag sensor assembly and front air bag sensor.

## DIAGNOSTIC CODE 22, AIR BAG/SRS WARNING LIGHT SYS MALFUNCTION

### Circuit Description

AIR BAG/SRS warning light is located in instrument cluster. When SRS is normal, air bag warning light glows for approximately 6 seconds after ignition switch is turned from LOCK to ACC or ON position. Warning light will glow if a malfunction exists in SRS. Code 22 is code for AIR BAG/SRS warning light system malfunction. Code 22 usually cannot be accessed through AIR BAG/SRS warning light because warning light circuit is defective.

### Diagnosis (Warning Light Does Not Glow)

1) Remove ECU "B" fuse. Replace fuse as necessary. If fuse is bad, check harness between ECU "B" fuse and AIR BAG/SRS warning light, and between ECU "B" fuse and center air bag sensor assembly. If fuse is okay, check center air bag sensor assembly connector and repair as necessary.

2) If connector is okay, disconnect negative battery cable and wait at least 90 seconds. Remove steering wheel pad. See STEERING WHEEL PAD & SPIRAL CABLE under REMOVAL & INSTALLATION.

3) Disconnect center air bag sensor assembly connector. Reconnect negative battery cable. Measure voltage between center air bag sensor harness side connector terminal LA and ground. See Fig. 11. If battery voltage is present, go to next step. If battery voltage is not present, repair air bag warning light circuit.

4) Disconnect negative battery cable. Reconnect center air bag sensor connector. Reconnect negative battery cable. Turn ignition switch to ACC or ON position, and check operation of AIR BAG/SRS warning light. If AIR BAG/SRS warning light glows, it is okay. If warning light does not glow, check terminal LA of center air bag sensor assembly and electrical connector check mechanism. See Fig. 11. If terminal and connector check mechanism are okay, replace center air bag sensor assembly.

### Diagnosis (Warning Light On)

Clear malfunction code stored in memory. Turn ignition switch to LOCK position, and wait at least 2 seconds. Turn ignition switch to ACC or ON position, and wait at least 20 seconds. Using Diagnosis Check Wiring (09843-18020), jump DLC terminals Tc and E1. See Fig. 9. If Code 22 is still present, replace center air bag sensor assembly. If no code is present, see AIR BAG/SRS WARNING LIGHT INTERMITTENT.

## DIAGNOSTIC CODE 31, CENTER AIR BAG SENS ASSEMBLY MALFUNCTION

### Circuit Description

Center air bag sensor assembly consists of a center air bag sensor, safing sensors, ignition control, drive circuit and diagnosis circuit. It receives signals from air bag sensors and determines whether air bag must be activated. Code 31 is recorded when a malfunction in center air bag sensor assembly is detected.

### Diagnosis

1) If a malfunction code other than Code 31 is displayed at same time as Code 31, repair the malfunction indicated by code other than Code 31.

2) If Code 31 is displayed again, clear malfunction Code 41 stored in memory. Turn ignition switch to LOCK position, and wait at least 20 seconds. Turn ignition switch to ACC or ON position, and wait

at least 20 seconds. Repeat this operation at least 5 times. Using Diagnosis Check Wiring (09843-18020), jump DLC terminals Tc and E1. If Code 31 is present, replace center air bag sensor assembly. If Code 31 is not present, see AIR BAG/SRS WARNING LIGHT INTERMITTENT.

## DIAGNOSTIC CODE 41, MALFUNCTION STORED IN MEMORY

### Circuit Description

If a malfunction occurs in SRS, malfunction Codes 11 to 31 may be displayed. When battery is disconnected after malfunction is repaired, malfunction Codes 11 to 31 will be cleared, but Code 41 will be displayed.

### Diagnosis

1) Clear malfunction Code 41 stored in memory. See CLEARING MALFUNCTION CODES under SELF-DIAGNOSTIC SYSTEM under DIAGNOSIS & TESTING. Turn ignition switch to LOCK position, and wait at least 2 seconds. Turn ignition switch to ACC or ON position, and wait at least 20 seconds. If AIR BAG warning light turns off, system is okay.

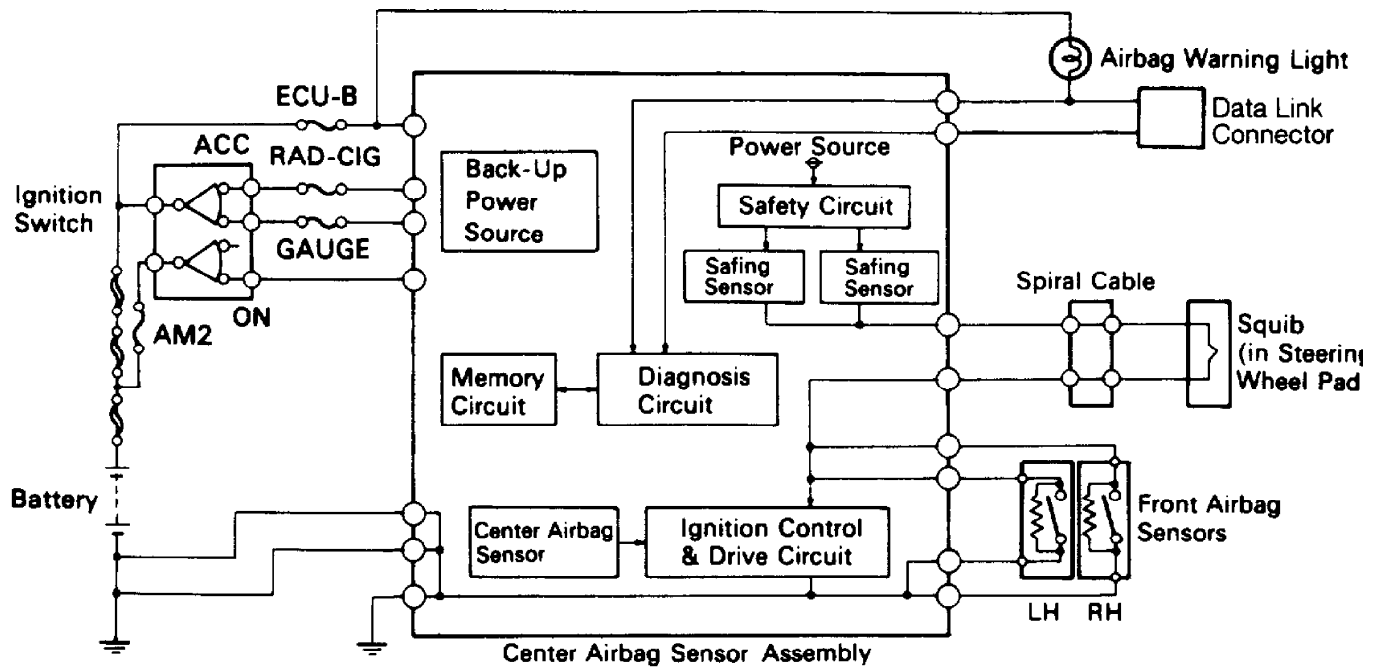
2) If warning light does not go off, turn ignition switch to ACC or ON position and wait at least 20 seconds. Using Diagnosis Check Wiring (09843-18020), jump DLC terminals Tc and E1. See Fig. 9 If Code 41 is present, check harness between ECU "B" fuse and center air bag sensor assembly. If harness is okay, replace center air bag sensor assembly. If Code 41 is not present, perform trouble shooting according to malfunction code displayed.

## POST-COLLISION AIR BAG SAFETY INSPECTION

POST-COLLISION AIR BAG SAFETY INSPECTION TABLE

|                                                                          |                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Replace After Deployment                                                 | <ul style="list-style-type: none"><li>* Air Bag Module</li><li>* Center Air Bag Sensor Assembly</li><li>* Front Air Bag Sensors</li></ul>                                                                         |
| Inspect & If Damaged, Replace Component (Even If Air Bag Did Not Deploy) | <ul style="list-style-type: none"><li>* Spiral Cable</li><li>* Steering Wheel</li><li>* Wiring Harness &amp; Connectors</li></ul>                                                                                 |
| Comments                                                                 | <ul style="list-style-type: none"><li>* If any components are damaged or bent, they must be replaced.</li><li>* Wiring for Front Air Bag Sensors can be repaired following manufacturer's instructions.</li></ul> |

## WIRING DIAGRAMS



93H75512

Fig. 14: SRS Wiring Diagram

Courtesy of Toyota Motor Sales, U.S.A., Inc.

# ALTERNATOR & REGULATOR

## 1993 Toyota Celica

1993 ELECTRICAL  
Toyota Alternators & Regulators

Celica

**WARNING:** Before performing service procedures on vehicle, disconnect negative battery cable. Wait at least one minute after disconnecting negative battery cable to avoid accidental air bag deployment on models so equipped.

### DESCRIPTION

The alternator is a small, high RPM, high performance type with an internal Integrated Circuit (IC) voltage regulator which controls charging system voltage. A transistor inside IC regulator controls alternator voltage output to maintain a constant voltage. Charging system voltage is maintained within an operating range of 13.2-15.0 volts. See ALTERNATOR REGULATED OUTPUT SPECIFICATIONS table for specific model.

When ignition is turned on, battery voltage flows from alternator terminal "L" through IC regulator to terminal "E" and to ground, causing discharge warning light to come on. When engine starts, alternator RPM increases, which increases alternator voltage output. When alternator output voltage is greater than battery voltage, voltage to recharge battery flows from terminal "B". At the same time voltage at terminal "L" increases and the potential difference between battery and terminal "L" ceases, causing discharge warning light to go off.

### TROUBLE SHOOTING

**NOTE:** See TROUBLE SHOOTING - BASIC PROCEDURES article in GENERAL INFORMATION section.

Check all fuses, fusible links, ignition switch, and appropriate relays (if equipped). Check alternator output test. See NO-LOAD TEST under ON-VEHICLE TESTING.

### ADJUSTMENTS

**NOTE:** New Belt refers to a belt which has been used 5 minutes or less on a running engine. Used Belt refers to a belt which has been used more than 5 minutes on a running engine. After installing new belt(s), run engine for at least 5 minutes and recheck tension.

### BELT TENSION

BELT TENSION SPECIFICATIONS TABLE (1)

| Application               | New Belt Lbs. (kg) | Used Belt Lbs. (kg) |
|---------------------------|--------------------|---------------------|
| 1.6L 4A-FE                |                    |                     |
| Alternator .....          | (2) 160 (73)       | (2) 130 (59)        |
| A/C .....                 | (2) 160 (73)       | (2) 100 (45)        |
| Power Steering .....      | (3) 125 (57)       | (2) 80 (36)         |
| 2.0L 3S-GTE               |                    |                     |
| Alternator With A/C ..... | (4) 165 (75)       | (2) 115 (52)        |

|                            |     |          |       |     |          |
|----------------------------|-----|----------|-------|-----|----------|
| Alternator Without A/C ... | (2) | 150 (68) | ..... | (3) | 130 (59) |
| Power Steering .....       | (3) | 125 (57) | ..... | (2) | 80 (36)  |
| 2.2L 5S-FE                 |     |          |       |     |          |
| Alternator With A/C .....  | (4) | 165 (75) | ..... | (4) | 110 (50) |
| Alternator Without A/C ... | (3) | 125 (57) | ..... | (2) | 95 (43)  |
| Power Steering .....       | (3) | 125 (57) | ..... | (2) | 80 (36)  |

- (1) - Measure belt tension with Burroughs (BT-33-73F) tension gauge.  
(2) - Plus or minus 20 lbs. (9 kg).  
(3) - Plus or minus 25 lbs. (11 kg).  
(4) - Plus or minus 10 lbs. (5 kg).

## ON-VEHICLE TESTING

### NO-LOAD TEST

1) Disconnect battery-to-alternator terminal "B" wire. See Fig. 1. Using an ammeter and voltmeter, connect negative ammeter lead to disconnected alternator terminal "B" wire end, and connect positive ammeter lead to alternator terminal "B".

2) Connect voltmeter positive lead to alternator terminal "B" and negative lead to ground. See Fig. 1.

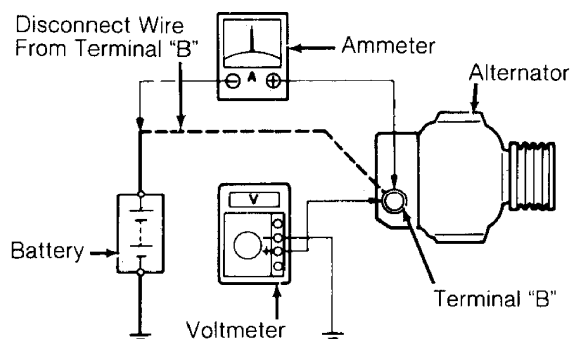
3) Start engine and increase engine speed to 2000 RPM. Both meters should read within specification. See ALTERNATOR REGULATED OUTPUT SPECIFICATIONS table.

4) If voltage is more than specified, replace IC regulator. If voltage is less than specified, ground "F" (full field) terminal. See Fig. 2. If voltage is more than specified range, replace IC regulator. If voltage is less than specified range, repair or replace alternator.

#### ALTERNATOR REGULATED OUTPUT SPECIFICATIONS (1)

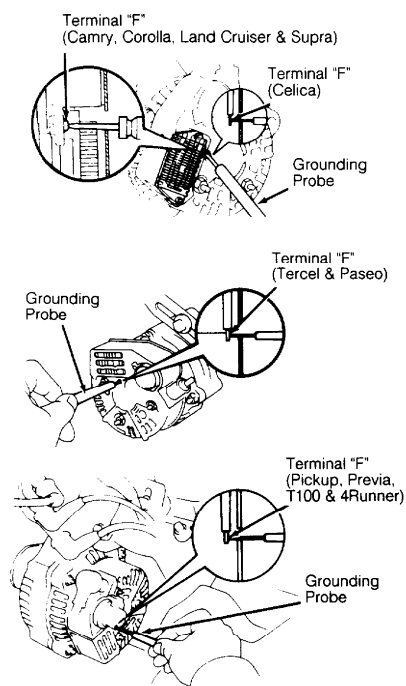
| Application  | Amps             | (2) Volts     |
|--------------|------------------|---------------|
| Celica ..... | 10 Or Less ..... | (2) 14.0-15.0 |
| Celica ..... | 10 Or Less ..... | (3) 13.5-14.3 |

- (1) - Specification given is with engine speed at 2000 RPM.  
(2) - Correct reading with temperature at 77°F (25°C).  
(3) - Correct reading with temperature at 239°F (115°C).



93F02148  
Fig. 1: Testing Charging Circuit  
Courtesy of Toyota Motor Sales, U.S.A., Inc.





93H82153

Fig. 2: Testing Alternator Full Field Output  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## LOAD TEST

**NOTE:** If battery is fully charged, disable ignition system. Crank engine for about 15 seconds to partially discharge battery.

1) Connect an ammeter as described in NO-LOAD TEST, step 1). See Fig. 1. Start engine. Turn on high beam headlights and place heater control on HI. Increase engine speed to 2000 RPM.

2) Check ammeter reading. Ammeter should read 30 amps or more. If amperage is less than specified, repair or replace alternator.

## ALTERNATOR RATED AMPERE OUTPUT SPECIFICATIONS

| Application       | (1) Amperes |
|-------------------|-------------|
| With A/C .....    | 80          |
| Without A/C ..... | 70          |

(1) - Rated output is measured with 12 volts supplied to alternator.

## BENCH TESTING

### BRUSHES

Brushes should slide smoothly in holders. Replace brushes if damaged or worn. New brush exposed length should be .413" (10.5 mm). Minimum exposed length should be more than .059" (1.5 mm). There are 2 different brush holders used, one has replaceable brushes, the other brush holder is replaced as an assembly. If exposed brush length is

less than minimum replace brushes or brush holder assembly. Install new brush springs when replacing brushes.

## ROTOR

1) Check rotor for open field windings by using an ohmmeter across slip rings. Rotor resistance should be 2.8-3.0 ohms.

2) Check rotor for shorts to ground by connecting ohmmeter between slip ring and rotor shaft. Ohmmeter should indicate no continuity. Check slip rings for wear or pitting. Standard slip ring diameter is .559"-.567" (14.2-14.4 mm) Turn slip rings on lathe if necessary. Minimum slip ring diameter is .504" (12.8 mm).

## STATOR

Connect ohmmeter between 2 stator leads. Continuity should exist between all stator leads. Connect ohmmeter between each stator lead and metal core. Continuity should not exist. If stator does not test as indicated, replace stator.

## DIODES

1) With diode/rectifier assembly removed and on bench, contact positive diode plate terminal with one ohmmeter probe. Using other ohmmeter probe, contact each of 3 diode leads in same plate. Note ohmmeter reading. Reverse ohmmeter probes, and repeat tests for all diodes.

2) All diodes should show continuity in one direction and no continuity in opposite direction. If any diode is defective, replace diode/rectifier assembly.

## OVERHAUL

### DISASSEMBLY

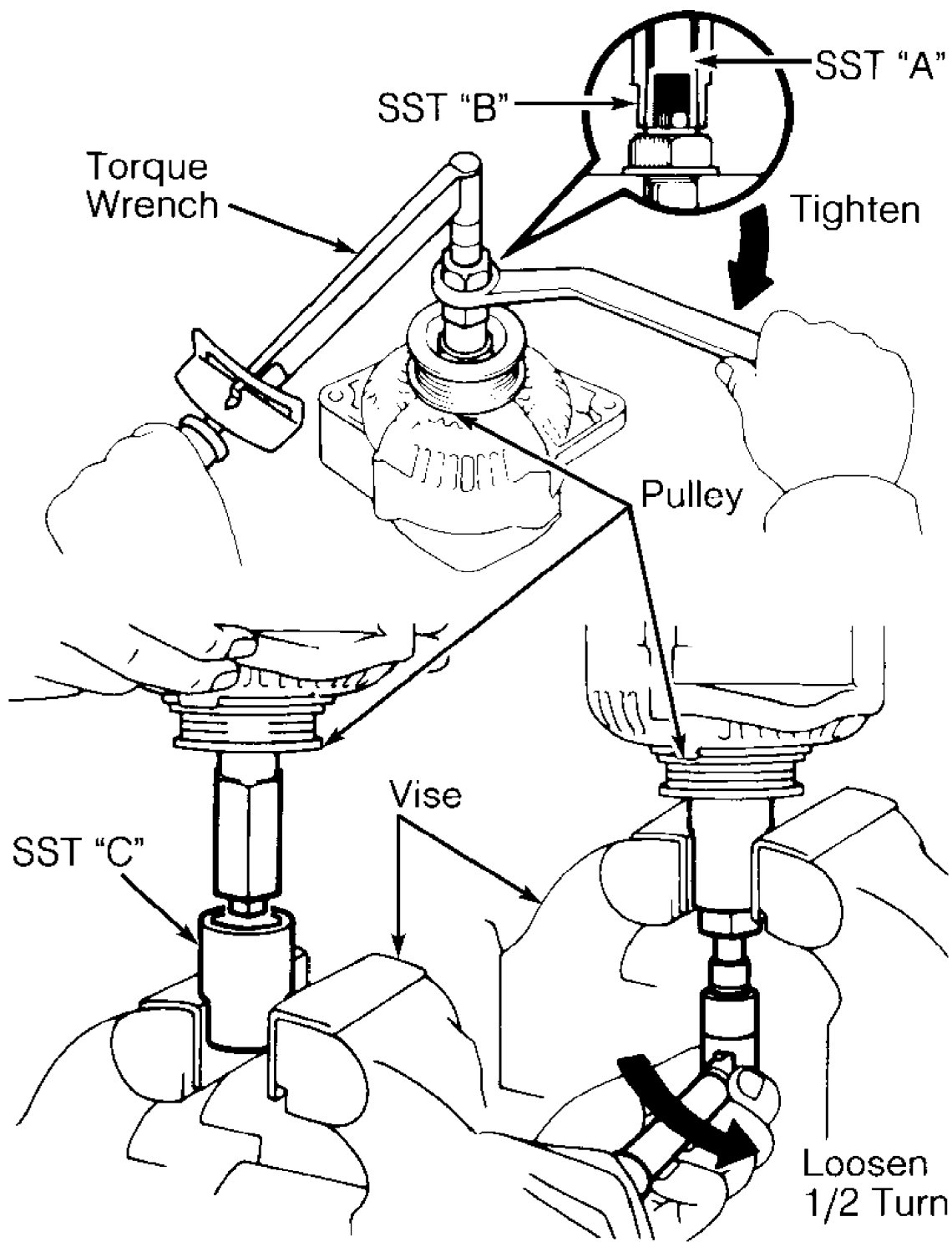
1) Remove dust cover (if equipped). Remove brush holder and IC regulator. Remove diode assembly. Remove rubber insulators or seal plate (if equipped). Use Alternator Pulley Set Nut Wrench Set (SST 09820-63010) to remove alternator pulley.

2) To remove pulley, install SST "A" and "B" to rotor shaft and tighten SST "B" clockwise to 29 ft. lbs. (39 N.m). Place SST "C" securely into a vise. Verify that SST "A" is secured to rotor shaft and install SST "A" and "B" and alternator into SST "C". See Fig. 3. Turn SST "A" in direction shown in Fig. 3 to loosen pulley nut. To prevent damage to rotor shaft, Do Not loosen pulley nut more than 1/2 turn. Remove SST "A" and "B", and alternator from SST "C". Remove SST "A" and "B" from rotor shaft and remove pulley nut and pulley.

3) Remove rectifier and frame using appropriate pulley. Remove alternator washer (if equipped). Remove rotor from drive end frame (stator).

### REASSEMBLY

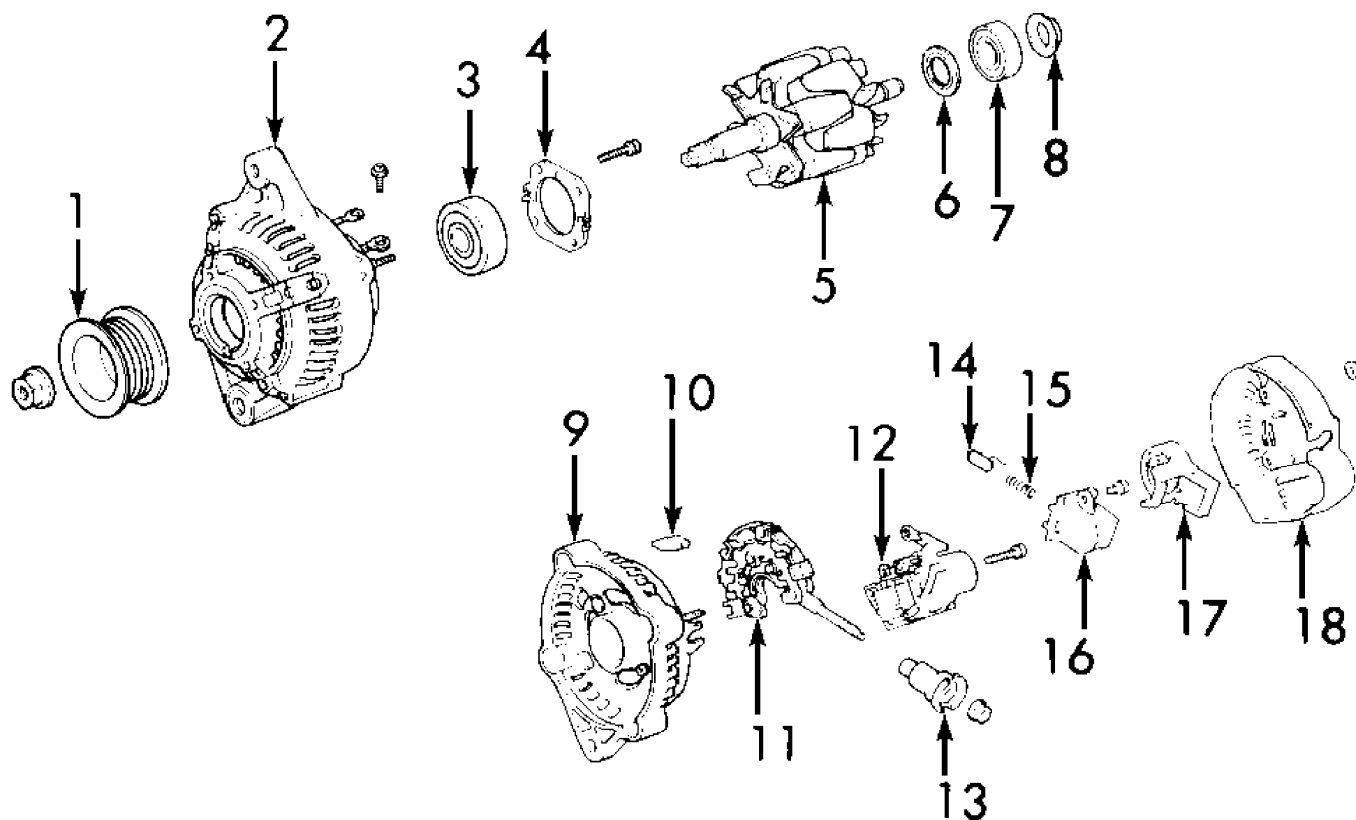
To assembly reverse disassembly procedures. Use Alternator Pulley Set Nut Wrench Set (SST 09820-63010) to install alternator pulley. Place SST "C" securely into a vise. Install SST "A" and "B" and alternator into SST "C". See Fig. 3. Tighten pulley nut to 81 ft. lbs. (110 N.m) in the opposite direction shown in Fig. 3. After completing assembly, verify rotor turns smoothly



93J82155

Fig. 3: Removing Alternator Pulley  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

NOTE: See Figs. 4-7 for exploded views of alternators which are similar.

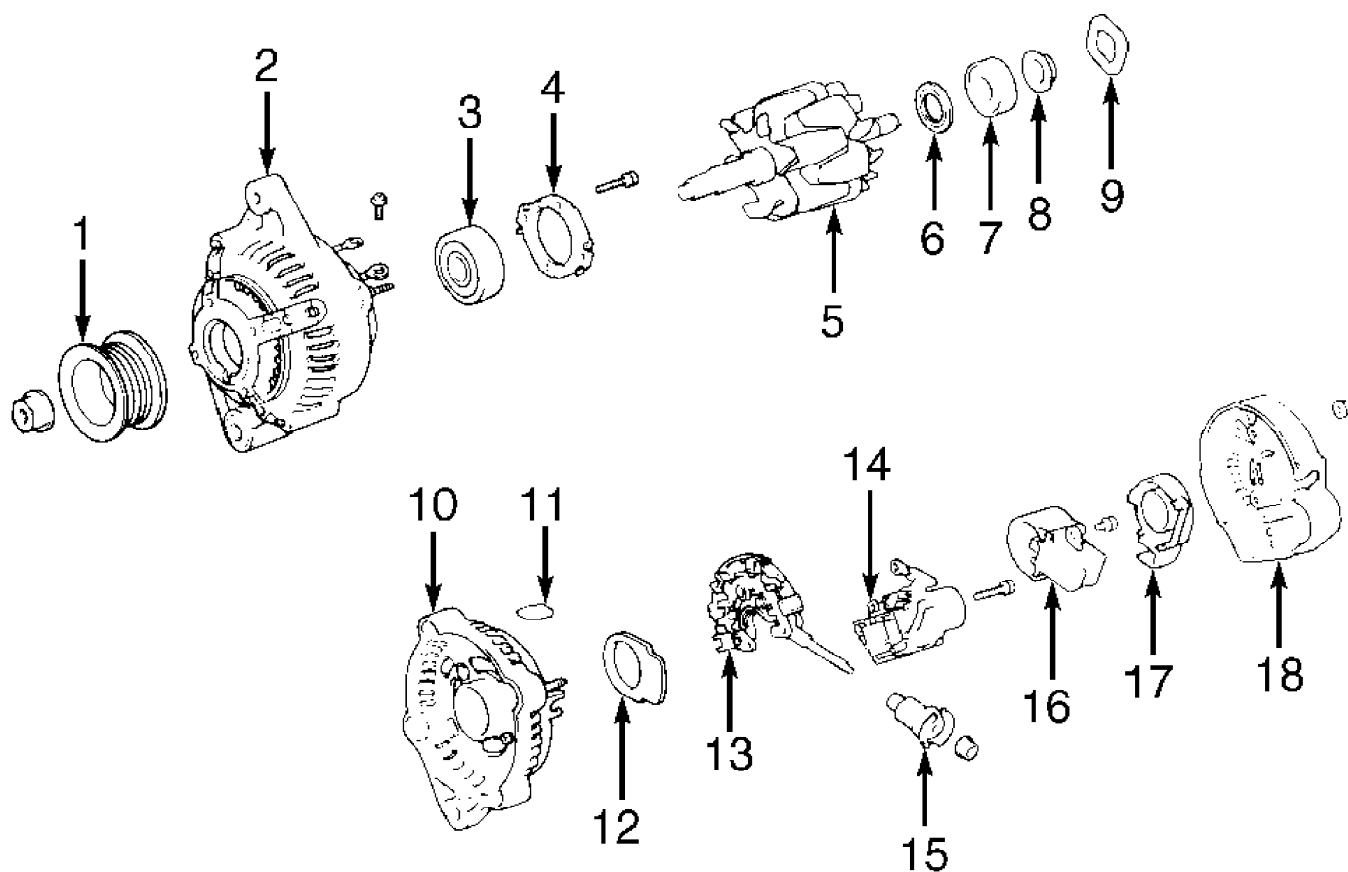


1. Pulley
2. Drive End Frame (Stator)
3. Front Bearing
4. Retainer
5. Rotor
6. Bearing Cover
7. Rear Bearing
8. Bearing Cover
9. Diode End Frame

10. Rubber Insulator
11. Diode Assembly
12. IC Regulator
13. Terminal Insulator
14. Brush
15. Spring
16. Brush Holder
17. Brush Holder Cover
18. Rear End Cover

93A82156

Fig. 4: Exploded View Of Alternator (4A-FE M/T)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

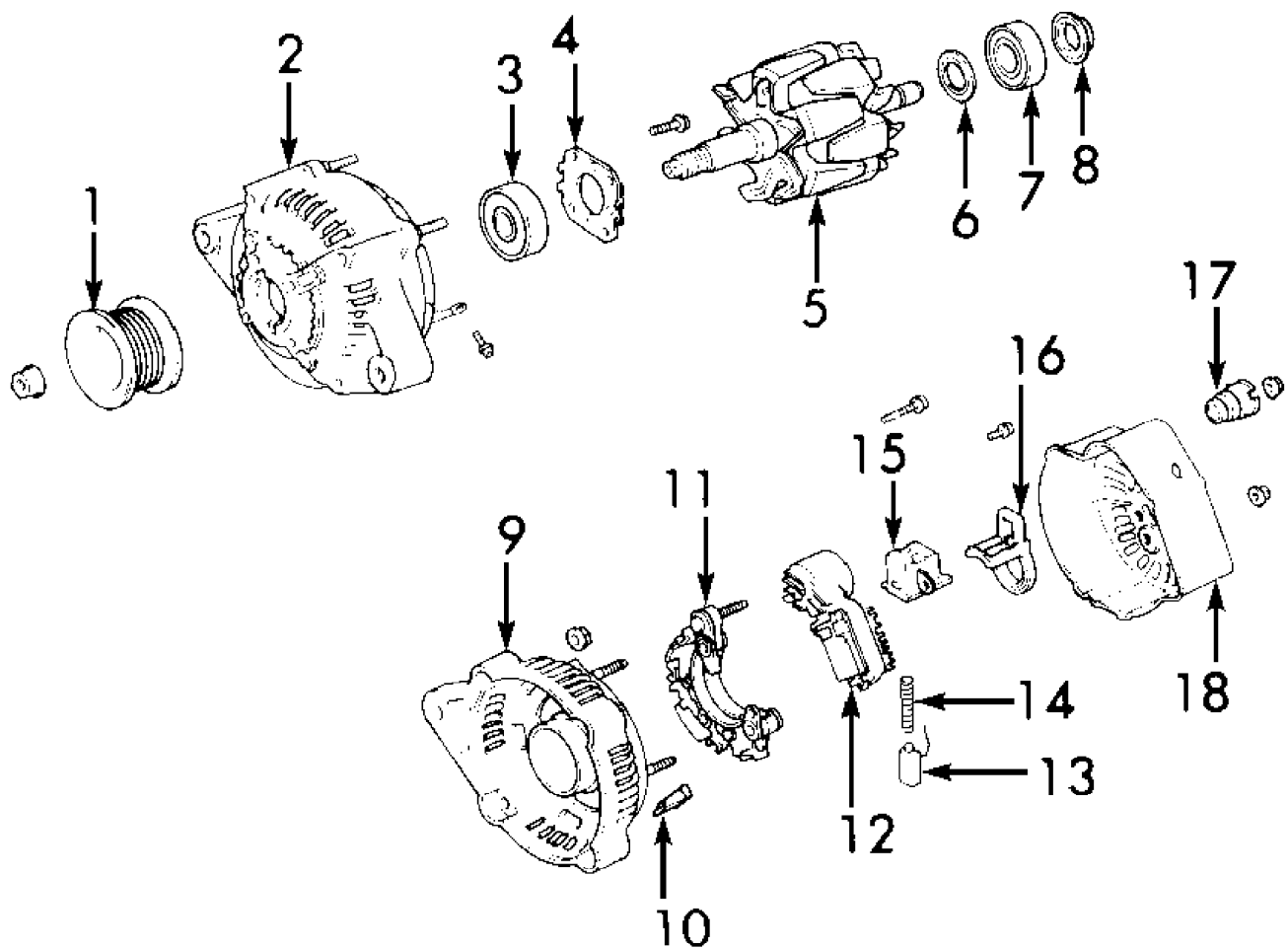


1. Pulley
2. Drive End Frame (Stator)
3. Front Bearing
4. Retainer
5. Rotor
6. Bearing Cover
7. Rear Bearing
8. Bearing Cover
9. Thrust Washer

10. Diode End Frame
11. Rubber Insulator
12. Seal Plate
13. Diode Assembly
14. IC Regulator
15. Terminal Insulator
16. Brush Holder & Brushes
17. Brush Holder Cover
18. Rear End Cover

93B82157

Fig. 5: Exploded View Of Alternator (4A-FE A/T)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

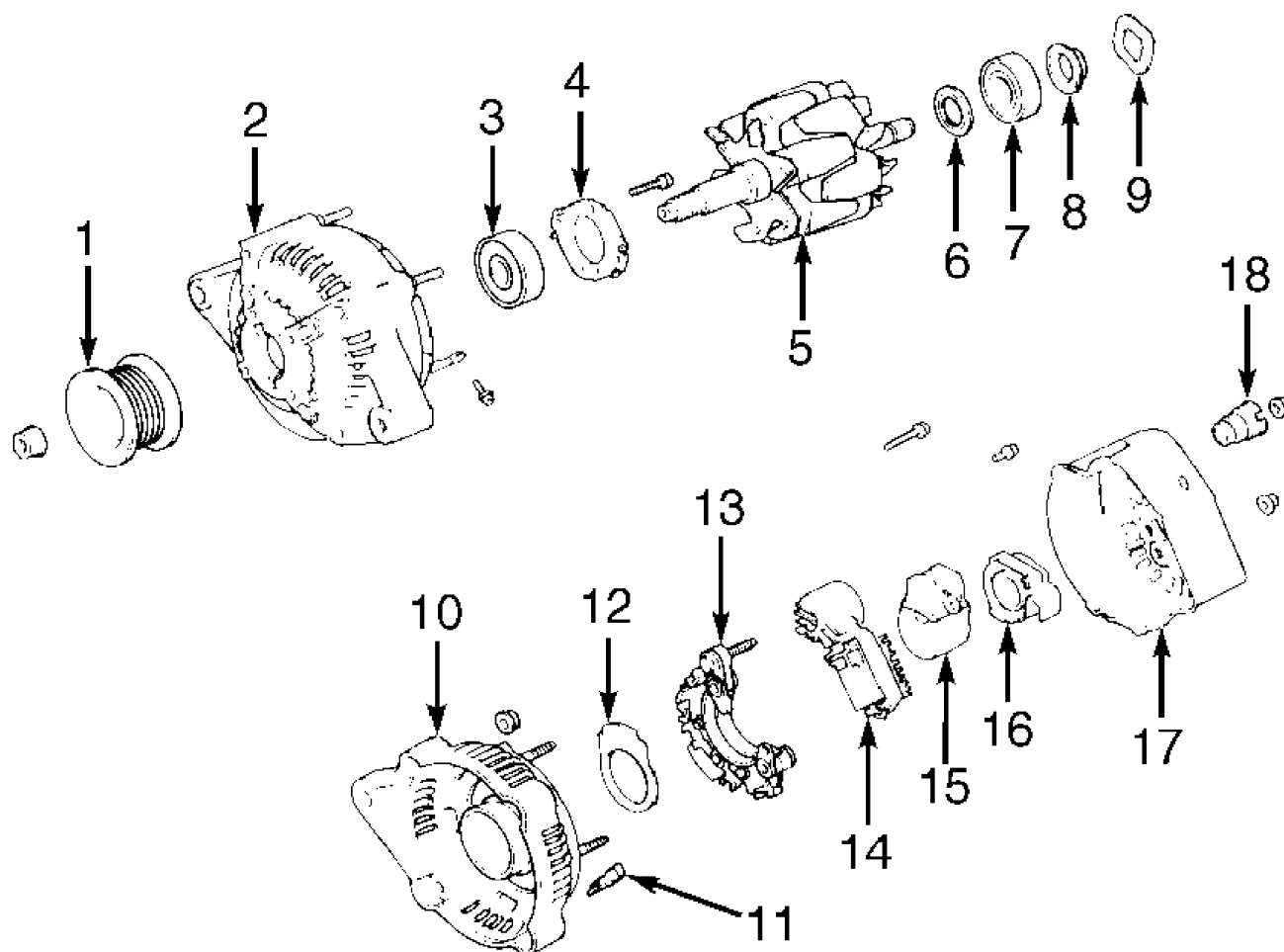


1. Pulley
2. Drive End Frame (Stator)
3. Front Bearing
4. Retainer
5. Rotor
6. Bearing Cover
7. Rear Bearing
8. Bearing Cover
9. Diode End Frame

10. Rubber Insulator
11. Diode Assembly
12. IC Regulator
13. Brush
14. Spring
15. Brush Holder
16. Brush Holder Cover
17. Terminal Insulator
18. Rear End Frame

93C82158

Fig. 6: Exploded View Of Alternator (5S-FE M/T)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.



- |                             |                            |
|-----------------------------|----------------------------|
| 1. Pulley                   | 10. Diode End Frame        |
| 2. Drive End Frame (Stator) | 11. Rubber Insulator       |
| 3. Front Bearing            | 12. Seal Plate             |
| 4. Retainer                 | 13. Diode Assembly         |
| 5. Rotor                    | 14. IC Regulator           |
| 6. Bearing Cover            | 15. Brush Holder & Brushes |
| 7. Rear Bearing             | 16. Brush Holder Cover     |
| 8. Bearing Cover            | 17. Rear End Cover         |
| 9. Thrust Washer            | 18. Terminal Insulator     |

93D82159

Fig. 7: Exploded View Of Alternator (3S-GTE & 5S-FE A/T)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS

See appropriate chassis WIRING DIAGRAMS article in WIRING  
DIAGRAMS section.



# ANTI-LOCK BRAKE SYSTEM

1993 Toyota Celica

1993 BRAKES  
Toyota Anti-Lock  
Celica

## DESCRIPTION

The Anti-Lock Brake System (ABS) consists of an ABS Electronic Control Unit (ECU), control relay, actuator, deceleration sensor (All-Trac), and 4 speed sensors. See Fig. 1.

An ABS indicator light is located on the instrument panel. This light comes on for 3 seconds as a bulb test when ignition is first turned on. A primary check is performed after each engine start, and initial vehicle speed exceeds 4 MPH. If brake pedal is pressed before vehicle exceeds 4 MPH, primary check will not occur until brake pedal is released.

NOTE: For more information on brake system, see BRAKE SYSTEM article in this section.

## OPERATION

Under normal driving conditions, the ABS functions as a standard brake system. With detection of wheel lock-up, short pedal pulsations occurring in rapid succession will be felt in brake pedal. Pedal pulsation will continue until there is no longer a need for ABS function.

CAUTION: See ANTI-LOCK BRAKE SAFETY PRECAUTIONS in this article.

## ANTI-LOCK BRAKE SAFETY PRECAUTIONS

- \* NEVER open a bleeder valve or loosen a hydraulic line while ABS is pressurized
- \* NEVER disconnect or reconnect any electrical connectors while ignition is on. Damage to ABS control unit may result.
- \* DO NOT attempt to bleed hydraulic system without first referring to the appropriate article.
- \* Only use specially designed brake hoses/lines on ABS-equipped vehicles.
- \* DO NOT tap on speed sensor components (sensor, sensor rings). Speed rings must be pressed, NOT hammered into hubs. Striking these components can cause demagnetization or a loss of polarization, affecting the accuracy of the speed signal returning to the ABS control unit.
- \* DO NOT mix tire sizes. Increasing the width, as long as tires remain close to the original diameter, is acceptable. Rolling diameter must be identical for all 4 tires. Some manufacturers recommend tires of the same brand, style and type. Failure to follow this precaution may cause inaccurate wheel speed readings.
- \* DO NOT contaminate speed sensor components with grease. Only use recommended anti-corrosion coating.
- \* When speed sensor components have been removed, ALWAYS check sensor-to-ring air gaps when applicable. These specifications can be found in each appropriate article.
- \* ONLY use recommended brake fluids. DO NOT use silicone brake fluids in an ABS-equipped vehicle.

- \* When installing transmitting devices (CB's, telephones, etc.) on ABS-equipped vehicles, DO NOT locate the antenna near the ABS control unit (or any control unit).
- \* Disconnect all on-board computers, when using electric welding equipment.
- \* DO NOT expose the ABS control unit to prolonged periods of high heat (185°F/85°C for 2 hours is generally considered a maximum limit).

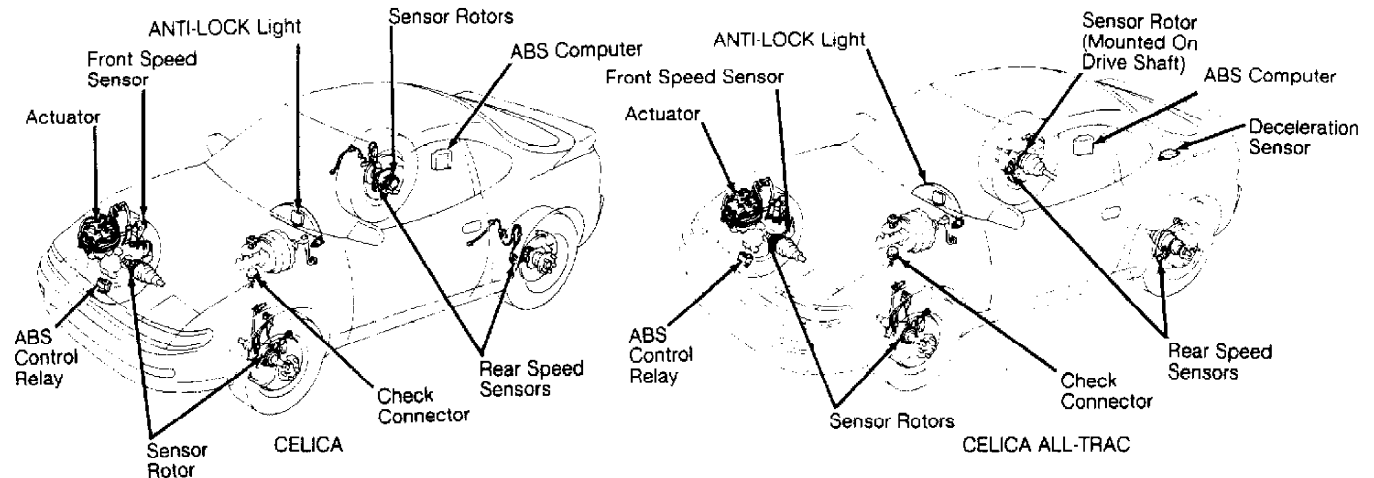


Fig. 1: Locating ABS Components  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## BLEEDING BRAKE SYSTEM

**CAUTION:** Brake fluid will damage painted surfaces. If brake fluid gets on a painted surface, wipe off immediately and clean with alcohol. Use only DOT 3 brake fluid from a sealed container. Do not mix brake fluid with any other type.

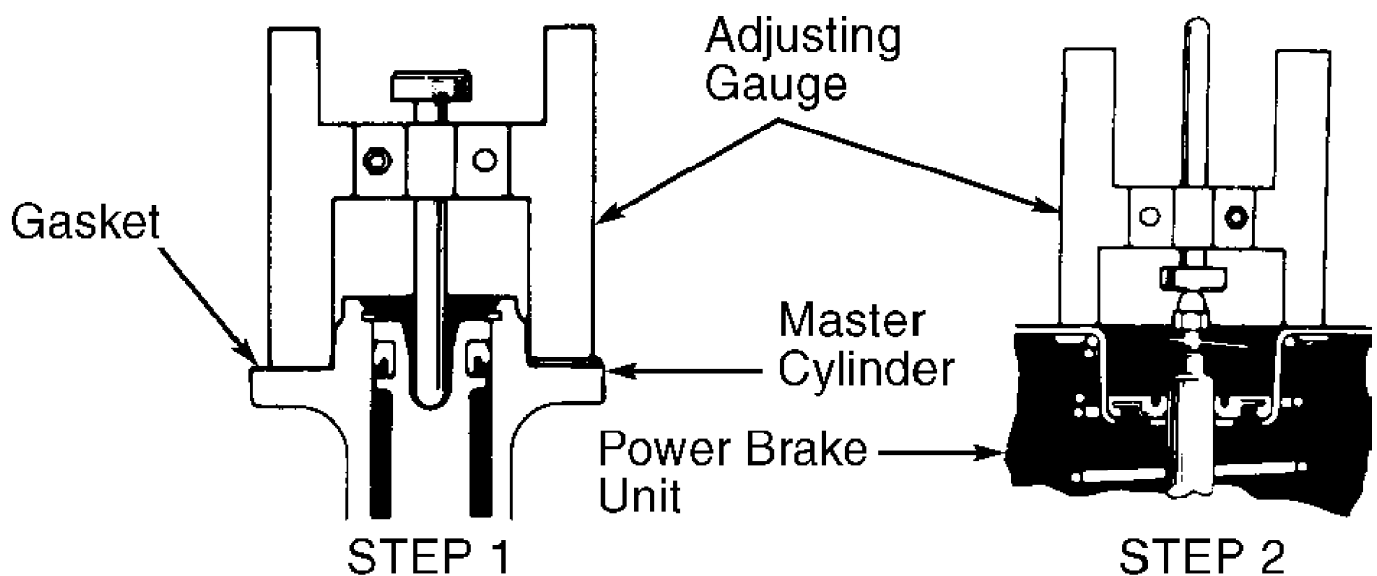
Brake bleeding procedure is same procedure used to bleed non-ABS systems. If master cylinder was rebuilt or reservoir ran empty, bleed master cylinder first. Bleed remaining wheels, starting with brake having longest hydraulic line, working to brake with shortest hydraulic line.

## ADJUSTMENTS

### MASTER CYLINDER PUSH ROD

1) Install Adjusting Gauge (09737-00010) onto master cylinder, with master cylinder gasket in place. Lower gauge pin until it just touches master cylinder piston. See Fig. 2 (STEP 1). Invert gauge, then install onto power brake unit (STEP 2).

2) Measure clearance between brake unit push rod and head of adjusting gauge. Clearance should be zero. If clearance is not zero, adjust brake unit push rod length until push rod just touches head of gauge pin.



92A01556

Fig. 2: Adjusting Master Cylinder Push Rod  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### BRAKE PEDAL HEIGHT

1) Measure brake pedal height from face of brake pedal pad to rib of floor panel. See Fig. 3. Brake pedal height should be 6.7-7.0" (169-178 mm). To adjust brake pedal height, loosen stoplight switch and lock nut on brake pedal push rod.

2) Adjust pedal height by rotating push rod. After adjusting brake pedal height, tighten lock nut. Adjust stoplight switch. See STOPLIGHT SWITCH. Check and adjust brake pedal free play. See BRAKE PEDAL FREE PLAY.

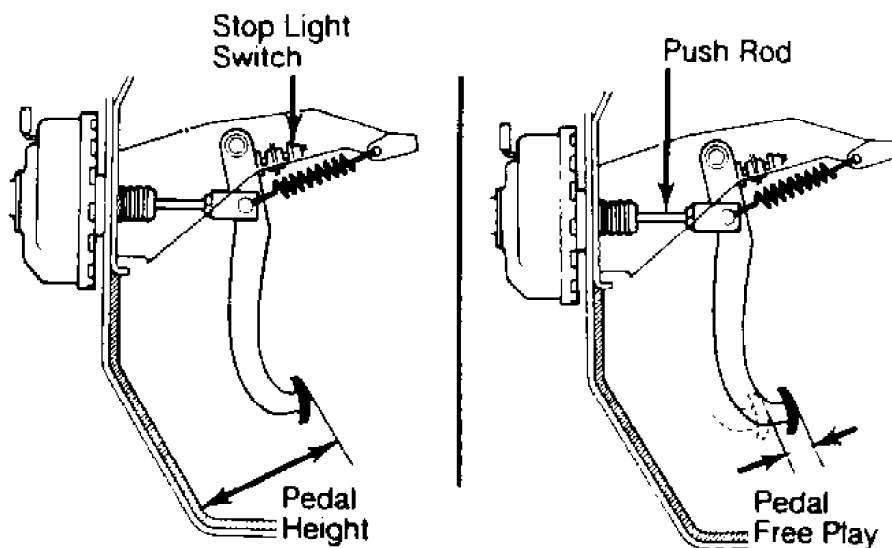


Fig. 3: Measuring Brake Pedal Height & Free Play  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### BRAKE PEDAL FREE PLAY

1) Brake pedal free play is distance brake pedal travels (with engine stopped) before encountering resistance. To measure brake pedal free play, press brake pedal several times to exhaust vacuum from power brake unit. Press brake pedal and measure travel until initial resistance occurs.

2) Brake pedal free play should be 0.04-0.24" (1.0-6.0 mm). If free play is not within specification, adjust by rotating push rod. See Fig. 3. After adjusting brake pedal free play, check brake pedal height. See BRAKE PEDAL HEIGHT.

## **BRAKE PEDAL RESERVE**

Measure brake pedal reserve from face of brake pedal pad to asphalt sheet under carpet, with engine running and force of 110 lbs. (50 kg) applied to brake pedal. Minimum brake pedal reserve should be 3.54" (90.0 mm). If distance is less than specified, inspect brake system.

## **STOPLIGHT SWITCH**

Stoplight switch is located above brake pedal. See Fig. 3. To adjust stoplight switch, loosen stoplight switch lock nuts and rotate stoplight switch until clearance between pedal stop and threaded end of switch is 0.02-0.09" (0.5-2.4 mm). Tighten lock nut. Check stoplight operation.

## **PARKING BRAKE SHOES**

Raise and support vehicle. Remove wheels. Temporarily install lug nuts to hold brake rotor in place. Remove hole plug to gain access to adjuster. Turn adjuster to expand shoes until brake rotor locks. Back off adjuster 8 notches. Install hole plug.

## **PARKING BRAKE**

**NOTE:** Service brake on rear drum brakes and parking brake shoe clearance on rear disc brakes must be adjusted before adjusting parking brake cable.

Parking brake lever stroke should be 4-7 notches (clicks) with a pull force of 44 lbs. (20 kg). To adjust stroke, remove console box. Loosen parking brake cable lock nut. Rotate adjuster nut until parking brake lever travel is as specified. Tighten lock nut. Install console box.

## **TROUBLE SHOOTING**

### **ANTI-LOCK WARNING LIGHT**

ABS Light Comes On

1) Unplug service connector. See Fig. 4. Connect jumper wire between Data Link Connector (DLC) terminals Tc and E1. See Fig. 5. Turn ignition on. If ABS light flashes codes, see DIAGNOSTIC CODES under DIAGNOSIS & TESTING. If ABS light stays on constantly or flashes 4 times per second, go to next step.

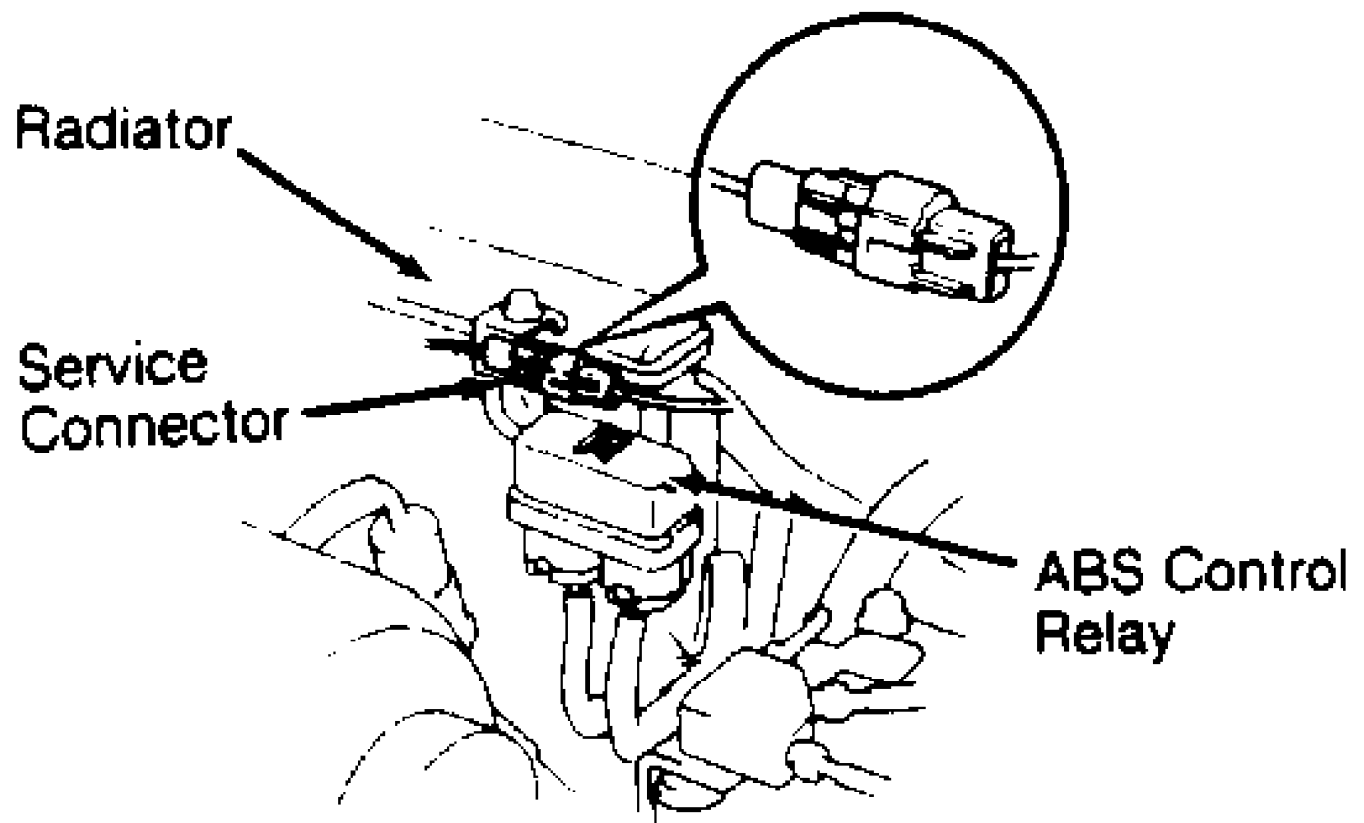
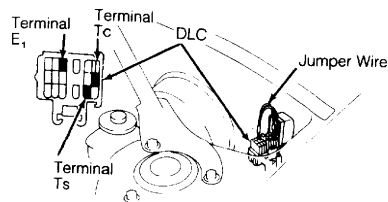


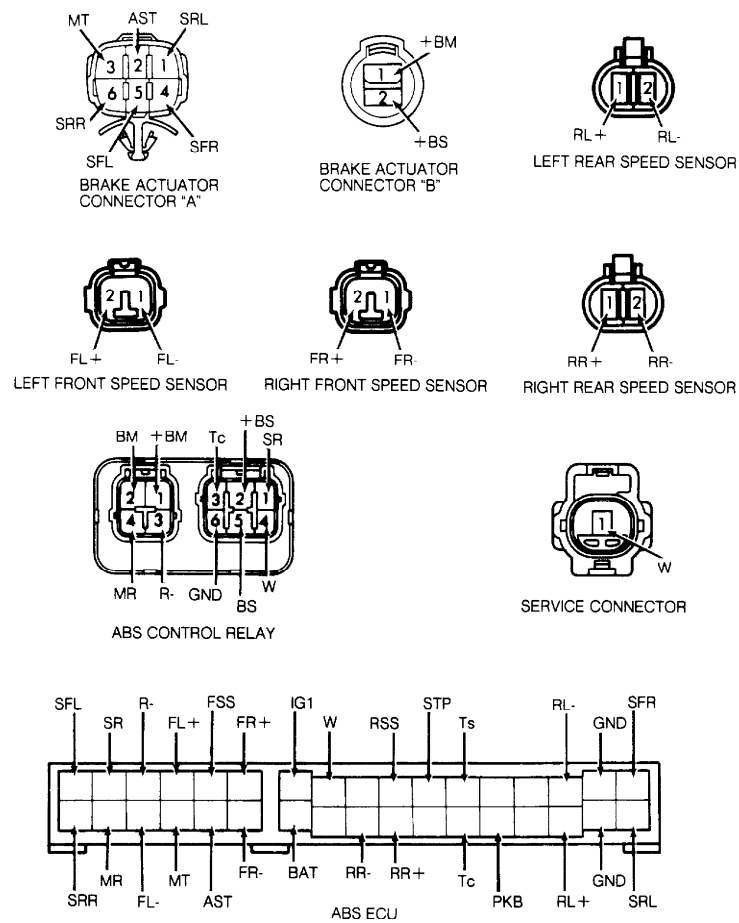
Fig. 4: Locating Service Connector  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93D83991  
Fig. 5: Locating Data Link Connector DLC  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

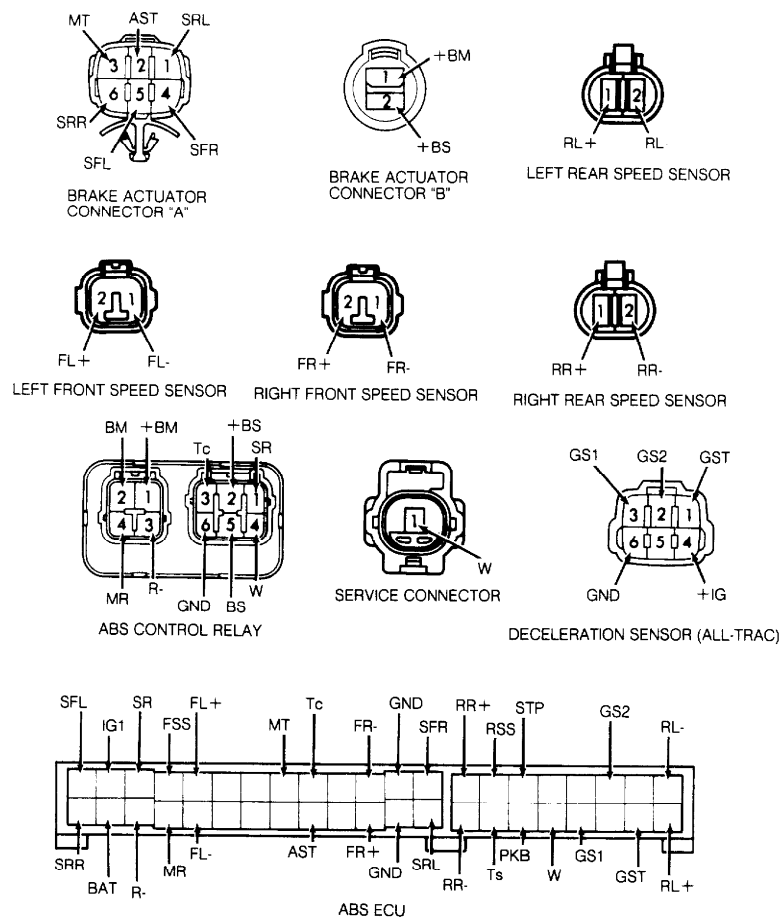
2) Inspect ABS ECU connector for secure attachment and undamaged terminals. Repair connector as necessary. Turn ignition on. Check for battery voltage between ABS ECU connector terminal IG1 (Black/Red wire) and ground. See Fig. 6 or 7. If battery voltage exists, go to next step. If battery voltage does not exist, repair faulty Black/Red wire.

3) Turn ignition off. Unplug ABS ECU connector. Turn ignition on. If ABS light comes on, repair short circuit between 6-pin ABS control relay connector terminal No. 4 (Green wire) and ABS ECU connector terminal "W" (Gray wire). If ABS light does not come on, temporarily substitute known good ABS ECU. Retest system.



93E83992

Fig. 6: Identifying ABS System Connector Terminals (FWD)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.



93F83993

Fig. 7: Identifying ABS System Connector Terminals (All-Trac)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

ABS Light Does Not Come On For 3 Seconds After Ignition Is Turned On

1) Unplug service connector. See Fig. 4. Connect service connector terminal "W" on harness (female) side to ground. Turn ignition on. If ABS light comes on, go to next step. If ABS light does not come on, replace ABS bulb, or repair open circuit between ABS light and service connector terminal "W". See Fig. 6 or 7.

2) Turn ignition off. Unplug ABS ECU connector and ABS control relay connectors. Connect ABS ECU connector terminal "W" (Gray wire) on wire harness side to ground. Turn ignition on. If ABS light comes on, go to next step. If ABS light does not come on, repair open circuit between ABS ECU connector terminal "W" (Gray wire) and ABS light.

3) Turn ignition off. Unplug ABS control relay connector. Check for continuity between 6-pin ABS control relay connector terminals No. 4 and 5. Transpose ohmmeter leads. Again check for continuity.

4) If continuity exists with ohmmeter connected only one way, temporarily substitute known good ABS ECU, then retest system. If continuity exists with ohmmeter connected both ways, or continuity does not exist at all, see ABS CONTROL RELAY under COMPONENT TESTING.

5) If short in ABS control relay diode exists, a malfunction at ABS ECU terminal "W" (Gray wire) will occur. When inspecting terminal, reconnect ABS wiring. Unplug control relay connectors and

DLC. Turn ignition on. If ABS light comes on, ABS ECU terminal "W" (Gray wire) is okay.

#### ABS Light Comes On Intermittently

Inspect for a short in wiring harness between DLC terminals Tc and E1 or between terminals Ts and E1. See Fig. 5.

## SYMPTOM DIAGNOSIS

### Symptoms:

- \* Brakes Pull
- \* Braking Inefficient
- \* ABS Operates During Normal Braking
- \* ABS Operates Just Before Stopping During Normal Braking
- \* Brake Pedal Pulsates Abnormally With ABS Functioning

### Diagnosis

1) Unplug service connector. See Fig. 4. Connect jumper wire between DLC terminals Tc and E1. See Fig. 5. Turn ignition on. If ABS light flashes 2 times per second, go to next step. If ABS light does not flash 2 times per second, refer to DIAGNOSTIC CODES under DIAGNOSIS & TESTING.

2) On FWD vehicles, go to next step. On All-Trac vehicles, enter deceleration sensor diagnostics. See DECELERATION SENSOR DIAGNOSTICS (ALL-TRAC) under DIAGNOSIS & TESTING. If deceleration sensor operation is okay, go to next step. If deceleration sensor operation is not as specified, inspect sensor for proper installation. If sensor is installed properly, replace sensor.

3) Check actuator operation. See ACTUATOR CHECK under DIAGNOSIS & TESTING. If actuator operation is okay, temporarily substitute known good ECU. Retest system. See ABS ECU under REMOVAL & INSTALLATION. If actuator operation is not as specified, replace faulty actuator.

4) Enter speed sensor diagnostics. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING. If speed sensor signal level is okay, go to next step. If speed sensor signal level is not okay, inspect sensor and sensor rotor. Replace as necessary.

5) Inspect all speed sensors for proper installation, and repair as necessary. Clean all foreign material and metal chips from speed sensor tip.

6) Turn ignition off. Unplug ABS ECU connector. Ensure continuity exists between each speed sensor connector and ABS ECU connector. If continuity readings change when wiring harness is twisted or moved, repair faulty wiring harness between speed sensor and ABS ECU connector.

### Symptom:

- \* ABS Works Inefficiently

### Diagnosis

1) Unplug service connector. See Fig. 4. Connect jumper wire between DLC terminals Tc and E1. See Fig. 5. Turn ignition on. If ABS light flashes 2 times per second, go to next step. If ABS light does not flash 2 times per second, refer to DIAGNOSTIC CODES under DIAGNOSIS & TESTING.

2) Check for battery voltage between ABS ECU connector terminal STP (Green/White wire) and ground when brake pedal is pressed. If battery voltage does not exist, repair open circuit in stoplight switch or wiring harness. Check actuator operation. See ACTUATOR CHECK under DIAGNOSIS & TESTING.



## DIAGNOSIS & TESTING

NOTE: DO NOT start engine when retrieving diagnostic codes.

### RETRIEVING CODES

1) Ensure battery voltage is about 12 volts. Turn ignition on. ABS light should come on, then go out after 3 seconds. If warning light does not come on, check fuse, bulb, and wiring harness.

2) With ignition on, unplug DLC. See Fig. 4. Connect jumper wire between DLC terminals Tc and E1. See Fig. 5. If a malfunction is detected, 4 seconds will elapse, then ABS light will begin to flash a 2-digit code. First series of flashes indicates first digit of code. After a 1.5-second pause, second series of flashes indicates second digit of code.

3) If 2 or more codes are stored, there will be a 2.5-second pause between each code. After all codes are displayed, a 4-second pause will occur, then all codes will repeat. If ABS system is functioning properly, ABS light will flash 2 times every second. For code interpretation, see DIAGNOSTIC CODES.

4) After replacing or repairing malfunctioning components, clear diagnostic codes. If a battery cable was disconnected during repairs, all codes will be erased. If battery cable was not disconnected during repairs, see CLEARING CODES.

### DIAGNOSTIC CODES

Check suspect components in order given. Checks consist mainly of a visual inspection and continuity tests.

#### Code 11

Open in solenoid relay circuit. Check actuator wiring harness, ABS control relay (solenoid relay), solenoid relay wiring harness, and solenoid relay connector. See ABS CONTROL RELAY under COMPONENT TESTING.

#### Code 12

Short in solenoid relay circuit. Check actuator wiring harness, ABS control relay (solenoid relay), solenoid relay wiring harness, and solenoid relay connector. See ABS CONTROL RELAY under COMPONENT TESTING.

#### Code 13

Open in pump motor relay circuit. Check actuator wiring harness, ABS control relay (pump motor relay), pump motor relay wiring harness, and pump motor relay connector. See ABS CONTROL RELAY under COMPONENT TESTING.

#### Code 14

Short in pump motor relay circuit. Check actuator wiring harness, ABS control relay (pump motor relay), pump motor relay wiring harness, and pump motor relay connector. See ABS CONTROL RELAY under COMPONENT TESTING.

#### Code 21

Open or short circuit in solenoid for right front wheel. Check actuator solenoid, wiring harness, and connector. See ACTUATOR CHECK under DIAGNOSIS & TESTING.

#### Code 22

Open or short circuit in solenoid for left front wheel. Check actuator solenoid, wiring harness, and connector. See ACTUATOR CHECK

under DIAGNOSIS & TESTING.

Code 23

Open or short circuit in solenoid for right rear wheel. Check actuator solenoid, wiring harness, and connector. See ACTUATOR CHECK under DIAGNOSIS & TESTING.

Code 24

Open or short circuit in solenoid for left rear wheel. Check actuator solenoid, wiring harness, and connector. See ACTUATOR CHECK under DIAGNOSIS & TESTING.

Code 31

Malfunction of right front wheel speed sensor signal. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 32

Malfunction of left front wheel speed sensor signal. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 33

Malfunction of right rear wheel speed sensor signal. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 34

Malfunction of left rear wheel speed sensor signal. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 35

Open in left front or right rear wheel speed sensor circuit. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 36

Open in right front or left rear wheel speed sensor circuit. Check speed sensor, sensor rotor, wiring harness, and connector. See SPEED SENSOR DIAGNOSTICS under DIAGNOSIS & TESTING.

Code 37 (FWD)

Wrong rear axle hubs on both sides. Check rear sensor rotors.

Code 41

Battery voltage is less than 9.5 volts or more than 16.2 volts. Check battery and voltage regulator.

Code 43 (All-Trac)

Deceleration sensor malfunction. Check deceleration sensor, deceleration sensor installation, wiring harness, and connector. See DECELERATION SENSOR DIAGNOSTICS (ALL-TRAC) under DIAGNOSIS & TESTING.

Code 44 (All-Trac)

Open or short circuit in deceleration sensor. Check deceleration sensor, deceleration sensor installation, wiring harness, and connector. See DECELERATION SENSOR DIAGNOSTICS (ALL-TRAC) under DIAGNOSIS & TESTING.

Code 51

Actuator pump motor is locked or pump motor circuit open. Check pump motor, ABS control relay (pump motor relay), vehicle

battery, actuator wiring harness, connectors, actuator pump motor circuit, and actuator ground bolt. See ACTUATOR CHECK under DIAGNOSIS & TESTING. See ABS CONTROL RELAY under COMPONENT TESTING.

#### ABS Light Always On

Malfunction of ABS ECU. Inspect ABS ECU connector for proper installation and undamaged terminals. Repair as necessary. If connector is okay, temporarily substitute known good ABS ECU. Retest system.

### CLEARING CODES

Turn ignition on. Connect jumper wire between DLC terminals Tc and E1. See Fig. 5. With vehicle stopped, press brake pedal 8 or more times within 3 seconds. Codes will be erased. Turn ignition on. Verify ABS light goes out after 3 seconds. Verify ABS light flashes a normal code. See RETRIEVING CODES under DIAGNOSIS & TESTING.

### ACTUATOR CHECK

1) Turn ignition off. Unplug actuator and control relay electrical connectors. See Fig. 6 or 7. Connect Test Harness (09990-00200) and Actuator Checker (09990-00150) to vehicle according to manufacturer's instructions.

2) Place Sheet "A" (09990-00163) onto actuator checker. Start and idle engine. Set SELECTOR switch to FRONT RH position. Press and hold MOTOR switch for a few seconds. Press and hold brake pedal.

NOTE: DO NOT press POWER switch for longer than 10 seconds.

3) Press POWER switch. When POWER switch is pressed, brake pedal should not go down. Release POWER switch. Brake pedal should go down. Press and hold MOTOR switch for a few seconds. Brake pedal should return. Release MOTOR switch. Release brake pedal.

4) Press and hold MOTOR switch for a few seconds. Release MOTOR switch. Press and hold brake pedal for about 15 seconds. As brake pedal is held, press MOTOR switch for a few seconds. Brake pedal should not pulsate.

5) Repeat steps 2) through 4) for FRONT LH, REAR RH, and REAR LH by setting SELECTOR switch to appropriate positions. When checking REAR LH position, press REAR LH switch instead of POWER switch.

6) After checking remaining wheels, press and hold MOTOR switch for a few seconds. Remove test harnesses and actuator checker.

7) If any actuator solenoid does not operate as specified, measure actuator solenoid resistance. Measure resistance of each actuator solenoid between 6-pin connector terminal No. 4 (White wire) and appropriate actuator solenoid connector. Resistance should be 1.2 ohms.

8) Replace actuator if resistance of any actuator solenoid is not as specified. Reconnect wiring harnesses to actuator and control relay. Clear diagnostic codes. See CLEARING CODES.

### SPEED SENSOR DIAGNOSTICS

NOTE: While diagnosing speed sensors, brake system functions as a conventional system.

1) Ensure battery voltage is about 12 volts. Turn ignition on. ABS light should come on, then go out after about 3 seconds. If ABS light does not come on, check fuse, bulb, and wiring harness.

2) Turn ignition off. DO NOT remove short pin from DLC. Connect DLC terminals Ts and E1. See Fig. 5. Engage parking brake. Start engine. Verify ABS light flashes 4 times per second.

3) Drive vehicle straight ahead at speed greater than 50 MPH. Verify ABS light flashes while vehicle is traveling at less than 28 MPH. ABS light will stop flashing at more than 28 MPH, and will then flash once at 50 MPH. If ABS light flashes as described, check is complete. If ABS light does not flash as indicated, stop vehicle.

4) Connect jumper between DLC terminals Tc and E1. See Fig. 5. If a malfunction is detected, 4 seconds will elapse, then ABS light will begin to flash a 2-digit code. First series of flashes indicates first digit of code. After a 1.5 second pause, second series of flashes indicates second digit of code.

5) If 2 or more codes are stored, there will be a 2.5-second pause between each code. After all codes are flashed, there will be a 4-second pause, then all codes will repeat.

6) Record diagnostic codes. Turn ignition off. Repair as necessary. See SPEED SENSOR DIAGNOSTIC CODES. Remove jumper wire from DLC. Clear diagnostic codes. See CLEARING CODES.

## **SPEED SENSOR DIAGNOSTIC CODES**

### **Code 71**

Low voltage of right front speed sensor signal. Check right front speed sensor and sensor installation.

### **Code 72**

Low voltage of left front speed sensor signal. Check left front speed sensor and sensor installation.

### **Code 73**

Low voltage of right rear speed sensor signal. Check right rear speed sensor and sensor installation.

### **Code 74**

Low voltage of left rear speed sensor signal. Check left rear speed sensor and sensor installation.

### **Code 75**

Abnormal signal from right front speed sensor. Check right front sensor rotor.

### **Code 76**

Abnormal signal from left front speed sensor. Check right front sensor rotor.

### **Code 77**

Abnormal signal from right rear speed sensor. Check right rear sensor rotor.

### **Code 78**

Abnormal signal from left rear speed sensor. Check left rear sensor rotor.

ABS Light Blinks 4 Times A Second

All speed sensors and sensor rotors are normal.

## **DECELERATION SENSOR DIAGNOSTICS (ALL-TRAC)**

**NOTE:** While diagnosing deceleration sensor, brake system functions as a conventional system.

### **System Inspection**

1) Ensure battery voltage is about 12 volts. Turn ignition on. ABS light should come on, then go out after 3 seconds. If ABS light does not come on, check fuse, bulb, and wiring harness.

2) Turn ignition off. Connect jumper wire between DLC terminals E1 and Ts. See Fig. 5. Engage parking brake. Start engine. DO NOT press brake pedal.

3) If ABS flashes 4 times per second, inspect sensor detection point and sensor operation. If ABS light does not flash, inspect parking brake switch, stoplight switch, DLC Ts terminal, deceleration sensor installation, and ABS ECU.

#### Sensor Detection Point

1) Raise rear of vehicle 26-28", measured from bottom of rear bumper to ground. Verify ABS light is off. Completely lower rear of vehicle. As vehicle is lowered, ABS light will start flashing. Raise front of vehicle 21-24", measured from lower body or spoiler edge of vehicle to ground.

2) Verify ABS light is off. As vehicle is lowered, ABS light will start flashing. If ABS light stays on steadily, inspect deceleration sensor installation. If sensor installation is okay, replace deceleration sensor.

#### Sensor Operation

1) Drive vehicle straight ahead at speed greater than 50 MPH. Verify ABS light flashes while vehicle is traveling at less than 28 MPH. ABS light will stop flashing at more than 28 MPH, and then will flash once at 50 MPH. If ABS light flashes as described, check is complete. If ABS light does not flash as indicated, stop vehicle.

2) Connect jumper between DLC terminals Tc and E1 (do not disconnect Ts and E1) of DLC. See Fig. 5. If a malfunction is detected, 4 seconds will elapse, then ABS light will begin to flash a 2-digit code. First series of flashes indicates first digit of code. After a 1.5-second pause, second series of flashes indicates second digit of code.

3) If 2 or more codes are stored, there will be a 2.5-second pause between each code. After all codes are flashed, there will be a 4-second pause, then all codes will repeat.

4) Record diagnostic codes. Turn ignition off. Repair as necessary. See DIAGNOSTIC CODES. Remove jumper wires from DLC. Clear diagnostic codes. See CLEARING CODES.

## COMPONENT TESTING

#### ABS ECU Wiring Harness

1) Remove ABS ECU, leaving connectors attached. See ABS ECU under REMOVAL & INSTALLATION. Backprobe each terminal to measure voltage or resistance at each terminal, as specified in table. See STEP 1

See Fig. 8 or 9 and 10. If circuit values are not as specified, check and repair or replace indicated component.

2) Unplug ABS ECU connectors. Measure resistance on wire harness side of ABS ECU connector terminals, as specified in table. See STEP 2

See Fig. 8 or 10. If circuit values are not as specified, check and repair or replace indicated component.

| <table border="1"> <tr> <td>SFL</td><td>SR</td><td>R-</td><td>FL+</td><td>FSS</td><td>FR+</td><td>IG1</td><td>W</td><td>RSS</td><td>STP</td><td>TS</td><td></td><td>RL-</td><td>GND</td><td>SFR</td></tr> <tr> <td>SRR</td><td>MR</td><td>FL-</td><td>MT</td><td>AST</td><td>FR-</td><td>BAT</td><td>D/G</td><td>RR-</td><td>RR+</td><td>TC</td><td>PKB</td><td>RL+</td><td>GND</td><td>SRL</td></tr> </table> |            | SFL                                                  | SR  | R-              | FL+                                          | FSS | FR+ | IG1 | W   | RSS | STP | TS  |     | RL- | GND | SFR | SRR | MR | FL- | MT | AST | FR- | BAT | D/G | RR- | RR+ | TC | PKB | RL+ | GND | SRL |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------|-----|-----------------|----------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|--|--|--|--|--|--|--|--|--|--|--|--|--|
| SFL                                                                                                                                                                                                                                                                                                                                                                                                            | SR         | R-                                                   | FL+ | FSS             | FR+                                          | IG1 | W   | RSS | STP | TS  |     | RL- | GND | SFR |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SRR                                                                                                                                                                                                                                                                                                                                                                                                            | MR         | FL-                                                  | MT  | AST             | FR-                                          | BAT | D/G | RR- | RR+ | TC  | PKB | RL+ | GND | SRL |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Tester Connection                                                                                                                                                                                                                                                                                                                                                                                              | Check Item | Condition                                            |     | Specified Value | Trouble Part                                 |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SFR                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | Actuator                                     |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| RL-                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| TS                                                                                                                                                                                                                                                                                                                                                                                                             | Voltage    | IG switch on and check connector Ts-E, not connected |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and check connector Ts-E, connected     |     | About 0V        |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| STP                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch off and brake pedal depressed              |     | Battery voltage | Stoplight switch<br>Stoplight                |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                | Continuity | IG switch off and brake pedal returned               |     | Continuity      |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| RSS                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| W                                                                                                                                                                                                                                                                                                                                                                                                              | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | ABS ECU<br>"ABS" warning light               |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| IG1                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on                                         |     | Battery voltage | ECU-IG Fuse                                  |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SRL                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | Actuator                                     |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| GND                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | Wiring harness                               |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PKB                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and PKB lever pulled                    |     | About 0V        | Parking brake switch<br>Level warning switch |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and PKB lever returned                  |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| TC                                                                                                                                                                                                                                                                                                                                                                                                             | Voltage    | IG switch on and check connector Tc-E, not connected |     | Battery voltage | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and check connector Tc-E, connected     |     | About 0V        |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| RR-                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| D/G                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and check connector Ts-E, not connected |     | About 0V        |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| BAT                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch off                                        |     | Battery voltage | DOVE Fuse                                    |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| FSS                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| R-                                                                                                                                                                                                                                                                                                                                                                                                             | Continuity | IG switch off                                        |     | Continuity      |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SR                                                                                                                                                                                                                                                                                                                                                                                                             | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SFL                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | Actuator                                     |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| FR-                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| AST                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | Actuator                                     |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| FL-                                                                                                                                                                                                                                                                                                                                                                                                            | Continuity | IG switch off                                        |     | Continuity      | ABS ECU                                      |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SRR                                                                                                                                                                                                                                                                                                                                                                                                            | Voltage    | IG switch on and "ABS" warning light goes on         |     | About 0V        | Actuator                                     |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                |            | IG switch on and "ABS" warning light goes off        |     | Battery voltage |                                              |     |     |     |     |     |     |     |     |     |     |     |     |    |     |    |     |     |     |     |     |     |    |     |     |     |     |  |  |  |  |  |  |  |  |  |  |  |  |  |

STEP 1 - VOLTAGE & CONTINUITY CHECKS WITH CONNECTORS ATTACHED

| Tester Connection | Check Item | Specified Value | Trouble Part          | Tester Connection | Check Item | Specified Value | Trouble Part  |
|-------------------|------------|-----------------|-----------------------|-------------------|------------|-----------------|---------------|
| SFR ↔ AST         | Resistance | About 6 Ω       | Actuator              | SR ↔ R-           | Resistance | 60 ~ 100 Ω      | Control relay |
| SRL ↔ AST         | Resistance | About 6 Ω       | Actuator              | SFL ↔ AST         | Resistance | About 6 Ω       | Actuator      |
| RL+ ↔ RL-         | Resistance | 1.1 ~ 1.7 kΩ    | Rear LH speed sensor  | AST ↔ Body ground | Resistance | About 5 Ω       | Actuator      |
| RR+ ↔ RR-         | Resistance | 1.1 ~ 1.7 kΩ    | Rear RH speed sensor  | MT ↔ Body ground  | Continuity | Continuity      | Actuator      |
| FR+ ↔ FR-         | Resistance | 0.8 ~ 1.3 kΩ    | Front RH speed sensor | MR ↔ R-           | Resistance | 50 ~ 80 Ω       | Control relay |
| FL+ ↔ FL-         | Resistance | 0.8 ~ 1.3 kΩ    | Front LH speed sensor | SRR ↔ AST         | Resistance | About 6 Ω       | Actuator      |

STEP 2 - RESISTANCE CHECKS WITH CONNECTORS UNPLUGGED

93E00574

Fig. 8: Testing Specifications Table For ABS ECU Wiring Harness - FWD  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

|     |     |    |     |     |  |    |     |  |     |     |     |     |     |     |   |     |     |     |     |
|-----|-----|----|-----|-----|--|----|-----|--|-----|-----|-----|-----|-----|-----|---|-----|-----|-----|-----|
| SFL | IG1 | SR | FSS | FL+ |  | MT | TC  |  | FR- | GND | SFR | RR+ | RSS | STP |   | D/G | GS2 |     | RL- |
| SRR | BAT | R- | MR  | FL- |  |    | AST |  | FR+ | GND | SRL | RR- | TS  | PKB | W | GS1 |     | GST | RL+ |

R01275

| Tester Connection | Check Item | Condition                                            | Specified Value | Trouble Part                  |
|-------------------|------------|------------------------------------------------------|-----------------|-------------------------------|
| RL-               | Continuity | IG switch off                                        | Continuity      | ABS ECU                       |
| GS2               | Voltage    | IG switch on                                         | 4 ~ 6 V         | Deceleration Sensor           |
| D/G               | Voltage    | IG switch on and check connector Ts-E, not connected | About 0V        | ABS ECU                       |
| STP               | Voltage    | IG switch off and brake pedal depressed              | Battery voltage | Stoplight switch<br>Stoplight |
|                   | Continuity | IG switch off and brake pedal returned               | Continuity      |                               |
| RSS               | Continuity | IG switch off                                        | Continuity      | ABS ECU                       |
| GS1               | Voltage    | IG switch on                                         | 4 ~ 6 V         | Deceleration Sensor           |
| W                 | Voltage    | IG switch on and "ABS" warning light goes on         | About 0V        | ABS ECU                       |
|                   |            | IG switch on and "ABS" warning light goes off        | Battery voltage | "ABS" warning light           |
| PKB               | Voltage    | IG switch on and PKB lever pulled                    | About 0V        | Parking brake switch          |
|                   |            | IG switch on and PKB lever returned                  | Battery voltage | Level warning switch          |
| TS                | Voltage    | IG switch on and check connector Ts-E, not connected | Battery voltage | ABS ECU                       |
|                   |            | IG switch on and check connector Ts-E, connected     | About 0V        |                               |
| RR-               | Continuity | IG switch off                                        | Continuity      |                               |
| SFR               | Voltage    | IG switch on and "ABS" warning light goes on         | About 0V        | Actuator                      |
|                   |            | IG switch on and "ABS" warning light goes off        | Battery voltage |                               |
| GND               | Continuity | IG switch off                                        | Continuity      | Wiring harness                |
| FR-               | Continuity | IG switch off                                        | Continuity      |                               |
| TC                | Voltage    | IG switch on and check connector Tc-E, not connected | Battery voltage | ABS ECU                       |
|                   |            | IG switch on and check connector Tc-E, connected     | About 0V        |                               |
| FSS               | Continuity | IG switch off                                        | Continuity      |                               |
| SR                | Voltage    | IG switch on and "ABS" warning light goes on         | About 0V        |                               |
|                   |            | IG switch on and "ABS" warning light goes off        | Battery voltage |                               |
| IG1               | Voltage    | IG switch on                                         | Battery voltage | ECU-IG Fuse                   |

STEP 1 - VOLTAGE & CONTINUITY CHECKS WITH CONNECTORS ATTACHED

93F00575  
Fig. 9: Testing Specs Table For ABS ECU Wir Harness All-Trac (1 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

| Tester Connection | Check Item | Condition                                     | Specified Value | Trouble Part |
|-------------------|------------|-----------------------------------------------|-----------------|--------------|
| SFL               | Voltage    | IG switch on and "ABS" warning light goes on  | About 0V        | Actuator     |
|                   |            | IG switch on and "ABS" warning light goes off | Battery voltage |              |
| SRL               | Voltage    | IG switch on and "ABS" warning light goes on  | About 0V        |              |
|                   |            | IG switch on and "ABS" warning light goes off | Battery voltage |              |
| AST               | Voltage    | IG switch on and "ABS" warning light goes on  | About 0V        | Actuator     |
|                   |            | IG switch on and "ABS" warning light goes off | Battery voltage |              |
| FL-               | Continuity | IG switch off                                 | Continuity      | ABS ECU      |
| R-                | Continuity | IG switch off                                 | Continuity      |              |
| BAT               | Voltage    | IG switch off                                 | Battery voltage | DOME Fuse    |
| SRR               | Voltage    | IG switch on and "ABS" warning light goes on  | About 0V        | Actuator     |
|                   |            | IG switch on and "ABS" warning light goes off | Battery voltage |              |

STEP 1 - VOLTAGE & CONTINUITY CHECKS WITH CONNECTORS ATTACHED (Cont.)

| Tester Connection | Check Item | Specified Value | Trouble Part          | Tester Connection | Check Item | Specified Value | Trouble Part          |
|-------------------|------------|-----------------|-----------------------|-------------------|------------|-----------------|-----------------------|
| RR+ ↔ RR-         | Resistance | 1.1 ~ 1.7 kΩ    | Rear RH speed sensor  | SFL ↔ AST         | Resistance | About 6 Ω       | Actuator              |
| RL+ ↔ RL-         | Resistance | 1.1 ~ 1.7 kΩ    | Rear LH speed sensor  | SRL ↔ AST         | Resistance | About 6 Ω       | Actuator              |
| SFR ↔ AST         | Resistance | About 6 Ω       | Actuator              | FR+ ↔ FR-         | Resistance | 0.8 ~ 1.3 kΩ    | Front RH speed sensor |
| MT ↔ Body ground  | Continuity | Continuity      | Actuator              | AST ↔ Body ground | Resistance | About 5 Ω       | Actuator              |
| FL+ ↔ FL-         | Resistance | 0.8 ~ 1.3 kΩ    | Front LH speed sensor | MR ↔ R-           | Resistance | 50 ~ 80 Ω       | Control relay         |
| SR ↔ R-           | Resistance | 60 ~ 100 Ω      | Control relay         | SRR ↔ AST         | Resistance | About 6 Ω       | Actuator              |

STEP 2 - RESISTANCE CHECKS WITH CONNECTORS UNPLUGGED

93G00576  
Fig. 10: Testing Specs Table For ABS ECU Wir Harness All-Trac (2 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

ABS Control Relay

1) Remove ABS control relay. See Fig. 1. Using ohmmeter capable of testing diodes, check for continuity between ABS control relay connector terminals A3 and A4. See Fig. 11. Transpose ohmmeter leads. Repeat measurement.

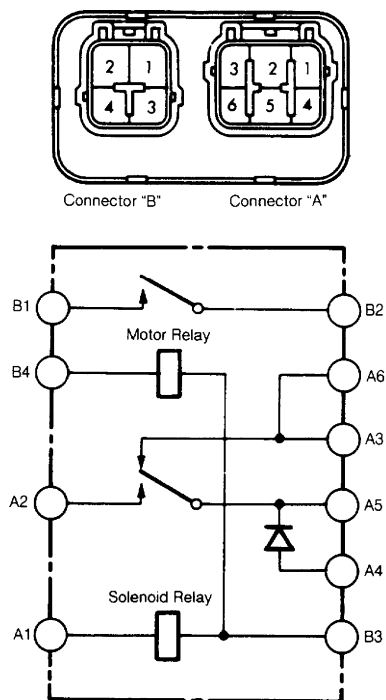
2) If continuity exists or circuit is open for both measurements, actuator diode is defective. Repair or replace relay, then go to next step. If continuity exists for only one measurement, temporarily substitute known good ABS ECU. Retest system.

3) Continuity should exist between terminals A3 and A6, between terminals A3 and A5, and between terminals A5 and A6.

4) Continuity should not exist between terminals A2 and A3, or between terminals B1 and B2.

5) Using fused jumper wire, connect positive battery terminal to terminal B3. Connect negative battery terminal to terminal B4. Continuity should not exist between terminals B1 and B2.

6) Using fused jumper wire, connect positive battery terminal to terminal B3. Connect negative battery terminal to terminal A1. Continuity should exist between terminals A2 and A5. Continuity should not exist between terminals A3 and A5. Replace relay if operation is not as specified.



93J83997

Fig. 11: Identifying ABS Control Relay Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Front Speed Sensors

1) Remove harness connector clamp. Unplug connector from front wheel speed sensor. Measure resistance between speed sensor terminals. Resistance should be 800-1300 ohms.

2) Measure insulation resistance between each sensor terminal and sensor body. Continuity should not exist. If resistance and continuity are not as specified, replace speed sensor.

#### Rear Speed Sensors

1) Remove rear seat cushion. Unplug wheel speed sensor



connector. Measure resistance between speed sensor terminals. On All-Trac models, resistance should be 800-1500 ohms. On FWD models, resistance should be 1100-1700 ohms.

2) Measure insulation resistance between each sensor terminal and sensor body. Continuity should not exist. If resistance and continuity are not as specified, replace speed sensor.

#### Sensor Rotors

Visually inspect sensor rotor serrations for scratches, cracks, missing teeth, or warping. Replace front drive shaft or rear hub as necessary if rotor is damaged.

## REMOVAL & INSTALLATION

**WARNING:** Hydraulic system may be under high pressure. Use caution when opening hydraulic system.

### ABS ECU

#### Removal & Installation

1) Turn ignition off. Disconnect negative battery cable. ABS ECU is located on right side of luggage compartment (left side on convertible models). Remove screws attaching ABS ECU to vehicle.

2) Remove wire harness from ECU bracket clamp. Unplug ABS ECU connector. Remove ABS ECU from vehicle. To install, reverse removal procedure.

### ABS CONTROL RELAY

#### Removal & Installation

Remove connector from wire harness clamp. Remove relay bolt. Remove relay from vehicle. See Fig. 1. To install, reverse removal procedure.

### ACTUATOR

#### Removal & Installation

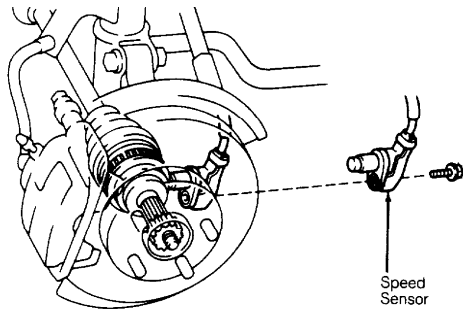
Turn ignition off. Remove actuator cover. Unplug electrical connectors. Remove cover bracket and bolt. Disconnect brake lines. Remove actuator from bracket. To install, reverse removal procedure. Bleed brake system. See BLEEDING BRAKE SYSTEM.

### FRONT SENSOR ROTOR

Front speed sensor rotor is an integral part of outboard CV joint. See appropriate DRIVE AXLES article in the POWERTRAIN section.

### FRONT WHEEL SPEED SENSORS

See Fig. 12.



93F83976

Fig. 12: Removing & Installing Front Speed Sensor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### REAR SENSOR ROTOR (ALL TRAC)

Rear speed sensor rotor is an integral part of outboard CV joint. See appropriate DRIVE AXLES article in the POWERTRAIN section.

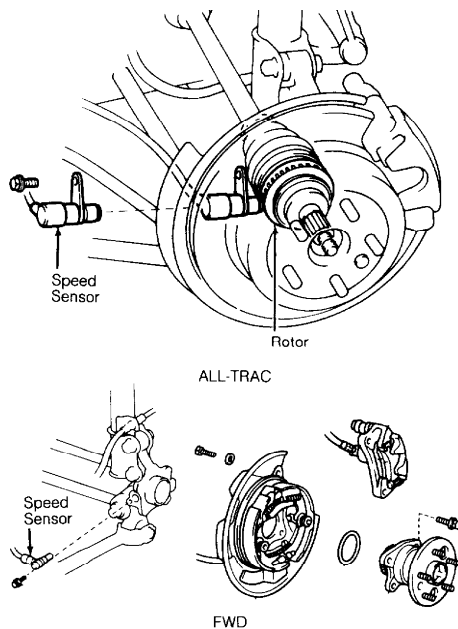
### REAR SENSOR ROTOR (FWD)

#### Removal & Installation

Rear speed sensor rotor is an integral part of rear hub. Remove rear hub. See SUSPENSION - REAR article in the SUSPENSION section.

### REAR WHEEL SPEED SENSORS

See Fig. 13.



93F83977

Fig. 13: Removing & Installing Rear Speed Sensor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## OVERHAUL

DO NOT attempt to overhaul or disassemble actuator assembly.  
If actuator is defective, replace entire assembly.

**TORQUE SPECIFICATIONS**

TORQUE SPECIFICATIONS TABLE

| Application                                        | Ft. Lbs. (N.m)  |
|----------------------------------------------------|-----------------|
| Brakeline Nuts .....                               | 11 (15)         |
| Power Steering Reservoir Tank Mounting Bolts ..... | 14 (19)         |
| Wheel Lug Nuts .....                               | 76 (103)        |
| 3-Way Union Bolt .....                             | 12 (16)         |
|                                                    | INCH Lbs. (N.m) |
| ABS ECU Mounting Screws .....                      | 27 (3)          |
| Actuator Mounting Nuts .....                       | 44 (5)          |
| Deceleration Sensor Mounting Bolts .....           | 27 (3)          |
| Proportioning Valve Bolts .....                    | 80 (9)          |
| Wheel Speed Sensor Mounting Bolt .....             | 71 (8)          |

# ANTI-LOCK BRAKE SAFETY PRECAUTIONS

1993 Toyota Celica

## GENERAL INFORMATION

Anti-Lock Brake Safety Precautions

### \* PLEASE READ THIS FIRST \*

This article is intended for general information purposes only. This information may not apply to all makes and models. If vehicle is equipped with Anti-Lock Brake System (ABS), refer to appropriate ANTI-LOCK BRAKE SYSTEM article in the BRAKES section for description, operation, depressurizing, testing, system bleeding, trouble shooting and servicing of specific system.

WARNING: Failure to depressurize ABS could lead to physical injury.

### ANTI-LOCK BRAKE SAFETY PRECAUTIONS

WARNING: Failure to depressurize ABS could lead to physical injury.

- \* NEVER open a bleeder valve or loosen a hydraulic line while ABS is pressurized.
- \* NEVER disconnect or reconnect any electrical connectors while ignition is on. Damage to ABS control unit may result.
- \* DO NOT attempt to bleed hydraulic system without first referring to the appropriate ANTI-LOCK BRAKE SYSTEM article in the BRAKES section.
- \* Only use specially designed brake hoses/lines on ABS equipped vehicles.
- \* DO NOT tap on speed sensor components (sensor, sensor rings). Sensor rings must be pressed into hubs, NOT hammered into hubs. Striking these components can cause demagnetization or a loss of polarization, affecting the accuracy of the speed signal returning to the ABS control unit.
- \* DO NOT mix tire sizes. Increasing the width, as long as tires remain close to the original diameter, is acceptable. Rolling diameter must be identical for all 4 tires. Some manufacturers recommend tires of the same brand, style and type. Failure to follow this precaution may cause inaccurate wheel speed readings.
- \* DO NOT contaminate speed sensor components with grease. Only use recommended coating, when system calls for an anti-corrosion coating.
- \* When speed sensor components have been removed, ALWAYS check sensor-to-ring air gaps when applicable. These specifications can be found in each appropriate article.
- \* ONLY use recommended brake fluids. DO NOT use silicone brake fluids in an ABS equipped vehicle.
- \* When installing transmission devices (CB's, telephones, etc.) on ABS equipped vehicles, DO NOT locate the antenna near the ABS control unit (or any control unit).
- \* Disconnect all on-board computers, when using electric welding equipment.
- \* DO NOT expose the ABS control unit to prolonged periods of high heat (185 °F/85°C for 2 hours is generally considered a maximum limit).

# CRUISE CONTROL SYSTEM

1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Cruise Control Systems

Celica

## DESCRIPTION

Cruise control system consists of Cruise Control Electronic Control Unit (CC ECU), actuator and associated cables, speed sensor(s), parking brake switch, cruise control switch, stoplight switch, park/neutral switch (A/T), clutch switch (M/T) and related wiring. See Fig. 1. System allows vehicle to cruise at a desired speed greater than 25 MPH.

Speed control will cancel when brake pedal or clutch pedal is depressed, CANCEL switch is activated or automatic transmission shift lever is moved to "N" position. If vehicle speed falls below 25 MPH or drops 10 MPH less than preset speed, will be canceled. If a malfunction is detected in either actuator, speed sensor or speed control switch circuits cruise control function, speed control will also be canceled.

## OPERATION

Pressing cruise control CRUISE ON-OFF to ON position, activates system. CRUISE indicator light in instrument cluster comes on to indicate activation of system. To set desired speed, press control SET/COAST switch and release switch. Vehicle speed will now be maintained. To increase speed, depress accelerator pedal enough to exceed set speed. When accelerator pedal is released, speed will return to speed previously set.

To cancel set speed, press cruise control CANCEL switch, depress brake pedal, depress clutch pedal or place shift lever in "N" position. If vehicle speed falls to less than 25 MPH, set speed will automatically cancel. If vehicle speed falls below set speed by 10 MPH, set speed will also automatically cancel.

Pressing cruise control RES/ACC switch, allows vehicle to return to set speed before cancellation. Pressing cruise control RES/ACC position and keeping it there gradually increases vehicle speed. Pressing cruise control SET/COAST position and keeping it there gradually decreases vehicle speed.

## ACTUATOR (MOTOR)

Actuator consists of a motor, safety magnetic clutch, control arm and position sensor. When actuator receives a signal from CC ECU, it engages safety magnetic clutch and activates motor. Motor causes control arm to move, opening or closing engine throttle valve.

When motor rotates forward, control arm also rotates via safety magnetic clutch, gears and drive shaft. Control arm pulls a cable connected to engine throttle valve and opens the valve accordingly. When motor rotates in a reverse direction, control arm also rotates in a reverse direction and engine throttle valve closes.

Safety magnetic clutch disengages motor from control arm and shuts throttle valve when vehicle speed approaches 10 MPH or more above set speed during cruise control operation, when motor malfunctions or when a circuit problem occurs. Position sensor detects rotary angle of control arm and constantly sends a signal to CC ECU.

## ACTUATOR (VACUUM)

Actuator consists of a vacuum diaphragm, control cable and vacuum control valve. When cruise control is activated, vacuum actuator receives an input signal from CC ECU, to increase vehicle speed. Signal from CC ECU causes vacuum control valve to close atmospheric intake port introducing a vacuum to diaphragm. Control cable is pulled and throttle valve is opened, increasing vehicle speed to preselected speed. To close throttle valve, signal from CC ECU is decreased, causing vacuum control valve to reduce vacuum and increase atmospheric pressure. Control cable is released and throttle valve is closed.

## CRUISE CONTROL SWITCH

### CRUISE ON-OFF Switch

Cruise control CRUISE ON-OFF switch is power switch for cruise control system. When ignition is turned off, cruise control CRUISE ON-OFF switch is also turned off. Switch remains off even when ignition is turned on again.

### SET/COAST Position

With cruise control CRUISE ON-OFF switch turned on and vehicle speed greater than 25 MPH, press cruise control switch in SET/COAST position and then release. CC ECU will store vehicle speed and control that speed constantly.

While in cruise control mode, if cruise control switch is pressed and held in SET/COAST position, actuator motor will be energized. Engine throttle valve will close, and vehicle will decelerate until switch is released. From then on, CC ECU will store new vehicle speed and control that speed constantly.

### RES/ACC Position

If cruise control system is canceled by any of various cancellation methods, the previously set speed can be resumed by moving cruise control switch to RES/ACC position and then releasing. Set speed, however, cannot be resumed if vehicle speed drops to less than 25 MPH, as CC ECU memory will be cleared.

While in cruise control mode, if cruise control switch is pressed and held in RES/ACC position, actuator motor will be energized. Engine throttle valve will open, and vehicle will accelerate until switch is released. From then on, CC ECU stores new vehicle speed and controls that speed constantly.

### CANCEL Position

When cruise control switch is turned to CANCEL position, a cancellation signal is sent from cruise control switch to CC ECU.

## CRUISE CONTROL ELECTRONIC CONTROL UNIT (CC ECU)

CC ECU constantly monitors and compares set speed with actual vehicle speed from input sensors. When vehicle speed is different from set speed, CC ECU activates actuator motor to change engine throttle valve, changing vehicle speed.

CC ECU includes a self-diagnostic function. If cruise control system is canceled by any condition other than driver operation, CC ECU assumes a malfunction has occurred and may set a corresponding fault code.

## SPEED SENSOR

Speed sensors are located in combination meter and/or on

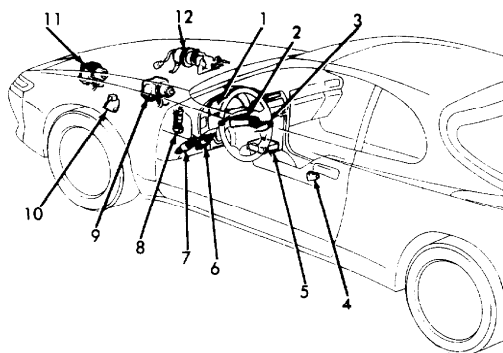
extension housing of Electronically Controlled Transmission (ECT) or M/T. Speed sensor rotor shaft is driven by a gear on speedometer output shaft. For each shaft rotation, speed sensor sends a 20-pulse signal to instrument cluster. This signal is converted inside instrument cluster to a 4-pulse signal which is sent to CC ECU. CC ECU calculates vehicle speed from this pulse frequency.

## SELF-DIAGNOSTIC SYSTEM

When vehicle is in cruise control mode, system will cancel due to a malfunction in either actuator, speed sensor or speed control switch circuit. When cruise control functions are canceled, CRUISE light will blink 5 times, indicating 2-digit trouble code(s) are stored in CC ECU memory. See CC ECU TROUBLE CODE DEFINITION table under SELF-DIAGNOSTICS. Two digit trouble codes(s) will be stored in CC ECU memory until ignition is turned off. See SELF-DIAGNOSTICS.

If a fault or symptom is present, but no trouble codes were set, a circuit function test can be performed. Circuit function test will display a one digit function code if circuit tested is okay. See CRUISE CONTROL CIRCUIT FUNCTION TEST under TROUBLE SHOOTING.

**NOTE:** Intermittent failures may cause CRUISE indicator light to flicker or come on. Light will go out after fault goes away. Fault may or may not be present at time of testing; however, a corresponding trouble code may be stored in CC ECU memory. See SELF-DIAGNOSTICS.



1. Combination Meter
  - Cruise Control Indicator
  - No. 1 Vehicle Speed Sensor (A/T)
2. Ignition Switch
3. Control Switch
4. Parking Brake Switch
5. Cruise Control ECU
6. Stoplight Switch
7. Clutch Switch (M/T)
8. No. 1 Vehicle Speed Sensor (M/T)
9. Motor Type Actuator (3S-GTE)
10. Park/Neutral Position Switch (A/T)
11. Vacuum Type Actuator (Except 3S-GTE)
12. Motor Type Actuator (Except 3S-GTE)

93H83391

Fig. 1: Locating Cruise Control Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## SELF-DIAGNOSTICS

**\* PLEASE READ THIS FIRST \***

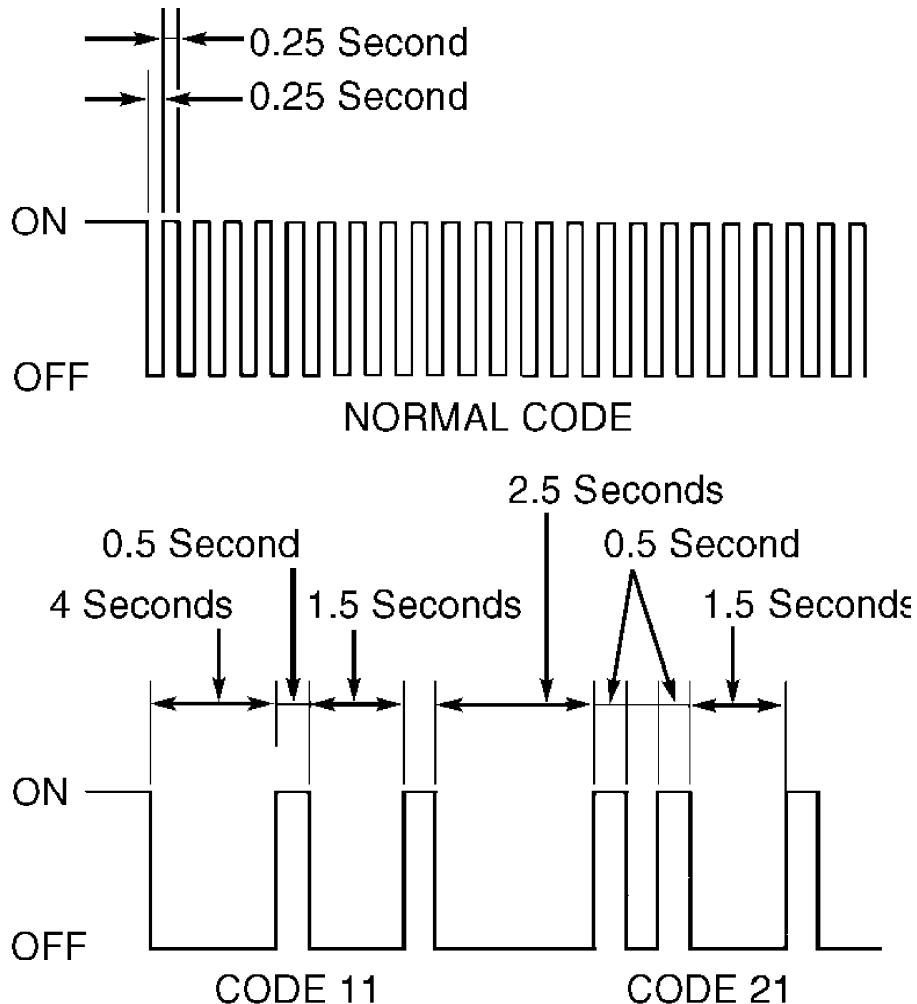
**CAUTION:** Celica is equipped with a Supplemental Restraint System (SRS). SRS wiring harness is routed close to instrument cluster, steering wheel and related components. All SRS wiring harnesses and connectors are Yellow. DO NOT use electrical test equipment on these circuits. Before working

on cruise control components, disable air bag system. Refer to AIR BAG RESTRAINT SYSTEM article in ACCESSORIES/SAFETY EQUIPMENT section.

**WARNING:** Wait at least 90 seconds after disabling SRS. Back-up power circuit, capacitor, maintains system voltage for about 90 seconds after battery is disconnected. Servicing cruise control system before 90 seconds may cause accidental air bag deployment and possible personal injury.

## READING TROUBLE CODES

Trouble codes are displayed as flashes of CRUISE indicator light. All trouble codes are 2-digit numbers. CC ECU outputs trouble codes from lowest to highest. These codes indicate current faults in system and should be serviced in order of appearance. Pay careful attention to length of pauses in order to read codes correctly. See Fig. 2.



G93C02769

Fig. 2: Reading Service Codes  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## RETRIEVING TROUBLE CODES



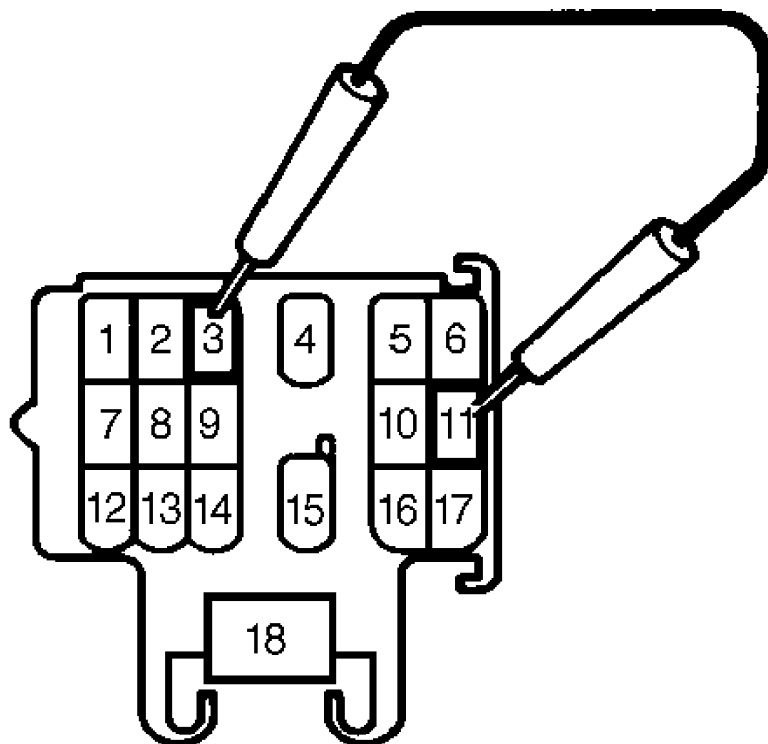
1) Codes from CC ECU self-diagnostic system are retrieved through self-test diagnostic Data Link Connector (DLC). Test drive vehicle to allow fault codes to set in CC ECU memory. If CRUISE indicator light begins to flash 5 times while driving or cruise control will not set or operate, check for trouble codes. Go to next step.

2) Stop vehicle and leave ignition switch in ON position. If ignition switch is turned to OFF position, any stored trouble codes will be erased from CC ECU memory. Connect jumper wire between DLC self-diagnostic terminals. See CC ECU CODE RETRIEVAL table. See Fig. 3.

3) If any code is present, perform test(s) in order given. See CC ECU FAULT CODE DEFINITION table. If no codes are present and CRUISE indicator light begins flashing on and off every .25 seconds, perform NORMAL code tests in order given. See CC ECU FAULT CODE DEFINITION table. If no codes are present and CRUISE indicator light does not flash on and off every .25 seconds, and cruise control system malfunction still exists, perform CRUISE CONTROL CIRCUIT FUNCTION TEST under TROUBLE SHOOTING.

#### CC ECU CODE RETRIEVAL TABLE

| Application  | DLC Self-Test<br>Terminals |
|--------------|----------------------------|
| Celica ..... | 3 & 11                     |



93A02768

Fig. 3: DLC Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### CC ECU FAULT CODE DEFINITION

## CC ECU FAULT CODE DEFINITION TABLE

| Code No. | Diagnosis Problem                                                                            | (1) Perform Test No.             |
|----------|----------------------------------------------------------------------------------------------|----------------------------------|
| Normal   | Indicator Light Flashes 5 Times,<br>CC System Can Not Be Set Or<br>Does Not Operate          | 3, 5, 6, 8,<br>9, 7, 4, (2), (3) |
| 11       | Excessive Voltage To Motor<br>Acceleration Side                                              | 4, (2), (3)                      |
| 11       | Over Current/Short In Motor Circuit                                                          | 4, (2), (3)                      |
| 12       | Open In Magnet Clutch Circuit                                                                | 4, 6, (2), (3)                   |
| 12       | Overcurrent/Short In<br>Magnet Clutch Circuit                                                | 4, 6, (2), (3)                   |
| 13 (4)   | Position Sensor Detects<br>Abnormal Voltage                                                  | 4, (2)                           |
| 13 (4)   | Open In Actuator Motor Circuit                                                               | 4, (2)                           |
| 13 (4)   | Position Sensor Signal Value<br>Does Not Change When Motor Operates                          | 4, (2)                           |
| 21       | Speed Sensor Signal Not Sent<br>To CC ECU For 140<br>Milliseconds Or Longer                  | 5, (2)                           |
| 23 (5)   | Vehicle Speed Decreased 10 MPH<br>Or More Below Speed Set During<br>Cruise Control Operation | (6), 5, 4, 13                    |
| 31 (4)   | RES/ACC Mode Is On At All Times<br>When MAIN Switch Is On                                    | 3, (2)                           |
| 32       | Short In Control Switch Circuit                                                              | 3, (2)                           |
| 34       | Control Does Not Turn Off<br>Before Switching                                                | 3, (2)                           |
| 41       | Faulty CC ECU                                                                                | (2)                              |

- (1) - Perform test numbers in order given.
- (2) - Replace CC ECU, and retest system.
- (3) - Check wire harness and connectors.
- (4) - Except with vacuum type actuator.
- (5) - If speed set can be maintained when speed control switch is again set to SET/COAST position, there is no malfunction.
- (6) - Check speed control cable and control link function.

## CLEARING CODES

CAUTION: Do not disconnect vehicle battery to clear codes.

To clear codes from CC ECU memory, turn ignition off. This procedure erases fault codes from CC ECU memory. If problem has not been corrected or fault is still present, code will be reset in CC ECU memory.

## TROUBLE SHOOTING

NOTE: Before TROUBLE SHOOTING BY SYMPTOM, perform SELF-DIAGNOSTICS. TROUBLE SHOOTING BY SYMPTOM should only be performed if no self-diagnostic trouble codes are present.

### CRUISE CONTROL CIRCUIT FUNCTION TEST

Retrieving 1-Digit Normal Function Code

Turn ignition on. Push cruise control switch to SET/COAST or RES/ACC position and hold. Set CRUISE ON-OFF (MAIN) switch to ON position. Ensure CRUISE indicator light comes on in combination meter and flashes after 3 seconds. Release cruise control switch from SET/COAST or RES/ACC position. Activate each switch circuit in order given. If other than a normal function code is displayed, go to TROUBLE SHOOTING BY SYMPTOM. If normal code is displayed and cruise control is still malfunctioning, go to TROUBLE SHOOTING BY SYMPTOM. Normal function code will be displayed by CRUISE indicator light, as each circuit is activated as follows:

#### \* SET/COAST

Set cruise control switch to SET/COAST position and hold.

Indicator light will flash normal Code 2, if circuit function is normal. SET/COAST circuit normal function code will repeat normal Code 2.

#### \* RES/ACC

Set cruise control switch to RES/ACC position and hold.

Indicator light will flash normal Code 3, if circuit function is normal. RES/ACC circuit normal function code will repeat normal Code 3.

#### \* CANCEL Switches

Raise vehicle and support drive wheels off ground. Perform

CANCEL switch test with engine running. Press CANCEL switch to ON and OFF positions. Depress and release brake pedal. Apply and release parking brake. On A/T models, move shift lever from "D" to "N" position. On M/T models, depress and release clutch pedal. If circuit is normal, indicator light will come on when switch is in OFF position and go off when switch is in ON position.

#### \* Vehicle Speed Sensor

Raise vehicle and support drive wheels off ground. Start engine and slowly depress accelerator pedal until specified speed is reached. Turn CRUISE ON-OFF (MAIN) switch to ON position. If circuit is normal, indicator light will flash on and off every .25 second at vehicle speeds greater than 25 MPH. At vehicle speeds of 25 MPH or less, if circuit is normal, indicator light will come on and stay on. On models with Electrically Controlled Transmission (ECT), drive vehicle at 25 MPH or less. If circuit is normal, indicator light will come on and stay on. If indicator light functions as described, this would be considered a normal code.

NOTE: When 2 or more function codes are sent to CC ECU, only the lowest numbered function code will be displayed.

### CLEARING FUNCTION CODES

CAUTION: DO NOT disconnect vehicle battery to clear codes.

Normal function code display will be canceled when CRUISE ON-OFF switch is set to OFF position.

## TROUBLE SHOOTING BY SYMPTOM

NOTE: Perform tests No. 1 and No. 2 to ensure power source and indicator light circuits are okay. Perform following tests in order listed.

Vehicle Speed Does Not Decrease When Speed Control Switch Set To SET/COAST

When normal function code is displayed, check speed control cable function and perform TEST 4 and 5. When other than normal function code is displayed, perform TEST 3.

Cruise Control System Can Not Be Set, Does Not Operate Or Indicator Light Flashes 5 Times

When normal function code is displayed, check wire harness, speed control cable function and perform TEST 3, 6, 8, 9, 7, and 4. When other than normal function code is displayed, perform TEST 5.

Vehicle Speed Fluctuates When Speed Control Switch Turned To SET/COAST Position

Check speed control cable function and perform TEST 5 and 4.

Large Speed Drop When Cruise Control Switch Turned To SET/COAST

Check speed control cable function and perform TEST 12 and 4.

Vehicle Speed Does Not Increase When Control Set To RES/ACC - Perform TEST 4, 5, 11, and 10. When other than normal function code is displayed, perform TEST 3.

Set Speed Changes On High Or Low Side

When normal function code is displayed, check speed control cable function and perform TEST 5 and 4. On models with motor type actuator perform TEST 12 only. When other than normal function code is displayed, perform TEST 5.

Acceleration Response Is Sluggish When Control Is Set To RES/ACC

On models with vacuum type actuator, perform TEST 3, 4, 11, and 10. On models with motor type actuator, perform TEST 3, 4, 11, 10, and 12.

Vehicle Does Not Return To Memorized Speed When Control Set To RES/ACC

When normal function code is displayed, check speed control cable function and perform TEST 4 and 5. When other than normal function code is displayed, perform TEST 3.

Set Speed Does Not Cancel When Brake Pedal Depressed

When normal function code is displayed, perform TEST 4 and 6. When other than normal function code is displayed, perform TEST 6.

Set Speed Does Not Cancel When Parking Brake Applied

When normal function code is displayed, perform TEST 4. When other than normal function code is displayed, perform TEST 7.

Speed Does Not Cancel When Shifted to "N" (A/T) Or Clutch Pedal Depressed (M/T)

When normal function code is displayed, perform TEST 4. When

other than normal function code is displayed, perform TEST 8 (A/T) or 9 (M/T).

Set Speed Does Not Cancel When Control Set To CANCEL

When normal function code is displayed, perform TEST 4. When other than normal function code is displayed, perform TEST 3.

Set Speed Can Be Set Below Approximately 25 MPH

When normal function code is displayed, perform TEST 4. When other than normal function code is displayed, perform TEST 5.

Set Speed Will Not Disengage At Approximately 25 MPH Or Less

When normal function code is displayed, perform TEST 4. When other than normal function code is displayed, perform TEST 5 and check speed control cable function.

## TESTING & DIAGNOSIS

### TEST 1: POWER SOURCE CIRCUIT

1) Check CC ECU-IG fuse and replace if faulty. If fuse is okay, go to next step. If fuse failure is continuous, turn ignition off and disconnect Cruise Control Electronic Control Unit (CC ECU) wire harness connector. Install NEW fuse. Turn ignition on and check fuse. If fuse fails, check for short in wire harness between CC ECU-IG fuse and CC ECU. If fuse is okay, replace CC ECU and retest.

2) Turn ignition off and disconnect CC ECU wire harness connector. Ensure continuity exists between CC ECU ground terminal and body ground. See CC ECU CONNECTOR GROUND & POWER TERMINAL IDENTIFICATION table. If continuity is not present, repair open in circuit. If continuity is present, go to next step.

3) Turn ignition on. Measure voltage between power terminal and ground. See CC ECU CONNECTOR GROUND & POWER TERMINAL IDENTIFICATION table. If battery voltage is not present, repair open in circuit. If battery voltage is present, replace CC ECU and retest system.

CC ECU CONNECTOR GROUND & POWER TERMINAL IDENTIFICATION TABLE

| Application           | Ground Term.<br>(Wire Color) | Power Term.<br>(Wire Color) |
|-----------------------|------------------------------|-----------------------------|
| Motor Actuator .....  | A9 (BRN) .....               | A1 (BLK/RED)                |
| Vacuum Actuator ..... | 15 (BRN) .....               | 12 (BLK/RED)                |

### TEST 2: CRUISE CONTROL INDICATOR LIGHT CIRCUIT

1) Turn ignition off. Ensure GAUGE fuse is okay. If fuse is okay, go to next step. If fuse is not okay replace fuse and turn ignition on. If fuse fails, check for short on harness side and repair as necessary.

2) Remove combination meter and disconnect connector(s). Turn ignition on and ensure battery voltage exists between ground and connector terminal on GAUGE fuse side of combination meter. If battery voltage is present, go to next step. If battery voltage is not present, go to step 5).

3) Remove CRUISE indicator light and ensure continuity. Replace if necessary. If bulb is okay install bulb, connect combination meter connector and go to next step.

4) Turn ignition off. Disconnect CC ECU wire harness connector(s). Ensure continuity exists between ground and CC ECU

indicator light terminal. If continuity is not present, go to next step. If continuity is present, repair short in indicator light circuit between CC ECU connector and combination meter or between combination meter circuit plate and indicator light. See appropriate vehicle under step 5) for terminal identification.

5) Turn ignition on. Ensure battery voltage exists between ground and CC ECU indicator light terminal. If battery voltage is present, replace CC ECU and retest system. If battery voltage is not present, repair open in indicator light circuit between CC ECU connector and combination meter or open in combination meter circuit plate between indicator light terminal and indicator bulb or between GAUGE fuse terminal and indicator bulb.

6) Test between GAUGE fuse and combination meter 12-pin connector "B" terminal No. B7 (Red/Blue wire). Test between combination meter Brown 10-pin connector "C" terminal No. C2 and CC ECU connector terminal No. B6 with motor actuator or No. 4 with vacuum actuator, (Light Green wire). Test between combination meter 12-pin connector "B" terminal No. B7 (Red/Blue wire) and indicator light to combination meter Brown 10-pin connector "C" terminal No. C2 (Light Green wire).

### TEST 3: CONTROL SWITCH CIRCUIT

NOTE: Ensure main switch and control switch operations are normal. See COMPONENT TESTING.

1) Disconnect control switch connector(s), (6-pin with Supplemental Restraint System (SRS) or 14-pin connector "A" and Black 20-pin connector "B" without SRS). Ensure continuity exists between ground and control switch connector terminal No. 3 or B17 (Brown wire). If continuity is present, go to next step. If continuity is not present, repair faulty ground or open in circuit between ground and control switch terminal No. 3 or B17.

3) Disconnect CC ECU connector. Ensure continuity exists between main switch connector terminal and ground terminal or resistance between control switch connector terminal and ground terminal at specified switch positions. See CRUISE CONTROL SWITCH TERMINALS and CRUISE CONTROL SWITCH TESTING tables. Replace cruise control switch if results are not as specified. If results are as specified, go to step 8).

CRUISE CONTROL SWITCH TERMINALS TABLE

| Main Switch<br>Terminal<br>(Wire Color) | Control Switch<br>Terminal<br>(Wire Color) | Ground<br>Terminal<br>(Wire Color) |
|-----------------------------------------|--------------------------------------------|------------------------------------|
| 4 (WHT) (1) .....                       | 5 (YEL/BLK) .....                          | 3 (BRN)                            |
| B15 (YEL/BLK) (2) .....                 | B5 (WHT) .....                             | B17 (BRN)                          |

(1) - With Supplemental Restraint System.  
(2) - Without Supplemental Restraint System.

CRUISE CONTROL SWITCH TESTING TABLE

| Switch Position       | Result                |
|-----------------------|-----------------------|
| Main OFF .....        | No Continuity         |
| Main ON .....         | Continuity            |
| Control OFF .....     | No Continuity         |
| Control RES/ACC ..... | Approximately 68 Ohms |

|                         |                        |
|-------------------------|------------------------|
| Control SET/COAST ..... | Approximately 198 Ohms |
| Control CANCEL .....    | Approximately 418 Ohms |

---

#### SLIP RING TESTING TABLE

| 5-Pin Slip Ring<br>Terminal No. | 6-Pin Slip Ring<br>Terminal No. |
|---------------------------------|---------------------------------|
| 2 .....                         | 6                               |
| 3 .....                         | 4                               |
| 4 .....                         | 5                               |

---

4) Turn ignition off and connect control switch connector. Disconnect CC ECU connector(s). Turn CRUISE ON-OFF (MAIN) switch to OFF position and ensure no continuity exists between ground and CC ECU harness side connector terminal. See CC ECU-TO-CRUISE ON-OFF (MAIN) SWITCH HARNESS TESTING table. If continuity is not present, go to next step. If continuity is present, repair short in circuit between CC ECU terminal and control switch terminal.

5) If continuity is not present, turn CRUISE ON-OFF (MAIN) switch to ON position and ensure continuity exists between ground and CC ECU connector terminal. If continuity is not present, repair open in circuit between CC ECU terminal and control switch terminal. See CC ECU-TO-CRUISE ON-OFF (MAIN) SWITCH HARNESS TESTING table. If continuity is present, go to next step.

#### CC ECU-TO-CRUISE ON-OFF (MAIN) SWITCH HARNESS TESTING TABLE

| CC ECU<br>Terminal                | Control Switch<br>Terminal |
|-----------------------------------|----------------------------|
| A8 (YEL/BLK) Motor Actuator ..... | 5 (W/ SRS), 15 (W/O SRS)   |
| 6 (YEL/BLK) Vacuum Actuator ..... | 5 (W/ SRS), 15 (W/O SRS)   |

---

6) Turn control switch to OFF position and ensure no continuity exists between ground and CC ECU harness side connector terminal. See CC ECU-TO-CRUISE ON-OFF (MAIN) SWITCH HARNESS TESTING table. If continuity is present, repair short in circuit between CC ECU terminal and control switch terminal.

7) Turn control switch to 3 operating positions. Measure resistance between ground and CC ECU terminal at each switch position. See CC ECU-TO-CONTROL SWITCH HARNESS TESTING table. If resistance is not as specified, repair open in harness between CC ECU terminal and control switch terminal. If resistance is as specified, replace cruise control switch.

#### CC ECU-TO-CONTROL SWITCH HARNESS TESTING TABLE

| CC ECU<br>Terminal             | Control Switch<br>Terminal |
|--------------------------------|----------------------------|
| B8 (WHT) Motor Actuator .....  | 4 (W/ SRS), B5 (W/O SRS)   |
| 19 (WHT) Vacuum Actuator ..... | 4 (W/ SRS), B5 (W/O SRS)   |

---

### TEST 4A: ACTUATOR CKT VAC TYPE ACTUATOR (W/ VAC ACTUATOR)

1) Inspect actuator vacuum hoses and ensure no cracks or other damage is present. Replace vacuum hose as necessary.

2) Disconnect 3-pin connector from vacuum actuator. Ensure

continuity exists between ground and actuator harness side connector terminal No. 3 (White/Black wire). If continuity is present, verify actuator operation. See ACTUATOR OPERATION - WITH VACUUM TYPE ACTUATOR under COMPONENT TESTING. If actuator is okay, go to step 5). Replace actuator if it is faulty and retest system. If continuity is not present, go to next step.

3) Disconnect CC ECU connector and ensure continuity exists between ground and CC ECU harness side connector terminal No. 15 (Brown wire). If continuity is present, go to next step. If continuity is not present, repair faulty ground or open in harness between CC ECU terminal No. 15 and ground.

4) Ensure continuity exists between CC ECU side connector terminals No. 15 (Brown wire) and No. 16 (White/Black wire). If continuity is not present, replace CC ECU and retest. If continuity is present, repair open in harness between CC ECU terminal No. 16 and actuator connector terminal No. 3.

5) Ensure correct installation and operation of stoplight switch. See step 2) under TEST 4B: ACTUATOR CIRCUIT MOTOR TYPE ACTUATOR.

6) Connect actuator connector and disconnect stoplight switch 4-pin connector. Measure resistance between stoplight switch connector terminal No. 4 (Red/Yellow wire) and ground. If approximately 68 ohms are present, go to next step. If approximately 68 ohms are not present, repair open or short in harness between stoplight connector terminal No. 4 and actuator connector terminal No. 1 (Red/Yellow wire).

7) Connect stoplight switch 4-pin connector. Turn ignition off and disconnect CC ECU connector. On harness side measure resistance between CC ECU terminals No. 5 (Red/Green wire) and No. 16 (White/Black wire). If approximately 30 ohms are present, go to next step. If approximately 30 ohms are not present, repair open or short in harness between CC ECU connector terminal No. 5 and actuator connector No. 2 (Red/Green wire).

8) Ensure brake pedal is released. On harness side measure resistance between CC ECU connector terminals No. 3 (Green/Black wire) and No. 16 (White/Black wire). If approximately 68 ohms are present, replace CC ECU and retest system. If approximately 68 ohms are not present, repair open or short in harness between CC ECU connector terminal No. 5 (Red/Green wire) and stoplight connector terminal No. 3 (Green/Black wire).

## TEST 4B: ACTUATOR CIRCUIT MOTOR TYPE ACTUATOR

1) Disconnect actuator connector. Ensure continuity exists on harness side between actuator connector No. 4 (White/Black wire) and ground. See Fig. 4. If continuity is present, verify actuator operation. See ACTUATOR OPERATION - MOTOR TYPE under COMPONENT TESTING. Replace actuator if faulty and retest system. If actuator is okay, go to next step. If continuity is not present, repair faulty ground or open in harness between actuator terminal No. 4 and ground.

2) Ensure stoplights come on when brake pedal is depressed. If stoplights come on, go to step 5). If stoplights do not come on, ensure .02-.09" (.5-2.3 mm) clearance between stoplight switch and brake pedal stop. Loosen switch lock nut to adjust clearance. If stoplights do not come on after adjusting switch when brake pedal is depressed, disconnect 4-pin stoplight switch connector and go to next step.

3) Depress brake pedal (stoplight switch pin free). Ensure continuity exists between stoplight switch connector terminals No. 1 (Green/White wire) and No. 2 (Green/Red wire). If continuity is not as specified, replace stoplight switch and retest system. If stoplight switch is okay, go to next step.

4) Release brake pedal (stoplight switch pin pushed in).



Ensure continuity exists between stoplight switch connector terminals No. 3 (Green/Black wire) and No. 4 (Red/Yellow wire). If continuity is not as specified, replace stoplight switch and retest system. If stoplight switch is okay, go to next step.

5) Connect actuator connector and disconnect stoplight switch connector. Measure resistance between ground and stoplight switch terminal No. 4 (Red/Yellow wire). If resistance is approximately 38.5 ohms, go to next step. If resistance is not approximately 38.5 ohms, repair open or short in wire harness between stoplight connector terminal No. 4 and actuator connector terminal No. 5 (Red/Yellow wire).

NOTE: Magnet clutch and motor circuits include a diode. If circuit shows no continuity or incorrect resistance, reverse positive and negative test leads and retest circuit.



93F83406

Fig. 4: Motor Type Actuator Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

6) Connect stoplight switch connector and disconnect CC ECU connector(s). Measure resistance between ground and CC ECU connector terminal. See CC ECU-TO-MOTOR TYPE ACTUATOR HARNESS TESTING table. If resistance is approximately 38.5 ohms, go to next step. If resistance is not approximately 38.5 ohms, repair open or short in harness between CC ECU connector terminal and actuator connector terminal.

7) Ensure no continuity exists between ground and CC ECU connector terminals. If continuity is not present, go to next step. If continuity is present, repair short in harness between CC ECU terminal and actuator terminal.

8) Move actuator arm to a position other than fully open or fully closed. Ensure continuity exists between CC ECU connector terminals. If continuity is present, go to next step. If continuity is not present, repair open in harness between CC ECU connector terminal(s) and actuator terminal(s). See CC ECU-TO-MOTOR TYPE ACTUATOR HARNESS TESTING table.

9) Ensure no continuity exists between ground and CC ECU connector terminal No. A12. If continuity is not present, go to next step. If continuity is present, repair short in harness between CC ECU terminal No. 24 or No. A12 and actuator connector terminal No. 1. See CC ECU-TO-MOTOR TYPE ACTUATOR HARNESS TESTING table.

10) Measure resistance between CC ECU connector terminals No. 24 and No. A10 and No. A12. If approximately 2000 ohms resistance are present, go to next step. If approximately 2000 ohms are not present, repair open in harness between CC ECU connector terminal No. A10 and actuator connector terminal No. 3 or between CC ECU connector terminal No. A12 and actuator connector terminal No. 1. See CC ECU-TO-MOTOR TYPE ACTUATOR HARNESS TESTING table.

11) Move position sensor arm from closed to open position and back again. Ensure resistance change, as arm is moved, between CC ECU connector terminals No. A10 and No. A11. If resistance changes evenly, replace CC ECU and retest system. If resistance does not change evenly, repair open or short in harness between CC ECU terminal No. A11 and actuator terminal No. 2. See CC ECU-TO-MOTOR TYPE ACTUATOR HARNESS TESTING table.

| CC ECU<br>Terminal      | Actuator Terminal<br>Or To Ground |
|-------------------------|-----------------------------------|
| Magnet Clutch Circuit   |                                   |
| B3 (GRN/BLK) .....      | 5 (RED/YEL)                       |
| Motor Circuit           |                                   |
| B4 (PNK/BLK) .....      | 6 (PNK/BLK)                       |
| B10 (RED/GRN) .....     | 7 (RED/GRN)                       |
| Position Sensor Circuit |                                   |
| A10 (GRY/BLK) .....     | 3 (GRY/BLK)                       |
| A11 (GRY/RED) .....     | 2 (GRY/RED)                       |
| A12 (GRY) .....         | 1 (GRY)                           |

---

## TEST 5B: NO. 1 VEHICLE SPEED SENSOR CIRCUIT (A/T)

1) If speedometer fluctuates while driving at a steady speed, replace speedometer cable and retest system. If speedometer does not fluctuate, go to next step.

2) Remove combination meter to gain access to back. Disconnect 12-pin connector "B". Ensure continuity exists between ground and connector "B" terminal No. B5, Brown wire. If continuity is present, verify No. 1 vehicle speed sensor operation. See NO. 1 VEHICLE SPEED SENSOR (A/T) under COMPONENT TESTING. If speed sensor is okay, go to step 3), for models with motor actuator, go to step 4) for models with vacuum actuator see step 5). If continuity is not present, repair faulty ground and/or open in harness between connector "B" terminal No. B5 and ground.

3) On models with motor actuator, connect combination meter connector "B" and disconnect CC ECU 12-pin connector "A". Ensure repeated continuity exists between CC ECU connector terminal No. A7 (Blue/White wire) and ground. If continuity repeats, replace CC ECU and retest system. If continuity does not repeat, repair open or short in harness between CC ECU connector terminal No. A7 and combination meter terminal No. B6, or an open or short in circuit plate between terminal No. B6 and speed sensor, or an open in circuit plate between combination meter terminal No. B5 and speed sensor.

4) On models with vacuum actuator, connect combination meter connector "B" and disconnect CC ECU connector. Ensure repeated continuity exists between CC ECU connector terminal No. 8 (5S-FE, Blue/White wire; except 5S-FE, Blue wire) and ground. If continuity repeats, replace CC ECU and retest system. If continuity does not repeat, repair open or short in harness between CC ECU connector terminal No. 8 and combination meter terminal No. B6, or an open or short in circuit plate between terminal No. B6 and speed sensor, or an open in circuit plate between combination meter terminal No. B5 and speed sensor.

## TEST 5C: NO. 1 VEHICLE SPEED SENSOR CIRCUIT (WITH M/T)

1) Locate speed sensor on manual transaxle and disconnect 3-pin connector. Ensure continuity exists between connector terminal No. 3 (Brown wire) and ground. If continuity is not present, repair faulty ground or open in harness between connector terminal No. 3 and ground. If continuity is present, verify No. 1 vehicle speed sensor operation. See NO. 1 VEHICLE SPEED SENSOR (M/T) under COMPONENT TESTING. If speed sensor is okay, go to step 2) for models with motor actuator or step 3) for models with vacuum actuator.

2) Connect combination meter connector "B" and disconnect CC ECU 12-pin connector "A". Ensure repeated continuity exists between CC ECU connector terminal No. A7 (Blue/White wire) and ground as speed sensor is rotated. If continuity repeats, replace CC ECU and retest system. If continuity does not repeat, repair open or short in harness

between CC ECU connector terminal No. A7 and speed sensor terminal No. 2 (Blue wire).

3) Connect combination meter connector "B" and disconnect CC ECU connector. Ensure repeated continuity exists between CC ECU connector terminal No. 8 (Blue wire) and ground as speed sensor is rotated. If continuity repeats, replace CC ECU and retest system. If continuity does not repeat, repair open or short in harness between CC ECU connector terminal No. 8 and speed sensor terminal No. 2 (Blue wire).

## TEST 6: STOPLIGHT SWITCH

1) Verify normal operation of stoplight system. If stoplight system operates correctly, go to step 9). If stoplight system does not operate correctly, ensure STOP fuse is okay. If STOP fuse is okay, go to step 4). Replace fuse if needed. If fuse fails again, go to step 2). If fuse does not fail again, go to step 3).

2) If STOP fuse fails after replacement, disconnect CC ECU connector and replace fuse. If fuse fails again repair short in harness between stoplight switch terminal and STOP fuse or between CC ECU terminal and STOP fuse. See STOPLIGHT SWITCH TO STOP FUSE CIRCUIT table.

### STOPLIGHT SWITCH TO STOP FUSE CIRCUIT TABLE

| Stoplight Switch<br>Terminal | CC ECU<br>Terminal |
|------------------------------|--------------------|
| Motor Actuator               |                    |
| 2 (GRN/RED) .....            | A2 (GRN/RED)       |
| Vacuum Actuator              |                    |
| 2 (GRN/RED) .....            | 18 (GRN/RED)       |

3) If STOP fuse does not fail after replacement, depress brake pedal and inspect STOP fuse. If fuse does not fail, replace CC ECU and retest system. If fuse fails again, repair short in harness between stoplight terminal and stoplights or between CC ECU terminal and stoplights. See STOPLIGHT SWITCH TO STOPLIGHTS CIRCUIT table.

### STOPLIGHT SWITCH TO STOPLIGHTS CIRCUIT TABLE

| Stoplight Switch<br>Terminal | CC ECU<br>Terminal |
|------------------------------|--------------------|
| Motor Actuator               |                    |
| 1 (GRN/WHT) .....            | A3 (GRN/WHT)       |
| Vacuum Actuator              |                    |
| 1 (GRN/WHT) .....            | 17 (GRN/WHT)       |

4) Ensure stoplights come on when brake pedal is depressed. If stoplights do not come on, ensure .02-.09" (.5-2.3 mm) clearance between stoplight switch and brake pedal stop. Loosen switch lock nut to adjust clearance. If stoplights do not come on after adjusting switch when brake pedal is depressed, disconnect 4-pin stoplight switch connector and go to next step.

5) Depress brake pedal (stoplight switch pin free). Ensure continuity exists between stoplight switch connector terminals No. 1 (Green/White wire) and No. 2 (Green/Red wire).

6) Release brake pedal (stoplight switch pin pushed in). Ensure continuity exists between stoplight switch connector terminals No. 3 (Green/Black wire) and No. 4 (Red/Yellow wire). If continuity is

not as specified, replace stoplight switch and retest system. If stoplight switch is okay, go to next step.

7) Disconnect stoplight switch connector. Ensure battery voltage between stoplight connector terminal No. 2 (Green/Red wire) and STOP fuse. If battery voltage is present, go to next step. If battery voltage is not present, repair open in harness between stoplight connector terminal No. 2 and STOP fuse.

8) Using a jumper wire, connect stoplight connector terminals No. 1 (Green/White wire) and No. 2 (Green/Red wire). If brake lights come on, go to next step. If brake lights do not come on, repair open in harness between stoplight connector terminal No. 1 and ground.

9) If stoplight system is operating normally, start testing with this step. Connect stoplight connector and disconnect CC ECU connector (12-pin connector "A", models with motor actuator). Ensure battery voltage between ground and CC ECU connector terminal No. A2 (with motor actuator, Green/Red wire) or No. 18 (with vacuum actuator, Green/Red wire). If battery voltage is present, go to next step. If battery voltage is not present, repair open in harness between terminal No. 1 or No. A2 or No. 18 and STOP fuse.

10) Ensure battery voltage, when brake pedal is depressed, between ground and CC ECU terminal No. A3 (with motor actuator) or No. 17 (with vacuum actuator), Green/White wire. If battery voltage is present, replace CC ECU and retest system. If battery voltage is not present, repair open in harness between CC ECU connector terminal No. 16 or No. A3 or No. 17 and stoplight switch terminal No. 1, Green/White wire.

## TEST 7: PARKING BRAKE SWITCH

1) Disconnect parking brake switch. Ensure continuity exists between switch terminal and ground with switch in ON position (switch pin released). Ensure no continuity exists between switch terminal and ground with switch in OFF position (switch pin pushed in). If continuity is not as specified replace switch. If continuity is as specified, go to next step.

2) Connect parking brake switch connector and disconnect CC ECU connector (12-pin connector "A", models with motor actuator). Release parking brake and ensure no continuity exists between ground and CC ECU terminal No. A5 (with motor actuator, Red/Green wire) or No. 14 (with vacuum actuator, Red/Green wire). If continuity is present, repair short in harness between parking brake switch and CC ECU terminal No. 3 or No. A5 or No. 14. If continuity is not present, go to next step.

3) Ensure continuity exists, with parking brake set, between ground and CC ECU terminal No. A5 (with motor actuator) or No. 14 (with vacuum actuator). If continuity is not present, repair open in harness between parking brake switch and terminal No. 3 or No. A5 or No. 14. If continuity is present replace CC ECU and retest system.

## TEST 8: PARK/NEUTRAL POSITION SWITCH

NOTE: Park/Neutral position switch includes a diode. If circuit shows no continuity, reverse positive and negative test leads and retest circuit.

1) Ensure starting system is operating normally. Disconnect CC ECU connector (12-pin connector "A", models with motor actuator). Set shift lever to "P" or "N" position. Ensure continuity exists between ground and CC ECU connector terminal No. A4 (with motor actuator, Black wire) or No. 13 (with vacuum actuator, Black wire). If continuity is not present, repair open in harness between CC ECU terminal No. 2 or No. A4 or No. 13 and park/neutral position switch connector terminal No. 2 (Black/White wire).

2) Set shift lever to "L", 2, 3, and "R" positions. Ensure no continuity exists between ground and CC ECU connector terminal No. A4 (with motor actuator) or No. 13 (with vacuum actuator). If continuity is not present, replace CC ECU and retest system. If continuity is present, repair short in harness between terminal No. 2 or No. A4 or No. 13 and park/neutral position switch connector terminal No. 2 (Black/White wire).

## **TEST 9: CLUTCH SWITCH (M/T)**

1) Ensure engine does not start when clutch pedal is released. If engine starts, ensure .176-.216" (4.5-5.5 mm) clearance between clutch switch and clutch pedal stop. Adjust clutch switch as necessary.

2) Disconnect 2-pin clutch switch connector. Ensure continuity exists between terminals when clutch pedal is depressed (switch pin free). Ensure no continuity when clutch pedal is released (switch pin pushed in). If continuity is not as specified, replace clutch switch.

3) Disconnect clutch switch connector. Ensure continuity exists between ground and clutch switch terminal No. 1. If continuity is present, go to next step. If continuity is not present, repair faulty ground and/or open in harness between terminal No. 1 or No. 2 and ground.

4) Connect clutch switch connector and disconnect CC ECU connector (12-pin connector "A", with motor actuator). Release clutch pedal and ensure no continuity exists between ground and CC ECU connector terminal A4 (with motor actuator) or No. 13 (with vacuum actuator), Black wire. If continuity is not present, go to next step. If continuity is present, repair short in harness between CC ECU connector terminal No. 2 or No. A4 or No. 13 and clutch switch terminal No. 2 or No. 1.

5) Depress clutch pedal and ensure continuity exists between ground and CC ECU connector terminal No. A4 (with motor actuator), or No. 13, (models with vacuum actuator), Black wire. If continuity is present, replace CC ECU and retest system. If continuity is not present, repair open in harness between CC ECU connector terminal No. 2 or No. A4 or No. 13 and clutch switch terminal No. 2 or No. 1.

**NOTE:** Ensure Electrical Controlled Transmission (ECT) system is operating normally. Repair system as necessary.

## **TEST 10A: NO. 2 SOLENOID ELECTRICAL CONTROLLED TRANSMISSION**

1) Disconnect CC ECU connector (10-pin connector "B", models with motor actuator). Measure resistance between ground and CC ECU connector terminal No. B1 (with motor actuator, Brown/Yellow wire) or No. 9 (5S-FE A/T with vacuum actuator, Brown/Yellow wire). If approximately 13 ohms are present, replace CC ECU and retest system.

2) If approximately 13 ohms are not present, repair open or short in harness between No. 2 solenoid and CC ECU connector terminal No. B1 (with motor actuator, Brown/Yellow wire) or No. 9 (5S-FE A/T with vacuum actuator, Brown/Yellow wire).

**NOTE:** Ensure Electrical Controlled Transmission (ECT) system is operating normally. Repair system as necessary.

## **TEST 11: O/D OFF CIRCUIT (A/T)**

**NOTE:** Ensure overdrive off system is operating normally. Repair system as necessary.

1) Disconnect CC ECU connector (10-pin connector "B" with

motor actuator). Set overdrive main switch to ON position. Ensure continuity exists between ground and CC ECU connector terminal No. B2 (with motor actuator) or No. 7 (with vacuum actuator), (Pink wire). If continuity is present, go to next step. If continuity is not present, repair open in harness between CC ECU connector terminal No. B2 (with motor actuator) or No. 7 (with vacuum actuator), (Pink wire) and overdrive main switch.

2) Set overdrive main switch to OFF position. Ensure no continuity exists between ground and CC ECU connector terminal No. B2 (with motor actuator) or No. 7 (with vacuum actuator), (Pink wire). If continuity is present, replace CC ECU and retest system. If continuity is not present, repair short in harness between CC ECU connector terminal No. B2 (with motor actuator) or No. 7 (with vacuum actuator), (Pink wire) and overdrive main switch.

## TEST 12: THROTTLE POSITION SENSOR IDL SIGNAL

**NOTE:** Ensure throttle position sensor adjustment and operation are normal. See G - TESTS W/CODES article in the ENGINE PERFORMANCE section. Adjust or replace throttle position sensor as necessary.

1) On models with motor actuator, connect Throttle Position Sensor (TPS) connector and disconnect CC ECU connector (10-pin connector "B", models with motor actuator). Release accelerator pedal and ensure continuity exists between ground and CC ECU connector terminal No. B9 (with motor actuator, Pink wire).

2) If continuity is present, go to next step. If continuity is not present, repair open in harness between CC ECU connector terminal No. 23 or No. B9 and terminal No. 2/IDL of 4-pin throttle position sensor connector (Pink wire).

3) Depress accelerator pedal and ensure no continuity exists between ground and CC ECU connector terminal No. B9 (with motor actuator, Pink wire). If continuity is not present, replace CC ECU and retest system. If continuity is present, repair short in harness between CC ECU connector terminal No. 23 or No. B9 and terminal No. 2 of 4-pin throttle position sensor connector.

## COMPONENT TESTING

### Actuator Operation - With Vacuum Type Actuator

1) Disconnect 3-pin actuator connector. Resistance between actuator harness connector terminals No. 1 (Red/Yellow wire) and No. 3 (White/Black wire) should be approximately 68 ohms. Resistance between actuator terminals No. 2 (Red/Green wire) and No. 3 (White/Black wire) should be 30 ohms. If resistance is as specified, go to next step. If resistance is not as specified, replace actuator.

2) Connect positive battery lead to actuator terminals No. 1 and No. 2. Connect negative battery lead to actuator terminal No. 3. Slowly apply vacuum from 0-11.81 in. Hg. Control cable should pull smoothly as vacuum increases approximately 1.57" (40 mm). With vacuum stabilized control cable should not return. Disconnect positive battery from terminals No. 1 and 2. Control cable should return to original position as vacuum returns to zero. If vacuum actuator operation is not as specified, replace actuator and retest system.

**NOTE:** As vacuum is applied and held, drawn-in diaphragm may return. This does not indicate a malfunction as actuator leakage is allowable.

### Actuator Operation - With Motor Type Actuator

1) Disconnect motor actuator linkage and 7-pin actuator

connector. Ensure magnet clutch moves smoothly by moving arm by hand. Connect positive battery lead to actuator terminal No. 5 (Red/Yellow wire) and negative battery lead to actuator terminal No. 4

(White/Black wire). This activates magnet clutch and arm should not move by hand. If operation is not as specified, replace actuator motor. If operation is as specified, go to next step. See Fig. 4.

2) Do not disconnect battery leads from actuator terminals No. 5 and No. 4. With magnet clutch activated, connect positive battery lead to actuator terminal No. 6 (Pink/Black wire). Connect negative battery lead to actuator terminal No. 7 (Red/Green wire). Motor operation should start and arm should move smoothly to open position (acceleration side). When arm reaches open position, motor operation should stop.

3) With magnet clutch activated, reverse positive and negative leads on actuator terminals No. 6 and No. 7. Motor operation should start and arm should move smoothly to closed position (deceleration side). When arm reaches closed position, motor operation should stop. If motor operation is not as specified, replace actuator motor and retest system. If operation is as specified, disconnect battery leads from 7-pin actuator connector.

#### CRUISE CONTROL SWITCH TESTING TABLE

| Switch Position         | Result                 |
|-------------------------|------------------------|
| Control OFF .....       | No Continuity          |
| Control RES/ACC .....   | Approximately 68 Ohms  |
| Control SET/COAST ..... | Approximately 198 Ohms |
| Control CANCEL .....    | Approximately 418 Ohms |

#### Position Sensor Operation

Measure resistance between actuator terminals No. 1 (Gray wire) and No. 3 (Gray/Black wire). Resistance should be approximately 2000 ohms. Check resistance between actuator connector terminals No. 2 (Gray/Red wire) and No. 3 while moving arm from closed to open position. Resistance should increase from approximately 500 ohms to 1800 ohms. If operation is not as specified, replace actuator motor.

#### No. 1 Vehicle Speed Sensor (A/T)

Remove combination meter and locate speedometer/speed sensor at back of meter under 12-pin connector in middle of meter. Locate terminals No. "A" and No. "B" on right side of speedometer/speed sensor shaft. Ensure continuity exists between terminals No. "A" and No. "B" 4 times for each revolution of speedometer shaft. Replace speed sensor if operation is not as specified and retest system.

#### No. 1 Vehicle Speed Sensor (M/T)

Locate speed sensor on manual transaxle and remove. Disconnect 3-pin connector. Connect positive battery lead to connector terminal No. 1 (Black/Red wire) and negative lead to terminal No. 3 (Brown wire). Connect voltmeter positive lead to terminal No. 2 (Blue wire) and negative tester lead to battery negative terminal. Ensure approximately a 0-5 volt, change 4 times per each revolution of speed sensor shaft. Replace speed sensor if operation is not as specified.

### CRUISE CONTROL ECU CIRCUIT TESTING CHARTS

NOTE: CC ECU circuit testing charts are provided to pinpoint a malfunctioning circuit. Checking pin voltages at CC ECU connector will help determine if CC ECU is receiving and sending proper voltage signals. Using test charts may also help in determining if there is a short or open in harness

or connectors. Test circuit continuity, resistance and voltages by backprobing CC ECU harness connector.

NOTE: Unless stated otherwise in testing procedures, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance. Voltage readings may vary slightly due to battery condition or charging rate.



93F83357

Fig. 5: CC ECU Harness & Conn. Testing (W/ Vacuum Actuator - 1 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

**CC ECU HARNESS & CONNECTOR TESTING <sup>2</sup> (Celica W/ Vacuum Actuator)**

| Test Lead<br>(Red) | Pin<br>No. | Component Description                                                                    | <sup>1</sup> Specified Value |        | Pin<br>No. | Common Lead<br>(Black) |
|--------------------|------------|------------------------------------------------------------------------------------------|------------------------------|--------|------------|------------------------|
|                    |            |                                                                                          | ON                           | OFF    |            |                        |
| WHT/BLU            | 1          | DLC 1 Circuit (Connector Terminals No. 3 & 11 Connected)                                 | < 1Ω                         | N/A    | N/A        | GROUND                 |
| WHT/BLU            | 1          | DLC 1 Circuit (Connector Terminals Disconnected)                                         | N/A                          | ≥ 1 MΩ | N/A        | GROUND                 |
| YEL/BLK            | 6          | Main Control Switch (ON - Switch In ON Position)                                         | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| PNK <sup>3</sup>   | 7          | Engine Coolant Temperature & O/D Switch (Engine Cold)                                    | < 1Ω                         | N/A    | N/A        | GROUND                 |
| PNK <sup>3</sup>   | 7          | Engine Coolant Temperature & O/D Switch (Engine Hot)<br>(ON - O/D Switch In ON Position) | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| BLK                | 13         | Park/Neutral Switch A/T (ON - "N" Or "P" Position)                                       | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| BLK                | 13         | Clutch Switch M/T (ON - Depressed)                                                       | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| RED/GRN            | 14         | Parking Brake Switch (ON - Parking Brake Applied)                                        | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| BRN                | 15         | Cruise Control ECU Ground                                                                | < 1Ω                         | N/A    | N/A        | GROUND                 |
| GRN/BLK            | 3          | Actuator Release Valve (ON - Brake Pedal Released)                                       | 68Ω                          | ≥ 1 MΩ | 16         | WHT/BLK                |
| RED/GRN            | 5          | Actuator Control Valve                                                                   | 30Ω                          | N/A    | 16         | WHT/BLK                |
| BRN/YEL            | 9          | Electrical Controlled Transaxle No. 2 Solenoid Valve                                     | 13Ω                          | N/A    | N/A        | GROUND                 |
| WHT                | 19         | Control Switch (ON - Control Switch OFF Position)                                        | ≥ 1 MΩ                       | N/A    | N/A        | GROUND                 |
| WHT                | 19         | Control Switch (ON - Control Switch RES/ACC Position)                                    | 68Ω                          | N/A    | N/A        | GROUND                 |
| WHT                | 19         | Control Switch (ON - Control Switch SET/COAST Position)                                  | 198Ω                         | N/A    | N/A        | GROUND                 |
| WHT                | 19         | Control Switch (ON - Control Switch CANCEL Position)                                     | 418Ω                         | N/A    | N/A        | GROUND                 |

<sup>1</sup> - Symbol definitions: < means less than; > means greater than; ≤ means equal to or less than;  
≥ means equal to or greater than; B+ means battery voltage;  
Ω means Ohms; MΩ means Mega (Million) Ohms.

<sup>2</sup> - Disconnect Cruise Control ECU connector and test on harness side as shown. If results are not as specified, replace CC ECU and retest.

<sup>3</sup> - ECU Terminal No. 7 is Pink wire on 5S-FE A/T with Electronic Controlled Transmission, and Yellow/Black wire on 4A-FE A/T without ECT.

93G83358

Fig. 6: CC ECU Harness & Conn. Testing (W/ Vacuum Actuator - 2 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93F83357

Fig. 7: CC ECU Harness Voltage Testing (W/ Vacuum Actuator - 1 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



### CC ECU VOLTAGE TESTING <sup>2</sup> (Celica W/ Vacuum Actuator)

| Test Lead<br>(Red) | Pin<br>No. | Component Description                                                                                          | <sup>1</sup> Specified Value |      | Pin<br>No. | Common Lead<br>(Black) |
|--------------------|------------|----------------------------------------------------------------------------------------------------------------|------------------------------|------|------------|------------------------|
|                    |            |                                                                                                                | ON                           | OFF  |            |                        |
| LT GRN →           | 4          | Cruise Control Indicator Light<br>(ON - Ignition Switch In ON Position)                                        | B +                          | Zero | N/A        | GROUND                 |
| →                  | 8          | No. 1 Vehicle Speed Sensor<br>Ignition In ON Position, Speedometer Shaft<br>Or No. 1 Speed Sensor Shaft Turned | *                            | N/A  | N/A        | GROUND                 |
| BLK/RED →          | 12         | Cruise Control ECU Power Source ECU-IG Fuse<br>(ON - Ignition Switch In ON Position)                           | B +                          | Zero | N/A        | GROUND                 |
| GRN/WHT →          | 17         | Stoplight (ON - Brake Pedal Depressed)                                                                         | B +                          | Zero | N/A        | GROUND                 |
| GRN/RED →          | 18         | Stoplight Power Source STOP Fuse                                                                               | B +                          | Zero | N/A        | GROUND                 |

<sup>1</sup> - Symbol definitions: < means less than; > means greater than; ≤ means equal to or less than;  
≥ means equal to or greater than; B+ means battery voltage;  
Ω means Ohms; MΩ means Mega (Million) Ohms.

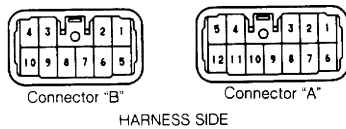
<sup>2</sup> - Disconnect Cruise Control ECU connector and test on harness side as shown. If results are not as specified, replace CC ECU and retest.

<sup>3</sup> - ECU terminal No. 8 is Blue wire. Except 5S-FE A/T, it is Blue/White wire.

<sup>4</sup> - Voltage will change repeatedly.

93H83359

Fig. 8: CC ECU Harness Voltage Testing (W/ Vacuum Actuator - 2 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93A83360

Fig. 9: CC ECU Harness Voltage Testing (W/ Motor Actuator - 1 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### CC ECU VOLTAGE TESTING <sup>2</sup> (Celica W/ Motor Actuator)

| Test Lead<br>(Red) | Pin<br>No. | Component Description                                                                                          | <sup>1</sup> Specified Value |      | Pin<br>No. | Common Lead<br>(Black) |
|--------------------|------------|----------------------------------------------------------------------------------------------------------------|------------------------------|------|------------|------------------------|
|                    |            |                                                                                                                | ON                           | OFF  |            |                        |
| BLK/RED →          | A1         | Cruise Control ECU Power Source ECU-IG Fuse<br>(ON - Ignition Switch In ON Position)                           | B +                          | Zero | N/A        | GROUND                 |
| GRN/RED →          | A2         | Stoplight Power Source STOP Fuse                                                                               | B +                          | N/A  | N/A        | GROUND                 |
| GRN/WHT →          | A3         | Stoplight (ON - Brake Pedal Depressed)                                                                         | B +                          | Zero | N/A        | GROUND                 |
| →                  | A7         | No. 1 Vehicle Speed Sensor<br>Ignition In ON Position, Speedometer Shaft<br>Or No. 1 Speed Sensor Shaft Turned | *                            | N/A  | N/A        | GROUND                 |

<sup>1</sup> - Symbol definitions: < means less than; > means greater than; ≤ means equal to or less than;  
≥ means equal to or greater than; B+ means battery voltage;  
Ω means Ohms; MΩ means Mega (Million) Ohms.

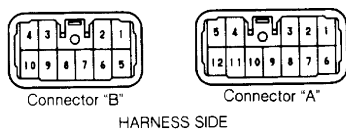
<sup>2</sup> - Disconnect Cruise Control ECU connector and test on harness side as shown. If results are not as specified, replace CC ECU and retest.

<sup>3</sup> - A/T is Blue/White wire. M/T is Blue wire.

<sup>4</sup> - Voltage will change repeatedly.

93B83361

Fig. 10: CC ECU Harness Voltage Testing (W/ Motor Actuator - 2 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93A83360

Fig. 11: CC ECU Harness & Conn. Testing (W/ Motor Actuator - 1 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## CC ECU HARNESS & CONNECTOR TESTING <sup>2</sup> (Celica W/ Motor Actuator)

| Test Lead<br>(Red) | Pin<br>No. | Component Description                                                                    | <sup>1</sup> Specified Value |        | Pin<br>No. | Common Lead<br>(Black) |
|--------------------|------------|------------------------------------------------------------------------------------------|------------------------------|--------|------------|------------------------|
|                    |            |                                                                                          | ON                           | OFF    |            |                        |
| BLK                | A4         | Park/Neutral Switch A/T (ON – "N" Or "P" Position)                                       | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| BLK                | A4         | Clutch Switch M/T (ON – Depressed)                                                       | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| RED/GRN            | A5         | Parking Brake Switch (ON – Parking Brake Applied)                                        | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| YEL/BLK            | A8         | Control Switch (ON – Switch In ON Position)                                              | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| BRN                | A9         | Cruise Control ECU Ground                                                                | < 1Ω                         | N/A    | N/A        | GROUND                 |
| GRY/BLK            | A10        | Actuator Position Sensor                                                                 | 2000Ω                        | N/A    | A12        | GRY                    |
| GRY/BLK            | A10        | Actuator Position Sensor (Actuator Arm Turned)                                           | <sup>4</sup>                 | N/A    | A11        | GRY/RED                |
| BRN/YEL            | B1         | Electrical Controlled Transaxle No. 2 Solenoid Valve                                     | 13Ω                          | N/A    | N/A        | GROUND                 |
| PNK                | B2         | Engine Coolant Temperature & O/D Switch (Engine Cold)                                    | < 1Ω                         | N/A    | N/A        | GROUND                 |
| PNK                | B2         | Engine Coolant Temperature & O/D Switch (Engine Hot)<br>(ON – O/D Switch In ON Position) | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |
| GRN/BLK            | B3         | Actuator Magnet Clutch (ON – Brake Pedal Released)                                       | 38.5Ω                        | ≥ 1 MΩ | N/A        | GROUND                 |
| PNK/BLK            | B4         | Actuator Motor (ON – Actuator Arm MAX. OPEN Position)                                    | < 1Ω                         | N/A    | B10        | RED/GRN                |
| RED/GRN            | B10        | Actuator Motor (ON – Actuator Arm MAX. CLOSE Position)                                   | < 1Ω                         | N/A    | B4         | PNK/BLK                |
| PNK/BLK            | B4         | Actuator Motor (ON – Actuator Arm In Any Position<br>Except OPEN Or CLOSE) <sup>3</sup>  | < 1Ω                         | N/A    | B10        | RED/GRN                |
| WHT/BLU            | B7         | DLC 1 Circuit (Connector Terminals No. 3 & 11 Connected)                                 | < 1Ω                         | N/A    | N/A        | GROUND                 |
| WHT/BLU            | B7         | DLC 1 Circuit (Connector Terminals Disconnected)                                         | N/A                          | ≥ 1 MΩ | N/A        | GROUND                 |
| WHT                | B8         | Control Switch (ON – Control Switch OFF Position)                                        | ≥ 1 MΩ                       | N/A    | N/A        | GROUND                 |
| WHT                | B8         | Control Switch (ON – Control Switch RES/ACC Position)                                    | 68Ω                          | N/A    | N/A        | GROUND                 |
| WHT                | B8         | Control Switch (ON – Control Switch SET/COAST Position)                                  | 198Ω                         | N/A    | N/A        | GROUND                 |
| WHT                | B8         | Control Switch (ON – Control Switch CANCEL Position)                                     | 418Ω                         | N/A    | N/A        | GROUND                 |
| PNK                | B9         | Throttle Position Sensor IDL (ON – Gas Pedal Released)                                   | < 1Ω                         | ≥ 1 MΩ | N/A        | GROUND                 |

<sup>1</sup> - Symbol definitions: < means less than; > means greater than; ≤ means equal to or less than;  
≥ means equal to or greater than; B+ means battery voltage;  
Ω means Ohms; MΩ means Mega (Million) Ohms.

<sup>2</sup> - Disconnect Cruise Control ECU connector and test on harness side as shown. If results are not as specified, replace CC ECU and retest.

<sup>3</sup> - Continuity should be present in either direction.

<sup>4</sup> - As actuator arm is turned resistance will change evenly.

93C83362

Fig. 12: CC ECU Harness & Conn. Testing (W/ Motor Actuator – 2 Of 2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS

Proceed to chassis WIRING DIAGRAMS article in WIRING DIAGRAMS section.

# DEFOGGER - REAR WINDOW

1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Rear Window Defoggers

Camry, Celica, Corolla, Land Cruiser, MR2, Paseo, Previa,  
Supra, Tercel, 4Runner

## DESCRIPTION & OPERATION

**NOTE:** Some systems use an integrated or multipurpose relay as defogger relay. Some systems use a timer between switch and heating grid, and some use only a switch and heating grid.

Rear window defogger systems use a heating wire grid bonded to the inside of window. Heat is regulated by a control switch and a relay/timer. Most systems have an indicator light to show system is operating.

Power to the control switch is through a fuse in the fuse block. Timer relay will keep power to the grid for 12-18 minutes, or until the ignition is turned off. On 4Runner, relay ground is through rear power window limit switch. Supra models are also available with an outside rearview mirror heater/defogger.

## TROUBLE SHOOTING

### DEFOGGER DOES NOT WORK

Blown fuse or poor contact. Defogger switch defective. Poor connections. Broken wire. Relay defective.

### INDICATOR LIGHT DOES NOT WORK

Bulb burned out. Open wire or poor connection.

## TESTING

### SYSTEM TESTING

1) Ensure all in-line fuses or circuit breakers are okay. Turn ignition and control switches to ON position. Glass should feel warm after a few minutes.

2) If glass is not warm, use a test light or voltmeter to check for battery voltage at grid feed wire. If voltage is not correct, check wiring harness, control switch and timer/relay.

### SWITCH TEST

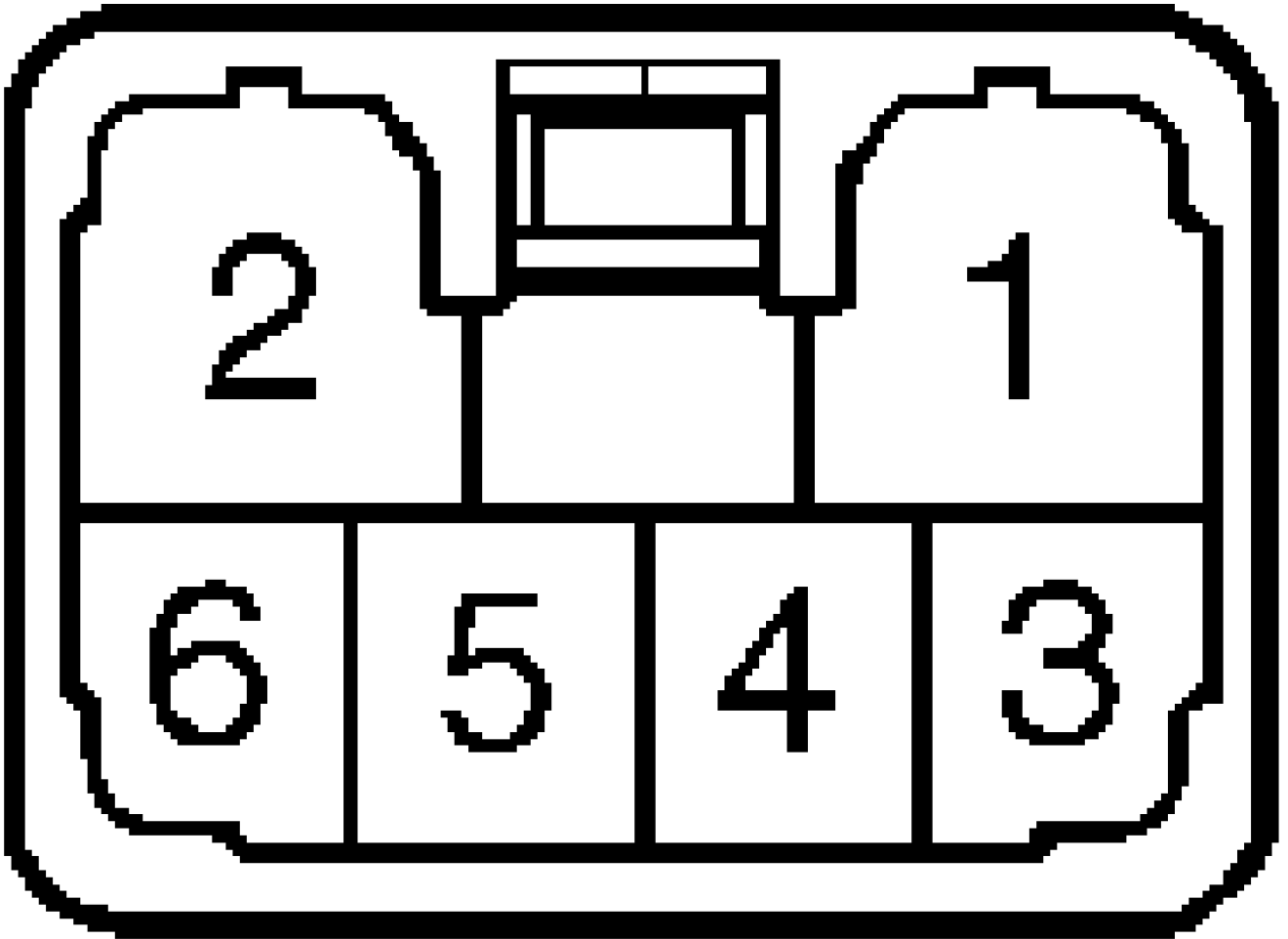
Camry

1) Remove defogger switch. Using an ohmmeter, check for continuity between terminals No. 3 and No. 4 of switch. See Fig. 1. Continuity should exist at all times.

2) With defogger switch on, check for continuity between terminals No. 1 and No. 5 of switch. Continuity should exist between terminals. With defogger switch off, no continuity should exist. If switch continuity is not as specified, replace switch.

3) To check switch indicator light operation, connect battery positive lead to terminal No. 3 of switch, and battery negative lead to terminal No. 4. Ensure defogger switch indicator light is on. If

switch indicator light is not on, replace switch.



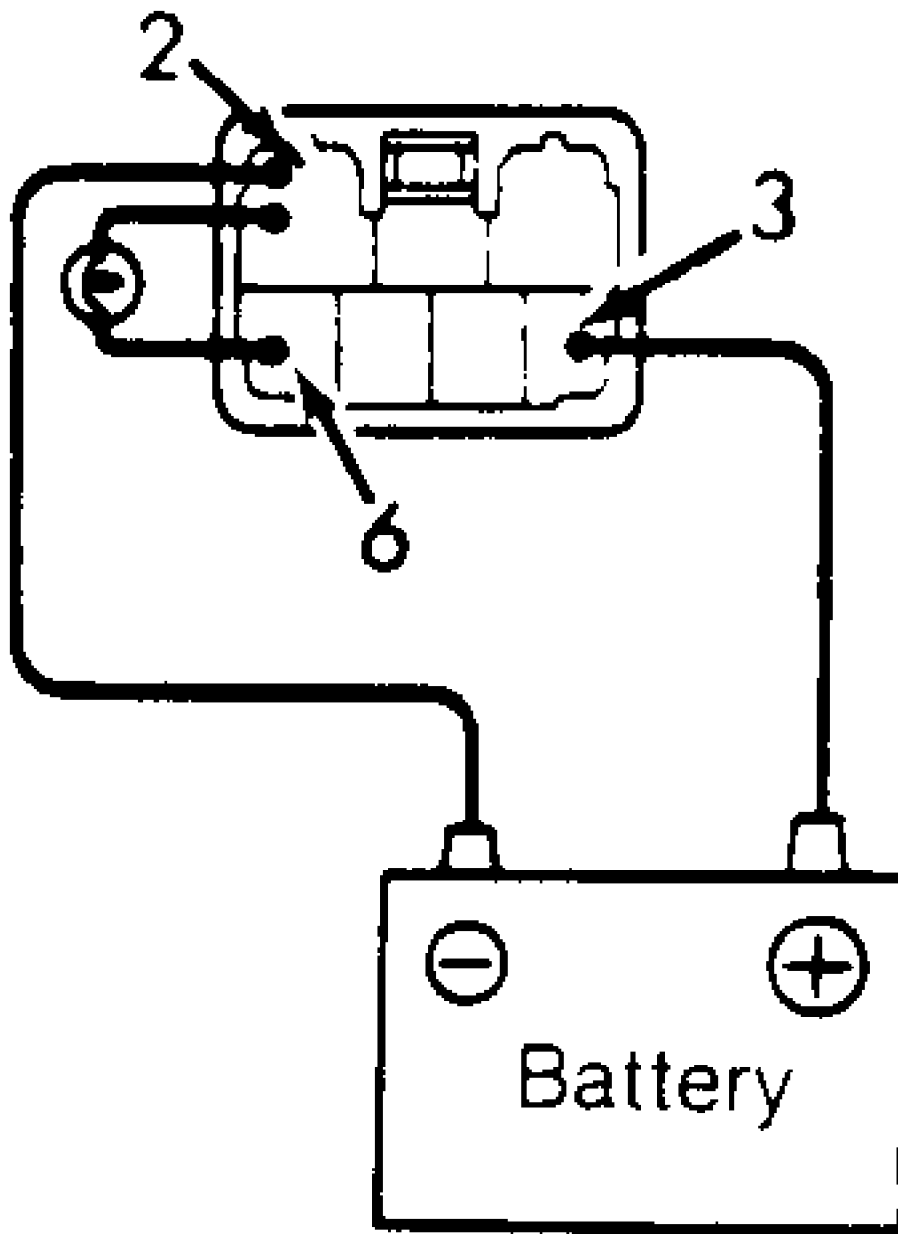
93D02543

Fig. 1: Defogger Switch Terminal ID (Except MR2 & Supra)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Celica

1) To test defogger switch without timer, ensure continuity exists between terminals No. 2, No. 3, and No. 6 with switch on. See Fig. 1. With switch off, continuity should not exist between terminals No. 2, No. 3, and No. 6. Check for continuity between terminals No. 4 and No. 5 (light bulb). Continuity should exist at all times. If continuity is not as specified, replace switch.

2) To test defogger switch with timer, connect battery positive lead to terminal No. 3, and battery negative lead to terminal No. 2. See Fig. 2. Connect 3.4-watt test light between terminals No. 2 and No. 6. Push defogger switch. Ensure test light lights for 12-18 minutes and then goes out. If switch does not operate as specified, replace switch.



WITH TIMER  
**93F02544**

Fig. 2: Testing Defogger Sw. W/ Timer (Celica Shown; Tercel Similiar)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Corolla

1) With defogger switch on, ensure continuity exists between

terminals No. 4 (White/Black wire) and No. 6 (Black/Blue wire). See Fig. 1. With switch off, continuity should not exist between terminals No. 4 and No. 6. Check for continuity between terminals No. 1 and No. 3 (light bulb). Continuity should exist at all times. If continuity is not as specified, replace switch.

2) To test defogger switch timer, connect battery positive lead to terminal No. 5 (Red/Blue wire) and battery negative lead to terminal No. 6 (Black/Blue wire). See Fig. 1. Connect 3.4-watt test light between terminals No. 4 (White/Black wire) and battery voltage. Push defogger switch to ON position. Ensure test light and indicator light, lights for 12-18 minutes and then goes out. If switch does not operate as specified, replace switch.

3) To test defogger timer circuit on harness side, disconnect defogger switch 6-pin connector. Ensure continuity between terminal No. 4 (Black/Blue wire) and ground all the time. Measure voltage between terminal No. 5 (Red/Blue wire) and ground and between terminal No. 6 (Black/Blue wire) and ground. Ensure battery voltage with ignition in ON position and zero voltage with ignition in OFF position. Using a jumper wire, connect connector terminals No. 4 and No. 6. With normal operation, defogger will come on. If results are as specified, replace defogger switch and retest system.

#### Land Cruiser, Previa & 4Runner

1) Remove rear defogger switch. Terminals No. 1 and No. 4 are for bulb illumination. Using an ohmmeter, check for continuity between terminals No. 1 and No. 4 of switch. See Fig. 1. Continuity should exist at all times.

2) With defogger switch on, check for continuity between terminals No. 2, No. 3, and No. 6 of switch. Continuity should exist between terminals. With defogger switch off, continuity should not exist between terminals No. 2, No. 3 and No. 6. If switch continuity is not as specified, replace switch.

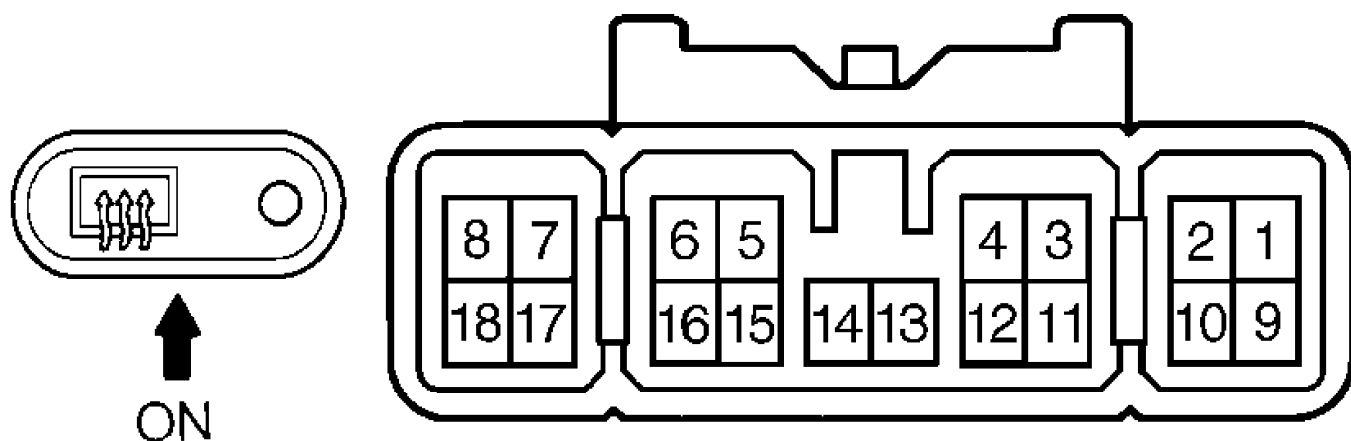
#### DEFOGGER SWITCH WIRE COLORS TABLE

| Terminal No. | Wire Color   |
|--------------|--------------|
| Land Cruiser |              |
| 2 .....      | Yellow       |
| 3 .....      | White/Black  |
| 6 .....      | Blue/Orange  |
| Previa       |              |
| 2 .....      | Red/Blue     |
| 3 .....      | White/Black  |
| 6 .....      | Blue         |
| 4Runner      |              |
| 2 .....      | Black/Orange |
| 3 .....      | Blue/Black   |
| 6 .....      | Black/Yellow |

#### MR2

1) On MR2, defogger switch is located in A/C control assembly. With defogger switch on, ensure continuity exists between terminals No. 10 (Red/Blue wire) and No. 11 (Red/Yellow wire) of switch. See Fig. 3. With switch off, continuity should not exist between terminals No. 10 and No. 11.

2) To check switch indicator light operation, connect battery positive lead to terminal No. 10 of switch, and battery negative lead to terminal No. 15 (White/Black wire). See Fig. 3. Ensure defogger switch indicator light is on. If switch indicator light is not on, replace switch.



93I02545

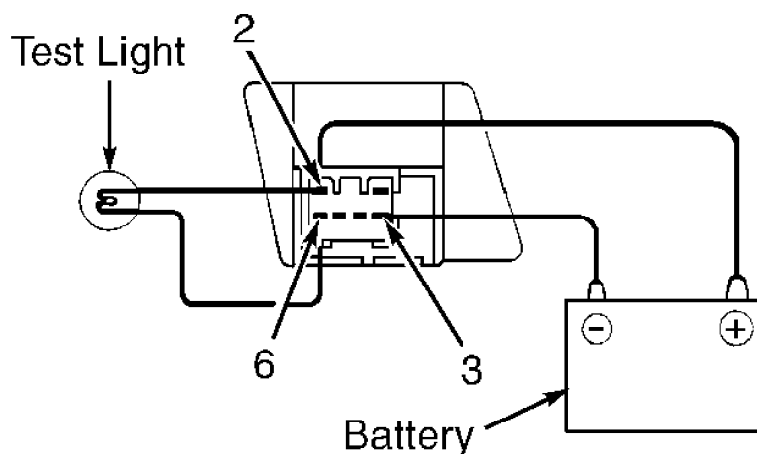
Fig. 3: Defogger Switch Terminal ID (MR2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Paseo

1) To test defogger switch with timer, remove defogger switch. Using an ohmmeter, check for continuity between terminals No. 1 and No. 4. See Fig. 1. If continuity does not exist, replace switch. Connect battery positive lead to terminal No. 2, and battery negative lead to terminal No. 3. See Fig. 4.

2) Connect 3.4-watt test light between terminals No. 2 and No. 6. Turn defogger on. Ensure test light lights for 12-18 minutes and then goes out. If switch does not operate as specified, replace switch.

3) To test defogger switch without timer, remove defogger switch. With switch on, ensure continuity exists between terminals No. 2, No. 4, and No. 6. See Fig. 1. With switch off, continuity should not exist between terminals No. 2, No. 4, and No. 6. Check for continuity between terminals No. 1 and No. 3 (light bulb). Continuity should exist at all times. If continuity is not as specified, replace switch.



93C02547

Fig. 4: Testing Defogger Switch (Paseo)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Supra

1) Defogger switch is part of heater control switch. Locate

A/C Heater Control switch Orange 14-pin "B" connector and ensure it is connected. See Fig. 5. Using voltmeter positive lead, backprobe connector terminal No. 7 (Pink/Black wire). Using voltmeter negative lead, backprobe "B" connector terminal No. 10 (Red/White wire). Ensure battery voltage with defogger switch in OFF position. Put switch to ON position, ensure indicator light is on and less than one volt between connector terminals No. 7 and No. 10. After 15 minutes, ensure defogger switch is off and battery voltage is again present. If voltage is not as specified, go to next step.

2) Disconnect A/C Heater Control switch Orange 14-pin connector. Put defogger switch to ON position. Ensure continuity between switch terminals No. 7 and No. 10. If continuity is not present, repair or replace A/C Heater Control switch.

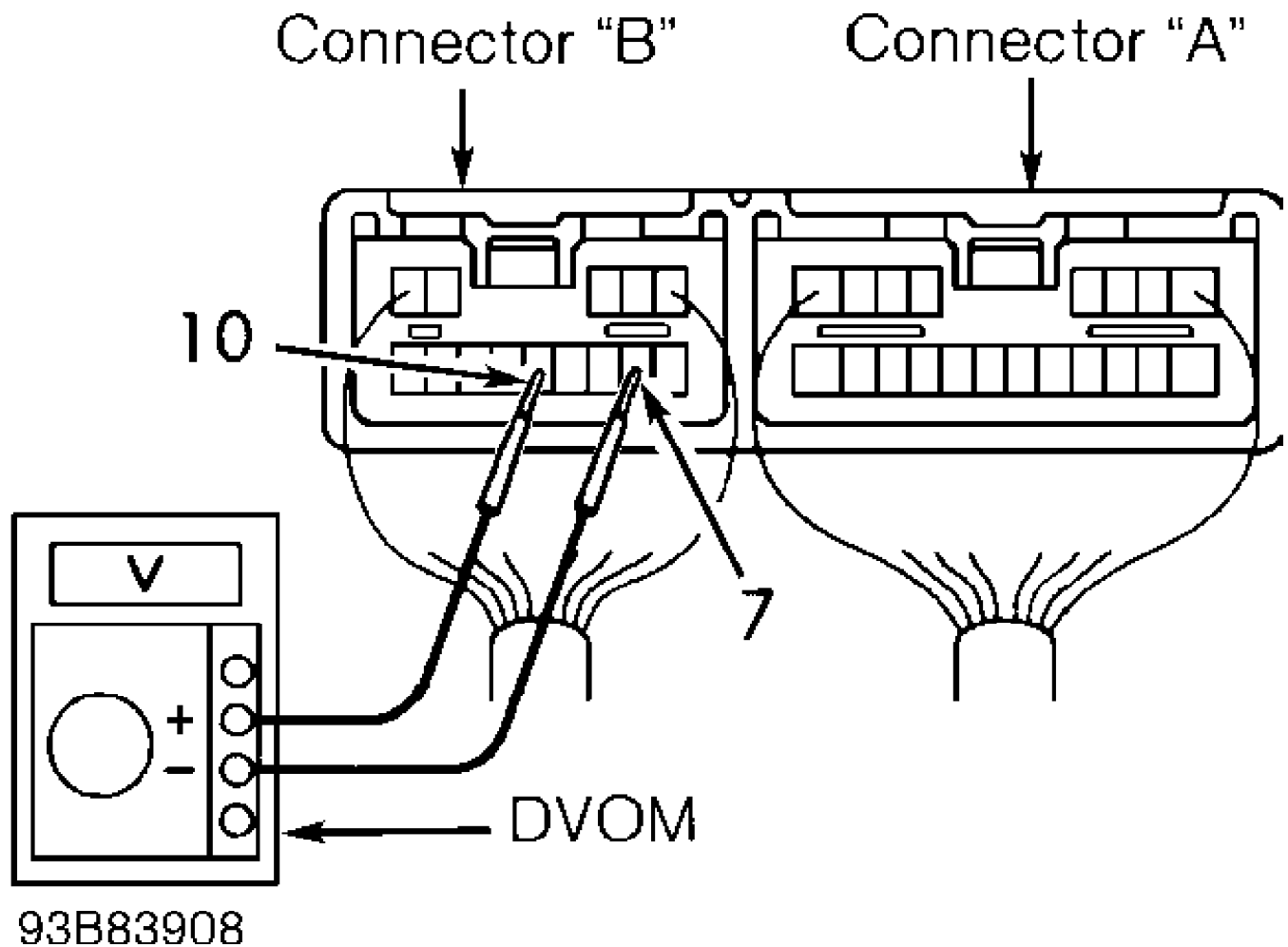


Fig. 5: Testing Defogger Switch (Supra)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Tercel

1) To test defogger switch without timer, remove defogger switch. Ensure continuity exists between switch connector terminals No. 2 (White/Black wire), No. 4 (Blue wire), and No. 6 (Black wire) with switch on. See Fig. 1. Check for continuity between switch connector terminals No. 2 and No. 6 (indicator bulb). Continuity should exist at all times. If continuity is not as specified, check switch indicator bulb and/or replace switch. With switch off, continuity should exist between switch connector terminals No. 1 and



No. 3. If continuity is not present, check illumination bulb.

2) To test defogger switch with timer, remove defogger switch. Ensure continuity between switch connector terminals No. 1 and No. 4. See Fig. 1. If continuity does not exist, replace illumination bulb.

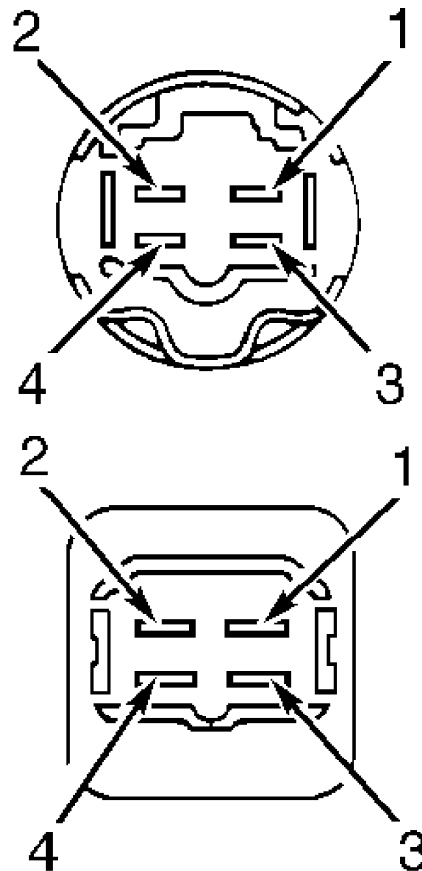
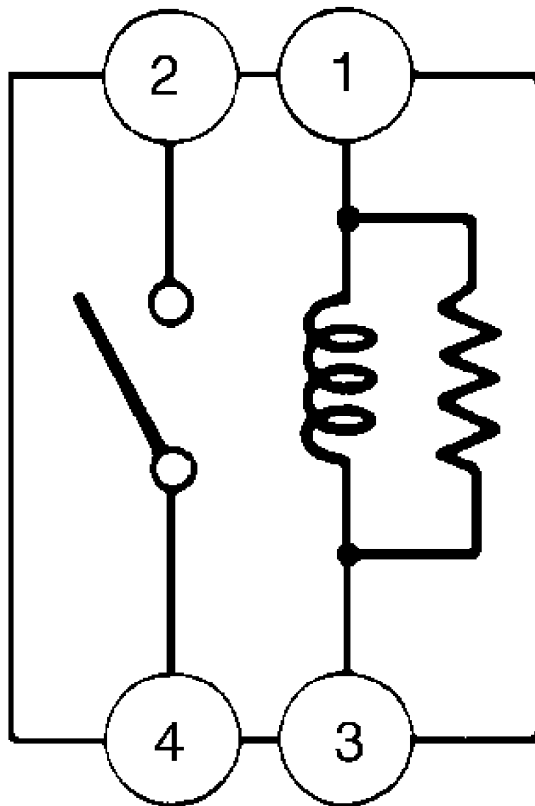
3) Connect battery positive lead to terminal No. 2 (Red/Blue wire), and battery negative lead to terminal No. 3 (White/Black wire). See Fig. 2. Connect 3.4-watt test light between terminals No. 2 and No. 6 (Green/White wire). Turn defogger on. Ensure indicator light and test light comes on for 12-18 minutes and then goes out. If defogger switch with timer, does not operate as specified, replace switch.

## RELAY TEST

Except Camry, MR2 & Tercel

1) Using an ohmmeter, ensure continuity exists between terminals No. 1 and No. 3. See Fig. 6. Continuity should not exist between terminals No. 2 and No. 4 and terminals No. 3 and No. 4. If continuity is not as specified, replace relay.

2) Connect battery positive lead to terminal No. 1 and battery negative lead to terminal No. 3. Continuity should exist between terminals No. 2 and No. 4. If operation is not as specified, replace relay.



91D00128

Fig. 6: Testing Defogger Relay (Except Camry, MR2 & Tercel)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Camry, MR2 & Tercel

1) Using an ohmmeter, ensure continuity exists between

terminals No. 1 and No. 2. See Fig. 7. Continuity should not exist between terminals No. 3 and No. 5. If continuity is not as specified, replace relay.

2) Connect battery positive lead to terminal No. 1 and battery negative lead to terminal No. 2. Continuity should exist between terminals No. 3 and No. 5. If operation is not as specified, replace relay.

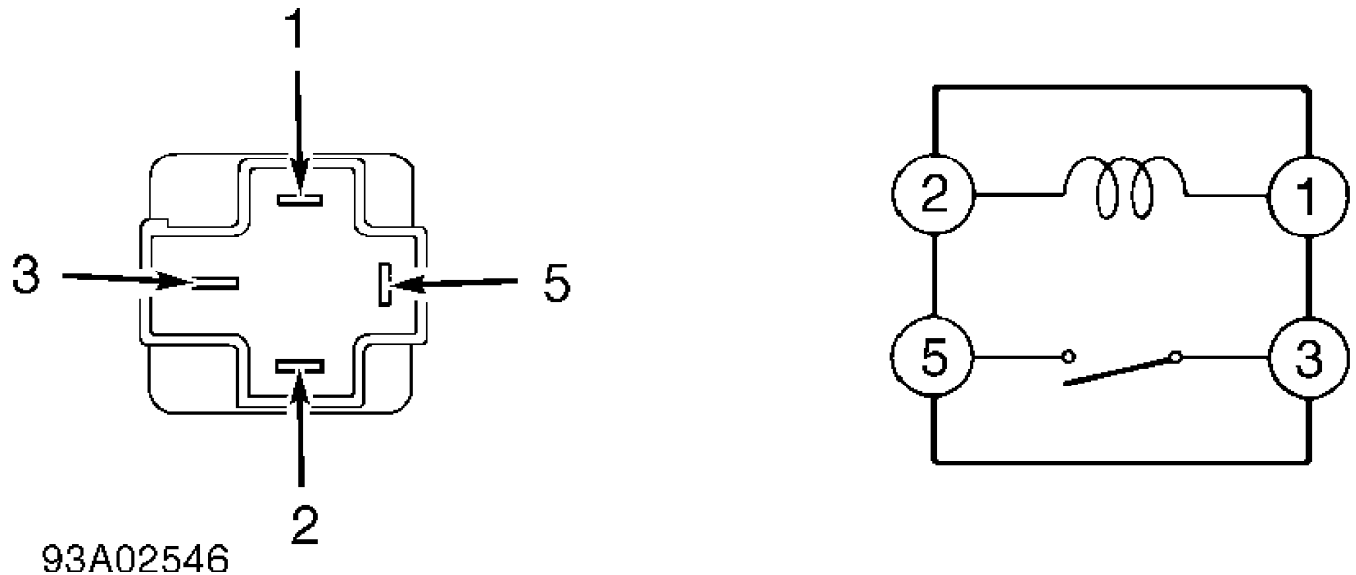


Fig. 7: Testing Defogger Relay (Camry, MR2 & Tercel)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## OUTSIDE REARVIEW MIRROR DEFOGGER TEST

### Rearview Mirror Defogger Operation (Supra)

Locate and disconnect outside rearview mirror 5-pin connector. Connect battery positive lead to terminal No. 2 (Blue/Red wire), and battery negative lead to terminal No. 1 (White/Black wire). After a short time, ensure mirror becomes warm. If mirror does not become warm, replace mirror assembly.

### Rearview Mirror Defogger Circuit (Supra)

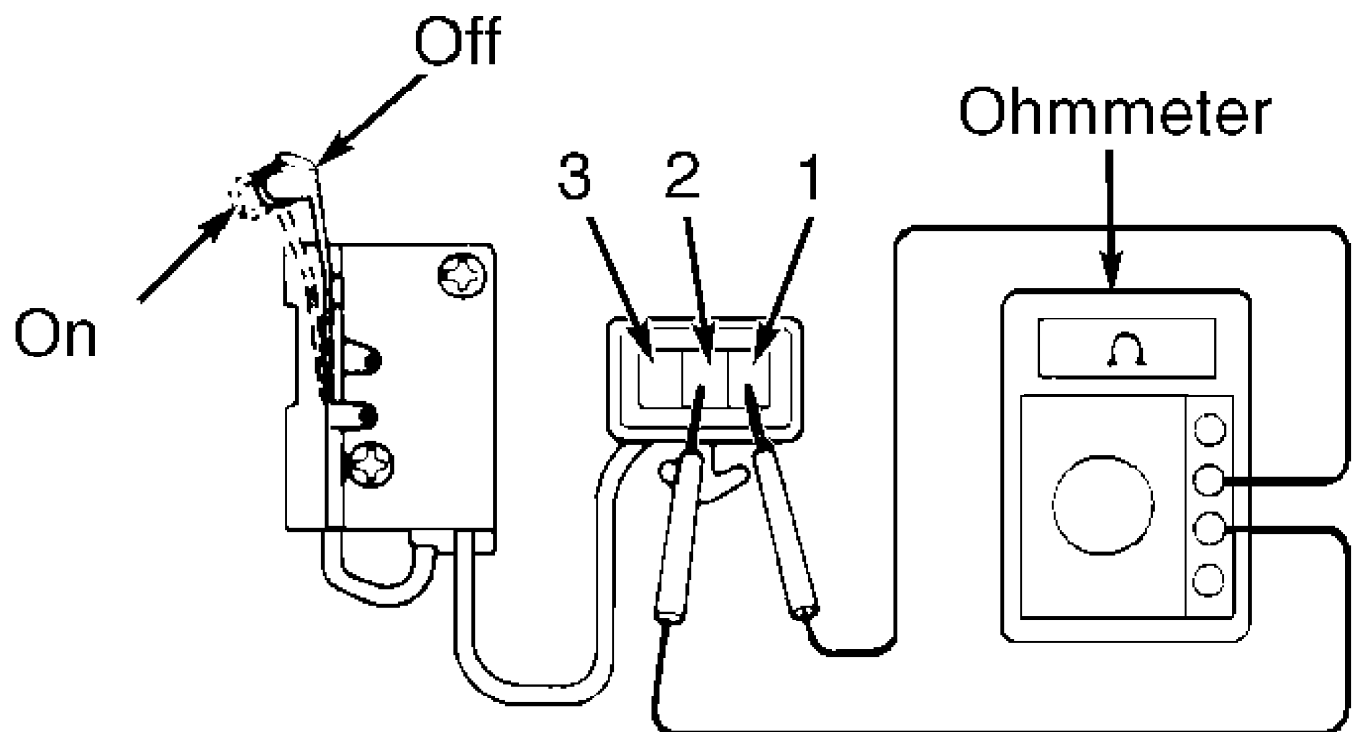
1) Locate and disconnect outside rearview mirror 5-pin connector. Ensure continuity between ground and 5-pin harness side connector terminal No. 1 (White/Black wire). If continuity is present, go to next step.

2) Turn ignition on and defogger switch off. Ensure no voltage between ground and 5-pin harness side connector terminal No. 2 (Blue/Red wire). Turn ignition switch on and defogger switch on. Ensure battery voltage between ground and 5-pin harness side connector terminal No. 2 (Blue/Red wire). If harness side circuit is not as specified, inspect other related components and/or harnesses.

## POWER WINDOW LIMIT SWITCH

### 4Runner

Power window limit switch is located behind trim panel and bottom of rear door. Using an ohmmeter, ensure continuity exists between terminals No. 1 (Red/Blue wire) and No. 2 (White/Black wire) when switch is turned to ON position. See Fig. 8. Continuity should not exist between any terminals when switch is pushed to OFF position. If continuity is not as specified, replace relay.



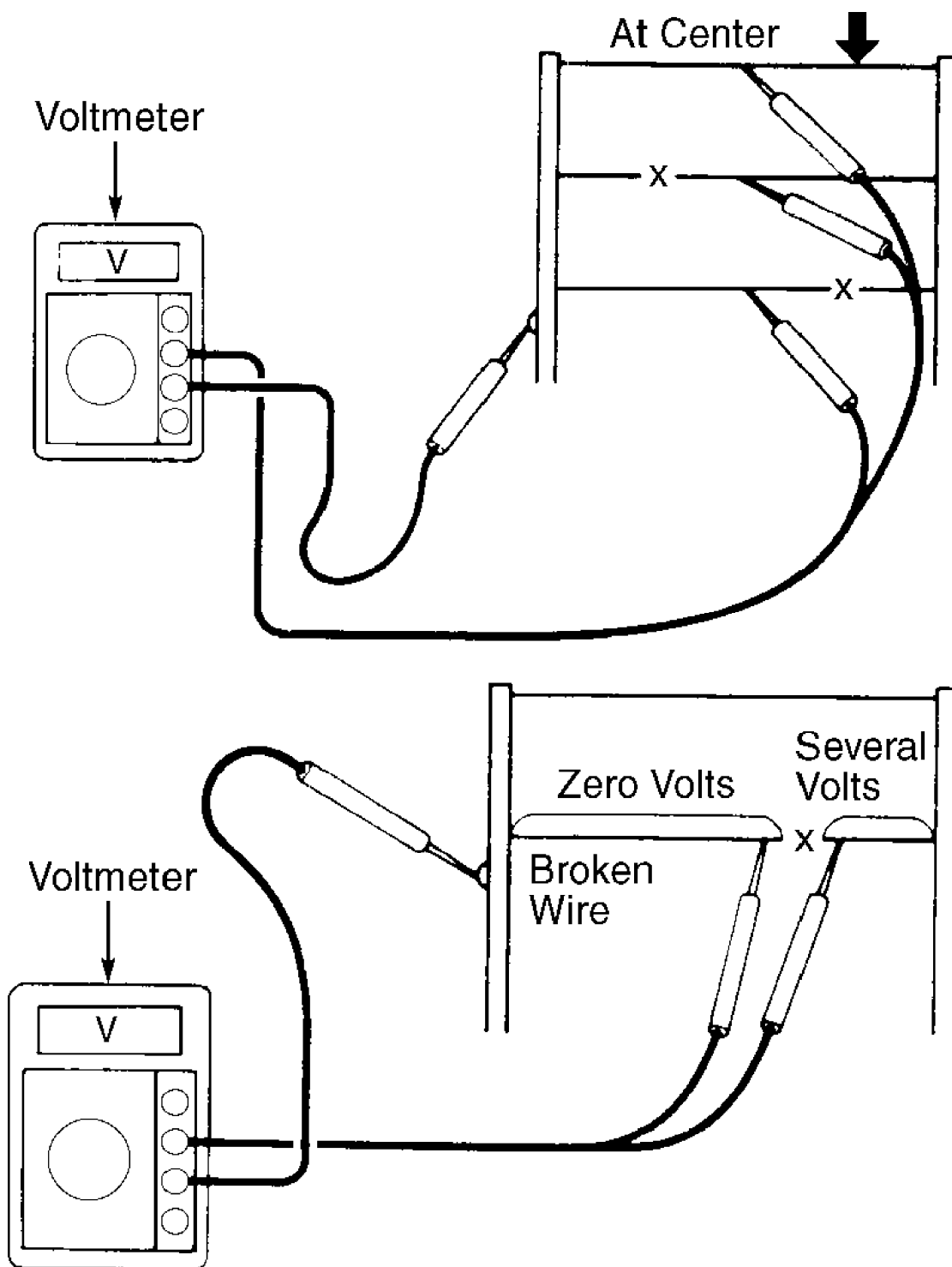
**91B03952**

Fig. 8: Testing Power Window Limit Switch (4Runner)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

### GRID FILAMENT TESTING

NOTE: When testing grid wires with voltmeter, wrap aluminum foil around end of test probe, then press foil to grid wire. This will prevent probe from damaging grid wire

1) To locate breaks in grid wire filaments, attach a voltmeter to middle portion of each filament. Attach other meter probe to vertical section of window grid. See Fig. 9.



91D03953

Fig. 9: Testing Grid Filament  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

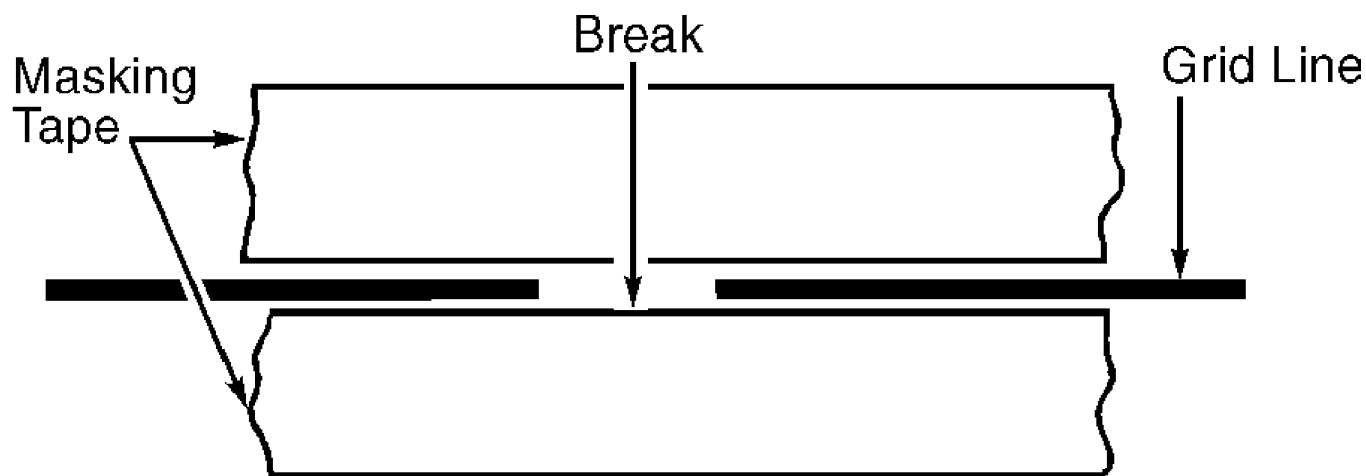
2) If a grid is broken, meter will register zero volts or about 10 volts, depending on if grid is broken between or outside test

leads. If wire is unbroken, meter will register about 5 volts. To locate break, move probe along wire until voltage changes abruptly.

## ON-VEHICLE SERVICE

### GRID FILAMENT REPAIR

Clean broken wire tips thoroughly. Place masking tape along both sides of broken wire. See Fig. 10. Apply Repair Paste (DuPont 4817) to broken section of grid. Remove masking tape after paste has dried. Wait 24 hours before using defogger.



G92A01033

Fig. 10: Repairing Rear Defogger Grid Filament  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS

Proceed to appropriate WIRING DIAGRAMS article listed below in WIRING DIAGRAMS section.

- \* WIRING DIAGRAMS (for Camry).
- \* WIRING DIAGRAMS (for Celica).
- \* WIRING DIAGRAMS (for Corolla).
- \* WIRING DIAGRAMS (for Land Cruiser).
- \* WIRING DIAGRAMS (for MR2).
- \* WIRING DIAGRAMS (for Paseo).
- \* WIRING DIAGRAMS (for Previa).
- \* WIRING DIAGRAMS (for Supra).
- \* WIRING DIAGRAMS (for Tercel).
- \* WIRING DIAGRAMS (for 4Runner).

# DOOR LOCKS - POWER

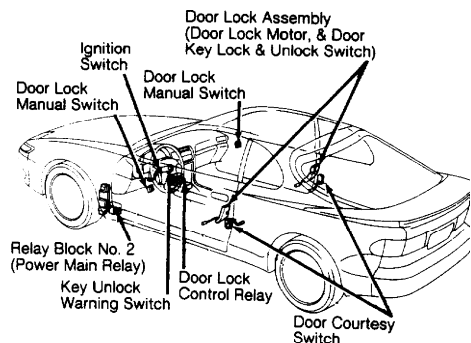
1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Power Door Locks

Celica

## DESCRIPTION & OPERATION

All doors can be locked or unlocked simultaneously using either front door. Turning driver door lock once will unlock driver door only, twice will unlock all doors. Door locks can be controlled by a switches on driver or passenger doors within vehicle, or by operating each door lock with key or lock knob. Front door(s) can not be manually locked when key is in ignition switch. See Fig. 1.



93184390

Fig. 1: Locating Power Door Lock Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TROUBLE SHOOTING

NOTE: Malfunctions that are most likely to occur are shown in order of their probability.

Door Lock System Does Not Operate.

- \* Check Fuse(s).
- \* Check Door Lock Switch Signal.
- \* Check Door Lock Motor Operation.
- \* Check Door Lock Control Relay. See POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS.
- \* Check Harness & Connectors.

Door Lock System Does Not Operate With Manual Switch.

- \* Check Door Lock Manual Switch.
- \* Check Door Lock Control Relay. See POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS.
- \* Check Door Lock Motor Operation.
- \* Check Harness & Connectors.

Door Lock System Does Not Operate With Door Key.

- \* Check Door Key Lock & Unlock Switch.
- \* Check Door Lock Control Relay. See POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS.
- \* Check Harness & Connectors.

- \* Check Door Lock Link Disconnected.

Driver Door 2-Key Turns, Key Unlock Function Does Not Operate.

- \* Check Door Key Lock & Unlock Switch.
- \* Check Harness & Connectors.
- \* Check Door Lock Control Relay. See POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS.

Key In Ignition Switch Warning, Does Not Operate.

- \* Check Key Unlock Warning Switch.
- \* Check Door Courtesy Switch.
- \* Check Door Lock Switch.
- \* Check Harness & Connectors.
- \* Check Door Lock Control Relay. See POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS.

Only One Door Lock Does Not Operate.

- \* Check Door Lock Motor Operation.
- \* Check Harness & Connectors.

## TESTING

### COMPONENT TESTING

#### Door Courtesy Switch

Locate door courtesy switch in each door next to power door lock assembly. Ensure continuity exists between terminal(s) and switch body with switch pin released (switch ON). Ensure no continuity exists between terminal(s) and switch body with switch pin pushed in (switch OFF). If continuity is not as specified, replace switch and retest system.

#### Door Key Lock & Unlock Switch

Locate door key lock and unlock switch connector behind door panel. Disconnect 7-pin connector. Ensure continuity exists between switch terminals No. 2 and No. 3, with switch in LOCK position. Ensure continuity exists between switch terminals No. 1 and No. 2, with switch in UNLOCK position. If continuity is not as specified, replace switch and retest system.

#### Door Lock Manual Switch

Locate door lock manual switch in front door. Disconnect 4-pin connector. Ensure continuity exists between switch terminals No. 2 (White/Black wire) and No. 4 (Blue/Black wire), with switch in lock position. Ensure continuity exists between switch terminals No. 2 and No. 3 (Blue wire), with switch in unlock position. Ensure no continuity in OFF position. If continuity is not as specified, replace switch and retest system.

#### Door Lock Motor Operation

Locate front or rear door lock motor and disconnect door lock motor 7-pin connector. Connect positive battery lead to terminal No. 7 (Blue/White wire) and negative battery lead to terminal No. 5 (Blue/Red wire). Ensure door lock link moves to LOCK position. Reverse battery leads and ensure door lock link moves to UNLOCK position. If door lock motor operation is not as specified, replace door lock assembly and retest system.

#### Door Unlock Detection Switch

Locate front door lock motor and disconnect door lock motor 7-pin connector. Ensure continuity exists between connector terminals No. 4 (Green wire) and terminal No. 6 (White/Black wire) with door unlock detection switch in UNLOCK position. Ensure no continuity with switch in LOCK position. If continuity is not as specified, replace door lock assembly and retest system.

#### Key Unlock Warning Switch

Locate ignition switch 10-pin connector. With key removed from switch, ensure continuity exists between connector terminals No. 1 and No. 5. If continuity is not present, replace key unlock warning switch.

#### Positive Temperature Coefficient (PTC) Thermistor Operation

1) Locate front or rear door lock motor in door. Disconnect door lock motor 7-pin connector. Connect positive battery lead to door lock motor terminal No. 7 (Blue/White wire). Connect ammeter positive lead to door lock motor terminal No. 5 (Blue/Red wire) and ammeter negative lead to battery negative terminal. Ensure current changes from 3.2 amps to less than 0.5 amps within 20-70 seconds. If current changes as specified, go to next step. If current does not change as specified, replace door lock assembly.

2) Disconnect test leads from terminals and wait at least 60 seconds. Connect positive battery lead to door lock motor terminal No. 5 and battery negative lead to terminal No. 7. Ensure door lock link moves to LOCK position. If operation is not as specified, replace door lock assembly and retest system.

### DOOR LOCK SWITCH SIGNAL TEST

**NOTE:** Ensure power door lock harness & connector circuits are okay before testing door lock switch signal. See appropriate POWER DOOR LOCK HARNESS & CONNECTOR CIRCUIT TESTING and POWER DOOR LOCK CIRCUIT VOLTAGE TESTING under POWER DOOR LOCKS CIRCUIT TESTING CHARTS.

#### Door Lock Switch Signal

Locate power door lock control relay. Ensure control relay 16-pin connector is connected. Using voltmeter positive lead backprobe connector terminal No. 3 (Blue/Red wire). Using voltmeter negative lead backprobe connector terminal No. 4 (Blue/White wire). Ensure voltage increases from zero to battery voltage for approximately 0.2 seconds with door lock manual switch in UNLOCK position. Reverse voltmeter leads and ensure voltage increases from zero to battery voltage for approximately 0.2 seconds with door lock manual switch in LOCK position. If voltage does not change as specified, replace power door lock control relay and retest system.

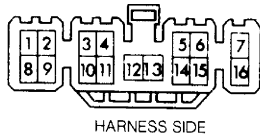
### POWER DOOR LOCK CONTROL RELAY CIRCUIT TESTING CHARTS

**NOTE:** Power Door Locks ECU or Control Relay Circuit Testing Charts are provided to pinpoint a malfunctioning circuit. Checking pin voltages at power door locks ECU or Control Relay connectors will help determine if power door locks ECU or Control Relay is receiving and sending proper voltage signals. Using test charts may also help in determining if there is a short or open in harness or connectors.

**NOTE:** Unless stated otherwise in testing procedures, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance. Voltage readings may vary



slightly due to battery condition or charging rate.



93H84407

Fig. 2: Power Door Lock Harness & Connector Circuit Testing  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### POWER DOOR LOCK HARNESS & CONNECTOR CIRCUIT TESTING (CELICA)

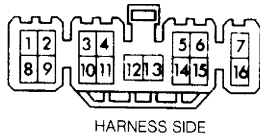
NOTE: Turn ignition off and disconnect power door lock relay 16-pin connector. Measure resistance (continuity) between terminal number shown and ground. If resistance is as specified, check door lock switch signal. See DOOR LOCK SWITCH SIGNAL under TESTING.

| Test Lead (Red) | Pin No. | Component Description                                                          | 1 Specified Value |                  | Pin No. | Common Lead (Black) |
|-----------------|---------|--------------------------------------------------------------------------------|-------------------|------------------|---------|---------------------|
|                 |         |                                                                                | ON                | OFF              |         |                     |
| RED/YEL         | 2       | Driver Door Courtesy Switch (ON - Door Open)                                   | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| GRN/YEL         | 5       | Pass. Door Unlock Detection Switch (ON - Door Unlocked)                        | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| GRN             | 6       | Driver Door Unlock Detection Switch (ON - Door Unlocked)                       | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| YEL             | 7       | Key Unlock Warning Switch (ON - Ignition Key Set)                              | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU/YEL         | 9       | Driver Door Key Lock & Unlock Switch (ON - Door Key Turned To Unlock Position) | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU/BLK         | 10      | Door Lock Manual Switch Position (ON - Lock Position)                          | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU             | 11      | Door Lock Manual Switch Position (ON - Unlock Position)                        | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU             | 11      | Pass. Door Key Lock & Unlock Switch Position (ON - Unlock Position)            | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU/ORG         | 12      | Door Key Lock & Unlock Switch Position (ON - Lock Position)                    | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| RED/GRN         | 14      | Pass. Door Courtesy Switch Position (ON - Door Open)                           | $< 1\Omega$       | $\geq 1 M\Omega$ | N/A     | GROUND              |
| BLU/YEL         | 15      | Power Main Relay Circuit (ON - Ign. Key Off)                                   | $70\Omega$        | N/A              | N/A     | GROUND              |
| WHT/BLK         | 16      | Power Door Lock Control Relay Ground                                           | $< 1\Omega$       | N/A              | N/A     | GROUND              |

1 - Symbol definitions:  $<$  means less than;  $>$  means greater than;  $\leq$  means equal to or less than;  $\geq$  means equal to or greater than; B+ means battery voltage;  $\Omega$  means Ohms;  $M\Omega$  means Mega (Million) Ohms.

93D84411

Fig. 3: Power Door Lock Harness & Connector Circuit Testing  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93H84407

Fig. 4: Power Door Lock Circuit Voltage Testing  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## POWER DOOR LOCK CIRCUIT VOLTAGE TESTING (CELICA)

NOTE: Turn ignition off and disconnect power door lock relay 16-pin connector. Measure voltage between terminal number shown and ground with ignition switch off or in specified position. If voltage is as specified, check door lock switch signal. See DOOR LOCK SWITCH SIGNAL under TESTING.

| Test Lead<br>(Red) | Pin<br>No. | Component Description                                                                        | 1 Specified Value |     | Pin<br>No. | Common Lead<br>(Black) |
|--------------------|------------|----------------------------------------------------------------------------------------------|-------------------|-----|------------|------------------------|
|                    |            |                                                                                              | ON                | OFF |            |                        |
| RED/BLU →          | 1          | Door Courtesy Switch (ON – Ign. In ON Position)                                              | B +               | N/A | N/A        | ← GROUND               |
| BLU/RED →          | 3          | Door Lock Control Switch (ON – Unlocked)                                                     | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| BLU/RED →          | 3          | Door Lock Control Switch (ON – Locked W/ Key<br>In Ignition & Driver Door Open) <sup>3</sup> | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| BLU/RED →          | 3          | Door Lock Knob (ON – Locked W/ Key<br>In Ignition & Driver Door Open) <sup>2</sup>           | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| BLU/RED →          | 3          | Driver/Pass. Door (ON – Unlocking Door Cylinder W/ Key)                                      | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| BLU/WHT →          | 4          | Door Lock Control Switch (ON – Switch Locked)                                                | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| BLU/WHT →          | 4          | Driver Or Pass. Door (ON – Locking Door Cylinder W/ Key)                                     | B + <sup>3</sup>  | N/A | N/A        | ← GROUND               |
| WHT/BLU →          | 8          | Constant Power Supply (30 Amp POWER Fuse)                                                    | B +               | N/A | N/A        | ← GROUND               |
| BLU/YEL →          | 9          | Driver Door Lock Cylinder (ON – Unlocked With Key)                                           | ZERO              | N/A | N/A        | ← GROUND               |
| BLU/BLK →          | 10         | Door Lock Control Switch (ON – Switch Locked)                                                | ZERO              | N/A | N/A        | ← GROUND               |
| BLU →              | 11         | ON – Door Lock Control Switch Unlocked Or<br>Pass. Door Lock Cylinder Unlocked W/ Key        | ZERO              | N/A | N/A        | ← GROUND               |
| BLU/ORG →          | 12         | Driver & Pass. Door Lock Cylinder (ON – Locked W/ Key)                                       | ZERO              | N/A | N/A        | ← GROUND               |
| BLU/YEL →          | 15         | Power Main Relay Circuit (ON – Ign. In ON Position)                                          | B +               | N/A | N/A        | ← GROUND               |

<sup>1</sup> - Symbol definitions: < means less than; > means greater than; ≤ means equal to or less than;  
≥ means equal to or greater than; B + means battery voltage;  
Ω means Ohms; MΩ means Mega (Million) Ohms.

<sup>2</sup> - Key in ignition switch, reminder function.

<sup>3</sup> - Battery voltage present for 0.2 seconds after switch operation.

93E84412

Fig. 5: Power Door Lock Circuit Voltage Testing  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS

Proceed to chassis WIRING DIAGRAMS article in WIRING DIAGRAMS section.

# ELECTRICAL COMPONENT LOCATOR

1993 Toyota Celica

1993 ELECTRICAL COMPONENT LOCATION  
Toyota Electrical Components

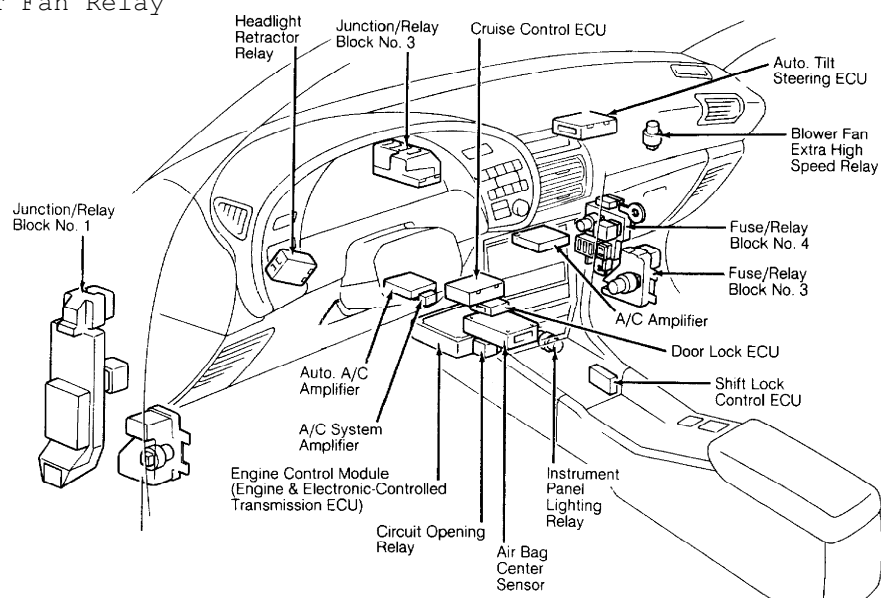
Toyota; Celica

## SAFETY PRECAUTION

**WARNING:** When working on vehicles equipped with Supplemental Restraint System (SRS), never apply electrical voltage to the system. This could cause the SRS (air bag) to be deployed. For complete Air Bag Safety precautions see AIR BAGS article in ACCESSORIES/SAFETY EQUIPMENT Section.

## BUZZERS, RELAYS & TIMERS

| Component                             | Component Location                                       |
|---------------------------------------|----------------------------------------------------------|
| ABS Control Relay                     | Mounted to top right side of radiator.                   |
| A/C Condenser Fan Relays<br>No. 2 & 3 | On relay block No. 5, right front of engine compartment. |
| A/C Magnetic Clutch Relay             | On relay block No. 5, right front of engine compartment. |
| Blower Fan Relay                      |                                                          |

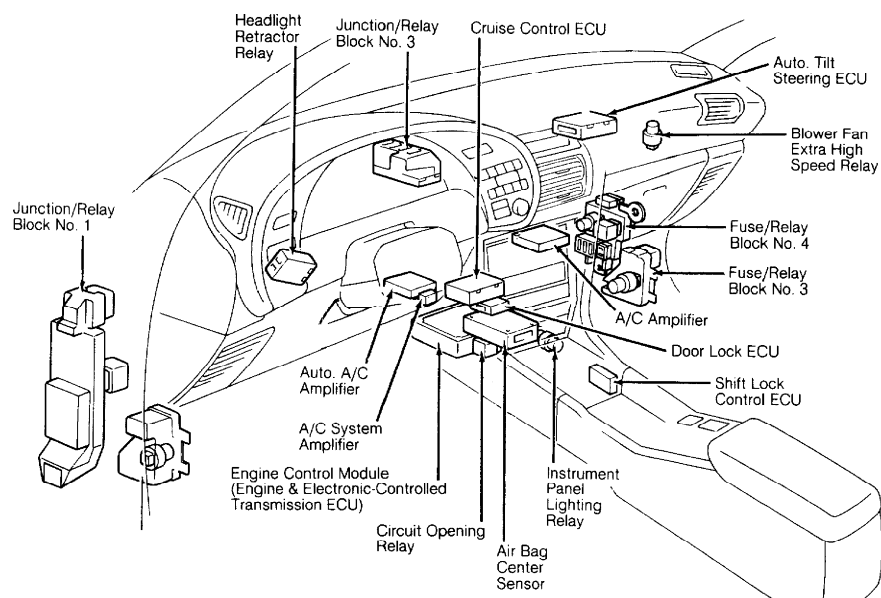


94A31468  
Component Location

Behind right side of dash,  
above glove box.

Bulb Check Relay

On rear of instrument cluster.



94A31468  
Circuit Opening Relay

Under center of dash,  
forward of engine control  
module.

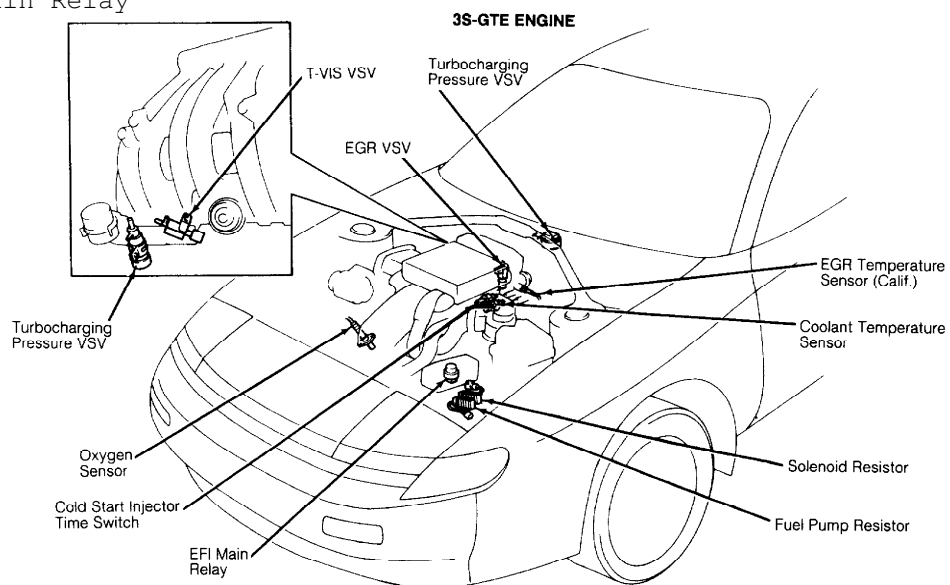
Defogger Relay

On junction/relay block No. 1,  
behind left kick panel.

Door Lock Relay

Mounted under center console,  
in front of gear selector.

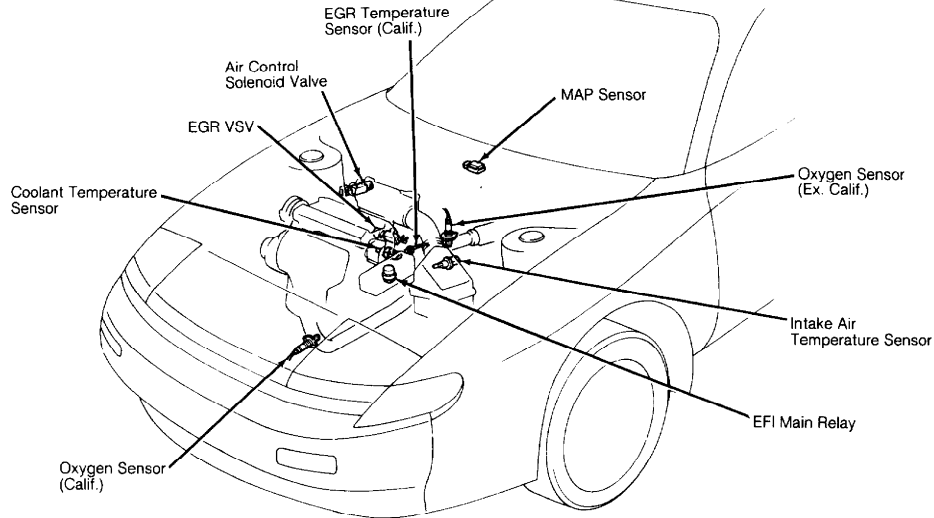
EFI Main Relay



94F31470  
(3S-GTE)

On junction/relay block  
No. 2, near battery.

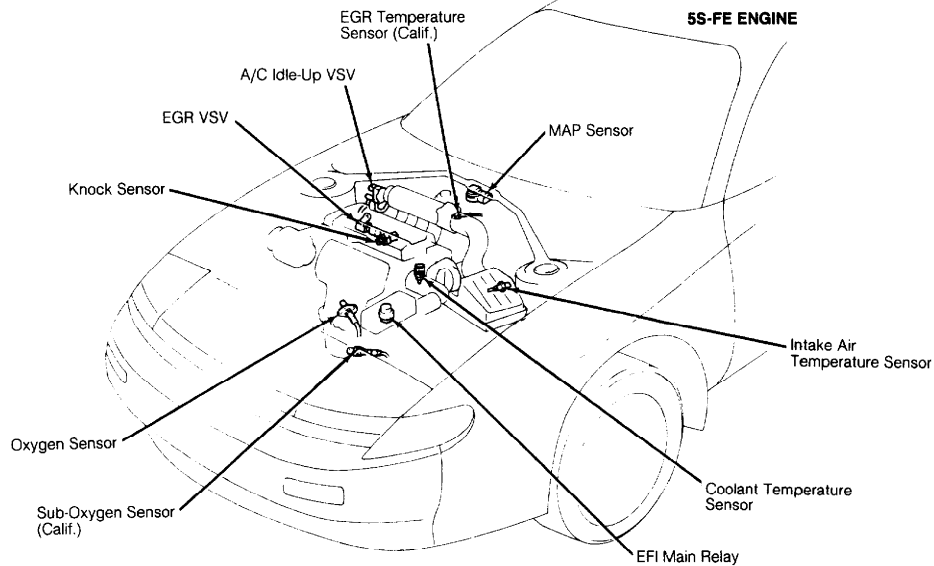
#### 4A-FE ENGINE



(4S-FE) 94F31471

On junction/relay block No. 2, near battery.

#### 5S-FE ENGINE



(5S-FE) 94G31472

On junction/relay block No. 2, near battery.

Engine Main Relay

On junction/relay block No. 2, near battery.

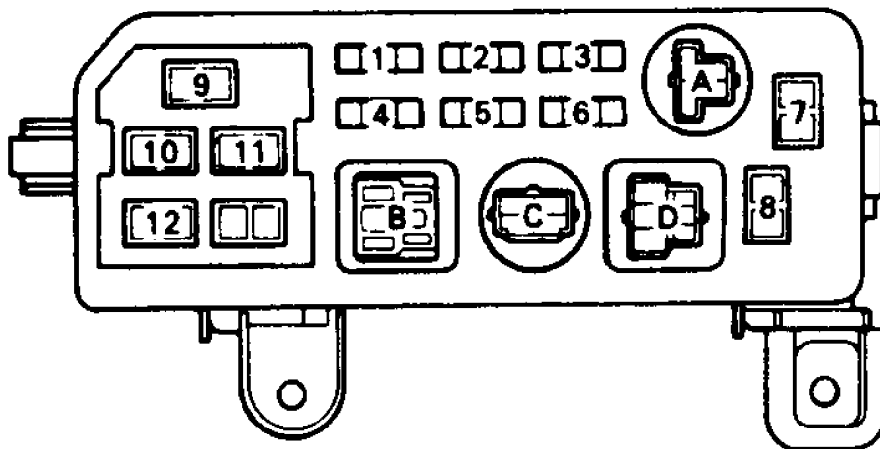
Foglight Relay

On fuse/relay block No. 3, right side kick panel.

Fuel Pump Relay (All-Track/4WD)

On relay block No. 5, right front of engine compartment.

**4A-FE & 5S-FE SHOWN; 3S-GTE SIMILAR**



**JUNCTION RELAY BLOCK NO. 2**

**FUSES & FUSIBLE LINKS (FL)**

1. Headlight (LH)
2. Headlight Retractor
3. Headlight (RH)
4. EFI
5. Dome
6. Hazard-Horn
7. Radiator Cooling Fan (FL)
8. Condenser Cooling Fan (FL)
9. ALT (FL)
10. AM2 (FL)
11. AM1 (FL)
12. ABS (FL)

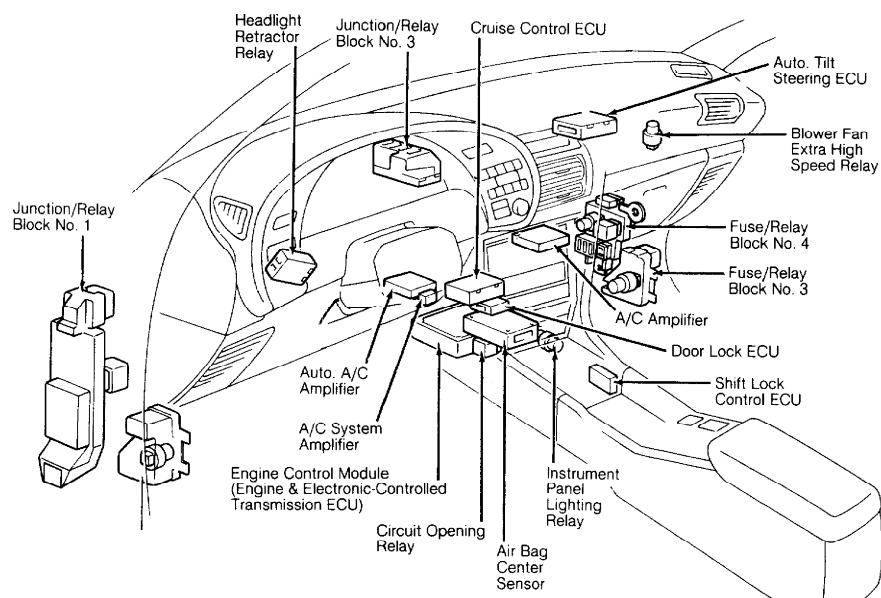
**RELAYS**

- A. A/C Fan Relay No. 1
- B. Headlight Control Relay
- C. EFI Main Relay
- D. Engine Main Relay

**93A27565**

Headlight Control Relay

On junction/relay block No.  
2, near battery.



94A31468  
Headlight Retractor Relay

Behind left side of dash.

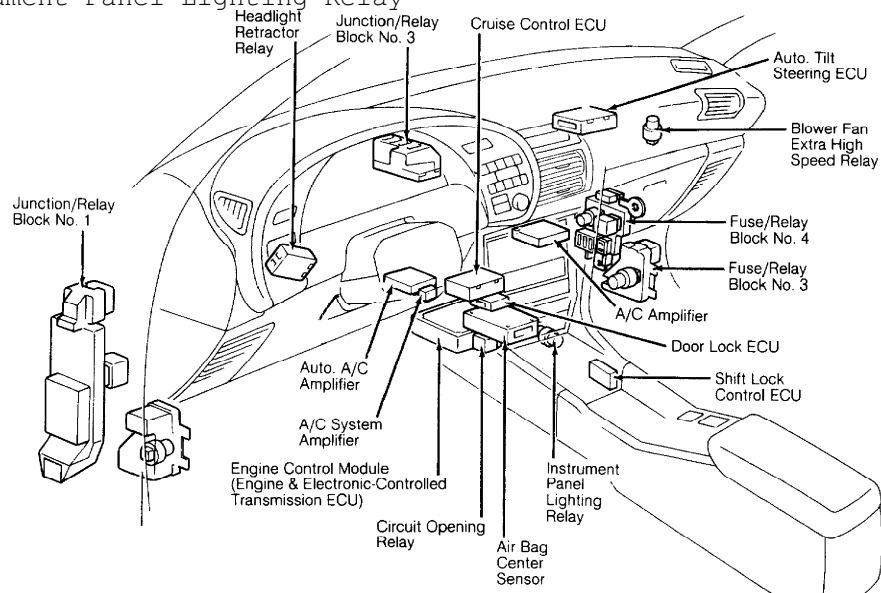
Heater Relay

On fuse/relay block No. 4,  
behind right kick panel.

Horn Relay

On relay block No. 5, right  
front of engine compartment.

Instrument Panel Lighting Relay



94A31468  
Component Location

Behind center of dash,  
mounted near right rear of  
ashtray bracket.

Integration Relay

On back side of junction/relay  
block No. 1.

Power Main Relay

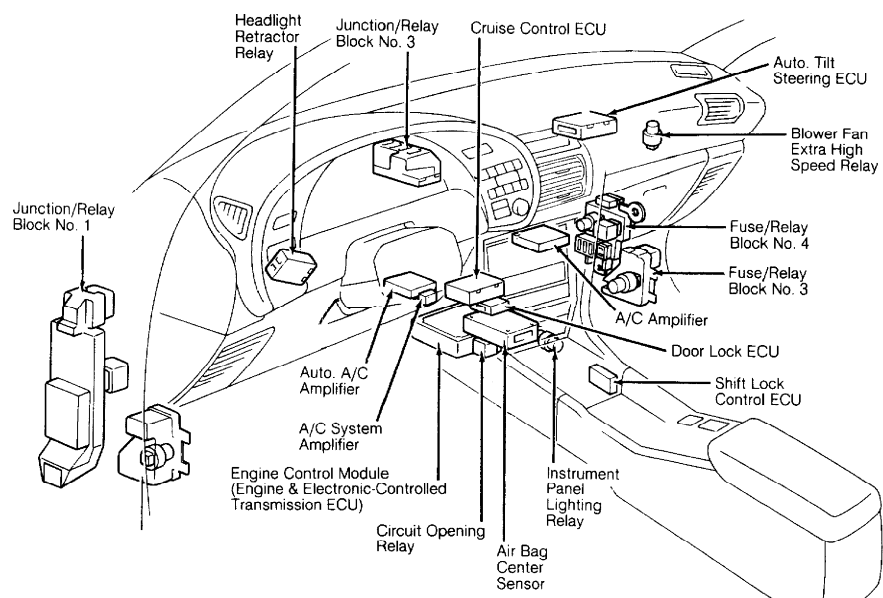
On fuse/relay block No. 2,

|                                  |                                                             |
|----------------------------------|-------------------------------------------------------------|
|                                  | behind left kick panel.                                     |
| Radiator Fan Relay No. 1         | On junction block No. 2,<br>near battery.                   |
| Rear Wiper Relay                 | Inside liftgate, on rear<br>window wiper motor.             |
| Starter Relay (M/T)              | On fuse/relay block No. 4,<br>behind right side kick panel. |
| Sun Roof Control Relay           | In front of sun roof opening.                               |
| Taillight Control Relay          | On junction/relay block No. 1,<br>behind left kick panel.   |
| Turn Signal/Hazard Flasher Relay | On junction/relay block No. 1,<br>behind left kick panel.   |
| Turn Signal/Hazard Flasher Relay | On junction/relay block No. 1,<br>behind left kick panel.   |

CIRCUIT PROTECTION DEVICES

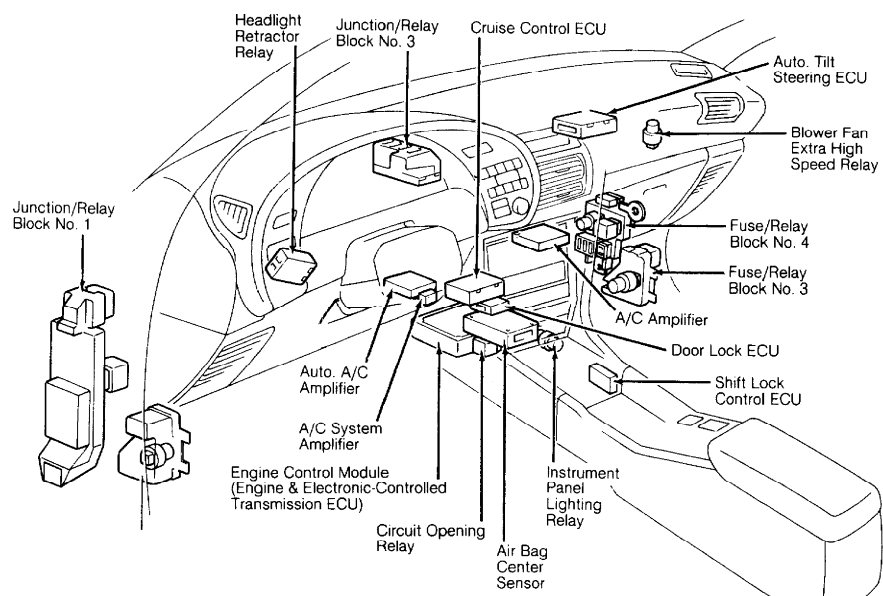
| Component         | Component Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fuse/Relay Blocks | <div data-bbox="175 1108 1042 1696"> <p>The diagram shows the engine compartment area with various electrical components labeled. On the left, 'Junction/Relay Block No. 1' is shown separately. In the center, 'Junction/Relay Block No. 3' is labeled. To its right is the 'Cruise Control ECU'. Further right is the 'Auto. Tilt Steering ECU'. Below these are 'Fuse/Relay Block No. 4' and 'Fuse/Relay Block No. 3'. Other components include 'Headlight Retractor Relay', 'Blower Fan Extra High Speed Relay', 'A/C Amplifier', 'A/C System Amplifier', 'Door Lock ECU', 'Shift Lock Control ECU', 'Engine Control Module (Engine &amp; Electronic-Controlled Transmission ECU)', 'Circuit Opening Relay', 'Air Bag Center Sensor', and 'Instrument Panel Lighting Relay'.</p> </div> |
| No. 2             | <div data-bbox="199 1707 285 1730"> <p>94A31468</p> </div> <div data-bbox="711 1724 1182 1780"> <p>Beside junction/relay block No. 1, behind left kick panel.</p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |





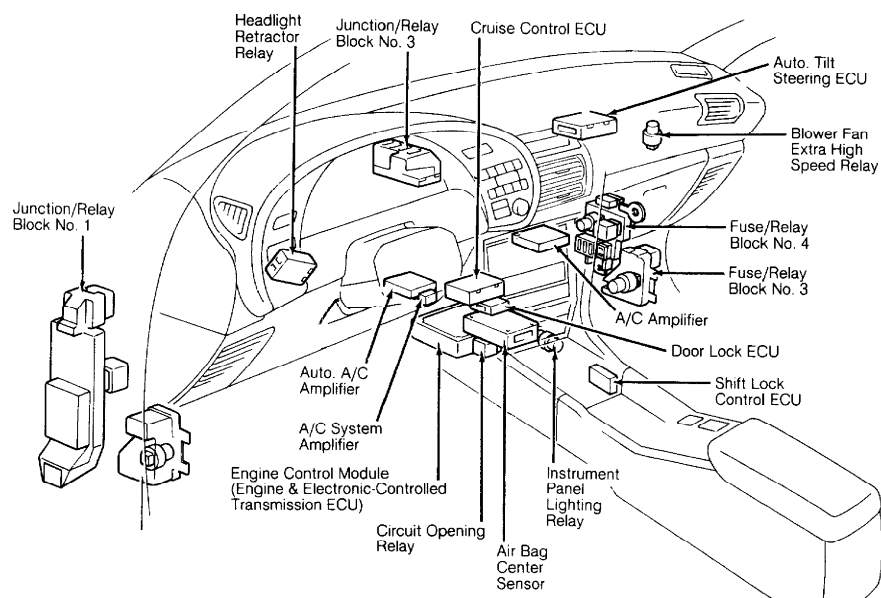
No. 3 94A31468

Behind right kick panel.



No. 4 94A31468

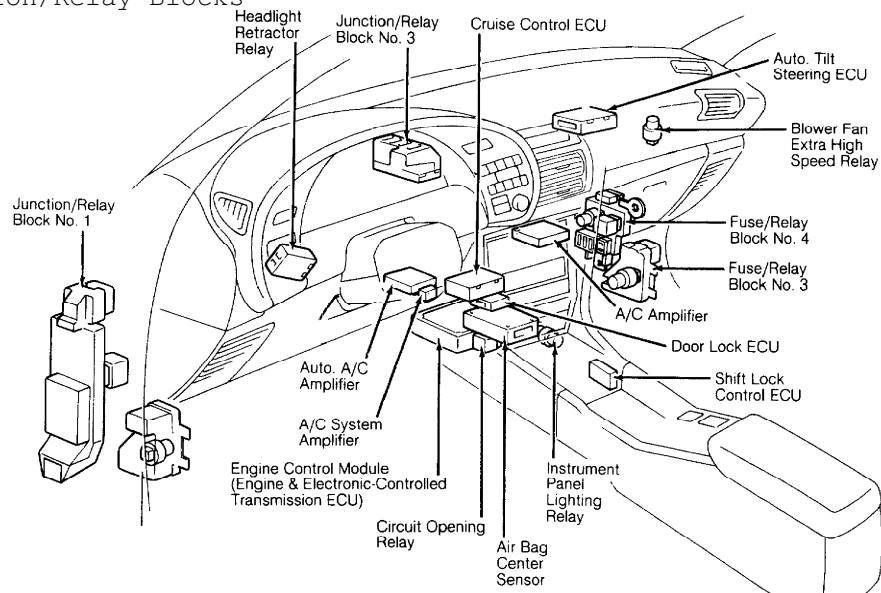
Behind right kick panel.



No. 5 94A31468

Right front corner of engine compartment.

#### Junction/Relay Blocks

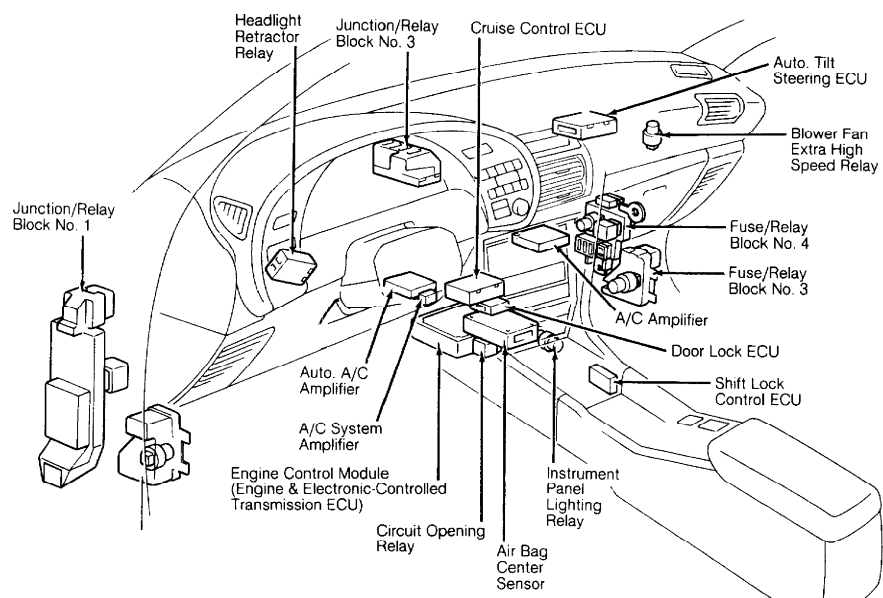


No. 1 94A31468

Behind left kick panel.

No. 4

In engine compartment, near battery.

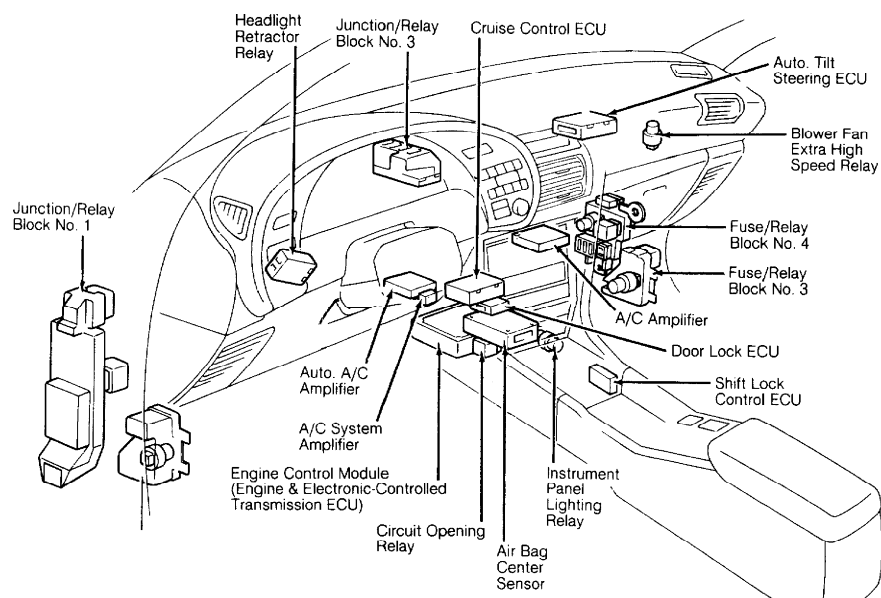


No. 3 94A31468

Behind left side of dash,  
behind instrument cluster.

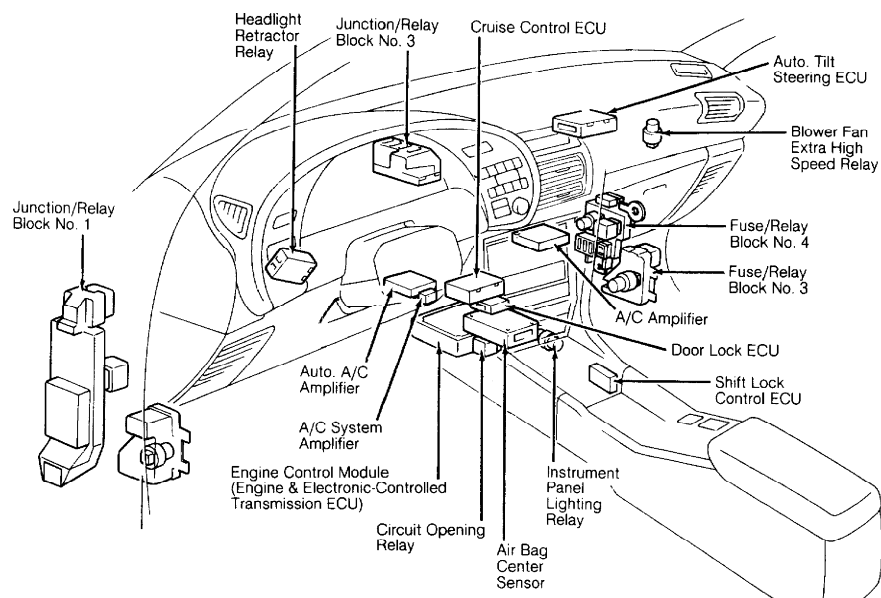
## CONTROL UNITS

| Component                    | Component Location                         |
|------------------------------|--------------------------------------------|
| ABS ECU (Convertible)        | In trunk, behind left rear quarter panel.  |
| ABS ECU (Except Convertible) | In trunk, behind right rear quarter panel. |



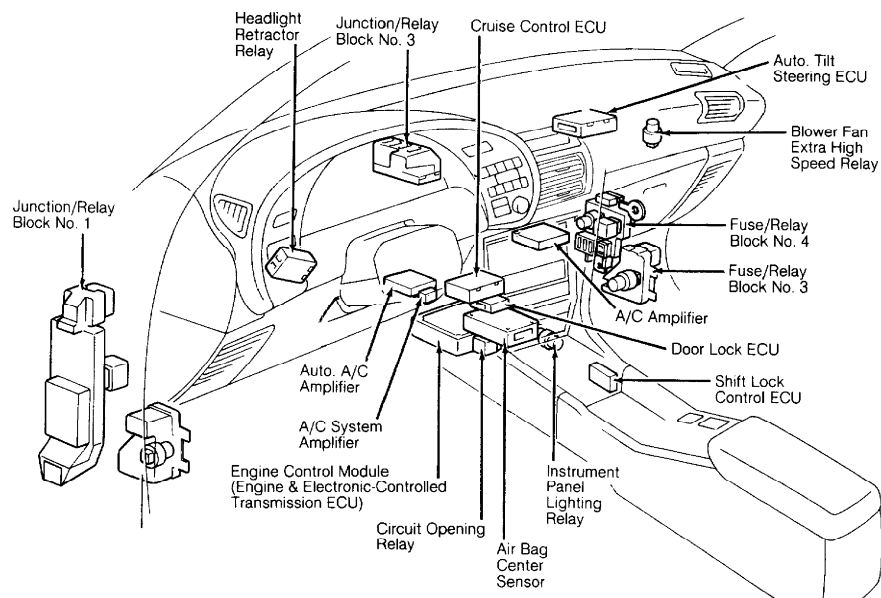
94A31468  
Auto. Tilt Steering ECU

Behind right side of dash,  
above glove box.



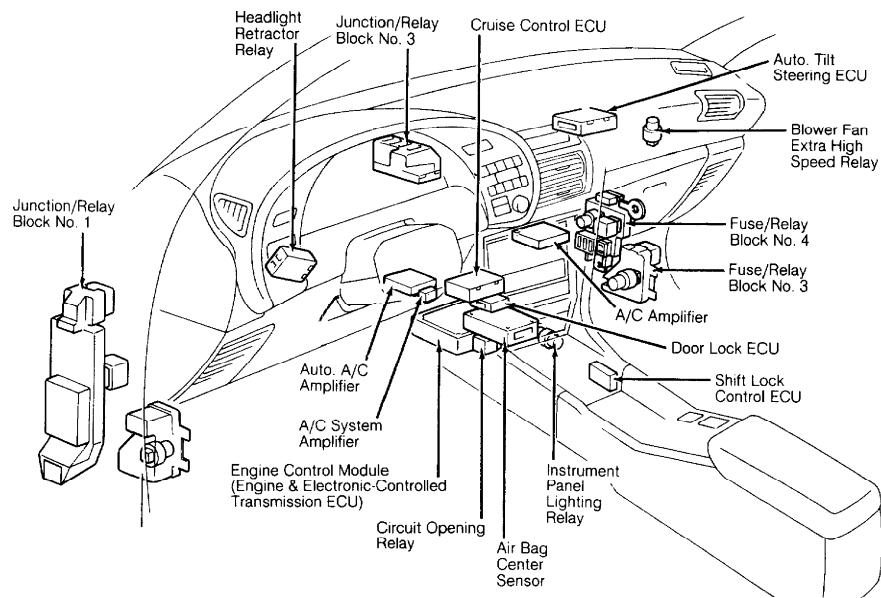
94A31468  
Cruise Control ECU

Behind center of dash, below  
radio.



94A31468  
Door Lock ECU

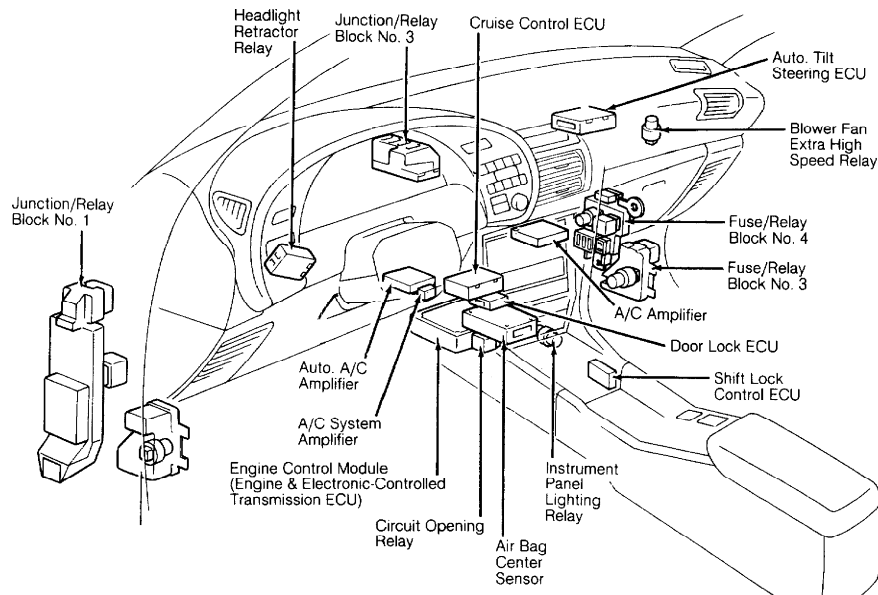
Behind center of dash behind  
ashtray.



94A31468  
Engine Control Module (ECM)

Behind front of center  
console.

Electronic-Controlled  
Transmission (ECU)

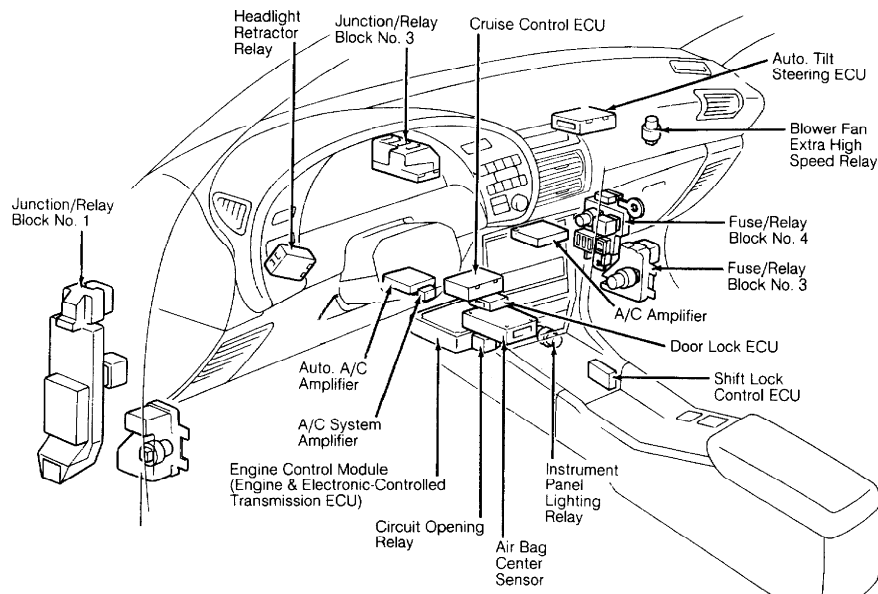


94A31468  
Component Location

Integrated within engine control module.

Engine Oil Level ECU

Behind rear seat, mounted to floor in trunk.



94A31468  
Shift Lock ECU

Under center floor console, at base of gear selector.

## MOTORS

Component

Component Location

ABS Actuator

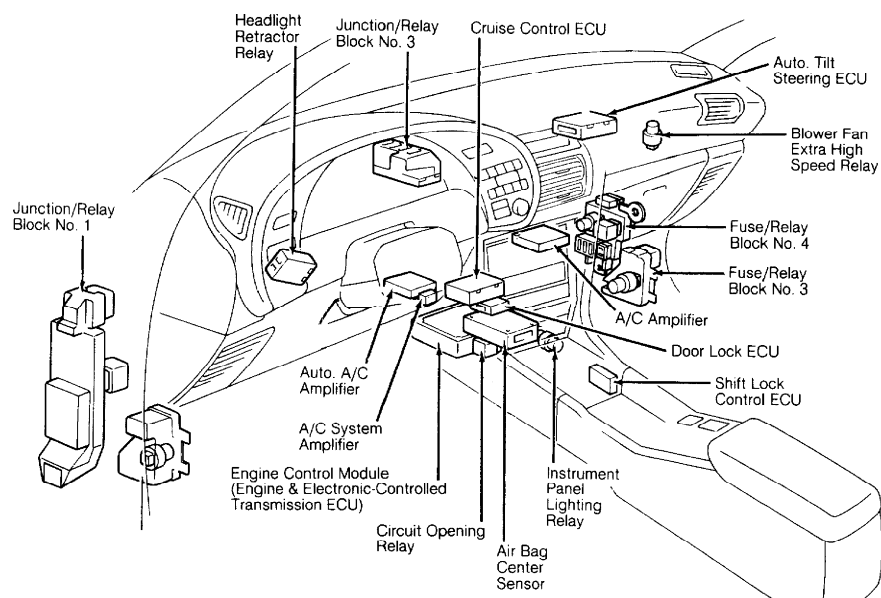
Right front corner of engine

|                                           |                                                                |
|-------------------------------------------|----------------------------------------------------------------|
|                                           | compartment.                                                   |
| A/C-Heater System<br>Air Inlet Servomotor | Behind glove box, on top left side of blower housing.          |
| Air Mix Control Servomotor                | Under center of dash, on front of A/C-heater core housing.     |
| Airflow Mode Control Servomotor           | Above accelerator pedal, mounted to A/C-heater housing.        |
| Auto. Tilt Steering Motor                 | Inside steering column tilt assembly.                          |
| Front Wiper Motor                         | On left rear corner of engine compartment, near brake booster. |
| Rear Wiper Motor                          | Inside liftgate.                                               |
| Sun Roof Motor                            | In front of sun roof opening, above control switch.            |
| Windshield Washer Motor                   | On right front corner of engine compartment.                   |

---

## SENDING UNITS & SENSORS

| Component                                 | Component Location                               |
|-------------------------------------------|--------------------------------------------------|
| ABS Deceleration Sensor (4WD)             | Behind rear seat, mounted to floor of trunk.     |
| ABS Speed Sensors                         | One sensor mounted to each wheel hub.            |
| A/C-Heater System<br>Ambient Temp. Sensor | On front grille of vehicle, near receiver-drier. |
| Coolant Temp. Sensor (Auto. A/C)          | On heater core housing.                          |
| In-Car Temp. Sensor (Auto. A/C)           | Behind A/C control panel.                        |
| Revolution Sensor                         | On rear of A/C compressor.                       |
| Solar Sensor (Auto. A/C)                  | On top right side of dash.                       |
| Air Bag                                   |                                                  |



94A31468  
Center Sensor

Mounted under center console,  
in front of gear selector.

Impact Sensors

On top of left and right inner  
fender panels.

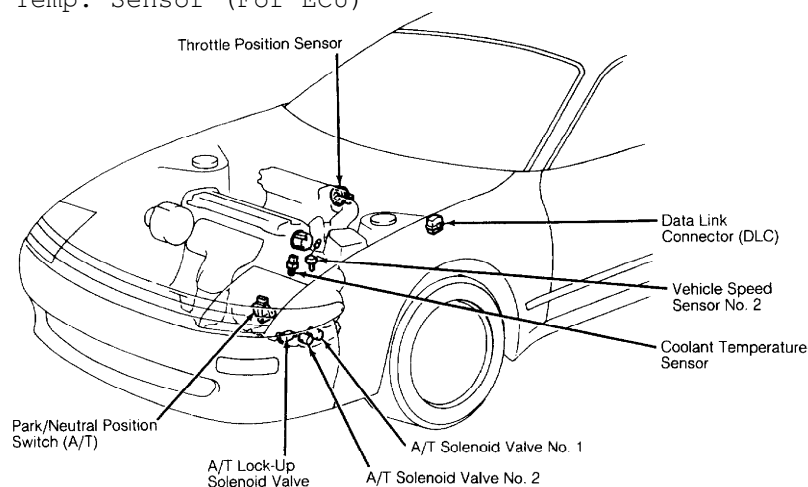
Coolant Temp. Gauge Sending Unit  
3S-GTE & 5S-FE

Threaded into under side of  
coolant outlet flange.

4A-FE

Threaded into top of coolant  
outlet flange.

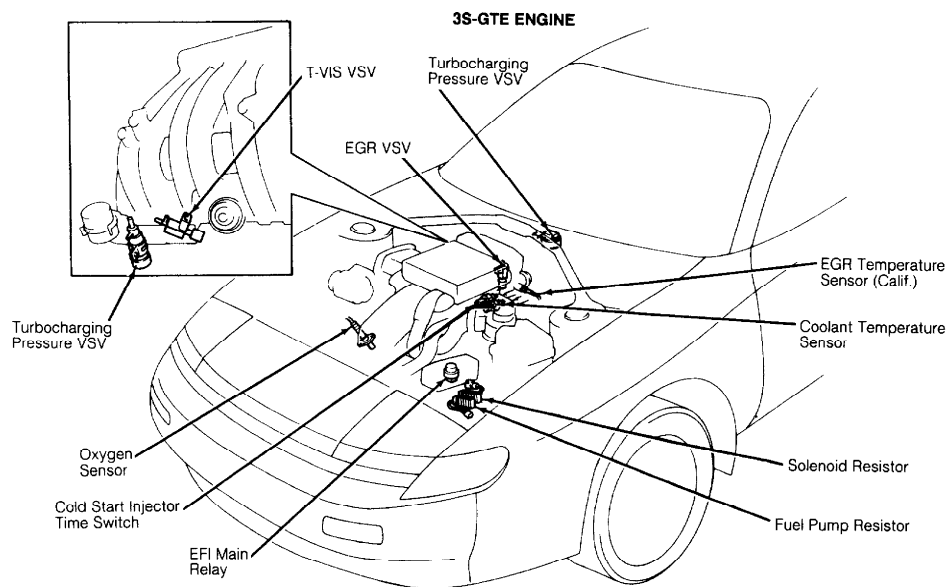
Coolant Temp. Sensor (For ECU)



94B31469  
(3S-GTE)

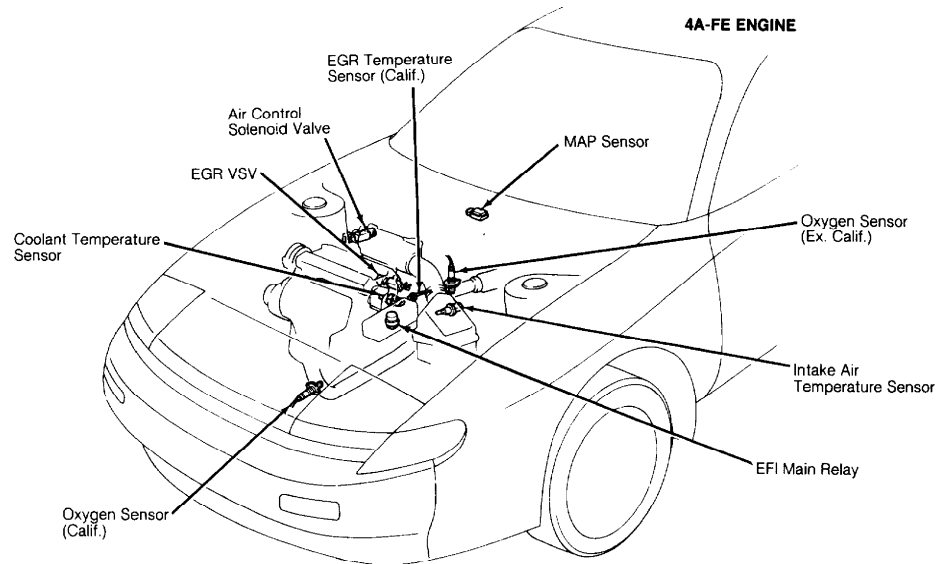
Threaded into top of coolant  
outlet flange nearest cylinder  
head.





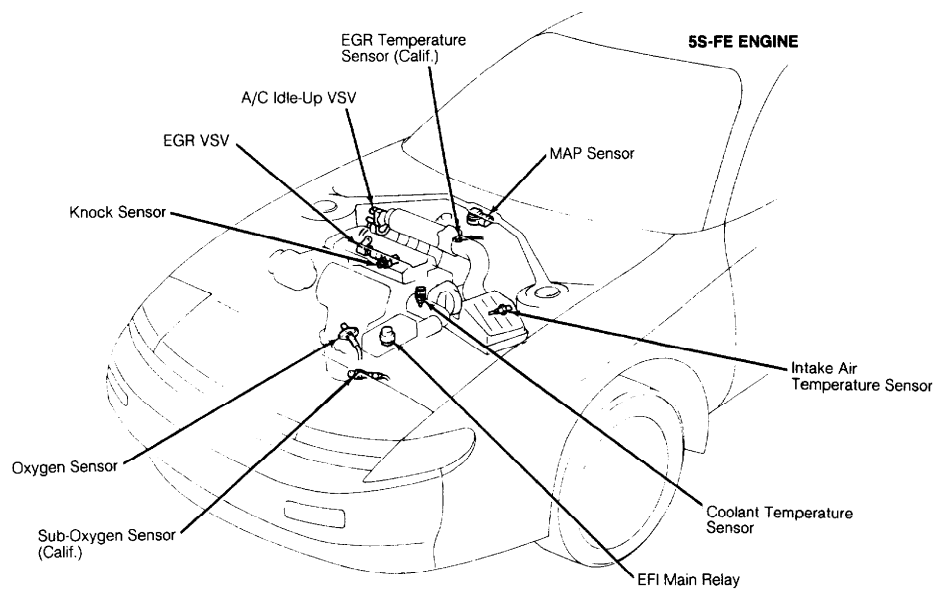
94F31470  
(3S-GTE)

Threaded into top of coolant outlet flange nearest cylinder head.



94F31471  
(4A-FE)

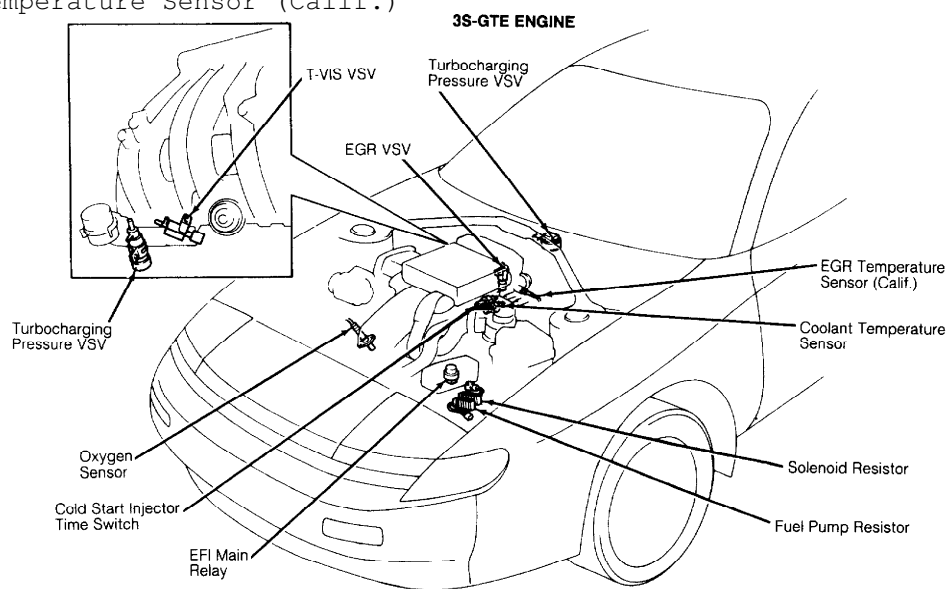
Threaded into top of coolant outlet flange nearest cylinder head.



94G31472  
(5S-FE)

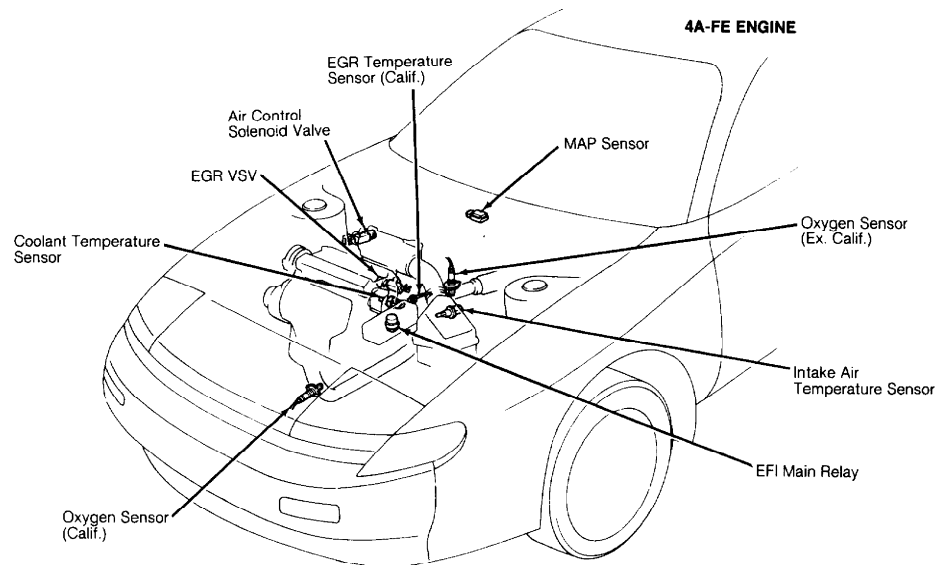
Threaded into top of coolant outlet flange nearest cylinder head.

EGR Temperature Sensor (Calif.)



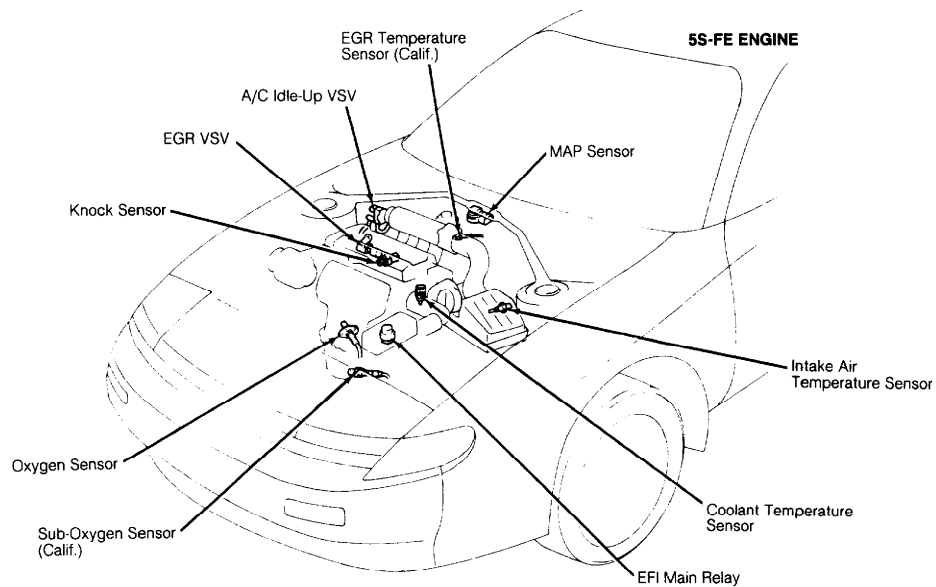
94F31470  
(3S-GTE)

Mounted on side of EGR valve.



(5S-FE) 94F31471

Mounted on side of EGR valve.

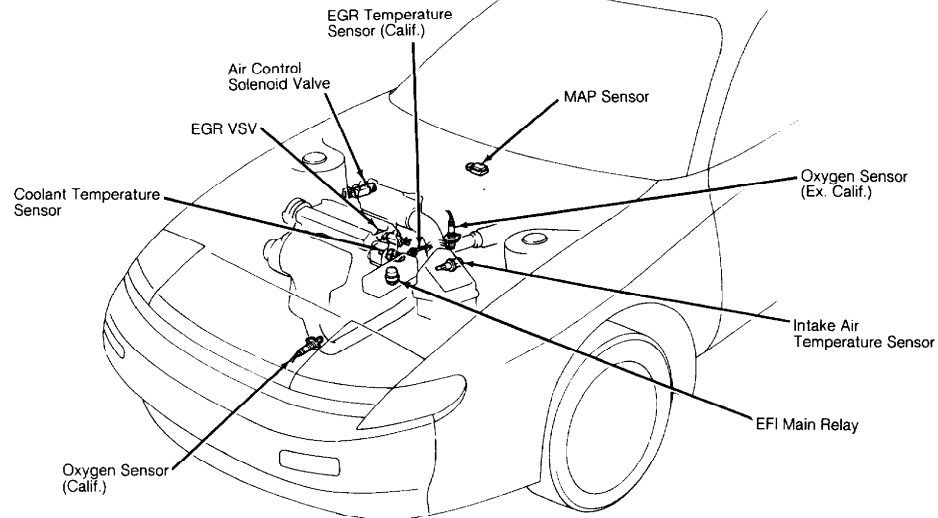


(5S-FE) 94G31472

Mounted on side of EGR valve.

Intake Air Temperature Sensor

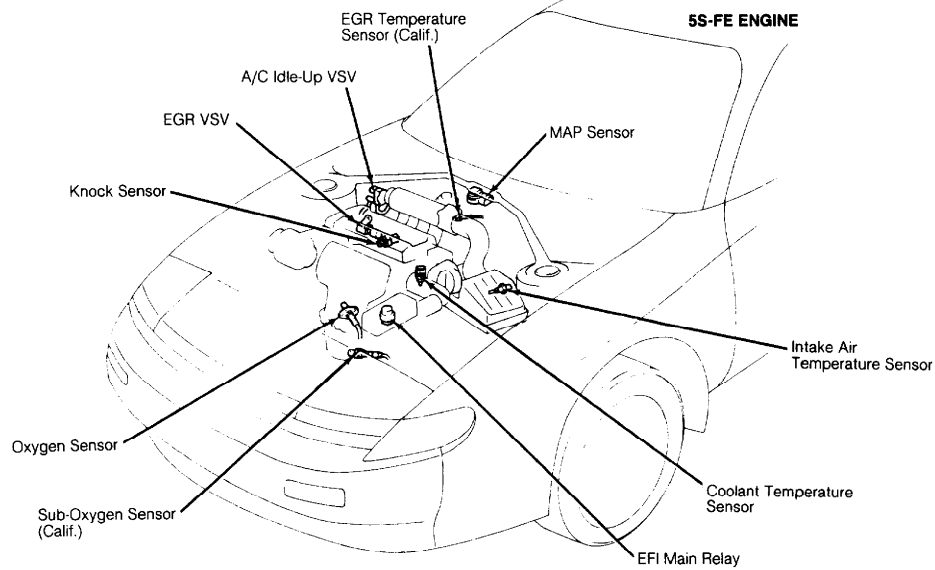
#### 4A-FE ENGINE



94F31471  
(4E-FE)

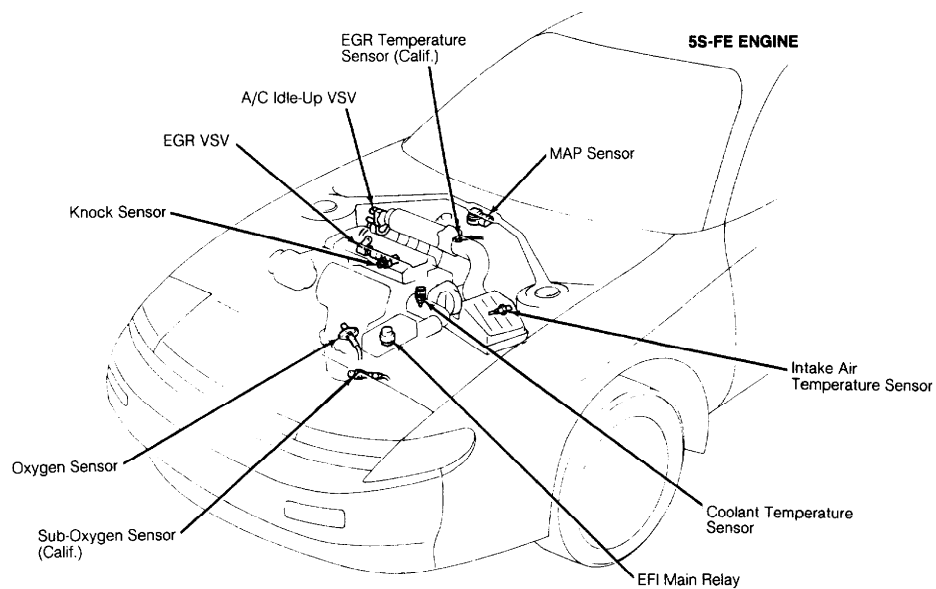
Mounted to air cleaner  
assembly or in air intake  
duct.

#### 5S-FE ENGINE



94G31472  
(5E-FE)

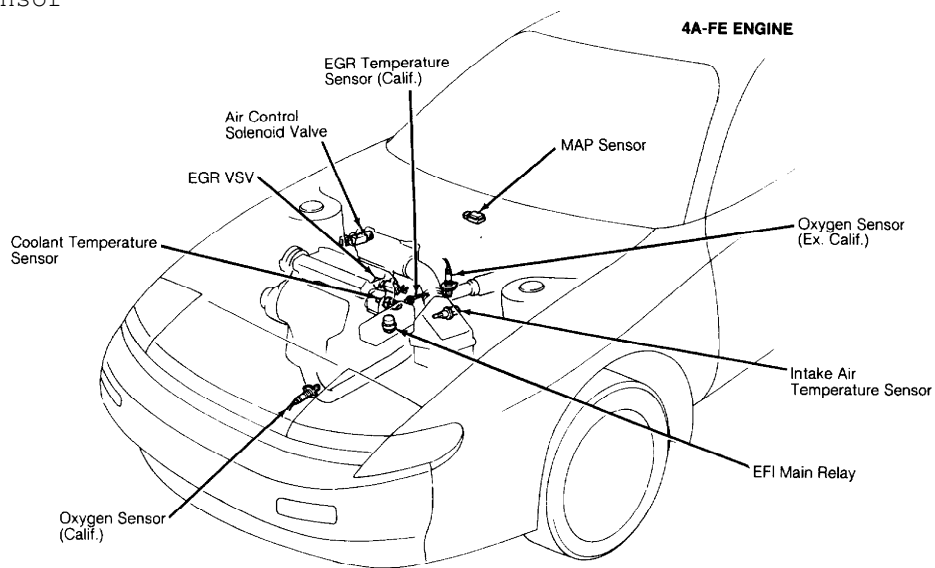
Mounted to air cleaner  
assembly or in air intake  
duct.



94G31472  
Knock Sensor (3S-GTE & 5S-FE)

On intake side of cylinder block, near No. 1 cylinder.

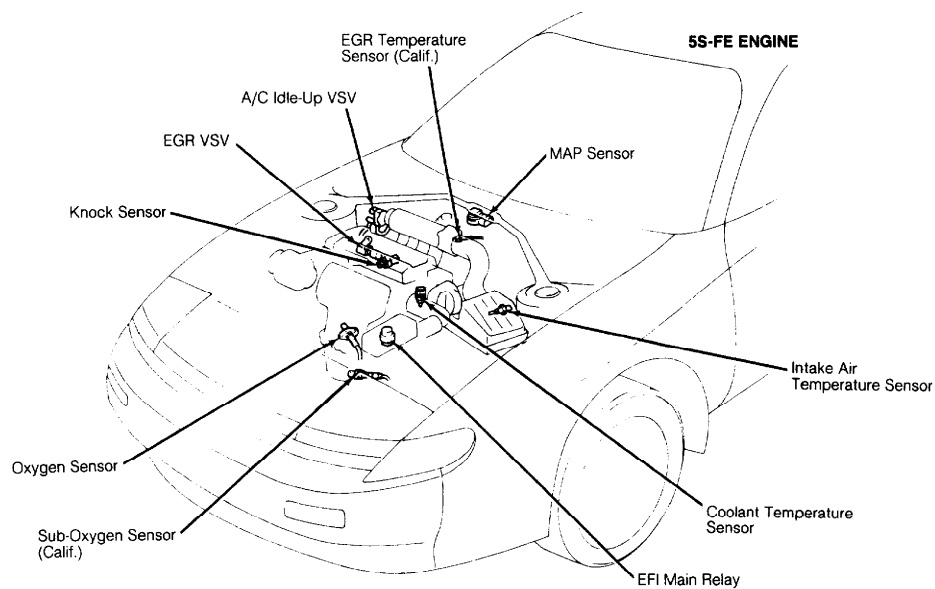
MAP Sensor



94F31471  
(4S-FE Shown 4S-FE Similar)

In engine compartment, near center of firewall.

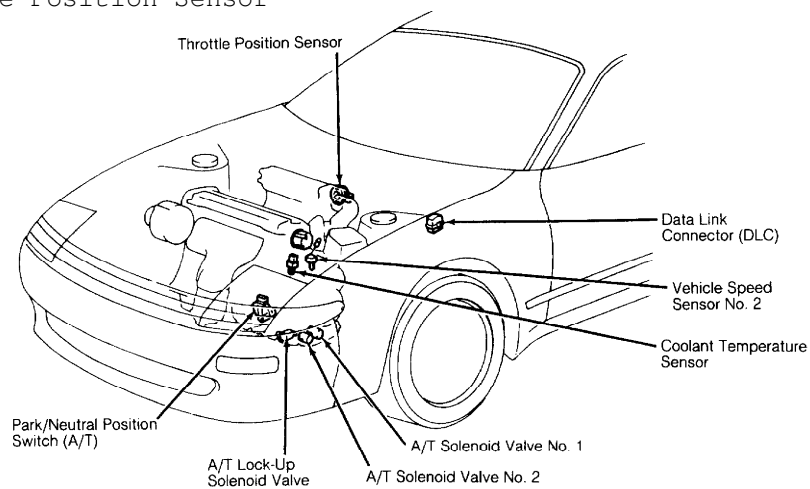
Speed Sensor



94G31472  
(5S-FE)

On rear of speedometer  
assembly.

#### Throttle Position Sensor



94B31469  
(5S-FE Shown 4S-FE Similar)

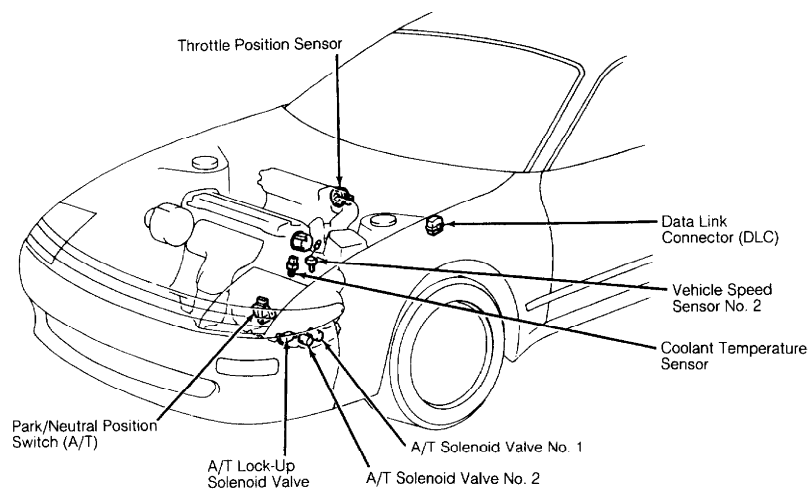
On throttle body housing.

#### Turbo Pressure Sensor (3S-GTE)

On center of firewall.

#### Vehicle Speed Sensors No. 1

On rear of speedometer  
assembly.



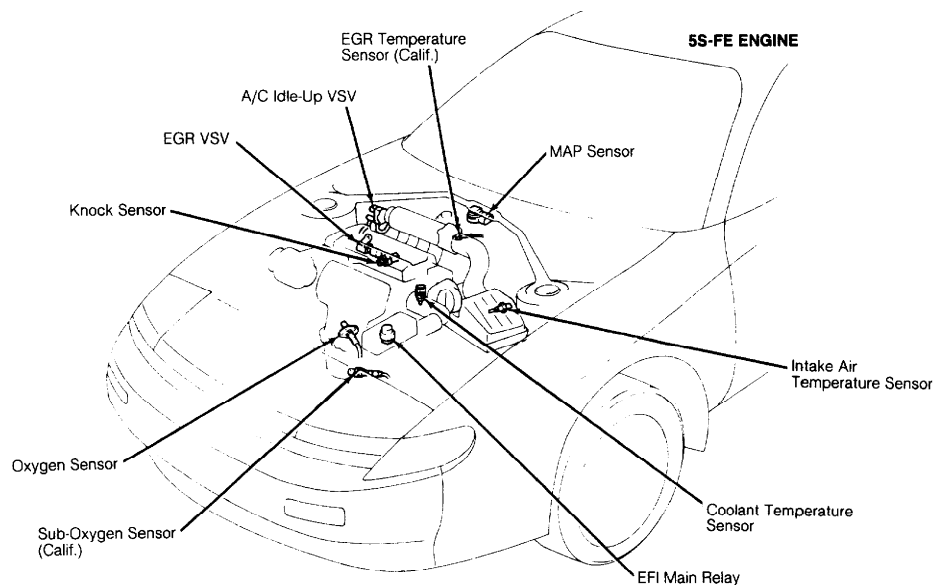
No. 2 94B31469

Mounted to top of transaxle.

## SOLENOIDS & SOLENOID VALVES

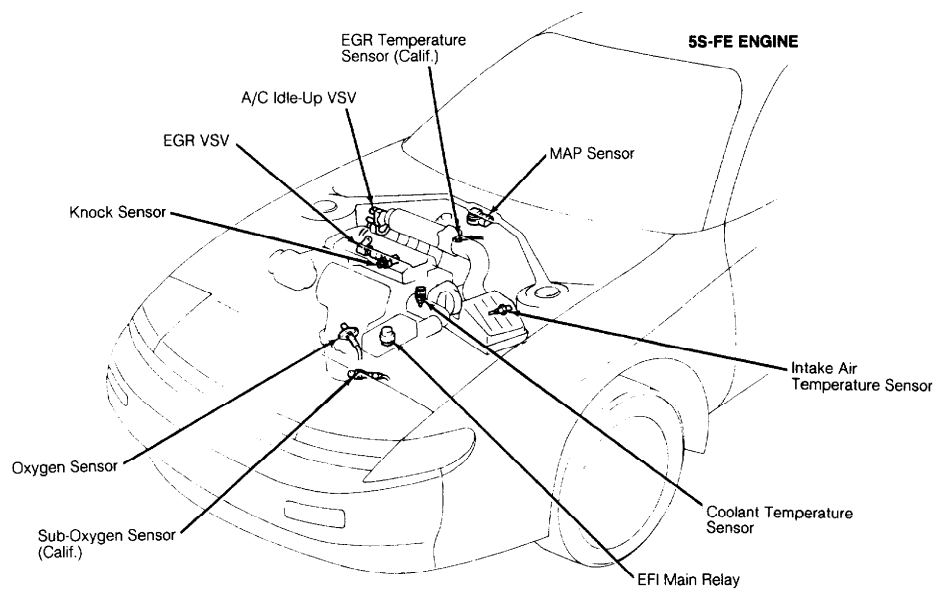
Component

Component Location



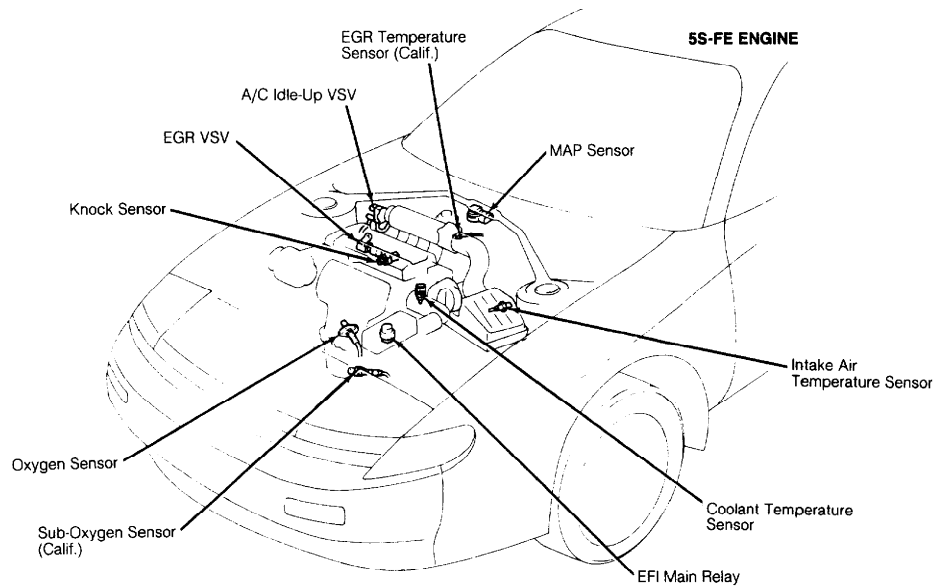
94G31472  
A/C Idle-Up Vacuum  
Switching Valve (VSV)

In engine compartment, on  
right side of firewall, behind  
shock tower.



94G31472  
Air Control Solenoid Valve

In right rear corner of engine compartment.



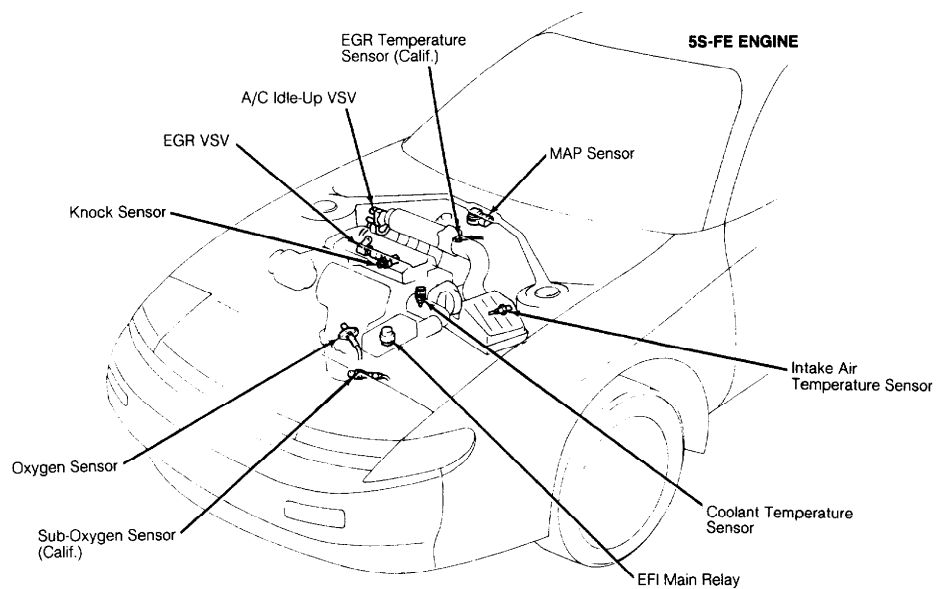
94G31472  
A/T Lock-Up Solenoid

On transaxle valve body.

A/T Overdrive Solenoid

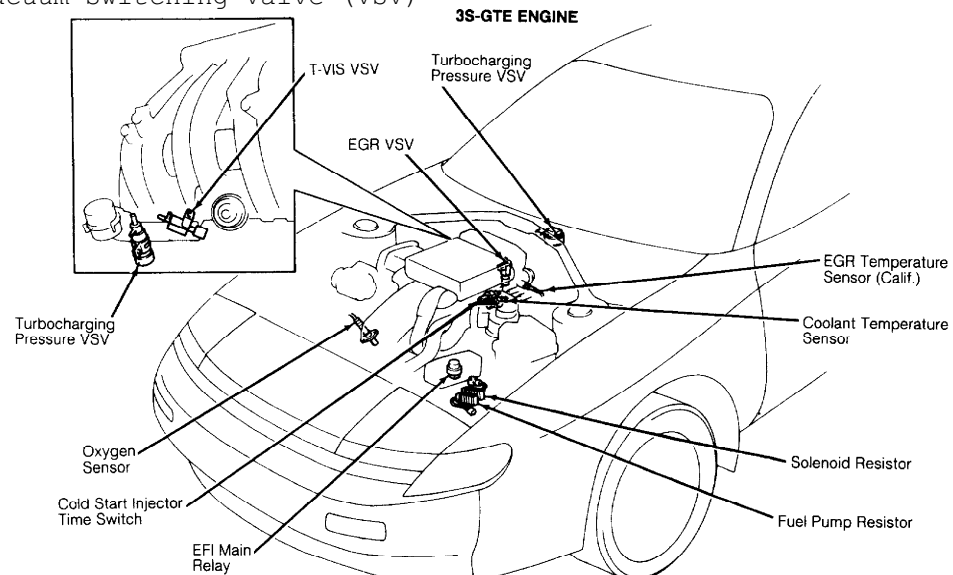
On transaxle valve body.



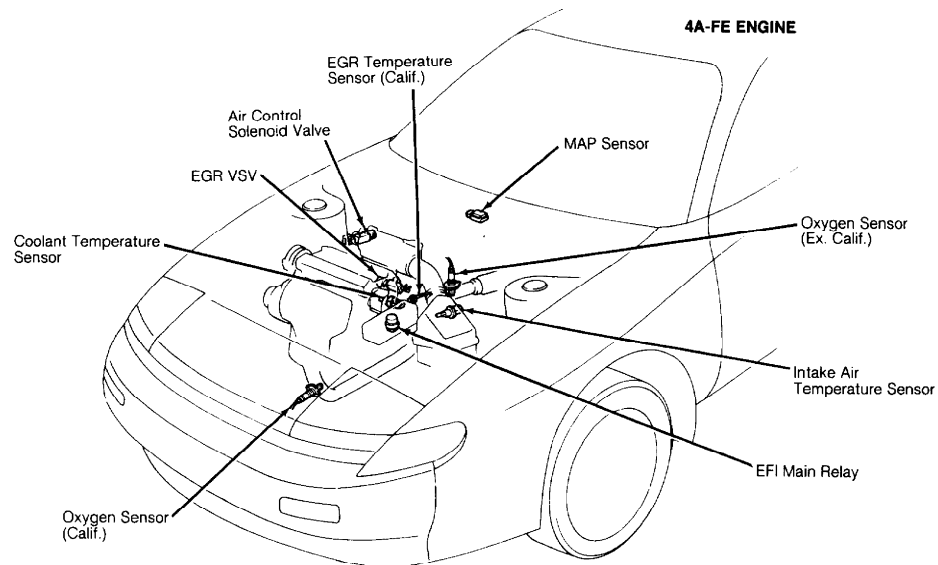


94G31472  
A/T Solenoid Valves No. 1 & 2 On transaxle valve body.

EGR Vacuum Switching Valve (VSV)

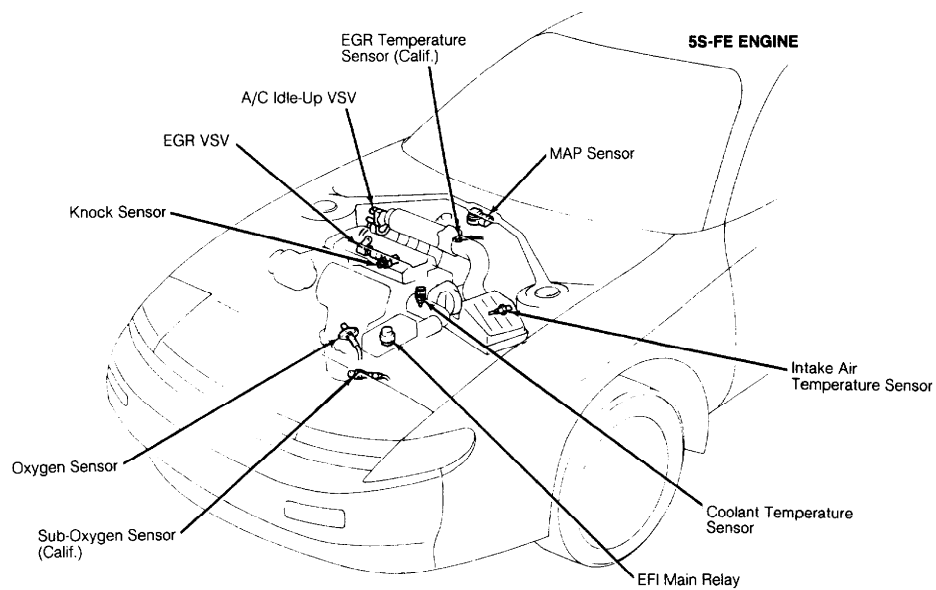


94F31470  
3S-GTE Near EGR Valve.



4A-FE 94F31471

Near EGR Valve.



5S-FE 94G31472

Near EGR Valve.

Idle Air Control (IAC) Valve  
(3S-GTE & 5S-FE)

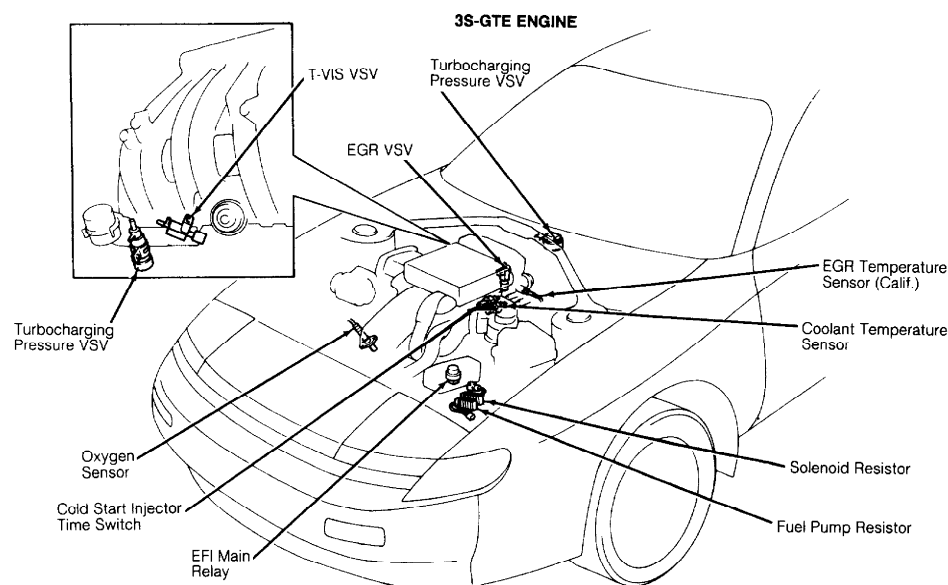
On throttle body.

Key Interlock Solenoid

On steering column, mounted to lock cylinder assembly.

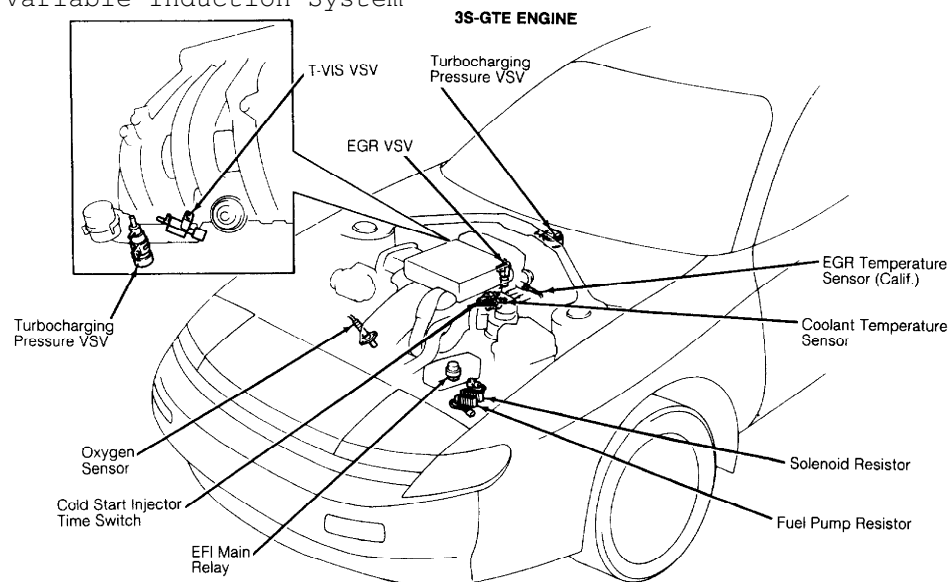
Shift Lock Solenoid (A/T)

Under center floor console at base of gear selector.



94F31470  
Turbocharging Pressure (3S-GTE) On center of firewall.

Turbo-Variable Induction System

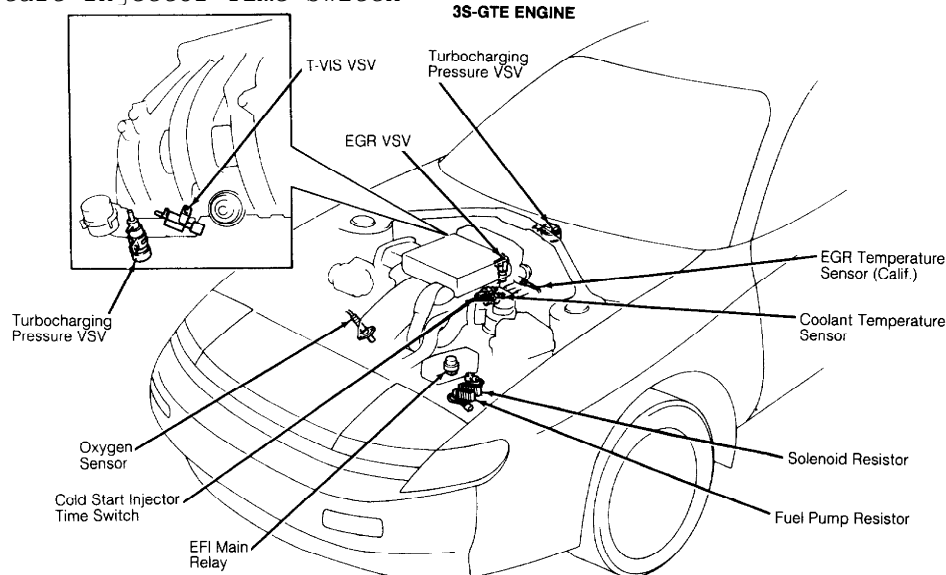


3S-GTE 94F31470  
On intake manifold, near turbocharging pressure VSV.

## SWITCHES

| Component                    | Component Location                                       |
|------------------------------|----------------------------------------------------------|
| A/C Cooling Fan Temp. Switch | On top of radiator.                                      |
| A/C Pressure Switch          | On A/C line, in right rear corner of engine compartment. |
| Back-Up Light Switch (M/T)   | On transaxle.                                            |

|                                 |                                  |
|---------------------------------|----------------------------------|
| Brake Fluid Warning Switch      | On cap of brake fluid reservoir. |
| Brake/Stoplight Switch          | On brake pedal bracket.          |
| Clutch Start Switch             | On clutch pedal bracket.         |
| Cold Start Injector Time Switch |                                  |

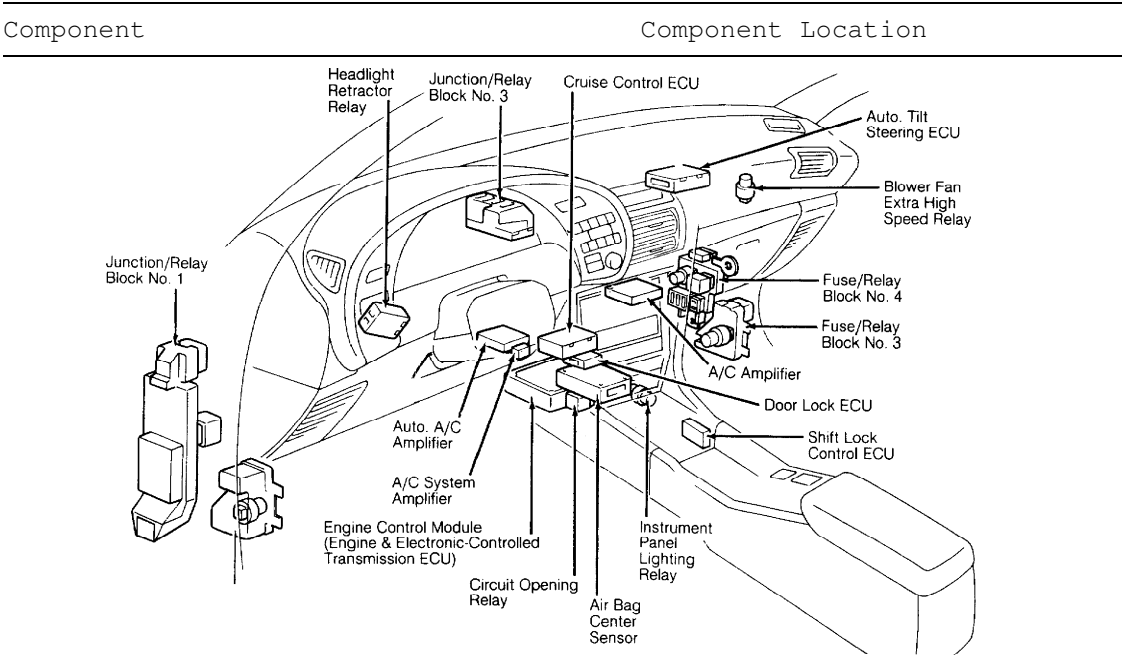


94F31470  
(3S-GTE)

On left side of engine,  
on coolant outlet.

|                                                       |                                                             |
|-------------------------------------------------------|-------------------------------------------------------------|
| Coolant Temperature Switch<br>3S-GTE & 5S-FE<br>4A-FE | In radiator.<br>Threaded into top of coolant outlet flange. |
| Key Unlock Warning Switch                             | On steering column, next to ignition switch.                |
| Low Oil Pressure Warning Switch<br>3S-GTE             | On rear of engine, near distributor.                        |
| 4A-FE                                                 | On left side of engine, near oil filter.                    |
| 5S-FE                                                 | On left rear of cylinder head.                              |
| Par/Neutral Position Switch (A/T)                     | On left side of transaxle.                                  |
| Parking Brake Switch                                  | At base of parking brake lever                              |
| Seat Belt Buckle Switch                               | At base of seat belt buckle.                                |
| Shift Lock Control Switch (A/T)                       | Under center floor console, at base of gear selector.       |
| Sun Roof Control & Limit Switches                     | In sun roof front opening rail, near sun roof motor.        |

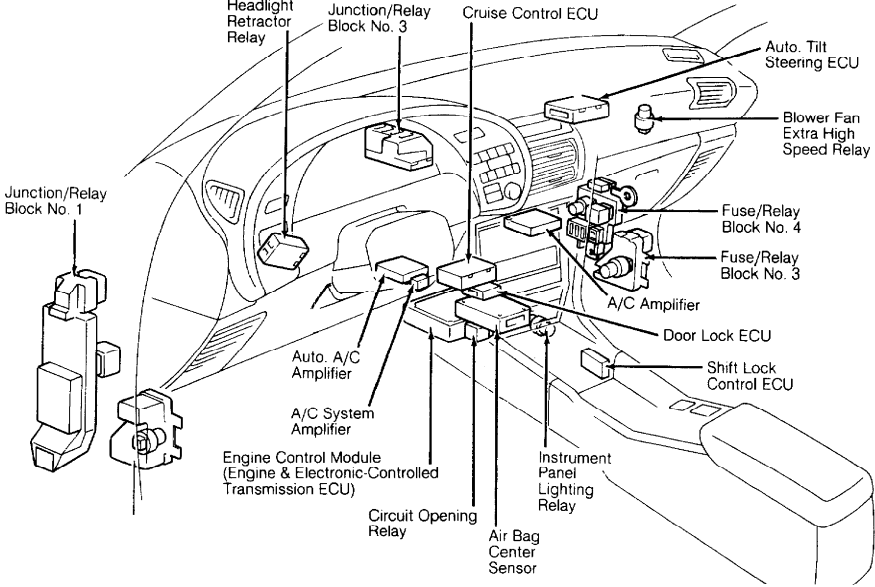
MISCELLANEOUS



94A31468  
A/C Amplifier

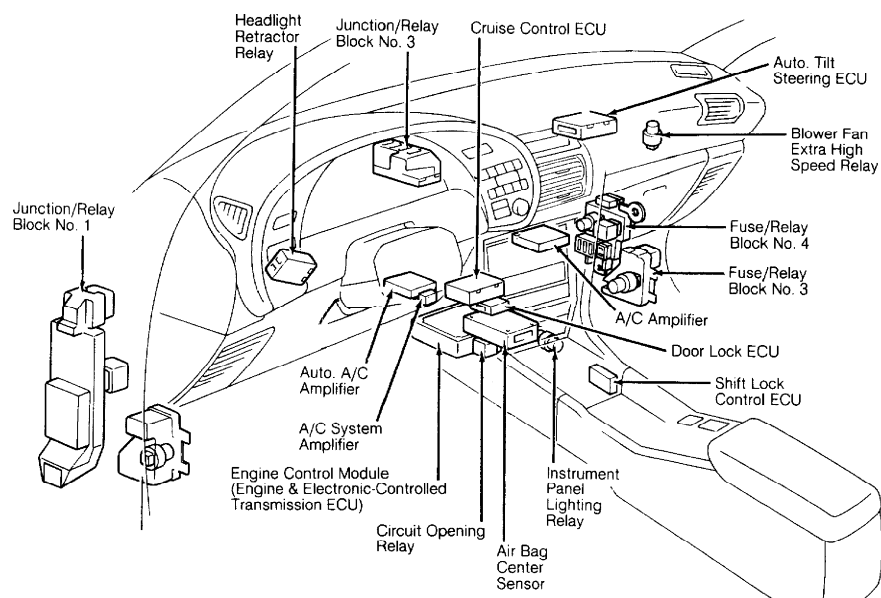
Behind glove box, attached to bottom of evaporator housing.

A/C System Amplifier



94A31468  
(Push-Button Control Switch)

Behind center of dash, in front of Auto. A/C Amplifier.



94A31468  
Auto. A/C Amplifier

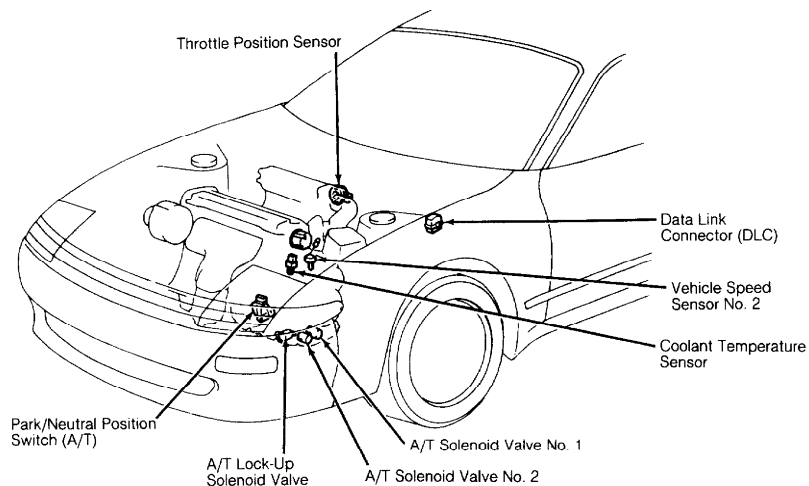
Behind center of dash, on  
bottom left of heater core  
housing.

Auto. A/C Power Transistor

Behind glove box, mounted to  
right rear side of evaporator  
housing.

Blower Resistor

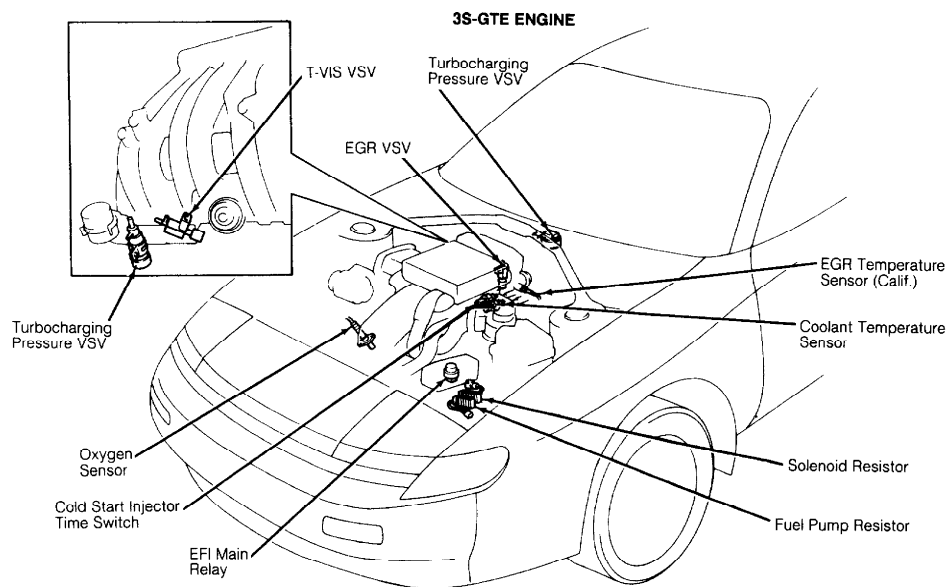
Behind glove box, mounted to  
right front side of evaporator  
housing.



94B31469  
Data Link Connector

In left rear of engine  
compartment, behind strut  
tower.

Fuel Pump Resistor



94F31470  
(3S-GTE)

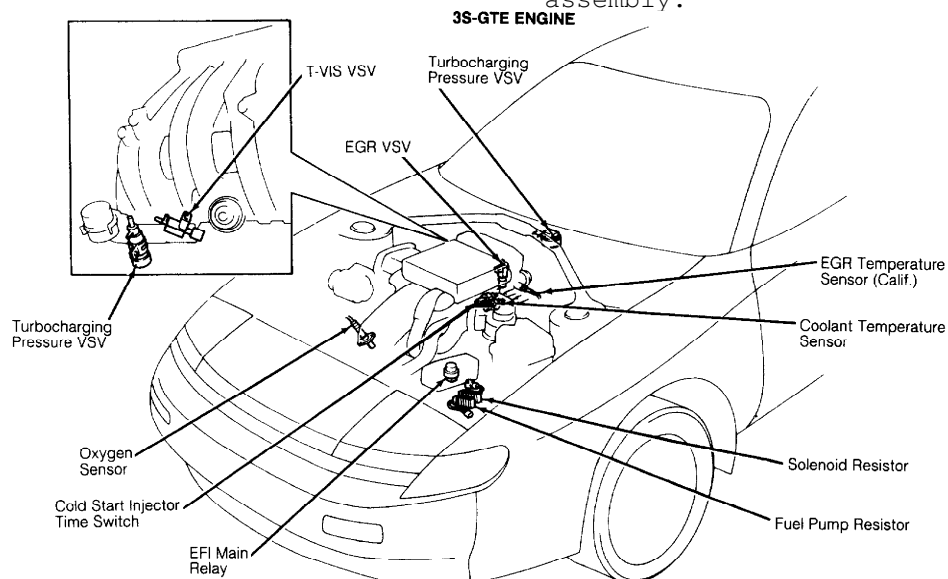
On left front corner of engine compartment.

Igniter  
3S-GTE

On left side of firewall.

4A-FE

Integrated within distributor assembly.



94F31470  
Solenoid Resistor

On left front corner of engine compartment.

# ENGINE COOLING FAN

## 1993 Toyota Celica

1993 ENGINE COOLING  
Toyota Specifications & Electric Cooling Fans  
Toyota; Celica

### SPECIFICATIONS

#### BELT ADJUSTMENT

Ensure belt tension is within specification. See appropriate BELT ADJUSTMENT table.

4-CYLINDER BELT ADJUSTMENT TABLE  
Tension In Lbs. (kg) Using Burroughs Tension Gauge

| Application          | New Belt | (1) Used Belt |
|----------------------|----------|---------------|
| Celica               |          |               |
| 1.6L (4A-FE)         |          |               |
| A/C .....            | 160 (73) | 100 (45)      |
| Alternator .....     | 160 (73) | 130 (59)      |
| Power Steering ..... | 125 (57) | 80 (36)       |
| 2.0L Turbo (3S-GTE)  |          |               |
| A/C .....            | 165 (75) | 84 (38)       |
| Alternator           |          |               |
| With A/C .....       | 175 (79) | 115 (52)      |
| Without A/C .....    | 150 (68) | 130 (59)      |
| Power Steering ..... | 125 (57) | 80 (36)       |
| 2.2L (5S-FE)         |          |               |
| A/C .....            | 165 (75) | 110 (50)      |
| Alternator           |          |               |
| With A/C .....       | 175 (79) | 130 (59)      |
| Without A/C .....    | 125 (57) | 95 (43)       |
| Power Steering ..... | 125 (57) | 80 (36)       |

(1) - Used belt is a belt in operation at least 5 minutes.

#### COOLING SYSTEM SPECIFICATIONS

4-CYLINDER COOLING SYSTEM SPECIFICATIONS TABLE

| Application                      | Specification             |
|----------------------------------|---------------------------|
| Coolant Replacement Interval     |                           |
| First Service Interval (1) ..... | 45,000 Miles Or 36 Months |
| Coolant Capacity                 |                           |
| Celica                           |                           |
| 1.6L (4A-FE)                     |                           |
| A/T .....                        | 5.9 Qts. (5.6L)           |
| M/T .....                        | 5.5 Qts. (5.2L)           |
| 2.0L Turbo (3S-GTE) .....        | 6.3 Qts. (6.0L)           |
| 2.2L (5S-FE)                     |                           |
| A/T .....                        | 7.0 Qts. (6.6L)           |
| M/T .....                        | 6.9 Qts. (6.5L)           |

(1) - After first service interval, replace coolant every 30,000 miles or 24 months.



## COOLING SYSTEM BLEEDING

No special cooling system bleeding procedure is required.

## ELECTRIC COOLING FAN

**NOTE:** Electric cooling fan may be used for the radiator or condenser. To verify electric cooling location and application, see ELECTRIC COOLING FAN IDENTIFICATION table. For condenser cooling fan testing, see appropriate A/C-HEATER SYSTEM article in the AIR CONDITIONING & HEAT section.

### ELECTRIC COOLING FAN IDENTIFICATION TABLE

| Application                 | Cooling Fan Identification         |
|-----------------------------|------------------------------------|
| Celica                      |                                    |
| Condenser Cooling Fan ..... | Passenger's Side Front Of Radiator |
| Radiator Cooling Fan .....  | Driver's Side Rear Of Radiator     |

## RADIATOR COOLING FAN SYSTEM TEST

**NOTE:** On A/C equipped models, a A/C high-pressure switch which may be used in conjunction with cooling fan relays for controlling of radiator cooling fan. See appropriate A/C-HEATER SYSTEM article in the AIR CONDITIONING & HEAT section.

1) Ensure engine coolant temperature is less than specified STEP 1 temperature. See RADIATOR COOLING FAN SYSTEM TESTING TEMPERATURE SPECIFICATIONS table. Turn ignition on. Ensure radiator cooling fan stops.

2) If radiator cooling fan stops, proceed to step 5). If radiator cooling fan operates, check cooling fan relay No. 1 and Engine Coolant Temperature (ECT) switch. See COOLING FAN RELAY NO. 1 and ENGINE COOLANT TEMPERATURE (ECT) SWITCH under COMPONENT TESTING.

3) If cooling fan relay No. 1 and ECT switch are okay, check for open circuit in wire between ECT switch and cooling fan relay No. 1. Consult appropriate wiring diagram if necessary. See WIRING DIAGRAMS.

4) Cooling fan relay No. 1 is located in fuse/relay box. See Fig. 1. For ECT switch locations, see ECT SWITCH LOCATIONS table.

5) With ignition on, disconnect electrical connector at Engine Coolant Temperature (ECT) switch. For ECT switch locations, see ECT SWITCH LOCATIONS table. Ensure radiator cooling fan operates.

6) If radiator cooling fan operates, proceed to step 9). If radiator cooling fan does not operate, check cooling fan relay No. 1, radiator cooling fan, engine or fan main relay and necessary fuses.

**NOTE:** Engine main relay is used on this model.

7) Consult appropriate wiring diagram for proper fuse applications. See WIRING DIAGRAMS. See COOLING FAN RELAY NO. 1, RADIATOR COOLING FAN, FAN MAIN RELAY and ENGINE MAIN RELAY under COMPONENT TESTING.

8) If all components are okay, check for short circuit in wire between cooling fan relay No. 1 and ECT switch. Consult appropriate wiring diagram if necessary. See WIRING DIAGRAMS.

9) Install electrical connector on ECT switch. Start engine.

Warm engine until coolant temperature is greater than specified STEP 2 temperature. See RADIATOR COOLING FAN SYSTEM TESTING TEMPERATURE SPECIFICATIONS table.

10) Ensure radiator cooling fan operates. Replace ECT switch if radiator cooling fan fails to operate and recheck operation.

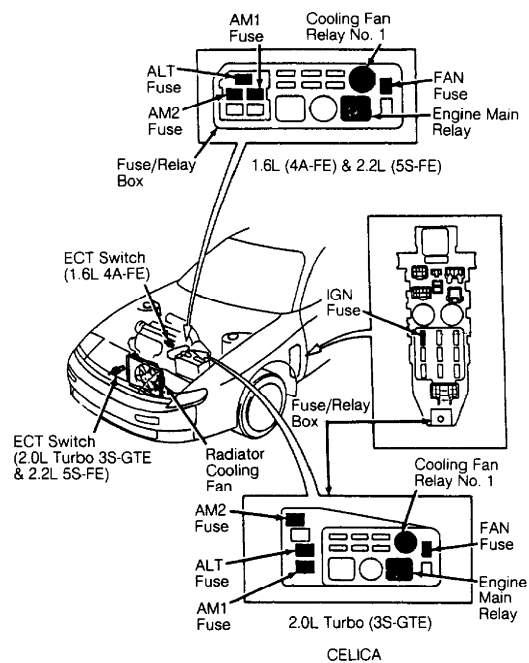
RADIATOR COOLING FAN SYSTEM TESTING TEMPERATURE SPECIFICATIONS TABLE

| Application              | Step 1 Temp.<br>°F (°C) | Step 2 Temp.<br>°F (°C) |
|--------------------------|-------------------------|-------------------------|
| Celica Without A/C ..... | 181 (83) .....          | 199 (93)                |

ECT SWITCH LOCATIONS TABLE

| Application                              | (1) Switch Location |
|------------------------------------------|---------------------|
| Celica                                   |                     |
| 1.6L (4A-FE) .....                       | (2)                 |
| 2.0L Turbo (3S-GTE) & 2.2L (5S-FE) ..... | Bottom Of Radiator  |

- (1) - For additional information on ECT switch locations, see Fig. 1.  
(2) - ECT switch is located in coolant housing below distributor and contains a single-pin connector with a Blue/Black wire.



93D84015  
Fig. 1: Identifying Radiator Cooling Fan & Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

COMPONENT TESTING

CAUTION: When battery is disconnected, vehicle computer and memory systems may lose memory data. Driveability problems may exist until computer systems have completed a relearn cycle.

A/C AMPLIFIER CIRCUIT TEST TABLE

Wiring Harness  
Terminals

Condition

Specification

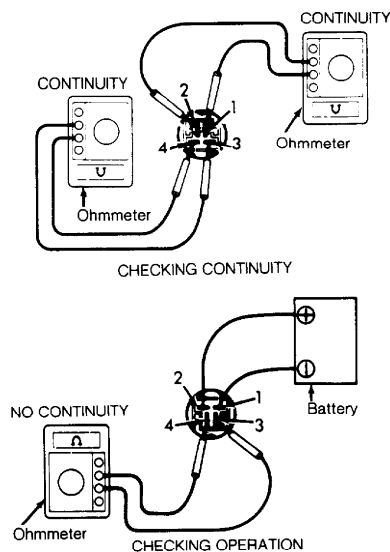
|                 |                        |                 |
|-----------------|------------------------|-----------------|
| No. 3 & Ground  | .....                  | Continuity      |
| No. 4 & Ground  | ..... Ignition On      | Battery Voltage |
| No. 9 & 15      | ..... 185°F (85°C) (1) | 1350 Ohms       |
| No. 9 & 15      | ..... 194°F (90°C) (1) | 1190 Ohms       |
| No. 9 & 15      | ..... 203°F (95°C) (1) | 1050 Ohms       |
| No. 10 & Ground | ..... Ignition On      | Battery Voltage |
| No. 13 & Ground | .....                  | Continuity      |

(1) - This is the coolant temperature.

Cooling Fan Relay No. 1

1) Disconnect negative battery cable. Remove cooling fan relay No. 1 relay from fuse/relay box. See Fig. 1. Using ohmmeter, ensure continuity exists between specified terminals. See Fig. 2.

2) To check relay operation, connect battery to specified terminals of cooling fan relay No. 1. See Fig. 2. Using ohmmeter, ensure no continuity exists between specified terminals. See Fig. 2. Replace cooling fan relay No. 1 if defective.



93E84024

Fig. 2: Testing Cooling Fan Relay No. 1 & Engine Compartment Cooling Fan Relay

Courtesy of Toyota Motor Sales, U.S.A., Inc.

Engine Coolant Temperature (ECT) Switch (Celica 2.0L 5S-FE)

1) Disconnect negative battery cable. Drain cooling system. Disconnect electrical connector and remove ECT switch. See ECT SWITCH LOCATIONS table. Place probe end of ECT switch in container of water.

2) Using ohmmeter, check for continuity and then no continuity between electrical terminals on ECT switch while heating water to specified temperatures. See ECT SWITCH CONTINUITY SPECIFICATIONS table.

3) Replace ECT switch if not as specified. Reinstall ECT switch using NEW "O" ring (if equipped). Fill cooling system. On MR2, air must be bled from cooling system. See COOLING SYSTEM BLEEDING.

Engine Coolant Temperature (ECT) Switch (Celica 1.6L 4A-FE)

1) Disconnect negative battery cable. Drain cooling system.

Disconnect electrical connector and remove ECT switch. See ECT SWITCH LOCATIONS table. Place probe end of ECT switch in container of water.

2) Using ohmmeter, check for continuity and then no continuity between electrical terminal on ECT switch and switch body while heating water to specified temperatures. See ECT SWITCH CONTINUITY SPECIFICATIONS table. Replace ECT switch if not as specified. Reinstall and fill cooling system.

ECT SWITCH LOCATIONS TABLE

| Application                                                                                                                  | (1) Switch Location |
|------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Celica                                                                                                                       |                     |
| 1.6L (4A-FE)                                                                                                                 | (2)                 |
| 2.0L Turbo (3S-GTE) & 2.2L (5S-FE)                                                                                           | Bottom Of Radiator  |
| (1) - For additional information on ECT switch locations, see Figs. 1 and 5.                                                 |                     |
| (2) - ECT switch is located in coolant housing below distributor and contains a single-pin connector with a Blue/Black wire. |                     |

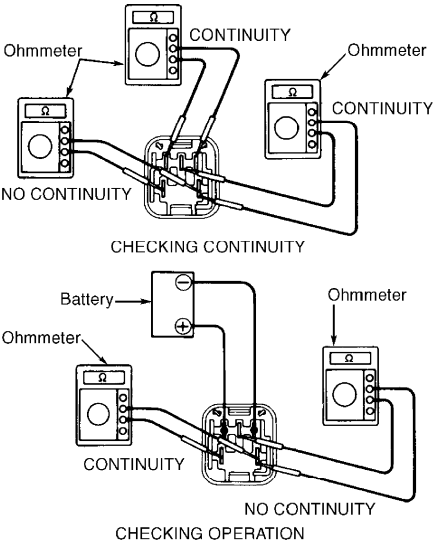
ECT SWITCH CONTINUITY SPECIFICATIONS TABLE

| Application   | Temperature               |
|---------------|---------------------------|
| Celica        |                           |
| Continuity    | Less Than 181°F (83°C)    |
| No Continuity | Greater Than 199°F (93°C) |

Engine Main Relay

1) Disconnect negative battery cable. Remove engine main relay from fuse/relay box. See Fig. 1. Using ohmmeter, ensure continuity and no continuity exists between specified terminals. See Fig. 3.

2) To check relay operation, connect battery to specified terminals of engine main relay. See Fig. 3. Using ohmmeter, ensure continuity and no continuity exists between specified terminals. See Fig. 3. Replace engine main relay if defective.



93C84030  
Fig. 3: Testing Engine Main Relay  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## Radiator Cooling Fan

1) Disconnect electrical connector from radiator cooling fan.  
Connect battery and ammeter to electrical connector on radiator  
cooling fan.

2) Radiator cooling fan should operate smoothly and amperage draw should be within specification. See RADIATOR COOLING FAN AMPERAGE DRAW SPECIFICATIONS table.

3) Replace radiator cooling fan if it fails to rotate smoothly or amperage draw is not within specification. Reinstall electrical connector.

## RADIATOR COOLING FAN AMPERAGE DRAW SPECIFICATIONS TABLE

| Application                   | Amps     |
|-------------------------------|----------|
| Celica                        |          |
| 1.6L 4A-FE & 2.2L 5S-FE ..... | 5.8-7.4  |
| 2.0L Turbo (3S-GTE) .....     | 8.8-10.8 |

## WIRING DIAGRAMS

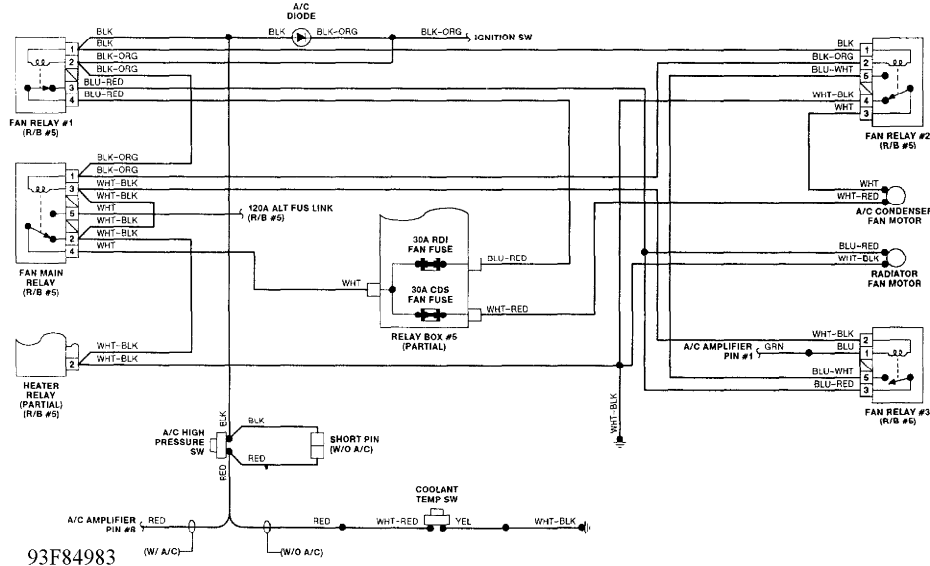


Fig. 4: Electric Cooling Fan Wiring Diagram

# FUSES & CIRCUIT BREAKERS

1993 Toyota Celica

FUSES & CIRCUIT BREAKERS  
1983-93 Toyota

Celica

## IDENTIFICATION

### FUSE PANEL LOCATION (1983-85)

There are 3 fuse panels on this vehicle. The driver's side fuse box is located behind the driver's kick panel. The passenger's side fuse box is behind the passenger's side kick panel. The main fuse and relay block is in the engine compartment.

### PASSENGER COMPARTMENT FUSE BOX IDENTIFICATION (1983-85)

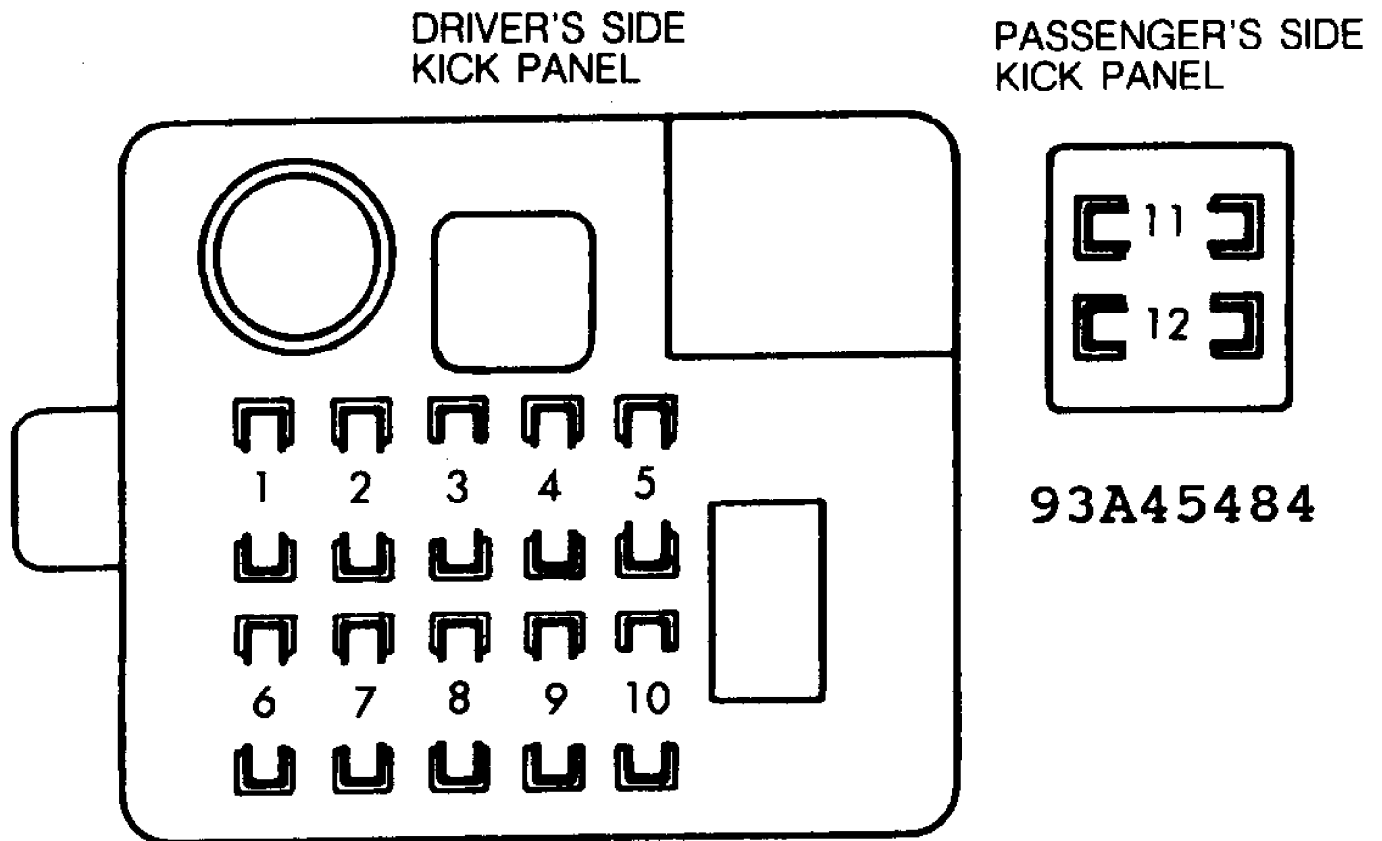


Fig. 1: Passenger Compartment Fuse Boxes Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

- \* 1 - 7.5 Amp  
Gauges & Warning Lights/Indicators, A/T Overdrive System &  
Back-Up Lights, Main Heater Relay Coil (1983)
- \* 2 - 7.5 Amp  
Radio & Cassette Player
- \* 3 - 15 Amp  
Trip Computer (Supra)
- \* 4 - 7.5 Amp  
Retractable Headlight System (1983-84), Outside Mirror

- Defogger (1985 Supra)
- \* 5 - 7.5 Amp  
Discharge Warning Light, Main Ignition Relay Coil, Main EFI Relay (1983 Supra)
- \* 6 - 7.5 Amp  
Interior, Personal & Door Courtesy Lights, Trunk, Ignition Switch & Door Warning Lights, Clock & Power Antenna (1984-85), Vanity Lights (Supra)
- \* 7 - 15 Amp  
Cigarette Lighter & Outside Mirror, Clock (1983), Trip Computer (1983 Supra)
- \* 8 - 15 Amp  
Cruise Control System, Electronically Controlled A/T System (1984-85 Supra)
- \* 9 - 7.5 Amp  
Turn Signal Lights
- \* 10 - 15 Amp  
Foglights (Supra)
- \* 11 - 10 Amp  
Air Conditioner
- \* 12 - Blank

#### ENGINE COMPARTMENT MAIN BLOCK IDENTIFICATION (1983-85)

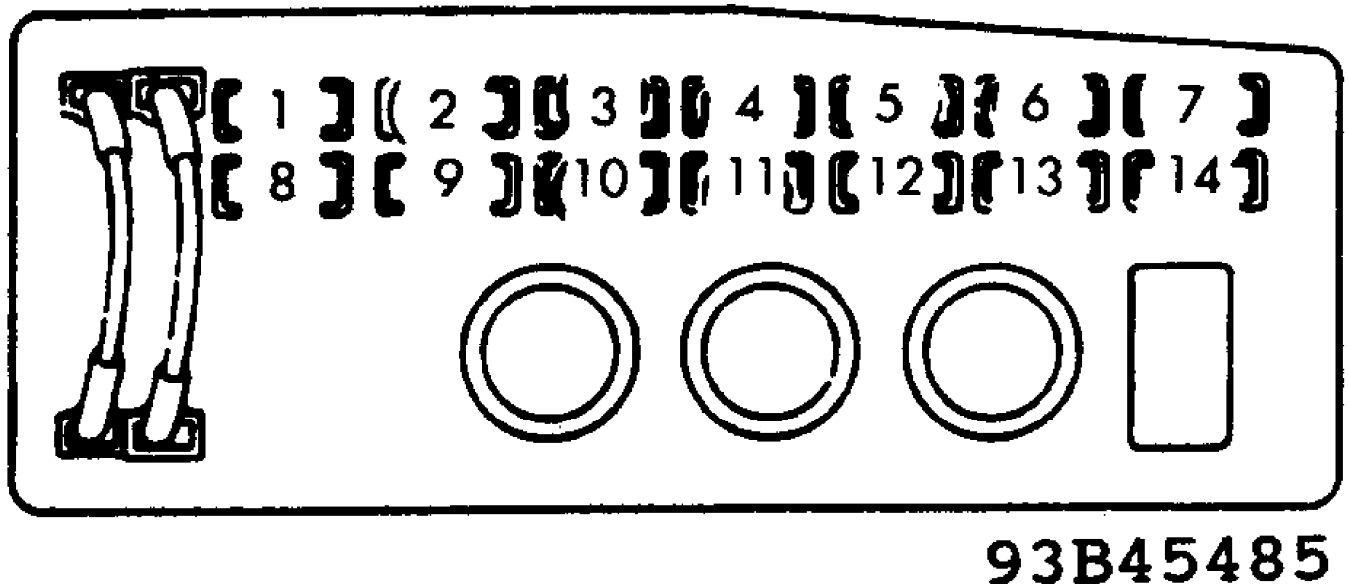


Fig. 2: Engine Compartment Main Fuse & Relay Block Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

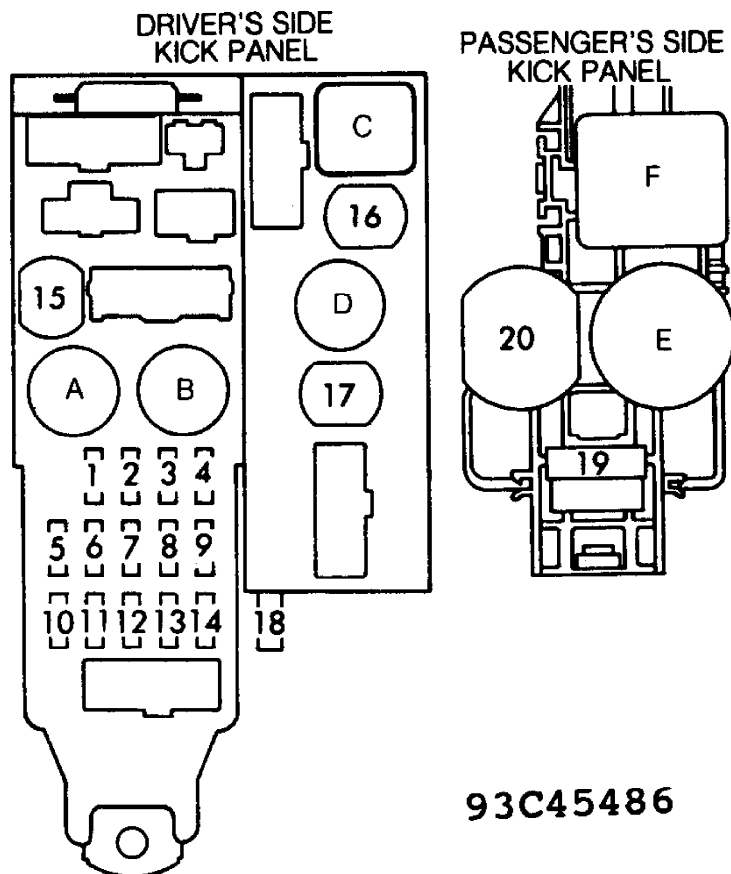
- \* 1 - Blank
- \* 2 - 7.5 Amp  
Discharge Warning Light
- \* 3 - 15 Amp  
Alternator With IC Regulator, Emission Control System (1984-85)
- \* 4 - 15 Amp  
Hazard Lights & Horns, Radio (1983), Retractable Headlights (1985)
- \* 5 - 15 Amp  
A/T Shift Position Indicator Light, Clock (Digital), Side Marker & License Lights, Park Lights, Taillights & Glove Box Light, Instrument Panel Lights, Foglight Relay (1983 Supra)
- \* 6 - 15 Amp

- Right Headlight, High Beam Indicator Light (1983)
- \* 7 - 15 Amp
- Left Headlight
- \* 8 - Blank
- \* 9 - 15 Amp
- EFI Control System
- \* 10 - 20 Amp
- Wipers & Washers, Headlight Cleaner Relay (1984-85 Supra)
- \* 11 - 15 Amp
- Stoplights, Cruise Control System (1983 Supra)
- \* 12 - Blank
- \* 13 - 15 Amp
- Right Low Beam Headlight (1983)
- \* 14 - 15 Amp
- Left Low Beam Headlight (1983)

### FUSE PANEL LOCATION (1986-89)

There are 3 fuse panels on this vehicle. The driver's side fuse box is located behind the driver's kick panel. The passenger's side fuse box is behind the passenger's side kick panel. The main fuse and relay block is in the engine compartment.

### PASSENGER COMPARTMENT FUSE BOX IDENTIFICATION (1986-89)



**93C45486**

Fig. 3: Passenger Compartment Fuse Boxes Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

- \* 1 - 15 Amp
- Gauges, Warning Lights & Buzzer, Back-Up Lights & A/T

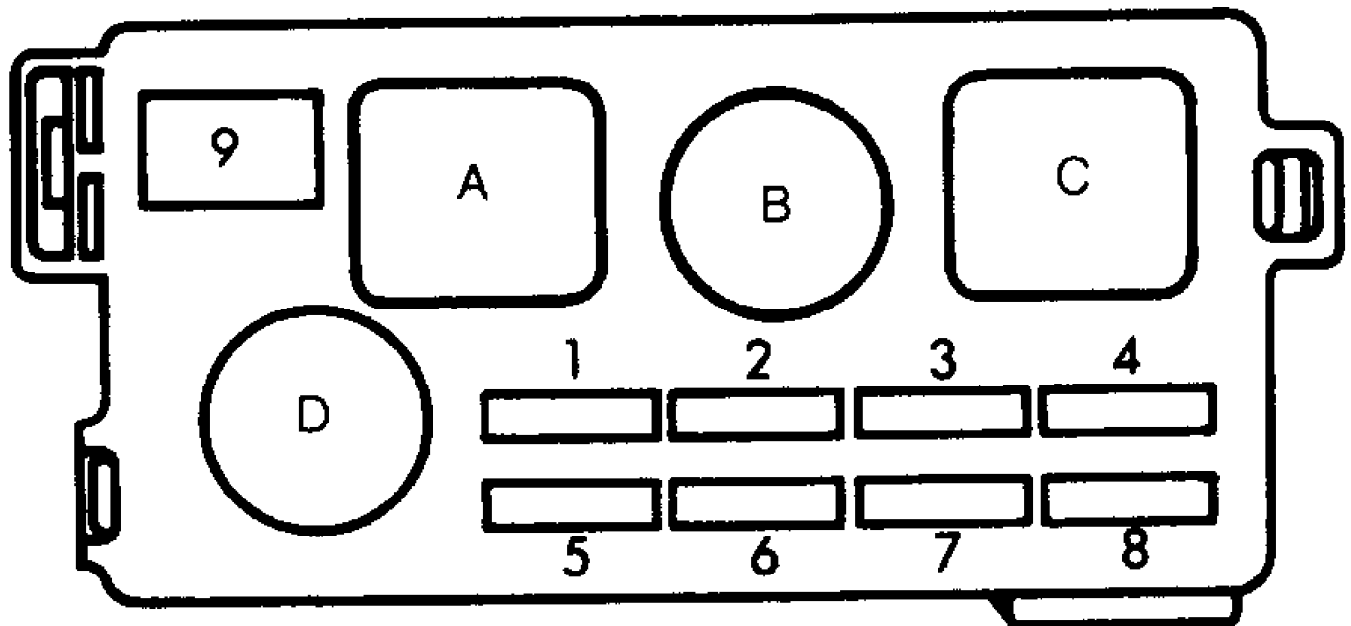


- Overdrive System, Rear Window Defogger & Power Antenna
- \* 2 - 7.5 Amp  
Clock & Open Door Warning Light (1987), Interior, Personal &  
Door Courtesy Lights (1987), Trunk & Ignition Switch Lights  
(1987)
- \* 3 - 15 Amp  
Stoplights & Cruise Control Cancel Device
- \* 4 - 15 Amp  
Tail, Park, License & Side Marker Lights, Glove Box Light &  
Instrument Illumination
- \* 5 - 15 Amp  
Cigarette Lighter & Digital Clock
- \* 6 - 7.5 Amp  
Radio & Cassette Player
- \* 7 - 10 Amp  
Turn Signal Lights
- \* 8 - 10 Amp  
Outside Mirror Defogger
- \* 9 - Blank
- \* 10 - 10 Amp  
Charging System
- \* 11 - 20 Amp  
Wipers & Washers
- \* 12 - 15 Amp  
Electric Sun Roof
- \* 13 - Blank
- \* 14 - 7.5 Amp  
Charging System & Discharge Warning Light, Electronic Fuel  
Injection & Electric Cooling Fans
- \* 15 - 30 Amp (Circuit Breaker)  
Defogger
- \* 16 - 14 Amp (Circuit Breaker 1986)
- \* 16 - 30 Amp (Circuit Breaker 1987-89)  
Power Door Locks
- \* 17 - 30 Amp (Circuit Breaker)  
Power Windows
- \* 18 - 15 Amp  
Anti-Lock Brake System & Foglights (1988-89)
- \* 19 - 10 Amp  
Air Conditioning System
- \* 20 - 30 Amp (Circuit Breaker)  
Heater

#### PASSENGER COMPARTMENT RELAY IDENTIFICATION (1986-89)

- \* A - Defogger
- \* B - Taillight Control
- \* C - Turn Signal Flasher
- \* D - Clutch Start
- \* E - Horn
- \* F - Heater

#### ENGINE COMPARTMENT BLOCK IDENTIFICATION (1986-89)



## ENGINE COMPARTMENT FUSE & RELAY BLOCK

**93D45487**

Fig. 4: Engine Compartment Fuse & Relay Block Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

- \* 1 - 15 Amp  
Headlights (Left)
- \* 2 - 15 Amp  
Emergency Flashers & Horns
- \* 3 - 15 Amp  
EFI System
- \* 4 - 7.5 Amp  
Charging System & Discharge Warning Light
- \* 5 - 15 Amp  
Headlights (Right)
- \* 6 - 30 Amp  
Retractable Headlight System
- \* 7 - 15 Amp (1986-87)
- \* 7 - 20 Amp (1988-89)  
Radio, Cassette Player & Power Antenna, Interior, Trunk,  
Personal & Courtesy Lights (1988-89), Ignition Switch & Door  
Warning Lights (1988-89), Clock (1988-89)
- \* 8 - Blank
- \* 9 - 30 Amp  
Fan

### ENGINE COMPARTMENT RELAY IDENTIFICATION (1986-89)

- \* A - Engine Main
- \* B - EFI Main
- \* C - Headlight Control
- \* D - Fan No. 1

### FUSE PANEL LOCATION (1990-93)

There are 3 fuse panels on this vehicle. The driver's side

fuse box is located behind the driver's kick panel. The passenger's side fuse box is behind the passenger's side kick panel. The main fuse and relay block is in the engine compartment.

#### PASSENGER COMPARTMENT FUSE BOX IDENTIFICATION (1990-93)

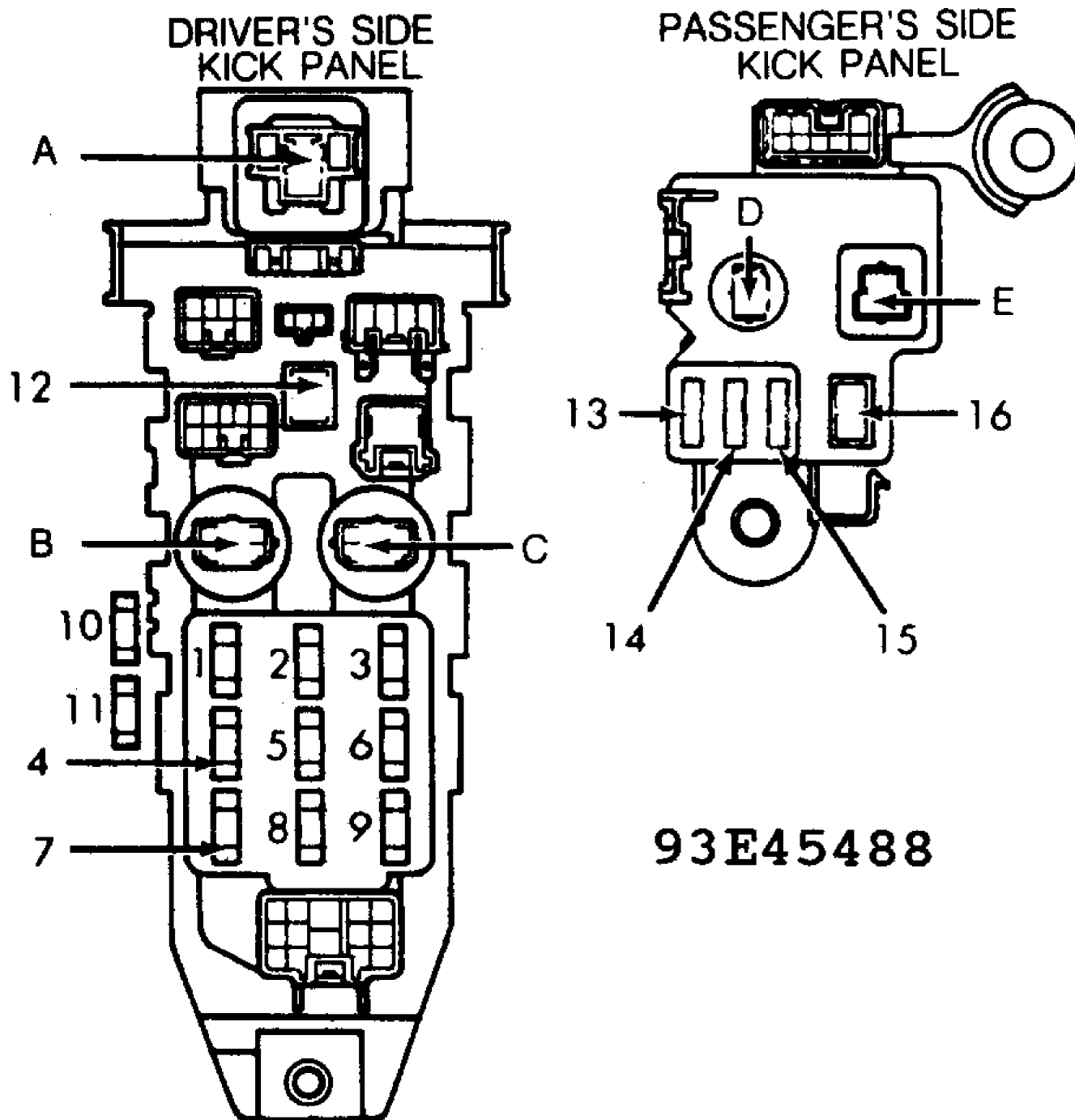


Fig. 5: Driver's Side & Passenger's Side Fuse Boxes Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

\* 1 - 7.5 Amp  
Charging System & Discharge Warning Light, Electric Cooling

- Fans & EFI System
- \* 2 - 15 Amp  
Cigarette Lighter & Digital Clock, Radio, Cassette Player & Compact Disc Player
- \* 3 - 15 Amp  
Stoplights & Cruise Control Cancel Device
- \* 4 - 15 Amp  
Gauges, Warning Lights & Buzzer, Back-Up Lights & A/T Overdrive System, Rear Window Defogger & Power Antenna, Tilt Steering, Convertible Top Control
- \* 5 - 10 Amp  
Turn Signal Lights
- \* 6 - 15 Amp  
Electric Sun Roof
- \* 7 - 20 Amp  
Wipers & Washers
- \* 8 - 15 Amp  
Anti-Lock Brake System
- \* 9 - 15 Amp  
Taillights, Parking Lights, Side Marker Lights, License Lights, Instrument Lights, Glove Box Light
- \* 10 - Blank
- \* 11 - 10 Amp  
Rear View Mirror
- \* 12 - 30 Amp (Circuit Breaker)  
Defogger
- \* 13 - Blank
- \* 14 - 20 Amp  
Front Foglights
- \* 15 - 10 Amp  
Air Conditioning
- \* 16 - 40 Amp (Circuit Breaker)  
Heater, A/C Control System

#### PASSENGER COMPARTMENT RELAY IDENTIFICATION (1990-93)

- \* A - Turn Signal Flasher
- \* B - Defogger
- \* C - Taillight Control
- \* D - Starter
- \* E - Heater Main

#### ENGINE COMPARTMENT MAIN BLOCK IDENTIFICATION (1990-93)

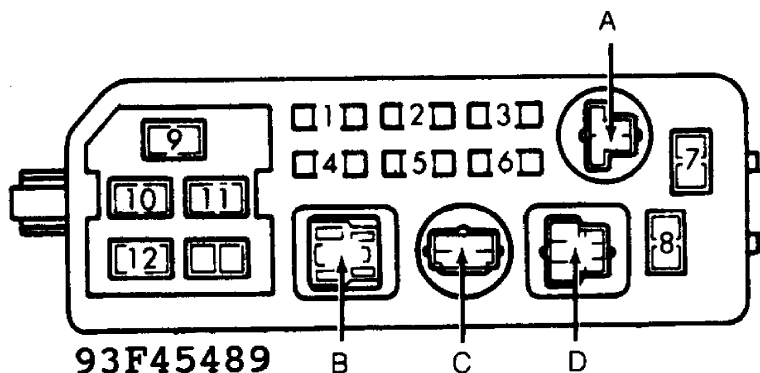


Fig. 6: Engine Compartment Main Fuse & Relay Block ID (Typical)  
Courtesy of Toyota Motor Sales, U.S.A., Inc

- \* 1 - 15 Amp  
Headlight (Left Side)

- \* 2 - 30 Amp  
Retractable Headlight System
- \* 3 - 15 Amp  
Headlight (Right Side)
- \* 4 - 15 Amp  
EFI System
- \* 5 - 20 Amp  
Radio, Cassette Player & Compact Disc Player, Power Antenna,  
Clock, Interior & Personal Lights, Door Courtesy Lights,  
Luggage Compartment Light, Trunk Light, Ignition Switch, Open  
Door Warning Lights
- \* 6 - 15 Amp  
Hazard Flashers & Horns
- \* 7 - 30 Amp (Circuit Breaker)  
Cooling Fan
- \* 8 - 30 Amp (Circuit Breaker)  
Condenser Cooling Fan
- \* 9 - 100 Amp (Circuit Breaker)  
Charging System
- \* 10 - 40 Amp (Circuit Breaker)  
AM2
- \* 10 - 30 Amp (1992-93)  
Electronic Fuel Injection, Ignition System
- \* 11 - 30 Amp (Circuit Breaker)  
AM1
- \* 11 - 40 Amp (1992-93)  
Charging System, Electric Underhood Cooling Fan
- \* 12 - 60 Amp (Circuit Breaker)  
Anti-Lock Brake System

## ENGINE COMPARTMENT MAIN BLOCK RELAY IDENTIFICATION (1990-93)

- \* A - Fan No. 1
- \* B - Headlight Control
- \* C - EFI System
- \* D - Engine Main

## CAUTIONS & WARNINGS

### SUPPLEMENTAL RESTRAINT SYSTEM (AIR BAG SYSTEM)

NOTE: For information on air bag DIAGNOSIS & TESTING or DISPOSAL PROCEDURES, see AIR BAGS article in the ACCESSORIES/SAFETY EQUIPMENT Section.

When servicing vehicles equipped with Supplemental Restraint Systems (SRS), always allow at least 20 seconds after turning ignition switch to LOCK position and disconnecting the negative terminal cable before beginning work on the SRS. The system has a back-up power supply and may deploy air bag if these precautions are not followed.

### RADIATOR FAN & CAP

To avoid injury, stay clear of the radiator fan when the ignition key is in the ON position. The fan is thermostatically controlled and may come on suddenly. DO NOT loosen or remove radiator cap when cooling system is hot.

# HEATER SYSTEM

1993 Toyota Celica

1993 Heater Systems

Celica

## DESCRIPTION

Heater assembly consists of heater core, control panel, blower motor, control cables (electric servomotors on push button-controlled models) and air ducts. All components are located under instrument panel.

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

**CAUTION:** When battery is disconnected, radio will go into anti-theft protection mode. Obtain radio anti-theft protection code from owner prior to servicing vehicle.

## OPERATION

### BLOWER SWITCH

Switch controls blower motor speed through blower resistor. Switch is operated by control lever, dial knob or push button.

### CONTROL PANEL

Lever-Controlled Models

Temperature and mode levers are cable-connected to heater coolant valve and air doors. All models have a fresh/recirculation lever to provide choice of outside air entry or inside air recirculation.

Push Button-Controlled Models

Air inlet (fresh/recirculation), mode control and air mix are controlled by electric servomotors. Temperature selection is controlled by slide lever or rotating dial knob.

## HEATER RELAY

A heater (or main) relay in heater circuit controls current flow through system. See WIRING DIAGRAM. For location of heater relay, see HEATER RELAY LOCATION table.

### HEATER RELAY LOCATION TABLE

| Models |       | Location                                  |
|--------|-------|-------------------------------------------|
| Celica | ..... | Relay/Fuse Block, Behind Right Kick Panel |

## ADJUSTMENTS

### AIR INLET DAMPER CABLE

#### Lever-Controlled Models

Set air inlet damper and control lever to fresh air position. Remove cable retaining clip, and ensure damper and cable are in full fresh position. Attach control cable using retaining clip, and check operation of air intake damper.

### AIR MIX DAMPER CABLE

#### Lever-Controlled Models

Set air door lever to warm position, and remove cable retaining clip. Ensure cable and damper are in full cool position. Install cable retaining clip. Check air mix damper operation.

### AIRFLOW MODE DAMPER CABLE

#### Lever-Controlled Models

Set control lever to defrost position. Remove cable retaining clip. Ensure airflow mode damper and cable are in full defrost position. Install cable retaining clip.

### WATER VALVE CONTROL CABLE

#### Lever-Controlled Models

Set control lever to warm position. Remove cable retaining clip. Ensure water valve control cable is in full warm position. Install cable retaining clip.

## TROUBLE SHOOTING

### BLOWER DOES NOT WORK

Check for open circuit breaker (some models), blown heater fuse and faulty heater relay. Also check for heater blower switch, heater blower resistor, heater blower motor, wiring or ground circuit fault.

### INCORRECT TEMPERATURE OUTPUT

Check for control cables broken or out of adjustment, heater hoses leaking or clogged and faulty water pump. Also check for broken air dampers, faulty servomotor (some models), clogged air ducts, leaking or clogged heater core and faulty heater control unit.

## TESTING

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

### AIR INLET CONTROL SERVOMOTOR

1) Disconnect air inlet control servomotor wiring harness connector. Ground terminal No. 3, and apply battery voltage to terminal No. 1. See Fig. 1. Ensure arm rotates smoothly to fresh (full counterclockwise) position.

2) Ground terminal No. 2, and apply battery voltage to terminal No. 1. Ensure arm rotates smoothly to recirculation (full

clockwise) position. If operation is not as specified, replace servomotor.

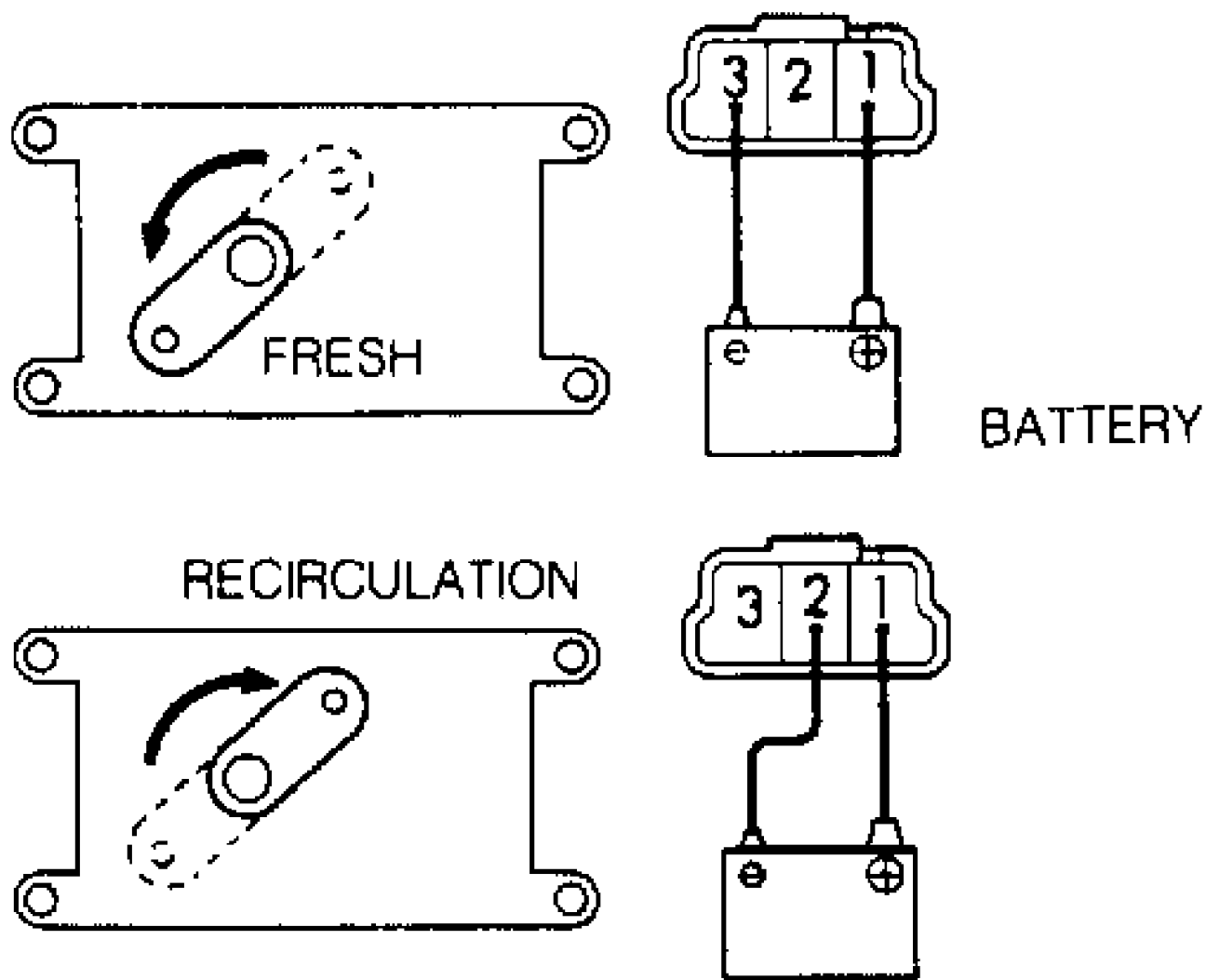


Fig. 1: Testing Air Inlet Control Servomotor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### AIR INLET CONTROL SWITCH

1) Disconnect wiring harness connector "A" of heater control assembly. See Fig. 2. With recirculation button depressed, continuity should exist between terminals No. 2 and 7.

2) With fresh button depressed, continuity should exist between terminals No. 2 and 8. Switch contains diodes. Before deciding switch is faulty, check for continuity in both directions. If continuity is not as specified, replace heater control assembly.



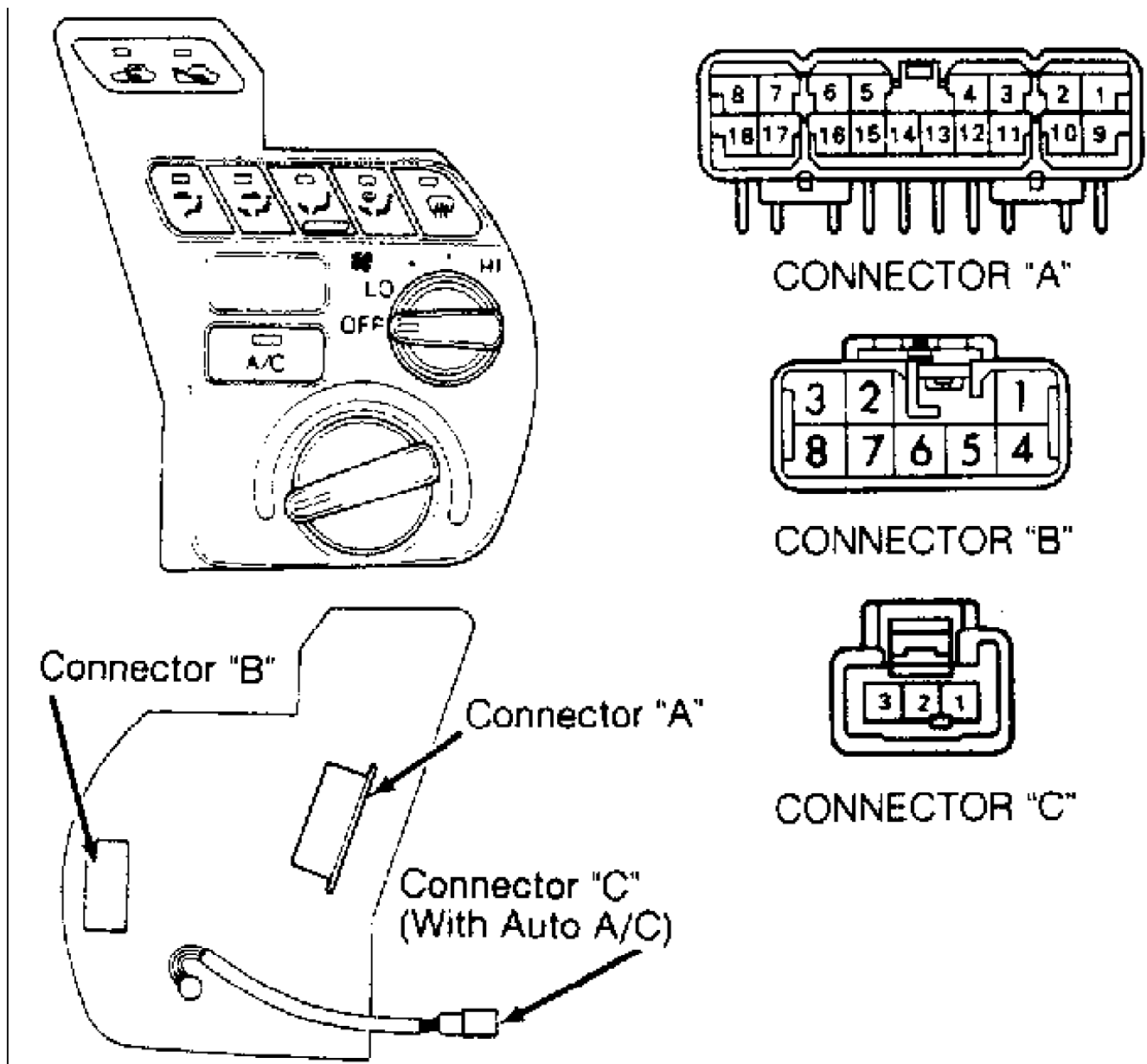


Fig. 2: Identifying Heater Control Assembly Connector Terminals  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

### AIRFLOW MODE CONTROL SWITCH

Disconnect wiring harness connector "A" of heater control assembly. See Fig. 2. Check for continuity at specified terminals. See TESTING AIRFLOW MODE CONTROL SWITCH table. If continuity is not as specified, replace heater control assembly.

TESTING AIRFLOW MODE CONTROL SWITCH TABLE

| Switch Position | Continuity Between Terminals |
|-----------------|------------------------------|
| Face .....      | 2 & 9                        |
| Bi-Level .....  | 2 & 10                       |
| Foot .....      | 2 & 11                       |

|              |        |
|--------------|--------|
| Foot/Defrost | 2 & 12 |
| Defrost      | 2 & 13 |

AIRFLOW MODE SERVOMOTOR

Disconnect servomotor connector. Ground terminal No. 6, and apply battery voltage to terminal No. 5. See Fig. 3. Ground each specified terminals, and ensure arm rotates smoothly to correct position. See TESTING AIRFLOW MODE SERVOMOTOR table. Replace servomotor if operation is not as specified.

TESTING AIRFLOW MODE SERVOMOTOR TABLE

| Ground Terminal No. | Arm Position |
|---------------------|--------------|
| 1                   | Vent         |
| 2                   | Bi-Level     |
| 3                   | Foot 2       |
| 4                   | Foot/Defrost |
| 7                   | Defrost      |
| 8                   | Foot 1       |

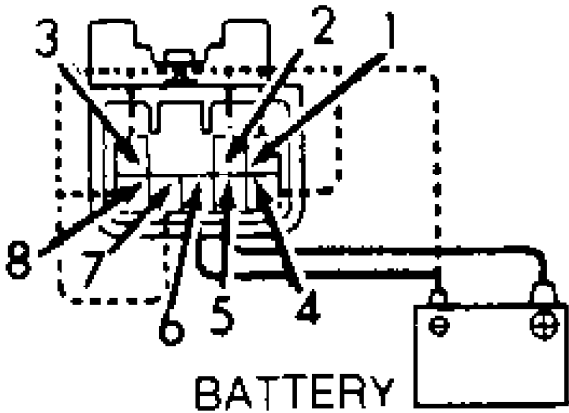
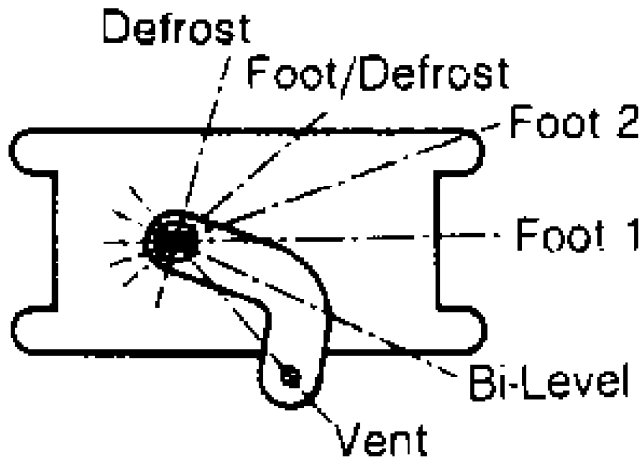


Fig. 3: Testing Airflow Mode Servomotor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

BLOWER FAN RELAY

Disconnect wiring harness connector. Test blower fan relay as specified. See TESTING BLOWER FAN RELAY table. See Fig. 4. If continuity is not as specified, replace relay.

TESTING BLOWER FAN RELAY TABLE

| Apply Battery Voltage<br>Between Terminal No. | (1) Continuity Should Exist<br>Between Terminal No. |
|-----------------------------------------------|-----------------------------------------------------|
| 5 & 6                                         | 1 & 3                                               |
| 5 & 7                                         | 3 & 4                                               |
| 5 & 8                                         | 2 & 3                                               |

(1) - When battery voltage is not applied, constant continuity is present between terminals No. 5 and 6,

5 and 7, and 5 and 8.

---

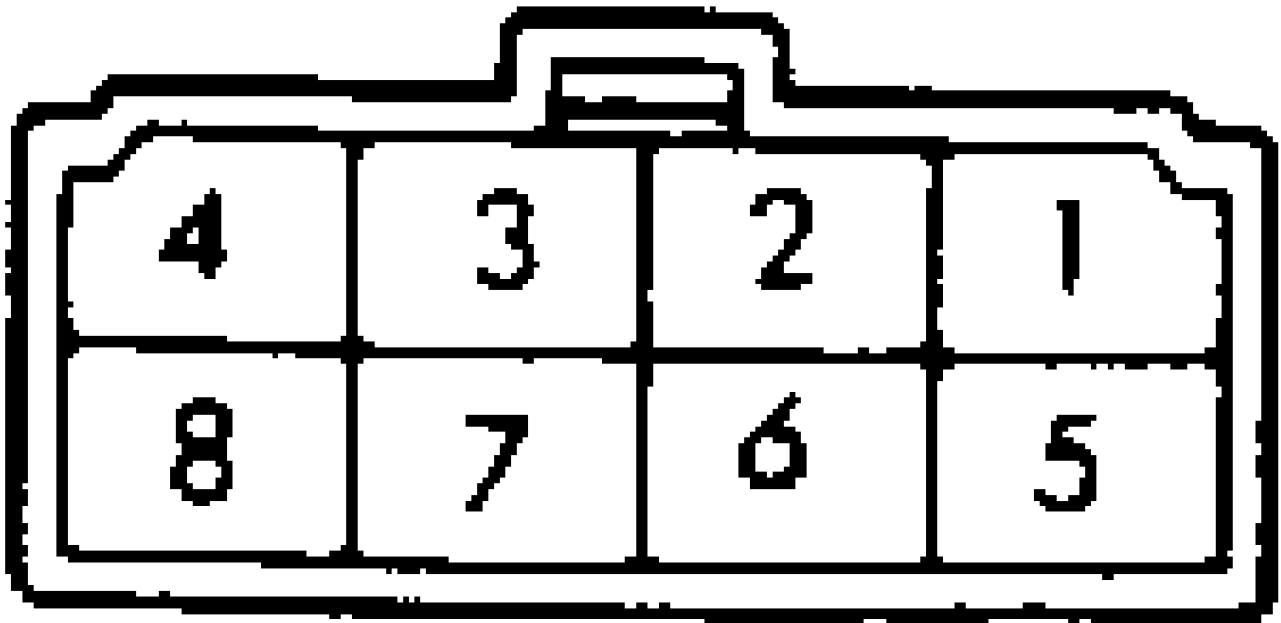
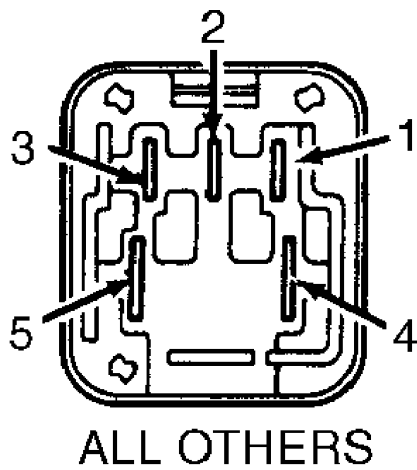


Fig. 4: Identifying Blower Fan Relay Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



94A10330

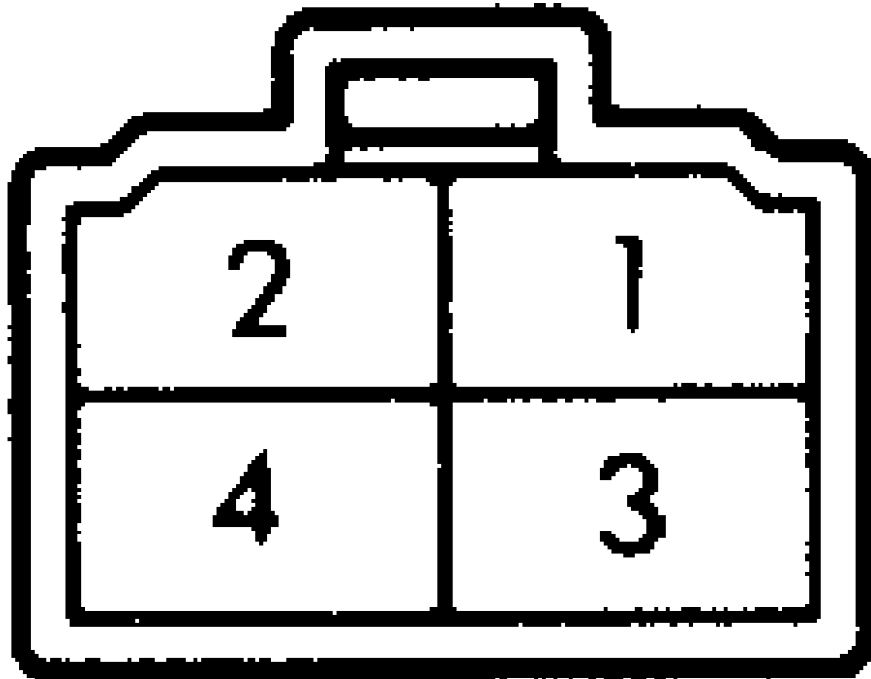
Fig. 5: Identifying Heater Relay Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

BLOWER MOTOR

Disconnect blower motor wiring harness connector. Apply battery voltage to motor side of connector. Motor should operate smoothly. If motor operation is not smooth, replace motor.

## BLOWER RESISTOR

Remove resistor from vehicle or disconnect resistor wiring. Using an ohmmeter, ensure continuity exists between blower resistor terminals No. 1 and 4. See Fig. 6. If continuity is not as specified, replace resistor.



# ALL OTHER MODELS

## 94E10334

Fig. 6: Identifying Heater Blower Resistor Connector Terminals  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## BLOWER SPEED CONTROL SWITCH

Disconnect wiring harness connector "B" of heater control panel. Using an ohmmeter, check continuity between specified terminals and switch positions. See Fig. 2. See appropriate TESTING BLOWER SPEED CONTROL SWITCH table. If continuity is not as specified, replace blower switch.

TESTING BLOWER SPEED CONTROL SWITCH TABLE

---

| Switch Position | Continuity Between Terminals |
|-----------------|------------------------------|
| OFF .....       | (1)                          |
| LO .....        | 1, 3 & 7                     |
| (i) (2) .....   | 2, 3 & 6                     |
| (i) (3) .....   | 2, 3 & 4                     |
| HI .....        | 2, 3 & 5                     |

- (1) - No continuity.  
 (2) - Square (i) closest to LO position.  
 (3) - Square (i) closest to HI position.
- 

## HEATER RELAY

### 4-Pin Type

- 1) Disconnect negative battery cable. Remove heater relay.

Using an ohmmeter, ensure continuity exists between heater relay terminals No. 1 and 3, and between terminals No. 2 and 4. See Fig. 5. Ensure continuity does not exist between terminals No. 4 and 5. If continuity is not as specified, replace relay.

- 2) Ground terminal No. 3, and apply battery voltage to terminal No. 1. Ensure continuity exists between terminals No. 4 and 5. If continuity is not as specified, replace relay.

## REMOVAL & INSTALLATION

**WARNING:** To avoid injury from accidental air bag deployment, read and carefully follow all SERVICE PRECAUTIONS and DISABLING & ACTIVATING AIR BAG SYSTEM procedures in appropriate AIR BAG RESTRAINT SYSTEM article in ACCESSORIES & ELECTRICAL section.

## BLOWER MOTOR

### Removal & Installation

Removal and installation procedures are not available.

Exploded views of heater systems are provided. See Fig. 7.

## HEATER ASSEMBLY

### Removal & Installation

Removal and installation procedures are not available.

Exploded views of heater systems are provided. See Fig. 7.

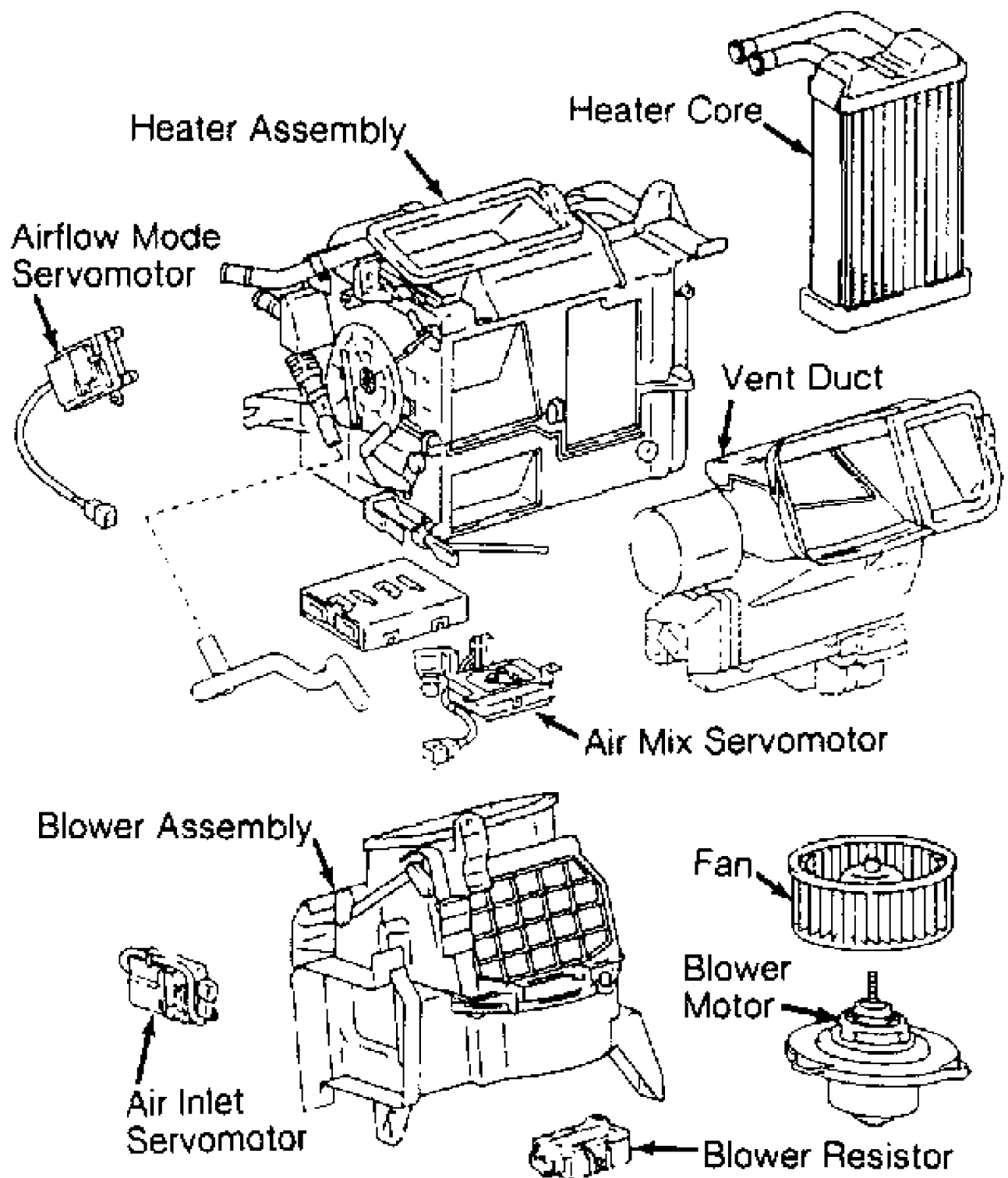


Fig. 7: Exploded View Of Heater Assembly  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAM

NOTE: For additional wiring diagrams, see appropriate A/C-HEATER SYSTEM - MANUAL article in the AIR CONDITIONING & HEAT section.

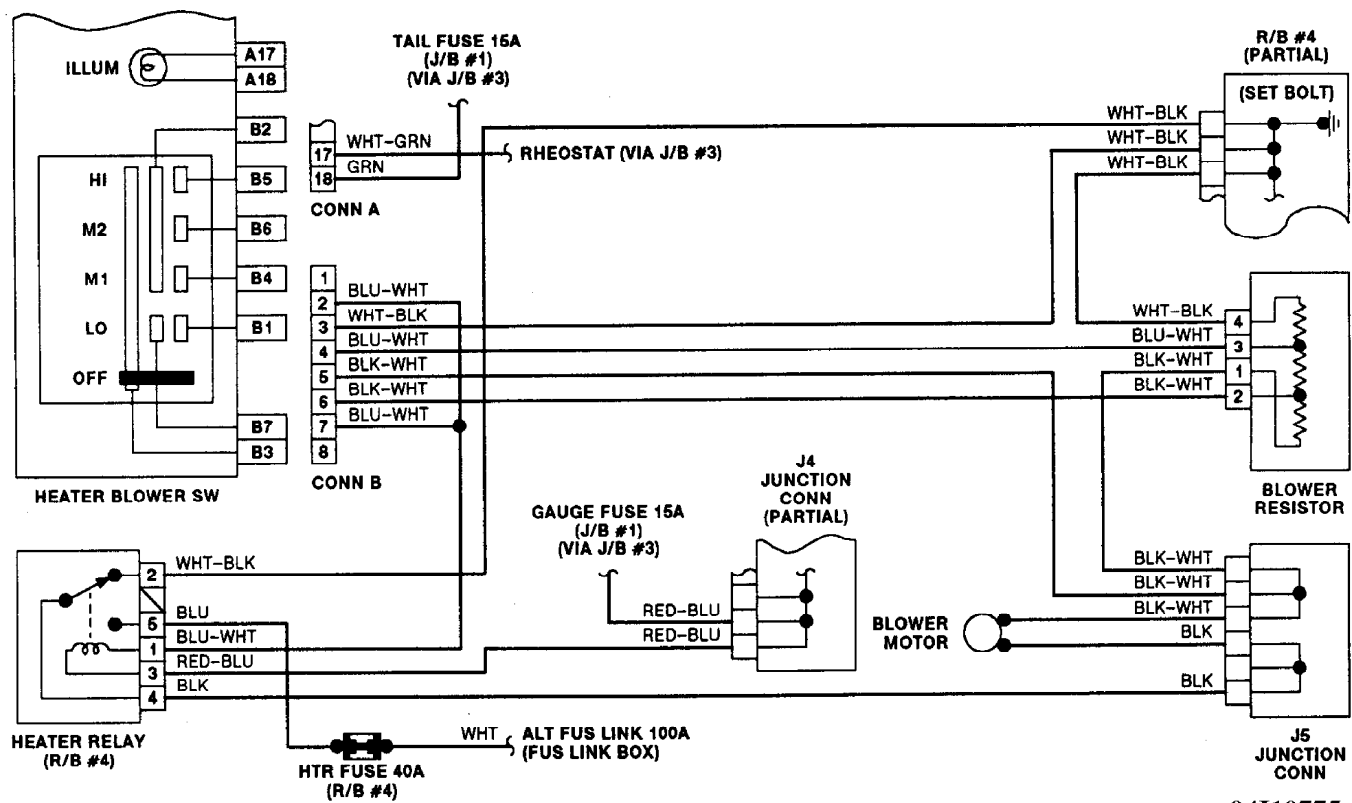


Fig. 8: Heater System Wiring Diagram

94H10775

# HOW TO USE SYSTEM WIRING DIAGRAMS

1993 Toyota Celica

GENERAL INFORMATION  
Using Wiring Diagrams

All Models

## INTRODUCTION

This CD obtains wiring diagrams and technical service bulletins, containing wiring diagram changes from the domestic and import manufacturers. These are checked for accuracy and are all redrawn into a consistent format for easy use.

In the past, when cars were simpler, diagrams were simpler. All components were connected by wires and diagrams seldom exceeded 4 pages in length. Today, some wiring diagrams require more than 16 pages. It would be impractical to expect a service technician to trace a wire from page 1 across every page to page 16.

Removing some of the wiring maze reduces eyestrain and time wasted searching across several pages. Today the majority of these diagrams follow a much improved format, which permits space for internal switch details.

Wiring diagrams are drawn in a "top-down" format. The diagrams are drawn with the power source at the top of the diagram and the ground point at the bottom of the diagram. Components locations are identified on the wiring diagrams. Any wires that don't connect directly to a component are identified on the diagram to indicate where they go.

## COLOR ABBREVIATIONS

COLOR ABBREVIATIONS TABLE

| Color             | Normal       | Optional |
|-------------------|--------------|----------|
| Black .....       | BLK .....    | BK       |
| Blue .....        | BLU .....    | BU       |
| Brown .....       | BRN .....    | BN       |
| Clear .....       | CLR .....    | CR       |
| Dark Blue .....   | DK BLU ..... | DK BU    |
| Dark Green .....  | DK GRN ..... | DK GN    |
| Green .....       | GRN .....    | GN       |
| Gray .....        | GRY .....    | GY       |
| Light Blue .....  | LT BLU ..... | LT BU    |
| Light Green ..... | LT GRN ..... | LT GN    |
| Orange .....      | ORG .....    | OG       |
| Pink .....        | PNK .....    | PK       |
| Purple .....      | PPL .....    | PL       |
| Red .....         | RED .....    | RD       |
| Tan .....         | TAN .....    | TN       |
| Violet .....      | VIO .....    | VI       |
| White .....       | WHT .....    | WT       |
| Yellow .....      | YEL .....    | YL       |

## IDENTIFYING WIRING DIAGRAM ABBREVIATIONS

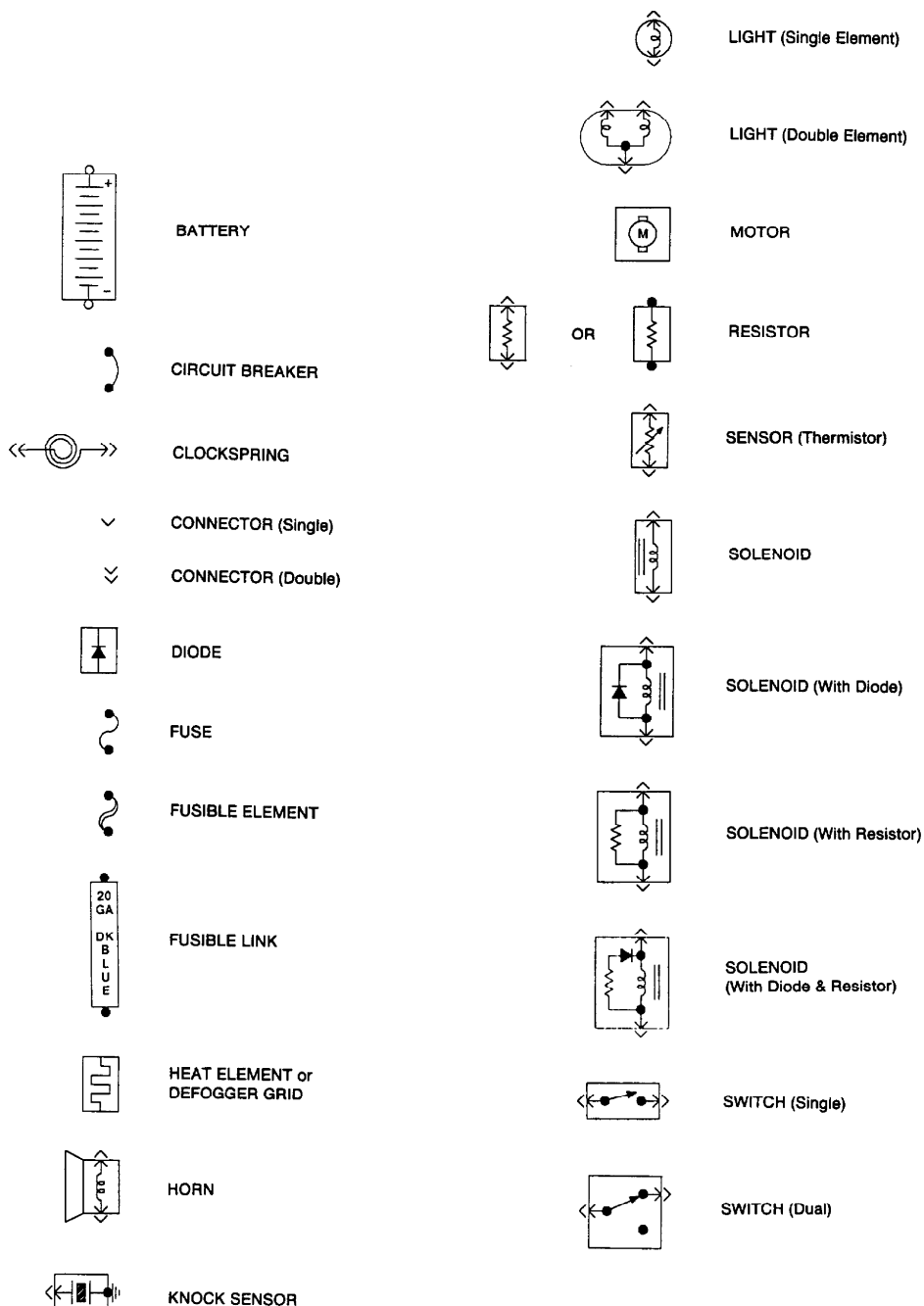
NOTE: Abbreviations used on these diagrams are normally self-explanatory. If necessary see ABBREVIATIONS



article in GENERAL INFORMATION.

## IDENTIFYING WIRING DIAGRAM SYMBOLS

**NOTE:** Standard wiring symbol are used in these diagrams. The illustration below will help clarify any symbols that are not easily understood at a glance. Most components are labeled "Motor", "Switch" or "Relay" in addition to being drawn with the standard symbol.



50E03637

Fig. 1: Identifying Wiring Diagram Symbols

## WIRING DIAGRAM COMPONENT LOCATIONS

When trying to locate a component in a wiring diagram and you don't know the specific system where it is located, use this handy component locator to find the system wiring diagram in which the component is located. Then, go to that system and locate the component within the wiring diagram.

For example, if you don't know the specific system in which the ignition switch is located, look up ignition switch in the wiring diagram component location tables and go to the appropriate wiring diagram(s) which contain either full or partial views of the ignition switch. The full view of the ignition switch is located in Power Distribution.

The first listing for the component will be the full or most complete view of the component. Additional listings will be partial views of the component. Not all components are used on all models.

All components will have a partial view in Ground Distribution and Power Distribution. Data Link Connectors show connecting circuits between modules. Alternate names for components may be listed in wiring diagram component locations tables.

WIRING DIAGRAM COMPONENT LOCATIONS TABLE

| Component                               | Wiring Diagram                                                                                                                     |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| ABS Electronic Control Unit .....       | Anti-Lock Brakes<br>Data Link Connectors                                                                                           |
| ABS Hydraulic Unit .....                | Anti-Lock Brakes                                                                                                                   |
| Acceleration Sensor .....               | Anti-Lock Brakes                                                                                                                   |
| Accessory Delay Relay .....             | Power Windows                                                                                                                      |
| A/C Compressor Clutch Relay .....       | Engine Performance                                                                                                                 |
| A/C Sensor .....                        | Engine Performance                                                                                                                 |
| A/C Pressure Switch .....               | Engine Performance                                                                                                                 |
| Adaptive Lamp Control Module .....      | Exterior Lights                                                                                                                    |
| Air Bag(s) .....                        | Air Bag Restraint System                                                                                                           |
| Air Bag Module .....                    | Air Bag Restraint System                                                                                                           |
| Air Bag Sensor(s) .....                 | Air Bag Restraint System                                                                                                           |
| Air Injection Pump Relay .....          | Engine Performance                                                                                                                 |
| Air Temperature Sensor .....            | Overhead Console                                                                                                                   |
| Alternator (Generator) .....            | Generators & Regulators                                                                                                            |
| Anti-Theft Control Module .....         | Anti-Theft System<br>Starters                                                                                                      |
| Autolamp Control Relay .....            | Headlight Systems<br>Daytime Running Lights                                                                                        |
| Automatic Shutdown (ASD) Relay .....    | Engine Performance<br>Generators & Regulators                                                                                      |
| Autostick Switch .....                  | Engine Performance                                                                                                                 |
| Auxiliary Battery Relay .....           | Generators & Regulators                                                                                                            |
| Back-Up Lights .....                    | Back-Up Lights<br>Exterior Lights                                                                                                  |
| Barometric (BARO) Pressure Sensor ..... | Engine Performance                                                                                                                 |
| Battery .....                           | Power Distribution                                                                                                                 |
| Battery Temperature Sensor .....        | Engine Performance                                                                                                                 |
| Body Control Module .....               | Body Control Computer<br>Anti-Theft System<br>Daytime Running Lights<br>Engine Performance<br>Headlight Systems<br>Warning Systems |
| Boost Control Solenoid .....            | Engine Performance                                                                                                                 |
| Boost Sensor .....                      | Engine Performance                                                                                                                 |
| Brake Fluid Level Switch .....          | Analog Instrument Panels                                                                                                           |

|                                               |                              |
|-----------------------------------------------|------------------------------|
| Brake On/Off (BOO) Switch .....               | Cruise Control Systems       |
|                                               | Engine Performance           |
|                                               | Shift Interlock Systems      |
| Buzzer Module .....                           | Warning Systems              |
| Camshaft Position (CMP) Sensor .....          | Engine Performance           |
| Central Control Module .....                  | Anti-Theft System            |
| Clockspring .....                             | Air Bag Restraint System     |
|                                               | Cruise Control Systems       |
|                                               | Steering Column Switches     |
| Clutch Pedal Position Switch .....            | Starters                     |
| Clutch Start Switch .....                     | Starters                     |
| Combination Meter .....                       | Analog Instrument Panels     |
| Constant Control Relay Module (CCRM) .....    | Engine Performance           |
|                                               | Electric Cooling Fans        |
| Convenience Center .....                      | Power Distribution           |
|                                               | Illumination/Interior Lights |
| Convertible Top Motor .....                   | Power Convertible Top        |
| Convertible Top Switch .....                  | Power Convertible Top        |
| Crankshaft Position (CKP) Sensor .....        | Engine Performance           |
| Cruise Control Module .....                   | Cruise Control Systems       |
| Cruise Control Switch .....                   | Cruise Control Systems       |
| Condenser Fan Relay(s) .....                  | Electric Cooling Fans        |
| Data Link Connector (DLC) .....               | Engine Performance           |
| Daytime Running Lights Module .....           | Daytime Running Lights       |
|                                               | Exterior Lights              |
| Defogger Relay .....                          | Rear Window Defogger         |
| Diagnostic Energy Reserve Module (DERM) ...   | Air Bag Restraint System     |
| Discriminating Sensor (Air Bag) .....         | Air Bag Restraint System     |
| Distributor .....                             | Engine Performance           |
| Door Lock Actuators .....                     | Power Door Locks             |
|                                               | Remote Keyless Entry         |
| Door Lock Relay(s) .....                      | Power Door Locks             |
| Electrochromic Mirror .....                   | Power Mirrors                |
| Electronic Level Control (ELC)                |                              |
| Height Sensor .....                           | Electronic Suspension        |
| Electronic Level Control (ELC) Module .....   | Electronic Suspension        |
| Engine Coolant Temperature (ECT)              |                              |
| Sending Unit .....                            | Analog Instrument Panels     |
| Engine Coolant Temperature (ECT) Sensor ..... | Engine Performance           |
| Engine Control Module .....                   | Engine Performance           |
|                                               | Generators & Regulators      |
|                                               | Starters                     |
| ETACS ECU .....                               | Warning Systems              |
|                                               | Power Windows                |
|                                               | Remote Keyless Entry         |
| Evaporative (EVAP) Emissions Canister .....   | Engine Performance           |
| EVAP Canister Purge Solenoid .....            | Engine Performance           |
| EVAP Canister Vent Solenoid .....             | Engine Performance           |
| Exhaust Gas Recirculation (EGR) Valve .....   | Engine Performance           |
| Fuel Tank Vacuum Sensor .....                 | Engine Performance           |
| Fog Lights .....                              | Headlight Systems            |
|                                               | Daytime Running Lights       |
| Fog Light Relay .....                         | Headlight Systems            |
|                                               | Daytime Running Lights       |
| Fuel Door Release Solenoid .....              | Power Fuel Door Release      |
| Fuel Gauge Sending Unit .....                 | Analog Instrument Panels     |
| Fuel Injectors .....                          | Engine Performance           |
| Fuel Pump .....                               | Engine Performance           |
| Fuel Pump Relay .....                         | Engine Performance           |
|                                               | Power Distribution           |
| Fuse/Relay Block .....                        | Power Distribution           |
| Fusible Links .....                           | Power Distribution           |
|                                               | Generators & Regulators      |

|                                               |                              |
|-----------------------------------------------|------------------------------|
|                                               | Starters                     |
| Generator .....                               | Generators & Regulators      |
|                                               | Engine Performance           |
|                                               | Power Distribution           |
| Generic Electronic Module (GEM) .....         | Body Control Modules         |
|                                               | Electronic Suspension        |
| Glow Plug Relay .....                         | Engine Performance           |
| Glow Plugs .....                              | Engine Performance           |
| Grounds .....                                 | Ground Distribution          |
| Headlight Door Module .....                   | Headlight Doors              |
| Headlight Relay .....                         | Headlight Systems            |
|                                               | Daytime Running Lights       |
| Headlights .....                              | Headlight Systems            |
|                                               | Daytime Running Lights       |
| Heated Oxygen Sensor(s) (HO2S) .....          | Engine Performance           |
| Heated Windshield Control Module .....        | Heated Windshields           |
| Height Sensor .....                           | Electronic Suspension        |
| Horns .....                                   | Steering Column Switches     |
| Horn Relay .....                              | Steering Column Switches     |
| Idle Air Control (IAC) Motor/Valve .....      | Engine Performance           |
| Ignition Coil(s) .....                        | Engine Performance           |
| Ignition Key Lock Cylinder .....              | Anti-Theft System            |
| Ignition Module .....                         | Engine Performance           |
| Ignition Switch .....                         | Power Distribution           |
|                                               | Engine Performance           |
|                                               | Generators & Regulators      |
|                                               | Starters                     |
| Illuminated Entry Module .....                | Illumination/Interior Lights |
| Illumination Lights .....                     | Illumination/Interior Lights |
| Impact Sensor .....                           | Air Bag Restraint System     |
| Inertia Fuel Shutoff Switch .....             | Engine Performance           |
| Inhibit Relay .....                           | Starters                     |
| Instrument Cluster .....                      | Analog Instrument Panels     |
| Intake Air Temperature (IAT) Sensor .....     | Engine Performance           |
| Interior Lights .....                         | Illumination/Interior Lights |
| Interlock Switch .....                        | Starters                     |
| Junction Block .....                          | Power Distribution           |
| Keyless Entry Receiver .....                  | Remote Keyless Entry         |
| Key Reminder Switch .....                     | Starters                     |
| Knock Sensor .....                            | Engine Performance           |
| Lamp Control Module .....                     | Exterior Lights              |
| License Plate Lamp .....                      | Exterior Lights              |
| Lighting Control Module .....                 | Lighting Control Modules     |
|                                               | Anti-Theft System            |
|                                               | Daytime Running Lights       |
|                                               | Headlight Systems            |
| Lower Relay .....                             | Power Convertible Top        |
| Malfunction Indicator Light (MIL) .....       | Engine Performance           |
|                                               | Instrument Panels            |
| Manifold Absolute Pressure (MAP) Sensor ..... | Engine Performance           |
| Mass Airflow (MAF) Sensor .....               | Engine Performance           |
| Mega Fuse .....                               | Generators & Regulators      |
| Memory Seat/Mirror Module .....               | Memory Systems               |
| Mirror Defogger .....                         | Rear Window Defogger         |
| Moon Roof Motor .....                         | Power Moon Roof              |
| Moon Roof Relay .....                         | Power Moon Roof              |
| Multi-Function Control Module .....           | Warning Systems              |
| Neutral Safety Switch .....                   | Starters                     |
| Oil Level Switch .....                        | Engine Performance           |
| Oil Pressure Switch/Sending Unit .....        | Analog Instrument Panels     |
|                                               | Engine Performance           |
| Overhead Console .....                        | Overhead Console             |
| Oxygen Sensor(s) (O2S) .....                  | Engine Performance           |

|                                                  |                          |
|--------------------------------------------------|--------------------------|
| Parking Brake Switch .....                       | Analog Instrument Panels |
| Park Lights .....                                | Exterior Lights          |
| Park/Neutral Position Switch .....               | Starters                 |
|                                                  | Engine Performance       |
|                                                  | Anti-Theft System        |
|                                                  | Body Control Module      |
| Perimeter Lighting Control Relay .....           | Exterior Lights          |
| Power Amplifier .....                            | Power Antennas           |
| Power Antenna Module .....                       | Power Antennas           |
| Power Antenna Motor .....                        | Power Antennas           |
| Power Distribution Center .....                  | Power Distribution       |
|                                                  | Generators & Regulators  |
|                                                  | Starters                 |
| Power Door Lock Motors .....                     | Power Door Locks         |
| Power Mirror Motors .....                        | Power Mirrors            |
|                                                  | Memory Systems           |
| Power Sliding Door Controller .....              | Power Sliding Side Door  |
| Power Seat Motors .....                          | Power Seats              |
|                                                  | Memory Systems           |
| Power Steering Pressure Switch .....             | Engine Performance       |
| Power Top Motor .....                            | Power Convertible Top    |
| Power Top Relay(s) .....                         | Power Convertible Top    |
| Powertrain Control Module .....                  | Engine Performance       |
|                                                  | Analog Instrument Panels |
|                                                  | Cruise Control Systems   |
|                                                  | Data Link Connectors     |
|                                                  | Generators & Regulators  |
|                                                  | Starters                 |
| Power Window Motors .....                        | Power Windows            |
| Power Window Relay(s) .....                      | Power Windows            |
| Radiator Fan Motor(s) .....                      | Electric Cooling Fans    |
| Radiator Fan Relay(s) .....                      | Engine Performance       |
|                                                  | Electric Cooling Fans    |
| Rainsense Module .....                           | Wiper/Washer Systems     |
| Raise Relay .....                                | Power Convertible Top    |
| Remote Anti-Theft Personality (RAP) Module ..... | Anti-Theft System        |
|                                                  | Starters                 |
|                                                  | Warning Systems          |
| Seat Belt Pretensioners .....                    | Air Bag Restraint System |
| Seat Belt Retractor Solenoid .....               | Passive Restraints       |
| Seat Belt Switch .....                           | Air Bag Restraint System |
|                                                  | Passive Restraints       |
| Shift Interlock Solenoid .....                   | Shift Interlock Systems  |
| Shift Lock Actuator .....                        | Shift Interlock Systems  |
| Side Marker Lights .....                         | Exterior Lights          |
| SIR Coil Assembly (Clockspring) .....            | Air Bag Restraint System |
| Slip Ring (Clockspring) .....                    | Air Bag Restraint System |
|                                                  | Steering Column Switches |
| SRS Control Module .....                         | Air Bag Restraint System |
| Starter Motor .....                              | Starters                 |
| Starter Interrupt Relay .....                    | Starters                 |
| Starter Solenoid .....                           | Starters                 |
| Starter Relay .....                              | Starters                 |
| Steering Wheel Position Sensor .....             | Anti-Lock Brakes         |
| Stoplights .....                                 | Exterior Lights          |
| Stoplight Switch .....                           | Engine Performance       |
|                                                  | Cruise Control Systems   |
|                                                  | Anti-Lock Brakes         |
| Sun Roof ECU .....                               | Power Sun Roof           |
| Sun Roof Motor .....                             | Power Sun Roof           |
| Sun Roof Position Sensor .....                   | Power Sun Roof           |
| Taillights .....                                 | Exterior Lights          |
| Throttle Position (TP) Sensor .....              | Engine Performance       |

|                                               |                          |
|-----------------------------------------------|--------------------------|
| Torque Converter Clutch Solenoid/Switch ..... | Engine Performance       |
| Traction Control Switch .....                 | Anti-Lock Brakes         |
| Trailer Tow Connector .....                   | Exterior Lights          |
| Trailer Tow Relay .....                       | Exterior Lights          |
| Transmission/Transaxle .....                  | Engine Performance       |
| Transmission Control Module (TCM) .....       | Engine Performance       |
|                                               | Starters                 |
| Transmission Range Sensor .....               | Starters                 |
|                                               | Back-Up Lights           |
|                                               | Engine Performance       |
| Transmission Range Switch .....               | Back-Up Lights           |
|                                               | Engine Performance       |
|                                               | Anti-Theft System        |
| Turn Signal Flasher .....                     | Exterior Lights          |
| Turn Signal Lights .....                      | Exterior Lights          |
| Twilight Sentinel Switch .....                | Headlight Systems        |
|                                               | Daytime Running Lights   |
| Vapor Canister Leak Detection Pump .....      | Engine Performance       |
| Vehicle Control Module (VCM) .....            | Engine Performance       |
| Vehicle Dynamic Module .....                  | Electronic Suspension    |
| Vehicle Speed Control Servo .....             | Cruise Control Systems   |
| Vehicle Speed Sensor .....                    | Data Link Connectors     |
|                                               | Analog Instrument Panels |
|                                               | Cruise Control Systems   |
|                                               | Electronic Suspension    |
| Voltage Regulator .....                       | Generators & Regulators  |
| Water-In-Fuel Sensor .....                    | Engine Performance       |
|                                               | Analog Instrument Panels |
| Wheel Speed Sensors .....                     | Anti-Lock Brakes         |
| Window Timer Module .....                     | Power Convertible Top    |
| Windshield Intermittent Wiper Relay .....     | Wiper/Washer Systems     |
| Windshield Washer Motor .....                 | Wiper/Washer Systems     |
| Wiper Motor .....                             | Wiper/Washer Systems     |

---

# INSTRUMENT PANEL

## 1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Instrument Panels

Celica

### **\* PLEASE READ THIS FIRST \***

**WARNING:** Celica is equipped with a driver-side air bag; use extreme caution while working around steering column. To disable air bag system, ensure ignition switch is in LOCK position and negative battery terminal is disconnected for at least one minute before attempting any repair. DO NOT apply electrical power to any instrument panel connector without disconnecting air bag control unit. Information labels are attached to air bag components. Follow all notices on labels. Use only DVOM (volt/ohmmeter) with minimum of 10-k/ohm impedance to check ANY circuit.

### **DESCRIPTION & OPERATION**

#### **GAUGES**

Standard instrument clusters contain fuel and temperature gauges with tell-tale warning lights. Some optional instrument panels are equipped with a tachometer, oil pressure gauge and voltmeter. Gauge internal operating components use either a 2-terminal bimetallic strip type, or a 3-terminal coil type. The 2-terminal type gauges are generally used on clusters without tachometers.

#### **SWITCHES**

Celica contains a hazard warning switch and cruise control main ON/OFF switch on instrument panel. Celica uses a combination switch for headlight, turn signal, wiper/washer, and cruise control switches. Combination switch is mounted on steering column. For testing and/or removal and installation procedures for combination switch components, see STEERING COLUMN SWITCHES article in the ACCESSORIES/SAFETY EQUIPMENT section.

### **TESTING - GAUGES**

#### **FUEL GAUGE & WARNING LIGHT**

Fuel Gauge And Wiring Harness Operational Test

1) Unplug fuel tank sending unit connector. Turn ignition on. If fuel gauge indicates EMPTY, go to next step. If fuel gauge does not indicate EMPTY, repair short circuit in wiring harness. See appropriate chassis WIRING DIAGRAMS article in WIRING DIAGRAMS section.

2) Connect a 12-volt, 3.4-watt test light between appropriate terminals of sending unit wiring harness connector. See FUEL GAUGE & HARNESS TEST table. See Fig. 1.

3) With ignition on, test light should flash and gauge needle should move toward FULL. If test light does not flash and gauge needle does not move, check wiring harness for open circuit. Repair or replace as necessary. If wiring harness checks good, replace fuel gauge.

FUEL GAUGE & HARNESS TEST TABLE

| Model  | Sending Unit Harness<br>Connector Terminals |
|--------|---------------------------------------------|
| Celica | 3 & 4                                       |



93F84553  
Fig. 1: Fuel Sending Unit Harness Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

FUEL SENDING UNIT TESTS

Fuel Sending Unit Resistance Test

- 1) Turn ignition off. Remove fuel sending unit from tank. Connect ohmmeter to appropriate sending unit terminals. See FUEL SENDING UNIT CONNECTOR TERMINALS table. Sending unit connector terminals are located opposite harness connector terminals. See Fig. 1.
- 2) Move sender arm and ensure resistance is within specifications. See FUEL SENDING UNIT RESISTANCE SPECIFICATIONS table. After a short delay, gauge pointer should move when sender is connected and float arm is moved.

FUEL SENDING UNIT CONNECTOR TERMINALS TABLE

| Model  | Sending Unit<br>Connector Terminals |
|--------|-------------------------------------|
| Celica | 3 & 4                               |

FUEL SENDING UNIT RESISTANCE SPECIFICATIONS TABLE

| Float Position | Ohms |
|----------------|------|
| Full           | 3    |
| Empty          | 110  |

Low Fuel Warning Light Sensor Operational Test

- 1) Remove fuel sending unit from gas tank. Prepare a battery to connect voltage to warning light sensor terminals of sending unit connector. See LOW FUEL WARNING LIGHT SENSOR TERMINALS table.



2) Connect a 12-volt, 3.4-watt test light between positive battery terminal and one warning light sensor terminal of sending unit connector. Connect other warning light sensor terminal to negative battery terminal.

3) With sending unit float/sensor dry, test light should come on within about 40 seconds. With sending unit float/sensor submerged in gasoline or water, test light should not come on. If test light does not function as described, reverse wire connections at battery terminals and retest. If test light still does not function as described, replace sensor or complete sending unit.

#### LOW FUEL WARNING LIGHT SENSOR TERMINALS TABLE

| Model               | Sending Unit<br>Connector Terminals |
|---------------------|-------------------------------------|
| 3S-GTE .....        | 2 & 4                               |
| 4A-FE & 5S-FE ..... | 1 & 4                               |

#### Fuel Gauge Resistance Test

Remove instrument cluster. Unplug cluster connector(s). Using an ohmmeter, check fuel gauge resistance by measuring across appropriate terminals. See Fig. 2. See FUEL GAUGE RESISTANCE SPECIFICATIONS table. Replace fuel gauge if not within specifications.

#### FUEL GAUGE RESISTANCE SPECIFICATIONS TABLE

| Application & Terminals   | Ohms |
|---------------------------|------|
| 3S-GTE                    |      |
| Terminals "A" & "B" ..... | 101  |
| Terminals "A" & "C" ..... | 252  |
| Terminals "B" & "C" ..... | 151  |
| 4A-FE & 5S-FE             |      |
| Terminals "A" & "B" ..... | 86   |
| Terminals "A" & "C" ..... | 274  |
| Terminals "B" & "C" ..... | 188  |

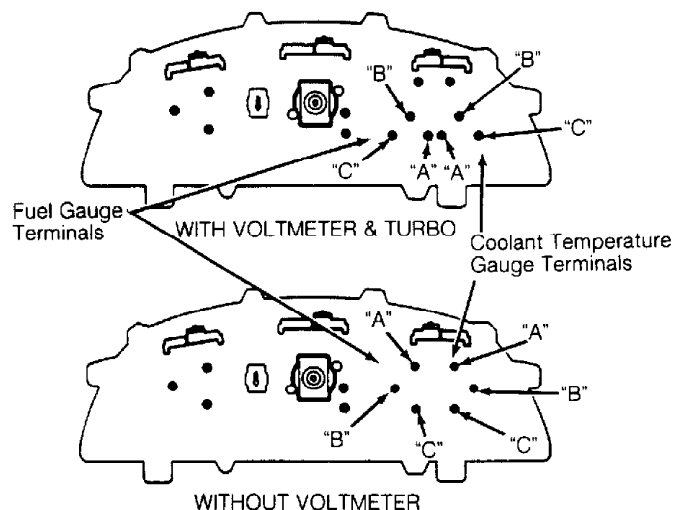


Fig. 2: Gauge Test Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### TEMPERATURE GAUGE & SENDER

#### Wiring Harness Operational Test

1) Unplug connector at coolant temperature sender. Turn ignition on. Temperature gauge should indicate COOL. Turn ignition off. Connect a 12-volt, 3.4-watt test light between coolant temperature sender harness connector terminal and ground.

2) Turn ignition on. Test light should glow and temperature gauge should slowly move to HOT. If gauge functions as described, replace sending unit. If gauge does not function as described, perform TEMPERATURE GAUGE RESISTANCE TEST.

#### Temperature Gauge Resistance Test

Remove instrument cluster. Using ohmmeter, check gauge resistance across appropriate terminals. See Fig. 2. Ensure ignition is off and harness connector is unplugged from instrument cluster. See TEMPERATURE GAUGE RESISTANCE SPECIFICATIONS table. Replace gauge if not within specification. If gauge is within specification, repair open or short circuit in wiring harness.

TEMPERATURE GAUGE RESISTANCE SPECIFICATIONS TABLE

| Application               | Ohms |
|---------------------------|------|
| Terminals "A" & "B" ..... | 182  |
| Terminals "A" & "C" ..... | 131  |
| Terminals "B" & "C" ..... | 51   |

## TACHOMETER TEST

Connect a tune-up test tachometer and start engine. Compare vehicle tachometer RPM reading against test tachometer. If vehicle tachometer reading is outside allowable range, replace tachometer. See TACHOMETER TEST table.

TACHOMETER TEST TABLE

| Vehicle RPM Reading | Allowable Range |
|---------------------|-----------------|
| 700 .....           | 610-750         |
| 3000 .....          | 2800-3200       |
| 5000 .....          | 4800-5200       |
| 7000 .....          | 6700-7300       |

## TURBO GAUGE

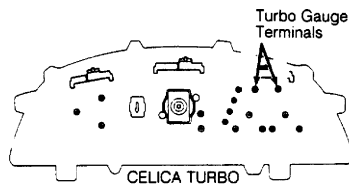
#### Turbo Gauge & Harness Operational Test

1) Disconnect turbo pressure sensor harness connector located on right rear of firewall. Turn ignition on. Turbo gauge needle should move to top of gauge.

2) Jump terminal No. 2 of turbo pressure sensor harness connector to ground. See Fig. 4. Turbo gauge needle should move to bottom of gauge. If gauge needle functions as described, test turbo pressure sensor and meter drive circuit. If gauge needle does not function as described, test turbo gauge resistance.

#### Turbo Gauge Resistance

Remove instrument cluster enough to leave harness connectors connected and still access turbo gauge terminals on rear of cluster. Using ohmmeter, check turbo gauge resistance. See Fig. 3. If resistance is not 72 ohms, replace turbo gauge. If turbo gauge resistance is 72 ohms, test turbo pressure sensor.



93A84566

Fig. 3: Turbo Gauge Test Terminal ID (Turbo)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Turbo Pressure Sensor (Voltage Test)

1) Connect three 1.5-volt dry cell batteries in series. Disconnect turbo pressure sensor connector located on right rear of firewall. Connect positive wire from batteries to terminal No. 3 of pressure sensor connector. See Fig. 4.

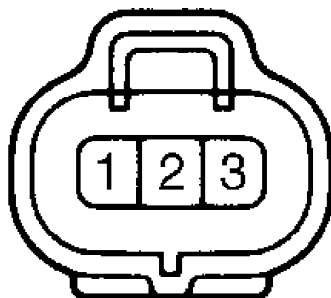
2) Connect negative wire of batteries to terminal No. 1 of pressure sensor connector. Using DVOM, check voltage between terminals No. 1 and 2. If voltage is not as specified, replace pressure sensor. See TURBO PRESSURE SENSOR VOLTAGE SPECIFICATIONS table. If voltage is within specification, go to next step.

#### TURBO PRESSURE SENSOR VOLTAGE SPECIFICATIONS TABLE

| Application | Approximate Voltage |
|-------------|---------------------|
| Turbo ..... | 2.4                 |

3) Apply 8 in. Hg (vacuum) to pressure sender. Check DVOM voltage reading to drop below specification. Using SST 09992-00241, apply 7 psi (.5 kg/cm<sup>2</sup>) air pressure to pressure sender. Ensure DVOM voltage reading increases slightly more than specification. See TURBO PRESSURE SENSOR VOLTAGE SPECIFICATIONS table.

4) If voltage readings do not change as described, replace turbo pressure sensor. If turbo pressure sensor tests okay, check wiring harness. Repair as necessary. For more information, see appropriate I - SYSTEM/COMPONENT TESTS article in the ENGINE PERFORMANCE section.



#### WIRING HARNESS CONNECTOR SIDE SHOWN

92A01603

Fig. 4: Turbo Pressure Sensor Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Turbo Meter Drive Circuit

1) Disconnect turbo pressure sensor connector. Remove

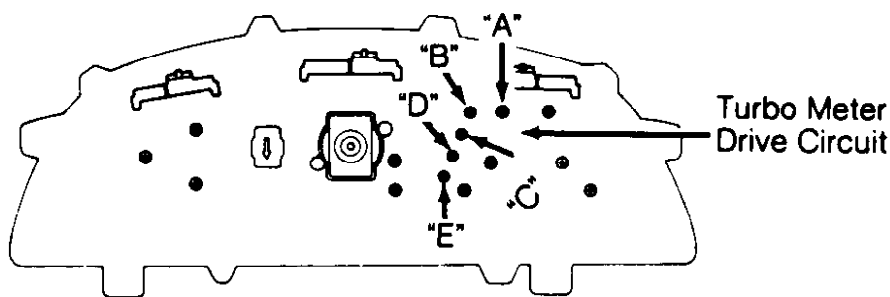
instrument cluster. See INSTRUMENT CLUSTER under REMOVAL & INSTALLATION. Inspect cluster wiring harness connectors/terminals for damage or poor connections.

2) Using DVOM, check turbo meter drive circuit terminals for correct values. See Fig. 5. See TURBO METER DRIVE CIRCUIT SPECIFICATIONS table. If values are not as specified, replace turbo meter drive circuit.

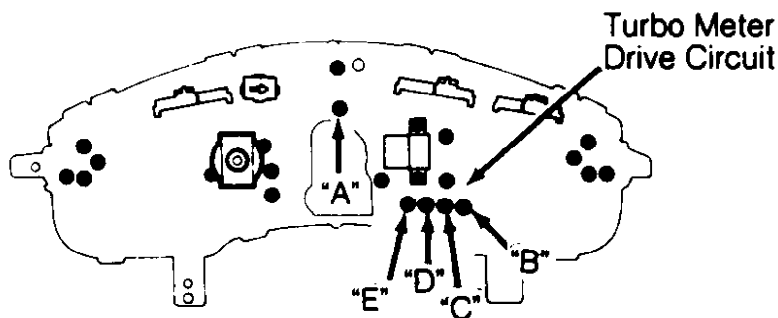
TURBO METER DRIVE CIRCUIT SPECIFICATIONS TABLE

| Connector-Terminal   | Ignition  | Specified Value |
|----------------------|-----------|-----------------|
| Resistance Check     |           |                 |
| "A" - "B" .....      | Off ..... | Continuity      |
| "C" - 2 (1) .....    | Off ..... | Continuity      |
| "D" - Ground .....   | Off ..... | Continuity      |
| 1 (1) - Ground ..... | Off ..... | Continuity      |
| Voltage Check        |           |                 |
| "E" - Ground .....   | Off ..... | Zero Volts      |
| "E" - Ground .....   | On .....  | Battery Voltage |
| 3 (1) - Ground ..... | Off ..... | Zero Volts      |
| 3 (1) - Ground ..... | On .....  | Battery Voltage |

(1) - Terminal number is located on turbo pressure sensor wiring harness connector. See Fig. 4.



CELICA TURBO



MR2 TURBO

92C01604

Fig. 5: Turbo Meter Drive Circuit Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TESTING - SWITCHES

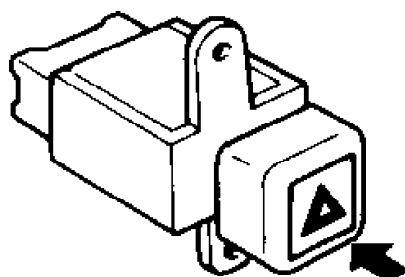
## HAZARD WARNING SWITCH

1) Ensure HAZARD-HORN and/or TURN fuses are good. Ensure flasher is good. See TURN SIGNAL FLASHER LOCATION table. With hazard warning switch removed from dash, ensure voltage is at switch wiring harness connector terminal No. 8. See Fig. 6. If voltage does not exist, check/repair fuses and wiring circuit. If voltage exists, disconnect switch from connector.

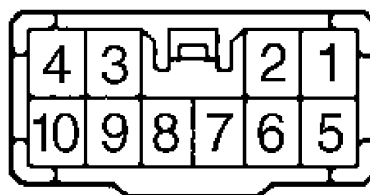
2) Using DVOM, ensure switch continuity exists between terminals listed, with switch in specified position. See HAZARD WARNING SWITCH CONTINUITY TEST table. Replace switch if continuity is not as specified. If switch is good, check wiring circuit. See appropriate chassis wiring diagram in WIRING DIAGRAMS.

TURN SIGNAL FLASHER LOCATION TABLE

| Model        | Location                                                           |
|--------------|--------------------------------------------------------------------|
| Celica ..... | Top Relay In Junction/Relay Block No. 1,<br>Behind Left Kick Panel |



93102705



Switch  
Terminals

Fig. 6: Hazard Warning Switch Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

HAZARD WARNING SWITCH CONTINUITY TEST TABLE (1)

| Switch<br>Position | Terminal<br>Numbers      | Continuity |
|--------------------|--------------------------|------------|
| Off .....          | 7 & 10 .....             | Yes        |
| On .....           | 4, 5, 6 & 9; 7 & 8 ..... | Yes        |

(1) - Terminals No. 2 and 3 are for switch illumination bulb.

## REMOVAL & INSTALLATION

\* PLEASE READ THIS FIRST \*

**WARNING:** Celica is equipped with a driver-side air bag; use extreme caution while servicing steering column. Ensure ignition switch is in LOCK position and negative battery terminal is disconnected for at least one minute before attempting any repair. DO NOT apply electrical power to any instrument panel connector without disconnecting air bag control unit. Information labels are attached to air bag components. Follow

all notices on labels. Use only DVOM (volt/ohmmeter) with minimum of 10-k/ohm impedance to check ANY circuit.

## HAZARD WARNING SWITCH

### Removal & Installation

Hazard warning switch is mounted to center air duct register, to left of clock. See Fig. 7. Using flat-blade screwdriver, pry top of register outward, away from dash opening and disconnect connectors. Remove switch from rear of register. To install, reverse removal procedure.

## INSTRUMENT CLUSTER

**CAUTION:** When removing steering wheel horn/pad, DO NOT pull on air bag wiring harness. When storing steering wheel horn/pad, ensure pad surface faces upward.

### Removal (With Air Bag)

1) Turn ignition switch to LOCK position. Disconnect negative battery cable. Wait at least one minute before continuing. Ensure front wheels are in straight-ahead position. Remove steering wheel. See STEERING WHEEL.

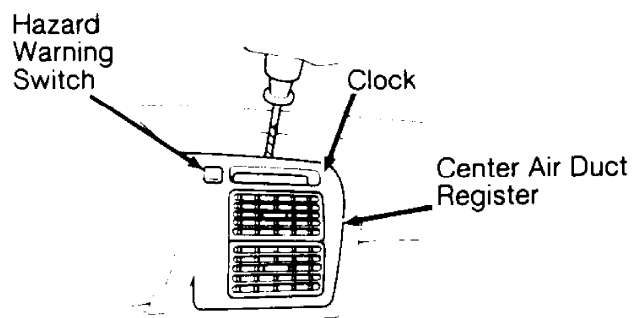
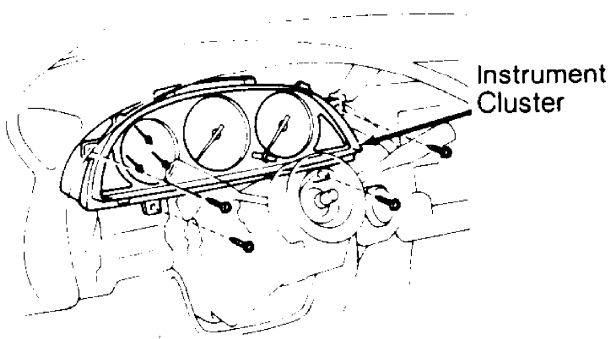
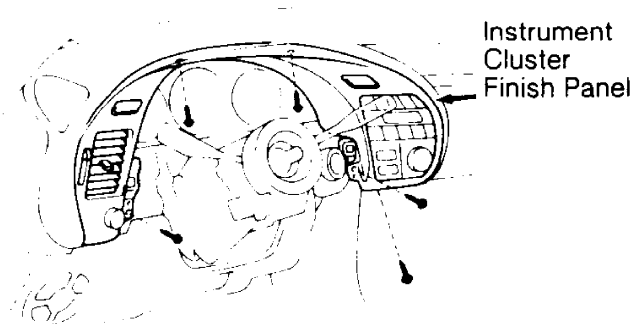
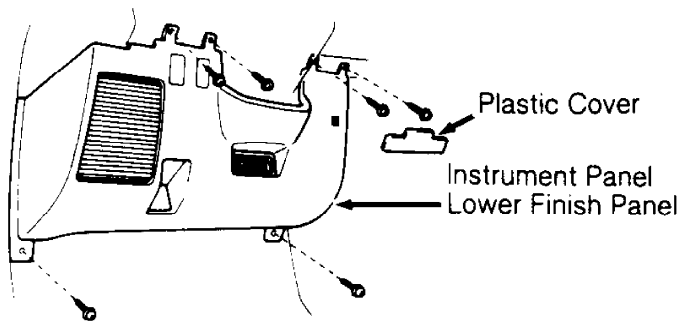
2) Remove steering column covers. Remove plastic screw covers from instrument panel lower finish panel and remove 6 retaining screws. See Fig. 7. Remove 5 bolts securing inner pad of lower finish panel and remove inner pad.

3) Pry off lower instrument cluster finish panel from around ignition key bezel and steering column. Remove 5 screws retaining instrument cluster finish panel. Pull cluster finish panel outward from heater controls and instrument cluster.

4) Remove instrument cluster retaining screws. Pull instrument cluster out far enough to disconnect harness connectors and speedometer cable (if equipped). Remove instrument cluster.

### Installation

To install, reverse removal procedure. Before installing steering wheel, center spiral cable.



93E02708

Fig. 7: Removing Instrument Cluster & Hazard Warning Switch  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

STEERING WHEEL

#### Removal (Without Air Bag)

1) On rear of steering wheel, locate and remove retaining screw securing horn pad. Pull horn pad out enough to disconnect horn electrical connector(s).

2) Remove steering wheel lock nut and washer. Make alignment mark on steering shaft and steering wheel for installation reference. Install suitable steering wheel puller and pull steering wheel from shaft.

#### Installation

Align reference marks on steering shaft and steering wheel. Tighten steering wheel lock nut to 25 ft. lbs. (34 N.m). Connect horn wiring and install horn pad.

#### Removal (With Air Bag)

1) Ensure front wheels are in straight-ahead position. Place ignition switch in LOCK position and remove key. Disconnect negative battery cable. Remove 2 screw covers from sides of steering wheel. See Fig. 8. Using Torx Wrench (T30), loosen 4 air bag pad retaining screws until screw head is snug against screw case.

2) Carefully pull air bag pad away from steering wheel enough to unlock and disconnect air bag electrical connector. DO NOT forcefully pull on electrical connector or wiring. Place air bag pad aside with pad facing upward.

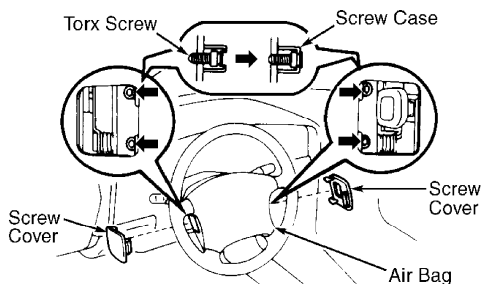
3) Remove steering wheel lock nut and washer from steering shaft. Make alignment mark on steering shaft and steering wheel for installation reference. Using appropriate steering wheel puller, pull steering wheel from shaft while guiding spiral cable wire through steering wheel opening.

#### Installation

1) Ensure front wheels are in straight-ahead position. Connect and install spiral cable to combination switch. Turn spiral cable counterclockwise by hand until it is hard to turn. Turn spiral cable clockwise about 2 1/2 turns and align Red mark at bottom with opening.

2) Guide spiral cable through steering wheel opening while installing steering wheel to shaft. Align reference marks on steering shaft and steering wheel. Tighten steering wheel lock nut to 25 ft. lbs. (34 N.m).

3) Connect air bag electrical connector and snap down connector lock. Ensure air bag pad Torx screws are retracted and snug against screw case. See Fig. 8. Install air bag pad to steering wheel ensuring wiring is not pinched and does not interfere with other moving parts. Tighten 4 Torx screws to 65 INCH lbs. (7.4 N.m). Install screw covers. Connect negative battery cable.



91G03959  
Fig. 8: Removing Air Bag From Steering Wheel  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS



Proceed to chassis WIRING DIAGRAMS article in WIRING DIAGRAMS  
section.

# MIRRORS - POWER

## 1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Power Mirrors

Camry, Celica, Corolla, Land Cruiser, MR2, Pickup,  
Previa, Supra, T100, 4Runner

### DESCRIPTION & OPERATION

Power mirrors are controlled by a dual control switch assembly, located on center console or instrument panel. Left/right control switch directs current to desired mirror. Horizontal/vertical control switch directs current to one of 2 motors located in mirror/motor assembly. Mirror and motors must be removed and serviced as an assembly.

### TROUBLE SHOOTING

#### POWER MIRROR

Both Power Mirrors Inoperative

Check RADIO fuse, power mirror switch, power mirror motor(s), wiring and ground. See TESTING.

Left Or Right Power Mirror Inoperative

Check power mirror switch, power mirror motor(s), wiring and ground. See TESTING.

Up/Down Or Left/Right Function Inoperative

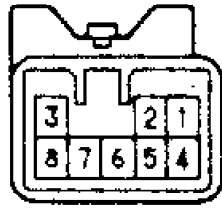
Check power mirror motor(s), wiring and ground. See POWER MIRROR MOTOR TEST under TESTING.

### TESTING

#### POWER MIRROR SWITCH TEST

NOTE: Mirror motor connectors may vary in configuration from vehicle to vehicle of the same model.

Remove power mirror switch. See POWER MIRROR SWITCH under REMOVAL & INSTALLATION. Using ohmmeter, check continuity between appropriate mirror switch terminals. See Figs. 1-6. See appropriate POWER MIRROR SWITCH CONTINUITY TEST table. If switch fails any test, replace switch.



MIRROR CONTROL SWITCH

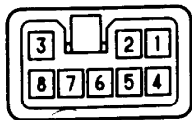


MIRROR MOTOR

Fig. 1: Power Mirror Switch & Motor Connector Terminal ID (Camry)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST (CAMRY)

| Application              | Terminals    |
|--------------------------|--------------|
| Switch In Left Position  |              |
| Up .....                 | 2 & 8; 3 & 5 |
| Down .....               | 2 & 5; 3 & 8 |
| Left .....               | 2 & 8; 5 & 6 |
| Right .....              | 2 & 5; 6 & 8 |
| Switch In Right Position |              |
| Up .....                 | 2 & 8; 5 & 7 |
| Down .....               | 2 & 5; 7 & 8 |
| Left .....               | 1 & 5; 2 & 8 |
| Right .....              | 1 & 8; 2 & 5 |



MIRROR CONTROL SWITCH



MIRROR MOTOR



93D02717

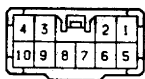
Fig. 2: Pwr. Mir. Sw. & Motor Conn. Term. ID (Celica, Corolla, Previa)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST TABLE (CELICA & PREVIA)

| Application              | Terminals    |
|--------------------------|--------------|
| Switch In Off Position   |              |
| Up .....                 | 2 & 3        |
| Down .....               | 1 & 2        |
| Left .....               | 2 & 3        |
| Right .....              | 1 & 2        |
| Switch In Left Position  |              |
| Up .....                 | 1 & 7; 2 & 3 |
| Down .....               | 1 & 2; 3 & 7 |
| Left .....               | 1 & 8; 2 & 3 |
| Right .....              | 1 & 2; 3 & 8 |
| Switch In Right Position |              |
| Up .....                 | 1 & 5; 2 & 3 |
| Down .....               | 1 & 2; 3 & 5 |
| Left .....               | 1 & 6; 2 & 3 |
| Right .....              | 1 & 2; 3 & 6 |

POWER MIRROR SWITCH CONTINUITY TEST TABLE (COROLLA)

| Application              | Terminals    |
|--------------------------|--------------|
| Switch In Left Position  |              |
| Up .....                 | 3 & 7; 4 & 8 |
| Down .....               | 3 & 4; 7 & 8 |
| Left .....               | 1 & 8; 3 & 7 |
| Right .....              | 1 & 3; 7 & 8 |
| Switch In Right Position |              |
| Up .....                 | 3 & 7; 6 & 8 |
| Down .....               | 3 & 6; 7 & 8 |
| Left .....               | 3 & 7; 5 & 8 |
| Right .....              | 3 & 5; 7 & 8 |



MIRROR CONTROL SWITCH



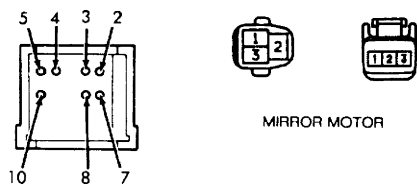
MIRROR MOTOR

93F02718

Fig. 3: Pwr. Mirror Sw. & Motor Connector Terminal ID (Land Cruiser)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST TABLE (LAND CRUISER)

| Application              | Terminals     |
|--------------------------|---------------|
| Switch In Off Position   |               |
| Up .....                 | 3 & 4         |
| Down .....               | 1 & 3         |
| Left .....               | 3 & 4         |
| Right .....              | 1 & 3         |
| Switch In Left Position  |               |
| Up .....                 | 1 & 10; 3 & 4 |
| Down .....               | 1 & 3; 4 & 10 |
| Left .....               | 1 & 9; 3 & 4  |
| Right .....              | 1 & 3; 4 & 9  |
| Switch In Right Position |               |
| Up .....                 | 1 & 6; 3 & 4  |
| Down .....               | 1 & 3; 4 & 6  |
| Left .....               | 1 & 2; 3 & 4  |
| Right .....              | 1 & 3; 2 & 4  |



MIRROR CONTROL SWITCH  
93H02719

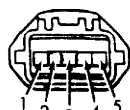
Fig. 4: Pwr. Mirror Sw. & Mtr. Conn. Term. ID (MR2, Pickup & 4Runner)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST TABLE (MR2, PICKUP & 4RUNNER)

| Application              | Terminals     |
|--------------------------|---------------|
| Switch In Off Position   |               |
| Up .....                 | 7 & 8         |
| Down .....               | 8 & 10        |
| Left .....               | 7 & 8         |
| Right .....              | 8 & 10        |
| Switch In Left Position  |               |
| Up .....                 | 2 & 10; 7 & 8 |
| Down .....               | 2 & 7; 8 & 10 |
| Left .....               | 3 & 10; 7 & 8 |
| Right .....              | 3 & 7; 8 & 10 |
| Switch In Right Position |               |
| Up .....                 | 5 & 10; 7 & 8 |
| Down .....               | 5 & 7; 8 & 10 |
| Left .....               | 4 & 10; 7 & 8 |
| Right .....              | 4 & 7; 8 & 10 |



MIRROR CONTROL SWITCH



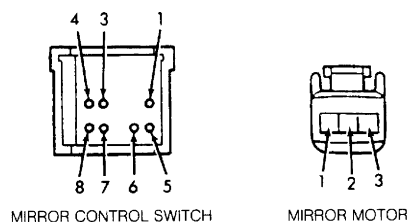
MIRROR MOTOR

93F84785

Fig. 5: Power Mirror Switch & Motor Connector Terminal ID (Supra)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST TABLE (SUPRA)

| Application              | Terminals    |
|--------------------------|--------------|
| Switch In Left Position  |              |
| Up .....                 | 2 & 5; 6 & 8 |
| Down .....               | 2 & 6; 5 & 8 |
| Left .....               | 1 & 8; 2 & 5 |
| Right .....              | 1 & 2; 5 & 8 |
| Switch In Right Position |              |
| Up .....                 | 2 & 5; 3 & 8 |
| Down .....               | 2 & 3; 5 & 8 |
| Left .....               | 2 & 5; 7 & 8 |
| Right .....              | 2 & 7; 5 & 8 |



93G84786  
 Fig. 6: Power Mirror Switch & Motor Connector Terminal ID (T100)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

POWER MIRROR SWITCH CONTINUITY TEST TABLE (T100)

| Application              | Terminals    |
|--------------------------|--------------|
| Switch In Left Position  |              |
| Up .....                 | 1 & 8; 3 & 4 |
| Down .....               | 1 & 3; 4 & 8 |
| Left .....               | 1 & 7; 3 & 4 |
| Right .....              | 1 & 3; 4 & 7 |
| Switch In Right Position |              |
| Up .....                 | 1 & 5; 3 & 4 |
| Down .....               | 1 & 3; 4 & 5 |
| Left .....               | 1 & 6; 3 & 4 |
| Right .....              | 1 & 3; 4 & 6 |

## POWER MIRROR MOTOR TEST

NOTE: Mirror motor connectors may vary in configuration from vehicle to vehicle of the same model.

Disconnect power mirror motor harness connector. Using 12-volt power source, connect positive and negative leads to specified pins or terminals. See appropriate POWER MIRROR MOTOR TEST table. See Figs. 1-6. If power mirror motor fails any test, replace power mirror assembly.

POWER MIRROR MOTOR TEST TABLE (CAMRY, CELICA & MR2)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 3 .....                      | 2 .....           | Up                  |
| 2 .....                      | 3 .....           | Down                |
| 1 .....                      | 2 .....           | Left                |
| 2 .....                      | 1 .....           | Right               |

POWER MIRROR MOTOR TEST TABLE (COROLLA)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 1 .....                      | 2 .....           | Up                  |
| 2 .....                      | 1 .....           | Down                |
| 3 .....                      | 2 .....           | Left                |
| 2 .....                      | 3 .....           | Right               |

POWER MIRROR MOTOR TEST TABLE (LAND CRUISER)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 2 .....                      | 3 .....           | Up                  |
| 3 .....                      | 2 .....           | Down                |
| 1 .....                      | 3 .....           | Left                |
| 3 .....                      | 1 .....           | Right               |

POWER MIRROR MOTOR TEST TABLE (PICKUP, PREVIA &amp; 4RUNNER)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 2 .....                      | 1 .....           | Up                  |
| 1 .....                      | 2 .....           | Down                |
| 3 .....                      | 1 .....           | Left                |
| 1 .....                      | 3 .....           | Right               |

POWER MIRROR MOTOR TEST TABLE (SUPRA)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 3 .....                      | 4 .....           | Up                  |
| 4 .....                      | 3 .....           | Down                |
| 5 .....                      | 4 .....           | Left                |
| 4 .....                      | 5 .....           | Right               |

POWER MIRROR MOTOR TEST TABLE (T100)

| Apply 12 Volts<br>To Pin No. | Ground<br>Pin No. | Mirror<br>Operation |
|------------------------------|-------------------|---------------------|
| 3 .....                      | 1 .....           | Up                  |
| 1 .....                      | 3 .....           | Down                |
| 2 .....                      | 1 .....           | Left                |
| 1 .....                      | 2 .....           | Right               |

## REMOVAL & INSTALLATION

### POWER MIRROR SWITCH

#### Removal & Installation (Celica & Supra)

Remove console box retaining screws and raise console box enough to disconnect mirror switch harness connector. Push on switch's connector end to remove switch from console box. To install, reverse removal procedure.

#### Removal & Installation (Camry, Corolla, Cressida & Land Cruiser)

Remove instrument panel lower trim cover and inner knee pad (if equipped). Remove retaining screws from bottom of power mirror switch. Using flat-blade screwdriver, carefully pry switch out of instrument panel. Disconnect harness connector. To install, reverse removal procedure.

Removal & Installation (Pickup, Previa, T100 & 4Runner)

Remove arm rest retaining screw caps, retaining screws and arm rest. Disconnect all electrical connectors. Remove power window master switch retaining screws. Remove power window master switch from arm rest. Push on power mirror switch connector end to remove switch from power window master switch. To install, reverse removal procedure.

## **POWER MIRROR ASSEMBLY**

Removal & Installation

Remove mirror inner trim panel on inside of the door. Disconnect power mirror motor harness connector. Support outside mirror assembly while removing 3 mirror retaining screws/bolts on inside of the door. To install, reverse removal procedure.

## **WIRING DIAGRAMS**

See appropriate chassis wiring diagram in WIRING DIAGRAMS section.



# J - PIN VOLTAGE CHARTS

## 1993 Toyota Celica

1993 ENGINE PERFORMANCE  
Toyota Pin Voltage Charts

Celica

### INTRODUCTION

Pin voltage charts are supplied to reduce diagnostic time. Checking pin voltages at the ECM determines whether it's receiving and transmitting proper voltage signals. Charts may also help determine if ECM harness is shorted or opened.

NOTE: Unless stated otherwise in testing procedures, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance. Voltage readings may vary slightly due to battery condition or charging rate.

### ECM LOCATION & IDENTIFICATION

ECM LOCATION TABLE

| Model  | Location                           |
|--------|------------------------------------|
| Celica | Bottom Center Of Dash, Under Radio |

ECM PIN CONNECTOR ABBREVIATION IDENTIFICATION TABLE

| Abbreviation      | Function                                               |
|-------------------|--------------------------------------------------------|
| ABS               | Anti-Lock Brake ECM                                    |
| ACA, ACT, ATS     | A/C Amplifier                                          |
| AC, AC1, A/C      | A/C Compressor Or Magnetic Switch                      |
| A/D               | Cruise Control Computer                                |
| AI                | Air Injection Vacuum Switching Valve                   |
| AS                | Air Suction Vacuum Switching Valve                     |
| BATT              | Battery                                                |
| BK                | Brake Switch                                           |
| B, +B, +B1        | EFI Main Relay                                         |
| CHK               | Sub-Oxygen Sensor                                      |
| C1                | Distributor                                            |
| DFG               | Defogger Relay                                         |
| DG                | Check Connector                                        |
| ECT               | Electronically Controlled Transmission/Transaxle (ECT) |
| EGR               | EGR Vacuum Switching Valve                             |
| ELS               | Headlight & Defogger Relay                             |
| E1, E2, E01, E02  | Engine Or Computer Ground                              |
| E11, E2, E21, E22 | Sensor Ground                                          |
| FC                | Circuit Opening Relay                                  |
| FP                | Fuel Pump Relay                                        |
| FPR               | Fuel Pump Relay                                        |
| FPU               | Fuel Pressure-Up Vacuum Switching Valve                |
| G, G1, G2         | Cam Position Sensor Or Distributor                     |
| G-, G+            | Cam Position Sensor Or Distributor (Crank Angle)       |
| HT                | Oxygen Sensor Heater                                   |
| HT1               | Oxygen Sensor Heater (Main)                            |
| HT2               | Oxygen Sensor Heater (Sub)                             |

|                        |       |                                                  |
|------------------------|-------|--------------------------------------------------|
| IDL                    | ..... | Throttle Position Sensor                         |
| IGDA, IGDB             | ..... | Ignitor                                          |
| IGF                    | ..... | Ignitor                                          |
| IG SW                  | ..... | Ignition Switch                                  |
| IGT                    | ..... | Ignitor (Primary Trigger)                        |
| ISCC, ISCO             | ..... | Idle Speed Control Valve                         |
| ISCV                   | ..... | A/C Idle-Up Vacuum Switching Valve               |
| ISC1, ISC2, ISC3, ISC4 | ..... | Idle Speed Control Valve                         |
|                        |       | Or Motor                                         |
| KNK, KNK1, KNK2        | ..... | Knock Sensor                                     |
| KS                     | ..... | Airflow Meter                                    |
| L                      | ..... | Neutral Start Switch Or Shift Position Switch    |
| LP                     | ..... | Headlight Relay                                  |
| L1, L2, L3, L4         | ..... | (1) ECT                                          |
| M                      | ..... | Neutral Start Switch Or Shift Position Switch    |
| M-REL                  | ..... | EFI Main Relay                                   |
| MS                     | ..... | Engine Oil Level Sensor                          |
| N                      | ..... | Neutral Start Switch Or Shift Position Switch    |
| N/C                    | ..... | Clutch Switch (M/T)                              |
| NE, NE-, NE+           | ..... | Cam Position Sensor Or Distributor (RPM)         |
| NSW                    | ..... | Neutral/Start Switch (A/T)                       |
| No. 1-4                | ..... | Injector                                         |
| No. 10, 20 & 30        | ..... | Injector                                         |
| OD, ODT                | ..... | Overdrive Solenoid                               |
| OD1                    | ..... | Overdrive Off Switch                             |
| OD2                    | ..... | Cruise Control Computer Or Overdrive Main Switch |
| O/G                    | ..... | Check Connector                                  |
| OIL                    | ..... | Oil Pressure Switch Or                           |
|                        |       | A/T Oil Temperature Warning Light                |
| OLS                    | ..... | Engine Oil Level Sensor                          |
| OMR                    | ..... | Engine Oil Feed Motor                            |
| OMT                    | ..... | Engine Oil Feed Motor Relay                      |
| OSFC                   | ..... | Overspeed Fuel Cut                               |
| OW                     | ..... | Engine Oil Warning Light                         |
| OX, OX1, OX+           | ..... | Oxygen Sensor                                    |
| OX2                    | ..... | Sub-Oxygen Sensor                                |
| P                      | ..... | Shift Position Switch Or Pattern Select Switch   |
| PS                     | ..... | Power Steering ECM                               |
| PSCT                   | ..... | Power Steering ECM                               |
| PSW                    | ..... | Throttle Position Sensor                         |
| PIM                    | ..... | Vacuum Sensor Or Turbocharger Pressure Sensor    |
| PSW                    | ..... | Throttle Position Sensor                         |
| PWR                    | ..... | Pattern Select Switch                            |
| R                      | ..... | Neutral Start Switch Or Shift Position Switch    |
| RSC                    | ..... | Idle Speed Control Valve                         |
| RSO                    | ..... | Idle Speed Control Valve                         |
| S                      | ..... | Neutral Start Switch Or Shift Position Switch    |
| SL, S1, S2             | ..... | (1) ECT Solenoid                                 |
| SPD, SPD1, SPD2        | ..... | Vehicle Speed Sensor                             |
| SP1, SPD1              | ..... | No. 1 Vehicle Speed Sensor, (1) ECT              |
|                        |       | In Instrument Cluster                            |
| SP2, SPD2              | ..... | No. 2 Vehicle Speed Sensor, (1) ECT              |
|                        |       | In Transaxle or Transmission                     |
| STA                    | ..... | Starter Signal                                   |
| STP                    | ..... | Stoplight Switch                                 |
| STJ                    | ..... | Cold Start Injector                              |
| S1, S2, S3, S4         | ..... | (1) ECT Solenoid                                 |
| T, TT                  | ..... | Check Connector Or (2) DLC                       |
| THA, THA1              | ..... | Air Temperature Sensor                           |
| THE                    | ..... | Evaporator Temperature Sensor                    |
| THG                    | ..... | EGR Gas Temperature Sensor                       |
| THO, THO1, THO2        | ..... | Transmission Oil Temperature Sensor              |
| THW                    | ..... | Coolant Temperature Sensor                       |

|              |       |                                                  |
|--------------|-------|--------------------------------------------------|
| TIL          | ..... | Turbo Indicator Light                            |
| TOV          | ..... | Throttle Opener Vacuum Switching Valve           |
| TPC, TPC1    | ..... | Turbocharger Pressure                            |
|              |       | Vacuum Switching Valve                           |
| TVIS         | ....  | Variable Induction System Vacuum Switching Valve |
| TWS          | ..... | Coolant Temperature Switch                       |
| T2           | ..... | Check Connector                                  |
| VC           | ..... | Airflow Meter, Vacuum Sensor Or                  |
|              |       | Throttle Position Sensor                         |
| VCC          | ..... | Throttle Position Sensor Or Vacuum Sensor        |
| VF, VF1, VF2 | ..... | Check Connector                                  |
| V-ISC        | ..... | Auxiliary Control Valve Or                       |
|              |       | Idle-Up System Vacuum Switching Valve            |
| VS           | ..... | Airflow Meter                                    |
| VSV1         | ..... | Air Control Vacuum Switching Valve               |
| VSV2         | ..... | Fuel Pressure-Up Vacuum Switching Valve          |
| VTA          | ..... | Throttle Position Sensor                         |
| W            | ..... | Malfunction Indicator Light                      |
| 2            | ..... | Neutral Start Switch Or Shift Position Switch    |

(1) - Electronically Controlled Transmission/Transaxle (ECT).  
(2) - Data Link Connector (DLC).

---

## PIN VOLTAGE CHARTS

### ECM PIN VOLTAGE TEST

CAUTION: Perform all voltage measurements with ECM harness connector installed. Use high-impedance digital voltmeter (10,000 ohm minimum). Verify battery voltage is greater than 11 volts.

NOTE: If voltage value is not as specified, proceed to test number indicated. Tests are located in G - TESTS W/CODES article in the ENGINE PERFORMANCE section.

| No. | Terminals                                | Condition                                                        |                                                                          | STD voltage (V) | TEST NO. |
|-----|------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------|----------|
| 1   | +B - E1<br>+B1                           | IG SW ON                                                         |                                                                          | 10 - 14         | ②        |
| 2   | BATT - E1                                | -                                                                |                                                                          | 10 - 14         | ①        |
| 3   | IDL - E2                                 | IG SW ON                                                         | Throttle valve open                                                      | 4.5 - 5.5       | ⑨        |
|     | VC - E2                                  |                                                                  | -                                                                        | 4.5 - 5.5       |          |
|     | VTA - E2                                 |                                                                  | Throttle valve fully closed<br>(Throttle opener must be cancelled first) | 0.1 - 1.0       |          |
|     |                                          |                                                                  | Throttle valve fully open                                                | 3.2 - 4.2       |          |
| 4   | VC - E2                                  | VS - E2                                                          | -                                                                        | 4.5 - 5.5       | ⑦        |
|     |                                          |                                                                  | Measuring plate fully closed                                             | 3.7 - 4.3       |          |
|     |                                          |                                                                  | Measuring plate fully open                                               | 0.2 - 0.5       |          |
|     |                                          |                                                                  | Idling                                                                   | 1.6 - 4.1       |          |
|     |                                          | 3,000 rpm                                                        | 1.0 - 2.0                                                                |                 |          |
| 5   | No.1<br>No.2 - E01<br>No.3 - E02<br>No.4 | IG SW ON                                                         |                                                                          | 10 - 14         | ⑬        |
| 6   | THA - E2                                 | IG SW ON                                                         | Intake air temp. 20°C (68°F)                                             | 1 - 3           | ⑥        |
| 7   | THW - E2                                 |                                                                  | Coolant temp. 80°C (176°F)                                               | 0.1 - 1.1       | ⑤        |
| 8   | STA - E1                                 | Cranking                                                         |                                                                          | 6 - 14          | ⑩        |
| 9   | IGT - E1                                 | Cranking or idling                                               |                                                                          | 0.8 - 1.2       | ③        |
| 10  | RSC - E1<br>RSO                          | IG SW ON                                                         | Engine ECU connectors disconnected                                       | 8 - 14          | ⑭        |
| 11  | W - E1                                   | No trouble ("CHECK" engine warning light off) and engine running |                                                                          | 10 - 14         | ⑮        |
| 12  | PIM - E2                                 | IG SW ON                                                         |                                                                          | 2.5 - 4.5       | ⑧        |
|     | VC - E2                                  |                                                                  |                                                                          | 4.5 - 5.5       |          |
| 13  | ACJ - E1                                 | IG SW ON                                                         | Air conditioning ON                                                      | 8 - 14          | ⑪        |

92G25839

Fig. 1: ECM Pin Voltage Test (3S-GTE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

| No. | Terminals              | Condition                                                        |                              | STD voltage (V) | TEST NO. |
|-----|------------------------|------------------------------------------------------------------|------------------------------|-----------------|----------|
| 1   | +B<br>+B1 - E1         | IG SW ON                                                         |                              | 10 - 14         | ②        |
| 2   | BATT - E1              | -                                                                |                              | 10 - 14         | ①        |
| 3   | IDL - E2               | IG SW ON                                                         | Throttle valve open          | 10 - 14         | ⑨        |
|     | PSW - E2               |                                                                  | Throttle valve fully closed  | 10 - 14         |          |
| 4   | PIM - E2               | IG SW ON                                                         |                              | 3.3 - 3.9       | ⑦        |
|     | VCC - E2               |                                                                  |                              | 4.5 - 5.5       |          |
| 5   | No.10 E01<br>No.20 E02 |                                                                  |                              | 10 - 14         | ⑬        |
| 6   | THA - E2               | IG SW ON                                                         | Intake air temp. 20°C (68°F) | 1 - 3           | ⑥        |
| 7   | THW - E2               |                                                                  | Coolant temp. 80°C (176°F)   | 2.0 - 2.8       | ⑤        |
| 8   | STA - E1               | Cranking                                                         |                              | 6 - 14          | ⑩        |
| 9   | IGT - E1               | Cranking or idling                                               |                              | 0.7 - 1.0       | ③        |
| 10  | W - E1                 | No trouble ("CHECK" engine warning light off) and engine running |                              | 10 - 14         | ⑮        |
| 11  | A/C - E1               | IG SW ON                                                         | Air conditioning ON          | 8 - 14          | ⑪        |

92A25841

Fig. 2: ECM Pin Voltage Test (4A-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

| No. | Terminals                  | Condition                                                        |                                                                          | STD voltage (V) | TEST NO. |
|-----|----------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------|----------|
| 1   | +B<br>+B1 - E1             | IG SW ON                                                         |                                                                          | 10 - 14         | ②        |
| 2   | BATT - E1                  | —                                                                |                                                                          | 10 - 14         | ①        |
| 3   | IDL - E2                   | IG SW ON                                                         | Throttle valve open                                                      | 8 - 14          | ⑨        |
|     | VC - E2                    |                                                                  | —                                                                        | 4.5 - 5.5       |          |
|     | VTA - E2                   |                                                                  | Throttle valve fully closed<br>(Throttle opener must be cancelled first) | 0.8 - 1.2       |          |
|     |                            |                                                                  | Throttle valve fully open                                                | 3.2 - 4.2       |          |
| 4   | PIM - E2                   | IG SW ON                                                         |                                                                          | 3.3 - 3.9       | ⑦        |
|     | VC - E2                    |                                                                  |                                                                          | 4.5 - 5.5       |          |
| 5   | No.10 - E01<br>No.20 - E02 |                                                                  |                                                                          |                 | 10 - 14  |
| 6   | THA - E2                   | IG SW ON                                                         | Intake air temp. 20°C (68°F)                                             | 1.9 - 2.9       | ⑥        |
| 7   | THW - E2                   |                                                                  | Coolant temp. 80°C (176°F)                                               | 0.1 - 1.1       | ⑤        |
| 8   | STA - E1                   | Cranking                                                         |                                                                          | 6 - 14          | ⑩        |
| 9   | IGT - E1                   | Cranking or idling                                               |                                                                          | 0.8 - 1.2       | ③        |
| 10  | ISCC - E1<br>ISCO - E1     | IG SW ON                                                         | Engine ECU connectors disconnected                                       | 8 - 14          | ⑭        |
| 11  | W - E1                     | No trouble ("CHECK" engine warning light off) and engine running |                                                                          | 10 - 14         | ⑮        |

92C25843

Fig. 3: ECM Pin Voltage Test (5S-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### ECM WIRING HARNESS RESISTANCE TEST

CAUTION: When measuring resistance at ECM wiring harness, DO NOT touch ECM terminals with ohmmeter. Turn ignition off and unplug ECM wiring harness connector. Tester probe should be inserted into the wiring connector from the WIRING side.

| Terminals              | Condition                    | STD resistance ( $\Omega$ ) |
|------------------------|------------------------------|-----------------------------|
| IDL - E2               | Throttle valve open          | Infinity                    |
|                        | Throttle valve fully closed  | 2,300 or less               |
| VTA - E2               | Throttle valve fully open    | 3,100 - 12,100              |
|                        | Throttle valve fully closed  | 470 - 6,100                 |
| VC - E2                | -                            | 3,900 - 9,000               |
| VS - E2                | Measuring plate fully closed | 200 - 600                   |
|                        | Measuring plate fully open   | 20 - 1,200                  |
| THA - E2               | Intake air temp. 20°C (68°F) | 2,000 - 3,000               |
| THW - E2               | Coolant temp. 80°C (176°F)   | 200 - 400                   |
| G1<br>G2 - G $\ominus$ | Cold                         | 125 - 190                   |
| NE - G $\ominus$       | Cold                         | 155 - 240                   |
| RSC +B<br>RSO - +B1    | -                            | 19.3 - 22.3                 |

92J25840

Fig. 4: ECM Wiring Harness Resistance Test (3S-GTE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

| Terminals              | Condition                          | STD resistance ( $\Omega$ ) |
|------------------------|------------------------------------|-----------------------------|
| IDL - E2               | Throttle valve open                | Infinity                    |
|                        | Throttle valve fully closed        | 0                           |
| PSW - E2               | Throttle valve fully open          | 0                           |
|                        | Throttle valve fully closed        | Infinity                    |
| THA - E2               | Intake air temperature 20°C (68°F) | 2,000 - 3,000               |
| THW - E2               | Coolant temperature 80°C (176°F)   | 200 - 400                   |
| G1<br>NE - G $\ominus$ | Cold                               | 185 - 265                   |

92B25842

Fig. 5: ECM Wiring Harness Resistance Test (4A-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### Resistance of ECU Wiring Connectors (5S-FE)

| Terminals           | Condition                                                                | STD resistance ( $\Omega$ ) |
|---------------------|--------------------------------------------------------------------------|-----------------------------|
| IDL - E2            | Throttle valve open                                                      | Infinity                    |
|                     | Throttle valve fully closed<br>(Throttle opener must be cancelled first) | 2,300 or less               |
| VTA - E2            | Throttle valve fully open                                                | 2,000 - 10,200              |
|                     | Throttle valve fully closed<br>(Throttle opener must be cancelled first) | 200 - 5,700                 |
| VC - E2             | —                                                                        | 2,500 - 5,900               |
| THA - E2            | Intake air temp. 20°C (68°F)                                             | 2,000 - 3,000               |
| THW - E2            | Coolant temp. 80°C (176°F)                                               | 200 - 400                   |
| G+ - G-             | Cold                                                                     | 185 - 265                   |
| NE+ - NE-           | Cold                                                                     | 370 - 530                   |
| ISCC +B<br>ISCO +B1 | —                                                                        | 19.3 - 22.3                 |

92D25844

Fig. 6: ECM Wiring Harness Resistance Test (5S-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### ECM TERMINAL IDENTIFICATION

|     |       |       |     |    |     |     |    |    |     |     |      |     |     |     |     |     |     |    |     |    |     |     |     |   |     |     |      |
|-----|-------|-------|-----|----|-----|-----|----|----|-----|-----|------|-----|-----|-----|-----|-----|-----|----|-----|----|-----|-----|-----|---|-----|-----|------|
| E01 | No. 1 | RSO   | RSC | HT | STJ | EGR | G2 | NE | IGF | TPC | TVIS | VF  | OX  | PIM | THW | THA | VS  | VC | STA | AC | SPD | ATS | FPR | W | STP | ELS | BATT |
| E02 | No. 3 | No. 4 |     |    | IGT |     | G1 | G⊖ |     |     | E1   | TE1 | TE2 | KNK | IDL | VTA | THG | E2 | ACT |    |     |     |     |   | FC  | +B1 | +B   |

93D79304

Fig. 7: ECM Terminal ID (3S-GTE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

|     |        |     |    |     |    |     |     |     |     |     |                |           |    |     |  |  |         |  |           |
|-----|--------|-----|----|-----|----|-----|-----|-----|-----|-----|----------------|-----------|----|-----|--|--|---------|--|-----------|
| E01 | No. 10 | STA | OX | G ⊖ | G  | IGF | IGT | THA | PIM | THW | NSW            | EGR       | T  | ACT |  |  | FC      |  | BATT + B1 |
| E02 | No. 20 | E1  |    | E21 | NE | THG | IDL | VCC | PSW | E2  | OD<br>or<br>HT | V-<br>ISC | VF | ODT |  |  | SPD A/C |  | W + B     |

92D25810

Fig. 8: ECM Terminal ID (4A-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



|     |        |        |      |      |      |   |      |      |     |    |    |     |     |     |     |     |     |     |    |     |     |     |     |   |     |     |     |      |
|-----|--------|--------|------|------|------|---|------|------|-----|----|----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|---|-----|-----|-----|------|
| E01 | No. 10 | No. 20 | ISCO | ISCC | ISCV | 2 | NE - | NE + | IGF | S1 | SL | VF  | TT  | OX1 | OX2 | THW | THA | PIM | VC | STA | SPD | ACA | OD2 | W | B/K | THE | ELS | BATT |
| E02 |        |        | EGR  | P    | IGT  | L | G +  | G -  | SP2 | S2 | E1 | E21 | TE1 | TE2 | KNK | IDL | VTA | THG | E2 | NSW | ACT | OD1 |     |   | ATS | FC  | +B1 | +B   |

92E25811

Fig. 9: ECM Terminal ID (5S-FE Automatic Transaxle)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

|     |        |      |      |      |     |      |     |     |      |     |     |     |     |     |     |     |     |     |     |     |      |     |
|-----|--------|------|------|------|-----|------|-----|-----|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|-----|
| E01 | No. 10 | ISCO | ISCC | ISCV | G + | NE + | IGF | STA | ISCV | VF  | TE2 | OX1 | KNK | THW | THA | PIM | OX2 | ACT | FC  | ACA | BATT | +B1 |
| E02 | No. 20 | E1   | EGR  | IGT  | G - | NE - |     |     |      | E21 | TE1 | THE | THG | IDL | VC  | VTA | E2  | ELS | SPD | ATS | W    | +B  |

92F25812

Fig. 10: ECM Terminal (5S-FE Manual Transaxle)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

# **POWER WINDOWS**

## **1993 Toyota Celica**

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Power Windows

Celica

### **DESCRIPTION & OPERATION**

System components consist of a power window relay, power window switches and power window motors for each door. With ignition switch in ON position, battery voltage is supplied through power window relay to power window switches. Power window switch supplies power and ground for power window motors.

Driver's side power window switch offers one-touch operation of driver's side window. Driver's side power window switch also includes a lock-out feature to prevent passengers from operating any of the other power window switches.

### **TROUBLE SHOOTING**

#### **ALL WINDOWS INOPERATIVE & LOCK-OUT IS INOPERATIVE**

Check for defective fuse (ALT, AM1, DOOR, GAUGE, POWER), ignition switch, door lock control relay, driver's side power window switch and wiring circuits.

#### **ALL WINDOWS INOPERATIVE & LOCK-OUT OPERATES NORMAL**

Check for defective fuse (GAUGE, POWER), ignition switch, door lock control relay, driver's side power window switch or power main relay.

#### **LOCK SWITCH INOPERATIVE**

Check driver's side power window lock switch.

#### **WINDOW LOCK ILLUMINATION DOES NOT OPERATE**

Check driver's side power window switch.

#### **ONE-TOUCH FEATURE IS INOPERATIVE**

Check driver's side power window switch.

#### **ONE WINDOW DOES NOT OPERATE**

Check driver's side power window switch for appropriate window, power window switch, power window motor and wiring circuits for appropriate window.

NOTE: For door lock control relay testing, see DOOR LOCKS - POWER article in the ACCESSORIES/SAFETY EQUIPMENT section. For ignition switch testing, see STEERING COLUMN SWITCHES article in the ACCESSORIES/SAFETY EQUIPMENT section.

### **TESTING**

## POWER WINDOW SWITCH ILLUMINATION TEST (DRIVER'S SIDE SWITCH)

NOTE: Illumination test is for all switches at driver's side power window switch.

1) Using a 12 volt battery, connect jumper wire from positive battery terminal to driver's side power window switch connector terminal No. 6. See Fig. 1. Connect jumper wire from negative battery terminal to connector terminal No. 3. Ensure power window switches illuminate.

2) Put power window lock switch in lock position. Ensure passenger side power window switch does not illuminate. If illumination is not as described, replace driver's side power window switch.

## POWER WINDOW SYSTEM CURRENT TEST

1) Disconnect connector from driver's side power window switch. Connect positive lead of ammeter to driver's side power window switch connector (harness side) terminal No. 5.

2) Connect ammeter negative lead to negative terminal of 12 volt battery. Connect jumper wire from positive battery terminal to driver's side power window switch connector terminal No. 1. See Fig. 1.

3) While lowering driver's side window, current should be about 7 amperes. When window operation is stopped, current should increase to about 14.5 amperes or more. If current is as described, replace driver's side power window switch. If current is not as described, check and repair appropriate circuit.

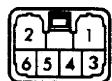
## POWER WINDOW SWITCH CONTINUITY TEST (DRIVER'S SIDE SWITCH)

Using an ohmmeter, check continuity between specified terminals with driver's side power window switch in specified position. See POWER WINDOW SWITCH CONTINUITY TEST table. See Fig. 1. If continuity does not exist at specified terminals, replace power window switch.

### POWER WINDOW SWITCH CONTINUITY TEST TABLE

| Application<br>& Position | (1) Terminal No. |
|---------------------------|------------------|
| Driver's Side Switch      |                  |
| Locked & Unlocked         |                  |
| UP                        | 1, 3, 5 & 6      |
| OFF                       | 1 & 3; 3 & 5     |
| DOWN                      | 1 & 6; 3 & 5     |
| Passenger Side Switch     |                  |
| Locked                    |                  |
| UP                        | 2 & 6            |
| OFF                       | 2 & 4            |
| DOWN                      | 4 & 6            |
| Unlocked                  |                  |
| UP                        | 2 & 6; 3 & 4     |
| OFF                       | 2 & 3; 3 & 4     |
| DOWN                      | 2 & 3; 4 & 6     |

(1) - See Fig. 1.



CELICA & PREVIA

93H84845

Fig. 1: Pwr. Win. Sw. Conn. Term. ID (Drvrs' Side Sw. - Harn Side)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## POWER WINDOW SW CONTINUITY TEST (PASSENGER SIDE & REAR SW'S)

Using an ohmmeter, check continuity between specified terminals with power window switch in specified position. See POWER WINDOW SWITCH CONTINUITY TEST (PASSENGER SIDE & REAR) table. See Fig. 2. If continuity does not exist at specified terminals, replace appropriate power window switch.

PWR WINDOW SW CONTINUITY TEST (PASSENGER SIDE & REAR) TABLE

| Application<br>& Position | (1) Terminal No. |
|---------------------------|------------------|
| UP .....                  | 1 & 2; 4 & 5     |
| OFF .....                 | 1 & 2; 3 & 4     |
| DOWN .....                | 1 & 5; 3 & 4     |

(1) - See Fig. 2.



CELICA

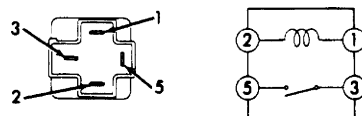
93J84847

Fig. 2: Pwr. Win. Conn. Term. ID (Pass. Side & Rear Sw's Harn. Side)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## POWER WINDOW MAIN RELAY

1) Remove power window main relay, located in junction block No. 2 at left side kick panel.

2) Using an ohmmeter, check continuity between terminals No. 1 and 2. Continuity should exist. See Fig. 3. Apply battery voltage to terminal No. 1 and ground terminal No. 2. Continuity should exist at terminals No. 3 and 5. If relay does not test as described, replace power window main relay.



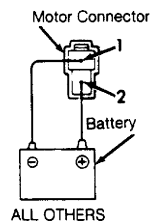
CAMRY, CELICA, COROLLA & SUPRA

93F84850

Fig. 3: Power Window Main Relay Terminal ID  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## POWER WINDOW MOTOR TEST

Using a 12 volt battery, connect jumper wire from positive battery terminal to terminal No. 2 of power window motor. See Fig. 4. Connect jumper wire from negative battery terminal to terminal No. 1 of power window motor. Motor should operate clockwise. Reverse jumper wire positions. Motor should operate counterclockwise. If motor does not test as described, replace motor.



93J84854

Fig. 4: Power Window Motor Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A. Inc.

## REMOVAL & INSTALLATION

### POWER WINDOW MOTOR

#### Removal & Installation

Remove door trim panel and waterproof shield. Remove glass retaining bolts and glass. Remove window regulator nuts and remove window regulator. Remove power window motor retaining screws and remove motor from window regulator. To install, reverse removal procedure.

### POWER WINDOW SWITCH

#### Removal & Installation

Disconnect negative battery cable. Pry out power window switch from door panel using flat screwdriver. Disconnect power window switch connectors and remove switch. To install, reverse removal procedure.

## WIRING DIAGRAMS

Proceed to chassis WIRING DIAGRAMS article in WIRING DIAGRAMS section.

# SEATS - POWER

## 1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Power Seats

Camry, Celica, Land Cruiser, Supra

### DESCRIPTION & OPERATION

#### CAMRY

Driver's seat cushion moves 6 ways (front up and down, rear up and down and slide forward and backward). See Fig. 1. Seat back reclines forward and backward. Switch controls power and ground circuits of power seat motors. Circuit breaker in power seat motor protects circuit.

#### CELICA

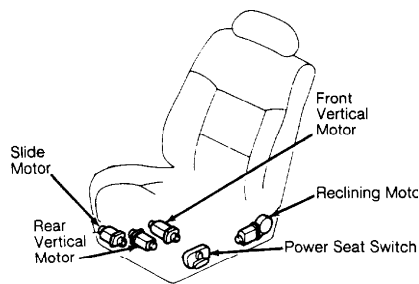
Driver's seat is equipped with an adjustable lumbar (lower back) support and side supports. See Fig. 2. System receives power through power main relay (near battery, in relay block No. 2). Switch controls power and ground circuits of power seat motors. Circuit breaker in power seat motor protects circuit.

#### LAND CRUISER

Both front seats move 6 ways (front up and down, rear up and down and slide forward and backward) and seat backs recline forward and backward. Driver's seat is also equipped with an adjustable lumbar (lower back) support. See Fig. 3. Switch controls power and ground circuits of power seat motors. Circuit breaker in power seat motor protects circuit.

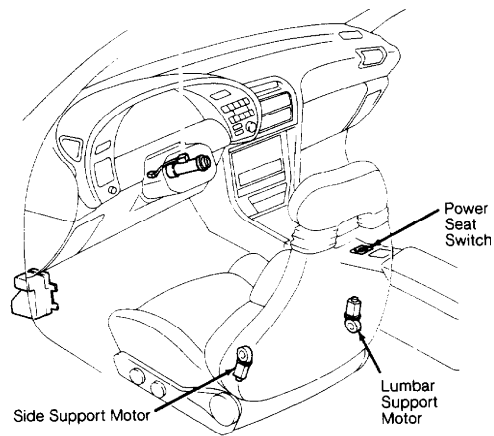
#### SUPRA

Driver's seat slides forward and backward and reclines forward and backward. Switch controls power and ground circuits of power seat motors. See Fig. 4. Circuit breaker in power seat motor protects circuit.



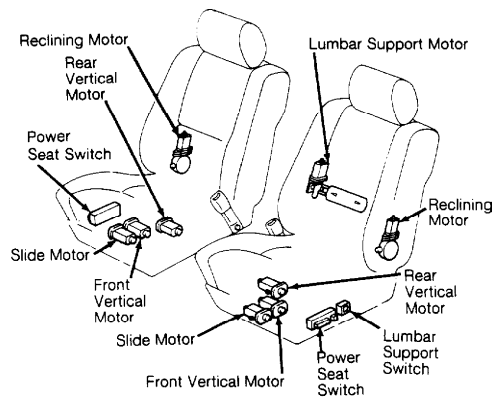
93184788

Fig. 1: Locating Power Seat Components (Camry)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



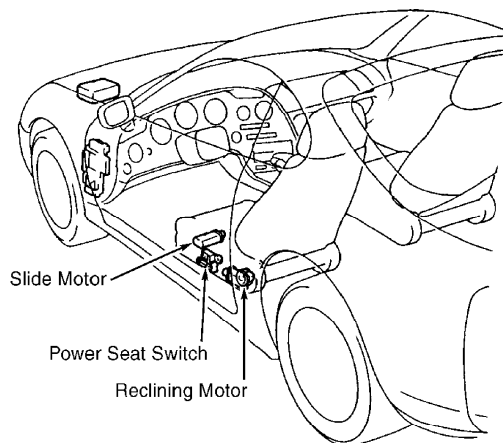
93J84789

Fig. 2: Locating Power Seat Components (Celica)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93C84790

Fig. 3: Locating Power Seat Components (Land Cruiser)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93D84791

Fig. 4: Locating Power Seat Components (Supra)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TROUBLE SHOOTING

### CAMRY & SUPRA (SEAT DOES NOT MOVE IN ANY DIRECTION)

Check fuse, power seat switch and wiring harness.

### CAMRY & SUPRA (SEAT MOVES IN AT LEAST ONE DIRECTION)

Check power seat switch. Check appropriate power seat motor. Check wiring harness.

### CELICA (LUMBAR SEAT DOES NOT MOVE IN ANY DIRECTION)

Check fuses and power main relay. Check power seat switch and wiring harness.

### CELICA (LUMBAR SEAT MOVES IN AT LEAST ONE DIRECTION)

Check movement of appropriate power seat motor. Check circuit breaker of appropriate power seat motor. Check power seat switch and wiring harness.

### LAND CRUISER (LUMBAR SEAT DOES NOT MOVE IN ANY DIRECTION)

Check fuses and power main relay. Check power seat switch and wiring harness.

### LAND CRUISER (LUMBAR SEAT MOVES IN AT LEAST ONE DIRECTION)

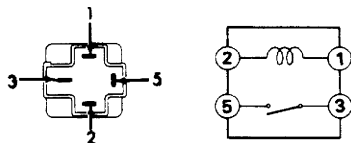
Check movement of appropriate power seat motor. Check circuit breaker of appropriate power seat motor. Check power seat switch and wiring harness.

## TESTING

### POWER MAIN RELAY TEST

Celica

Remove power main relay (near battery, in relay block No. 2). Check continuity between terminals No. 3 and 5. See Fig. 5. If there is continuity, replace relay. If there is no continuity, apply battery voltage across terminals No. 1 and 2. Check continuity between terminals No. 3 and 5. If there is no continuity, replace relay. If there is continuity, relay is okay.



93E84792

Fig. 5: Power Main Relay Terminal ID (Celica)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### POWER SEAT MOTOR TEST

1) Disconnect power seat motor connector. Apply battery voltage across power seat motor connector terminals. Reverse polarity to move seat in opposite direction. If power seat motor does not operate, replace power seat motor.



2) If power seat motor operates, move seat fully in one direction until it stops. With seat stopped, keep battery voltage applied to terminals until circuit breaker is heard operating (4-60 seconds). If circuit breaker does not operate within 60 seconds, replace power seat motor.

3) If circuit breaker operates within 60 seconds, reverse polarity and wait until seat moves in opposite direction. If seat moves in opposite direction within 60 seconds, power seat motor is okay. If seat does not move in opposite direction within 60 seconds, replace power seat motor.

## POWER SEAT SWITCH TEST

Disconnect power seat switch connector. Check continuity between specified terminals of power seat switch connector. See appropriate POWER SEAT SWITCH CONTINUITY TEST table. Replace switch if continuity is not as specified. See Figs. 6-8.

POWER SEAT SWITCH CONTINUITY TEST TABLE (CAMRY)

| Switch Position | (1) Terminals   |
|-----------------|-----------------|
| Slide           |                 |
| Forward .....   | 5 & 10; 8 & 9   |
| Off .....       | 5 & 9; 8 & 9    |
| Backward .....  | 5 & 9; 8 & 10   |
| Front Up/Down   |                 |
| Up .....        | 10 & 12; 4 & 11 |
| Off .....       | 4 & 11; 4 & 12  |
| Down .....      | 4 & 12; 10 & 11 |
| Rear Up/Down    |                 |
| Up .....        | 2 & 10; 6 & 7   |
| Off .....       | 2 & 7; 6 & 7    |
| Down .....      | 2 & 7; 6 & 10   |
| Recline         |                 |
| Forward .....   | 4 & 10; 7 & 9   |
| Off .....       | 4 & 9; 7 & 9    |
| Backward .....  | 4 & 9; 7 & 10   |

(1) - See Figs. 6-8.

POWER SEAT SWITCH CONTINUITY TEST TABLE (CELICA)

| Switch Position        | (1) Terminals |
|------------------------|---------------|
| Lumbar                 |               |
| Push (Outward) .....   | 2 & 7; 8 & 9  |
| Off .....              | 2, 8 & 9      |
| Release (Inward) ..... | 2 & 8; 7 & 9  |
| Side Support           |               |
| Spread (Outward) ..... | 3 & 8; 5 & 7  |
| Off .....              | 3, 5 & 8      |
| Close (Inward) .....   | 3 & 7; 5 & 8  |

(1) - See Figs. 6-8.

POWER SEAT SWITCH CONTINUITY TEST TABLE (LAND CRUISER)

| Switch Position | (1) Terminals |
|-----------------|---------------|
| Slide           |               |

|                   |                 |
|-------------------|-----------------|
| Forward           | 5 & 10; 8 & 9   |
| Off               | 5, 8 & 9        |
| Backward          | 5 & 9; 8 & 10   |
| Front Up/Down     |                 |
| Up                | 4 & 11; 10 & 12 |
| Off               | 4, 11 & 12      |
| Down              | 4 & 12; 10 & 11 |
| Rear Up/Down      |                 |
| Up                | 2 & 10; 6 & 7   |
| Off               | 2, 6 & 7        |
| Down              | 2 & 7; 6 & 10   |
| Recline           |                 |
| Forward           | 4 & 10; 7 & 9   |
| Off               | 4, 7 & 9        |
| Backward          | 4 & 9; 7 & 10   |
| Lumbar            |                 |
| Forward (Outward) | 1 & 9; 3 & 10   |
| Off               | 1, 3 & 9        |
| Backward (Inward) | 1 & 10; 3 & 9   |

(1) - See Figs. 6-8.

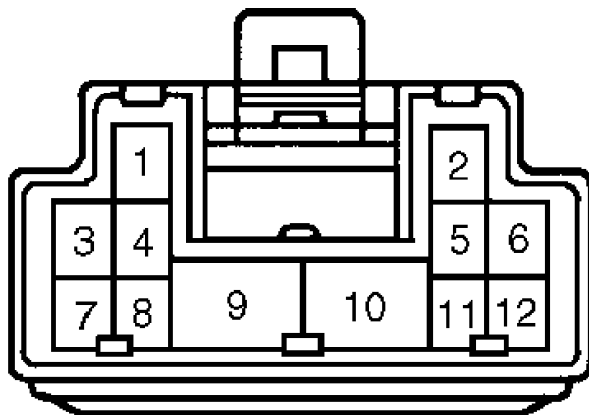
---

POWER SEAT SWITCH CONTINUITY TEST TABLE (SUPRA)

---

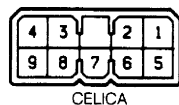
| Switch Position | (1) Terminals    |
|-----------------|------------------|
| Slide           |                  |
| Forward         | A1 & A3; A2 & A4 |
| Off             | A1 & A3; A1 & A4 |
| Backward        | A1 & A4; A2 & A3 |
| Recline         |                  |
| Forward         | A1 & B1; A2 & B2 |
| Off             | A1 & B1; A1 & B2 |
| Backward        | A1 & B2; A2 & B1 |

(1) - See Figs. 6-8.



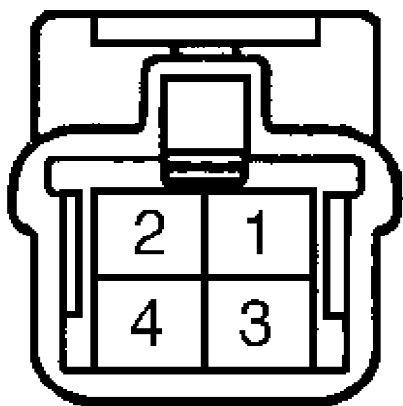
**93F84793**

Fig. 6: Power Seat Switch Connector Terminal ID  
 Courtesy of Toyota Motor Sales, U.S.A.



93G84794

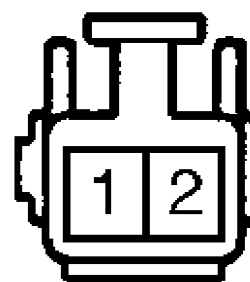
Fig. 7: Power Seat Switch Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A.



CONNECTOR "A"

93H84795

Fig. 8: Power Seat Switch Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A.



CONNECTOR "B"

## WIRING DIAGRAMS

See appropriate chassis wiring diagram in WIRING DIAGRAMS section.

# K - SENSOR RANGE CHARTS

## 1993 Toyota Celica

1993 ENGINE PERFORMANCE  
Toyota Sensor Operating Range Charts  
Celica

### INTRODUCTION

Sensor operating range information can help determine if a sensor is out of calibration. An out-of-calibration sensor may not set a trouble code, but it may cause driveability problems.

NOTE: Unless stated otherwise in testing procedure, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance.

### AIRFLOW METER

NOTE: Airflow meter may be referred to as Volume Airflow (VAF) meter or Mass Airflow (MAF) meter.

#### AIRFLOW METER RESISTANCE SPECIFICATIONS TABLE

| Application & Terminals (1)        | Ohms          |
|------------------------------------|---------------|
| Turbo                              |               |
| E2-VC .....                        | 200-400       |
| E2-THA                             |               |
| -4°F (20°C) .....                  | 10,000-20,000 |
| 32°F (0°C) .....                   | 4000-7000     |
| 68°F (20°C) .....                  | 2000-3000     |
| 104°F (40°C) .....                 | 900-1300      |
| 140°F (60°C) .....                 | 400-700       |
| E2-VS                              |               |
| Measuring Plate Fully Closed ..... | 200-600       |
| Measuring Plate Fully Opened ..... | 20-1200       |

(1) - For terminal identification and testing, see I - SYS /COMP TESTS article in the ENGINE PERFORMANCE section.

### AIR TEMPERATURE SENSOR

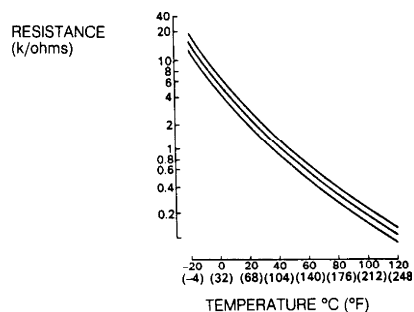
NOTE: Air temperature sensor may be referred to as intake air temperature sensor.

1.6L 4A-FE & 2.2L 5S-FE

Air temperature sensor is located in air intake, near air filter. Measure resistance between sensor terminals. For resistance specifications, see Fig. 1.

2.0L Turbo

Air temperature sensor is an integral part of airflow meter. Check resistance at terminals E2 and THA of airflow meter. See AIRFLOW METER RESISTANCE SPECIFICATIONS table.



CAMRY 2.2L, CELICA 1.6L 4A-FE & 2.2L 5S-FE,  
 PASEO & TERCEL 93B79583

Fig. 1: Checking Air Temperature Sensor (1.6L)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## COLD START INJECTOR TIME SWITCH

COLD START INJECTOR TIME SWITCH RESISTANCE SPECIFICATIONS TABLE

| Application &<br>Terminals (1) | Temperature<br>°F (°C)     | Ohms  |
|--------------------------------|----------------------------|-------|
| Turbo                          |                            |       |
| STA-STJ .....                  | Less Than 50 (10) .....    | 30-50 |
|                                | Greater Than 77 (25) ..... | 70-90 |
| STA-Ground .....               | .....                      | 30-90 |

(1) - For terminal identification and testing, see I - SYS/  
 COMP TESTS article in the ENGINE PERFORMANCE section.

## COOLANT TEMPERATURE SENSOR

NOTE: Coolant temperature sensor may also be referred to as Engine  
 Coolant Temperature (ECT) sensor.

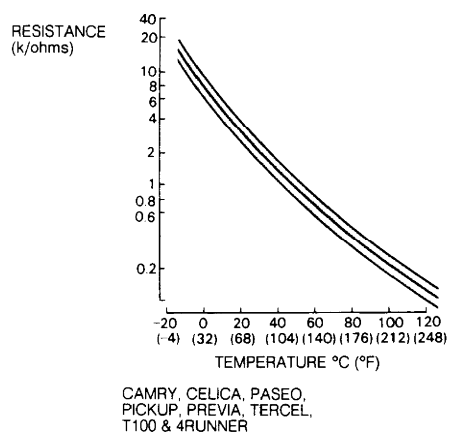
Measure coolant temperature sensor resistance between sensor  
 terminals. For resistance specifications, see Fig. 2.

## EGR GAS TEMPERATURE SENSOR

EGR GAS TEMPERATURE SENSOR SPECIFICATIONS TABLE (1)

| Temperature - °F (°C) | k/Ohms |
|-----------------------|--------|
| 122 (50) .....        | 69-89  |
| 212 (100) .....       | 11-15  |
| 302 (150) .....       | 2-4    |

(1) - For terminal identification and testing, see I - SYS/  
 COMP TESTS article in the ENGINE PERFORMANCE section.



93D79585  
Fig. 2: Checking Coolant Temperature Sensor  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## MANIFOLD ABSOLUTE PRESSURE SENSOR

NOTE: See VACUUM SENSOR section in this article.

## OXYGEN SENSOR HEATER

OXYGEN SENSOR HEATER RESISTANCE SPECIFICATIONS TABLE (1)

| Application  | Ohms                  |
|--------------|-----------------------|
| Celica ..... | 5.1-6.3 @ 68°F (20°C) |

(1) - Measure resistance between sensor terminals +B and HT.  
For terminal identification, see I - SYS/COMP TESTS  
article in the ENGINE PERFORMANCE section.

## THROTTLE POSITION SENSOR

NOTE: For terminal identification and testing procedures, see  
I - SYS/COMP TESTS article in the ENGINE PERFORMANCE section.

TPS RESISTANCE SPECIFICATIONS TABLE

| Application         | Clearance<br>In. (mm) | Terminal       | Ohmmeter<br>Reading |
|---------------------|-----------------------|----------------|---------------------|
| 1.6L 4A-FE .....    | .024 (.61) ....       | IDL & E2 ..... | Continuity          |
|                     |                       | PSW-E2 .....   | No Continuity       |
|                     |                       | IDL-E2 .....   | No Continuity       |
|                     | .031 (.79) ....       | PSW-E2 .....   | No Continuity       |
|                     |                       | IDL-E2 .....   | No Continuity       |
| 2.0L Turbo (1) ...  | Fully Open ....       | PSW-E2 .....   | Continuity          |
|                     |                       | VTA & E2 ..... | 470-6100            |
|                     | 0 (0) .....           | IDL & E2 ..... | 2300 Or Less        |
|                     | .020 (.51) ....       | IDL & E2 ....  | No Continuity       |
|                     | .028 (.71) ....       | VTA & E2 ..... | 3100-12,100         |
| 2.2L 5S-FE (1) .... | Fully Open ...        | VC & E2 .....  | 3900-9000           |
|                     |                       | VTA & E2 ..... | 200-5700            |
|                     | 0 (0) .....           | IDL & E2 ..... | 2300 Or Less        |
|                     | .020 (.51) ....       | IDL & E2 ....  | No Continuity       |
|                     | .028 (.71) ....       | VTA & E2 ..... | 2000-10,200         |
|                     | Fully Open ...        | VC & E2 .....  | 2500-5900           |

(1) - Apply vacuum to throttle opener before checking TPS.

---

## TURBOCHARGING PRESSURE SENSOR

TURBOCHARGING PRESSURE SENSOR OUTPUT VOLTAGE SPECIFICATIONS TABLE

| Applied Vacuum<br>In. Hg | Output Voltage<br>Drop |
|--------------------------|------------------------|
| 3.94 .....               | .15-.35                |
| 7.87 .....               | .40-.60                |
| 11.81 .....              | .65-.85                |
| 15.75 .....              | .90-1.10               |
| 19.69 .....              | 1.15-1.35              |

## VACUUM SENSOR

NOTE: Vacuum sensor may also be referred to as Manifold Absolute Pressure (MAP) sensor. Vacuum sensor is used on 1.6L 4A-FE and 2.2L 5S-FE. For testing procedures, see I - SYS/COMP TESTS article in the ENGINE PERFORMANCE section.

VACUUM SENSOR SUPPLY VOLTAGE SPECIFICATIONS TABLE

| Application                   | Voltage   |
|-------------------------------|-----------|
| 1.6L 4A-FE & 2.2L 5S-FE ..... | 4.50-5.50 |

VACUUM SENSOR OUTPUT VOLTAGE DROP SPECIFICATIONS TABLE

| Applied Vacuum<br>In. Hg | Output Voltage<br>Drop |
|--------------------------|------------------------|
| 3.94 .....               | .3-.5                  |
| 7.87 .....               | .7-.9                  |
| 11.81 .....              | 1.1-1.3                |
| 15.75 .....              | 1.5-1.7                |
| 19.69 .....              | 1.9-2.1                |

# SHIFT LOCK SYSTEM

1993 Toyota Celica

AUTOMATIC TRANSMISSIONS  
Toyota Shift Lock System

## APPLICATION

### APPLICATION

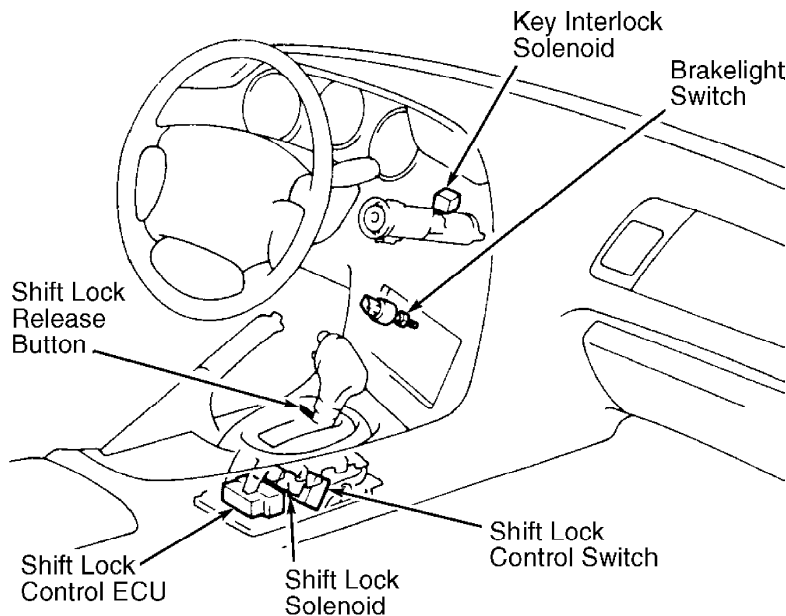
| Application  | Model                                                                             |
|--------------|-----------------------------------------------------------------------------------|
| Lexus .....  | ES300, GS300, LS400, SC300 & SC400                                                |
| Toyota ..... | Camry, Celica, Corolla, Land Cruiser, MR2, Paseo, Pickup, Supra, Tercel & 4Runner |

## DESCRIPTION

Transmission is equipped with a electronically controlled shift lock and key lock system. See Fig. 1. Shift lock system prevents shift lever from being moved from Park unless brake pedal is depressed. In case of a malfunction, shift lever can be released by depressing shift lock override button, located near shift lever. Key lock system prevents ignition key from being moved from ACC to LOCK position on ignition switch unless shift lever is in Park.

The system consists of a brakelight switch, key interlock solenoid, shift lock control switch, shift lock Electronic Control Unit (ECU), shift lock solenoid and shift lock override button. See Fig. 1.

NOTE: Shift lock Electronic Control Unit is referred to as ECU.



95J20536  
Fig. 1: Shift & Key Lock System (Typical)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## OPERATION



## SHIFT LOCK SYSTEM

With ignition on, when brake pedal is depressed, an input signal is sent from brakelight switch to the ECU. With shift lever in Park, an input signal from shift control switch is input to ECU, indicating shift lever is in Park. The ECU then operates shift lock solenoid, so shift lever can be moved from Park.

## KEY LOCK SYSTEM

With ignition in ON or ACC position and shift lever in Park, shift lock control switch opens and voltage from ECU to key interlock solenoid is turned off. When key interlock solenoid is turned off, ignition key can be turned from ACC to LOCK position on ignition switch.

NOTE: If ignition is left in ACC or ON position with shift lever in any gear range except Park for about one hour, ECU then operates to release the lock mechanism.

## TESTING

### ELECTRONIC CONTROL UNIT (ECU)

Access ECU. See Fig. 1. The ECU is mounted under the center console, in front or back of shifter. Turn ignition on. Backprobing ECU harness connector with DVOM, check voltage between designated terminals. Voltage should be as specified. Refer to the appropriate SHIFT LOCK SYSTEM PIN VOLTAGES table. For circuit identification, see appropriate wiring diagram in WIRING DIAGRAMS.

NOTE: Ground (GND) terminal is also referred to as "E" terminal.

### SHIFT LOCK SYSTEM PIN VOLTAGES

SHIFT LOCK SYSTEM PIN VOLTAGES (CAMRY, CELICA, COROLLA, ES300, GS300, LAND CRUISER, PASEO & TERCEL)

| Measure Between<br>Terminals: | Description                                                                  | Voltage |
|-------------------------------|------------------------------------------------------------------------------|---------|
| ECU                           |                                                                              |         |
| ACC-GND .....                 | Ignition Switch In ACC Position .....                                        | 10-14   |
| IG-GND .....                  | Ignition Switch In ON Position .....                                         | 10-14   |
| STP-GND .....                 | Depress Brake Pedal .....                                                    | 10-14   |
| KLS-GND ....                  | Ignition Switch In ACC, Shifter In "P" .....                                 | 0       |
|                               | Ignition Switch In ACC, Shifter Except "P" ....                              | 10-14   |
|                               | (1) Ignition Switch In ACC, Shifter Except "P" ....                          | 6-9     |
| Shift Lock Solenoid           |                                                                              |         |
| SLS+-SLS- ..                  | Ignition Switch In ACC, Shifter In "P" .....                                 | 0       |
|                               | Depress Brake Pedal .....                                                    | 8-13.5  |
|                               | Depress Brake Pedal (After 20 Seconds) ....                                  | 5.5-9.5 |
|                               | Ignition Switch In ACC, Shifter Except "P" .....                             | 0       |
| Shift Lock Control Switch     |                                                                              |         |
| P1-P .....                    | Ignition Switch In ON Position,<br>Shifter In "P", Depress Brake Pedal ..... | 0       |
|                               | Ignition Switch In ON, Shifter Except "P" ....                               | 9-13.5  |
| P2-P .....                    | Ignition Switch In ACC, Shifter In "P" .....                                 | 9-13.5  |
|                               | Ignition Switch In ACC, Shifter Except "P" .....                             | 0       |

(1) - Voltage measurement after 1 second.

---

#### SHIFT LOCK SYSTEM PIN VOLTAGES (LS400, SC300, SC400 & SUPRA)

---

| Measure Between<br>Terminals: | Description | Voltage |
|-------------------------------|-------------|---------|
|-------------------------------|-------------|---------|

##### ECU

|         |                                                |        |
|---------|------------------------------------------------|--------|
| ACC-GND | ..... Ignition Switch In ACC Position          | 10-14  |
| IG-GND  | ..... Ignition Switch In ON Position           | 10-14  |
| STP-GND | ..... Depress Brake Pedal                      | 10-14  |
| KLS-GND | .... Ignition Switch In ACC, Shifter In "P"    | 0      |
|         | Ignition Switch In ACC, Shifter Except "P"     | 7.5-11 |
|         | (1) Ignition Switch In ACC, Shifter Except "P" | 6-9.5  |

##### Shift Lock Solenoid

|           |                                            |         |
|-----------|--------------------------------------------|---------|
| SLS+-SLS- | .. Ignition Switch In ACC, Shifter In "P"  | 0       |
|           | Depress Brake Pedal                        | 8-13.5  |
|           | Depress Brake Pedal (After 20 Seconds)     | 5.5-9.5 |
|           | Ignition Switch In ACC, Shifter Except "P" | 0       |

##### Shift Lock Control Switch

|      |                                                                              |        |
|------|------------------------------------------------------------------------------|--------|
| P1-P | ..... Ignition Switch In ON Position,<br>Shifter In "P", Depress Brake Pedal | 0      |
|      | Ignition Switch In ON, Shifter Except "P"                                    | 9-13.5 |
| P2-P | ..... Ignition Switch In ACC, Shifter In "P"                                 | 9-13.5 |
|      | Ignition Switch In ACC, Shifter Except "P"                                   | 0      |

(1) - Voltage measurement after 1 second.

---

#### SHIFT LOCK SYSTEM PIN VOLTAGES (MR2)

---

| Measure Between<br>Terminals: | Description | Voltage |
|-------------------------------|-------------|---------|
|-------------------------------|-------------|---------|

##### ECU

|         |                                                |        |
|---------|------------------------------------------------|--------|
| ACC-GND | ..... Ignition Switch In ACC Position          | 10-14  |
| IG-GND  | ..... Ignition Switch In ON Position           | 10-14  |
| STP-GND | ..... Depress Brake Pedal                      | 10-14  |
| KLS-GND | .... Ignition Switch In ACC, Shifter In "P"    | 0      |
|         | Ignition Switch In ACC, Shifter Except "P"     | 7.5-11 |
|         | (1) Ignition Switch In ACC, Shifter Except "P" | 6-9    |

##### Shift Lock Solenoid

|          |                                            |        |
|----------|--------------------------------------------|--------|
| SLS+-GND | ... Ignition Switch In ACC, Shifter In "P" | 0      |
|          | Depress Brake Pedal                        | 8-13.5 |
|          | Ignition Switch In ACC, Shifter Except "P" | 0      |

##### Shift Lock Control Switch

|      |                                                                              |        |
|------|------------------------------------------------------------------------------|--------|
| P1-P | ..... Ignition Switch In ON Position,<br>Shifter In "P", Depress Brake Pedal | 0      |
|      | Ignition Switch In ON, Shifter Except "P"                                    | 9-13.5 |
| P2-P | ..... Ignition Switch In ACC, Shifter In "P"                                 | 0-13.5 |

(1) - Voltage measurement after 1 second.

---

#### SHIFT LOCK SYSTEM PIN VOLTAGES (PICKUP & 4RUNNER)

---

Measure Between

| Terminals:                                | Description                                                                  | Voltage |
|-------------------------------------------|------------------------------------------------------------------------------|---------|
| ECU                                       |                                                                              |         |
| ACC-GND .....                             | Ignition Switch In ACC Position .....                                        | 10-14   |
| IG-GND .....                              | Ignition Switch In ON Position .....                                         | 10-14   |
| STP-GND .....                             | Depress Brake Pedal .....                                                    | 10-14   |
| KLS-GND ....                              | Ignition Switch In ACC, Shifter In "P" .....                                 | 0       |
|                                           | Ignition Switch In ACC, Shifter Except "P" ....                              | 10-14   |
|                                           | (1) Ignition Switch In ACC, Shifter Except "P" ....                          | 6-9     |
| Shift Lock Solenoid                       |                                                                              |         |
| SLS+-SLS- ..                              | Ignition Switch In ACC, Shifter In "P" .....                                 | 0       |
|                                           | Depress Brake Pedal .....                                                    | 10-14   |
|                                           | Ignition Switch In ACC, Shifter Except "P" .....                             | 0       |
| Shift Lock Control Switch                 |                                                                              |         |
| P1-P .....                                | Ignition Switch In ON Position,<br>Shifter In "P", Depress Brake Pedal ..... | 0       |
|                                           | Ignition Switch In ON, Shifter Except "P" .....                              | 10-14   |
| P2-P .....                                | Ignition Switch In ACC, Shifter In "P" .....                                 | 10-14   |
|                                           | Ignition Switch In ACC, Shifter Except "P" .....                             | 0       |
| (1) - Voltage measurement after 1 second. |                                                                              |         |

## SHIFT LOCK SOLENOID

1) Disconnect electrical connector from shift lock solenoid. Using ohmmeter, measure resistance between electrical terminals of shift lock solenoid.

2) Replace shift lock solenoid if resistance is not within specification. See SHIFT LOCK SOLENOID RESISTANCE SPECIFICATIONS table.

3) Apply battery voltage between electrical terminals of shift lock solenoid. Replace shift lock solenoid if operating sound cannot be heard.

### SHIFT LOCK SOLENOID RESISTANCE SPECIFICATIONS

| Application                                                         | Ohms  |
|---------------------------------------------------------------------|-------|
| Camry, Celica, ES300, GS300 & MR2 .....                             | 21-27 |
| Corolla, Paseo & Tercel .....                                       | 30-35 |
| Land Cruiser, LS400, Pickup, SC300,<br>SC400, Supra & 4Runner ..... | 20-28 |
| Pickup & 4Runner .....                                              | 29-36 |

## KEY INTERLOCK SOLENOID

1) Disconnect electrical connector from key interlock solenoid. Using ohmmeter, measure resistance between electrical terminals of key interlock solenoid.

2) Replace key interlock solenoid if resistance is not 12-17 ohms. Apply battery voltage between electrical terminals of key interlock solenoid. Replace key interlock solenoid if operating sound cannot be heard.

## SHIFT LOCK CONTROL SWITCH

NOTE: Testing for Land Cruiser not available from manufacturer.

Disconnect electrical connector from shift lock control

switch. Using ohmmeter, check continuity between specified terminals in relation to shift lever. See SHIFT LOCK CONTROL SWITCH CONTINUITY table. Replace switch if continuity is not as specified.

NOTE:       Continuity must be checked in accordance with position of  
              release button on shift lever and shift lever position.

SHIFT LOCK CONTROL SWITCH CONTINUITY

| Shift Lever Position & Condition       | Terminals |
|----------------------------------------|-----------|
| Park & Release Button Not Pushed ..... | P & P1    |
| Any Other Gear Except Park .....       | P & P2    |

WIRING DIAGRAMS

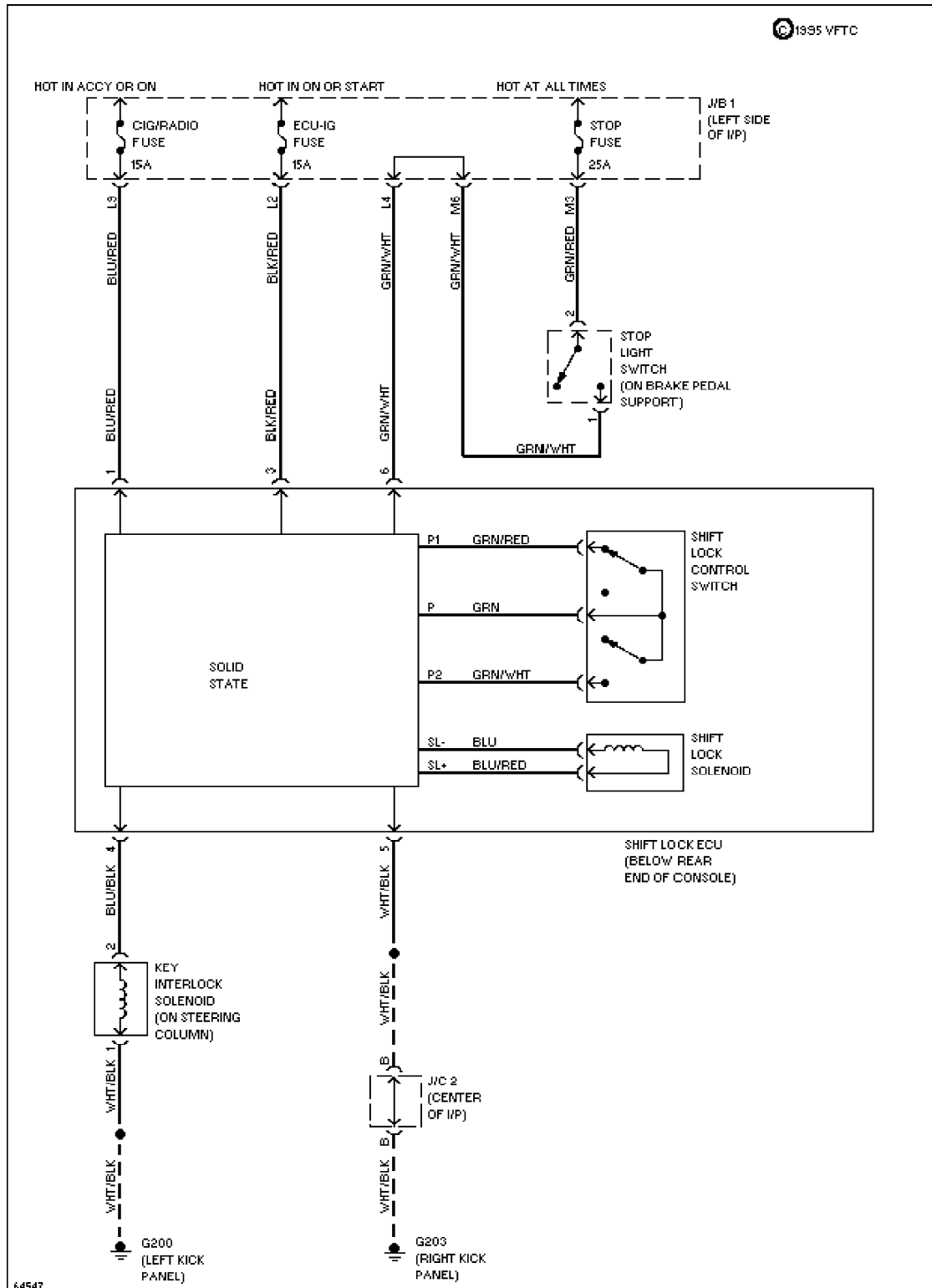


Fig. 2: Shift Interlock System Wiring Diagram (ES300)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

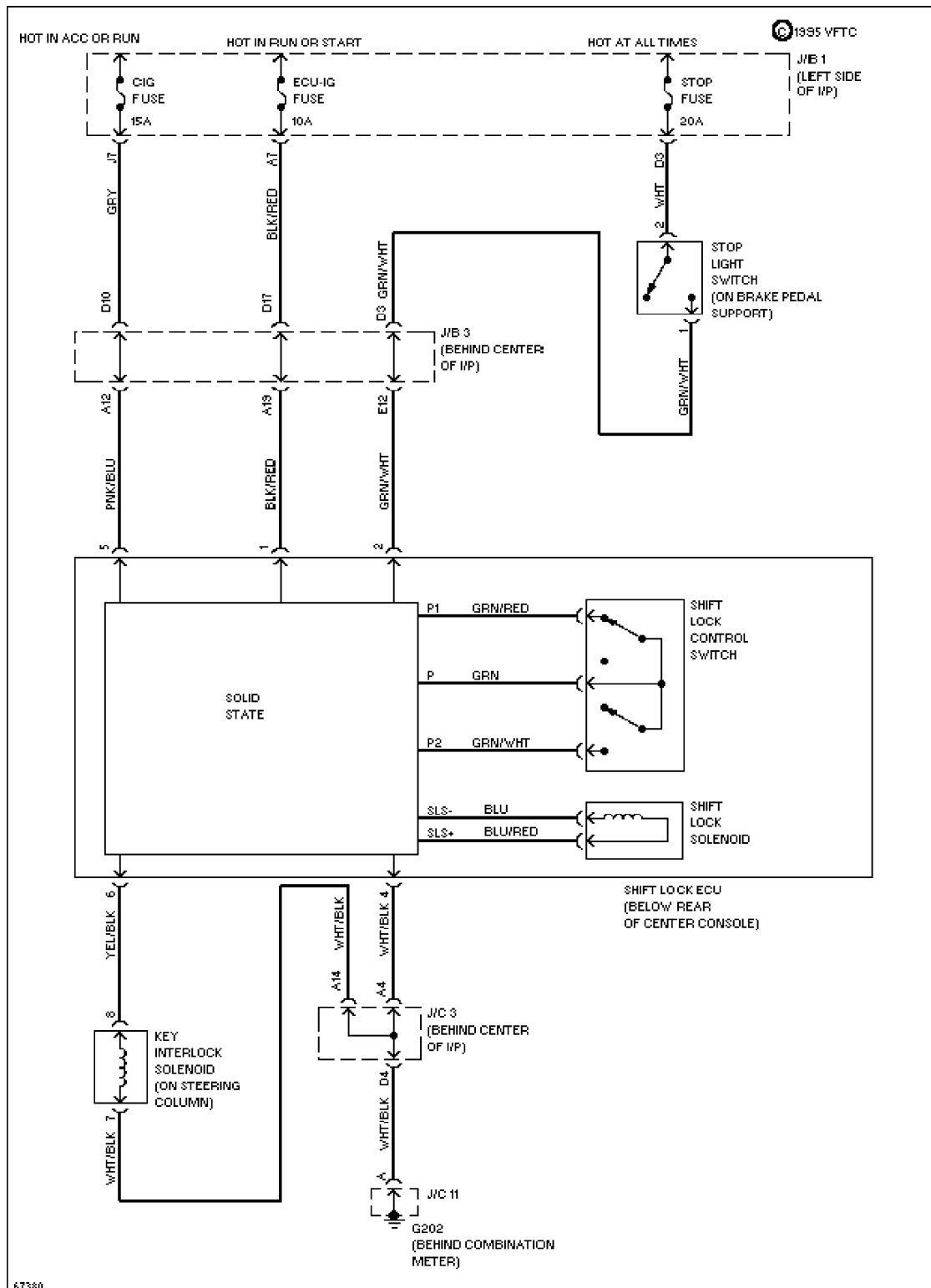


Fig. 3: Shift Interlock System Wiring Diagram (GS300)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

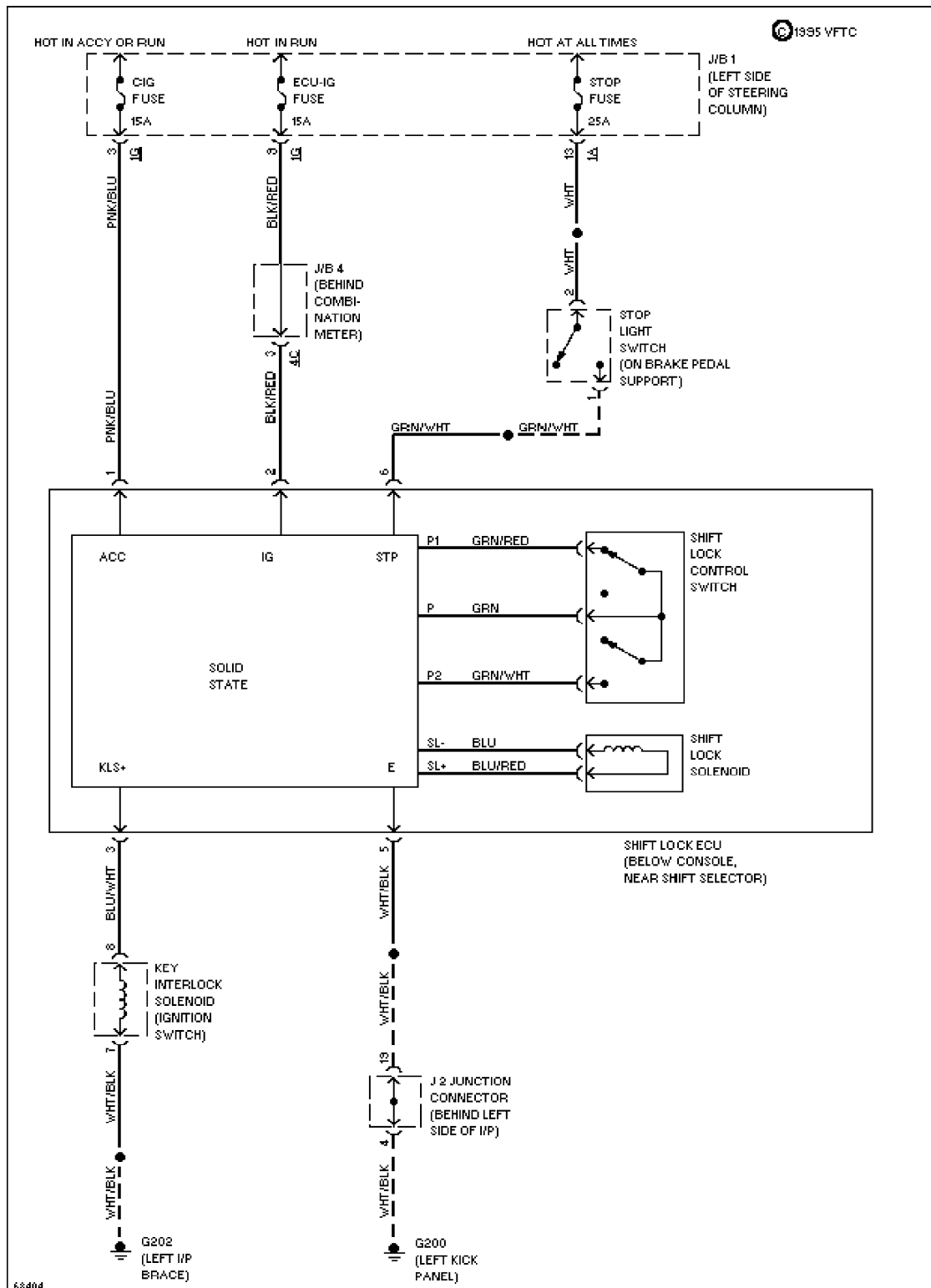


Fig. 4: Shift Interlock System Wiring Diagram (LS300)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

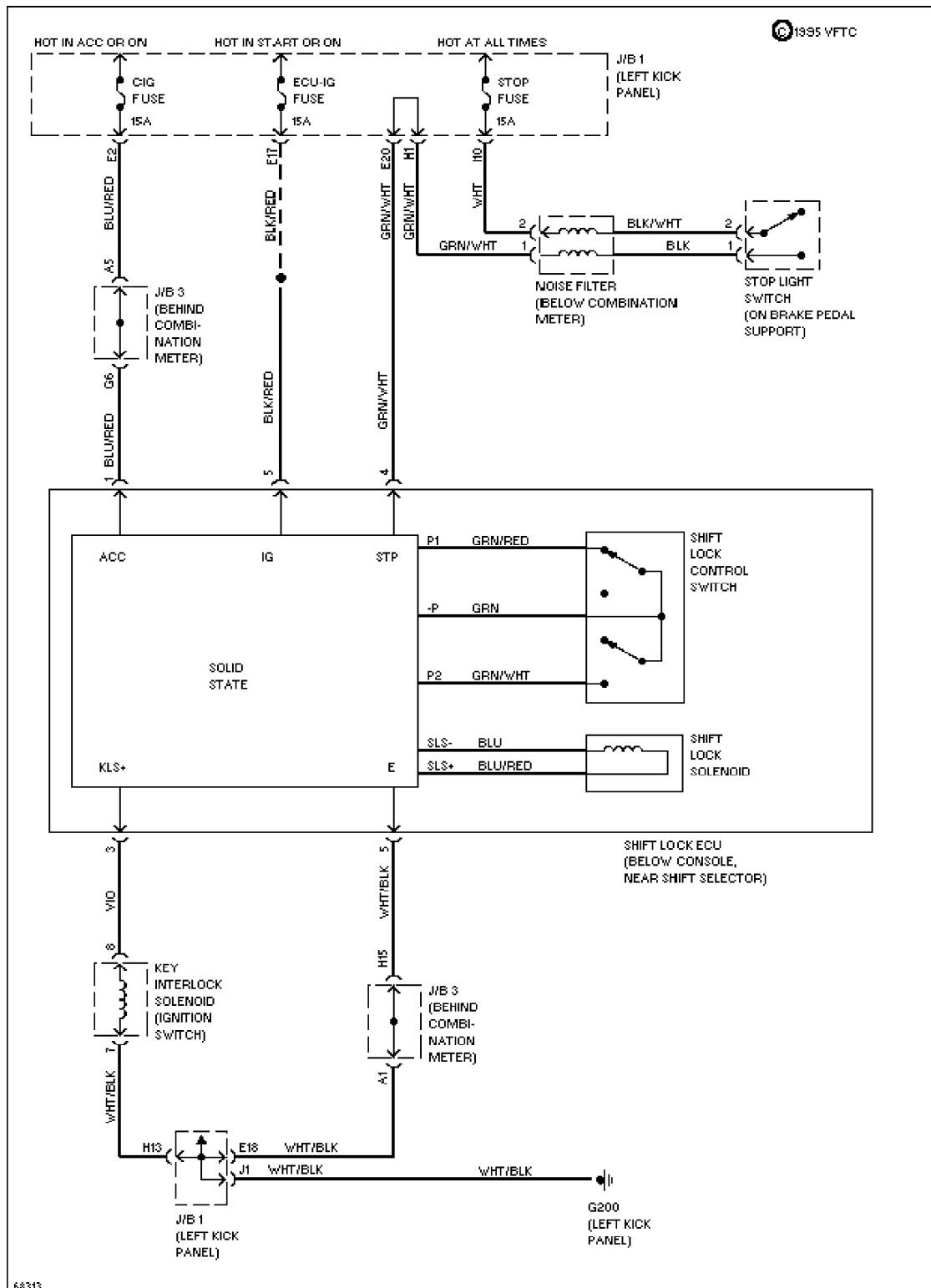
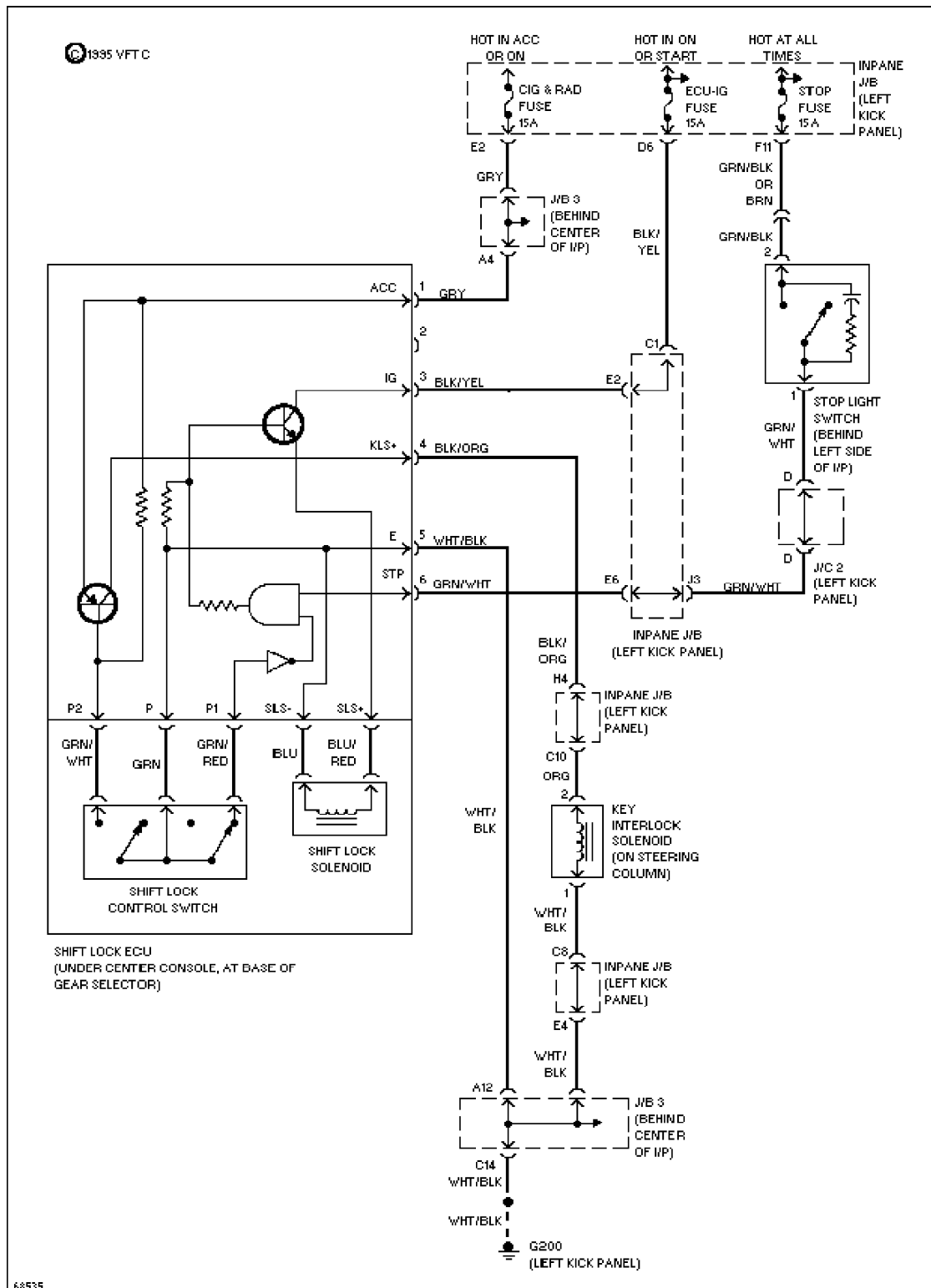


Fig. 5: Shift Interlock System Wiring Diagram (SC300 & SC400)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.





Courtesy of Toyota Motor Sales, U.S.A., Inc.



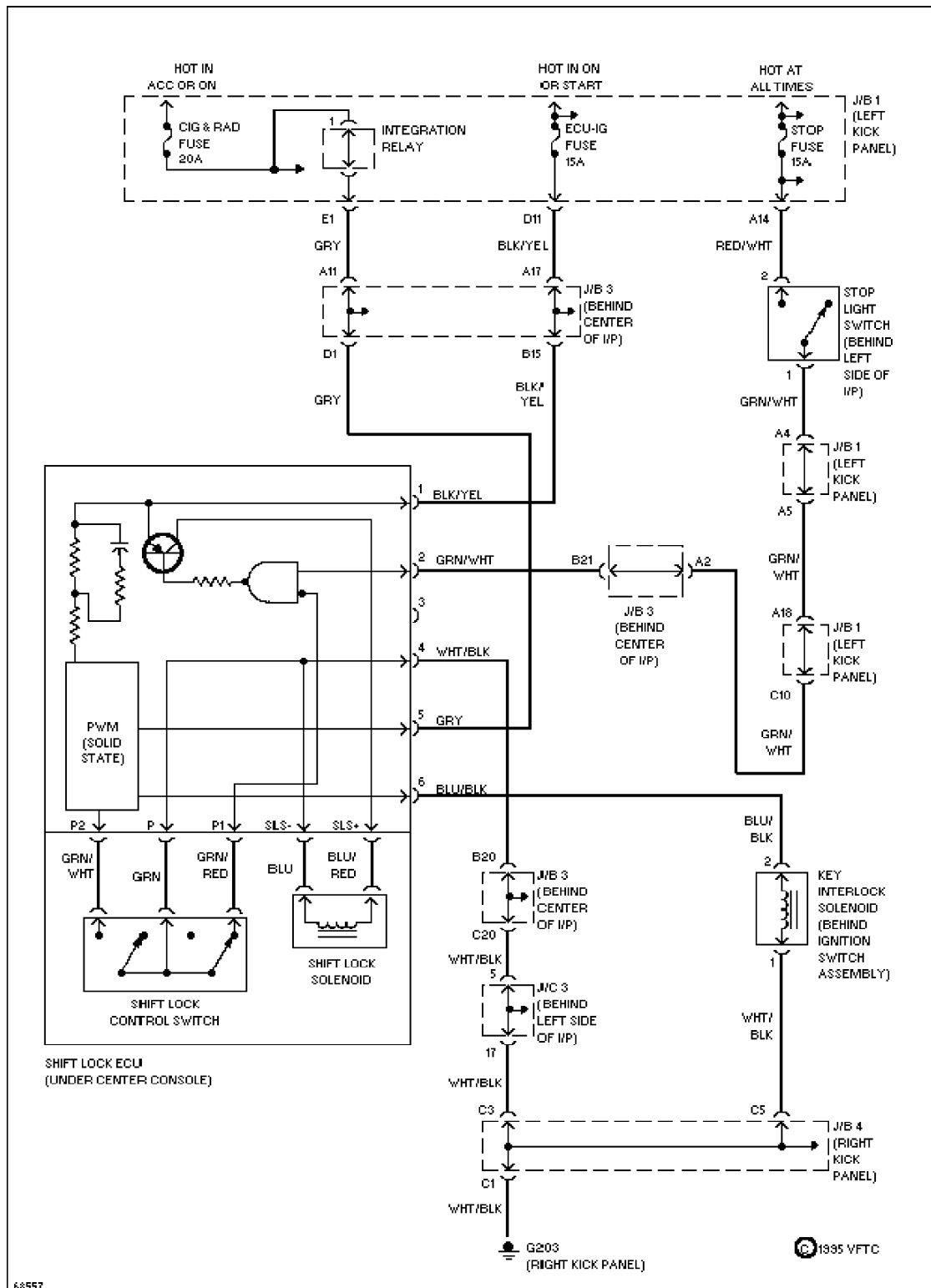


Fig. 8: Shift Interlock System Wiring Diagram (Corolla)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

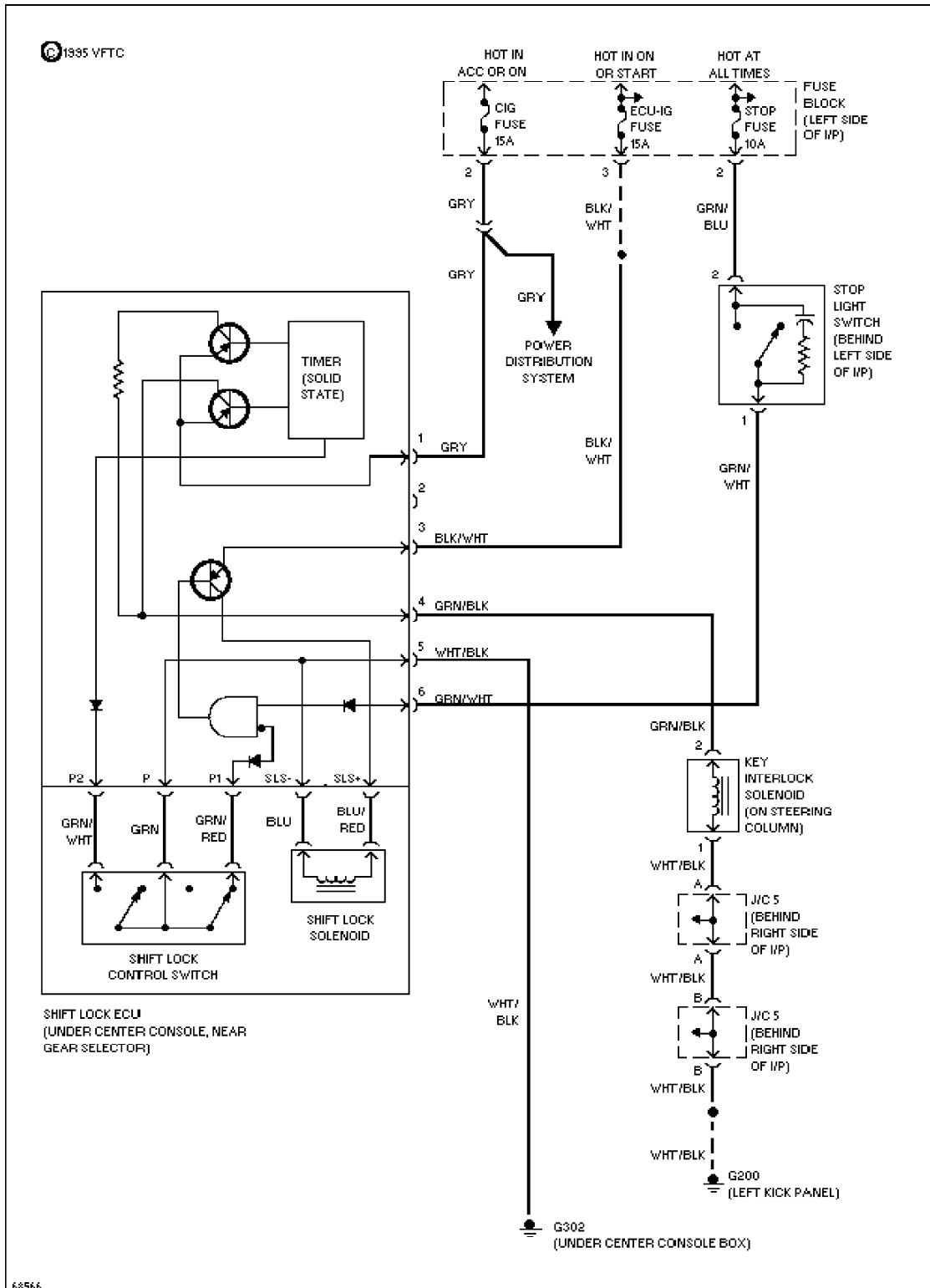


Fig. 9: Shift Interlock System Wiring Diagram (Land Cruiser)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

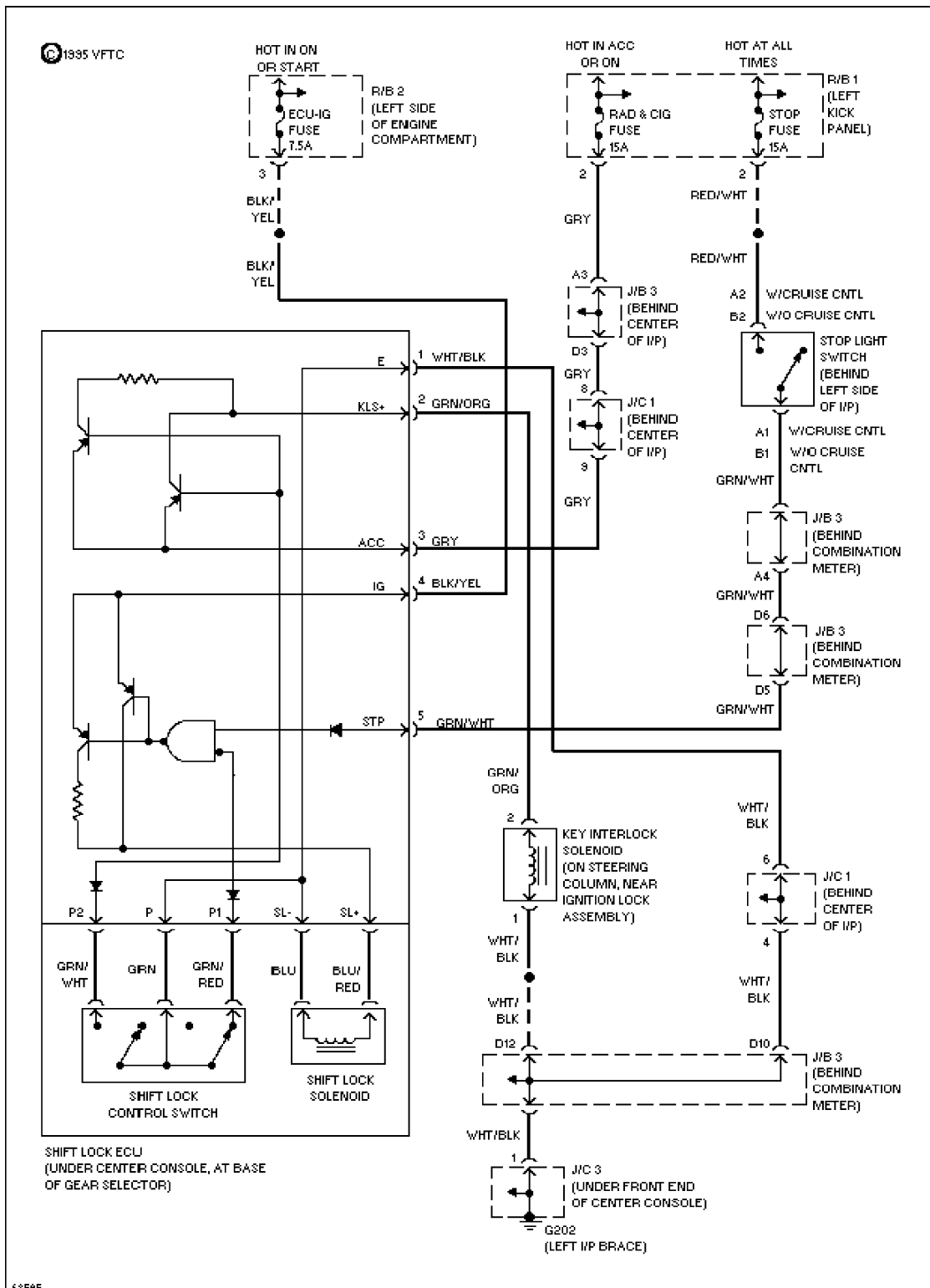


Fig. 10: Shift Interlock System Wiring Diagram (MR2)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

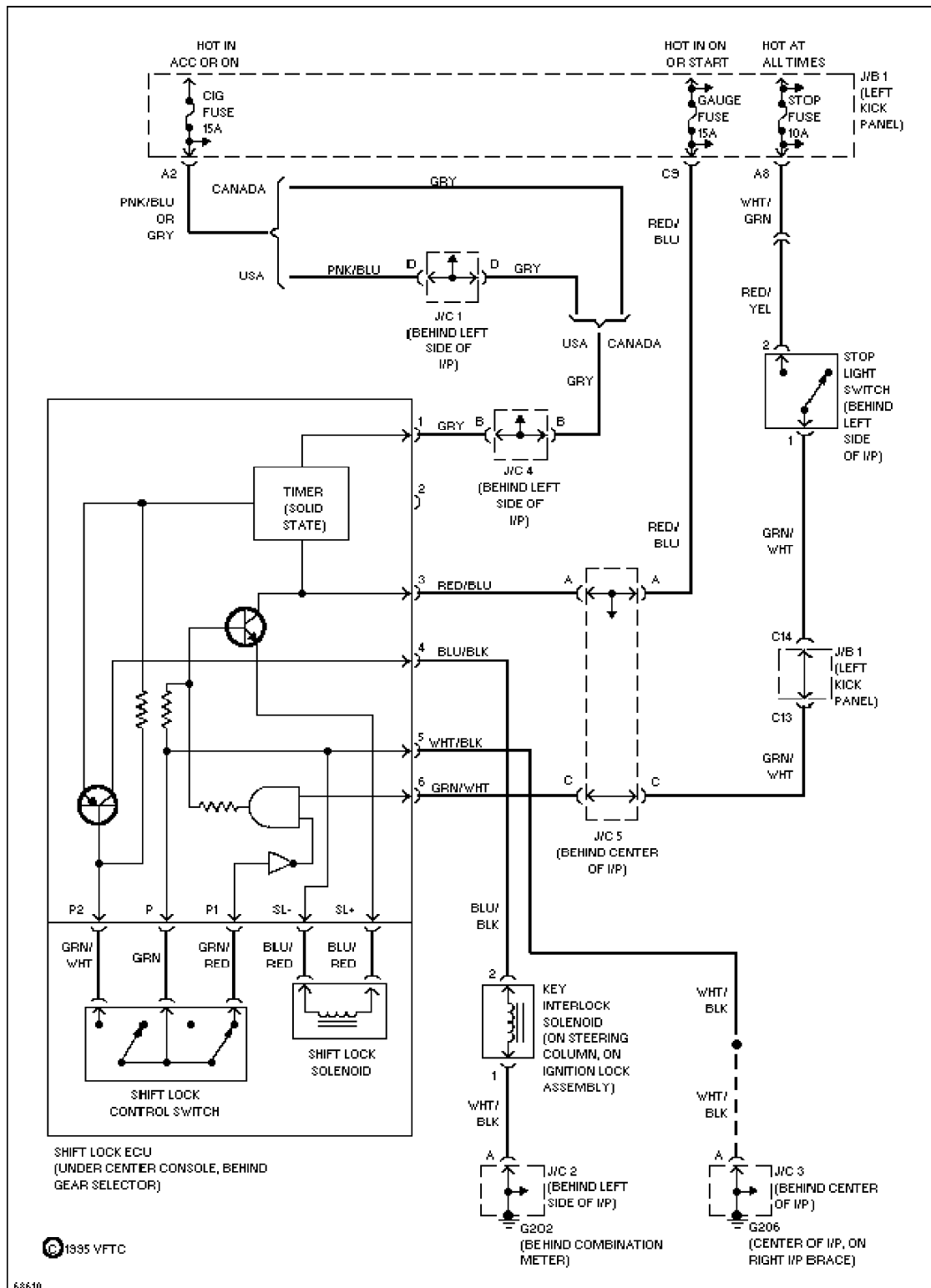


Fig. 11: Shift Interlock System Wiring Diagram (Paseo)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

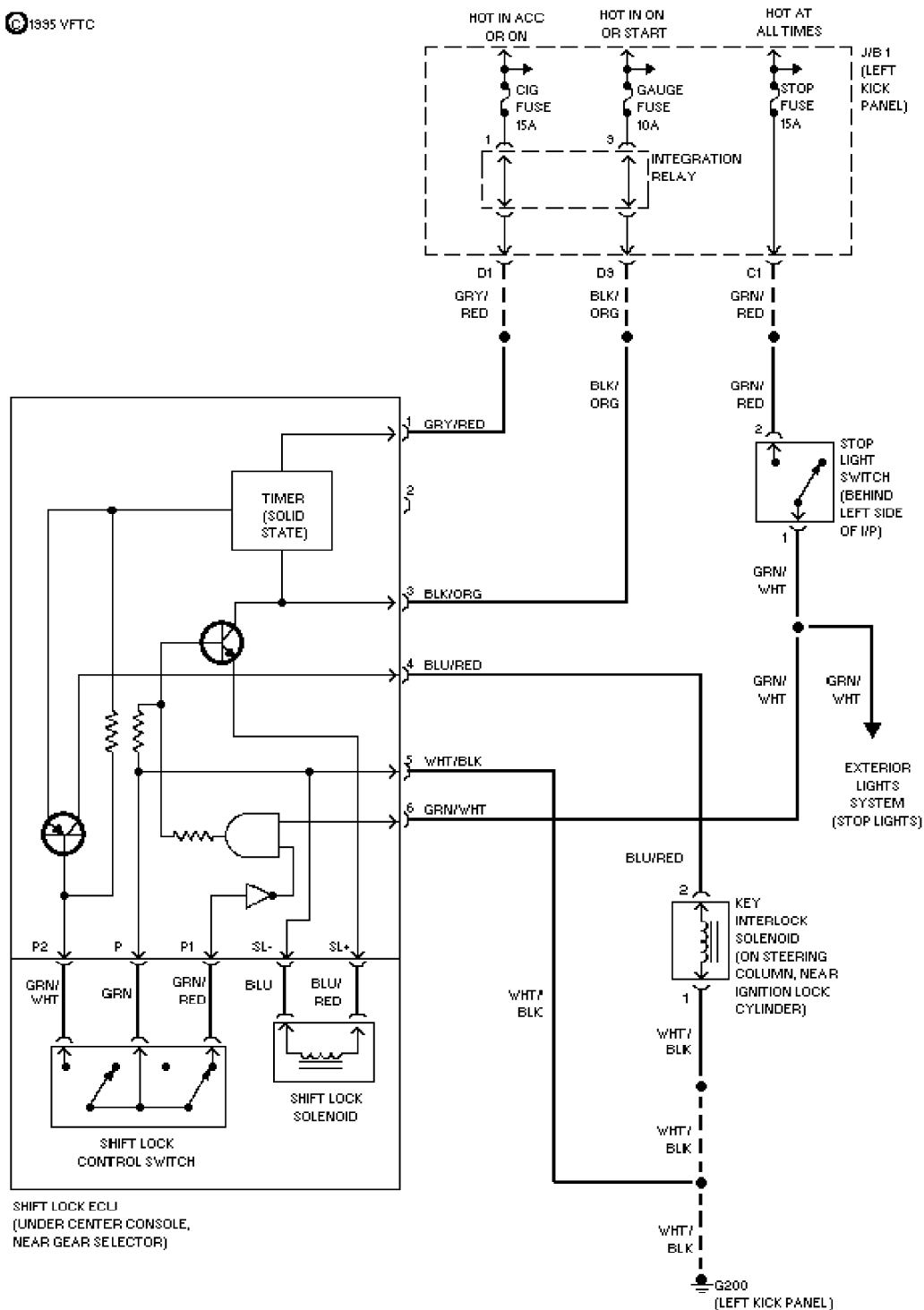


Fig. 12: Shift Interlock System Wiring Diagram (Pickup)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

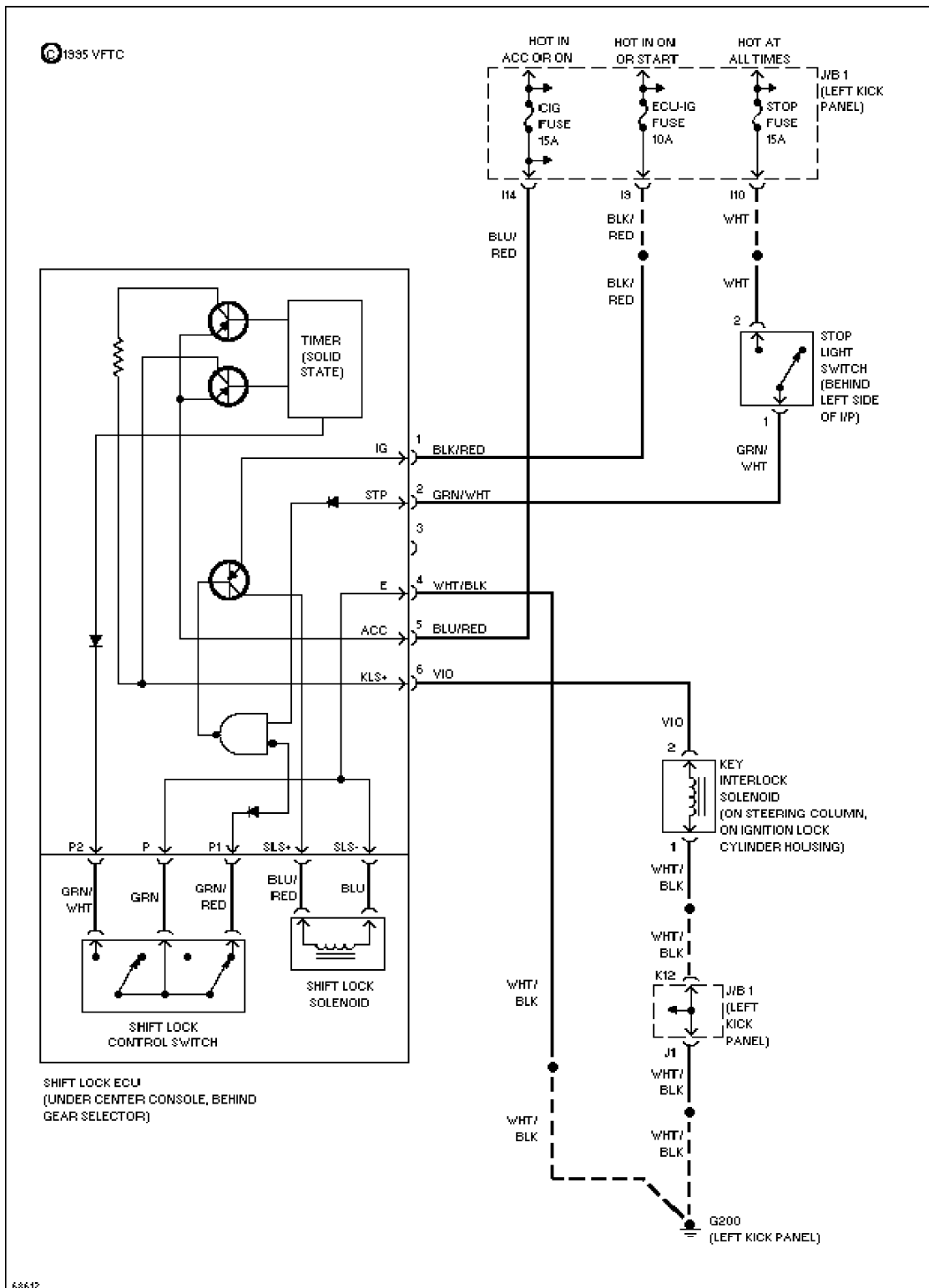


Fig. 13: Shift Interlock System Wiring Diagram (Supra)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.



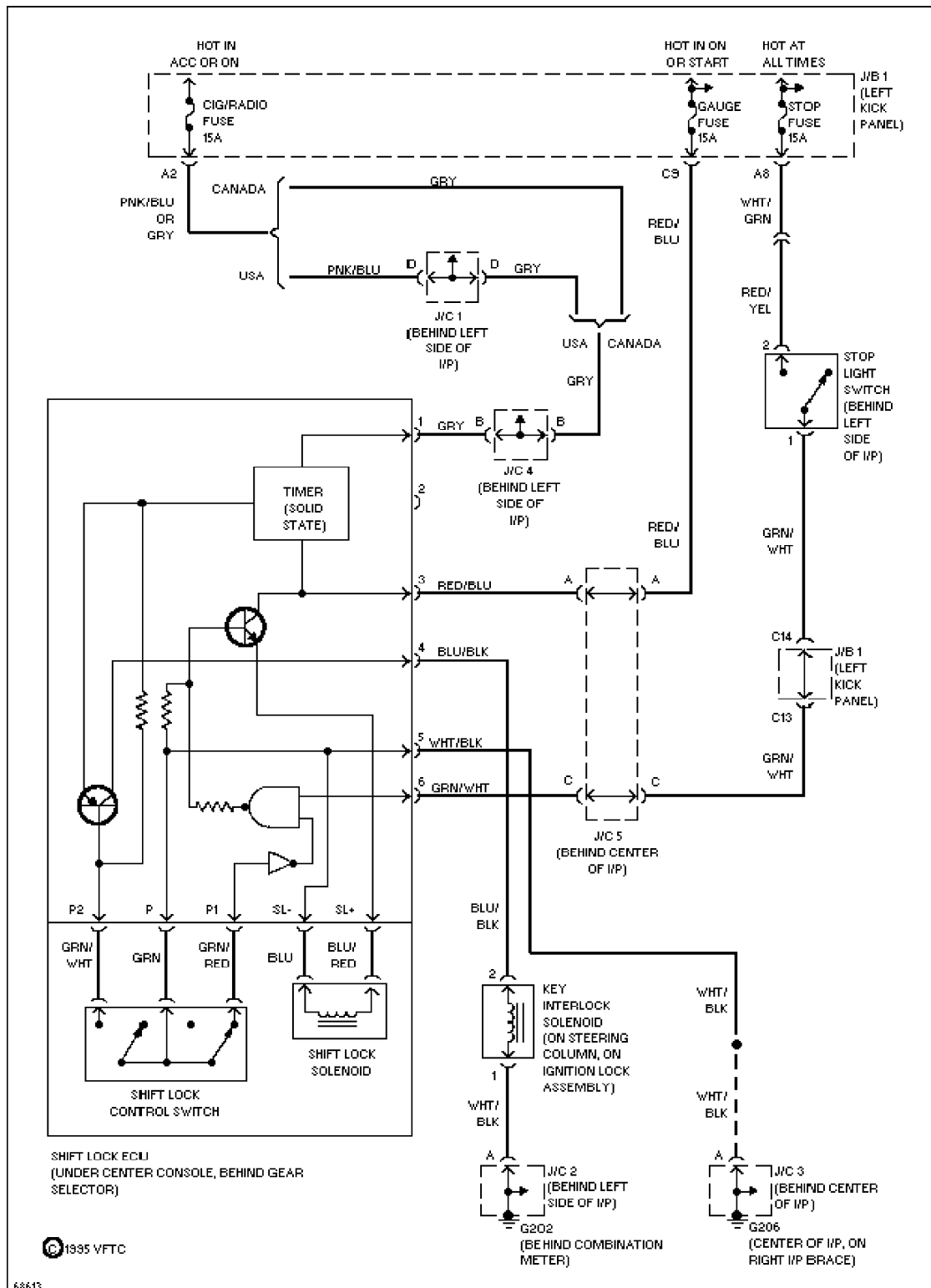


Fig. 14: Shift Interlock System Wiring Diagram (Tercel)  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

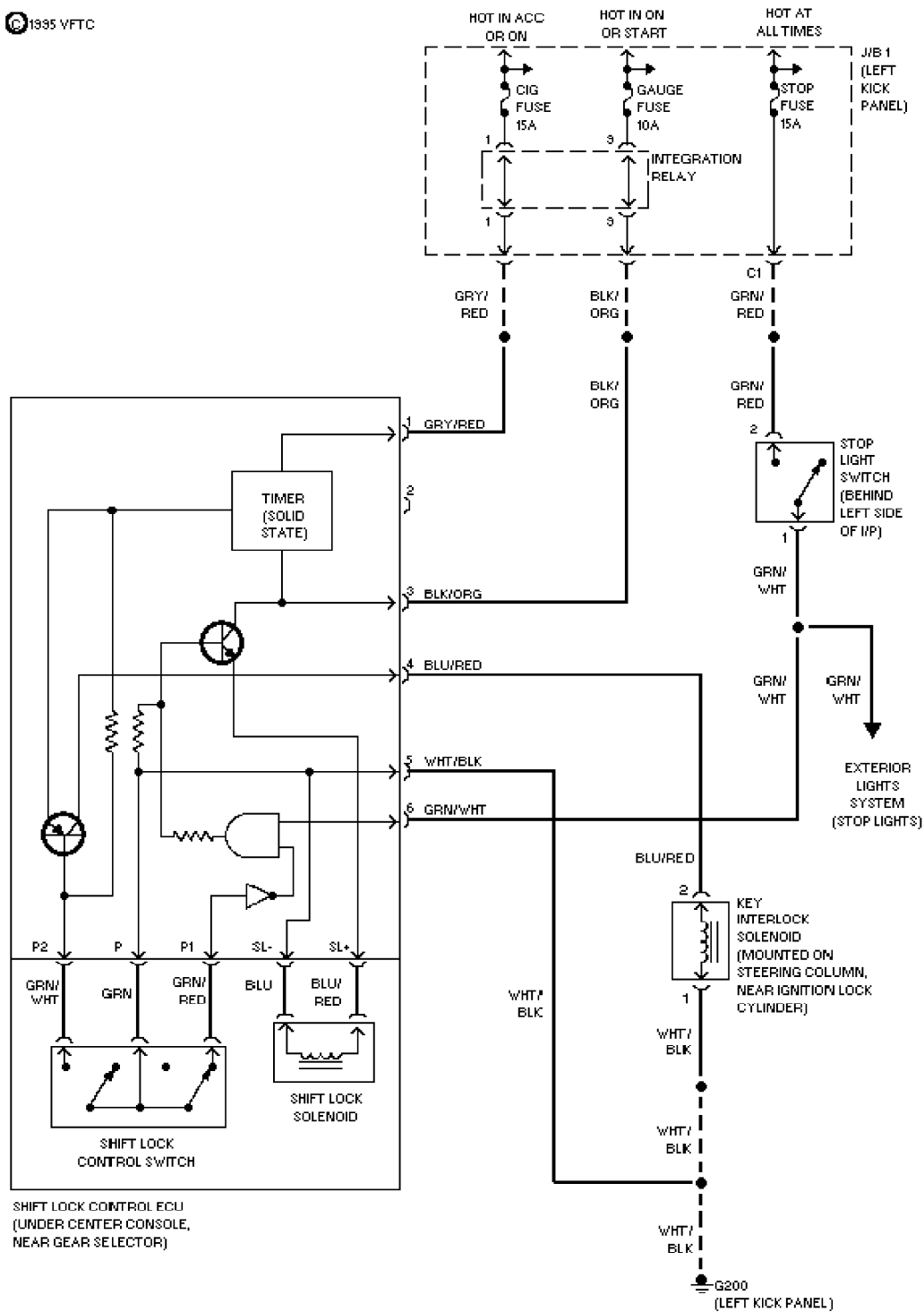


Fig. 15: Shift Interlock System Wiring Diagram (4Runner)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

# STARTER

## 1993 Toyota Celica

1993 ELECTRICAL  
Toyota Starters

Celica

### DESCRIPTION

Celica uses a Nippondenso 4-brush, solenoid-actuated, gear reduction-type starter, equipped with an overrunning clutch. The brush holder assembly retains 4 brushes and springs in the starter housing.

Reduction gear type starters contain an integral solenoid attached to drive housing, a reduction idler gear and bearing installed into starter housing, and a clutch drive assembly. The clutch drive assembly is mounted to starter housing and is driven by the reduction idler gear from armature shaft. The brush holder assembly retains 4 brushes and 4 springs in the end cover of field frame housing.

Manual transmission vehicles use a clutch start switch and automatic transmission vehicles use a park/neutral switch to energize starter relay. On models with theft deterrent system, theft deterrent system ECU provides ground for starter relay.

NOTE: Starter type and kilowatt (kW) rating can be found on a metal label attached to side of starter.

### TROUBLE SHOOTING

1) If a no-start condition exists and battery is known to be good, connect test light or voltmeter between starter solenoid terminal No. 50 and ground. See Fig. 8.

2) Turn ignition switch to START position. If test light or voltmeter does not indicate voltage, check main fusible links and large ampere main fuses in engine compartment relay box. If fusible links and fuses are okay, see IGNITION SWITCH CONTINUITY TEST and/or STARTER RELAY TEST under ON-VEHICLE TESTING.

### ON-VEHICLE TESTING

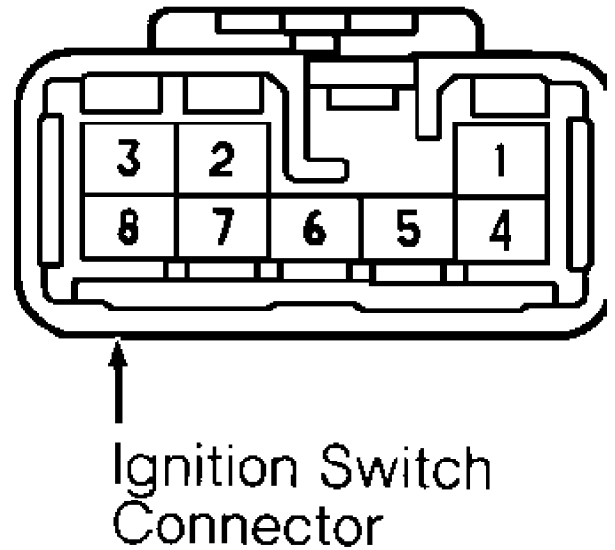
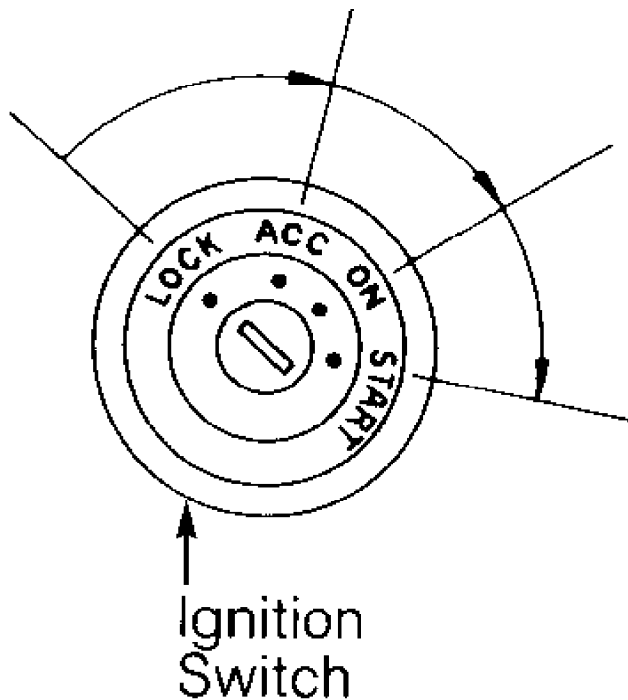
NOTE: Before testing, ensure battery is fully charged, battery cables and terminal ends are tight and clean, and engine grounds are secure.

#### CLUTCH START SWITCH TEST

1) Switch is located above clutch pedal on bracket. Disconnect wiring harness connector from switch.

2) Connect ohmmeter probes to switch terminals. Depress clutch pedal. If continuity does not exist, adjust or replace clutch start switch. If continuity exists, check circuit to starter relay for open, and check starter relay. See STARTER RELAY TEST.

#### IGNITION SWITCH CONTINUITY TEST

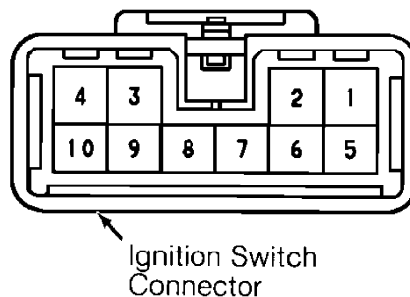
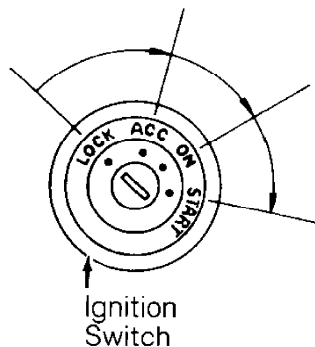


93G82269

Fig. 1: Ignition Switch 8-Pin Connector ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

1) Disconnect negative battery cable. Remove driver's lower instrument panel cover. Remove upper and lower steering column covers if needed. Locate ignition switch wiring harness 10-pin connector. See Fig. 2.

2) With ignition switch in LOCK position, there should be no continuity between any terminals. With ignition switch in ACC position, there should be continuity between terminals No. 3 and 4. With ignition switch in ON position, there should be continuity between terminals No. 2, 3 and 4, and between terminals No. 9 and 10. With ignition switch in START position, there should be continuity between terminals No. 2, 4 and 7, and between terminals No. 6, 9 and 10. If continuity is not as specified, replace switch.



93J82270

Fig. 2: Ignition Switch 10-Pin Connector ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

PARK/NEUTRAL SWITCH

NOTE: If vehicle will not start with shift lever in Park/Neutral position, verify correct park/neutral switch adjustment. If park/neutral switch is correctly adjusted, verify switch continuity.

Adjusting Park/Neutral Switch

Locate park/neutral switch at transmission or transaxle. Loosen park/neutral position switch bolt(s) and verify shift selector is in "N" position. Align switch shaft groove with neutral basic line on switch. Hold switch in position and tighten switch bolt(s) to 48 INCH lbs. (5.4 N.m).

Park/Neutral Switch Continuity Check

Disconnect electrical connector from park/neutral switch at transmission or transaxle. Using ohmmeter, check for continuity at specified terminals with gearshift in proper positions. See Figs. 3 and 4. See PARK/NEUTRAL SWITCH CONTINUITY table. Replace switch if defective.

PARK/NEUTRAL SWITCH CONTINUITY

| Gearshift Position |       | Check Between Terminals |
|--------------------|-------|-------------------------|
| 2.2L 5S-FE         |       |                         |
| Park               | ..... | 2 & 3, 1 & 6            |
| Reverse            | ..... | 5 & 6                   |
| Neutral            | ..... | 2 & 3, 6 & 7            |
| Drive              | ..... | 6 & 8                   |
| 2                  | ..... | 6 & 9                   |
| Low                | ..... | 4 & 6                   |
| 1.6L 4A-FE         |       |                         |
| Park               | ..... | "B" & "N"               |
| Reverse            | ..... | RB & RL                 |
| Neutral            | ..... | "B" & "N"               |

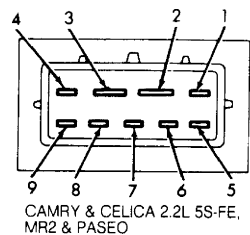


Fig. 3: Park/Neutral Switch Terminal ID (2.2L)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

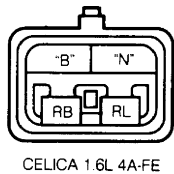


Fig. 4: Park/Neutral Switch Terminal ID (1.6L)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

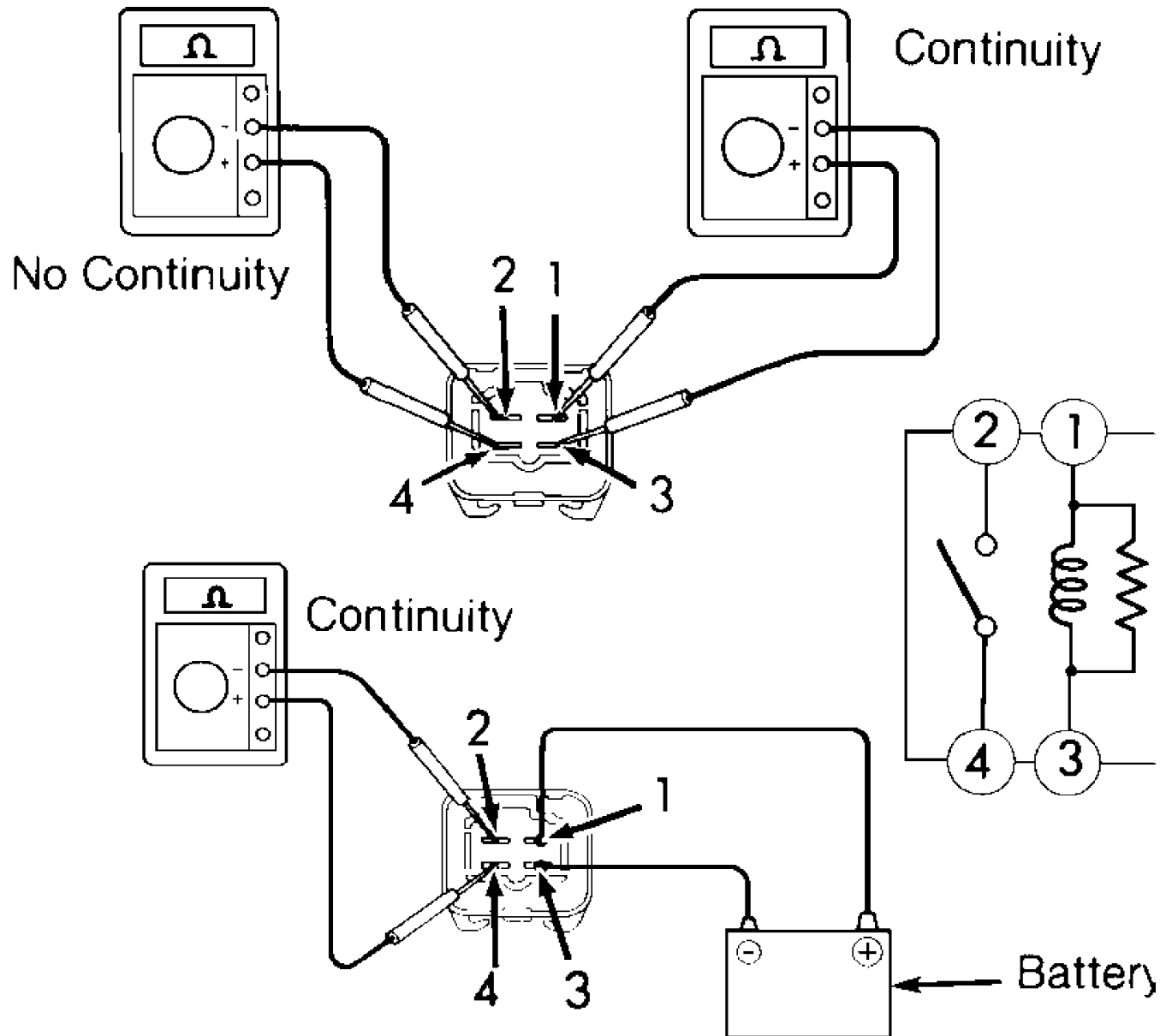
STARTER RELAY TEST

Starter Relay With Terminals No. 1, 2, 3 & 4

1) Locate and remove starter relay. Relay is located behind right kick panel, on relay box No. 4. See Fig. 7. Using ohmmeter,

verify continuity between relay terminals No. 1 and 3. See Fig. 5. Continuity should not exist between terminals No. 2 and 4. If continuity is not as indicated, replace relay.

2) Check relay operation by applying battery voltage through terminals No. 1 and 3. See Fig. 5. Continuity should now exist between terminals No. 2 and 4. If relay does not test as indicated, replace relay.



92F01587

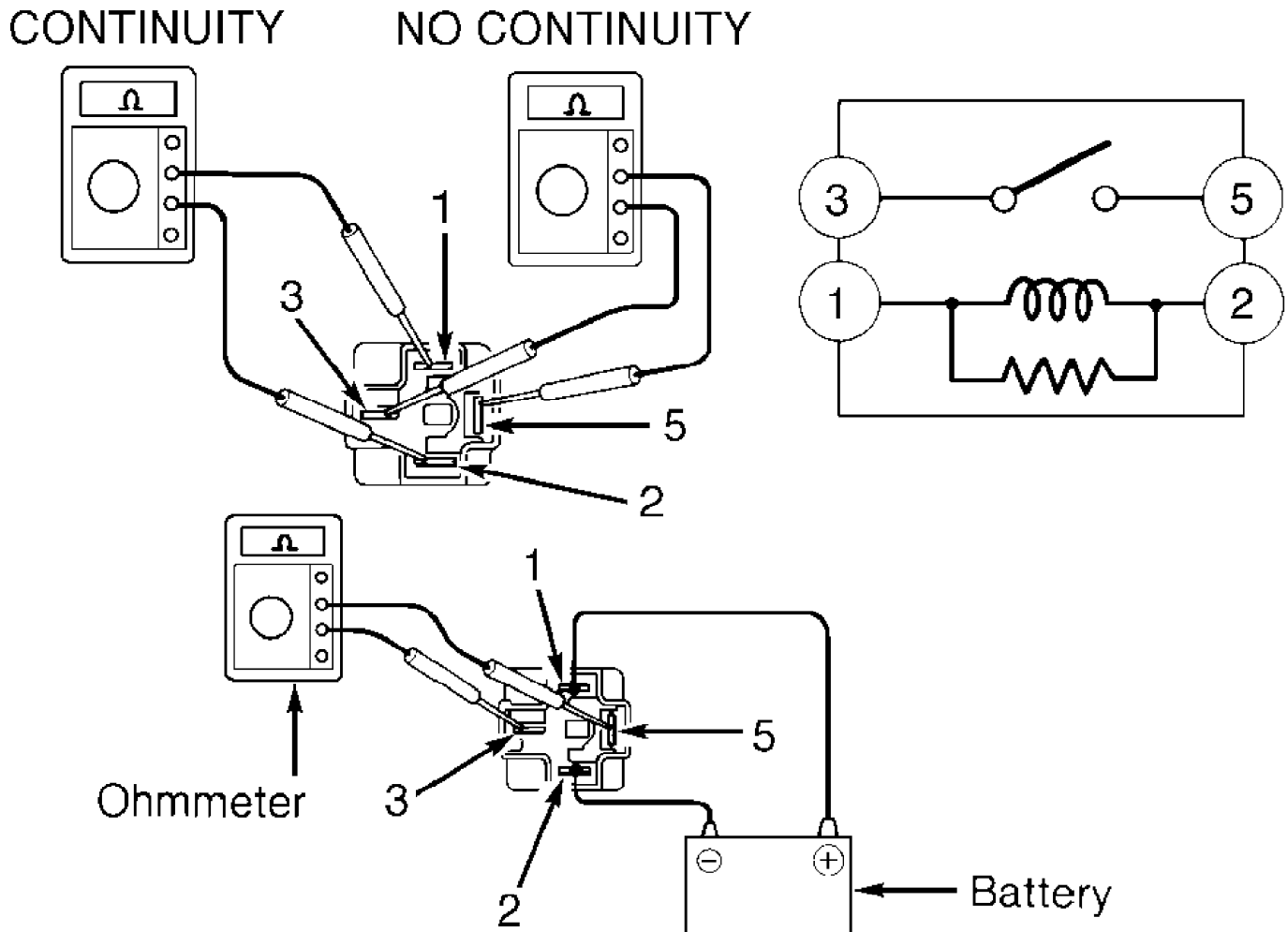
Fig. 5: Testing Starter Relay With Terminals No. 1, 2, 3 & 4  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Starter Relay With Terminals No. 1, 2, 3 & 5

1) Locate and remove starter relay. Relay is located behind

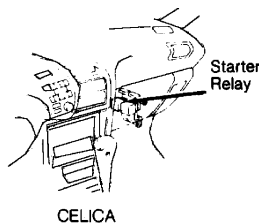
right kick panel, on relay box No. 4. See Fig. 7. Using ohmmeter, verify continuity between relay terminals No. 1 and 2. See Fig. 6. Continuity should not exist between terminals No. 3 and 5. If continuity is not as indicated, replace relay.

2) Check relay operation by applying battery voltage through terminals No. 1 and 2. See Fig. 6. Continuity should now exist between terminals No. 3 and 5. If relay does not test as indicated, replace relay.



93A82271

Fig. 6: Testing Starter Relay With Terminals No. 1, 2, 3 & 5  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93C02156

Fig. 7: Locating Starter Relay  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## BENCH TESTING

### NO-LOAD TEST

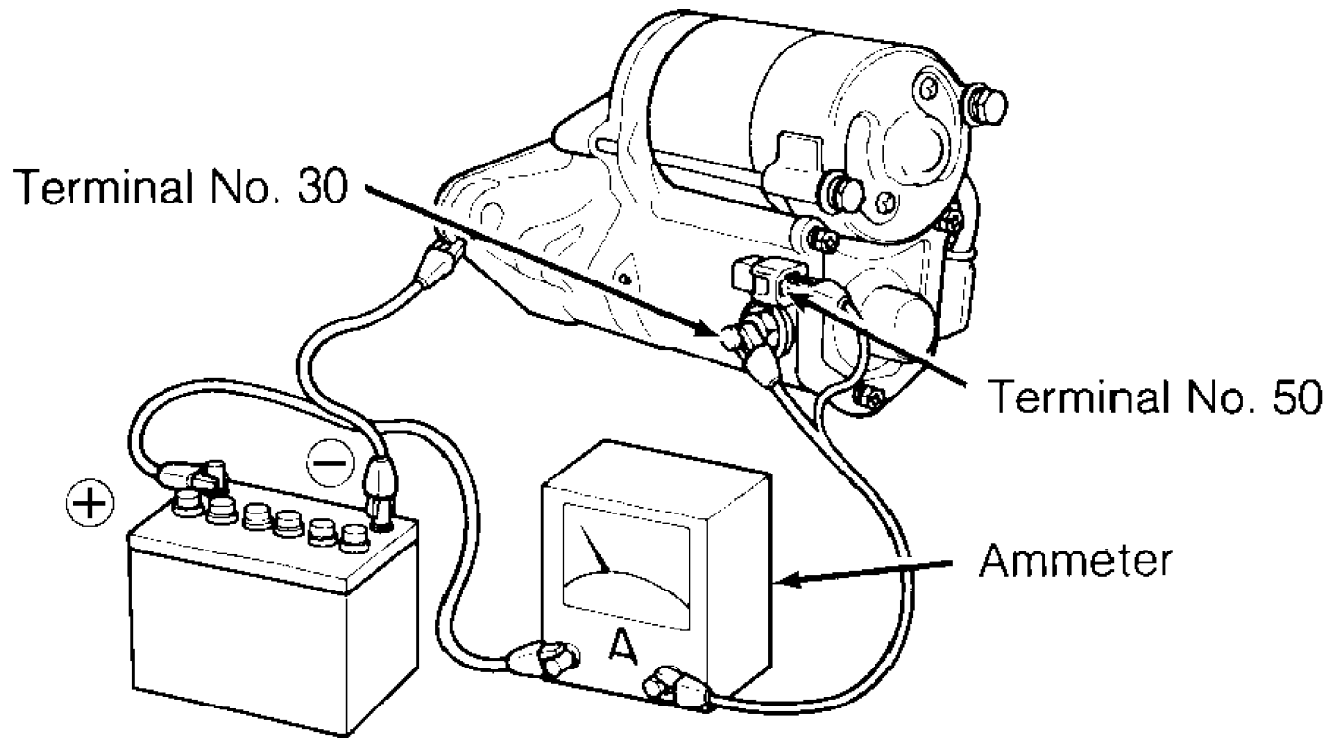
CAUTION: DO NOT engage starter solenoid for more than 5 seconds during testing or damage to coil winding will result.

1) Remove starter. Connect ammeter in series between starter motor terminal No. 30 (battery terminal) and a fully charged 12-volt battery. Connect battery negative to starter case ground. See Figs. 8 and 9. Connect voltmeter to battery to observe voltage draw readings.

2) Connect remote starter or jumper wire to terminal No. 30 and to terminal No. 50 to engage starter. Starter drive pinion gear should extend quickly and spin smoothly. Verify starter amperage draw and battery voltage draw to be within specifications. See NO-LOAD TEST SPECIFICATIONS table. Replace starter if not within specification.

#### NO-LOAD TEST SPECIFICATIONS

| Application          | Max. Amps @ (Volts) | RPM  |
|----------------------|---------------------|------|
| (1.0 kW) .....       | 90 (11.5) .....     | 3000 |
| (1.4 & 1.6 kW) ..... | 90 (11.5) .....     | 3500 |



93I02164

Fig. 8: Testing Starter No-Load (Reduction Gear)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

### SOLENOID TESTS



CAUTION: DO NOT engage starter solenoid for more than 5 seconds during testing or damage to coil winding will result.

#### Pull-In Coil Test

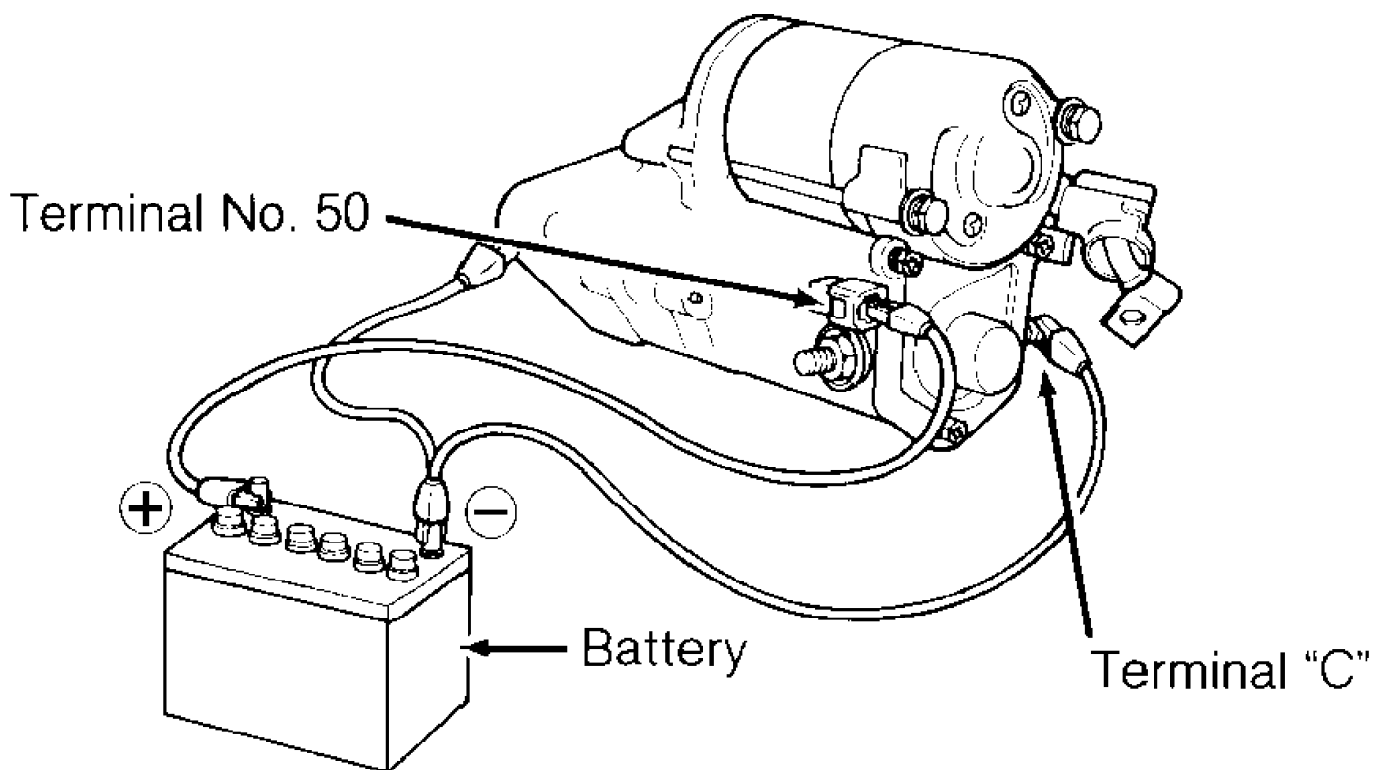
Disconnect field coil lead from terminal "C". Connect jumper wires from negative battery terminal to terminal "C" and to starter housing. When wire is connected from positive battery terminal to terminal No. 50, clutch pinion gear should extend fully. See Figs. 9-11. If clutch pinion gear does not move, replace solenoid. If clutch pinion gear does move, go to next test.

#### Hold-In Coil Test

With battery connected as in previous test and clutch pinion gear still extended, disconnect jumper wire from starter terminal "C". See Figs. 9-11. Clutch pinion gear should remain extended. If clutch pinion gear does not remain extended, replace solenoid. If clutch pinion gear does remain extended, go to next test.

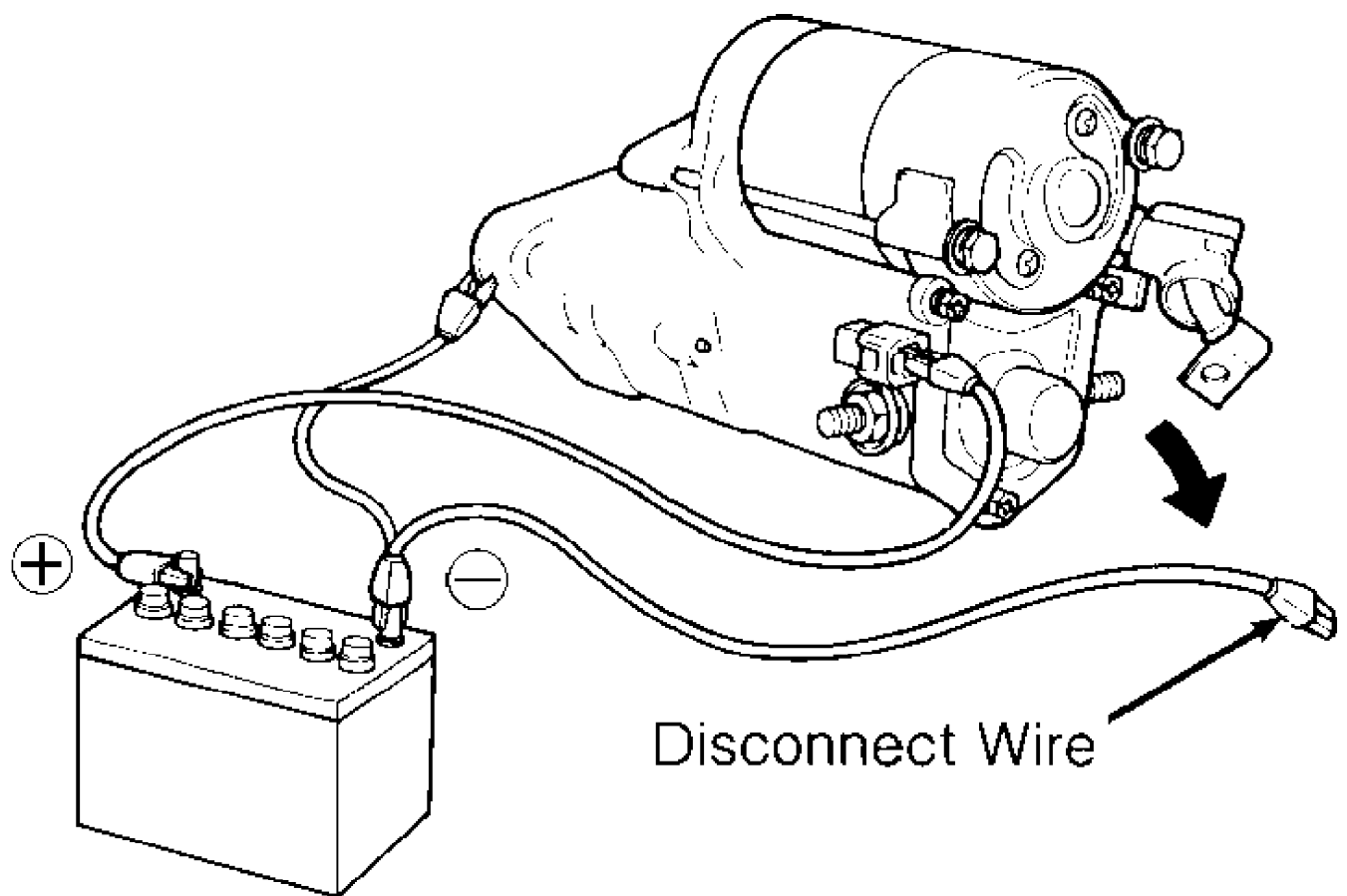
#### Drive Pinion Return Test

Disconnect jumper wire from negative battery terminal to starter housing. See Figs. 9-11. Pinion gear should now retract. If it does not retract, replace solenoid.



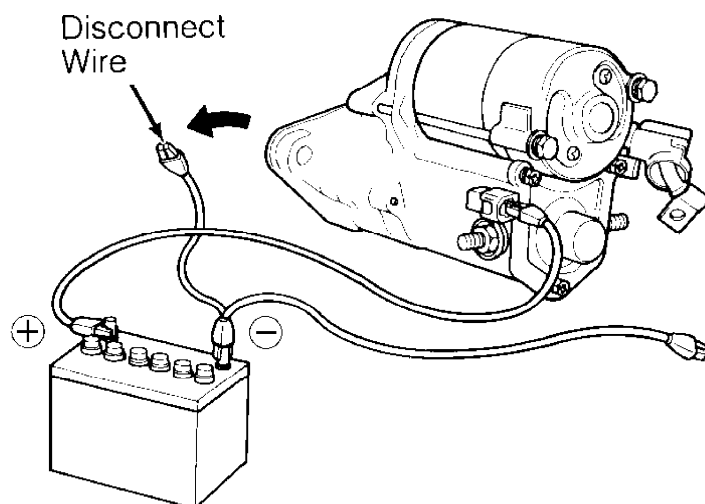
93B02165

Fig. 9: Testing Solenoid Pull-In Coil  
Courtesy of Toyota Motor Sales, U.S.A.Inc.



93D02166

Fig. 10: Testing Solenoid Hold-In Coil  
Courtesy of Toyota Motor Sales, U.S.A.Inc.



93F02167

Fig. 11: Testing Solenoid Drive Pinion Return  
Courtesy of Toyota Motor Sales, U.S.A.Inc.

## STARTER COMPONENT INSPECTION

#### Armature Coil

1) Using ohmmeter, check for continuity between armature coil core and insulation between commutator segments. If continuity is present, replace armature. Check armature for shorts using a growler. Replace armature as necessary.

2) Check for continuity between segments of commutator. If continuity is not present between any segment, replace armature.

#### Brushes & Springs

1) Check brush length. If length is less than specification, replace brushes. See NIPPONDENSO STARTER SPECIFICATIONS table under STARTER SPECIFICATIONS.

2) Check brush holders, springs, spring clip and insulation between positive and negative holders. Verify no continuity between positive and negative brush holders. Repair or replace components as needed.

#### Clutch Assembly & Gears

1) Inspect teeth on pinion gear, idler gear and clutch assembly for wear or damage. If damaged, replace gear or clutch assembly and inspect flywheel ring gear for wear or damage.

2) Inspect clutch pinion gear by rotating pinion gear. Depending on engine, pinion gear will rotate freely in one direction and lock when rotated in opposite direction. On 3S-GTE and 5S-FE, clutch pinion gear will lock when rotated in a clockwise direction. On all other models, lock in a counterclockwise direction. If necessary replace, clutch assembly.

#### Commutator

1) If commutator surface is dirty or burnt, it can be cleaned with No. 400 grit sandpaper or on a lathe. If commutator runout (out-of-round) is more than .002" (.05 mm), turn commutator on a lathe. Wear or cutting limit of commutator diameter is 1.06" (27 mm) for 0.8 kW starter, 1.14" (29.0 mm) for 1.2, 1.4 and 1.6 kW starter, and 1.34" (34 mm) for 2.2 kW starter. If commutator diameter is less than minimum, replace armature.

2) Ensure undercut depth between commutator segments are clean, free of debris, and that edges are smooth. Minimum undercut depth is .008" (.20 mm). If undercut depth is less than minimum use a hacksaw blade to correct to a depth of .008-.024" (.20-.60 mm).

#### Field Frame (Field Coil)

Verify continuity between lead wire and field coil brush lead. If continuity is not present, replace field coil. Verify there is no continuity between field coil end and field frame. If continuity exists, replace or repair field frame.

## REMOVAL & INSTALLATION

### STARTER MOTOR

**NOTE:** Only starters with specific removal procedures are listed. On models not listed, disconnect negative battery cable and remove starter. To install, reverse removal procedure. See TORQUE SPECIFICATIONS.

#### Removal & Installation (4A-FE)

Disconnect negative battery cable. Raise vehicle. Remove suspension lower crossmember to access lower starter bolt. Remove air cleaner cap to access upper starter bolt. Remove starter, then remove starter terminal/wire cover and remove starter wires. To install, reverse removal procedure. Tighten starter mounting bolts to 29 ft.

lbs. (39 N.m). Tighten suspension lower crossmember to 112 ft. lbs. (152 N.m).

#### Removal & Installation (3S-GTE)

1) Disconnect negative battery cable. Disconnect electrical connectors and hoses from air cleaner assembly. Remove air intake tube from throttle body. Remove air intake tube and air cleaner top cap as an assembly. Remove air filter and air cleaner case from engine compartment.

2) Remove engine relay box from battery bracket. Remove battery and tray. Remove starter terminal/wire cover and remove starter wires. Remove starter. To install, reverse removal procedure. Tighten starter mounting bolts to 29 ft. lbs. (39 N.m).

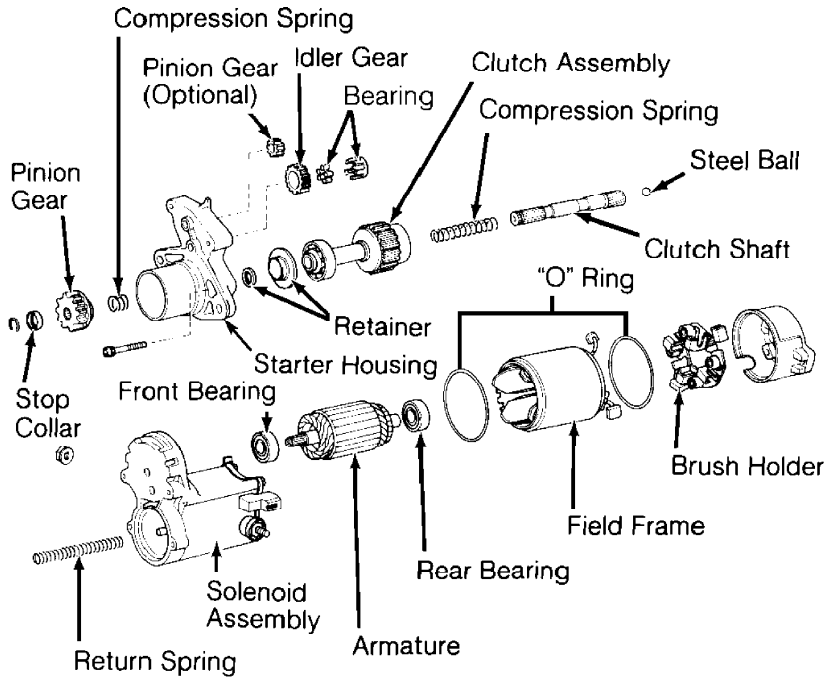
#### Removal & Installation (5S-FE)

1) Disconnect negative battery cable. Disconnect air intake temperature sensor connector from air cleaner assembly. Remove air intake tube from throttle body. Remove air intake tube and air cleaner from throttle body. Remove air filter and air cleaner case from engine compartment.

2) Remove engine relay box and locate out of way. Remove cruise control actuator on models equipped with ABS brakes. Remove starter terminal/wire cover and remove starter wires. Remove starter. To install, reverse removal procedure. Tighten starter mounting bolts to 29 ft. lbs. (39 N.m).

## OVERHAUL

NOTE: Overhaul procedures are not available from manufacturer. For exploded views of starters, see Fig. 12.



92J01589

Fig. 12: Exploded View of Gear Reduction Starter (3S-GTE, 5S-FE)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## STARTER SPECIFICATIONS

## NIPPONDENSO STARTER SPECIFICATIONS

| Application                       | Specification          |
|-----------------------------------|------------------------|
| Brush Minimum Length              |                        |
| 1.4 & 1.6 kW                      | .394" (10.0 mm)        |
| 1.0 kW (1)                        | .335" (8.5 mm)         |
| Brush Spring Load                 |                        |
| 1.4 & 1.6 kW                      | 3.9-5.3 Lbs. (18-24 N) |
| Commutator Minimum Diameter       |                        |
| 0.8 kW                            | 1.06" (27 mm)          |
| 1.0, 1.2, 1.4, 1.6 kW             | 1.14" (29.0 mm)        |
| 2.2 kW                            | 1.34" (34 mm)          |
| Commutator Minimum Undercut Depth | .008" (.2 mm)          |
| Commutator Runout                 | .002" (.05 mm)         |
| Armature                          |                        |
| Core Runout                       | .002" (.05 mm)         |
| End Play                          | .002" (.05 mm)         |

(1) Reduction gear.

## TORQUE SPECIFICATIONS

### TORQUE SPECIFICATIONS

| Application                         | Ft. Lbs. (N.m) |
|-------------------------------------|----------------|
| Lower Crossmember Bolts (4A-FE)     | 112 (152)      |
| Starter Mounting Bolts (All Others) | 29 (39)        |

# STEERING COLUMN SWITCHES

1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Steering Column Switches

Celica

## DESCRIPTION

**\* PLEASE READ THIS FIRST \***

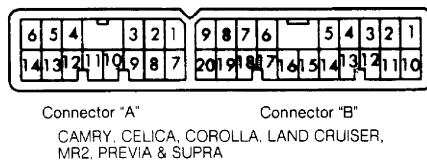
**WARNING:** Celica is equipped with a driver-side air bag. Before servicing steering column, disable air bag system to prevent air bag deployment. To disable air bag system, turn ignition switch to LOCK position. Disconnect negative battery cable. Wait at least 90 seconds to allow back-up power supply to discharge. For more information, see AIR BAG RESTRAINT SYSTEM article in the ACCESSORIES/SAFETY EQUIPMENT section.

## TESTING

**NOTE:** This article TABLE contains information on headlight and taillight system relays because they are part of exterior lighting system.

## COMBINATION SWITCH

**NOTE:** Combination switch includes switches for headlights, turn signals and windshield wipers. For testing, see appropriate component procedure.



**93J84664**

Fig. 1: Combination Switch Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A. Inc.

## HEADLIGHT, DIMMER & PASSING SWITCHES

With headlight switch in specified position, check continuity between specified terminals of combination switch connector. See HEADLIGHT SWITCH CONTINUITY TEST table and/or DIMMER & PASSING SWITCHES CONTINUITY TEST table. See Fig. 1. If continuity is not as specified, replace headlight switch or turn signal/dimmer switch as necessary.

### HEADLIGHT SWITCH CONTINUITY TEST TABLE

| Application &<br>Switch Position | Terminals<br>(1)         | Continuity<br>Present |
|----------------------------------|--------------------------|-----------------------|
| Off .....                        | A2, A11, A13 & B20 ..... | No                    |
| (Next Position) .....            | A11 & B20 .....          | Yes                   |
| Tail .....                       | A2, A11 & B20 .....      | Yes                   |
| Head .....                       | A2, A11 & A13 .....      | Yes                   |

(1) - See Fig. 1.

DIMMER & PASSING SWITCHES CONTINUITY TEST TABLE

| Application &<br>Switch Position | Terminals<br>(1) | Continuity<br>Present |
|----------------------------------|------------------|-----------------------|
| Celica .....                     | (2) .....        | (2)                   |

(1) - See Fig. 1.  
(2) - See HEADLIGHT SWITCH CONTINUITY TEST table.

HEADLIGHT CONTROL RELAY

Remove relay. See HEADLIGHT CONTROL RELAY LOCATION table.  
Apply battery voltage across specified terminals of relay connector.  
See HEADLIGHT CONTROL RELAY TEST table. See Fig. 2. Check continuity  
between other specified terminals. If there is no continuity, replace  
relay.

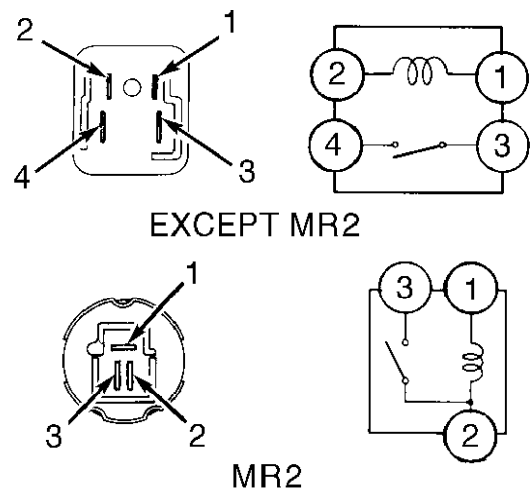
HEADLIGHT CONTROL RELAY LOCATION TABLE

| Application  | Location                                   |
|--------------|--------------------------------------------|
| Celica ..... | Left Side Engine Bay, In Relay Block No. 2 |

HEADLIGHT CONTROL RELAY TEST TABLE (1)

| Application<br>Type | Apply Voltage To<br>These Terminals | Continuity Between<br>These Terminals |
|---------------------|-------------------------------------|---------------------------------------|
| Celica .....        | 1 & 2 .....                         | 3 & 4                                 |

(1) - See Fig. 2.



93B02170  
Fig. 2: Headlight Control Relay Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

TAILLIGHT CONTROL RELAY

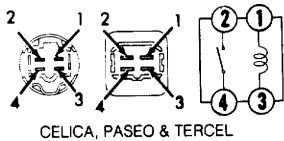
Remove relay. See TAILLIGHT CONTROL RELAY LOCATION table.  
Apply battery voltage across specified terminals of relay connector.  
See TAILLIGHT CONTROL RELAY TEST table. See Fig. 3. Check continuity  
between other specified terminals. If there is no continuity, replace  
relay.

TAILLIGHT CONTROL RELAY LOCATION TABLE

| Application                   | Location                                      |
|-------------------------------|-----------------------------------------------|
| Celica .....                  | Behind Left Side Of Dash In Relay Block No. 1 |
| (1) - Theft Deterrent System. |                                               |

TAILLIGHT CONTROL RELAY TEST TABLE (1)

| Application Type  | Apply Voltage To These Terminals | Continuity Between These Terminals |
|-------------------|----------------------------------|------------------------------------|
| Celica .....      | 1 & 3 .....                      | 2 & 4                              |
| (1) - See Fig. 3. |                                  |                                    |



93D84668  
Fig. 3: Taillight Control Relay Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

HORN SWITCH

- 1) Disconnect combination switch connector. See Fig. 6. With horn pad pressed and held down (horn ON), check continuity between appropriate wire terminal of combination switch connector. See Fig. 1. See appropriate wiring diagram.
- 2) If there is no continuity, repair wiring or replace components as necessary. Components include horn pad and contact pin, slip ring and spiral cable (with air bag). If there is continuity, check for open in horn circuit, faulty horn or faulty horn relay. See HORN RELAY LOCATION table.

HORN RELAY LOCATION TABLE

| Application  | Location                                    |
|--------------|---------------------------------------------|
| Celica ..... | Right Side Engine Bay, In Relay Block No. 5 |

IGNITION SWITCH

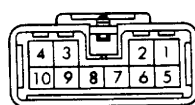
Disconnect ignition switch connector. With ignition switch in specified position, check continuity between specified terminals of ignition switch connector. See IGNITION SWITCH CONTINUITY TEST table. See Fig. 4. If continuity is not as specified, replace ignition switch.



# IGNITION SWITCH CONTINUITY TEST TABLE

| Application &<br>Switch Position | Terminals<br>(1)            | Continuity<br>Present |
|----------------------------------|-----------------------------|-----------------------|
| Lock .....                       | 2, 3, 4, 6, 7, 9 & 10 ..... | No                    |
| Accessory .....                  | 3 & 4 .....                 | Yes                   |
| On .....                         | 2, 3 & 4; 9 & 10 .....      | Yes                   |
| Start .....                      | 2, 4 & 7; 6, 9 & 10 .....   | Yes                   |

(1) - See Fig. 4.



CAMRY, CELICA,  
LAND CRUISER,  
MR2 & PREVIA

93184671

Fig. 4: Ignition Switch Connector Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TURN SIGNAL SWITCH

Disconnect combination switch connector. See Fig. 6. With turn signal switch in specified position, check continuity between specified terminals of combination switch connector. See TURN SIGNAL SWITCH CONTINUITY TEST table. See Fig. 1. If continuity is not as specified, replace turn signal switch.

# TURN SIGNAL SWITCH CONTINUITY TEST TABLE

| Application &<br>Switch Position | Terminals<br>(1)  | Continuity<br>Present |
|----------------------------------|-------------------|-----------------------|
| Left .....                       | A1 & A5 .....     | Yes                   |
| Neutral .....                    | A1, A5 & A8 ..... | No                    |
| Right .....                      | A1 & A8 .....     | Yes                   |

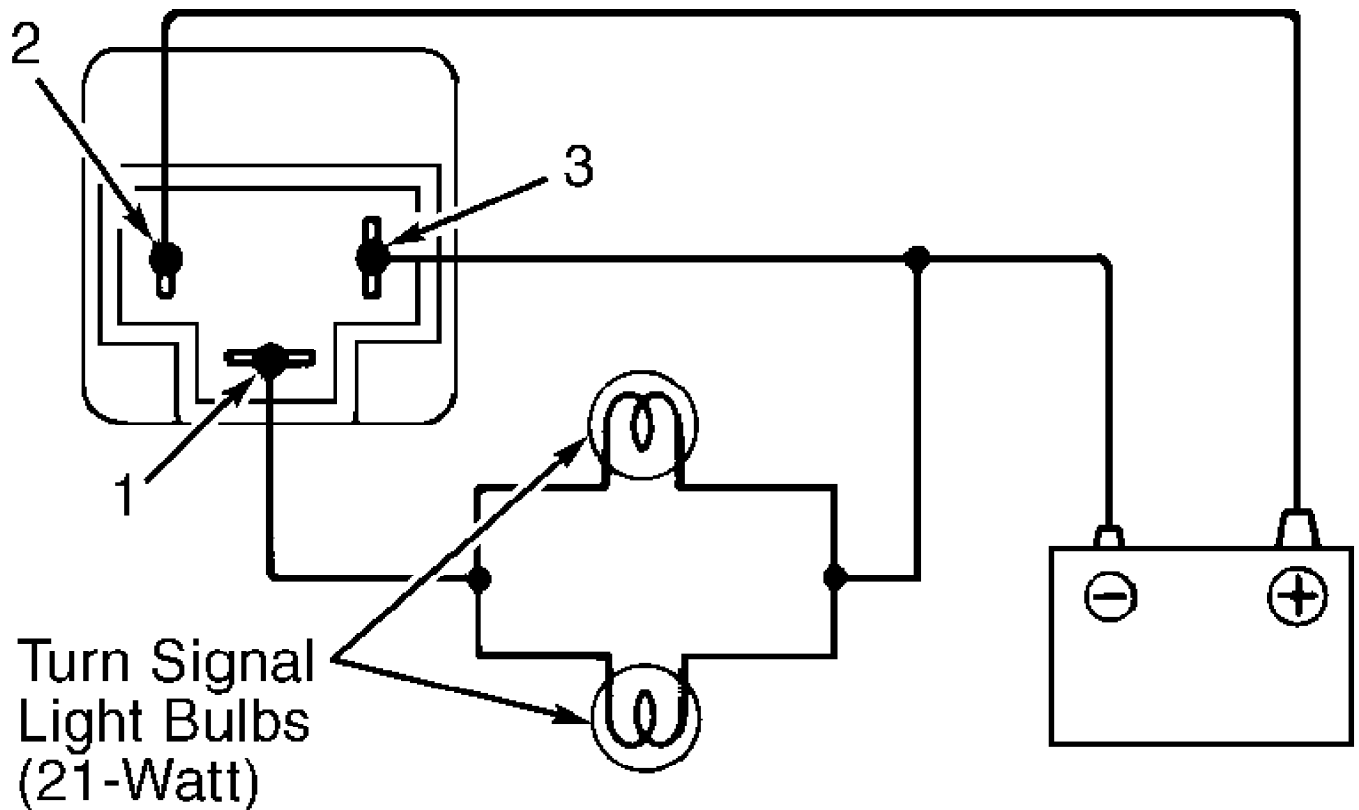
(1) - See Fig. 1.

## TURN SIGNAL FLASHER

Remove turn signal flasher. See TURN SIGNAL FLASHER LOCATION table. At turn signal flasher terminals, connect battery positive lead to terminal No. 2. See Fig. 5. Connect battery negative lead to terminal No. 3. Connect 2 turn signal bulbs in parallel between terminal No. 1 and battery negative terminal. Replace turn signal flasher if bulbs do not flash 60-120 times per minute.

# TURN SIGNAL FLASHER LOCATION TABLE

| Application  | Location                                     |
|--------------|----------------------------------------------|
| Celica ..... | Behind Left Kick Panel, In Relay Block No. 1 |



93A84673

Fig. 5: Testing Turn Signal Flasher  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIPER/WASHER SWITCH

NOTE: See appropriate WIPER/WASHER SYSTEM article in the ACCESSORIES/SAFETY EQUIPMENT section.

## REMOVAL & INSTALLATION

\* PLEASE READ THIS FIRST \*

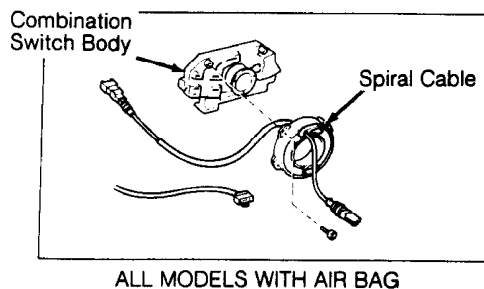
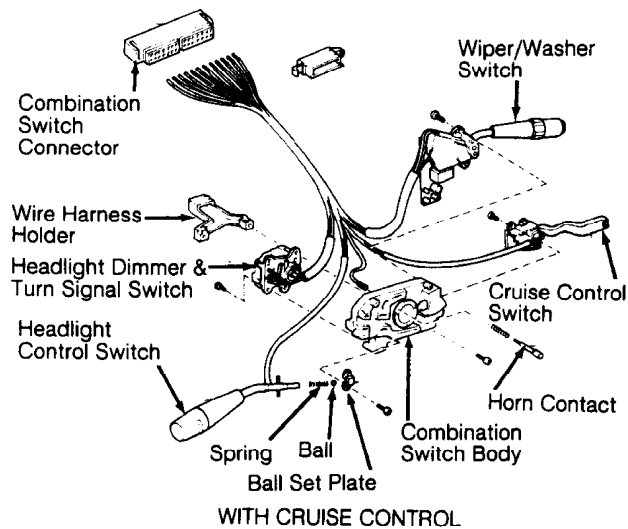
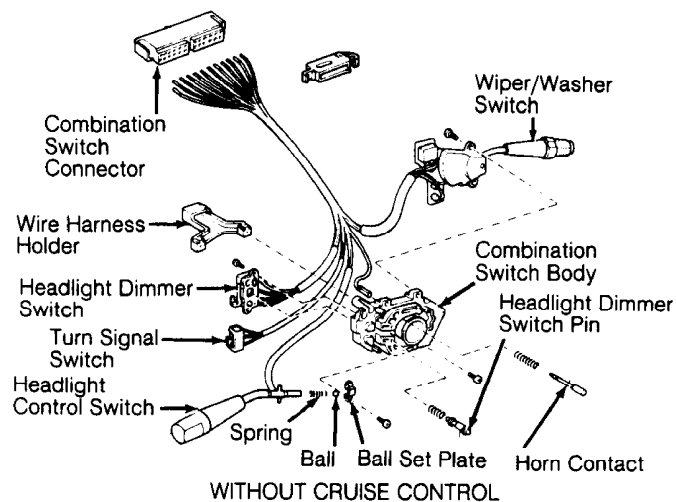
**WARNING:** Celica is equipped with a driver-side air bag. Before servicing steering column, disable air bag system to prevent air bag deployment. To disable air bag system, turn ignition switch to LOCK position. Disconnect negative battery cable. Wait at least 90 seconds to allow back-up power supply to discharge. For more information, see AIR BAG RESTRAINT SYSTEM article in the ACCESSORIES/SAFETY EQUIPMENT section.

## COMBINATION SWITCH

1) Remove steering wheel. See STEERING WHEEL. Remove upper and lower steering column covers (it may be necessary to first remove lower finish panel from instrument panel). Disconnect combination switch connector. See Fig. 6.

2) On vehicles with cruise control switches on steering wheel, disconnect cruise control electrical connector(s) from slip ring. Remove slip ring from combination switch.

3) Remove wire ties securing combination switch wiring harness to column. Remove screws securing combination switch to steering column. Remove combination switch. To install, reverse removal procedure.



93J02174

Fig. 6: Exploded View Of Combination Switch  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## IGNITION SWITCH

1) Disconnect negative battery cable. Remove upper and lower steering column covers (it may be necessary to first remove lower instrument panel trim panel). Remove outer plastic trim cover from lock cylinder assembly (if equipped).

2) Disconnect ignition switch electrical connector(s). Remove screw(s) retaining ignition switch to lock cylinder. Remove ignition switch from lock cylinder. To install, reverse removal procedure.

## STEERING WHEEL

### Removal

1) Ensure front wheels are in straight-ahead position. Turn ignition switch to LOCK position. Remove key. Disconnect negative battery cable. Remove 2 screw covers from outer sides of steering wheel. Using Torx Wrench (T30), loosen 4 horn pad screws until screw head is snug against screw case. See Fig. 7.

2) Carefully pull steering wheel pad away from steering wheel enough to unlock and disconnect air bag electrical connector. DO NOT pull on electrical connector or wiring. Store air bag pad aside with pad facing upward.

3) Remove steering wheel lock nut and washer from steering shaft. Make alignment mark on steering shaft and steering wheel for installation reference. Using appropriate steering wheel puller, pull steering wheel from shaft while guiding spiral cable wire through steering wheel opening.

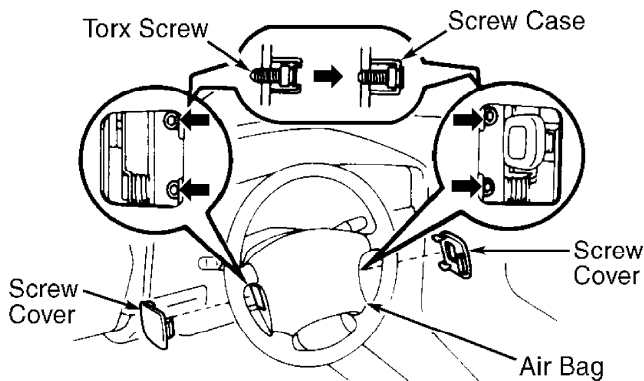
### Installation

1) Ensure front wheels are in straight-ahead position. Install spiral cable on combination switch body. Turn spiral cable counterclockwise by hand until it is hard to turn. Turn spiral cable clockwise about 2 1/2 turns, aligning marks on spiral cable.

2) Install steering wheel, guiding spiral cable through opening in steering wheel. Align reference marks on steering shaft and steering wheel. Tighten steering wheel lock nut to specification. See TORQUE SPECIFICATIONS.

3) Connect air bag module electrical connector. Close connector lock. Ensure air bag module Torx screws are retracted and snug against screw case. See Fig. 7.

4) Install air bag module, ensuring wiring is not pinched and does not interfere with other moving parts. Tighten air bag module Torx screws to specification. Install screw covers. Connect negative battery cable.



91G03959

Fig. 7: Removing Air Bag From Steering Wheel  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TURN SIGNAL/DIMMER SWITCH

NOTE: To remove turn signal/dimmer switch, it may be necessary to first remove headlight control switch, spring, ball and ball set plate. See Fig. 6.

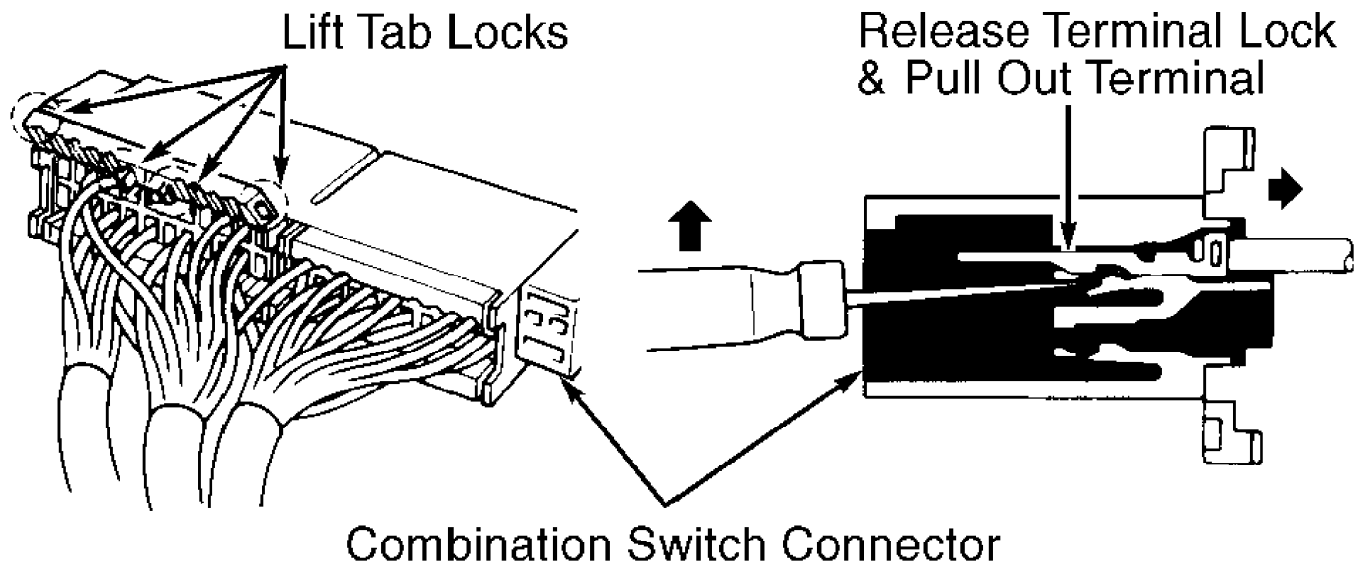
Removal

1) Disconnect negative battery cable. Remove steering column covers. At combination switch connector, lift tab locks and remove clamp. See Fig. 8. Identify turn signal/dimmer switch wire terminals in combination switch connector.

2) Using miniature screwdriver, release wire terminal lock inside connector. Pull wires from connector and from harness jacket tubing. Remove turn signal/dimmer switch screws from combination switch body. See Fig. 6. Remove turn signal/dimmer switch.

Installation

To install, reverse removal procedure. To install wire terminals into combination switch connector cavities, push wire terminal into cavity until terminal clicks into place.



93A02179

Fig. 8: Releasing Wire Terminals Inside Connector  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TORQUE SPECIFICATIONS

### TORQUE SPECIFICATIONS TABLE

| Application                      | Ft. Lbs. (N.m)  |
|----------------------------------|-----------------|
| Steering Wheel Nut .....         | 26 (35)         |
|                                  | INCH Lbs. (N.m) |
| Air Bag Module Torx Screws ..... | 65 (7.3)        |

# STEERING COLUMN - AUTOMATIC TILT WHEEL

1993 Toyota Celica

1993 STEERING

Toyota - Steering Columns - Automatic Tilt Wheel

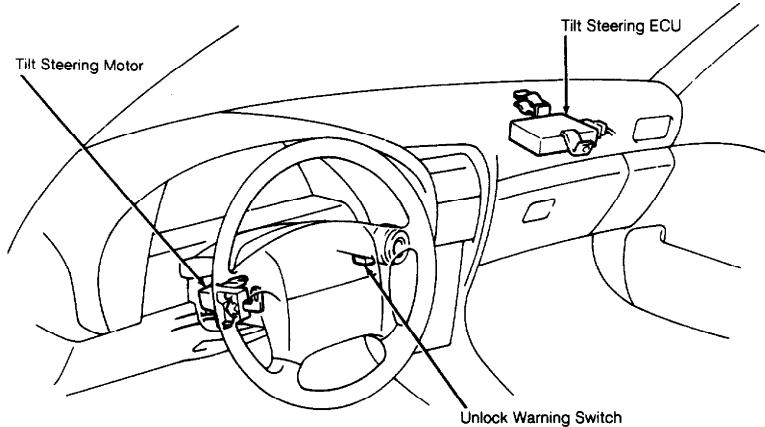
Celica All-Trac

## DESCRIPTION & OPERATION

Automatic tilt steering column uses a standard tilt steering column with a motor located in the column housing. The motor allows the column to automatically swing up when the ignition key is removed for easier exit and entry to the vehicle. When the key is inserted in the ignition and the steering wheel is pulled down, it will automatically return to the pre-set position.

The position of the wheel can be adjusted by pulling the lock release knob, placing the wheel in the desired position and releasing the knob. An unlock warning switch is located in the ignition key cylinder and a tilt steering ECU is located behind the instrument panel above the glove box.

NOTE: For Adjustments and Removal and Installation of steering wheel pad, combination switch, ignition switch and Overhaul of steering column see STEERING COLUMN - TILT article in the STEERING section.



93E84594

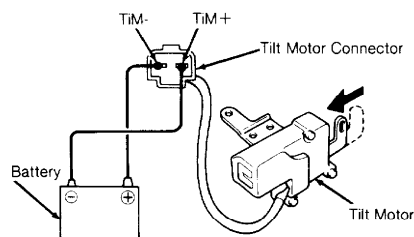
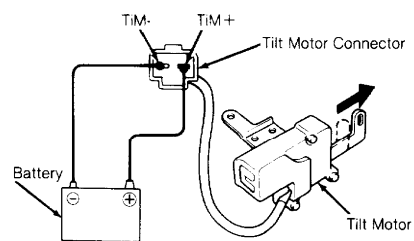
Fig. 1: Locating Celica All-Trac Auto Tilt Wheel Steer Col Components  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## TESTING

### COMPONENT TESTING

Tilt Steering Motor

1) Disconnect tilt motor electrical connector. Connect positive lead from battery to connector terminal TiM+ and negative lead to terminal TiM-. Motor arm should move away from housing. Reverse battery leads to terminals. Motor arm should move toward housing. If motor does operate as specified, replace motor. See Fig. 2.



93F84595

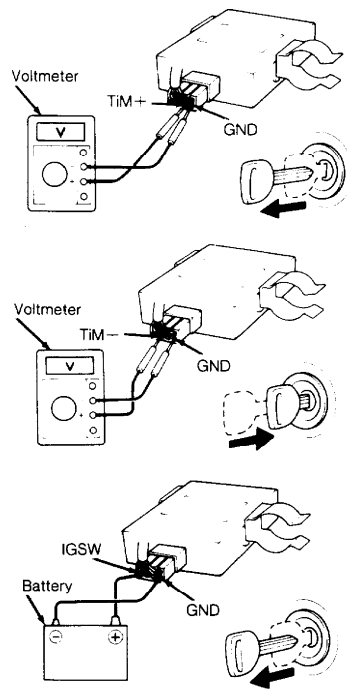
Fig. 2: Tilt Steering Motor Test  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Tilt Steering ECU

1) Place ignition key in ignition switch. Remove glove box to gain access to ECU. Connect positive voltmeter lead to ECU terminal TiM+ and negative lead to GRD terminal. See Fig. 3. Check that meter needle jumps once to approximately 10 volts when the key is removed from ignition.

2) Connect voltmeter positive lead to ECU terminal TiM- and negative lead to GRD terminal. Check that meter needle jumps once to approximately 10 volts when key is inserted in ignition.

3) Connect positive lead from battery to ECU terminal lead IGSW and negative lead to GRD terminal. Check that tilt-away feature does not operate when key is removed from ignition. If ECU operation is not as specified, replace ECU.



**93G84596**

Fig. 3: Tilt Steering ECU Test  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### Key Unlock Warning Switch

Using ohmmeter, check for continuity between terminals 1 and 5 of ignition switch. With ignition key removed, there should be no continuity. With ignition key installed there should be continuity. If continuity is not as specified, replace unlock switch.

### SYSTEM TESTING

1) Check battery voltage. If less than 8 volts, check battery and charging system. If more than 8 volts, check POWER fuse in fuse block. If fuse is blown, replace fuse and check auto-tilt operation. If system does not operate, check for short circuit in wire harness between fuse and tilt ECU terminal B+. If fuse is okay, but system does not operate, disconnect tilt ECU connector.

2) Using volt-ohmmeter, check for battery voltage between terminal B+ and body ground. See Fig. 4. If voltage is okay, go to next step. If voltage is not okay, check for open circuit between fuse and terminal B+

3) Using volt-ohmmeter, check for continuity between terminal GND and body ground. If continuity exists, go to next step. If no continuity, check body ground, or for open circuit between terminal GND and ground.

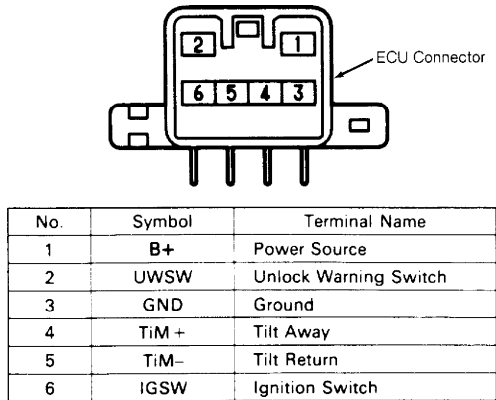
4) Check for continuity between terminal UWSW and GND. There should be no continuity with ignition key removed, and continuity with key inserted in ignition switch. If continuity is okay, go to next step. If continuity is not as specified, check for open circuit between terminal UWSW and unlock warning switch. If circuit is okay, replace unlock warning switch.

5) Check for voltage between terminals IGSW and GND. With ignition switch on, battery voltage should be present. With switch off, 0 volts should be present. If voltage is okay, go to next step.



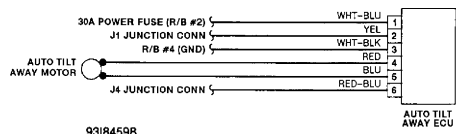
If voltage is not as specified, check for open circuit between terminal IGSW and ignition switch.

6) Check for continuity between terminals TiM+ and TiM-. If continuity exists, check ECU. See Tilt Steering ECU under COMPONENT TESTING. If ECU fails test, replace ECU. If continuity does not exist, check for open circuit between terminals TiM+ and TiM-. If circuit is okay, check tilt motor. See Tilt Steering Motor under COMPONENT TESTING. If motor fails test, replace motor.



93H84597  
Fig. 4: Tilt Steering ECU Connector Terminal Identification  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS



93I84598  
Fig. 5: Automatic Tilt Wheel Steering Column Wiring Diagram

# SUN ROOF - POWER

1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Power Sun Roof

Celica

## DESCRIPTION & OPERATION

The ignition switch must be in ON position for sun roof to operate. The sun roof has a tilt mode and sliding mode. When closing sun roof in sliding mode, sun roof will stop partway until sun roof switch is pressed a second time.

## ADJUSTMENTS

NOTE: Remove sun roof trim plates from top of sun roof opening before making adjustments.

### SUN ROOF HEIGHT

Close sun roof fully. Measure sun roof height. Roof panel should be even with glass weatherstrip within 0.04" (1.0 mm). If adjustment is required, install or remove shims between bracket and sun roof. Recheck sun roof height.

### SUN ROOF CLEARANCE & ALIGNMENT

#### Closing Clearance

If sun roof closing clearances are unequal between right and left sides, loosen sun roof slider mount nuts. Move either side forward or backward as required. Tighten sun roof slide mount nuts. Recheck sun roof alignment and operation.

#### Front-To-Rear Alignment

If front-to-rear position of sun roof is incorrect, loosen sun roof slider mount nuts. Move sun roof to front or rear as necessary. Tighten sun roof slider mount nuts.

#### Side-To-Side Alignment

If sun roof is mounted to right or left of center line in roof opening, loosen sun roof slider mount nuts. Move sun roof right or left as necessary. Tighten sun roof slider mount nuts.

## TROUBLE SHOOTING

NOTE: Ensure all component terminals and ground connections are clean and tight. Check possible faults in order listed. Repair or replace components and circuits as necessary.

#### Power Sun Roof Operates Abnormally

Defective GAUGE fuse. Defective POWER fuse. Defective power main relay. Defective power sun roof switch. See SUN ROOF SWITCH under TESTING. Defective limit switch. See LIMIT SWITCH under TESTING. Defective motor. See MOTOR under TESTING. Defective sliding roof control relay. See CONTROL RELAY under TESTING. Defective wire harness.

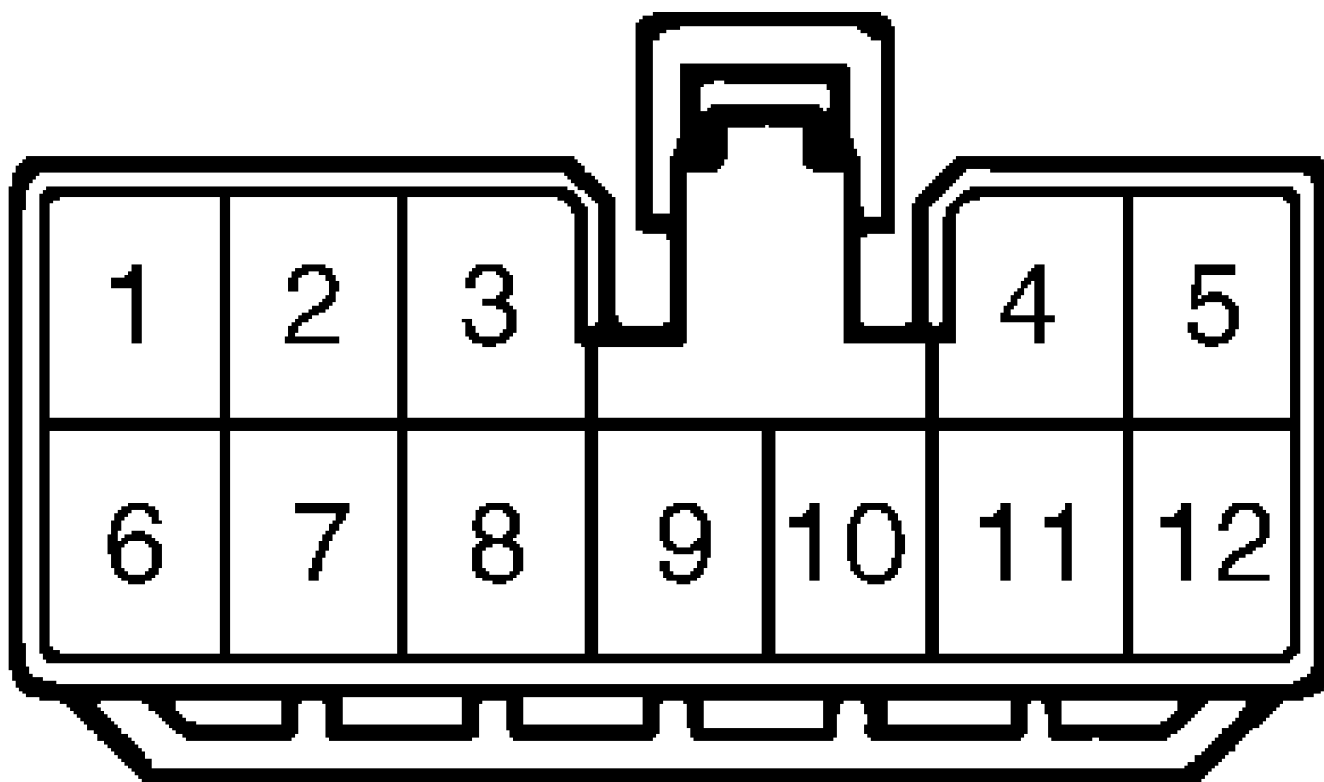
## TESTING

## CONTROL RELAY

See CONTROL RELAY LOCATION table for relay location. Unplug relay connector. Check for continuity or voltage as indicated at control relay connector terminals with sun roof switch or limit switch in specified positions. See appropriate CONTROL RELAY TEST table. If test results are as specified, replace control relay. If test results are not as specified, repair or replace appropriate switch or wire. See Fig. 1.

CONTROL RELAY LOCATION TABLE

| Model        | Location         |
|--------------|------------------|
| Celica ..... | To Left Of Motor |



93C84873

Fig. 1: Control Relay Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

CONTROL RELAY TEST TABLE

| Terminal No. &<br>Switch Position   | Condition     |
|-------------------------------------|---------------|
| 1 And 5 .....                       | Continuity    |
| 2 (1)<br>DOWN, OPEN, Or CLOSE ..... | No Continuity |
| Up .....                            | Continuity    |

|    |                             |                 |
|----|-----------------------------|-----------------|
| 4  | (1) OFF (Roof Open)         | No Continuity   |
| 4  | (1) ON (Roof closed)        | Continuity      |
| 5  | (1)                         | Continuity      |
| 6  | (1)                         |                 |
|    | Ignition Switch LOCK or ACC | No Voltage      |
|    | Ignition Switch ON          | Battery Voltage |
| 7  | (1)                         |                 |
|    | UP, DOWN, Or OPEN           | No Continuity   |
|    | CLOSE                       | Continuity      |
| 8  | (1)                         |                 |
|    | UP, DOWN, Or CLOSE          | No Continuity   |
|    | OPEN                        | Continuity      |
| 9  | (1)                         |                 |
|    | UP, OPEN, Or CLOSE          | No Continuity   |
|    | DOWN                        | Continuity      |
| 10 | (1)                         |                 |
|    | OFF                         |                 |
|    | Roof Tilted Up              | No Continuity   |
|    | Roof Partly Open            | No Continuity   |
|    | ON                          |                 |
|    | Except As Above             | Continuity      |
| 12 | (1)                         | Continuity      |

(1) - Terminal connected to ground.

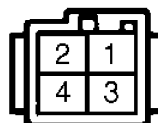
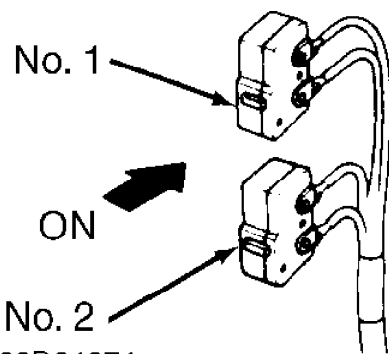
## MOTOR

Unplug sun roof motor connector. Using fused jumper wires, connect motor connector terminals to battery voltage. Motor should rotate smoothly. Transpose jumper wires. Motor should rotate smoothly in opposite direction. Replace motor if operation is not as specified.

## LIMIT SWITCH

1) Unplug switch connector. See Fig. 2. For limit switch No. 1, continuity should not exist between terminals No. 1 and 4 with switch pin released. With pin pressed in, continuity should exist between terminals No. 1 and 4.

2) For limit switch No. 2, continuity should not exist between terminals No. 2 and 4 with switch pin released. With pin pressed in, continuity should exist between terminals No. 2 and 4. Replace switch if continuity is not as specified.

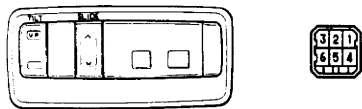


93D84874

Fig. 2: Limit Switch Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## SUN ROOF SWITCH

Remove power sun roof switch. Check for continuity between switch terminals with switch in specified positions. If continuity is not as specified, replace switch. See Fig. 3. See appropriate SUN ROOF SWITCH CONTINUITY TEST table.



**93B84880**  
 Fig. 3: Sun Roof Switch Terminal ID  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

| SUN ROOF SWITCH CONTINUITY TEST TABLE |       |                     |
|---------------------------------------|-------|---------------------|
| Position                              |       | Terminal Continuity |
| Slide                                 |       |                     |
| Off                                   | ..... | (1)                 |
| Open                                  | ..... | 3 & 4               |
| Close                                 | ..... | 4 & 6               |
| Tilt                                  |       |                     |
| Off                                   | ..... | (1)                 |
| Up                                    | ..... | 4 & 5               |
| Down                                  | ..... | 2 & 4               |

(1) - Continuity should not exist.

## REMOVAL & INSTALLATION

### DRIVE CABLES

Removal & Installation  
 See SUN ROOF.

### DRIVE MOTOR

Removal & Installation  
 See SUN ROOF.

### SUN ROOF SWITCH

**NOTE:** Tape the end of a flat blade screwdriver to protect switch and trim panels during removal.

Removal & Installation  
 Disconnect negative battery cable. Pry sun roof switch assembly from roof panel. Remove screws and switch from switch panel. Unplug switch connector. To install, reverse removal procedure.

### SUN ROOF

**WARNING:** Disable air bag (if equipped) before removing sun roof. See appropriate AIR BAG RESTRAINT SYSTEM article in the ACCESSORIES/SAFETY EQUIPMENT section.

**NOTE:** For installation reference, note number and position of shims when removing sun roof.

#### Removal & Installation

1) Tilt sun roof upward. Disconnect negative battery cable. Disable air bag. Remove map light. Remove sunvisors, rear view mirror, opening trim panels, and front assist grips. Remove front pillar trim moldings.

2) Unplug and remove drive motor. Remove sun roof mounting nuts and shims. Pull upward on sun roof glass to remove. Remove drip channel. Remove wind deflector. Remove cable guide. To install, reverse removal procedure.

### **WIND DEFLECTOR**

See SUN ROOF.

### **WIRING DIAGRAMS**

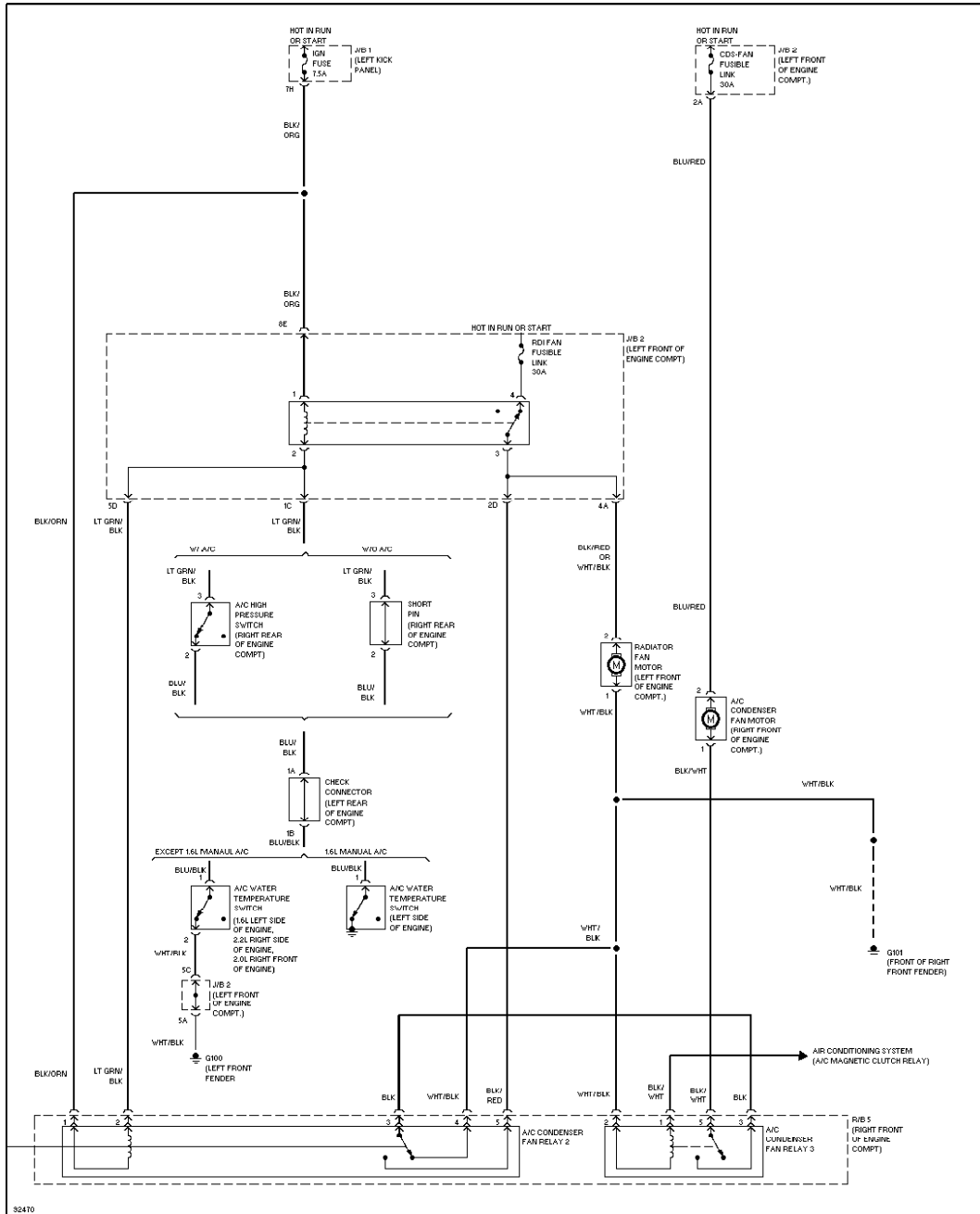
Proceed to chassis WIRING DIAGRAMS article in WIRING DIAGRAMS section.

# SYSTEM WIRING DIAGRAMS

1993 Toyota Celica

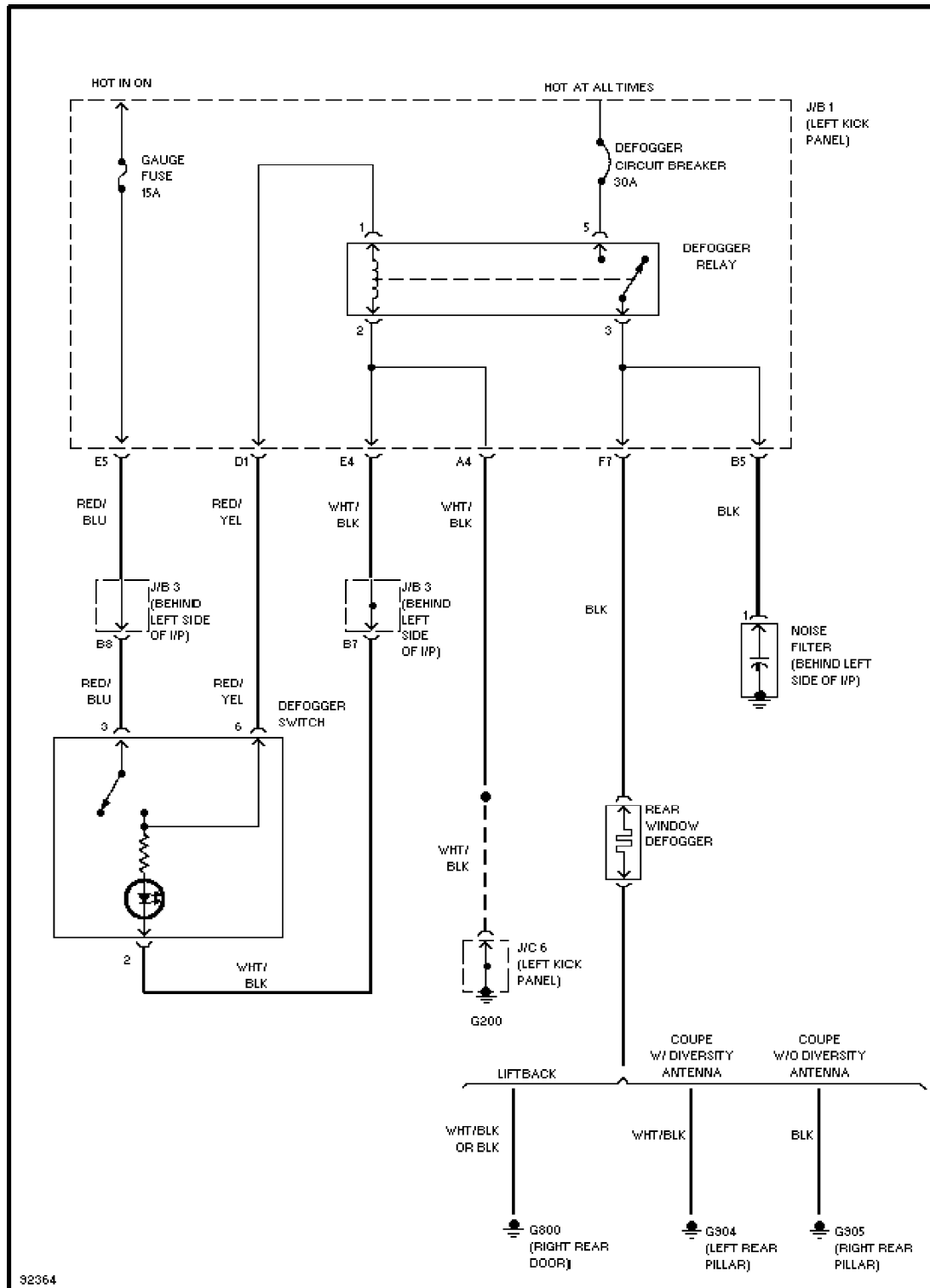
1993 System Wiring Diagrams  
Toyota - Celica

## COOLING FAN



Cooling Fan Circuit

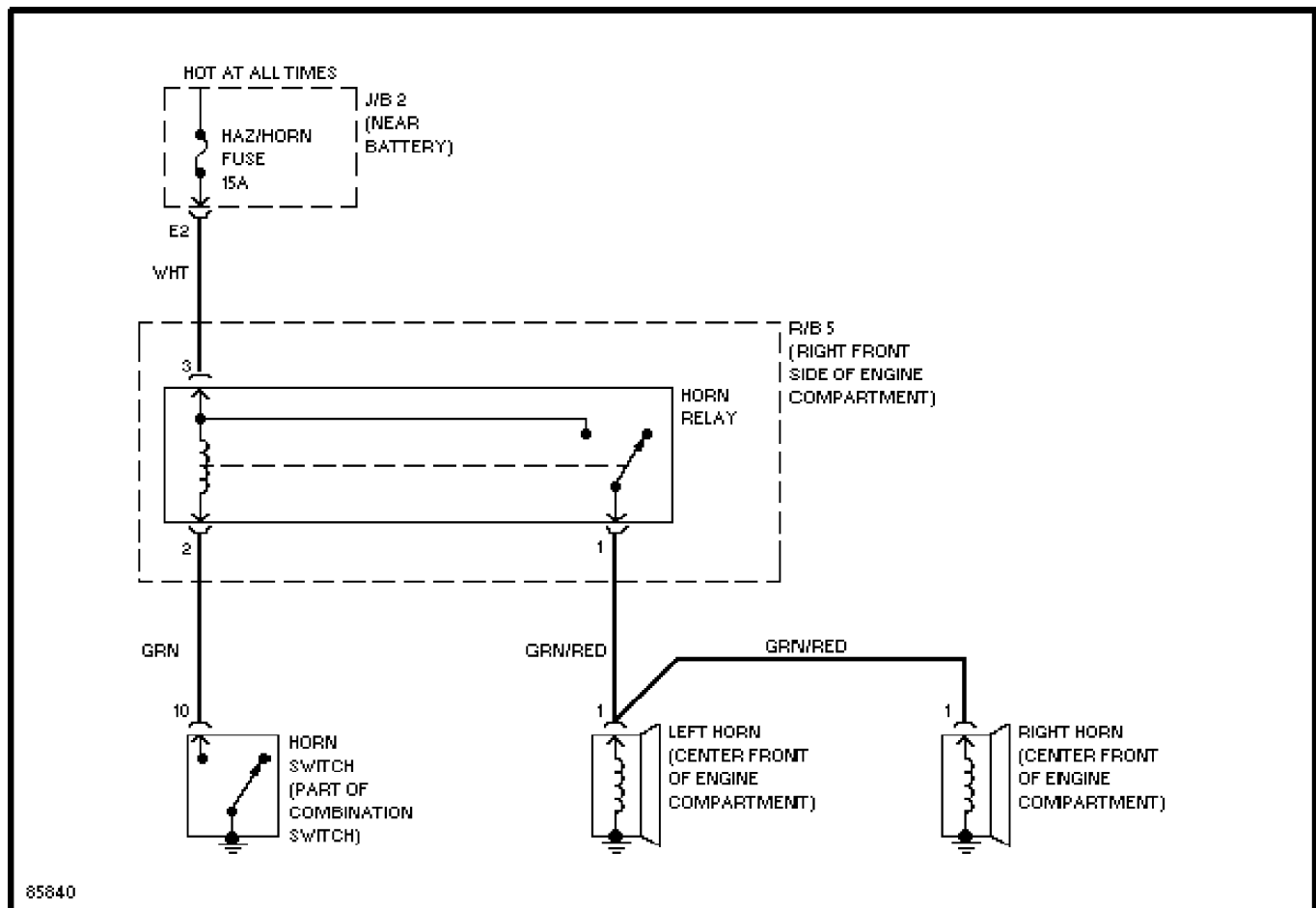
## DEFOGGERS



Defogger Circuit

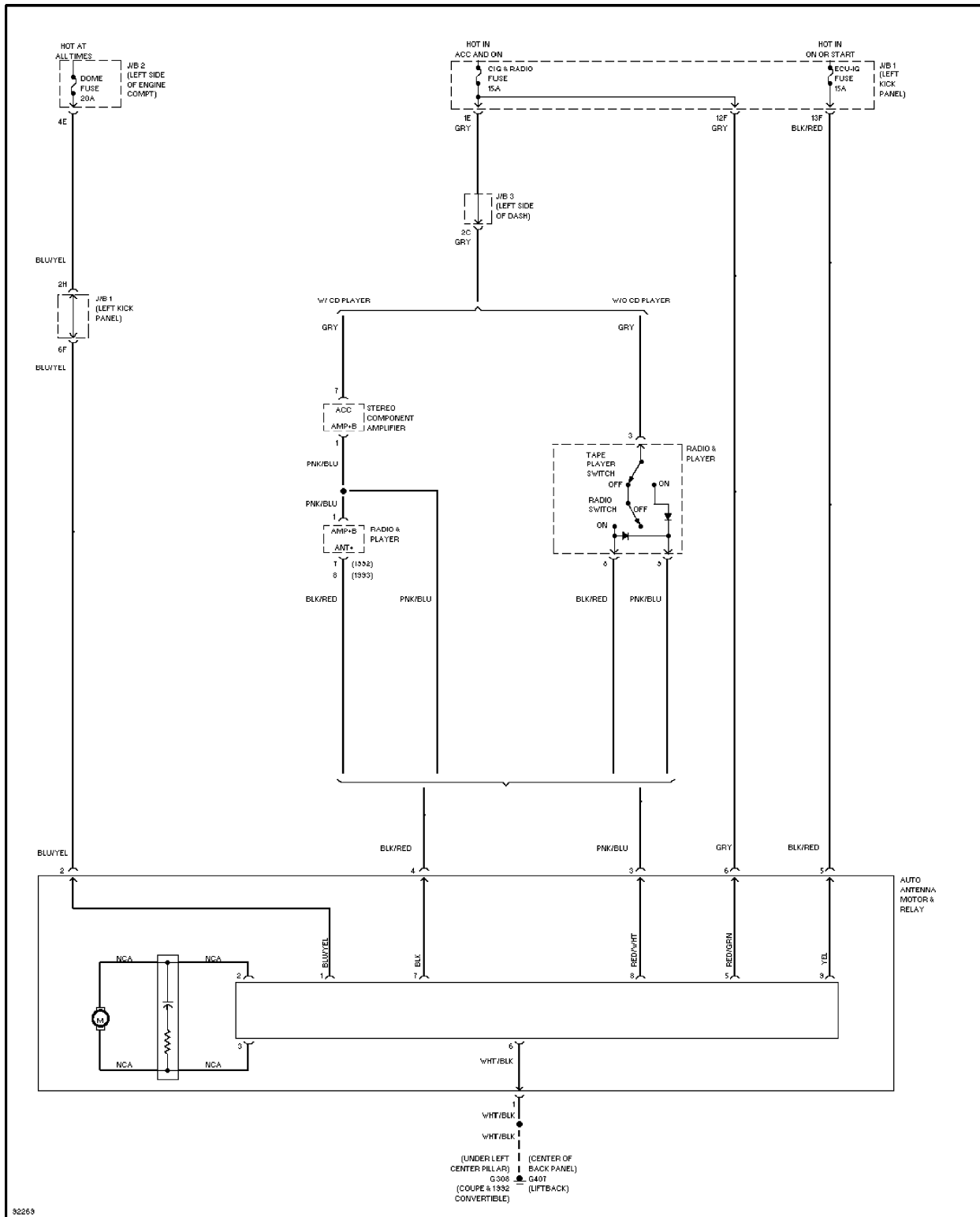
## HORN





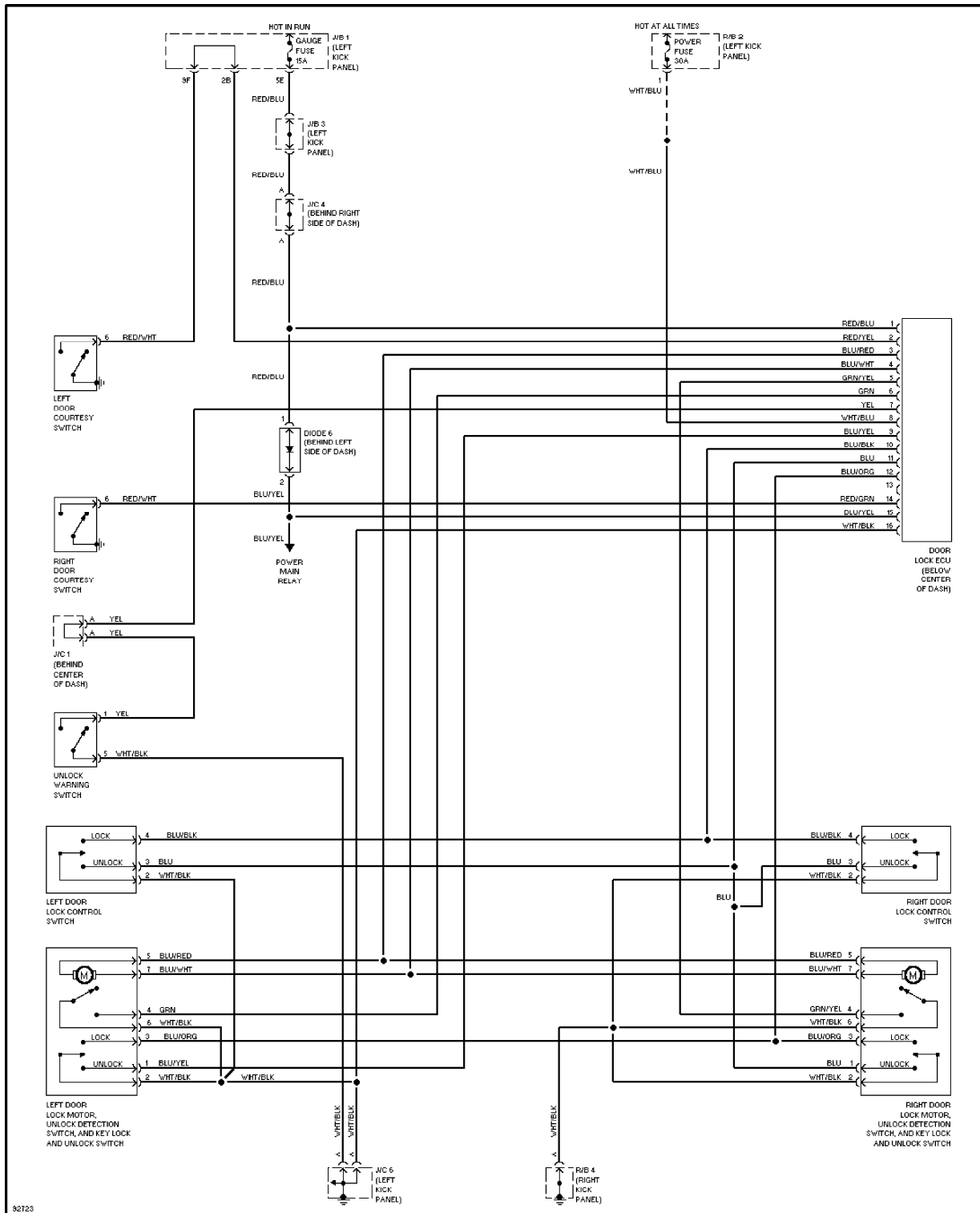
Horn Circuit

## POWER ANTENNA



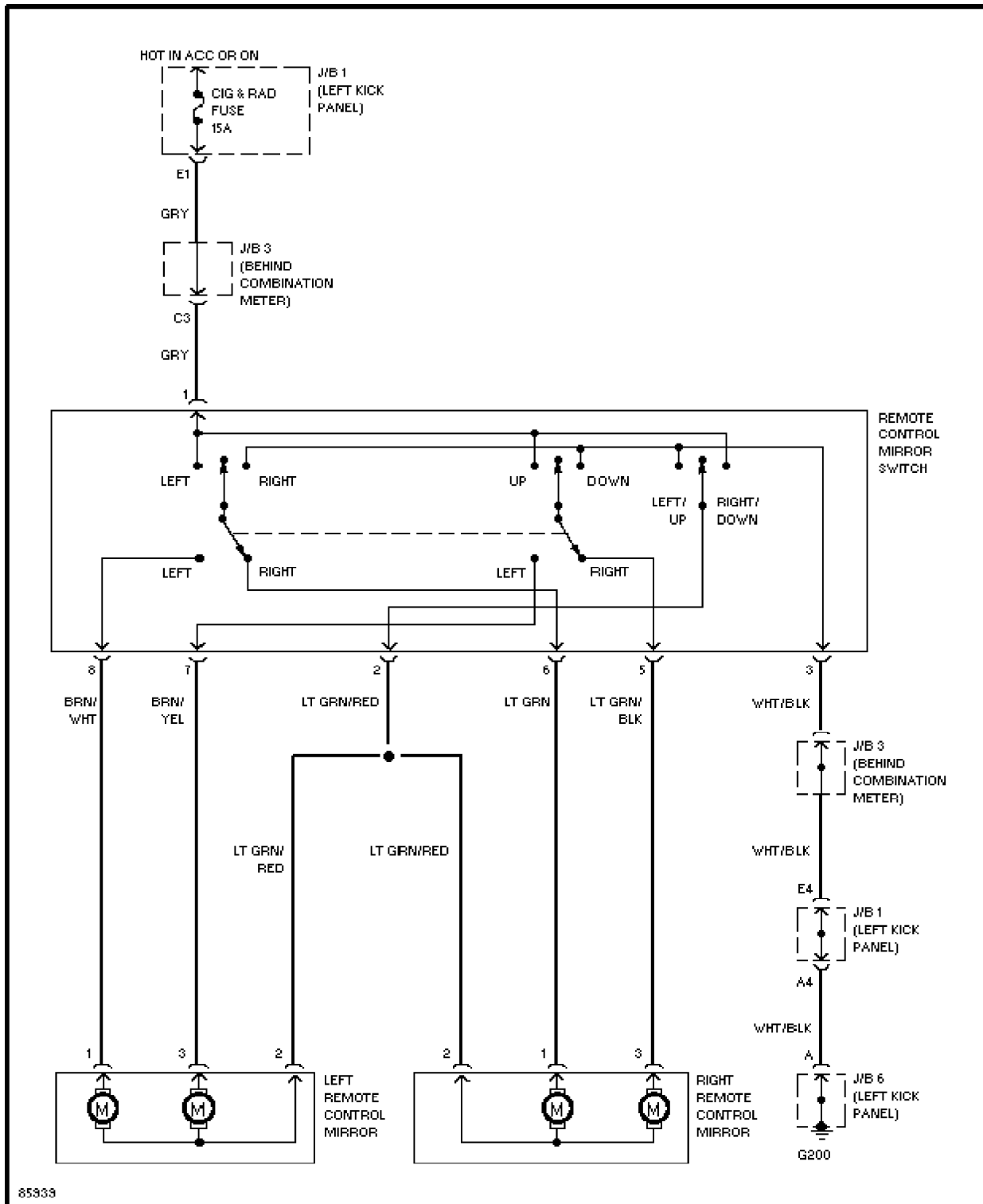
Power Antenna Circuit

## POWER DOOR LOCKS



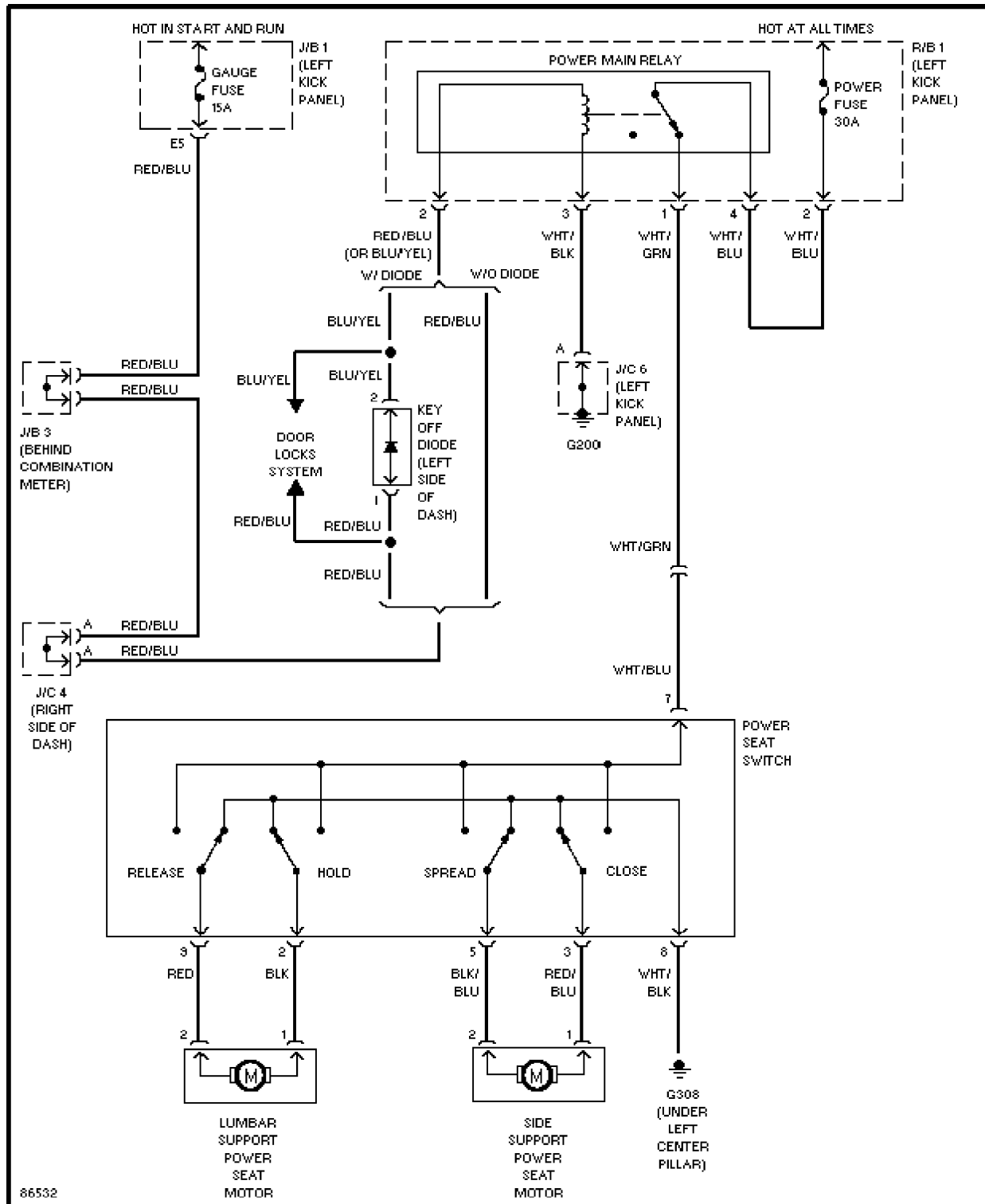
Power Door Lock Circuit

## POWER MIRRORS



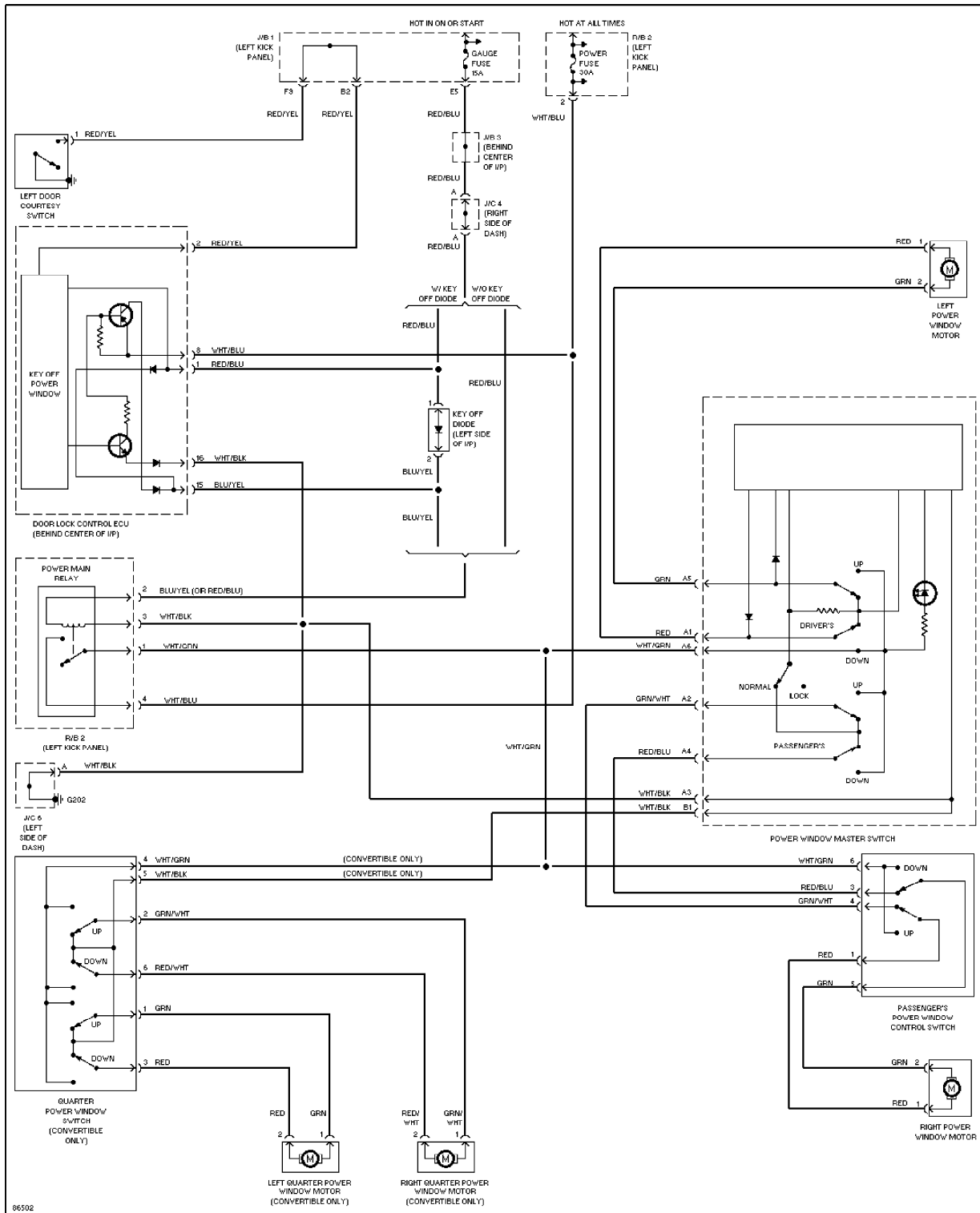
Power Mirror Circuit

## POWER SEATS



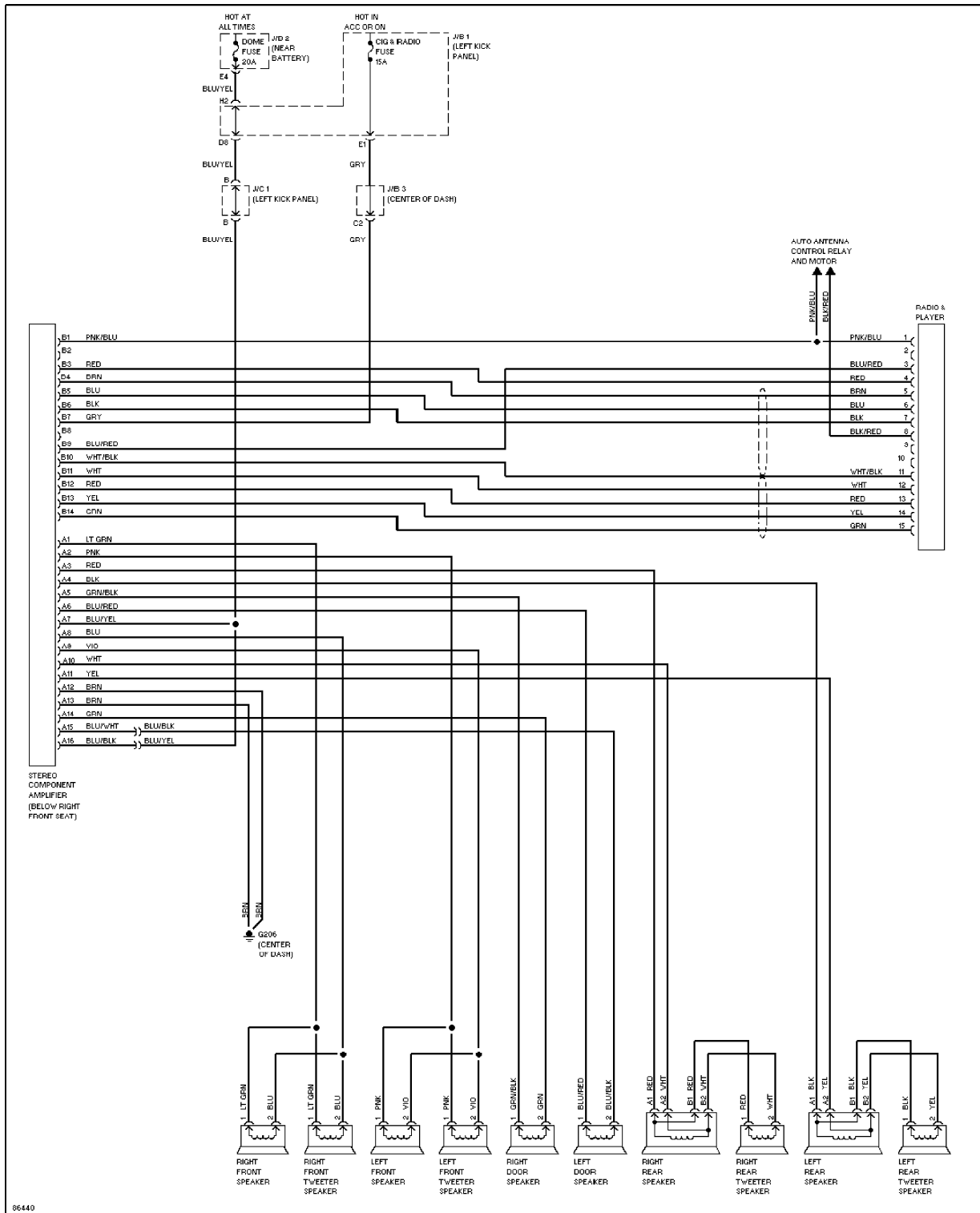
Power Seat Circuits

## POWER WINDOWS

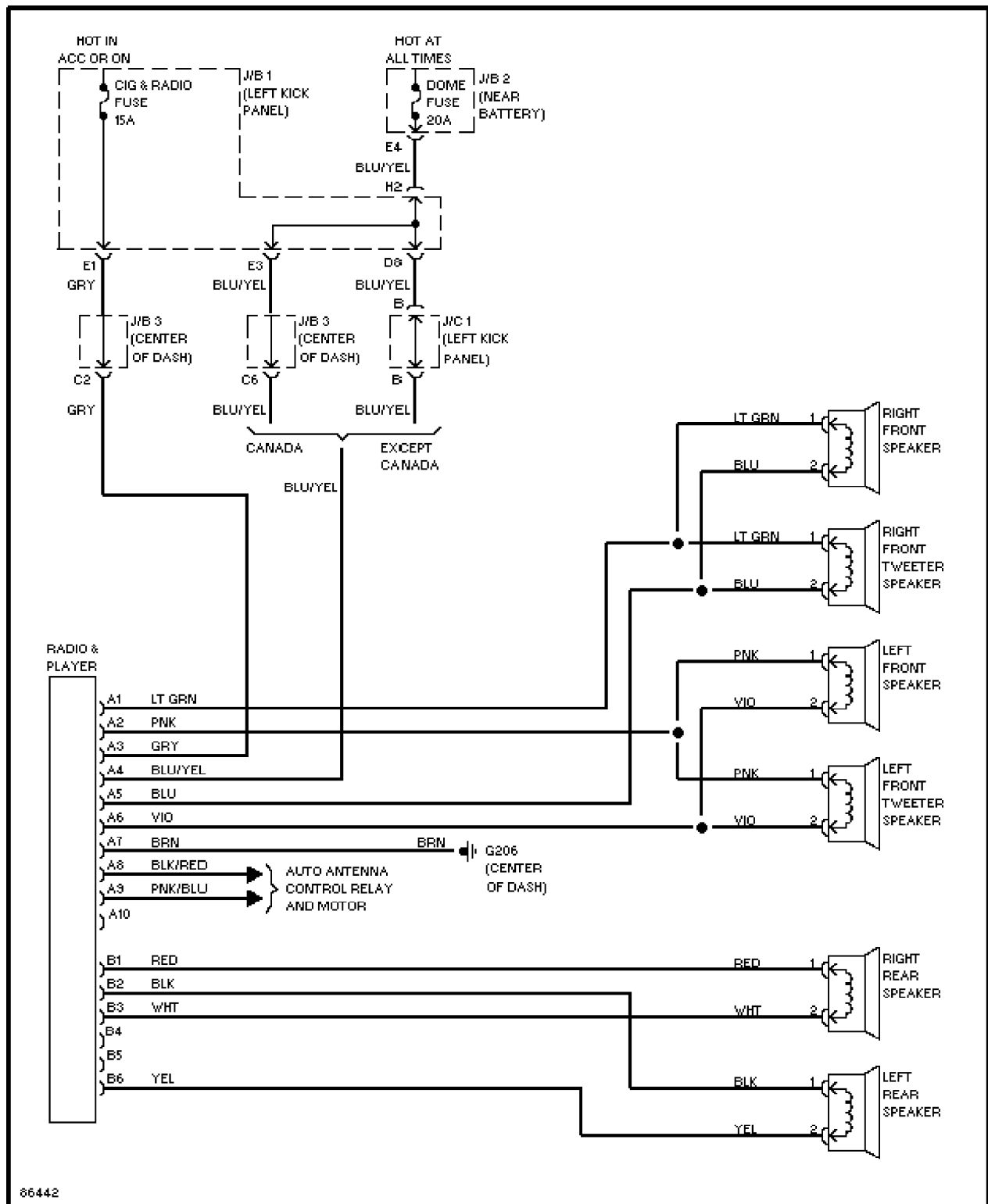


Power Window Circuit

## RADIO



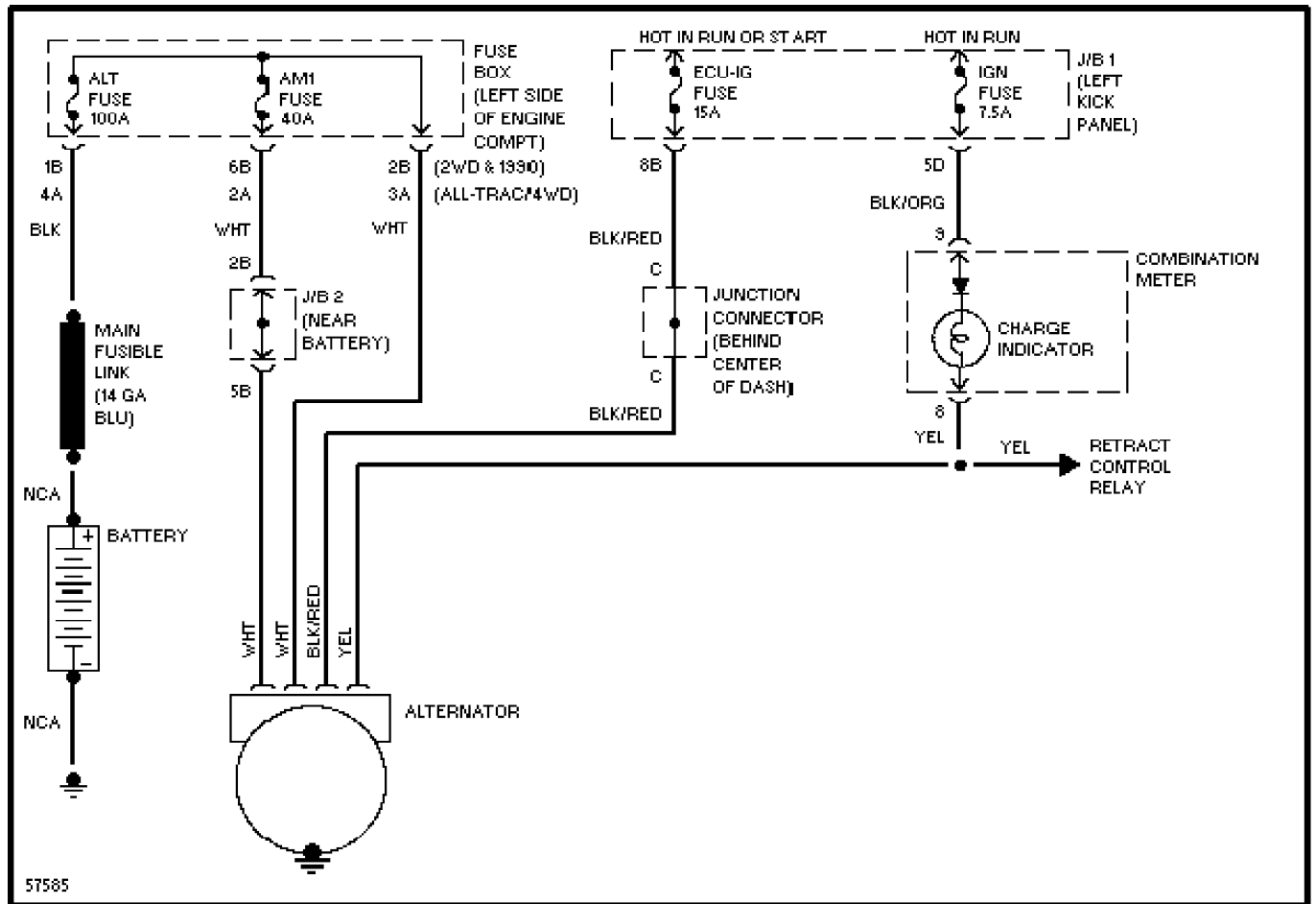
Radio Circuits, W/ CD Player



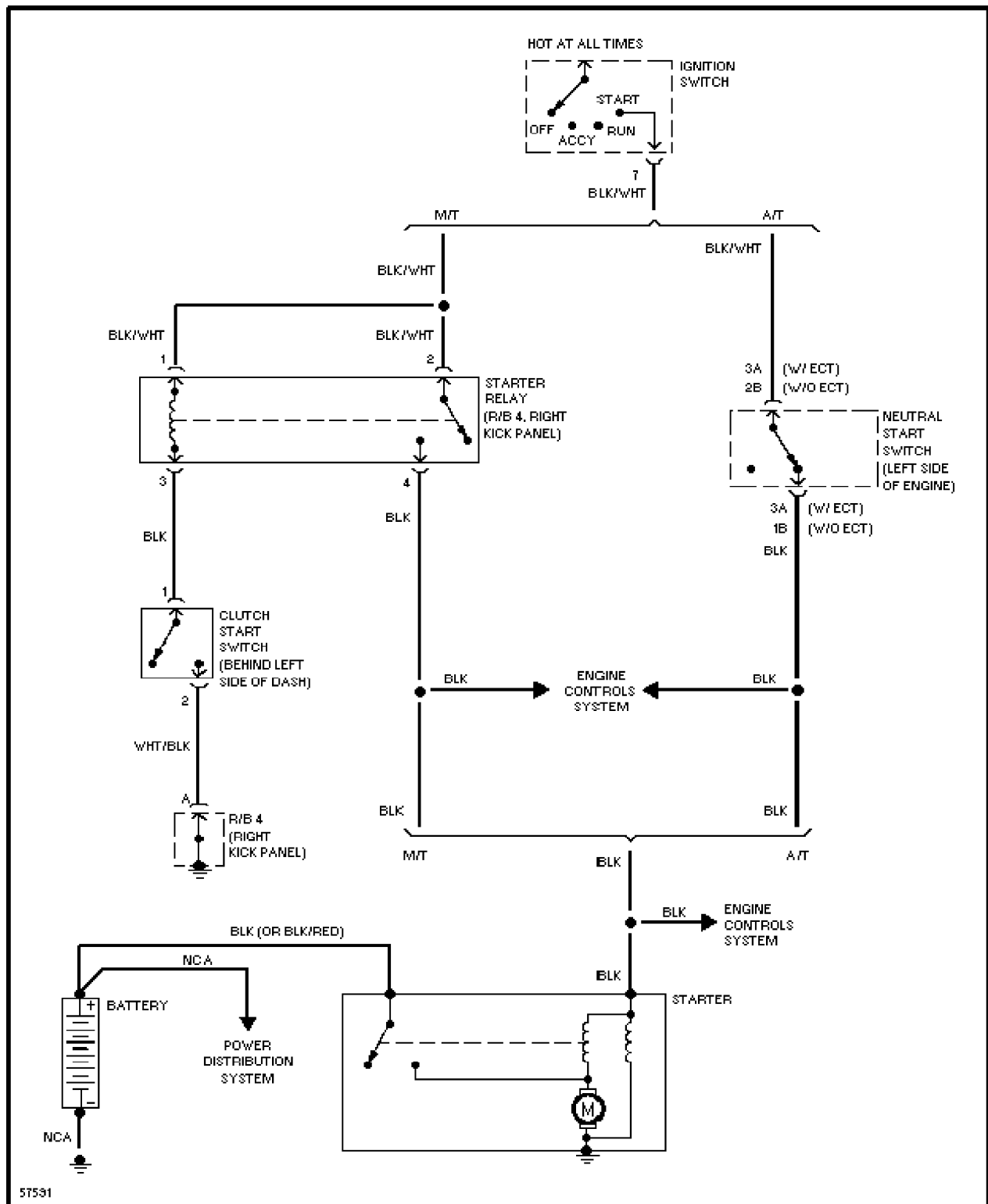
Radio Circuits, W/O CD Player

## STARTING/CHARGING





Charging Circuit

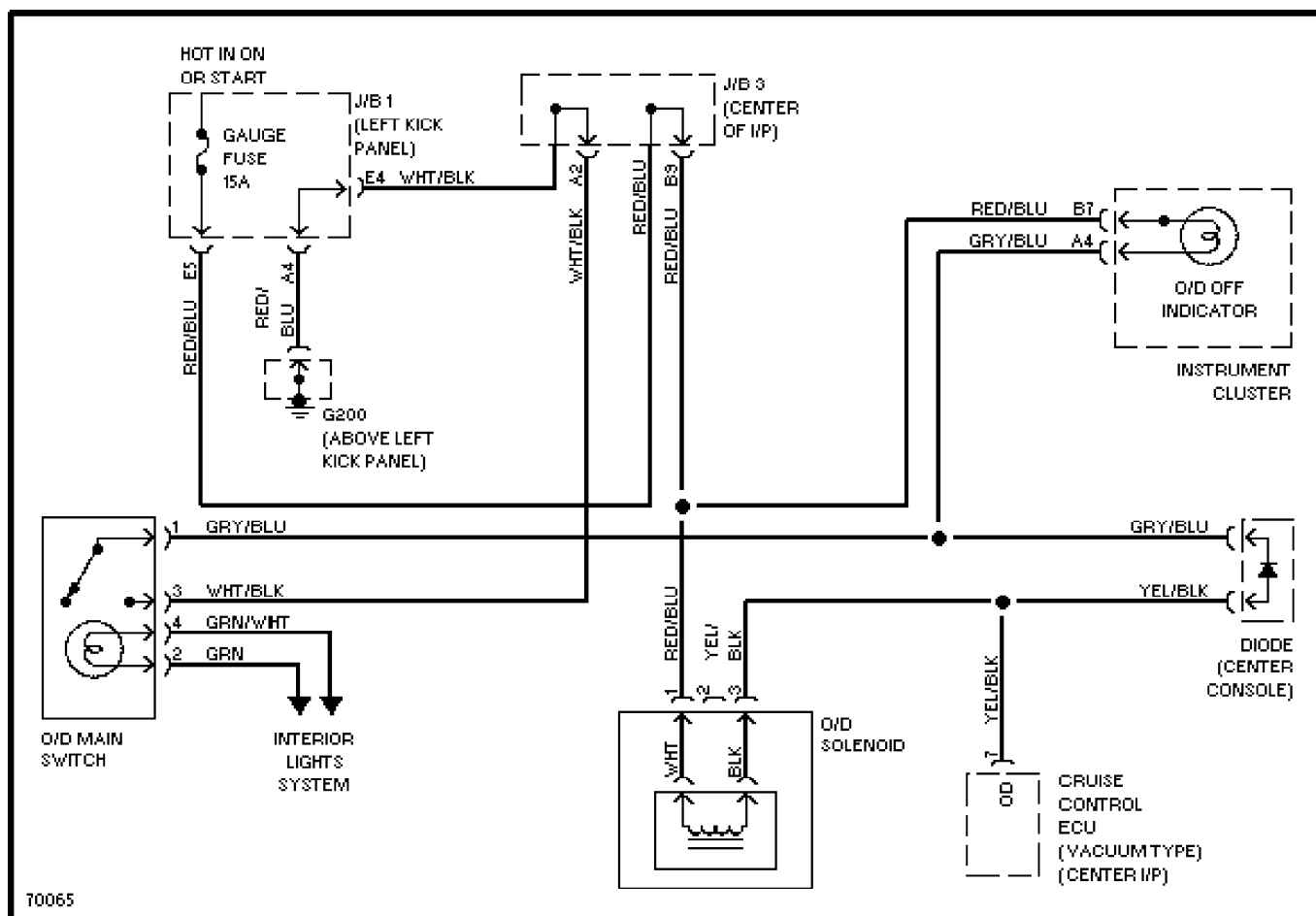


57531

Starting Circuit

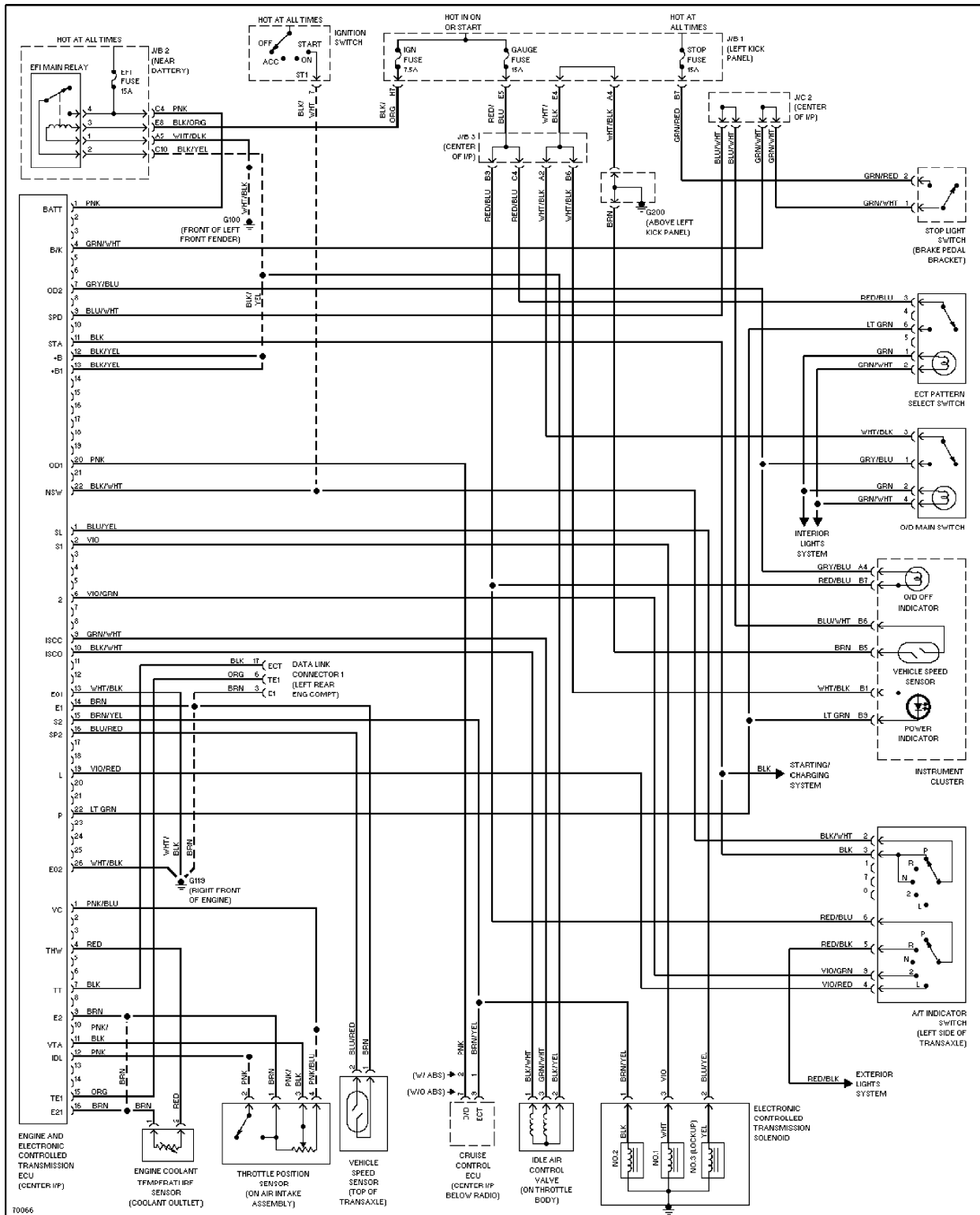
## TRANSMISSION

1.6L



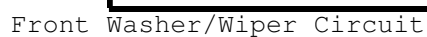
### 1.6L, Transmission Circuit

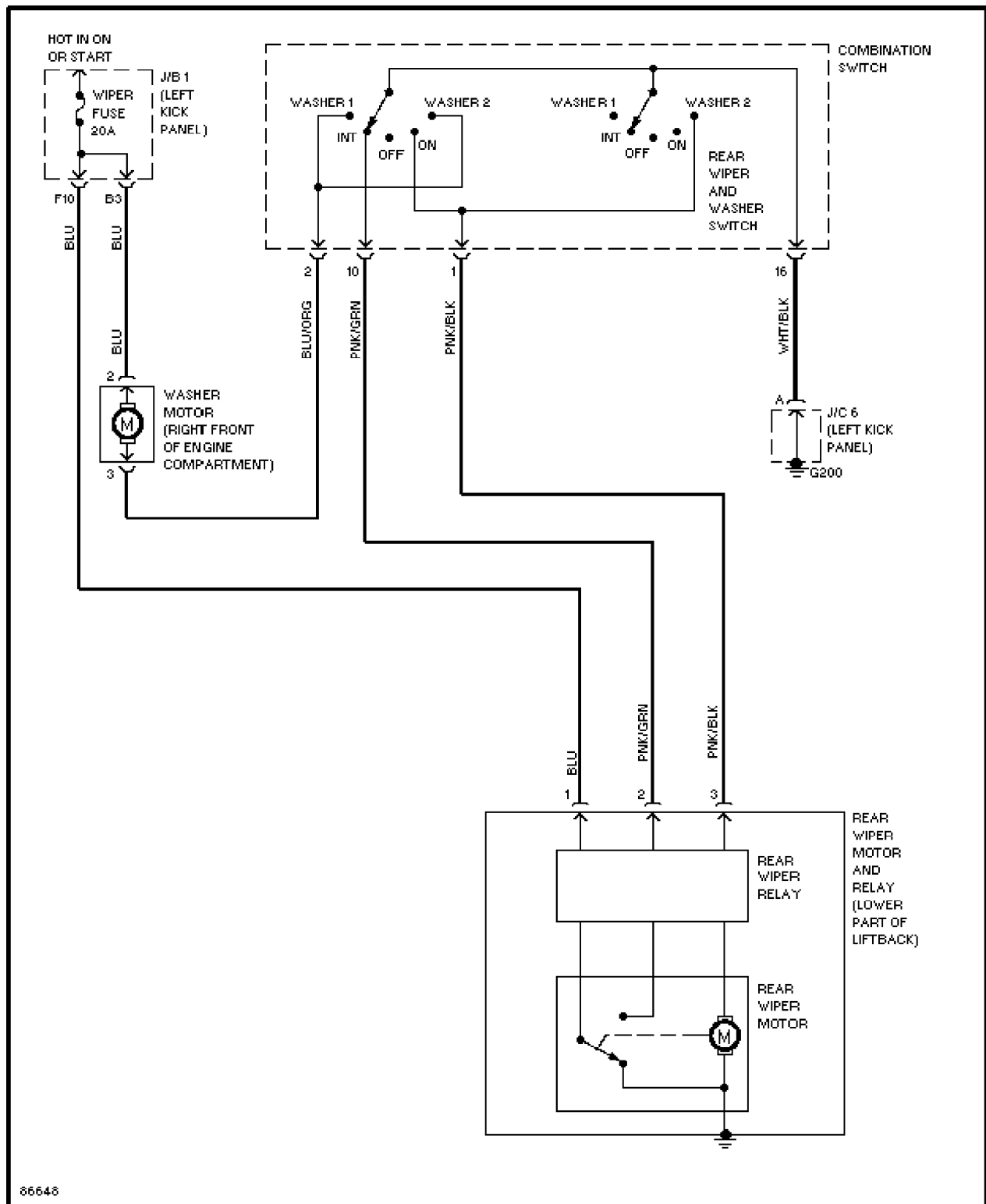
2.2L



2.2L, Transmission Circuit

## WIPER/WASHER





86648

Rear Wiper/Washer Circuit

# TROUBLE SHOOTING - BASIC PROCEDURES

1993 Toyota Celica

## GENERAL TROUBLE SHOOTING

### \* PLEASE READ THIS FIRST \*

NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

## ACCESSORIES & ELECTRICAL

### CHARGING SYSTEM TROUBLE SHOOTING

NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC CHARGING SYSTEM TROUBLE SHOOTING CHART

| CONDITION                                     | POSSIBLE CAUSE                         | CORRECTION                                                                                                  |
|-----------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Vehicle Will Not Start                        | Dead battery                           | Check battery cells, alternator belt tension and alternator output                                          |
|                                               | Loose or corroded battery connections  | Check all charging system connections                                                                       |
|                                               | Ignition circuit or switch malfunction | Check and replace as necessary                                                                              |
| Alternator Light Stays On With Engine Running | Loose or worn alternator drive belt    | Check alternator drive tension and condition, See Belt Adjustment in TUNE-UP article in the TUNE-UP section |
|                                               | Loose alternator wiring connections    | Check all charging system connections                                                                       |
|                                               | Short in alternator light wiring       | See Indicator Warning Lights in STANDARD INSTRUMENTS in the ACCESSORIES & EQUIPMENT section                 |
|                                               | Defective alternator stator or diodes  | See Bench Tests in ALTERNATOR article                                                                       |
|                                               | Defective regulator                    | See Regulator Check in                                                                                      |

|                                                             |                                                                |                                                                                                                                 |
|-------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|                                                             |                                                                | ALTERNATOR article                                                                                                              |
| Alternator<br>Light Stays Off<br>With Ignition<br>Switch ON | Blown fuse                                                     | See WIRING DIAGRAMS                                                                                                             |
|                                                             | Defective alternator                                           | See Testing in<br>ALTERNATOR article                                                                                            |
|                                                             | Defective indicator light<br>bulb or socket                    | See Indicator Warning<br>Lights in STANDARD<br>INSTRUMENTS in the<br>ACCESSORIES &<br>EQUIPMENT section                         |
| Alternator<br>Light Stays OFF<br>With Ignition<br>Switch ON | Short in alternator wiring                                     | See On-Vehicle Tests<br>in ALTERNATOR article                                                                                   |
|                                                             | Defective rectifier bridge                                     | See Bench Tests in<br>ALTERNATOR article                                                                                        |
| Lights or Fuses<br>Burn Out<br>Frequently                   | Defective alternator wiring                                    | See On-Vehicle Tests<br>in ALTERNATOR article                                                                                   |
|                                                             | Defective regulator                                            | See Regulator Check in<br>ALTERNATOR article                                                                                    |
|                                                             | Defective battery                                              | Check and replace as<br>necessary                                                                                               |
| Ammeter Gauge<br>Shows Discharge                            | Loose or worn drive belt                                       | Check alternator drive<br>belt tension and<br>condition. See Belt<br>Adjustment in TUNE-UP<br>article in the<br>TUNE-UP section |
|                                                             | Defective wiring                                               | Check all wires and<br>wire connections                                                                                         |
|                                                             | Defective alternator or<br>regulator                           | See Bench Tests and<br>On-Vehicle Tests in<br>ALTERNATOR article                                                                |
|                                                             | Defective ammeter, or<br>improper ammeter wiring<br>connection | See Testing in<br>STANDARD INSTRUMENTS<br>in the ACCESSORIES &<br>EQUIPMENT section                                             |
| Noisy<br>Alternator                                         | Loose drive pulley                                             | Tighten drive pulley<br>attaching nut                                                                                           |
|                                                             | Loose mounting bolts                                           | Tighten all alternator<br>mounting bolts                                                                                        |
|                                                             | Worn or dirty bearings                                         | See Bearing<br>Replacement<br>ALTERNATOR article                                                                                |

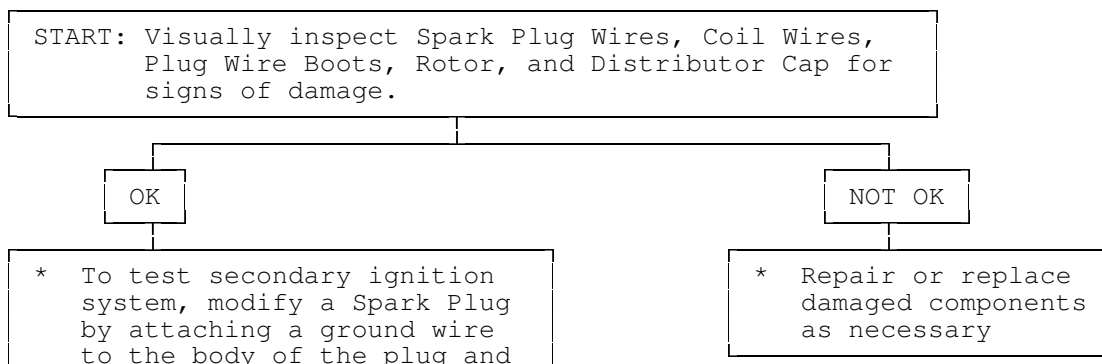


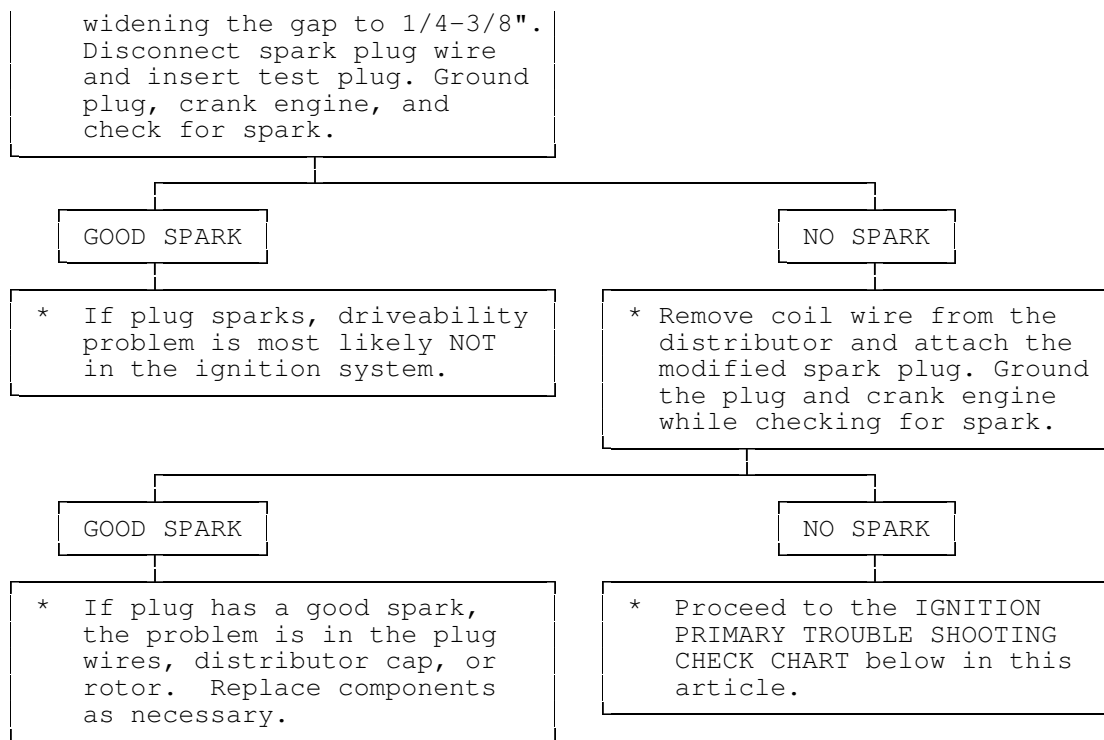
|                                          |                                                             |                                                                                                                              |
|------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                                          | Defective diodes or stator                                  | See Bench Test in ALTERNATOR article                                                                                         |
| Battery Does Stay Charged                | Loose or worn drive belt                                    | Check alternator drive belt tension and condition. See Belt Adjustment in appropriate TUNE-UP article in the TUNE-UP section |
|                                          | Loose or corroded battery connections                       | Check all charging system connections                                                                                        |
|                                          | Loose alternator connections                                | Check all charging system connections                                                                                        |
|                                          | Defective alternator or battery                             | See On-Vehicle Tests and Bench Tests in ALTERNATOR article                                                                   |
|                                          | Add-on electrical accessories exceeding alternator capacity | Install larger alternator                                                                                                    |
| Battery Overcharged- Uses Too Much Water | Defective battery                                           | Check alternator output and repair as necessary                                                                              |
|                                          | Defective alternator                                        | See On-Vehicle Test and Bench Tests in ALTERNATOR article                                                                    |
|                                          | Excessive alternator voltage                                | Check alternator output and repair as necessary                                                                              |

## IGNITION SYSTEM TROUBLE SHOOTING

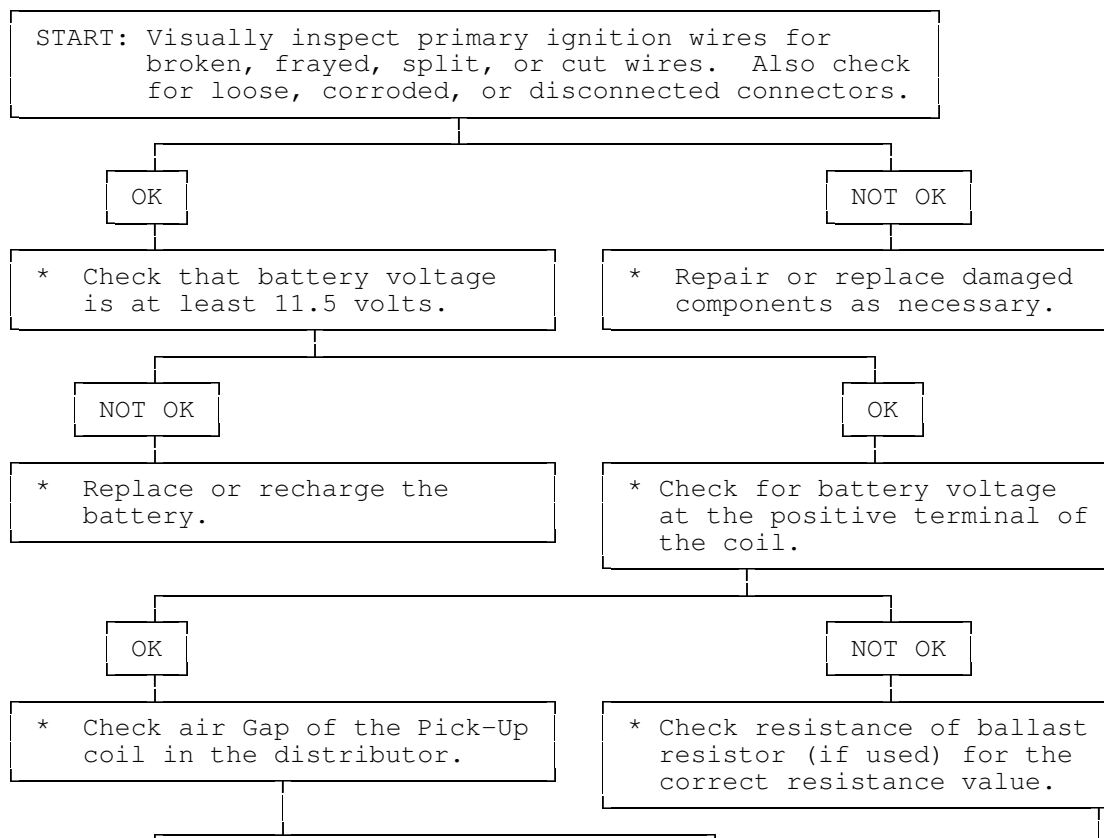
**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

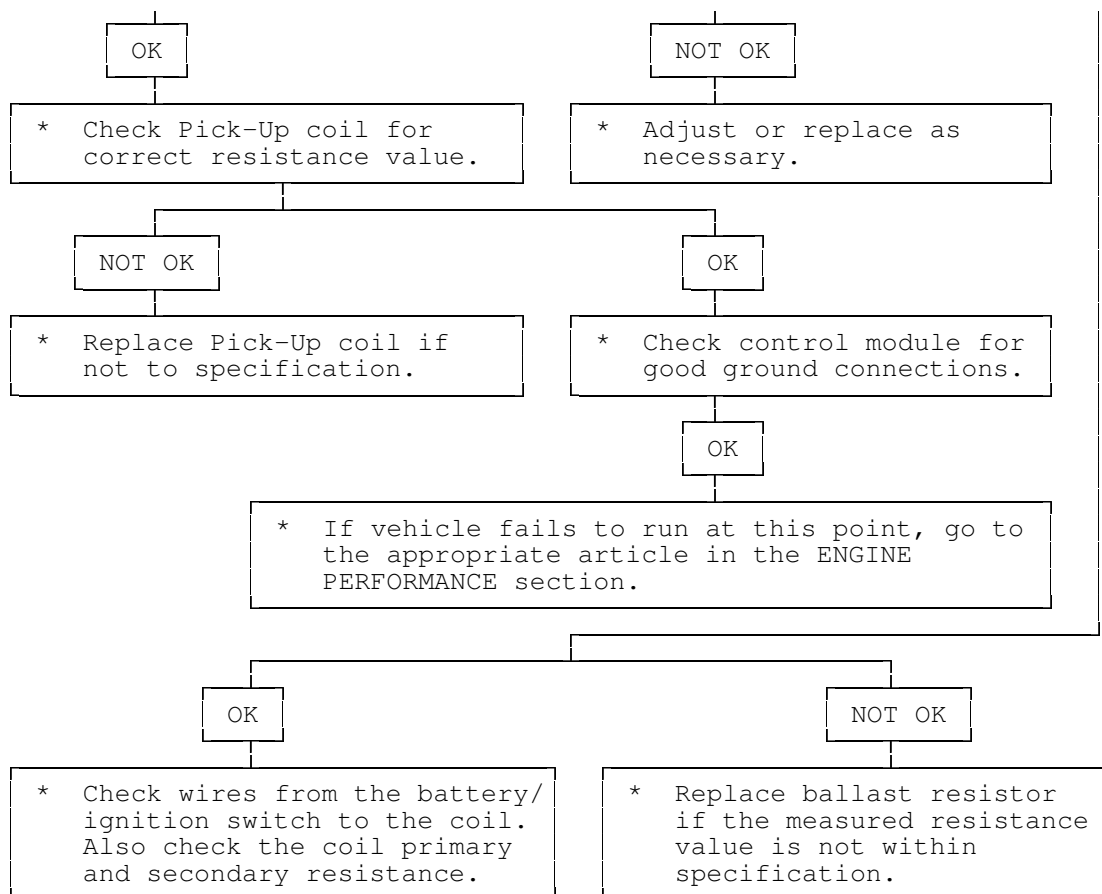
Ignition Secondary Trouble Shooting Chart





### Ignition Primary Trouble Shooting Chart





## STARTER TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC STARTER TROUBLE SHOOTING CHART

| CONDITION                | POSSIBLE CAUSE                                                         | CORRECTION                                                    |
|--------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|
| Starter Fails to Operate | Dead battery or bad connections between starter and battery            | Check battery charge and all wires and connections to starter |
|                          | Ignition switch faulty or misadjusted                                  | Adjust or replace ignition switch                             |
|                          | Open circuit between starter switch ignition terminal on starter relay | Check and repair wires and connections as necessary           |
|                          | Starter relay or starter defective                                     | See Testing in STARTER article                                |

|                                             |                                                         |                                                              |
|---------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|
|                                             | Open solenoid pull-in wire                              | See Testing in STARTER article                               |
| Starter Does Not Operate and Headlights Dim | Weak battery or dead cell                               | Charge or replace battery as necessary                       |
|                                             | Loose or corroded battery connections                   | Check that battery connections are clean and tight           |
|                                             | Internal ground in starter windings                     | See Testing in STARTER article                               |
|                                             | Grounded starter fields                                 | See Testing in STARTERS                                      |
|                                             | Armature rubbing on pole shoes                          | See STARTER article                                          |
| Starter Turns but Engine Does Not Rotate    | Starter clutch slipping                                 | See STARTER article                                          |
|                                             | Broken clutch housing                                   | See STARTER article                                          |
|                                             | Pinion shaft rusted or dry                              | See STARTER article                                          |
|                                             | Engine basic timing incorrect                           | See Ignition Timing in TUNE-UP article                       |
|                                             | Broken teeth on engine flywheel                         | Replace flywheel and check for starter pinion gear damage    |
| Starter Will Not Crank Engine               | Faulty overrunning clutch                               | See STARTER article                                          |
|                                             | Broken clutch housing                                   | See STARTER article                                          |
|                                             | Broken flywheel teeth                                   | Replace flywheel and check for starter pinion gear damage    |
|                                             | Armature shaft sheared or reduction gear teeth stripped | See STARTER article                                          |
|                                             | Weak battery                                            | Charge or replace battery as necessary                       |
|                                             | Faulty solenoid                                         | See On-Vehicle Tests in STARTER article                      |
|                                             | Poor grounds                                            | Check all ground connections for tight and clean connections |
|                                             | Ignition switch faulty or misadjusted                   | Adjust or replace ignition switch as necessary               |
| Starter Cranks Engine Slowly                | Battery weak or defective                               | Charge or replace battery as necessary                       |

|                                         |                                                         |                                                                |
|-----------------------------------------|---------------------------------------------------------|----------------------------------------------------------------|
|                                         | Engine overheated                                       | See ENGINE COOLING SYSTEM article                              |
|                                         | Engine oil too heavy                                    | Check that proper viscosity oil is used                        |
|                                         | Poor battery-to-starter connections                     | Check that all between battery and starter are clean and tight |
|                                         | Current draw too low or too high                        | See Bench Tests in STARTER article                             |
|                                         | Bent armature, loose pole shoes screws or worn bearings | See STARTER article                                            |
|                                         | Burned solenoid contacts                                | Replace solenoid                                               |
|                                         | Faulty starter                                          | Replace starter                                                |
| Starter Engages Engine Only Momentarily | Engine timing too far advanced                          | See Ignition Timing in TUNE-UP article                         |
|                                         | Overrunning clutch not engaging properly                | Replace overrunning clutch. See STARTER article                |
|                                         | Broken starter clutch                                   | See STARTER article                                            |
|                                         | Broken teeth on engine flywheel                         | Replace flywheel and check starter pinion gear for damage      |
|                                         | Weak drive assembly thrust spring                       | See STARTER article                                            |
|                                         | Weak hold-in coil                                       | See Bench Tests in STARTER article                             |
| Starter Drive Will Not Engage           | Defective point assembly                                | See Testing in STARTER article                                 |
|                                         | Poor point assembly ground                              | See Testing in STARTER article                                 |
|                                         | Defective pull-in coil                                  | Replace starter solenoid                                       |
| Starter Relay Does Not Close            | Dead battery                                            | Charge or replace battery as necessary                         |
|                                         | Faulty wiring                                           | Check all wiring and connections leading to relay              |
|                                         | Neutral safety switch faulty                            | Replace neutral safety switch                                  |
|                                         | Starter relay faulty                                    | Replace starter relay                                          |

|                                                                                             |                                                                                                |                                                                                       |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Starter Drive Will Not Disengage                                                            | Starter motor loose on mountings                                                               | Tighten starter attach bolts                                                          |
|                                                                                             | Worn drive end bushing                                                                         | See STARTER article                                                                   |
|                                                                                             | Damaged engine flywheel teeth                                                                  | Replace flywheel and starter pinion gear for damage                                   |
|                                                                                             | Drive yolk return spring broken or missing                                                     | Replace return spring                                                                 |
|                                                                                             | Faulty ignition switch                                                                         | Replace ignition switch                                                               |
|                                                                                             | Insufficient clearance between winding leads to solenoid terminal and main contact in solenoid | Replace starter solenoid                                                              |
|                                                                                             | Starter clutch not disengaging                                                                 | Replace starter clutch                                                                |
|                                                                                             | Ignition starter switch contacts sticking                                                      | Replace ignition switch                                                               |
| Starter Relay Operates but Solenoid Does Not                                                | Faulty solenoid switch, switch connections or                                                  | Check all wiring between relay and solenoid or replace relay or solenoid as necessary |
|                                                                                             | Broken lead or loose soldered connections                                                      | Repair wire or wire connections as necessary                                          |
| Solenoid Plunger Vibrates When Switch is Engaged                                            | Weak battery                                                                                   | Charge or replace battery as necessary                                                |
|                                                                                             | Solenoid contacts corroded                                                                     | Clean contacts or replace solenoid                                                    |
|                                                                                             | Faulty wiring                                                                                  | Check all wiring leading to solenoid                                                  |
|                                                                                             | Broken connections inside switch cover                                                         | Repair connections or replace solenoid                                                |
|                                                                                             | Open hold-in wire                                                                              | Replace solenoid                                                                      |
| Low Current Draw                                                                            | Worn brushes or weak                                                                           | Replace brushes or brush springs as necessary                                         |
| High Pitched Whine During Cranking Before Engine Fires but Engine Fires and Cranks Normally | Distance too great between starter pinion and flywheel                                         | Align starter or check that correct starter and flywheel are being used               |
| High Pitched Whine After Engine                                                             | Distance too small between starter pinion and flywheel                                         |                                                                                       |

Fires With Key released. Engine Fires and Cranks Normally Flywheel runout contributes to the intermittent nature

---

AIR CONDITIONING & HEAT

AIR CONDITIONING TROUBLE SHOOTING

WARNING: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

BASIC AIR CONDITIONING TROUBLE SHOOTING CHART

| CONDITION                                         | POSSIBLE CAUSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compressor Not Working                            | <ul style="list-style-type: none"><li>• Compressor clutch circuit open.</li><li>• Compressor clutch coil inoperative.</li><li>• Poor clutch ground connection.</li><li>• Fan belts loose.</li><li>• Thermostatic switch inoperative.</li><li>• Thermostatic switch not adjusted.</li><li>• Ambient temperature switch open.</li><li>• Superheat fuse blown.</li></ul>                                                                                                                         |
| Excessive Noise or Vibration                      | <ul style="list-style-type: none"><li>• Missing or loose mounting bolts.</li><li>• Bad idler pulley bearings.</li><li>• Fan belts not tightened correctly.</li><li>• Compressor clutch contacting body.</li><li>• Excessive system pressure.</li><li>• Compressor oil level low.</li><li>• Damaged clutch bearings.</li><li>• Damaged reed valves.</li><li>• Damaged compressor.</li></ul>                                                                                                    |
| Insufficient or No Cooling;<br>Compressor Working | <ul style="list-style-type: none"><li>• Expansion valve inoperative.</li><li>• Heater control valve stuck open.</li><li>• Low system pressure.</li><li>• Blocked condenser fins.</li><li>• Blocked evaporator fins.</li><li>• Vacuum system leak.</li><li>• Vacuum motors inoperative.</li><li>• Control cables improperly adjusted.</li><li>• Restricted air inlet.</li><li>• Mode doors binding.</li><li>• Blower motor inoperative.</li><li>• Temperature above system capacity.</li></ul> |

HEATER SYSTEM TROUBLE SHOOTING

NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to

problem symptoms. For model-specific Trouble Shooting, refer to DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC HEATER SYSTEM TROUBLE SHOOTING CHART

| CONDITION                            | POSSIBLE CAUSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Insufficient, Erratic, or No Heat    | <ul style="list-style-type: none"> <li>• Low Coolant Level</li> <li>• Incorrect thermostat.</li> <li>• Restricted coolant flow through heater core.</li> <li>• Heater hoses plugged.</li> <li>• Misadjusted control cable.</li> <li>• Sticking heater control valve.</li> <li>• Vacuum hose leaking.</li> <li>• Vacuum hose blocked.</li> <li>• Vacuum motors inoperative.</li> <li>• Blocked air inlet.</li> <li>• Inoperative heater blower motor.</li> <li>• Oil residue on heater core fins.</li> <li>• Dirt on heater core fins.</li> </ul> |
| Too Much Heat                        | <ul style="list-style-type: none"> <li>• Improperly adjusted cables.</li> <li>• Sticking heater control valve.</li> <li>• No vacuum to heater control valve.</li> <li>• Temperature door stuck open.</li> </ul>                                                                                                                                                                                                                                                                                                                                  |
| Air Flow Changes During Acceleration | <ul style="list-style-type: none"> <li>• Vacuum system leak.</li> <li>• Bad check valve or reservoir.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Air From Defroster At All Times      | <ul style="list-style-type: none"> <li>• Vacuum system leak.</li> <li>• Improperly adjusted control cables.</li> <li>• Inoperative vacuum motor.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |
| Blower Does Not Operate Correctly    | <ul style="list-style-type: none"> <li>• Blown fuse.</li> <li>• Blower motor windings open.</li> <li>• Resistors burned out.</li> <li>• Motor ground connection loose.</li> <li>• Wiring harness connections loose.</li> <li>• Blower motor switch inoperative.</li> <li>• Blower relay inoperative.</li> <li>• Fan binding or foreign object in housing.</li> <li>• Fan blades broken or bent.</li> </ul>                                                                                                                                       |

## BRAKES

### BRAKE SYSTEM TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.



## BRAKE SYSTEM TROUBLE SHOOTING CHART

| CONDITION                         | POSSIBLE CAUSE                         | CORRECTION                            |
|-----------------------------------|----------------------------------------|---------------------------------------|
| Brakes Pull Left or Right         | Incorrect tire pressure                | Inflate tires to proper pressure      |
|                                   | Front end out of alignment             | See WHEEL ALIGNMENT                   |
|                                   | Mismatched tires                       | Check tires sizes                     |
|                                   | Restricted brake lines or hoses        | Check hose routing                    |
|                                   | Loose or malfunctioning caliper        | See DISC BRAKES or BRAKE SYSTEM       |
|                                   | Bent shoe or oily linings              | See DRUM BRAKES or BRAKE SYSTEM       |
|                                   | Malfunctioning rear brakes             | See DRUM, DISC BRAKES or BRAKE SYSTEM |
|                                   | Loose suspension parts                 | See SUSPENSION                        |
| Noises Without Brakes Applied     | Front linings worn out                 | Replace linings                       |
|                                   | Dust or oil on drums or rotors         | See DRUM, DISC BRAKES or BRAKE SYSTEM |
| Noises With Brakes Applied        | Insulator on outboard shoe damaged     | See DISC BRAKES or BRAKE SYSTEM       |
|                                   | Incorrect pads or linings              | Replace pads or linings               |
| Brake Rough, Chatters or Pulsates | Excessive lateral runout               | Check rotor runout                    |
|                                   | Parallelism not to specifications      | Reface or replace rotor               |
|                                   | Wheel bearings not adjusted            | See SUSPENSION                        |
|                                   | Rear drums out-of-round                | Reface or replace drums               |
|                                   | Disc pad reversed, steel against rotor | Remove and reinstall pad              |
| Excessive Pedal Effort            | Malfunctioning power unit              | See POWER BRAKES or BRAKE SYSTEM      |
|                                   | Partial system failure                 | Check fluid and pipes                 |
|                                   | Worn disc pad or lining                | Replace pad or lining                 |
|                                   | Caliper piston stuck or                |                                       |

|                         |                                                                       |                                                           |
|-------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|
|                         | sluggish                                                              | See DISC BRAKES or BRAKE SYSTEM                           |
|                         | Master cylinder piston stuck                                          | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Brake fade due to incorrect pads for linings                          | Replace pads or linings                                   |
|                         | Linings or pads glazed                                                | Replace pads or linings                                   |
|                         | Worn drums                                                            | Reface or replace drums                                   |
| <hr/>                   |                                                                       |                                                           |
| Excessive Pedal Travel  | Partial brake system failure                                          | Check fluid and pipes                                     |
|                         | Insufficient fluid in master cylinder                                 | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Air trapped in system                                                 | See BRAKE BLEEDING or BRAKE SYSTEM                        |
|                         | Rear brakes not adjusted                                              | See Adjustments in DRUM BRAKES or BRAKE SYSTEM            |
|                         | Bent shoe or lining                                                   | See DRUM BRAKES or BRAKE SYSTEM                           |
|                         | Plugged master cylinder cap                                           | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Improper brake fluid                                                  | Replace brake fluid                                       |
| <hr/>                   |                                                                       |                                                           |
| Pedal Travel Decreasing | Compensating port plugged                                             | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Swollen cup in master cylinder                                        | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Master cylinder piston not returning                                  | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Weak shoe retracting springs                                          | See DRUM BRAKES or BRAKE SYSTEM                           |
|                         | Wheel cylinder piston sticking                                        | See DRUM BRAKES or BRAKE SYSTEM                           |
| <hr/>                   |                                                                       |                                                           |
| Dragging Brakes         | Master cylinder pistons not returning                                 | See MASTER CYLINDERS or BRAKE SYSTEM                      |
|                         | Restricted brake lines or hoses<br>Incorrect parking brake adjustment | Check line routing<br><br>See DRUM BRAKES or BRAKE SYSTEM |

|                                            |                                               |                                          |
|--------------------------------------------|-----------------------------------------------|------------------------------------------|
|                                            | Parking Brake cables frozen                   | See DRUM BRAKES<br>BRAKE SYSTEM          |
|                                            | Incorrect installation of<br>inboard disc pad | Remove and replace<br>correctly          |
|                                            | Power booster output<br>rod too long          | See POWER BRAKE UNITS<br>BRAKE SYSTEM    |
|                                            | Brake pedal not returning<br>freely           | See DISC, DRUM BRAKES<br>BRAKE SYSTEM    |
| Brakes Grab or<br>Uneven Braking<br>Action | Malfunction of combination<br>valve           | See CONTROL VALVE or<br>BRAKE SYSTEM     |
|                                            | Malfunction of power brake<br>unit            | See POWER BRAKE UNITS<br>or BRAKE SYSTEM |
|                                            | Binding brake pedal                           | See DISC, DRUM BRAKES<br>or BRAKE SYSTEM |
| Pulsation or<br>Roughness                  | Uneven pad wear caused by<br>caliper          | See DISC BRAKES or<br>BRAKE SYSTEM       |
|                                            | Uneven rotor wear                             | See DISC BRAKES or<br>BRAKE SYSTEM       |
|                                            | Drums out-of-round                            | Reface or replace drums                  |

## ENGINE MECHANICAL

### COOLING SYSTEM TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### COOLING SYSTEM TROUBLE SHOOTING

| CONDITION   | POSSIBLE CAUSE                     | CORRECTION                    |
|-------------|------------------------------------|-------------------------------|
| Overheating | Coolant Leak                       | Fill/Pressure Test<br>System  |
|             | A/C Condenser Fins Clogged         | Remove/Clean Condenser        |
|             | Radiator Fins Clogged              | Remove/Clean Radiator         |
|             | Thermostat Stuck Closed            | Replace Thermostat            |
|             | Clogged Cooling System<br>Passages | Clean/Flush Cooling<br>System |

|                             |                                         |                              |
|-----------------------------|-----------------------------------------|------------------------------|
|                             | Water Pump Malfunction                  | Replace Water Pump           |
|                             | Fan Clutch Malfunction                  | Replace Fan Clutch           |
|                             | Retarded Ignition Timing                | Reset Ignition Timing        |
|                             | Cooling Fan Malfunction                 | Test Cooling Fan/<br>Circuit |
|                             | Cooling Fan Motor<br>Malfunction        | Test Fan Motor               |
|                             | Cooling Fan Relay<br>Malfunction        | Test Fan Relay               |
|                             | Faulty Radiator Cap                     | Replace Radiator Cap         |
|                             | Broken/Slipping Fan Belt                | Replace Fan Belt             |
|                             | Restricted Exhaust                      | Repair Exhaust System        |
| Corrosion                   | Impurities In Coolant                   | Clean/Flush System           |
| Coolant Leakage             | Damaged hose                            | Replace Hose                 |
|                             | Leaky Water Pump                        | Replace Water Pump           |
|                             | Damaged Radiator Seam                   | Replace/Repair Radiator      |
|                             | Leaky Thermostat Cover                  | Replace Thermostat<br>Cover  |
|                             | Cylinder Head Problem                   | Check Head/Head Gasket       |
|                             | Leaky Freeze Plugs                      | Replace Freeze Plugs         |
| Recovery System Inoperative | Loose and/or Defective<br>Radiator Cap  | Replace Radiator Cap         |
|                             | Overflow Tube Clogged<br>and/or Leaking | Repair Tube                  |
|                             | Recovery Bottle Vent<br>Restricted      | Clean Vent                   |
| No Heater Core Flow         | Collapsed Heater Hose                   | Replace Heater Hose          |
|                             | Plugged Heater Core                     | Clean/Replace Heater<br>Core |
|                             | Faulty Heater Valve                     | Replace Heater Valve         |

## GASOLINE ENGINE - MECHANICAL TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available

in the section(s) you are accessing.

# BASIC GASOLINE ENGINE - MECHANICAL TROUBLE SHOOTING CHART

| CONDITION                   | POSSIBLE CAUSE                               | CORRECTION                                          |
|-----------------------------|----------------------------------------------|-----------------------------------------------------|
| Engine Lopes At Idle        | Intake manifold-to-head leaks                | Replace manifold gasket, See ENGINES                |
|                             | Blown head gasket                            | Replace head gasket, See ENGINES                    |
|                             | Worn timing gears, chain or sprocket         | Replace gears, chain or sprocket                    |
|                             | Worn camshaft lobes                          | Replace camshaft, See ENGINES                       |
|                             | Overheated engine                            | Check cooling system, See COOLING                   |
|                             | Blocked crankcase vent valve                 | Remove restriction                                  |
|                             | Leaking EGR valve                            | Repair leak and/or replace valve                    |
|                             | Faulty fuel pump                             | Replace fuel pump                                   |
| Engine Has Low Power        | Leaking fuel pump                            | Repair leak and/or replace fuel pump                |
|                             | Excessive piston-to-bore clearance           | Install larger pistons, See ENGINES                 |
|                             | Sticking valves or weak valve springs        | Check valve train components, See ENGINES           |
|                             | Incorrect valve timing                       | Reset valve timing, See ENGINES                     |
|                             | Worn camshaft lobes                          | Replace camshaft, See ENGINES                       |
|                             | Blown head gasket                            | Replace head gasket. See ENGINES.                   |
|                             | Clutch slipping                              | Adjust pedal and/or replace components, See ENGINES |
|                             | Engine overheating                           | Check cooling system, See COOLING                   |
|                             | Auto. Trans. pressure regulator valve faulty | Replace pressure regulator valve                    |
|                             | Auto. Trans. fluid level too low             | Add fluid as necessary                              |
|                             | Improper vacuum diverter valve operation     | Replace vacuum diverter valve                       |
|                             | Vacuum leaks                                 | Inspect vacuum system and repair as required        |
|                             | Leaking piston rings                         | Replace piston rings, See ENGINES                   |
| Faulty High Speed Operation | Low fuel pump volume                         | Replace fuel pump                                   |
|                             | Leaking valves or worn                       | Replace valves and/or springs, See ENGINES          |
|                             | Incorrect valve timing                       | Reset valve timing, See ENGINES                     |
|                             | Intake manifold restricted                   | Remove restriction                                  |
|                             | Worn distributor shaft                       | Replace distributor                                 |
| Faulty Acceleration         | Improper fuel pump stroke                    | Remove pump and reset pump stroke                   |

|                         |                                               |                                                     |
|-------------------------|-----------------------------------------------|-----------------------------------------------------|
|                         | Incorrect ignition timing                     | Reset ignition timing,<br>See TUNE-UP               |
|                         | Leaking valves                                | Replace valves, See<br>ENGINES                      |
|                         | Worn fuel pump diaphragm<br>or piston         | Replace diaphragm or<br>piston                      |
| Intake Backfire         | Improper ignition timing                      | Reset ignition timing,<br>See TUNE-UP               |
|                         | Faulty accelerator pump<br>discharge          | Replace accelerator<br>pump                         |
|                         | Improper choke operation                      | Check choke and adjust<br>as required               |
|                         | Defective EGR valve                           | Replace EGR valve                                   |
|                         | Fuel mixture too lean                         | Reset air/fuel mixture,<br>See TUNE-UP              |
|                         | Choke valve initial<br>clearance too large    | Reset choke valve<br>initial clearance              |
| Exhaust Backfire        | Vacuum leak                                   | Inspect and repair<br>vacuum system                 |
|                         | Faulty vacuum diverter<br>valve               | Replace vacuum diverter<br>valve                    |
|                         | Faulty choke operation                        | Check choke and adjust<br>as required               |
|                         | Exhaust system leak                           | repair exhaust system<br>leak                       |
| Engine<br>Detonation    | Ignition timing too far<br>advanced           | Reset ignition timing,<br>See TUNE-UP               |
|                         | Faulty ignition system                        | Check ignition timing,<br>See TUNE-UP               |
|                         | Spark plugs loose or<br>faulty                | Retighten or replace<br>plugs                       |
|                         | Fuel delivery system<br>clogged               | Inspect lines, pump and<br>filter for clog          |
|                         | EGR valve inoperative                         | Replace EGR valve                                   |
|                         | PCV system inoperative                        | Inspect and/or replace<br>hoses or valve            |
|                         | Vacuum leaks                                  | Check vacuum system and<br>repair leaks             |
|                         | Excessive combustion<br>chamber deposits      | Remove built-up<br>deposits                         |
|                         | Leaking, sticking or<br>broken valves         | Inspect and/or replace<br>valves                    |
| External Oil<br>Leakage | Fuel pump improperly seated<br>or worn gasket | Remove pump, replace<br>gasket and seat<br>properly |
|                         | Oil pan gasket broken or<br>pan bent          | Straighten pan and<br>replace gasket                |
|                         | Timing chain cover gasket<br>broken           | Replace timing chain<br>cover gasket                |
|                         | Rear main oil seal worn                       | Replace rear main oil<br>seal                       |
|                         | Oil pan drain plug not<br>seated properly     | Remove and reinstall<br>drain plug                  |
|                         | Camshaft bearing drain<br>hole blocked        | Remove restriction                                  |
|                         | Oil pressure sending<br>switch leaking        | Remove and reinstall<br>sending switch              |

|                           |                                                    |                                                      |
|---------------------------|----------------------------------------------------|------------------------------------------------------|
| Excessive Oil Consumption | Worn valve stems or guides                         | Replace stems or guides, See ENGINES                 |
|                           | Valve "O" ring seals damaged                       | Replace "O" ring seals, See ENGINES                  |
|                           | Plugged oil drain back holes                       | Remove restrictions                                  |
|                           | Improper PCV valve operation                       | Replace PCV valve                                    |
|                           | Engine oil level too high                          | Remove excess oil                                    |
|                           | Engine oil too thin                                | Replace thicker oil                                  |
|                           | Valve stem oil deflectors damaged                  | Replace oil deflectors                               |
|                           | Incorrect piston rings                             | Replace piston rings, See ENGINES                    |
|                           | Piston ring gaps not staggered                     | Reinstall piston rings, See ENGINES                  |
|                           | Insufficient piston ring tension                   | Replace rings, See ENGINES                           |
|                           | Piston ring grooves or oil return slots clogged    | Replace piston rings, See ENGINES                    |
|                           | Piston rings sticking in grooves                   | Replace piston rings, See ENGINES                    |
|                           | Piston ring grooves excessively worn               | Replace piston and rings, See ENGINES                |
|                           | Compression rings installed upside down            | Replace compression rings correctly, See ENGINES     |
|                           | Worn or scored cylinder walls                      | Rebore cylinders or replace block                    |
|                           | Mismatched oil ring expander and rail              | Replace oil ring expander and rail, See ENGINES      |
|                           | Intake gasket dowels too long                      | Replace intake gasket dowels                         |
|                           | Excessive main or connecting rod bearing clearance | Replace main or connecting rod bearings, See ENGINES |
| No Oil Pressure           | Low oil level                                      | Add oil to proper level                              |
|                           | Oil pressure sender or gauge broken                | Replace sender or gauge                              |
|                           | Oil pump malfunction                               | Remove and overhaul oil pump, See ENGINES            |
|                           | Oil pressure relief valve sticking                 | Remove and reinstall valve                           |
|                           | Oil pump passages blocked                          | Overhaul oil pump, See ENGINES                       |
|                           | Oil pickup screen or tube blocked                  | remove restriction                                   |
|                           | Loose oil inlet tube                               | Tighten oil inlet tube                               |
|                           | Loose camshaft bearings                            | Replace camshaft bearings, See ENGINES               |
| Low Oil Pressure          | Internal leakage at oil passages                   | Replace block or cylinder head                       |
|                           | Low engine oil level                               | Add oil to proper level                              |
|                           | Engine oil too thin                                | Remove and replace with thicker oil                  |
|                           | Excessive oil pump clearance                       | Reduce oil pump clearance, See ENGINES               |
|                           | Oil pickup tube or screen blocked                  | Remove restrictions                                  |
|                           | Main, rod or cam bearing clearance excessive       | Replace bearing to reduce clearance, See             |

## ENGINES

|                         |                                                |                                      |
|-------------------------|------------------------------------------------|--------------------------------------|
| High Oil Pressure       | Improper grade of oil                          | Replace with proper oil              |
|                         | Oil pressure relief valve stuck closed         | Eliminate binding                    |
|                         | Oil pressure sender or gauge faulty            | Replace sender or gauge              |
| Noisy Main Bearings     | Inadequate oil supply                          | Check oil delivery to main bearings  |
|                         | Excessive main bearing clearance               | Replace main bearings, See ENGINES   |
|                         | Excessive crankshaft end play                  | Replace crankshaft, See ENGINES      |
|                         | Loose flywheel or torque converter             | Tighten attaching bolts              |
|                         | Loose or damaged vibration damper              | Tighten or replace vibration damper  |
|                         | Crankshaft journals out-of-round               | Re-grind crankshaft journals         |
|                         | Excessive belt tension                         | Loosen belt tension                  |
| Noisy Connecting Rods   | Excessive bearing clearance or missing bearing | Replace bearing, See ENGINES         |
|                         | Crankshaft rod journal out-of-round            | Re-grind crankshaft journal          |
|                         | Misaligned connecting rod or cap               | Remove rod or cap and realign        |
|                         | Incorrectly tightened rod bolts                | Remove and re-tighten rod bolts      |
|                         |                                                |                                      |
| Noisy Pistons and Rings | Excessive piston-to-bore clearance             | Install larger pistons, See ENGINES  |
|                         | Bore tapered or out-of-round                   | Rebore block                         |
|                         | Piston ring broken                             | Replace piston rings, See ENGINES    |
|                         | Piston pin loose or seized                     | Replace piston pin, See ENGINES      |
|                         | Connecting rods misaligned                     | Realign connecting rods              |
|                         | Ring side clearance too loose or tight         | Replace with larger or smaller rings |
|                         | Carbon build-up on piston                      | Remove carbon                        |
|                         |                                                |                                      |
| Noisy Valve Train       | Worn or bent push rods                         | Replace push rods, See ENGINES       |
|                         | Worn rocker arms or bridged pivots             | Replace push rods, See ENGINES       |
|                         | Dirt or chips in valve lifters                 | Remove lifters and remove dirt/chips |
|                         | Excessive valve lifter leak-down               | Replace valve lifters, See ENGINES   |
|                         | Valve lifter face worn                         | Replace valve lifters, See ENGINES   |
|                         | Broken or cocked valve springs                 | replace or reposition springs        |
|                         | Too much valve stem-to-guide clearance         | Replace valve guides, See ENGINES    |
|                         | Valve bent                                     | Replace valve, See ENGINES           |
|                         | Loose rocker arms                              | Retighten rocker arms, See ENGINES   |
|                         | Excessive valve seat run-out                   | Reface valve seats, See ENGINES      |
|                         |                                                |                                      |
|                         |                                                |                                      |
|                         |                                                |                                      |



|                                   |                                                 |                                            |
|-----------------------------------|-------------------------------------------------|--------------------------------------------|
|                                   | Missing valve lock                              | Install new valve lock                     |
|                                   | Excessively worn camshaft lobes                 | Replace camshaft, See ENGINES              |
|                                   | Plugged valve lifter oil holes                  | Eliminate restriction or replace lifter    |
|                                   | Faulty valve lifter check ball                  | Replace lifter check ball, See ENGINES     |
|                                   | Rocker arm nut installed upside down            | Remove and reinstall correctly             |
|                                   | Valve lifter incorrect for engine               | Remove and replace valve lifters           |
|                                   | Faulty push rod seat or lifter plunger          | Replace plunger or push rod                |
| Noisy Valves                      | Improper valve lash                             | Re-adjust valve lash, See ENGINES          |
|                                   | Worn or dirty valve lifters                     | Clean and/or replace lifters               |
|                                   | Worn valve guides                               | Replace valve guides, See ENGINES          |
|                                   | Excessive valve seat or face run-out            | Reface seats or valve face                 |
|                                   | Worn camshaft lobes                             | Replace camshaft, See ENGINES              |
|                                   | Loose rocker arm studs                          | Re-tighten rocker arm studs, See ENGINES   |
|                                   | Bent push rods                                  | Replace push rods, See ENGINES             |
|                                   | Broken valve springs                            | Replace valve springs, See ENGINES         |
| Burned, Sticking or Broken Valves | Weak valve springs or warped valves             | Replace valves and/or springs, See ENGINES |
|                                   | Improper lifter clearance                       | Re-adjust clearance or replace lifters     |
|                                   | Worn guides or improper guide clearance         | Replace valve guides, See ENGINES          |
|                                   | Out-of-round valve seats or improper seat width | Re-grind valve seats                       |
|                                   | Gum deposits on valve stems, seats or guides    | Remove deposits                            |
|                                   | Improper spark timing                           | Re-adjust spark timing                     |
| Broken Pistons/Rings              | Undersize pistons                               | Replace with larger pistons, See ENGINES   |
|                                   | Wrong piston rings                              | Replace with correct rings, See ENGINES    |
|                                   | Out-of-round cylinder bore                      | Re-bore cylinder bore                      |
|                                   | Improper connecting rod alignment               | Remove and realign connecting rods         |
|                                   | Excessively worn ring grooves                   | Replace pistons, See ENGINES               |
|                                   | Improperly assembled piston pins                | Re-assemble pin-to-piston, See ENGINES     |
|                                   | Insufficient ring gap clearance                 | Install new rings, See ENGINES             |
|                                   | Engine overheating                              | Check cooling system                       |
|                                   | Incorrect ignition timing                       | Re-adjust ignition timing, See TUNE-UP     |
| Excessive Exhaust Noise           | Leaks at manifold to head, or to pipe           | Replace manifold or pipe gasket            |
|                                   | Exhaust manifold                                | Replace exhaust                            |

cracked or broken

manifold, See ENGINES

## ENGINE PERFORMANCE

### CARBURETOR TROUBLE SHOOTING:

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC COLD START SYMPTOMS TROUBLE SHOOTING CHART

| CONDITION                                                   | POSSIBLE CAUSE                      | CORRECTION                                  |
|-------------------------------------------------------------|-------------------------------------|---------------------------------------------|
| Engine Won't Start                                          | Choke not closing                   | Check choke operation, see FUEL SYSTEMS     |
|                                                             | Choke linkage bent                  | Check linkage, see FUEL SYSTEM              |
| Engine Starts, Then Dies                                    | Choke vacuum kick setting too wide  | Check setting and adjust see, FUEL SYSTEMS  |
|                                                             | Fast idle RPM too low               | Reset RPM to specification, see TUNE-UP     |
|                                                             | Fast idle cam index incorrect       | Reset fast idle cam index, see FUEL SYSTEMS |
|                                                             | Vacuum leak                         | Inspect vacuum system for leaks             |
|                                                             | Low fuel pump outlet                | Repair or replace pump, see FUEL SYSTEMS    |
|                                                             | Low carburetor fuel level           | Check float setting see FUEL SYSTEM         |
| Engine Quits Under Load                                     | Choke vacuum kick setting incorrect | Reset vacuum kick setting, see FUEL SYSTEMS |
|                                                             | Fast idle cam index incorrect       | Reset fast idle cam index, see FUEL SYSTEM  |
|                                                             | Incorrect hot fast idle speed RPM   | Reset fast idle RPM, see TUNE-UP            |
| Engine Starts, Runs Up, Then Idles, Slowly With Black Smoke | Choke vacuum kick set too narrow    | Reset vacuum kick, see FUEL SYSTEMS         |
|                                                             | Fast idle cam index                 | Reset fast idle cam                         |

incorrect

index, see FUEL  
SYSTEMS

Hot fast idle RPM too low

Reset fast idle RPM,  
see TUNE-UP

---

#### BASIC HOT START SYMPTOMS TROUBLE SHOOTING CHART

---

| CONDITION          | POSSIBLE CAUSE | CORRECTION              |
|--------------------|----------------|-------------------------|
| Engine Won't Start | Engine flooded | Allow fuel to evaporate |

---

---

#### BASIC COLD ENGINE DRIVEABILITY SYMPTOMS TROUBLE SHOOTING CHART

---

| CONDITION                 | POSSIBLE CAUSE                      | CORRECTION                                     |
|---------------------------|-------------------------------------|------------------------------------------------|
| Engine Stalls in Gear     | Choke vacuum kick setting incorrect | Reset choke vacuum kick, see FUEL SYSTEMS      |
|                           | Fast idle RPM incorrect             | Reset fast idle RPM, see TUNE-UP               |
|                           | Fast idle cam index incorrect       | Reset fast idle cam see FUEL SYSTEMS           |
| Acceleration Sag or Stall | Defective choke control switch      | Replace choke control switch                   |
|                           | Choke vacuum kick setting incorrect | Reset choke vacuum kick see, FUEL SYSTEMS      |
|                           | Float level incorrect (too low)     | Adjust float level, FUEL SYSTEMS               |
|                           | Accelerator pump defective          | Repair or replace pump see FUEL SYSTEMS        |
|                           | Secondary throttles not closed      | Inspect lockout adjustment, see FUEL SYSTEMS   |
| Sag or Stall After Warmup | Defective choke control switch      | Replace choke control switch, see FUEL SYSTEMS |
|                           | Defective accelerator pump          | Replace pump, see FUEL SYSTEMS                 |
|                           | Float level incorrect (too low)     | Adjust float level, see FUEL SYSTEMS           |
| Backfiring & Black Smoke  | Plugged heat crossover system       | Remove restriction                             |

---

---

#### BASIC WARM ENGINE DRIVEABILITY SYMPTOMS TROUBLE SHOOTING CHART

---

| CONDITION                                          | POSSIBLE CAUSE                             | CORRECTION                                    |
|----------------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Hesitation With Small Amount of Gas Pedal Movement | Vacuum leak                                | Inspect vacuum lines                          |
|                                                    | Accelerator pump weak or inoperable        | Replace pump, see FUEL SYSTEMS                |
|                                                    | Float level setting too low                | Reset float level, see, FUEL SYSTEMS          |
|                                                    | Metering rods sticking or binding          | Inspect and/or replace rods, see FUEL SYSTEMS |
|                                                    | Carburetor idle or transfer system plugged | Inspect system and remove restriction         |
|                                                    | Frozen or binding heated air inlet         | Inspect heated air door for binding           |
| Hesitation With Heavy Gas Pedal Movement           | Defective accelerator pump                 | Replace pump, see FUEL SYSTEMS                |
|                                                    | Metering rod carrier sticking or binding   | Remove restriction                            |
|                                                    | Large vacuum leak                          | Inspect vacuum system and repair leak         |
|                                                    | Float level setting too low                | Reset float level, see FUEL SYSTEMS           |
|                                                    | Defective fuel pump, lines or filter       | Inspect pump, lines and filter                |
|                                                    | Air door setting incorrect                 | Adjust air door setting, see FUEL             |

## DIESEL ENGINE TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

**NOTE:** Diesel engines mechanical diagnosis is the same as gasoline engines for items such as noisy valves, bearings, pistons, etc. The following trouble shooting covers only items pertaining to diesel engines.

### BASIC DIESEL ENGINE TROUBLE SHOOTING CHART

| CONDITION          | POSSIBLE CAUSE                            | CORRECTION                       |
|--------------------|-------------------------------------------|----------------------------------|
| Engine Won't Crank | Bad battery connections or dead batteries | Check connections and/or replace |

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | Bad starter connections<br>or bad starter                                                                                                                                                                                                                                                                                                                                                                                                 | batteries<br>Check connections<br>and/or replace<br>batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Engine Cranks<br>Slowly, Won't<br>Start                | Bad battery connections<br>or dead batteries<br><br>Engine oil too heavy                                                                                                                                                                                                                                                                                                                                                                  | Check connections<br>and/or replace<br>batteries<br>Replace engine oil                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Engine Cranks<br>Normally, But<br>Will Not Start       | Glow plugs not functioning<br><br>Glow plug control not<br>functioning<br>Fuel not injected into<br>cylinders<br>No fuel to injection pump<br><br>Fuel filter blocked<br>Fuel tank filter blocked<br><br>Fuel pump not operating<br><br>Fuel return system blocked<br><br>No voltage to fuel solenoid<br><br>Incorrect or contaminated<br>fuel<br>Incorrect injection pump<br>timing<br>Low compression<br><br>Injection pump malfunction | Check glow plug system,<br>see FUEL SYSTEMS<br>Check controller, see<br>FUEL SYSTEMS<br>Check fuel injectors,<br>see FUEL SYSTEMS<br>Check fuel delivery<br>system<br>Replace fuel filter<br>Replace fuel tank<br>filter<br>Check pump operation<br>and/or replace pump<br>Inspect system and<br>remove restriction<br>Check solenoid and<br>connections<br>Replace fuel<br><br>Re-adjust pump timing,<br>see FUEL SYSTEMS<br>Check valves, pistons,<br>rings, see ENGINES<br>Inspect and/or replace<br>injection pump |
| Engine Starts,<br>Won't Idle                           | Incorrect slow idle<br>adjustment<br>Fast idle solenoid<br>malfunctioning<br>Fuel return system blocked<br><br>Glow plugs go off too soon<br><br>Injection pump timing<br>incorrect<br>No fuel to injection pump<br><br>Incorrect or contaminated<br>fuel<br>Low compression<br><br>Injection pump malfunction<br><br>Fuel solenoid closes in RUN<br>position                                                                             | Reset idle adjustment,<br>see TUNE-UP<br>Check solenoid and<br>connections<br>Check system and remove<br>restrictions<br>See glow plug diagnosis<br>in FUEL SYSTEMS<br>Reset pump timing, see<br>FUEL SYSTEMS<br>Check fuel delivery<br>system<br>Replace fuel<br><br>Check valves, piston,<br>rings, see ENGINES<br>Replace injection pump,<br>see FUEL SYSTEMS<br>Check solenoid and<br>connections                                                                                                                  |
| Engines Starts/<br>Idles Rough W/out<br>Smoke or Noise | Incorrect slow idle<br>adjustment<br>Injection line fuel leaks<br><br>Fuel return system blocked                                                                                                                                                                                                                                                                                                                                          | Reset slow idle, see<br>TUNE-UP<br>Check lines and<br>connections<br>Check lines and<br>connections                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                                               |                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                               | Air in fuel system<br>Incorrect or contaminated fuel<br>Injector nozzle malfunction                                                                                                                                                                                                                                                             | Bleed air from system<br>Replace fuel<br>Check nozzles, see FUEL SYSTEMS                                                                                                                                                                                                                                                                                                 |
| Engines Starts and Idles Rough W/out Smoke or Noise, But Clears After Warm-Up | Injection pump timing incorrect<br>Engine not fully broken in<br>Air in system<br>Injector nozzle malfunction                                                                                                                                                                                                                                   | Reset pump timing, see FUEL SYSTEMS<br>Put more miles on engine<br>Bleed air from system<br>Check nozzles, see FUEL SYSTEMS                                                                                                                                                                                                                                              |
| Engine Idles Correctly, Misfires Above Idle                                   | Blocked fuel filter<br>Injection pump timing incorrect<br>Incorrect or contaminated fuel                                                                                                                                                                                                                                                        | Replace fuel filter<br>Reset pump timing, see FUEL SYSTEMS<br>Replace fuel                                                                                                                                                                                                                                                                                               |
| Engine Won't Return To Idle                                                   | Fast idle adjustment incorrect<br>Internal injection pump malfunction<br>External linkage binding                                                                                                                                                                                                                                               | Reset fast idle, see TUNE-UP<br>Replace injection pump, see FUEL SYSTEMS<br>Check linkage and remove binding                                                                                                                                                                                                                                                             |
| Fuel Leaks On Ground                                                          | Loose or broken fuel line<br>Internal injection pump seal leak                                                                                                                                                                                                                                                                                  | Check lines and connections<br>Replace injection pump, see FUEL SYSTEMS                                                                                                                                                                                                                                                                                                  |
| Cylinder Knocking Noise                                                       | Injector nozzles sticking open<br>Very low nozzle opening pressure                                                                                                                                                                                                                                                                              | Test injectors, see FUEL SYSTEMS<br>Test injectors and/or replace                                                                                                                                                                                                                                                                                                        |
| Loss of Engine Power                                                          | Restricted air intake<br>EGR valve malfunction<br>Blocked or damaged exhaust system<br><br>Blocked fuel tank filter<br>Restricted fuel filter<br><br>Block vent in gas cap<br><br>Tank-to-injection pump fuel supply blocked<br>Blocked fuel return system<br>Incorrect or contaminated fuel<br>Blocked injector nozzles<br><br>Low compression | Remove restriction<br>Replace EGR valve<br>Remove restriction and/or replace components<br>Replace filter<br>Remove restriction and/or replace filter<br>Remove restriction and/or replace cap<br>Check fuel lines and connections<br>Remove restriction<br>Replace fuel<br><br>Check nozzle for blockage, see FUEL SYSTEMS<br>Check valves, rings, pistons, see ENGINES |
| Loud Engine Noise With Black Smoke                                            | Basic timing incorrect<br>EGR valve malfunction<br>Internal injection pump malfunction                                                                                                                                                                                                                                                          | Reset timing, see FUEL SYSTEMS<br>Replace EGR valve<br>Replace injection pump, see FUEL SYSTEMS                                                                                                                                                                                                                                                                          |

|                       |                                                                        |                                                   |
|-----------------------|------------------------------------------------------------------------|---------------------------------------------------|
|                       | Incorrect injector pump housing pressure                               | Check pressure, see FUEL SYSTEMS                  |
| Engine Overheating    | Cooling system leaks                                                   | Check cooling system and repair leaks             |
|                       | Belt slipping or damaged                                               | Check tension and/or replace belt                 |
|                       | Thermostat stuck closed                                                | Remove and replace thermostat, see ENGINE COOLING |
|                       | Head gasket leaking                                                    | Replace head gasket                               |
| Oil Light on at Idle  | Low oil pump pressure                                                  | Check oil pump operation, see ENGINES             |
|                       | Oil cooler or line restricted                                          | Remove restriction and/or replace cooler          |
| Engine Won't Shut Off | Injector pump fuel solenoid does not return fuel valve to OFF position | Remove and check solenoid and replace if needed   |
| VACUUM PUMP DIAGNOSIS |                                                                        |                                                   |
| Excessive Noise       | Loose pump-to-drive assembly screws                                    | Tighten screws                                    |
|                       | Loose tube on pump assembly                                            | Tighten tube                                      |
|                       | Valves not functioning properly                                        | Replace valves                                    |
| Oil Leakage           | Loose end plug                                                         | Tighten end plug                                  |
|                       | Bad seal crimp                                                         | Remove and re-crimp seal                          |

## FUEL INJECTION TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC FUEL INJECTION TROUBLE SHOOTING CHART

| CONDITION                             | POSSIBLE CAUSE                       | CORRECTION                              |
|---------------------------------------|--------------------------------------|-----------------------------------------|
| Engine Won't Start (Crankes Normally) | Cold start valve inoperative         | Test valve and circuit                  |
|                                       | Poor connection;vacuum or wiring     | Check vacuum and electrical connections |
|                                       | Contaminated fuel                    | Test fuel for water or alcohol          |
|                                       | Defective fuel pump relay or circuit | Test relay and wiring                   |
|                                       | Battery too low                      | Charge and test battery                 |

|                           |                                          |                                                                                      |
|---------------------------|------------------------------------------|--------------------------------------------------------------------------------------|
|                           | Low fuel pressure                        | Test pressure regulator and fuel pump, check for restricted lines and filters        |
|                           | No distributor reference pulses          | Repair ignition system as necessary                                                  |
|                           | Open coolant temperature sensor circuit  | Test sensor and wiring                                                               |
|                           | Shorted W.O.T. switch in T.P.S.          | Disconnect W.O.T. switch, engine should start                                        |
|                           | Defective ECM                            | Replace ECM                                                                          |
|                           | Fuel tank residual pressure valve leaks  | Test for fuel pressure drop after shut down                                          |
| Hard Starting             | Disconnected hot air tube to air cleaner | Reconnect tube and test control valve                                                |
|                           | Defective Idle Air Control (IAC) valve   | Test valve operation and circuit                                                     |
|                           | Shorted, open or misadjusted T.P.S.      | Test and adjust or replace T.P.S.                                                    |
|                           | EGR valve open                           | Test EGR valve and control circuit                                                   |
|                           | Poor Oxygen sensor signal                | Test for shorted or circuit                                                          |
|                           | Incorrect mixture from PCV system        | Test PCV for flow, check sealing of oil filter cap                                   |
| Poor High Speed Operation | Low fuel pump volume                     | Faulty pump or restricted fuel lines or filters                                      |
|                           | Poor MAP sensor signal                   | Test MAP sensor, vacuum hose and wiring                                              |
|                           | Poor Oxygen sensor signal                | Test for shorted or open sensor or circuit                                           |
|                           | Open coolant temperature sensor circuit  | Test sensor and wiring                                                               |
|                           | Faulty ignition operation                | Check wires for cracks or poor connections, test secondary voltage with oscilloscope |
|                           | Contaminated fuel                        | Test fuel for water                                                                  |



|                               |                                       |                                                                                    |
|-------------------------------|---------------------------------------|------------------------------------------------------------------------------------|
|                               |                                       | or alcohol                                                                         |
|                               | Intermittent ECM ground               | Test ECM ground connection for resistance                                          |
|                               | Restricted air cleaner                | Replace air cleaner                                                                |
|                               | Restricted exhaust system             | Test for exhaust manifold back pressure                                            |
|                               | Poor MAF sensor signal                | Check leakage between sensor and manifold                                          |
|                               | Poor VSS signal                       | If tester for ALCL hook-up is available check that VSS reading matches speedometer |
| Ping or Knock on Acceleration | Poor Knock sensor signal              | Test for shorted or open sensor or circuit                                         |
|                               | Poor Baro sensor signal               | Test for shorted or open sensor or circuit                                         |
|                               | Improper ignition timing              | See VEHICLE EMISSION CONTROL LABEL (where applicable)                              |
|                               | Check for engine overheating problems | Low coolant, loose belts or electric cooling fan inoperative                       |

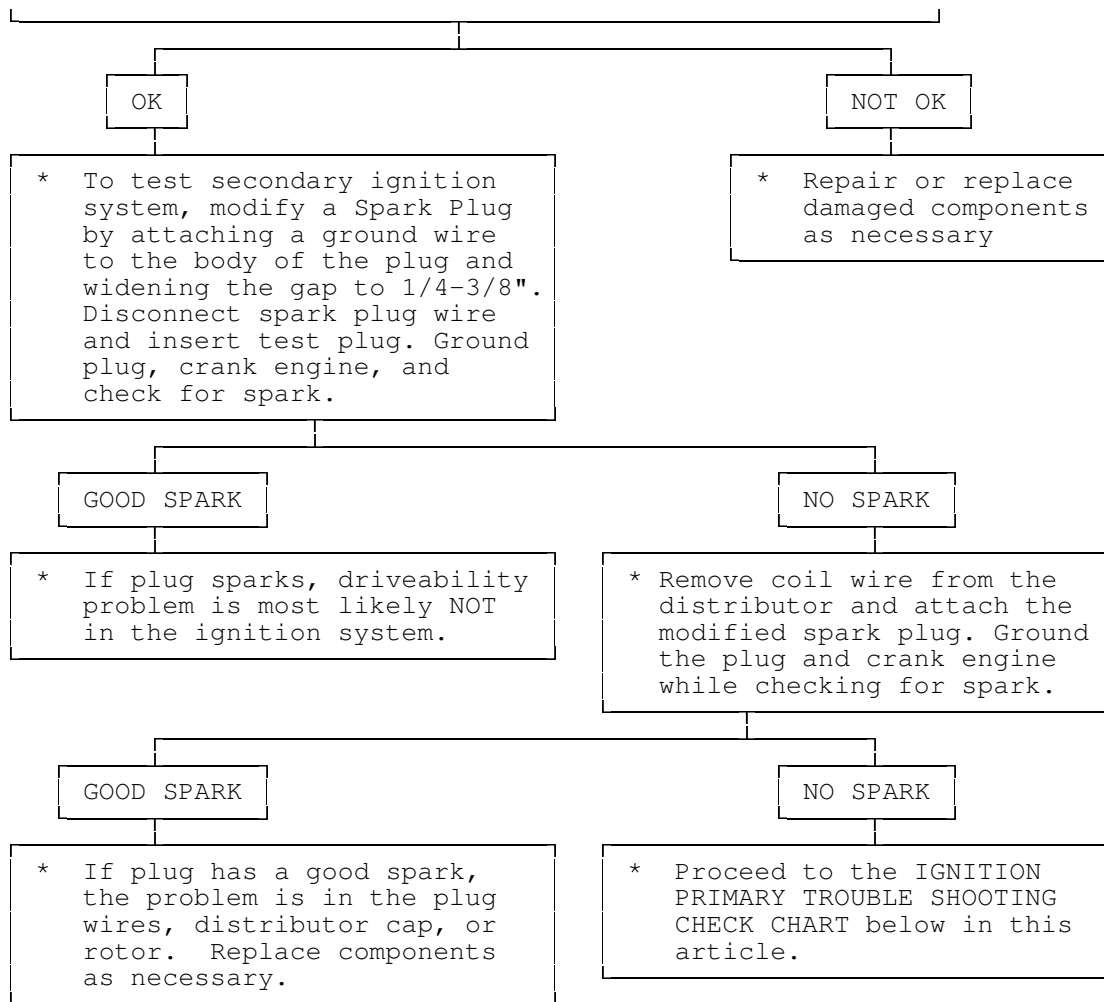
NOTE: For additional electronic fuel injection trouble shooting information, see the appropriate article in the ENGINE PERFORMANCE section (not all vehicles have Computer Engine Control articles). Information is provided there for diagnosing fuel system problems on vehicles with electronic fuel injection.

## IGNITION SYSTEM TROUBLE SHOOTING

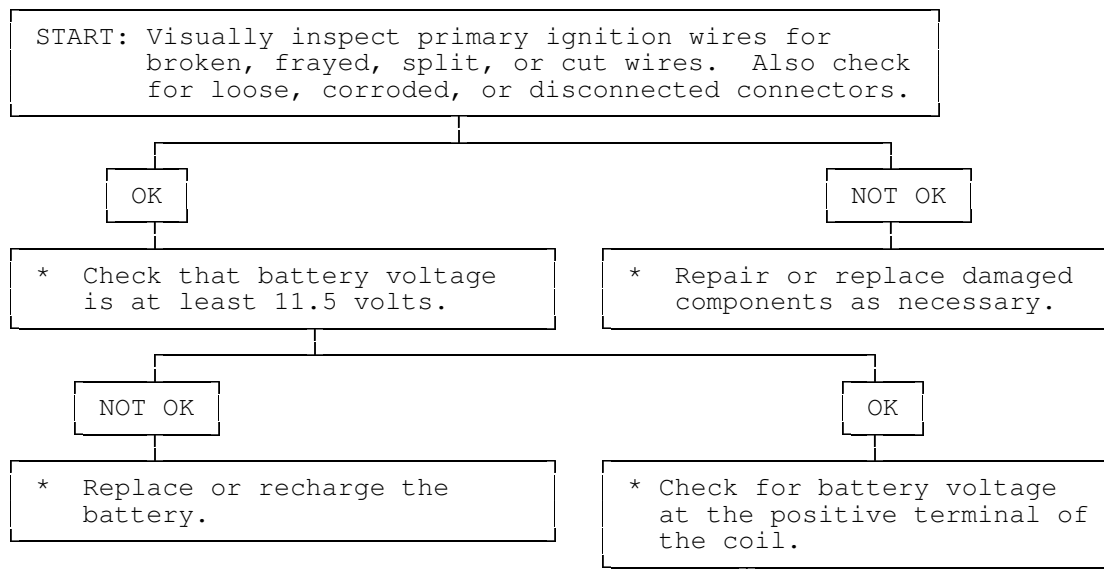
NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

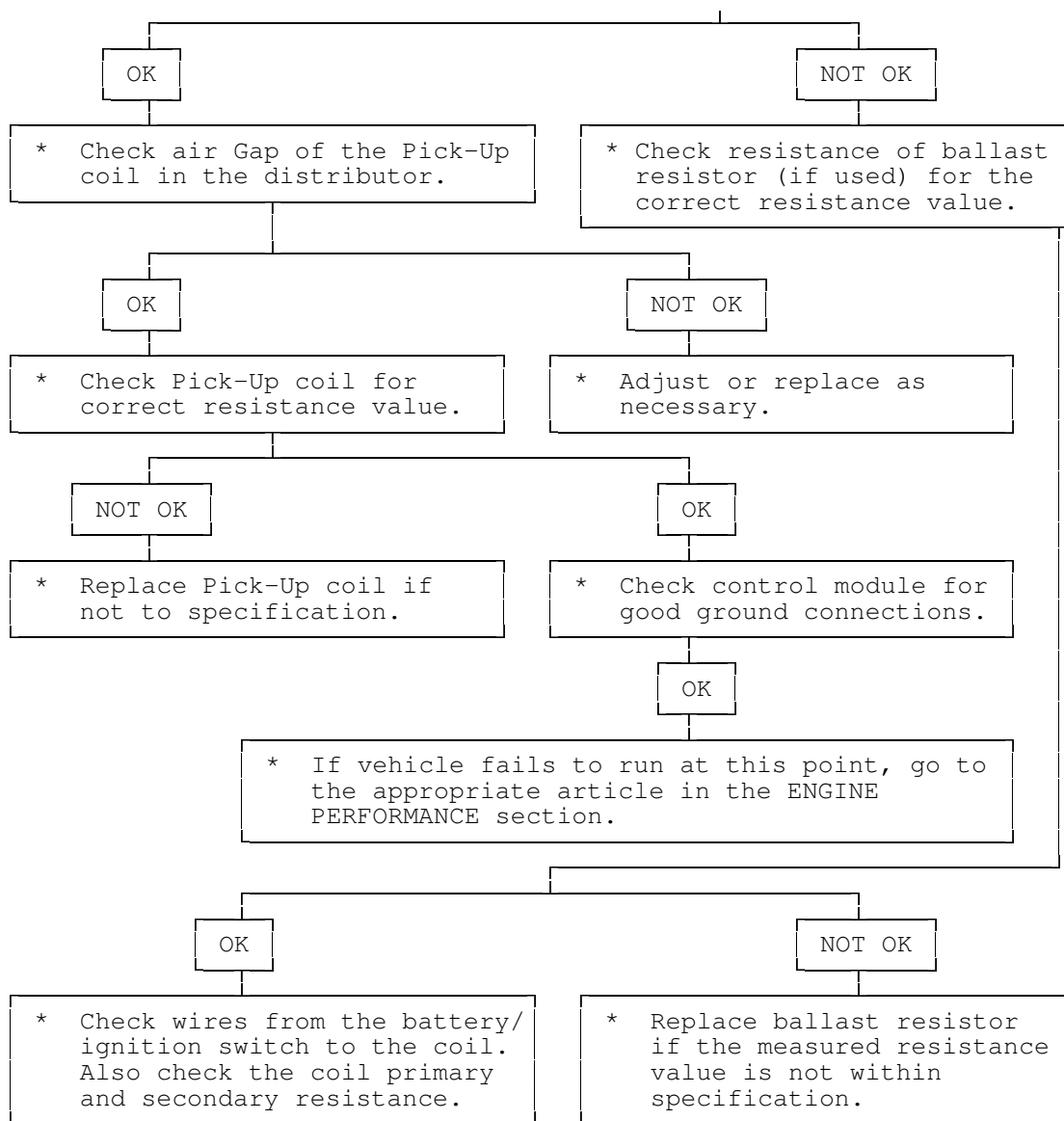
Ignition Secondary Trouble Shooting Chart

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| <p>START: Visually inspect Spark Plug Wires, Coil Wires, Plug Wire Boots, Rotor, and Distributor Cap for signs of damage.</p> |
|-------------------------------------------------------------------------------------------------------------------------------|



Ignition Primary Trouble Shooting Chart





## STARTER TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC STARTER TROUBLE SHOOTING CHART

| CONDITION                | POSSIBLE CAUSE                                              | CORRECTION                                                    |
|--------------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| Starter Fails to Operate | Dead battery or bad connections between starter and battery | Check battery charge and all wires and connections to starter |

|                                             |                                                                        |                                                           |
|---------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------|
|                                             | Ignition switch faulty or misadjusted                                  | Adjust or replace ignition switch                         |
|                                             | Open circuit between starter switch ignition terminal on starter relay | Check and repair wires and connections as necessary       |
|                                             | Starter relay or starter defective                                     | See Testing in STARTER article                            |
|                                             | Open solenoid pull-in wire                                             | See Testing in STARTER article                            |
| Starter Does Not Operate and Headlights Dim | Weak battery or dead cell                                              | Charge or replace battery as necessary                    |
|                                             | Loose or corroded battery connections                                  | Check that battery connections are clean and tight        |
|                                             | Internal ground in starter windings                                    | See Testing in STARTER article                            |
|                                             | Grounded starter fields                                                | See Testing in STARTERS                                   |
|                                             | Armature rubbing on pole shoes                                         | See STARTER article                                       |
| Starter Turns but Engine Does Not Rotate    | Starter clutch slipping                                                | See STARTER article                                       |
|                                             | Broken clutch housing                                                  | See STARTER article                                       |
|                                             | Pinion shaft rusted or dry                                             | See STARTER article                                       |
|                                             | Engine basic timing incorrect                                          | See Ignition Timing in TUNE-UP article                    |
|                                             | Broken teeth on engine flywheel                                        | Replace flywheel and check for starter pinion gear damage |
| Starter Will Not Crank Engine               | Faulty overrunning clutch                                              | See STARTER article                                       |
|                                             | Broken clutch housing                                                  | See STARTER article                                       |
|                                             | Broken flywheel teeth                                                  | Replace flywheel and check for starter pinion gear damage |
|                                             | Armature shaft sheared or reduction gear teeth stripped                | See STARTER article                                       |
|                                             | Weak battery                                                           | Charge or replace battery as necessary                    |
|                                             | Faulty solenoid                                                        | See On-Vehicle Tests in STARTER article                   |
|                                             | Poor grounds                                                           | Check all ground                                          |

|                                         |                                                         |                                                                |
|-----------------------------------------|---------------------------------------------------------|----------------------------------------------------------------|
|                                         |                                                         | connections for tight and clean connections                    |
|                                         | Ignition switch faulty or misadjusted                   | Adjust or replace ignition switch as necessary                 |
| Starter Cranks Engine Slowly            | Battery weak or defective                               | Charge or replace battery as necessary                         |
|                                         | Engine overheated                                       | See ENGINE COOLING SYSTEM article                              |
|                                         | Engine oil too heavy                                    | Check that proper viscosity oil is used                        |
|                                         | Poor battery-to-starter connections                     | Check that all between battery and starter are clean and tight |
|                                         | Current draw too low or too high                        | See Bench Tests in STARTER article                             |
|                                         | Bent armature, loose pole shoes screws or worn bearings | See STARTER article                                            |
|                                         | Burned solenoid contacts                                | Replace solenoid                                               |
|                                         | Faulty starter                                          | Replace starter                                                |
| Starter Engages Engine Only Momentarily | Engine timing too far advanced                          | See Ignition Timing in TUNE-UP article                         |
|                                         | Overrunning clutch not engaging properly                | Replace overrunning clutch. See STARTER article                |
|                                         | Broken starter clutch                                   | See STARTER article                                            |
|                                         | Broken teeth on engine flywheel                         | Replace flywheel and check starter pinion gear for damage      |
|                                         | Weak drive assembly thrust spring                       | See STARTER article                                            |
|                                         | Weak hold-in coil                                       | See Bench Tests in STARTER article                             |
| Starter Drive Will Not Engage           | Defective point assembly                                | See Testing in STARTER article                                 |
|                                         | Poor point assembly ground                              | See Testing in STARTER article                                 |
|                                         | Defective pull-in coil                                  | Replace starter solenoid                                       |
| Starter Relay                           | Dead battery                                            | Charge or replace                                              |

|                                                  |                                                                                                |                                                                                       |
|--------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Does Not Close                                   |                                                                                                | battery as necessary                                                                  |
|                                                  | Faulty wiring                                                                                  | Check all wiring and connections leading to relay                                     |
|                                                  | Neutral safety switch faulty                                                                   | Replace neutral safety switch                                                         |
|                                                  | Starter relay faulty                                                                           | Replace starter relay                                                                 |
| Starter Drive Will Not Disengage                 | Starter motor loose on mountings                                                               | Tighten starter attach bolts                                                          |
|                                                  | Worn drive end bushing                                                                         | See STARTER article                                                                   |
|                                                  | Damaged engine flywheel teeth                                                                  | Replace flywheel and starter pinion gear for damage                                   |
|                                                  | Drive yolk return spring broken or missing                                                     | Replace return spring                                                                 |
|                                                  | Faulty ignition switch                                                                         | Replace ignition switch                                                               |
|                                                  | Insufficient clearance between winding leads to solenoid terminal and main contact in solenoid | Replace starter solenoid                                                              |
|                                                  | Starter clutch not disengaging                                                                 | Replace starter clutch                                                                |
|                                                  | Ignition starter switch contacts sticking                                                      | Replace ignition switch                                                               |
| Starter Relay Operates but Solenoid Does Not     | Faulty solenoid switch, switch connections or                                                  | Check all wiring between relay and solenoid or replace relay or solenoid as necessary |
|                                                  | Broken lead or loose soldered connections                                                      | Repair wire or wire connections as necessary                                          |
| Solenoid Plunger Vibrates When Switch is Engaged | Weak battery                                                                                   | Charge or replace battery as necessary                                                |
|                                                  | Solenoid contacts corroded                                                                     | Clean contacts or replace solenoid                                                    |
|                                                  | Faulty wiring                                                                                  | Check all wiring leading to solenoid                                                  |
|                                                  | Broken connections inside switch cover                                                         | Repair connections or replace solenoid                                                |
|                                                  | Open hold-in wire                                                                              | Replace solenoid                                                                      |
| Low Current Draw                                 | Worn brushes or weak                                                                           | Replace brushes or brush springs as necessary                                         |

|                                                                                                            |                                                                                                                        |                                                                                  |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| High Pitched Whine<br>During Cranking<br>Before Engine<br>Fires but Engine<br>Fires and Cranks<br>Normally | Distance too great<br>between starter<br>pinion and flywheel                                                           | Align starter or check<br>that correct starter<br>and flywheel are being<br>used |
| High Pitched<br>Whine After Engine<br>Fires With Key<br>released. Engine<br>Fires and Cranks<br>Normally   | Distance too small between<br>starter pinion and flywheel<br>Flywheel runout contributes<br>to the intermittent nature |                                                                                  |

## TUNE-UP TROUBLE SHOOTING - GAS ENGINE VEHICLES

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC SPARK PLUG TROUBLE SHOOTING CHARTS

| CONDITION                          | POSSIBLE CAUSE                            | CORRECTION                                              |
|------------------------------------|-------------------------------------------|---------------------------------------------------------|
| Normal Spark<br>Plug Condition     | Light Tan or Gray deposits                | No Action                                               |
|                                    | Electrode not burned or fouled            | No Action                                               |
|                                    | Gap tolerance not changed                 | No Action                                               |
| Cold Fouling or<br>Carbon Deposits | Overrich air/fuel mixture                 | Adjust air/fuel mixture, see ENGINE PERFORMANCE section |
|                                    | Faulty choke                              | Replace choke assembly, see ENGINE PERFORMANCE section  |
|                                    | Clogged air filter                        | Clean and/or replace air filter                         |
|                                    | Incorrect idle speed or dirty carburetor  | Reset idle speed and/or clean carburetor                |
|                                    | Faulty ignition wires                     | Replace ignition wiring                                 |
|                                    | Prolonged operation at idle               | Shut engine off during long idle                        |
|                                    | Sticking valves or worn valve guide seals | Check valve train                                       |
| Wet Fouling<br>or Oil Deposits     | Worn rings and pistons                    | Install new rings and pistons                           |

|                                   |                                                            |                                                         |
|-----------------------------------|------------------------------------------------------------|---------------------------------------------------------|
|                                   | Excessive cylinder wear                                    | Rebore or replace block                                 |
|                                   | Excessive valve guide clearance                            | Worn or loose bearing                                   |
| Gap Bridged                       | Deposits in combustion chamber becoming fused to electrode | Clean combustion chamber of deposits                    |
| Blistered Electrode               | Engine overheating                                         | Check cooling system                                    |
|                                   | Wrong type of fuel                                         | Replace with correct fuel                               |
|                                   | Loose spark plugs                                          | Retighten spark plugs                                   |
|                                   | Over-advanced ignition timing                              | Reset ignition timing see ENGINE PERFORMANCE            |
| Pre-Ignition or Melted Electrodes | Incorrect type of fuel                                     | Replace with correct fuel                               |
|                                   | Incorrect ignition timing                                  | Reset ignition timing see ENGINE PERFORMANCE            |
|                                   | Burned valves                                              | Replace valves                                          |
|                                   | Engine Overheating                                         | Check cooling system                                    |
|                                   | Wrong type of spark plug, too hot                          | Replace with correct spark plug, see ENGINE PERFORMANCE |
| Chipped Insulators                | Severe detonation                                          | Check for over-advanced timing or combustion            |
|                                   | Improper gapping procedure                                 | Re-gap spark plugs                                      |
| Rust Colored Deposits             | Additives in unleaded fuel                                 | Try different fuel brand                                |
| Water In Combustion Chamber       | Blown head gasket or cracked head                          | Repair or replace head or head gasket                   |

NOTE: Before diagnosing an electronic ignition system, ensure that all wiring is connected properly between distributor, wiring connector and spark plugs. Ignition problem will show up either as: Engine Will Not Start or Engine Runs Rough.

#### BASIC ELECTRONIC IGNITION TROUBLE SHOOTING CHARTS

| CONDITION          | POSSIBLE CAUSE                                          | CORRECTION     |
|--------------------|---------------------------------------------------------|----------------|
| Engine Won't Start | Open circuit between distributor and bulkhead connector | Repair circuit |
|                    | Open circuit between bulkhead connector and             | Repair circuit |



|                   |                                                           |                                              |
|-------------------|-----------------------------------------------------------|----------------------------------------------|
|                   | ignition switch                                           |                                              |
|                   | Open circuit between ignition switch and starter solenoid | Repair circuit                               |
| Engine Runs Rough | Fuel lines leaking or clogged                             | Tighten fitting, remove restriction          |
|                   | Initial timing incorrect                                  | Reset ignition timing see ENGINE PERFORMANCE |
|                   | Centrifugal advance malfunction                           | Repair distributor advance                   |
|                   | Defective spark plugs or wiring                           | Replace plugs or plug wiring                 |
| Component Failure | Spark arc-over on cap, rotor or coil                      | Replace cap, rotor or or coil                |
|                   | Defective pick-up coil                                    | Replace pick-up coil                         |
|                   | Defective ignition coil                                   | Replace ignition coil                        |
|                   | Defective vacuum unit                                     | Replace vacuum unit                          |
|                   | Defective control module                                  | Replace control module                       |

#### BASIC ELECTRONIC IGNITION TROUBLE SHOOTING CHARTS - USING OSCILLOSCOPE PATTERNS

| CONDITION                                              | POSSIBLE CAUSE                         | CORRECTION                                            |
|--------------------------------------------------------|----------------------------------------|-------------------------------------------------------|
| Firing Voltage Lines are the Same, but Abnormally High | Retarded ignition timing               | Reset ignition timing, see ENGINE PERFORMANCE section |
|                                                        | Fuel mixture too lean                  | Readjust carburetor, see ENGINE PERFORMANCE           |
|                                                        | High resistance in coil wire           | Replace coil wire                                     |
|                                                        | Corrosion in coil tower terminal       | Clean and/or replace coil                             |
|                                                        | Corrosion in distributor coil terminal | Clean and/or replace distributor cap                  |
| Firing Voltage Lines are the Same but Abnormally Low   | Fuel mixture too rich                  | Readjust carburetor, see ENGINE PERFORMANCE           |
|                                                        | Breaks in coil wire causing arcing     | Replace coil wire                                     |
|                                                        | Cracked coil tower causing arcing      | Replace coil                                          |
|                                                        | Low coil output                        | Replace coil                                          |
|                                                        | Low engine compression                 | Determine cause and                                   |

|                                                                      |                                        |                                             |
|----------------------------------------------------------------------|----------------------------------------|---------------------------------------------|
|                                                                      |                                        | repair                                      |
| One or More, But Not All Firing Voltage Lines are Higher Than Others | Carburetor idle mixture not balanced   | Readjust carburetor, see ENGINE PERFORMANCE |
|                                                                      | EGR valve stuck open                   | Clean and/or replace valve                  |
|                                                                      | High resistance in spark plug wires    | Replace spark plug wires                    |
|                                                                      | Cracked or broken spark plug insulator | Replace spark plugs                         |
|                                                                      | Intake vacuum leak                     | Repair leak                                 |
|                                                                      | Defective spark plugs                  | Replace spark plugs                         |
|                                                                      | Corroded spark plug terminals          | Replace spark plugs                         |
| One or More, But Not All Firing Voltage Lines Are Lower Than Others  | Carb idle mixture not balanced         | Readjust carburetor, see ENGINE PERFORMANCE |
|                                                                      | Breaks in plug wires causing arcing    | Replace plug wires                          |
|                                                                      | Cracked coil tower causing arcing      | Replace coil                                |
|                                                                      | Low compression                        | Determine cause and repair                  |
|                                                                      | Defective spark plugs                  | Replace spark plugs                         |
| Cylinders Not Firing                                                 | Corroded spark plugs                   | Replace spark plugs                         |
|                                                                      | Cracked distributor cap terminals      | Replace distributor cap                     |
|                                                                      | Shorted spark plug wire                | Determine cause and repair                  |
|                                                                      | Mechanical problem in engine           | Determine cause and repair                  |
|                                                                      | Defective spark plugs                  | Replace spark plugs                         |
|                                                                      | Spark plugs fouled                     | Replace spark plugs                         |

#### BASIC DRIVEABILITY PROBLEMS TROUBLE SHOOTING TABLE

| CONDITION     | POSSIBLE CAUSE             | CORRECTION                      |
|---------------|----------------------------|---------------------------------|
| Hard Starting | Binding carburetor linkage | Eliminate binding               |
|               | Binding choke linkage      | Eliminate binding               |
|               | Binding choke piston       | Eliminate binding               |
|               | Restricted choke vacuum    | Check vacuum lines for blockage |

|                     |                                        |                                                              |
|---------------------|----------------------------------------|--------------------------------------------------------------|
|                     | Worn or dirty needle valve and seat    | Clean carburetor, see ENGINE PERFORMANCE                     |
|                     | Float sticking                         | Readjust or replace float see the ENGINE PERFORMANCE section |
|                     | Incorrect choke adjustment             | Reset choke adjustment see ENGINE PERFORMANCE                |
|                     | Defective coil                         | Replace coil                                                 |
|                     | Improper spark plug gap                | Regap spark plugs                                            |
|                     | Incorrect ignition timing              | Reset ignition timing see ENGINE PERFORMANCE                 |
| Detonation          | Over-advanced ignition timing          | Reset ignition timing see ENGINE PERFORMANCE                 |
|                     | Defective spark plugs                  | Replace spark plugs                                          |
|                     | Fuel lines clogged                     | Clean fuel lines                                             |
|                     | EGR system malfunction                 | Check and repair EGR system                                  |
|                     | PCV system malfunction                 | Repair PCV system                                            |
|                     | Vacuum leaks                           | Check and repair vacuum system                               |
|                     | Loose fan belts                        | Tighten or replace fan belts, see ENGINE PERFORMANCE         |
|                     | Restricted airflow                     | Remove restriction                                           |
|                     | Vacuum advance malfunction             | Check distributor operation                                  |
| Dieseling           | Binding carburetor linkage             | Eliminate binding                                            |
|                     | Binding throttle linkage               | Eliminate blinding                                           |
|                     | Binding choke linkage or fast idle cam | Eliminate binding                                            |
|                     | Defective idle solenoid                | Replace idle solenoid see ENGINE PERFORMANCE                 |
|                     | Improper base idle speed               | Reset idle speed, see see ENGINE PERFORMANCE                 |
|                     | Incorrect ignition timing              | Reset ignition timing see ENGINE PERFORMANCE                 |
|                     | Incorrect idle mixture setting         | Reset idle mixture, see ENGINE PERFORMANCE                   |
| Faulty Acceleration | Incorrect ignition timing              | Reset ignition timing see ENGINE PERFORMANCE                 |

|                             |                                           |                                                 |
|-----------------------------|-------------------------------------------|-------------------------------------------------|
|                             | Engine cold and choke too lean            | Adjust choke and allow engine to warm-up        |
|                             | Defective spark plugs                     | Replace spark plugs                             |
|                             | Defective coil                            | Replace coil                                    |
| Faulty Low Speed Operation  | Clogged idle transfer slots               | Clean idle transfer slots, see FUEL             |
|                             | Restricted idle air bleeds and passages   | Disassemble and clean carburetor, see FUEL      |
|                             | Clogged air cleaner                       | Replace air filter                              |
|                             | Defective spark plugs                     | Replace spark plugs                             |
|                             | Defective ignition wires                  | Replace ignition wire<br>see ENGINE PERFORMANCE |
|                             | Defective distributor cap                 | Replace distributor cap                         |
| Faulty High Speed Operation | Incorrect ignition timing                 | Reset ignition timing<br>see ENGINE PERFORMANCE |
|                             | Defective distributor centrifugal advance | Replace advance mechanism                       |
|                             | Defective distributor vacuum advance      | Replace advance unit                            |
|                             | Incorrect spark plugs or plug gap         | Check gap and/or replace spark plugs            |
|                             | Faulty choke operation                    | Check choke and repair as required              |
|                             | Clogged vacuum passages                   | Remove restrictions                             |
|                             | Improper size or clogged main jet         | Check jet size and clean, see FUEL              |
|                             | Restricted air cleaner                    | Check filter and replace as necessary           |
|                             | Defective distributor cap, rotor or coil  | Replace cap, rotor or coil                      |
| Misfire at All Speeds       | Defective spark plugs                     | Replace spark plugs                             |
|                             | Defective spark plug wires                | Replace spark plug wires                        |
|                             | Defective distributor cap, rotor, or coil | Replace cap, rotor, or coil                     |
|                             | Cracked or broken vacuum hoses            | Replace vacuum hoses                            |

|                                 |                                         |                                                                 |
|---------------------------------|-----------------------------------------|-----------------------------------------------------------------|
|                                 | Vacuum leaks                            | Repair vacuum leaks                                             |
|                                 | Fuel lines clogged                      | Remove restriction                                              |
| Hesitation                      | Cracked or broken vacuum hoses          | Replace vacuum hoses                                            |
|                                 | Vacuum leaks                            | Repair Vacuum leaks                                             |
|                                 | Binding carburetor linkage              | Eliminate binding                                               |
|                                 | Binding throttle linkage                | Eliminate binding                                               |
|                                 | Binding choke linkage or fast idle cam  | Eliminate binding                                               |
|                                 | Improper float setting                  | Readjust float setting, see FUEL                                |
|                                 | Cracked or broken ignition wires        | Replace ignition wires                                          |
| Rough idle, Missing or Stalling | Incorrect curb idle or fast idle speed  | Reset idle speed, see see ENGINE PERFORMANCE                    |
|                                 | Incorrect basic timing                  | Reset ignition timing see ENGINE PERFORMANCE                    |
|                                 | Improper idle mixture adjustment        | Reset idle mixture, see ENGINE PERFORMANCE                      |
|                                 | Improper feedback system operation      | Check feedback system see ENGINE PERFORMANCE                    |
|                                 | Incorrect spark plug gap                | Reset spark plug gap, see ENGINE PERFORMANCE                    |
|                                 | Moisture in ignition components         | Dry components                                                  |
|                                 | Loose or broken ignition wires          | Replace ignition wires                                          |
|                                 | Damaged distributor cap or or rotor     | Replace distributor cap or rotor                                |
|                                 | Faulty ignition coil                    | Replace ignition coil                                           |
|                                 | Fuel filter clogged or worn             | Replace fuel filter                                             |
|                                 | Damaged idle mixture screw              | Replace idle mixture screw, see FUEL                            |
|                                 | Improper fast idle cam adjustment       | Reset fast idle cam adjustment, see TUNE-see ENGINE PERFORMANCE |
|                                 | Improper EGR valve operation            | Replace EGR valve                                               |
|                                 | Faulty PCV valve air flow               | Replace PCV valve                                               |
|                                 | Choke binding or improper choke setting | Reset choke or eliminate binding                                |

|                     |                                                       |                                              |
|---------------------|-------------------------------------------------------|----------------------------------------------|
|                     | Vacuum leak                                           | Repair vacuum leak                           |
|                     | Improper float bowl fuel level                        | Reset float adjustment, see FUEL             |
|                     | Clogged air bleed or idle passages                    | Clean carburetor passages, see FUEL          |
|                     | Clogged or worn air cleaner filter                    | Replace air filter                           |
|                     | Faulty choke vacuum diaphragm                         | Replace diaphragm, see ENGINE PERFORMANCE    |
|                     | Exhaust manifold heat valve inoperative               | Replace heat valve                           |
|                     | Improper distributor spark advance                    | Check distributor operation                  |
|                     | Leaking valves or valve components                    | Check and repair valvetrain                  |
|                     | Improper carburetor mounting                          | Remove and remount carburetor                |
|                     | Excessive play in distributor shaft                   | Replace distributor                          |
|                     | Loose or corroded wiring connections                  | Repair or replace as required                |
| Engine Surges       | Improper PCV valve airflow                            | Replace PCV valve                            |
|                     | Vacuum leaks                                          | Repair vacuum leaks                          |
|                     | Clogged air bleeds                                    | Remove restriction                           |
|                     | EGR valve malfunction                                 | Replace EGR valve                            |
|                     | Restricted air cleaner filter                         | Replace air filter                           |
|                     | Cracked or broken vacuum hoses                        | Replace vacuum hoses                         |
|                     | Cracked or broken ignition wires                      | Replace ignition wires                       |
|                     | Vacuum advance malfunction                            | Check unit and replace as necessary          |
|                     | Defective or fouled spark plugs                       | Replace spark plugs                          |
| Ping or Spark Knock | Incorrect ignition timing                             | Reset ignition timing see ENGINE PERFORMANCE |
|                     | Distributor centrifugal or vacuum advance malfunction | Check operation and replace as necessary     |
|                     | Carburetor setting too lean                           | Readjust mixture                             |

|                       |                                 |                                             |
|-----------------------|---------------------------------|---------------------------------------------|
|                       |                                 | setting, see ENGINE PERFORMANCE             |
|                       | Vacuum leak                     | Eliminate vacuum leak                       |
|                       | EGR valve malfunction           | Replace EGR valve                           |
| Poor Gasoline Mileage | Cracked or broken vacuum hoses  | Replace vacuum hoses                        |
|                       | Vacuum leaks                    | Repair vacuum leaks                         |
|                       | Defective ignition wires        | Replace wires                               |
|                       | Incorrect choke setting         | Readjust setting, see ENGINE PERFORMANCE    |
|                       | Defective vacuum advance        | Replace vacuum advance                      |
|                       | Defective spark plugs           | Replace spark plugs                         |
|                       | Binding carburetor power piston | Eliminate binding                           |
|                       | Dirt in carburetor jets         | Clean and/or replace jets                   |
|                       | Incorrect float adjustment      | Readjust float setting, see FUEL            |
|                       | Defective power valve           | Replace power valve, see ENGINE PERFORMANCE |
|                       | Incorrect idle speed            | Readjust idle speed                         |
| Engine Stalls         | Improper float level            | Readjust float level                        |
|                       | Leaking needle valve and seat   | Replace needle valve and seat               |
|                       | Vacuum leaks                    | Eliminate vacuum leaks                      |

## VACUUM PUMP - DIESEL TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

**NOTE:** Diesel engines mechanical diagnosis is the same as gasoline engines for items such as noisy valves, bearings, pistons, etc. The following trouble shooting covers only items pertaining to diesel engines.

### VACUUM PUMP (DIESEL) TROUBLE SHOOTING CHART

| CONDITION | POSSIBLE CAUSE | CORRECTION |
|-----------|----------------|------------|
|-----------|----------------|------------|

|                 |                                     |                          |
|-----------------|-------------------------------------|--------------------------|
| Excessive Noise | Loose pump-to-drive assembly screws | Tighten screws           |
|                 | Loose tube on pump assembly         | Tighten tube             |
|                 | Valves not functioning properly     | Replace valves           |
| Oil Leakage     | Loose end plug                      | Tighten end plug         |
|                 | Bad seal crimp                      | Remove and re-crimp seal |

## MANUAL TRANSMISSION

### MANUAL TRANSMISSION TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### MANUAL TRANSMISSION/TRANSAXLE TROUBLE SHOOTING

| Condition                                                                            | Possible Cause                                                                                                                               |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Noisy In Forward Gears                                                               | .Low gear oil level,<br>.Loose bell housing bolts,<br>.Worn bearings or gears                                                                |
| Clunk On Deceleration<br>(FWD Only)                                                  | .Loose engine mounts,<br>.Worn inboard CV joints,<br>.Worn differential pinion shaft,<br>.Side gear hub counterbore<br>in case worn oversize |
| Gear Clash When Shifting<br>Forward Gears                                            | .Clutch Out Of Adjustment,<br>.Shift linkage damaged or<br>out of adjustment,<br>.Gears or synchronizers damaged,<br>.Low gear oil level     |
| Transmission Noisy When<br>Moving (RWD Only) Quiet In<br>Neutral With Clutch Engaged | .Worn rear outputshaft bearing                                                                                                               |
| Gear Rattle                                                                          | .Worn bearings,<br>.Wrong gear oil,<br>.Low gear oil,<br>.Worn gears                                                                         |
| Steady Ticking At Idle<br>(Increases With RPM)                                       | .Broken tooth on gear                                                                                                                        |
| Gear Clash When Shifting<br>Forward Gears                                            | .Worn or broken synchronizers                                                                                                                |
| Loud Whine In Reverse                                                                | .Normal condition (1)                                                                                                                        |



|                                                              |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noise When Stepping On Clutch                                | .Bad release bearing,<br>.Worn pilot bearing                                                                                                                                                                                                                                                               |
| Ticking Or Screeching As Clutch Is Engaged                   | .Faulty release bearing,<br>.Uneven pressure plate fingers                                                                                                                                                                                                                                                 |
| Click Or Snap When Clutch Is Engaged                         | .Worn clutch fork,<br>.Worn or broken front bearing retainer                                                                                                                                                                                                                                               |
| Transmission Shifts Hard                                     | .Clutch not releasing,<br>.Shift mechanism binding,<br>.Clutch installed backwards                                                                                                                                                                                                                         |
| Will Not Shift Into One Gear, Shifts Into All Others         | .Bent shift fork,<br>.Worn detent balls                                                                                                                                                                                                                                                                    |
| Locked Into Gear, Cannot Shift                               | .Clutch adjustment,<br>.Worn detent balls                                                                                                                                                                                                                                                                  |
| Transmission Jumps Out Of Gear                               | .Pilot bearing worn,<br>.Bent shift fork,<br>.Worn gear teeth or face<br>.Excessive gear train end play<br>.Worn synchronizers<br>.Missing detent ball spring<br>.Shift mechanism worn or out of adjustment<br>.Engine or transmission mount bolts loose or out of adjustment<br>.Transmission not aligned |
| Shift Lever Rattle                                           | .Worn shift lever or detents<br>.Worn shift forks<br>.Worn synchronizers sleeve                                                                                                                                                                                                                            |
| Shift Lever Hops Under Acceleration                          | .Worn engine or transmission mounts                                                                                                                                                                                                                                                                        |
| (1) - Most units use spur cut gears in reverse and are noisy |                                                                                                                                                                                                                                                                                                            |

## POWERTRAIN

### CLUTCH TROUBLE SHOOTING

NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC CLUTCH TROUBLE SHOOTING CHART

| CONDITION              | POSSIBLE CAUSE                              | CORRECTION                                      |
|------------------------|---------------------------------------------|-------------------------------------------------|
| Chattering or Grabbing | Incorrect clutch adjustment                 | Adjust clutch                                   |
|                        | Oil, grease or glaze on facings             | Disassemble and clean or replace                |
|                        | Loose "U" joint flange                      | See DRIVE AXLES article                         |
|                        | Worn input shaft spline                     | Replace input shaft                             |
|                        | Binding pressure plate                      | Replace pressure plate                          |
|                        | Binding release lever                       | See CLUTCH article                              |
|                        | Binding clutch disc hub                     | Replace clutch disc                             |
|                        | Unequal pressure plate contact              | Replace worn/misaligned components              |
|                        | Loose/bent clutch disc                      | Replace clutch disc                             |
|                        | Incorrect transmission alignment            | Realign transmission                            |
|                        | Worn pressure plate, disc or flywheel       | Replace damaged components                      |
|                        | Broken or weak pressure springs             | Replace pressure plate                          |
|                        | Sticking clutch pedal                       | Lubricate clutch pedal & linkage                |
|                        | Incorrect clutch disc facing                | Replace clutch disc                             |
|                        | Engine loose in chassis                     | Tighten all mounting bolts                      |
| Failure to Release     | Oil or grease on clutch facings             | Clean or replace clutch clutch disc             |
|                        | Incorrect release lever or pedal adjustment | See CLUTCH article                              |
|                        | Worn or broken clutch facings               | Replace clutch disc                             |
|                        | Bent clutch disc or pressure plate          | Replace damaged components                      |
|                        | Clutch disc hub binding on input shaft      | Clean or replace clutch disc and/or input shaft |
|                        | Binding pilot bearing                       | Replace pilot bearing                           |
|                        | Sticking release bearing sleeve             | Replace release bearing and/or sleeve           |
|                        | Binding clutch cable                        | See CLUTCH article                              |

|                          |                                            |                                             |
|--------------------------|--------------------------------------------|---------------------------------------------|
|                          | Defective clutch master                    | Replace master cylinder                     |
|                          | Defective clutch slave                     | Replace slave cylinder                      |
|                          | Air in hydraulic system                    | Bleed hydraulic system                      |
| Rattling                 | Weak or broken release lever spring        | Replace spring and check alignment          |
|                          | Damaged pressure plate                     | Replace pressure plate                      |
|                          | Broken clutch return spring                | Replace return spring                       |
|                          | Worn splines on clutch disc or input shaft | Replace clutch disc and/or input shaft      |
|                          | Worn clutch release bearing                | Replace release bearing                     |
|                          | Dry or worn pilot bearing                  | Lubricate or replace pilot bearing          |
|                          | Unequal release lever contact              | Align or replace release lever              |
|                          | Incorrect pedal free play                  | Adjust free play                            |
|                          | Warped or damaged clutch disc              | Replace damaged components                  |
| Slipping                 | Pressure springs worn or                   | Release pressure plate                      |
|                          | Oily, greasy or worn facings               | Clean or replace clutch disc                |
|                          | Incorrect clutch alignment                 | Realign clutch assembly                     |
|                          | Warped clutch disc or pressure plate       | Replace damaged components                  |
|                          | Binding release levers or clutch pedal     | Lubricate and/or replace release components |
| Squeaking                | Worn or damaged release                    | Replace release bearing                     |
|                          | Dry or worn pilot or release bearing       | Lubricate or replace assembly               |
|                          | Pilot bearing turning in crankshaft        | Replace pilot bearing and/or crankshaft     |
|                          | Worn input shaft bearing                   | Replace bearing and seal                    |
|                          | Incorrect transmission alignment           | Realign transmission                        |
|                          | Dry release fork between pivot             | Lubricate release fork and pivot            |
| Heavy and/or Stiff Pedal | Sticking release bearing sleeve            | Replace release bearing and/or sleeve       |

|                          |                                   |                                       |
|--------------------------|-----------------------------------|---------------------------------------|
|                          | Dry or binding clutch pedal hub   | Lubricate and align components        |
|                          | Floor mat interference with pedal | Lay mat flat in proper area           |
|                          | Dry or binding ball/fork pivots   | Lubricate and align components        |
|                          | Faulty clutch cable               | Replace clutch cable                  |
| Noisy Clutch Pedal       | Faulty interlock switch           | Replace interlock switch              |
|                          | Self-adjuster ratchet noise       | Lubricate or replace self-adjuster    |
|                          | Speed control interlock switch    | Lubricate or replace interlock switch |
| Clutch Pedal Sticks Down | Binding clutch cable              | See CLUTCH article                    |
|                          | Springs weak in pressure plate    | Replace pressure plate                |
|                          | Binding in clutch linkage         | Lubricate and free linkage            |
| Noisy                    | Dry release bearing               | Lubricate or replace release bearing  |
|                          | Dry or worn pilot bearing         | Lubricate or replace bearing          |
|                          | Worn input shaft bearing          | Replace bearing                       |
| Transmission Click       | Weak springs in pressure plate    | Replace pressure plate                |
|                          | Release fork loose on ball stud   | Replace release fork and/or ball stud |
|                          | Oil on clutch disc damper         | Replace clutch disc                   |
|                          | Broken spring in slave cylinder   | Replace slave cylinder                |

## DRIVE AXLE - NOISE DIAGNOSIS

### Unrelated Noises

Some driveline trouble symptoms are also common to the engine, transmission, wheel bearings, tires, and other parts of the vehicle. Ensure cause of trouble actually is in the drive axle before adjusting, repairing, or replacing any of its parts.

### Non-Drive Axle Noises

A few conditions can sound just like drive axle noise and have to be considered in pre-diagnosis. The 4 most common noises are exhaust, tires, CV/universal joints and wheel trim rings.

In certain conditions, the pitch of the exhaust gases may sound like gear whine. At other times, it may be mistaken for a wheel bearing rumble.

Tires, especially radial and snow, can have a high-pitched tread whine or roar, similar to gear noise. Also, some non-standard tires with an unusual tread construction may emit a roar or whine.

Defective CV/universal joints may cause clicking noises or excessive driveline play that can be improperly diagnosed as drive axle problems.

Trim and moldings also can cause a whistling or whining noise. Ensure none of these components are causing the noise before disassembling the drive axle.

#### Gear Noise

A "howling" or "whining" noise from the ring and pinion gear can be caused by an improper gear pattern, gear damage, or improper bearing preload. It can occur at various speeds and driving conditions, or it can be continuous.

Before disassembling axle to diagnose and correct gear noise, make sure that tires, exhaust, and vehicle trim have been checked as possible causes.

#### Chuckle

This is a particular rattling noise that sounds like a stick against the spokes of a spinning bicycle wheel. It occurs while decelerating from 40 MPH and usually can be heard until vehicle comes to a complete stop. The frequency varies with the speed of the vehicle.

A chuckle that occurs on the driving phase is usually caused by excessive clearance due to differential gear wear, or by a damaged tooth on the coast side of the pinion or ring gear. Even a very small tooth nick or a ridge on the edge of a gear tooth is enough the cause the noise.

This condition can be corrected simply by cleaning the gear tooth nick or ridge with a small grinding wheel. If either gear is damaged or scored badly, the gear set must be replaced. If metal has broken loose, the carrier and housing must be cleaned to remove particles that could cause damage.

#### Knock

This is very similar to a chuckle, though it may be louder, and occur on acceleration or deceleration. Knock can be caused by a gear tooth that is damaged on the drive side of the ring and pinion gears. Ring gear bolts that are hitting the carrier casting can cause knock. Knock can also be due to excessive end play in the axle shafts.

#### Clunk

Clunk is a metallic noise heard when an automatic transmission is engaged in Reverse or Drive, or when throttle is applied or released. It is caused by backlash somewhere in the driveline, but not necessarily in the axle. To determine whether driveline clunk is caused by the axle, check the total axle backlash as follows:

- 1) Raise vehicle on a frame or twinpost hoist so that drive wheels are free. Clamp a bar between axle companion flange and a part of the frame or body so that flange cannot move.

- 2) On conventional drive axles, lock the left wheel to keep it from turning. On all models, turn the right wheel slowly until it is felt to be in Drive condition. Hold a chalk marker on side of tire about 12" from center of wheel. Turn wheel in the opposite direction until it is again felt to be in Drive condition.

- 3) Measure the length of the chalk mark, which is the total

axle backlash. If backlash is one inch or less, drive axle is not the source of clunk noise.

#### Bearing Whine

Bearing whine is a high-pitched sound similar to a whistle. It is usually caused by malfunctioning pinion bearings. Pinion bearings operate at drive shaft speed. Roller wheel bearings may whine in a similar manner if they run completely dry of lubricant. Bearing noise will occur at all driving speeds. This distinguishes it from gear whine, which usually comes and goes as speed changes.

#### Bearing Rumble

Bearing rumble sounds like marbles being tumbled. It is usually caused by a malfunctioning wheel bearing. The lower pitch is because the wheel bearing turns at only about 1/3 of drive shaft speed.

#### Chatter On Turns

This is a condition where the entire front or rear of vehicle vibrates when vehicle is moving. The vibration is plainly felt as well as heard. Extra differential thrust washers installed during axle repair can cause a condition of partial lock-up that creates this chatter.

#### Axle Shaft Noise

Axle shaft noise is similar to gear noise and pinion bearing whine. Axle shaft bearing noise will normally distinguish itself from gear noise by occurring in all driving modes (Drive, cruise, coast and float), and will persist with transmission in Neutral while vehicle is moving at problem speed.

If vehicle displays this noise condition, remove suspect axle shafts, replace wheel seals and install a new set of bearings. Re-evaluate vehicle for noise before removing any internal components.

#### Vibration

Vibration is a high-frequency trembling, shaking or grinding condition (felt or heard) that may be constant or variable in level and can occur during the total operating speed range of the vehicle.

The types of vibrations that can be felt in the vehicle can be divided into 3 main groups:

- \* Vibrations of various unbalanced rotating parts of the vehicle.
- \* Resonance vibrations of the body and frame structures caused by rotating of unbalanced parts.
- \* Tip-in moans of resonance vibrations from stressed engine or exhaust system mounts or driveline flexing modes.

## DRIVE AXLE - RWD TROUBLE SHOOTING

NOTE: This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing. For definitions of listed noises or sounds, see DRIVE AXLE - NOISE DIAGNOSIS under POWERTRAIN.

### DRIVE AXLE (RWD) TROUBLE SHOOTING

---

| CONDITION | POSSIBLE CAUSE | CORRECTION |
|-----------|----------------|------------|
|-----------|----------------|------------|

---

---

**Knocking or Clunking**

|                                  |                                     |
|----------------------------------|-------------------------------------|
| Differential Side Gear Clearance | Check Clearance                     |
| Worn Pinion Shaft                | Replace Pinion Shaft                |
| Axle Shaft End Play              | Check End Play                      |
| Missing Gear Teeth               | Check Differential/<br>Replace Gear |
| Wrong Axle Backlash              | Check Backlash                      |
| Misaligned Driveline             | Realign Driveline                   |

---

**Clinking During Engagement**

|                          |                       |
|--------------------------|-----------------------|
| Side Gear Clearance      | Check Clearance       |
| Ring and Pinion Backlash | Check Backlash        |
| Worn/Loose Pinion Shaft  | Replace Shaft/Bearing |
| Bad "U" Joint            | Replace "U" Joint     |
| Sticking Slip Yoke       | Lube Slip Yoke        |
| Broken Rear Axle Mount   | Replace Mount         |
| Loose Drive Shaft Flange | Check Flange          |

---

**Click/Chatter On Turns**

|                                  |                       |
|----------------------------------|-----------------------|
| Differential Side Gear Clearance | Check Clearance       |
| Wrong Turn On Plates (1)         | Replace Clutch Plates |
| Wrong Differential Lubricant (1) | Change Lubricant      |

---

**Knock Or Click**

|                                 |                       |
|---------------------------------|-----------------------|
| Flat Spot on Rear Wheel Bearing | Replace Wheel Bearing |
|---------------------------------|-----------------------|

---

**Low Vibration At All Speeds**

|                         |                       |
|-------------------------|-----------------------|
| Faulty Wheel Bearing    | Replace Wheel Bearing |
| Faulty "U" Joint        | Replace "U" Joint     |
| Faulty Drive Shaft      | Balance Drive Shaft   |
| Faulty Companion Flange | Replace Flange        |
| Faulty Slip Yoke Flange | Replace Flange        |

(1) - Limited slip differential only.

---

## FWD AXLE SHAFTS & CV JOINTS TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to

problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC FWD AXLE SHAFTS & CV JOINTS TROUBLE SHOOTING CHART

| CONDITION                            | POSSIBLE CAUSE                                                |
|--------------------------------------|---------------------------------------------------------------|
| Grease Leaks                         | CV boot torn or cracked                                       |
| Clicking Noise on Cornering          | Damaged outer CV                                              |
| Clunk Noise on Acceleration          | Damaged inner CV                                              |
| Vibration or Shudder on Acceleration | Sticking, damaged or worn CV<br>Misalignment or spring height |

## STEERING & SUSPENSION

### MANUAL STEERING GEAR TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

#### BASIC MANUAL STEERING GEAR TROUBLE SHOOTING CHART

| CONDITION                                   | POSSIBLE CAUSE                          | CORRECTION                       |
|---------------------------------------------|-----------------------------------------|----------------------------------|
| Rattle or Chucking Noise in Rack and Pinion | Rack and pinion mounting bracket loose  | Tighten all mounting bolts       |
|                                             | Lack of/or incorrect lubricant          | Correct as necessary             |
|                                             | Steering gear mounting bolts loose      | Tighten all mounting bolts       |
| Excessive Play                              | Front wheel bearing improperly adjusted | See FRONT SUSPENSION article     |
|                                             | Loose or worn steering linkage          | See STEERING LINKAGE article     |
|                                             | Loose or worn steering gear shift       | See MANUAL STEERING GEAR article |
|                                             | Steering arm loose on gear shaft        | See MANUAL STEERING GEAR article |
|                                             | Steering gear housing bolts loose       | Tighten all mounting bolts       |
|                                             | Steering gear adjustment too loose      | See MANUAL STEERING GEAR article |
|                                             | Steering arms loose on                  | Tighten and check                |



|                            |                                            |                                             |
|----------------------------|--------------------------------------------|---------------------------------------------|
|                            | knuckles                                   | steering linkage                            |
|                            | Rack and pinion mounting loose             | Tighten all mounting bolts                  |
|                            | Rack and pinion out of adjustment          | See adjustment in STEERING article          |
|                            | Tie rod end loose                          | Tighten and check steering linkage          |
|                            | Excessive Pitman shaft-to-ball nut lash    | Repair as necessary                         |
| Poor Returnability         | Lack of lubricant in ball joint or linkage | Lubricate and service systems               |
|                            | Binding in linkage or ball joints          | See STEERING LINKAGE and SUSPENSION article |
|                            | Improper front end alignment               | See WHEEL ALIGNMENT article                 |
|                            | Improper tire pressure                     | Inflate to proper pressure                  |
|                            | Tie rod binding                            | Inflate to proper pressure                  |
|                            | Shaft seal rubbing shaft                   | See STEERING COLUMN article                 |
| Excessive Vertical Motion  | Improper tire pressure                     | Inflate to proper pressure                  |
|                            | Tires, wheels or rotors out of balance     | Balance tires then check wheels and rotors  |
|                            | Worn or faulty shock absorbers             | Check and replace if necessary              |
|                            | Loose tie rod ends or steering             | Tighten or replace if necessary             |
|                            | Loose or worn wheel bearings               | See SUSPENSION article                      |
| Steering Pulls to One Side | Improper tire pressure                     | Inflate to proper pressure                  |
|                            | Front tires are different sizes            | Rotate or replace if necessary              |
|                            | Wheel bearings not adjusted properly       | See FRONT SUSPENSION article                |
|                            | Bent or broken suspension components       | See FRONT SUSPENSION article                |
|                            | Improper wheel alignment                   | See WHEEL ALIGNMENT article                 |

|             |                                 |                                   |
|-------------|---------------------------------|-----------------------------------|
|             | Brakes dragging                 | See BRAKES article                |
| Instability | Low or uneven tire pressure     | Inflate to proper pressure        |
|             | Loose or worn wheel bearings    | See FRONT SUSPENSION article      |
|             | Loose or worn idler arm bushing | See FRONT SUSPENSION article      |
|             | Loose or worn strut bushings    | See FRONT SUSPENSION article      |
|             | Incorrect front wheel alignment | See WHEEL ALIGNMENT article       |
|             | Steering gear not centered      | See MANUAL STEERING GEARS article |
|             | Springs or shock                | Check and replace if necessary    |
|             | Improper cross shaft            | See MANUAL STEERING GEARS article |

## POWER STEERING TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC POWER STEERING TROUBLE SHOOTING CHART

| CONDITION                | POSSIBLE CAUSE                                  | CORRECTION                      |
|--------------------------|-------------------------------------------------|---------------------------------|
| Rattle or Chucking Noise | Pressure hoses touching engine parts            | Adjust to proper clearance      |
|                          | Loose Pitman shaft                              | Adjust or replace if necessary  |
|                          | Tie rods ends or Pitman arm loose               | Tighten and check system        |
|                          | Rack and pinion mounts loose                    | Tighten all mounting bolts      |
|                          | Free play in worm and                           | See POWER STEERING GEAR article |
|                          | Loose sector shaft or thrust bearing adjustment | See POWER STEERING GEAR         |
|                          | Free play in pot coupling                       | See STEERING COLUMN article     |

|                           |                                       |                                                   |
|---------------------------|---------------------------------------|---------------------------------------------------|
|                           | Worn shaft serrations                 | See STEERING COLUMN article                       |
| Growl in Steering Pump    | Excessive pressure in hoses           | Restricted hoses, see POWER STEERING GEAR article |
|                           | Scored pressure plates                | See POWER STEERING GEAR article                   |
|                           | Scored thrust plates or rotor         | See POWER STEERING GEAR article                   |
|                           | Extreme wear of cam ring              | See POWER STEERING GEAR article                   |
| Rattle in Steering Pump   | Vanes not installed                   | See POWER STEERING PUMP article                   |
|                           | Vanes sticking in rotor               | See POWER STEERING PUMP article                   |
| Swish noise in Pump       | Defective flow control valve          | See POWER STEERING PUMP article                   |
| Groan in Steering Pump    | Air in fluid                          | See POWER STEERING PUMP article                   |
|                           | Poor pressure hose connection         | Tighten and check, replace if necessary           |
| Squawk When Turning       | Damper "O" ring on valve spool cut    | See POWER STEERING PUMP article                   |
| Moan or Whine in Pump     | Pump shaft bearing scored             | Replace bearing and fluid                         |
|                           | Air in fluid or fluid level low       | See POWER STEERING PUMP article                   |
|                           | Hose or column grounded               | Check and replace if necessary                    |
|                           | Cover "O" ring missing or damaged     | See POWER STEERING PUMP article                   |
|                           | Valve cover baffle missing or damaged | See POWER STEERING PUMP article                   |
|                           | Interference of components in pump    | See POWER STEERING PUMP article                   |
|                           | Loose or poor bracket alignment       | Correct or replace if necessary                   |
| Hissing When Parking      | Internal leakage in steering gear     | Check valved assembly first                       |
| Chirp in Steering Pump    | Loose or worn power steering belt     | Adjust or replace if necessary                    |
| Buzzing When Not Steering | Noisy pump                            | See POWER STEERING PUMP article                   |

|                        |                                        |                                                |
|------------------------|----------------------------------------|------------------------------------------------|
|                        | Free play in steering shaft bearing    | See STEERING COLUMN article                    |
|                        | Bearing loose on shaft serrations      | See STEERING COLUMN article                    |
| Clicking Noise in Pump | Pump slippers too long                 | See POWER STEERING PUMP article                |
|                        | Broken slipper springs                 | See POWER STEERING PUMP article                |
|                        | Excessive wear or nicked rotors        | See POWER STEERING PUMP article                |
|                        | Damaged cam contour                    | See POWER STEERING PUMP article                |
| Poor Return of Wheel   | Wheel rubbing against turn signal      | See STEERING COLUMN SWITCHES article           |
|                        | Flange rubbing steering gear adjuster  | See STEERING COLUMN article                    |
|                        | Tight or frozen steering shaft bearing | See STEERING COLUMN article                    |
|                        | Steering gear out of adjustment        | See POWER STEERING GEAR article                |
|                        | Sticking or plugged spool valve        | See POWER STEERING PUMP article                |
|                        | Improper front end alignment           | See WHEEL ALIGNMENT article                    |
|                        | Wheel bearings worn or loose           | See FRONT SUSPENSION article                   |
|                        | Ties rods or ball joints binding       | Check and replace if necessary                 |
|                        | Intermediate shaft joints binding      | See STEERING COLUMN article                    |
|                        | Kinked pressure hoses                  | Correct or replace if necessary                |
|                        | Loose housing head spanner nut         | See POWER STEERING GEAR article                |
|                        | Damaged valve lever                    | See POWER STEERING GEAR article                |
|                        | Sector shaft adjusted too tight        | See ADJUSTMENTS in POWER STEERING GEAR article |
|                        | Worm thrust bearing adjusted too tight | See ADJUSTMENTS in POWER STEERING GEAR article |
|                        | Reaction ring sticking in cylinder     | See POWER STEERING GEAR article                |

|                                                                                                               |                                        |                                 |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------|
|                                                                                                               | Reaction ring sticking in housing head | See POWER STEERING GEAR article |
|                                                                                                               | Steering pump internal leakage         | See POWER STEERING PUMP article |
|                                                                                                               | Steering gear-to-column misalignment   | See STEERING COLUMN article     |
|                                                                                                               | Lack of lubrication in linkage         | Service front suspension        |
|                                                                                                               | Lack of lubrication in ball joints     | Service front suspension        |
| Increased Effort When Turning Wheel Fast Foaming, Milky Power Steering Fluid, Low Fluid Level or Low Pressure | High internal pump leakage             | See POWER STEERING PUMP article |
|                                                                                                               | Power steering pump belt slipping      | Adjust or replace if necessary  |
|                                                                                                               | Low fluid level                        | Check and fill to proper level  |
|                                                                                                               | Engine idle speed too low              | Adjust to correct setting       |
|                                                                                                               | Air in pump fluid system               | See POWER STEERING PUMP article |
|                                                                                                               | Pump output low                        | See POWER STEERING PUMP article |
|                                                                                                               | Steering gear malfunctioning           | See POWER STEERING GEAR article |
| Wheel Surges or Jerks                                                                                         | Low fluid level                        | Check and fill to proper level  |
|                                                                                                               | Loose fan belt                         | Adjust or replace if necessary  |
|                                                                                                               | Insufficient pump pressure             | See POWER STEERING PUMP article |
|                                                                                                               | Sticky flow control valve              | See POWER STEERING PUMP article |
|                                                                                                               | Linkage hitting oil pan at full turn   | Replace bent components         |
| Kick Back or Free Play                                                                                        | Air in pump fluid system               | See POWER STEERING PUMP article |
|                                                                                                               | Worn poppet valve in steering gear     | See POWER STEERING PUMP article |
|                                                                                                               | Excessive over center lash             | See POWER STEERING GEAR article |
|                                                                                                               | Thrust bearing out of adjustment       | See POWER STEERING GEAR article |

|                       |                                         |                                      |
|-----------------------|-----------------------------------------|--------------------------------------|
|                       | Free play in pot coupling               | See POWER STEERING PUMP article      |
|                       | Steering gear coupling loose on shaft   | See POWER STEERING PUMP article      |
|                       | Steering disc mounting bolts loose      | Tighten or replace if necessary      |
|                       | Coupling loose on worm shaft            | Tighten or replace if necessary      |
|                       | Improper sector shaft adjustment        | See POWER STEERING GEAR article      |
|                       | Excessive worm piston side play         | See POWER STEERING GEAR article      |
|                       | Damaged valve lever                     | See POWER STEERING GEAR article      |
|                       | Universal joint loose                   | Tighten or replace if necessary      |
|                       | Defective rotary valve                  | See POWER STEERING GEAR article      |
| No Power When Parking | Sticking flow control valve             | See POWER STEERING PUMP article      |
|                       | Insufficient pump pressure output       | See POWER STEERING PUMP article      |
|                       | Excessive internal pump leakage         | See POWER STEERING PUMP article      |
|                       | Excessive internal gear leakage         | See POWER STEERING PUMP article      |
|                       | Flange rubs against gear adjust plug    | See STEERING COLUMN article          |
|                       | Loose pump belt                         | Adjust or replace if necessary       |
|                       | Low fluid level                         | Check and add proper amount of fluid |
|                       | Engine idle too low                     | Adjust to correct setting            |
|                       | Steering gear-to-column misaligned      | See STEERING COLUMN article          |
| No Power, Left Turn   | Left turn reaction seal "O" ring worn   | See POWER STEERING GEAR article      |
|                       | Left turn reaction seal damaged/missing | See POWER STEERING GEAR article      |
|                       | Cylinder head "O" ring damaged          | See POWER STEERING PUMP article      |

|                                      |                                                                 |                                             |
|--------------------------------------|-----------------------------------------------------------------|---------------------------------------------|
| No Power,<br>Right Turns             | Column pot coupling<br>bottomed                                 | See STEERING COLUMN<br>article              |
|                                      | Right turn reaction seal<br>"O" ring worn                       | See POWER STEERING GEAR<br>article          |
|                                      | Right turn reaction seal<br>damaged                             | See POWER STEERING GEAR<br>article          |
|                                      | Internal leakage through<br>piston end plug                     | See POWER STEERING GEAR<br>article          |
|                                      | Internal leakage through<br>side plugs                          | See POWER STEERING GEAR<br>article          |
| Lack of Effort<br>in Turning         | Left and/or right<br>reaction seal sticking<br>in cylinder head | Replace, see POWER<br>STEERING GEAR article |
| Wanders to One<br>Side               | Front end alignment<br>incorrect                                | See WHEEL ALIGNMENT<br>article              |
|                                      | Unbalanced steering<br>gear valve                               | See POWER STEERING GEAR<br>article          |
| Low Pressure Due<br>to Steering Pump | Flow control valve stuck<br>or inoperative                      | See POWER STEERING PUMP<br>article          |
|                                      | Pressure plate not flat<br>against cam ring                     | See POWER STEERING PUMP<br>article          |
|                                      | Extreme wear of cam ring                                        | Replace and check<br>adjustments            |
|                                      | Scored plate, thrust<br>plate or rotor                          | See POWER STEERING PUMP<br>article          |
|                                      | Vanes not installed<br>properly                                 | See POWER STEERING PUMP<br>article          |
|                                      | Vanes sticking in rotor<br>slots                                | See POWER STEERING PUMP<br>article          |
|                                      | Cracked/broken thrust or<br>pressure plate                      | See POWER STEERING PUMP<br>article          |

## STEERING COLUMN TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC STEERING COLUMN TROUBLE SHOOTING CHART

| CONDITION            | POSSIBLE CAUSE        | CORRECTION                      |
|----------------------|-----------------------|---------------------------------|
| Noise in<br>Steering | Coupling pulled apart | See STEERING COLUMNS<br>article |

|                            |                                        |                                 |
|----------------------------|----------------------------------------|---------------------------------|
|                            | Column not correctly aligned           | See STEERING COLUMNS article    |
|                            | Broken lower joint                     | Replace joint                   |
|                            | Horn contact ring not                  | See STEERING COLUMN article     |
|                            | Bearing not lubricated                 | See STEERING COLUMN article     |
|                            | Shaft snap ring not properly seated    | Reseat or replace snap ring     |
|                            | Plastic spherical joint not lubricated | See STEERING COLUMN article     |
|                            | Shroud or housing loose                | Tighten holding screws          |
|                            | Lock plate retaining ring not seated   | See STEERING COLUMN article     |
|                            | Loose sight shield                     | Tighten holding screws          |
| High Steering Shaft Effort | Column assembly misaligned             | See STEERING COLUMN article     |
|                            | Improperly installed dust shield       | Adjust or replace               |
|                            | Tight steering universal joint         | See STEERING COLUMN article     |
| High Shift Effort          | Column is out of alignment             | See STEERING COLUMN article     |
|                            | Improperly installed dust shield       | Adjust or replace               |
|                            | Seals or bearings not lubricated       | See STEERING COLUMNS article    |
|                            | Mounting bracket screws too long       | Replace with new shorter screws |
|                            | Burrs on shift tube                    | Remove burrs or replace tube    |
|                            | Lower bowl bearing assembled wrong     | See STEERING COLUMN article     |
|                            | Shift tube bent or broken              | Replace as necessary            |
| Improper Trans. Shifting   | Improper adjustment of shift levers    | See STEERING COLUMN article     |
|                            | Sheared shift tube joint               | Replace as necessary            |
|                            | Sheared lower shaft lever              | Replace as necessary            |
|                            | Improper shift lever adjustment        | See STEERING COLUMN article     |



|                        |                                      |                                 |
|------------------------|--------------------------------------|---------------------------------|
|                        | Improper gate plate adjustment       | See STEERING COLUMN article     |
| Excess Play in Column  | Instrument panel bracket bolts loose | Tighten bolts and check bracket |
|                        | Broken weld nut on jacket            | See STEERING COLUMN article     |
|                        | Instrument bracket capsule sheared   | See STEERING COLUMN article     |
|                        | Column bracket/jacket bolts loose    | Tighten bolts and check bracket |
| Steering Locks in Gear | Release lever mechanism              | See STEERING COLUMN article     |

## SUSPENSION TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC SUSPENSION TROUBLE SHOOTING CHART

| CONDITION                               | POSSIBLE CAUSE                     | CORRECTION                                                    |
|-----------------------------------------|------------------------------------|---------------------------------------------------------------|
| Front End Noise                         | Loose or worn wheel                | See Wheel Bearing Adjustment in SUSPENSION                    |
|                                         | Worn shocks or shock mountings     | Replace struts or strut mountings                             |
|                                         | Worn struts or strut mountings     | Replace struts or strut mountings                             |
|                                         | Loose or worn lower control arm    | See SUSPENSION                                                |
|                                         | Loose steering gear-to-frame bolts | See STEERING                                                  |
|                                         | Worn control arm bushings          | See SUSPENSION                                                |
|                                         | Ball joints not lubricated         | Lubricate ball joints & see Ball Joint Checking in SUSPENSION |
| Front Wheel Shake, Shimmy, or Vibration | Tires or wheels out of balance     | Check tire balance                                            |
|                                         | Incorrect wheel alignment          | See WHEEL ALIGNMENT                                           |
|                                         | Drive shaft unbalanced             | Check drive shaft balance                                     |

|                       |                                 |                                       |
|-----------------------|---------------------------------|---------------------------------------|
|                       | Loose or worn wheel bearings    | See WHEEL ALIGNMENT                   |
|                       | Loose or worn tie rod ends      | See SUSPENSION                        |
|                       | Worn upper ball joints          | See Ball Joint Checking in SUSPENSION |
|                       | Worn shock absorbers            | Replace shock absorbers               |
|                       | Worn strut bushings             | Replace strut bushings                |
| Car Pulls to One Side | Mismatched or uneven tires      | Check tire condition                  |
|                       | Broken or sagging springs       | See SUSPENSION                        |
|                       | Loose or worn strut bushings    | See SUSPENSION                        |
|                       | Improper wheel alignment        | See WHEEL ALIGNMENT                   |
|                       | Improper rear axle alignment    | Check rear axle alignment             |
|                       | Power steering gear unbalanced  | See STEERING                          |
|                       | Front brakes dragging           | See BRAKES                            |
| Abnormal Tire Wear    | Unbalanced tires                | Check tire balance & rotation         |
|                       | Sagging or broken springs       | See SUSPENSION                        |
|                       | Incorrect front end alignment   | See WHEEL ALIGNMENT                   |
|                       | Faulty shock absorbers          | Replace chock absorbers               |
| Scuffed Tires         | Toe-In incorrect                | See WHEEL ALIGNMENT                   |
|                       | Suspension arm bent or twisted  | See appropriate SUSPENSION article    |
| Springs Bottom or Sag | Bent or broken springs          | See SUSPENSION                        |
|                       | Leaking or worn shock absorbers | Replace shock absorbers               |
|                       | Frame misalignment              | Check frame for damage                |
| Spring Noises         | Loose "U" Bolts                 | See SUSPENSION                        |
|                       | Loose or worn bushings          | See SUSPENSION                        |
|                       | Worn or missing interliners     | See SUSPENSION                        |
| Shock Absorber Noise  | Loose shock mountings           | Check & tighten mountings             |
|                       | Worn bushings                   | Replace bushings                      |

|                               |                                      |                             |
|-------------------------------|--------------------------------------|-----------------------------|
|                               | Air in system                        | Bleed air from system       |
|                               | Undercoating on shocks               | Remove undercoating         |
| Car Leans or Sways on Corners | Loose stabilizer bar                 | See SUSPENSION              |
|                               | Faulty shocks or mountings           | Replace shocks or mountings |
|                               | Broken or sagging springs            | See SUSPENSION              |
| Shock Absorbers Leaking       | Worn seals or reservoir tube crimped | See SUSPENSION              |
| Broken Springs                | Loose "U" bolts                      | See SUSPENSION              |
|                               | Inoperative shock absorbers          | Replace shock absorbers     |

## WHEEL ALIGNMENT TROUBLE SHOOTING

**NOTE:** This is GENERAL information. This article is not intended to be specific to any unique situation or individual vehicle configuration. The purpose of this Trouble Shooting information is to provide a list of common causes to problem symptoms. For model-specific Trouble Shooting, refer to SUBJECT, DIAGNOSTIC, or TESTING articles available in the section(s) you are accessing.

### BASIC WHEEL ALIGNMENT TROUBLE SHOOTING CHART

| CONDITION           | POSSIBLE CAUSE                   | CORRECTION                                     |
|---------------------|----------------------------------|------------------------------------------------|
| Premature Tire Wear | Improper tire inflation          | Check tire pressure                            |
|                     | Front alignment out of tolerance | See ALIGNMENT SPECS in WHEEL ALIGNMENT section |
|                     | Suspension components worn       | See SUSPENSION section                         |
|                     | Steering system components worn  | See STEERING section                           |
|                     | Improper standing height         | See WHEEL ALIGNMENT                            |
|                     | Uneven or sagging springs        | See SUSPENSION section                         |
|                     | Bent wheel                       | See WHEEL ALIGNMENT                            |
|                     | Improper torsion bar adjustment  | See SUSPENSION section                         |
|                     | Loose or worn wheel bearings     | See WHEEL BEARING ADJ. in SUSPENSION section   |
|                     | Worn or defective shock          | Replace shock absorbers                        |
| Pulls to One Side   | Tires out of balance             | Check tire balance                             |
|                     | Improper tire inflation          | Check tire pressure                            |
|                     | Brake dragging                   | See BRAKE section                              |

|                   |                                       |                                          |
|-------------------|---------------------------------------|------------------------------------------|
|                   | Mismatched tires                      | See WHEEL ALIGNMENT                      |
|                   | Broken or sagging spring              | See SUSPENSION section                   |
|                   | Broken torsion bar                    | See SUSPENSION section                   |
|                   | Power steering valve not centered     | See STEERING section                     |
|                   | Front alignment out of tolerance      | See WHEEL ALIGNMENT section              |
|                   | Defective wheel bearing               | See WHEEL BEARINGS in SUSPENSION section |
|                   | Uneven sway bar links                 | See SUSPENSION section                   |
|                   | Frame bent                            | Check for frame damage                   |
|                   | Steering system bushing worn          | See STEERING section                     |
| Hard Steering     | Idler arm bushing too tight           | See STEERING LINKAGE in STEERING section |
|                   | Ball joint tight or seized            | See SUSPENSION section                   |
|                   | Steering linkage too tight            | See STEERING LINKAGE in STEERING section |
|                   | Power steering fluid low              | Add proper amount of fluid               |
|                   | Power steering drive belt loose       | See STEERING section                     |
|                   | Power steering pump defective         | See STEERING section                     |
|                   | Steering gear out of adjustment       | See STEERING section                     |
|                   | Incorrect wheel alignment             | See WHEEL ALIGNMENT                      |
|                   | Damaged steering gear                 | See STEERING section                     |
|                   | Damaged suspension                    | See SUSPENSION section                   |
| Vehicle "Wanders" | Bent steering knuckle or supports     | See SUSPENSION section                   |
|                   | Strut rod or control arm bushing worn | See SUSPENSION section                   |
|                   | Loose or worn wheel bearings          | See WHEEL BEARINGS in SUSPENSION section |
|                   | Improper tire inflation               | Check tire pressure                      |
|                   | Stabilizer bar missing or defective   | See SUSPENSION section                   |

|                       |                                         |                                              |
|-----------------------|-----------------------------------------|----------------------------------------------|
|                       | Wheel alignment out of tolerance        | See Adjustment in WHEEL ALIGNMENT section    |
|                       | Broken spring                           | See SUSPENSION section                       |
|                       | Defective shock absorbers               | Replace shock absorbers                      |
|                       | Worn steering & suspension components   | See SUSPENSION section                       |
| Front End Shimmy      | Tire out of balance/round               | Check tire balance                           |
|                       | Excessive wheel runout                  | See WHEEL ALIGNMENT                          |
|                       | Insufficient or improper caster         | See WHEEL ALIGNMENT section                  |
|                       | Worn suspension or steering components  | See SUSPENSION section                       |
|                       | Defective shock absorbers               | Replace shock absorber                       |
|                       | Wheel bearings worn or loose            | See WHEEL BEARING ADJ. in SUSPENSION section |
|                       | Power steering reaction Bracket loose   | See STEERING section                         |
|                       | Steering gear box (rack) mounting loose | See STEERING section                         |
|                       | Steering gear adjustment loose          | See STEERING section                         |
|                       | Worn spherical joints                   | See SUSPENSION section                       |
| Toe-In Not Adjustable | Lower control arm bent                  | See SUSPENSION section                       |
|                       | Frame bent                              | Check frame for damage                       |
| Camber Not Adjustable | Control arm bent                        | See SUSPENSION section                       |
|                       | Frame bent                              | Check frame for damage                       |
|                       | Hub & bearing not seated properly       | See SUSPENSION section                       |

# WIPER/WASHER SYSTEM

## 1993 Toyota Celica

1993 ACCESSORIES/SAFETY EQUIPMENT  
Toyota Wiper/Washer Systems

Camry, Celica, Corolla, Land Cruiser, MR2, Paseo, Pickup,  
Previa, Supra, Tercel, T100, 4Runner

**WARNING:** Camry, Celica, Corolla, MR2, Paseo, Previa, Supra and Tercel are equipped with a driver's side air bag. Use extreme caution while servicing components on steering column. Ensure ignition switch is in LOCK position and negative battery cable is disconnected for at least one minute before starting service. DO NOT apply electrical power to any component on steering column without disconnecting air bag control unit. Information labels are attached to air bag components. Follow all notices on labels. Use only DVOM with minimum of 10-k/ohms impedance to check ANY circuit.

### DESCRIPTION & OPERATION

All models are equipped with 2-speed wiper motors. An intermittent wiper system is optional. Front wiper motors are protected by an internal circuit breaker. Celica, Corolla, Land Cruiser, Previa, Supra and 4Runner are equipped with rear wipers.

### TESTING

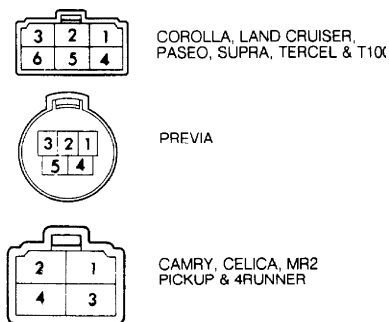
#### FRONT WIPER MOTOR TEST

Camry, Celica, MR2, Pickup & 4Runner

1) Using 12-volt battery, connect jumper wire from positive battery terminal to wiper motor terminal No. 2. See Fig. 1. Connect another jumper wire between negative battery terminal and wiper motor body or motor ground wire location at mounting bolt hole. Motor should operate at low speed.

2) Remove jumper wire from terminal No. 2. Reconnect jumper wire to wiper motor terminal No. 1. Motor should operate at high speed. Remove jumper wire from terminal No. 1 and reconnect jumper wire to wiper motor terminal No. 2. With motor running at low speed, stop motor at any position except PARK position by disconnecting jumper wire from terminal No. 2. Install another jumper wire between wiper motor terminals No. 2 and 3.

3) Ensure motor body ground is still connected to negative battery terminal. Connect positive battery terminal jumper wire to wiper motor terminal No. 4. Motor should start running and then stop at PARK position. Replace motor if operation is not as described.



93F84660

Fig. 1: Front Wiper Motor Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

#### All Other Models

1) Using 12-volt battery, connect jumper wire from positive battery terminal to wiper motor terminal No. 3. See Fig. 1. Connect another jumper wire between negative battery terminal and wiper motor terminal No. 1. Motor should operate at low speed.

2) Remove jumper wire from terminal No. 3. Reconnect positive battery terminal jumper wire to wiper motor terminal No. 2. Motor should operate at high speed.

3) Remove jumper wire from terminal No. 2 and reconnect jumper wire to wiper motor terminal No. 3. With motor running at low speed, stop motor at any position except PARK position by disconnecting jumper wire from terminal No. 3. Install another jumper wire between wiper motor terminals No. 3 and 4 on Previa, or terminals No. 3 and 5 on all other models.

4) Ensure motor terminal No. 1 (ground terminal) is still connected to battery negative terminal. Connect positive battery terminal jumper wire to wiper motor terminal No. 5 on Previa, or terminal No. 6 on all other models. Motor should start running and then stop at the PARK position. Replace motor if operation is not as described.

## FRONT WIPER RELAY TEST

#### Previa

1) Ensure wiper motor operation is correct. See FRONT WIPER MOTOR TEST. If motor operates correctly when tested separately from switch, check wiper switch and front wiper relay. See FRONT WIPER SWITCH TEST. For front relay location, see Fig. 2.

2) Check front wiper relay. Using an ohmmeter, check continuity between relay terminals No. 1 and 4. See Fig. 3. No continuity should exist. Connect ohmmeter leads to relay terminals No. 1 and 3. Continuity should exist. If continuity is not as described, replace relay.

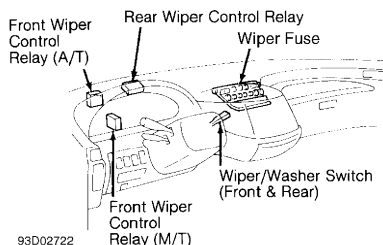
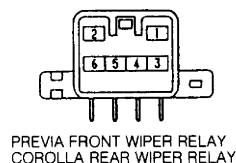


Fig. 2: Locating Wiper Relays (Previa)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93G84661

Fig. 3: Front & Rear Wiper Relay Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## FRONT WIPER SWITCH TEST

With wiper switch in specified position, ensure continuity exists between terminals listed. See WIPER SWITCH CONTINUITY TEST table. See Fig. 4. Replace wiper switch assembly if continuity is not as specified.

WIPER SWITCH CONTINUITY TEST TABLE

| Switch Position             | (1) Terminals |
|-----------------------------|---------------|
| Camry, Celica & MR2         |               |
| Off (Normal) .....          | B4 & B7       |
| Off (Mist) .....            | B7 & B18      |
| Intermittent (Normal) ..... | B4 & B7       |
| Intermittent (Mist) .....   | B7 & B18      |
| Low (Normal) .....          | B7 & B18      |
| Low (Mist) .....            | B7 & B18      |
| High (Normal) .....         | B13 & B18     |
| High (Mist) .....           | B7, B13 & B18 |
| Corolla                     |               |
| Mist .....                  | B7 & B18      |
| Off .....                   | B4 & B7       |
| Intermittent .....          | B4 & B7       |
| Low .....                   | B7 & B18      |
| High .....                  | B13 & B18     |
| Land Cruiser                |               |
| Off .....                   | B4 & B7       |
| Intermittent .....          | B4 & B7       |
| Low .....                   | B7 & B18      |
| High .....                  | B13 & B18     |
| Paseo & Tercel              |               |
| Intermittent .....          | 1 & 6         |
| Mist .....                  | 1 & 5         |
| Off .....                   | 1 & 6         |
| Low .....                   | 1 & 5         |
| High .....                  | 2 & 5         |
| Pickup, T100 & 4Runner      |               |
| Intermittent .....          | A7 & A8       |
| Mist .....                  | A4 & A8       |
| Off .....                   | A7 & A8       |
| Low .....                   | A4 & A8       |



|                     |       |                     |
|---------------------|-------|---------------------|
| High                | ..... | A4 & A9             |
| Previa              |       |                     |
| Off                 | ..... | B4 & B7             |
| Intermittent        | ..... | B4 & B7; B12 & B16  |
| Low                 | ..... | B7 & B18            |
| High                | ..... | B13 & B18           |
| Supra               |       |                     |
| Off                 | ..... | B4 & B7             |
| Off & Mist          | ..... | B7 & B18            |
| Intermittent        | ..... | B4 & B7             |
| Low                 | ..... | B7 & B18            |
| Low & Mist          | ..... | B7 & B18            |
| Intermittent & Mist | ..... | B7 & B18            |
| High                | ..... | B13 & B18           |
| High & Mist         | ..... | B7 & B13; B13 & B18 |

(1) - See Fig. 4.

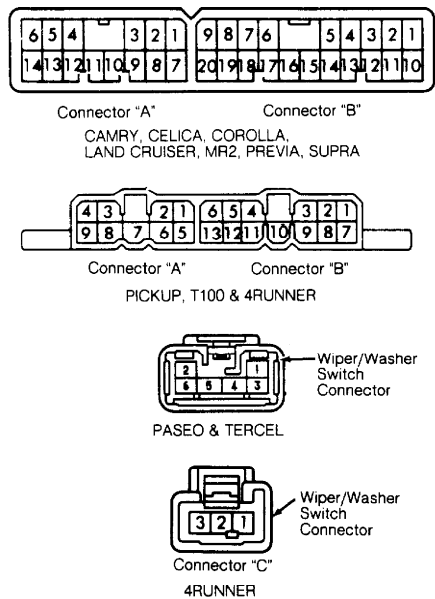


Fig. 4: Combination Switch Connectors For Wiper ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## FRONT WASHER SWITCH TEST

With washer switch in specified position, ensure continuity exists between terminals listed. See WASHER SWITCH CONTINUITY TEST table. See Fig. 4. Replace complete wiper switch assembly if continuity is not as specified.

### WASHER SWITCH CONTINUITY TEST TABLE

| Switch Position           | (1) Terminals     | Continuity |
|---------------------------|-------------------|------------|
| Paseo & Tercel            |                   |            |
| Off                       | 1, 2, 3, 4, 5 & 6 | No         |
| On                        | 3 & 4             | Yes        |
| Pickup, T100<br>& 4Runner |                   |            |

|            |                |     |
|------------|----------------|-----|
| Off .....  | A1 & A2 .....  | No  |
| On .....   | A1 & A2 .....  | Yes |
| All Others |                |     |
| Off .....  | B8 & B16 ..... | No  |
| On .....   | B8 & B16 ..... | Yes |

(1) - See Fig. 4.

---

## REAR WIPER MOTOR TEST

### Celica & Supra

1) Disconnect wiring connector from wiper motor. Using 12-volt battery, connect jumper wire from positive battery terminal to wiper motor terminal No. 1. See Fig. 5. Using another jumper wire, connect wiper motor terminal No. 3 and motor body to negative battery terminal. Motor should operate.

2) To check automatic PARK operation, with motor running, motor should stop by disconnecting jumper wire from terminal No. 3. To check intermittent operation, connect jumper wire from negative battery terminal to wiper motor terminal No. 2. Motor should operate intermittently for about 9-15 seconds. Replace motor and relay as a complete assembly if operation is not as described.

### Corolla

Disconnect wiring connector from wiper motor. Using 12-volt battery, connect jumper wire from positive battery terminal to wiper motor terminal No. 3. See Fig. 5. Using another jumper wire, connect wiper motor body (ground) to negative battery terminal. Motor should operate. Replace motor if operation is not as described.

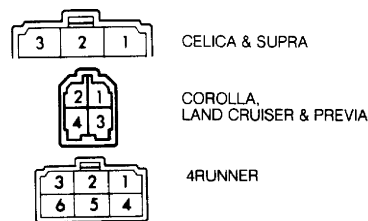
### Land Cruiser & Previa

1) Disconnect wiring connector from wiper motor. Using 12-volt battery, connect jumper wire from positive battery terminal to wiper motor terminal No. 3. See Fig. 5. Using another jumper wire, connect wiper motor terminal No. 2 to negative battery terminal. Motor should operate.

2) To check automatic PARK operation, with motor running, stop motor operation anywhere except PARK position by removing power from terminal No. 3. Install another jumper wire between terminals No. 3 and 4. Connect jumper wire from positive battery terminal to wiper motor terminal No. 1. Ensure motor operates and stops at PARK position. Replace motor if operation is not as described.

### 4Runner

Disconnect wiring connector from wiper motor. Using 12-volt battery, connect jumper wire from negative battery terminal to wiper motor terminal No. 3. See Fig. 5. Using another jumper wire, connect wiper motor terminal No. 1 to positive battery terminal. Motor should operate in clockwise direction. Reverse jumper wires. Motor should operate in counterclockwise direction. Replace motor if operation is not as described.



93184663

Fig. 5: Rear Wiper Motor Terminal ID  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## REAR WIPER RELAY TEST

**NOTE:** Testing information is not available from manufacturer for all models. For relay locations, see REAR WIPER RELAY LOCATION table.

### Corolla

Using an ohmmeter, check continuity between relay terminals No. 3 and 4. See Fig. 3. No continuity should exist. Connect ohmmeter leads to relay terminals No. 4 and 5. Continuity should exist. If continuity is not as described, replace relay.

### Land Cruiser & Previa

Using an ohmmeter, check continuity between relay terminals No. 1 and 3. See Fig. 3. No continuity should exist. Connect ohmmeter leads to relay terminals No. 2 and 3. Continuity should exist. If continuity is not as described, replace relay.

## REAR WIPER RELAY LOCATION TABLE

| Model          | Location                                             |
|----------------|------------------------------------------------------|
| Celica & Supra | Inside Back Door At Rear Wiper Motor                 |
| Corolla        | Inside Back Door, To Right Of Motor                  |
| Land Cruiser   | Inside Back Door, To Left Of Motor                   |
| Previas        | Behind Center Of Instrument Cluster                  |
| 4Runner        | Behind Interior Left Rear Side Panel, Near Rear Door |

## REAR WIPER SWITCH TEST

With wiper switch in specified position, ensure continuity exists between terminals listed. See REAR WIPER/WASHER SWITCH CONTINUITY TEST table. See Fig. 4. Replace complete wiper switch if continuity is not as specified.

## REAR WIPER/WASHER SWITCH CONTINUITY TEST TABLE

| Switch Position               | Terminals        | Continuity |
|-------------------------------|------------------|------------|
| Celica, Corolla, Land Cruiser |                  |            |
| Supra & Previa (1)            |                  |            |
| Off                           | B1, B2, B10, B16 | No         |
| Intermittent                  | B10 & B16        | Yes        |
| On                            | B1 & B16         | Yes        |
| Wash                          |                  |            |

|                       |                       |     |
|-----------------------|-----------------------|-----|
| W/Wiper On .....      | B2 & B16 .....        | Yes |
| W/Intermittent On ... | B1, B2 & B16 .....    | Yes |
| 4Runner (1)           |                       |     |
| Off .....             | A1, C1, C2 & C3 ..... | No  |
| Intermittent .....    | A1 & C2 .....         | Yes |
| On .....              | A1 & C1 .....         | Yes |
| Wash W/Wiper On ..... | A1, C1 & C3 .....     | Yes |
| Wash Only .....       | A1 & C3 .....         | Yes |

(1) - See Fig. 4.

---

## REMOVAL & INSTALLATION

**WARNING:** Camry, Celica, Corolla, MR2, Paseo, Previa, Supra and Tercel are equipped with a driver's side air bag. Use extreme caution while servicing components on steering column. Ensure ignition switch is in LOCK position and negative battery cable is disconnected for at least one minute before starting service. DO NOT apply electrical power to any component on steering column without disconnecting air bag control unit. Information labels are attached to air bag components. Follow all notices on labels. Use only DVOM with minimum of 10-k/ohms impedance to check any circuit.

### FRONT WIPER SWITCH

**NOTE:** Wiper switch is component of combination switch, mounted to steering column. For further information and/or illustrations on removal procedure, see COMBINATION SWITCH under REMOVAL & INSTALLATION in appropriate STEERING COLUMN SWITCHES article listed below in the ACCESSORIES/SAFETY EQUIPMENT section.

- \* STEERING COLUMN SWITCHES (for Camry).
- \* STEERING COLUMN SWITCHES (for Celica).
- \* STEERING COLUMN SWITCHES (for Corolla).
- \* STEERING COLUMN SWITCHES (for Land Cruiser).
- \* STEERING COLUMN SWITCHES (for MR2).
- \* STEERING COLUMN SWITCHES (for Paseo).
- \* STEERING COLUMN SWITCHES (for Pickup).
- \* STEERING COLUMN SWITCHES (for Previa).
- \* STEERING COLUMN SWITCHES (for Supra).
- \* STEERING COLUMN SWITCHES (for T100).
- \* STEERING COLUMN SWITCHES (for Tercel).
- \* STEERING COLUMN SWITCHES (for 4Runner).

#### Removal & Installation (All Models)

1) Disconnect negative battery cable. To access combination switch connector, remove instrument panel lower finish panel under steering column. See INSTRUMENT CLUSTER under REMOVAL & INSTALLATION in appropriate INSTRUMENT PANELS article listed below in the ACCESSORIES/SAFETY EQUIPMENT section. Remove steering column upper and lower covers. Remove steering wheel. See STEERING WHEEL under REMOVAL & INSTALLATION.

- \* INSTRUMENT PANEL (for Camry).
- \* INSTRUMENT PANEL (for Celica).
- \* INSTRUMENT PANEL (for Corolla).
- \* INSTRUMENT PANEL (for Land Cruiser).
- \* INSTRUMENT PANEL (for MR2).
- \* INSTRUMENT PANEL (for Paseo).

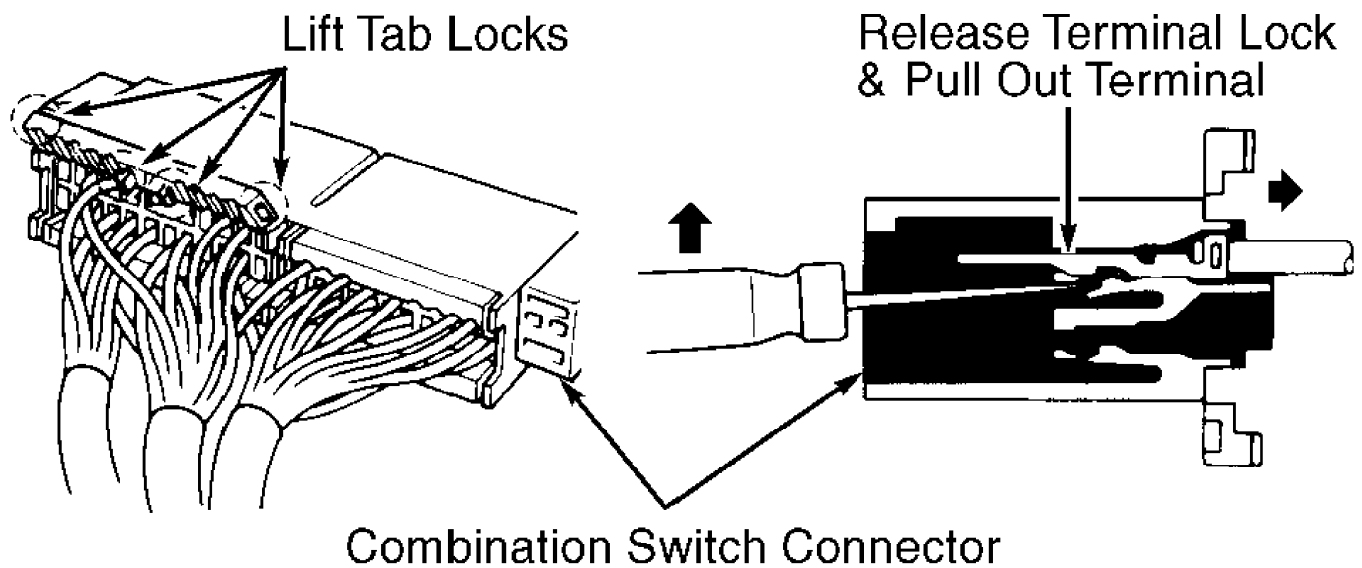
- \* INSTRUMENT PANEL (for Pickup).
- \* INSTRUMENT PANEL (for Previa).
- \* INSTRUMENT PANEL (for Supra).
- \* INSTRUMENT PANEL (for T100).
- \* INSTRUMENT PANEL (for Tercel).
- \* INSTRUMENT PANEL (for 4Runner).

2) Locate wiper/washer switch wire terminal positions installed into combination switch harness connector. See Fig. 4. Lift tab locks on combination switch connector. See Fig. 6.

3) Using miniature screwdriver, release wire terminal securing tab inside connector. See Fig. 6. Pull wire(s) from connector and through harness jacket tubing. Remove wiper switch retaining screws from rear of combination switch body and remove switch. See Fig. 7.

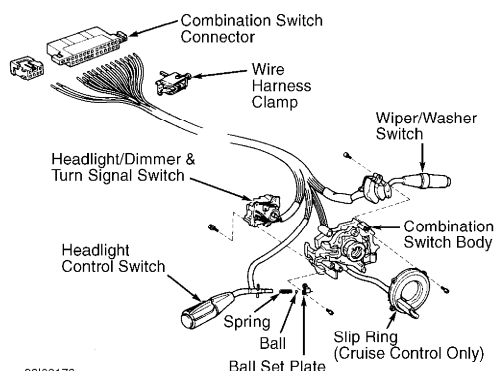
#### Installation

To install, reverse removal procedure. When installing wiper switch wire terminals to harness connector, push inward on wire terminal until terminal clicks into place.



93A02179

Fig. 6: Releasing Wire Terminal Inside Combination Connector  
Courtesy of Toyota Motor Sales, U.S.A., Inc.



93102178

Fig. 7: Combination Switch Component ID (Typical)  
Courtesy of Toyota Motor Sales, U.S.A., Inc.

## REAR WIPER SWITCH

### Removal & Installation (All Models)

Switch is integral part of wiper switch mounted to combination switch on steering column. See FRONT WIPER SWITCH under REMOVAL & INSTALLATION.

## STEERING WHEEL

### Removal (Without Air Bag)

1) On rear of steering wheel, locate and remove retaining screw securing horn pad. Remove horn pad enough to disconnect horn electrical connector(s).

2) Remove steering wheel lock nut and washer. Make alignment mark on steering shaft and steering wheel for installation reference. Install suitable steering wheel puller and pull steering wheel from shaft.

### Installation

Align reference marks on steering shaft and steering wheel. Tighten steering wheel lock nut to 25 ft. lbs. (34 N.m). Install horn pad.

### Removal (With Air Bag)

1) Ensure front wheels are in straight ahead position. Place ignition switch in LOCK position and remove key. Disconnect negative battery cable. Remove 2 screw covers from outer sides of steering wheel. Using Torx Wrench (T30), loosen 4 air bag pad retaining screws until screw head is snug against screw case. See Fig. 8.

2) Carefully pull air bag pad away from steering wheel enough to unlock and disconnect air bag electrical connector. DO NOT pull on electrical connector or wiring. Store air bag pad aside with pad facing upward.

3) Remove steering wheel lock nut and washer from steering shaft. Make alignment mark on steering shaft and steering wheel for installation reference. Using appropriate steering wheel puller, pull steering wheel from shaft while guiding spiral cable wire through steering wheel opening.

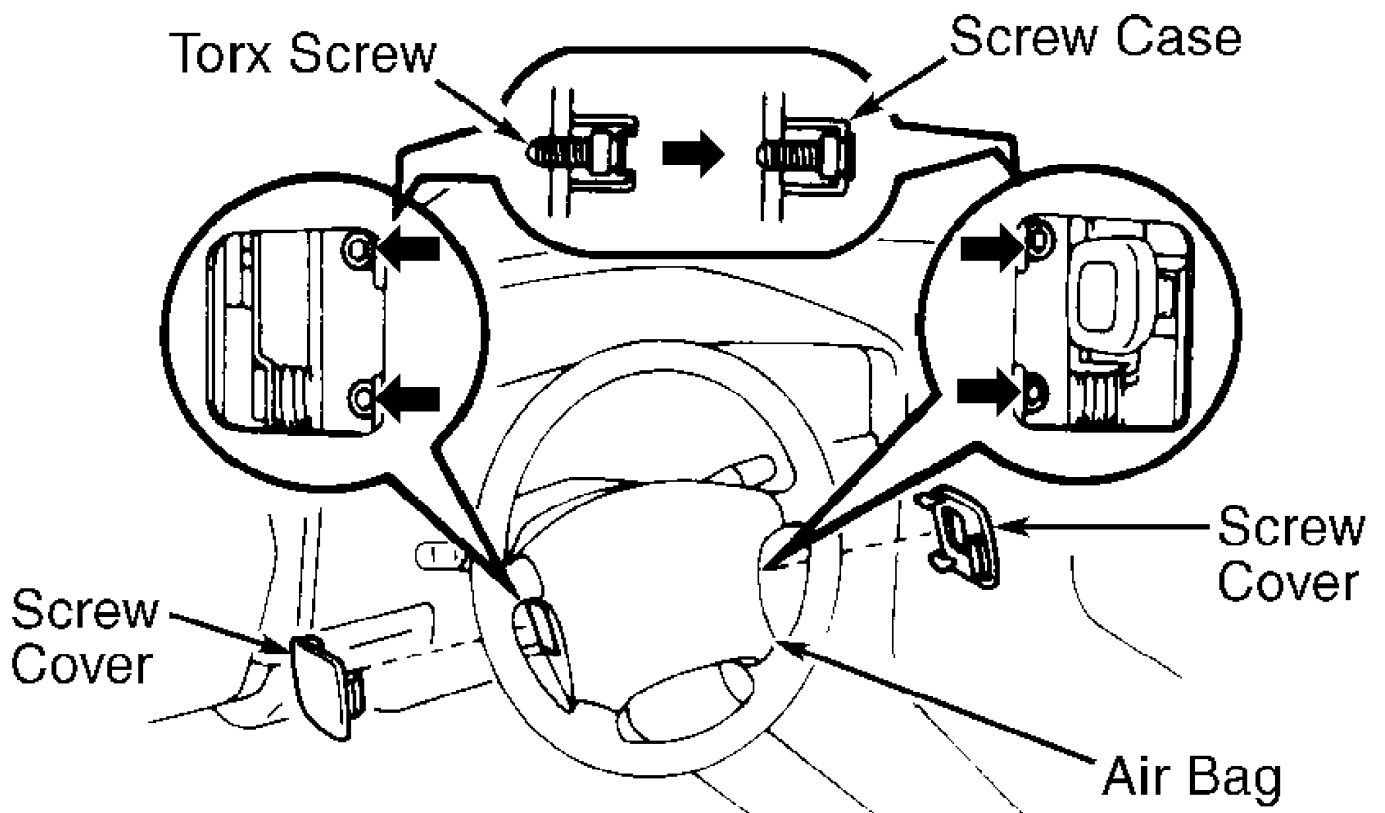
### Installation

1) Ensure front wheels are in straight ahead position. Connect and install spiral cable to combination switch. Turn spiral cable counterclockwise by hand until it is hard to turn. Turn spiral cable clockwise about 2 1/2 turns and align Red mark at bottom with opening.

2) Guide spiral cable through steering wheel opening while installing steering wheel to shaft. Align reference marks on steering shaft and steering wheel. Tighten steering wheel lock nut to 25 ft. lbs. (34 N.m).

3) Connect air bag electrical connector and snap down connector lock. Before installing air bag pad, ensure Torx screws are retracted and snug against screw case. See Fig. 8.

4) Install air bag pad to steering wheel ensuring all wiring is NOT pinched and DOES NOT interfere with moving parts. Tighten 4 Torx screws to 65 INCH lbs. (7.4 N.m). Install screw covers. Connect negative battery cable.



91G03959

Fig. 8: Removing Air Bag From Steering Wheel  
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

## WIRING DIAGRAMS

Proceed to appropriate WIRING DIAGRAMS article listed below  
 in WIRING DIAGRAMS section.

- \* WIRING DIAGRAMS (for 4Runner).
- \* WIRING DIAGRAMS (for Camry).
- \* WIRING DIAGRAMS (for Celica).
- \* WIRING DIAGRAMS (for Corolla).
- \* WIRING DIAGRAMS (for Land Cruiser).
- \* WIRING DIAGRAMS (for MR2).
- \* WIRING DIAGRAMS (for Paseo).
- \* WIRING DIAGRAMS (for Pickup).
- \* WIRING DIAGRAMS (for Previa).
- \* WIRING DIAGRAMS (for Supra).
- \* WIRING DIAGRAMS (for T100).
- \* WIRING DIAGRAMS (for Tercel).

# WIRING DIAGRAMS

## 1993 Toyota Celica

1993 WIRING DIAGRAMS  
Toyota Wiring Diagrams

Toyota; Celica

### IDENTIFICATION

### COMPONENT LOCATION MENU

#### COMPONENT LOCATIONS TABLE

| Component                                | Figure No. (Location) |
|------------------------------------------|-----------------------|
| A/C COMPRESSOR CLUTCH RELAY (AUTO A/C)   | 11 (D 41)             |
| A/C CONDENSER FAN MOTOR (AUTO A/C)       | 11 (E 40)             |
| A/C CONDENSER FAN RELAY #2 (AUTO A/C)    | 11 (D 40)             |
| A/C POWER TRANSISTOR (AUTO A/C)          | 11 (D 41)             |
| A/C SYSTEM (AUTO)                        | 9 (A-E 32-35)         |
| A/C SYSTEM (MANUAL)                      | 10 (A-E 36-39)        |
| AUTOMATIC TRANSMISSION                   |                       |
| INDICATOR SWITCH                         | 14 (E 53-54)          |
| ANTI-LOCK BRAKE SYSTEM (ABS)             | 13 (A-E 48-51)        |
| ALTERNATOR                               | 1 (B 3)               |
| AUTO ANTENNA                             | 14 (E 55)             |
| AUTO TILT AWAY ECU                       | 12 (E 47)             |
| BACK-UP LIGHT SWITCH                     | 16 (D 63)             |
| BATTERY                                  | 1 (A 2)               |
| BLOWER RESISTOR (AUTO A/C)               | 11 (D 43)             |
| CENTER AIRBAG SENSOR ASSEMBLY            | 12 (E 44)             |
| CHECK CONNECTOR (1.6L)                   | 3 (D 11)              |
| CHECK CONNECTOR (2.0L)                   | 4 (C-D 15)            |
| CHECK CONNECTOR (2.2L A/T)               | 5 (D-E 19)            |
| CHECK CONNECTOR (2.2L M/T)               | 6 (D-E 23)            |
| CIRCUIT OPENING RELAY (1.6L)             | 3 (A 10)              |
| CIRCUIT OPENING RELAY (2.0L)             | 4 (A 13)              |
| CIRCUIT OPENING RELAY (2.2L A/T)         | 5 (A 18)              |
| CIRCUIT OPENING RELAY (2.2L M/T)         | 6 (A 22)              |
| COLD START INJECTOR                      | 1 (B 2)               |
| COMBINATION SWITCH                       | 15 (C-E 59)           |
| CRUISE CONTROL SYSTEM                    | 12 (A-D 44-47)        |
| DEFOGGER SWITCH                          | 14 (A 52)             |
| DIRECTIONAL FLASHER                      | 8 (D 30)              |
| DOOR LOCK SYSTEM                         | 16 (D-E 60-62)        |
| ECM (1.6L)                               | 3 (B-D 8)             |
| ECM (2.0L)                               | 4 (A-E 12)            |
| ECM (2.2L A/T)                           | 5 (A-E 16)            |
| ECM (2.2L M/T)                           | 6 (B-E 20)            |
| ELECTRONICALLY CONTROLLED                |                       |
| TRANSMISSION (ECT) PATTERN SELECT SWITCH | 14 (C 52)             |
| EXTRA HIGH SPEED RELAY (AUTO A/C)        | 11 (D 41)             |
| FOG LIGHT SWITCH                         | 1 (C-D 2)             |
| FUEL SENDER                              | 15 (A-B 59)           |
| FUSIBLE LINK BOX                         | 2 (C 5-6)             |
| HAZARD SWITCH                            | 15 (E 56)             |
| HEATER RELAY (AUTO A/C)                  | 11 (D 41)             |
| HORNS                                    | 1 (C 1)               |
| IGNITER (1.6L)                           | 3 (E 9-10)            |
| IGNITER (2.0L)                           | 3 (D-E 15)            |
| IGNITER (2.2L A/T)                       | 5 (E 18)              |



|                                        |                  |
|----------------------------------------|------------------|
| IGNITION KEY CYLINDER LIGHT .....      | 11 (B 40)        |
| IGNITION NOISE FILTER (1.6L) .....     | 3 (D 10)         |
| IGNITION NOISE FILTER (2.0L) .....     | 4 (E 14)         |
| IGNITION NOISE FILTER (2.2L A/T) ..... | 5 (E 17)         |
| IGNITION NOISE FILTER (2.2L M/T) ..... | 6 (E 21)         |
| IGNITION SWITCH .....                  | 7 (A 24)         |
| J1 JUNCTION CONN .....                 | 11 (B-C 43)      |
| J2 JUNCTION CONN .....                 | 12 (B-C 44)      |
| J3 JUNCTION CONN .....                 | 7 (E 24)         |
| J4 JUNCTION CONN (AUTO A/C) .....      | 11 (C-D 43)      |
| J5 JUNCTION CONN (AUTO A/C) .....      | 11 (E 43)        |
| J6 JUNCTION CONN .....                 | 7 (A 27)         |
| JUNCTION BLOCK #1 (J/B #1) .....       | 7, 8 (B-D 25-30) |
| JUNCTION BLOCK #2 (J/B #2) .....       | 2 (B-E 5-6)      |
| JUNCTION BLOCK #3 (J/B #3) .....       | 14 (A-B 53-54)   |
| LEFT DOOR COURTESY SWITCH .....        | 11 (B 43)        |
| LUGGAGE COMPARTMENT LIGHT .....        | 11 (A 42)        |
| OVERDRIVE (O/D) MAIN SWITCH .....      | 14 (C-D 52)      |
| OVERDRIVE (O/D) SOLENOID .....         | 14 (D 55)        |
| POWER MIRROR SWITCH .....              | 16 (B 60)        |
| POWER SEAT SWITCH .....                | 16 (A 60)        |
| POWER WINDOW MASTER SWITCH .....       | 16 (C-D 62)      |
| RADIATOR FAN MOTOR (AUTO A/C) .....    | 11 (E 40)        |
| REAR WIPER MOTOR & RELAY .....         | 15 (C-D 57)      |
| RELAY BLOCK # 2 (R/B #2) .....         | 8 (A 28)         |
| RELAY BLOCK # 3 (R/B #3) .....         | 1 (C 3)          |
| RELAY BLOCK # 4 (R/B #4) .....         | 8 (A-B 29-31)    |
| RELAY BLOCK # 5 (R/B #5) .....         | 2 (A 5-6)        |
| RETRACT CONTROL RELAY .....            | 1 (D 3)          |
| RHEOSTAT .....                         | 14 (C 55)        |
| RIGHT DOOR COURTESY SWITCH .....       | 11 (A 43)        |
| SHIFT LOCK ECU .....                   | 14 (D-E 52)      |
| STARTER .....                          | 1 (A 3)          |
| STOP LIGHT SWITCH .....                | 12 (B 47, C 47)  |
| SUNROOF SYSTEM .....                   | 16 (A-B 61-62)   |
| TURBOCHARGING PRES SENSOR (2.0L) ..... | 4 (B 15)         |
| UNLOCK WARNING SWITCH .....            | 7 (A 24)         |
| WASHER MOTOR .....                     | 1 (E 2)          |
| WIPER MOTOR .....                      | 1 (E 3)          |

---

## WIRING DIAGRAMS

# COMPONENT LOCATOR:

|                                  |            |
|----------------------------------|------------|
| A/C COMPRESSOR CLUTCH            | D 41       |
| RELAY (AUTO A/C)                 | A-E 32-35  |
| A/C CONDENSER FAN                | E 40       |
| MOTOR (AUTO A/C)                 | A-E 48-51  |
| RELAY #2 (AUTO A/C)              | D 40       |
| A/C POWER TRANSISTOR             | D 41       |
| (AUTO A/C)                       | A-E 32-35  |
| A/C SYSTEM (AUTO)                | A-E 36-39  |
| A/T IND SW                       | E 53-54    |
| ABS SYSTEM                       | A-E 48-51  |
| ALTERNATOR                       | B-3        |
| AUTO ANTENNA                     | E 55       |
| AUTO TILT AWAY ECU               | E 47       |
| BACK-UP LT SW                    | E 63       |
| BATTERY                          | A 2        |
| BLOWER RESISTOR (AUTO A/C)       | D 43       |
| CENTER AIRBAG SENSOR ASSEMBLY    | E 44       |
| CHECK CONN (1.6L)                | D 11       |
| CHECK CONN (2.0L)                | C-D 15     |
| CHECK CONN (2.2L A/T)            | D-E 19     |
| CHECK CONN (2.2L M/T)            | D-E 23     |
| CIRCUIT OPENING RELAY (1.6L)     | A 10       |
| CIRCUIT OPENING RELAY (2.0L)     | A 13       |
| RELAY (2.2L A/T)                 | A 18       |
| RELAY (2.2L M/T)                 | A 22       |
| COLD START INJECTOR              | B 2        |
| COMBINATION SW                   | C-E 59     |
| CRUISE CONTROL SYSTEM            | A-D 44-47  |
| DEFOG SW                         | A 52       |
| DIR FLASHER                      | D 30       |
| DOOR LOCK SYSTEM                 | D-E 60-62  |
| ECM (1.6L)                       | B-D 8      |
| ECM (2.0L)                       | A-E 12     |
| ECM (2.2L A/T)                   | A-E 16     |
| ECM (2.2L M/T)                   | B-E 20     |
| ECT PATTERN SELECT SW            | C 52       |
| EXTRA HIGH SPEED                 | D 41       |
| RELAY (AUTO A/C)                 | C-D 2      |
| FUEL SENDER                      | A 59, B 59 |
| FUSE LINK BOX                    | C 5-6      |
| HAZARD SW                        | E 56       |
| HEATER RELAY (AUTO A/C)          | D 41       |
| HORN                             | C 1        |
| IGNITER (1.6L)                   | E 9-10     |
| IGNITER (2.0L)                   | D-E 15     |
| IGNITER (2.2L A/T)               | E 18       |
| IGNITION KEY CYLINDER LT         | B 40       |
| IGNITION NOISE FILTER (1.6L)     | D 10       |
| IGNITION NOISE FILTER (2.0L)     | E 14       |
| IGNITION NOISE FILTER (2.2L A/T) | E 17       |
| IGNITION SW                      | E 21       |
| J1 JUNCTION CONN                 | B-C 43     |
| J2 JUNCTION CONN                 | B-C 44     |
| J3 JUNCTION CONN                 | E 24       |
| J4 JUNCTION CONN (AUTO A/C)      | C-D 43     |
| J5 JUNCTION CONN (AUTO A/C)      | E 43       |
| J6 JUNCTION CONN                 | A 27       |
| JUNCTION BLOCK #1 (J/B #1)       | B-D 25-30  |
| JUNCTION BLOCK #2 (J/B #2)       | B-E 5-6    |
| JUNCTION BLOCK #3 (J/B #3)       | A-B 53-54  |
| LEFT DOOR COURTESY SW            | B 43       |
| LUGGAGE COMPARTMENT LT           | A 42       |
| O/D MAIN SW                      | C-D 52     |
| O/D SOLENOID                     | D 55       |
| POWER MIRROR SW                  | B 60       |
| RADIATOR FAN MOTOR (AUTO A/C)    | E 40       |
| REAR WIPER MOTOR & RELAY         | C-D 57     |
| RELAY BLOCK #2 (R/B #2)          | C 3        |
| RELAY BLOCK #3 (R/B #3)          | A-B 29-31  |
| RELAY BLOCK #4 (R/B #4)          | A 5-6      |
| RETRACT CONTROL RELAY            | D 3        |
| RHEOSTAT                         | C 85       |
| RIGHT DOOR COURTESY SW           | A 43       |
| SHIFT LOCK ECU                   | D-E 62     |
| STARTER                          | A 3        |
| STOP LT SW                       | B 47, C 47 |
| SUNROOF SYSTEM                   | A-R 61-69  |
| TURBOCHARGING PRES               | B 15       |
| SENSOR (2.0L)                    | A 24       |
| UNLOCK WARNING SW                | E 2        |
| WASHER MOTOR                     | E 3        |

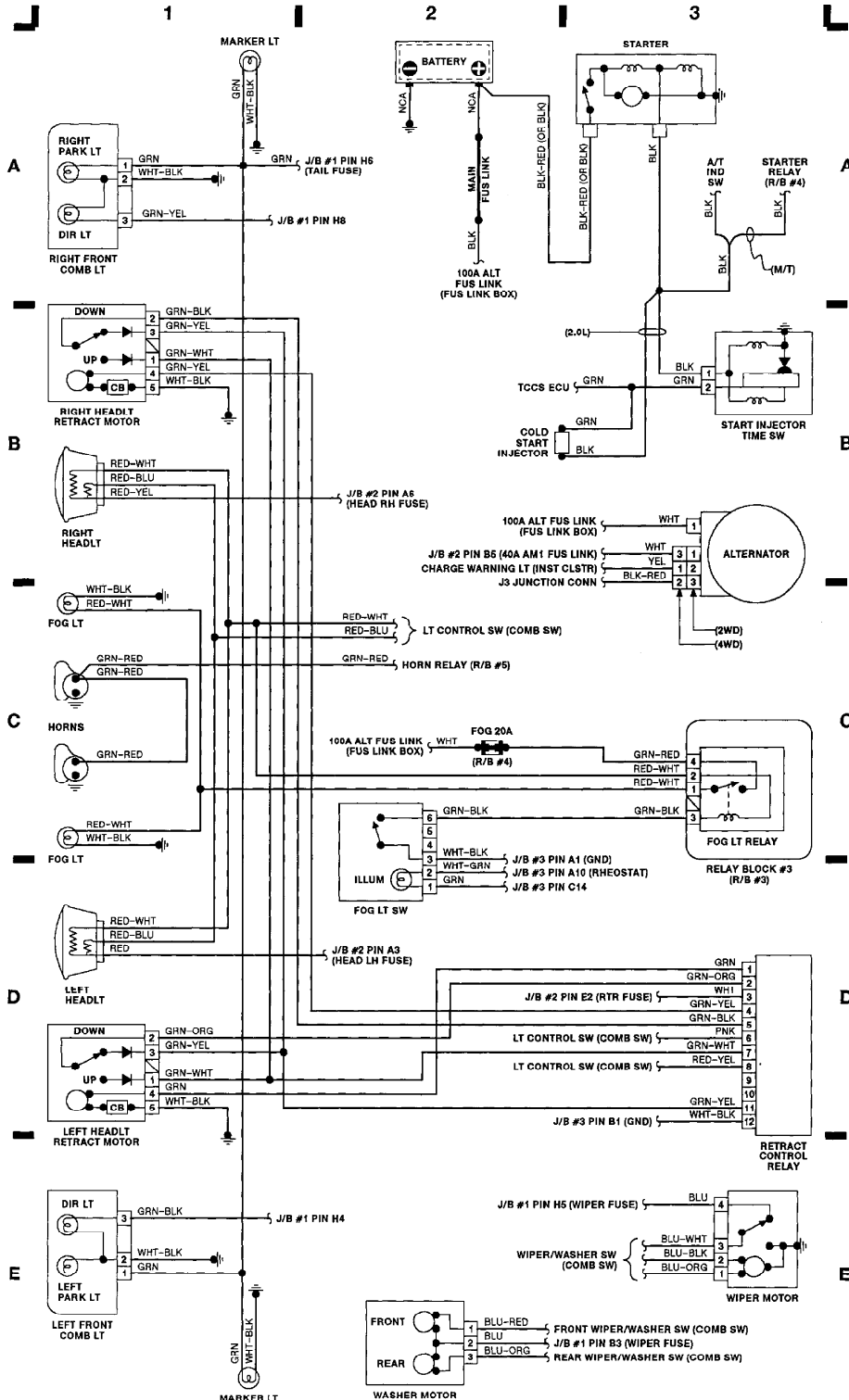


Fig. 1: Engine Compartment, Headlights, Starter (Grid 1-3)





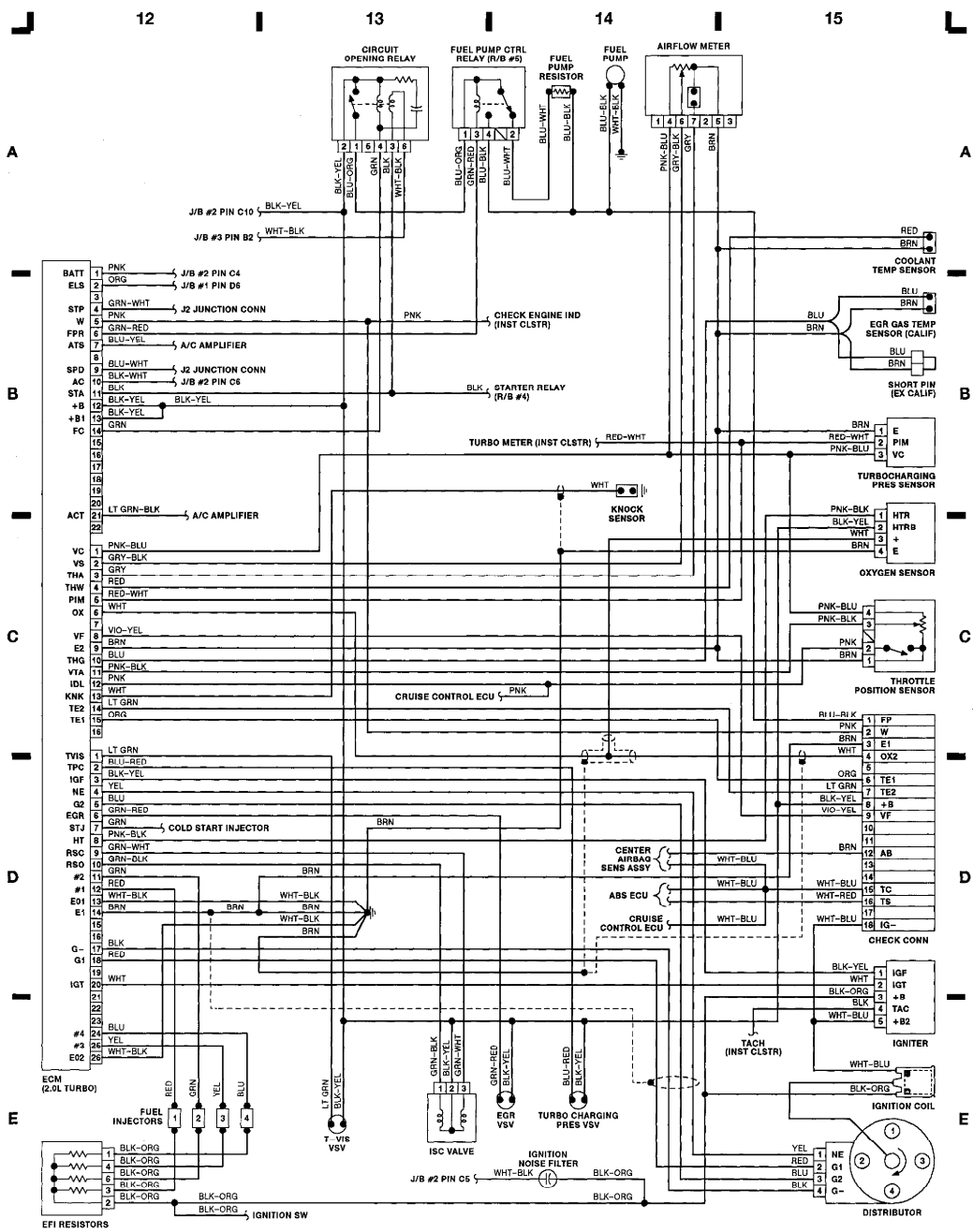


Fig. 4: ECM ECU & Engine Sensors (2.0L Turbo) (Grid 12-15)



94G30458

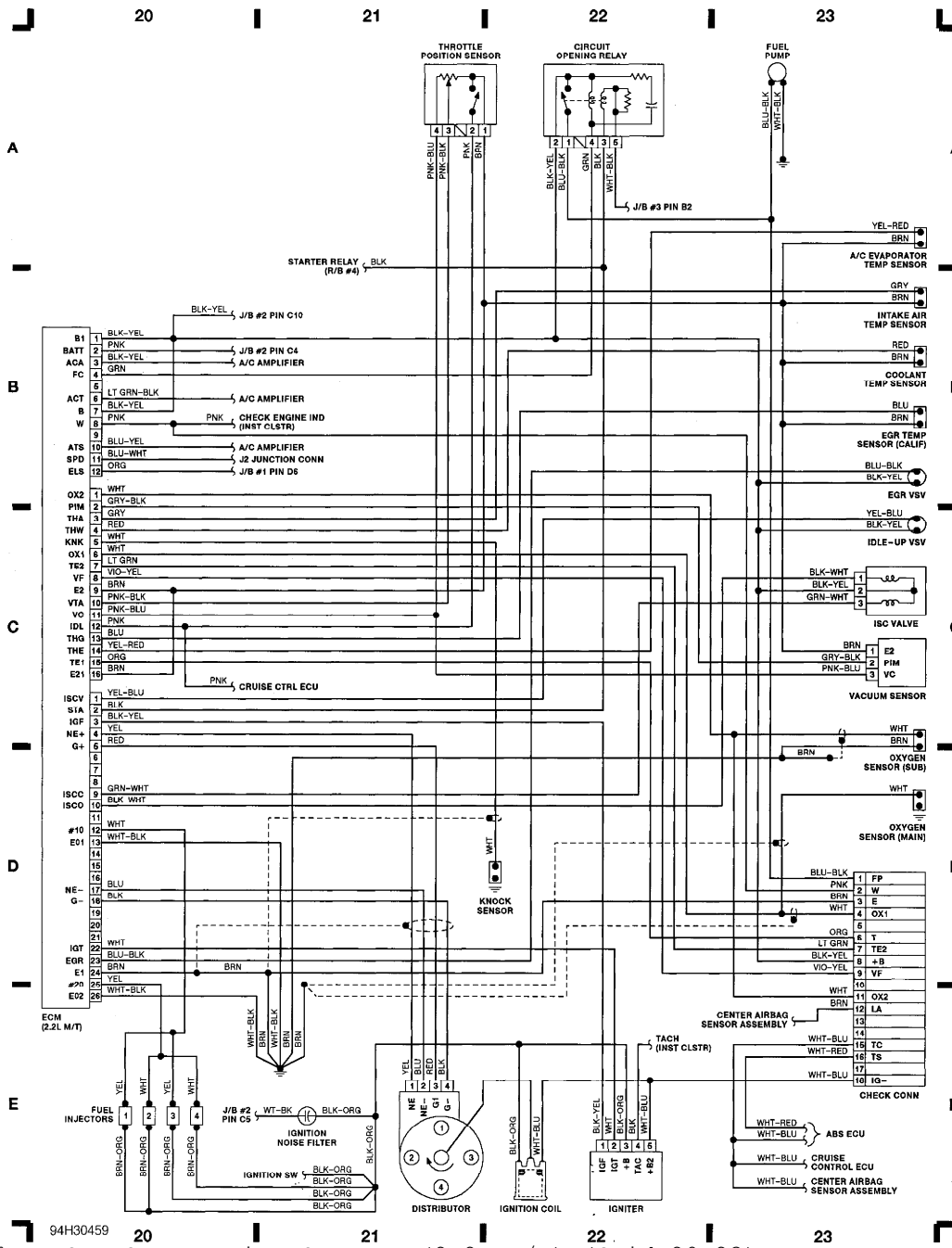


Fig. 6: ECM ECU & Engine Sensors (2.2L M/T) (Grid 20-23)

Fig. 7: Junction Block #1 (Partial), Ignition SW (Grid 24-27)





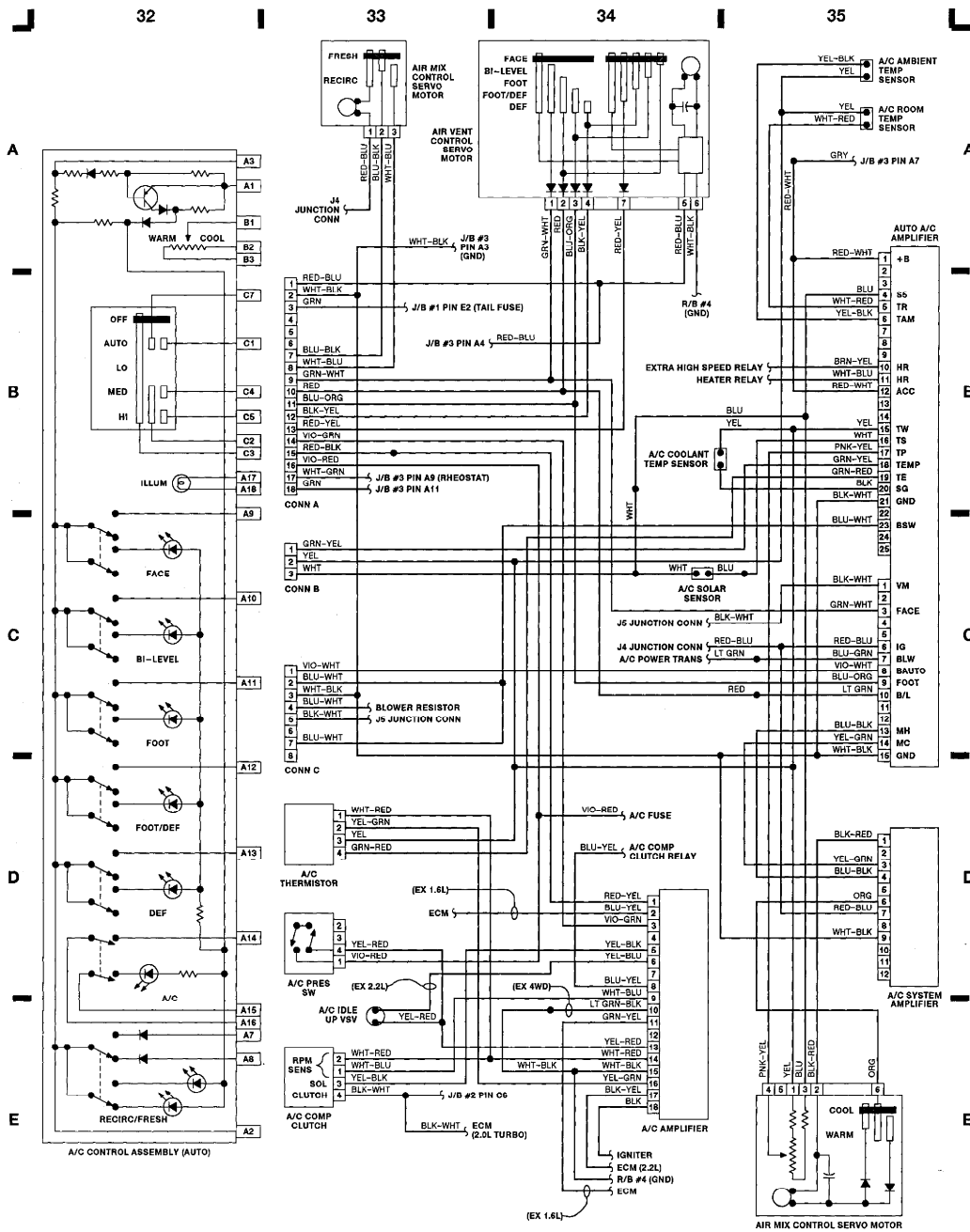


Fig. 9: Auto A/C & Heater System (Grid 32-35)



Fig. 10: Manual A/C & Heater System (Grid 36-39)

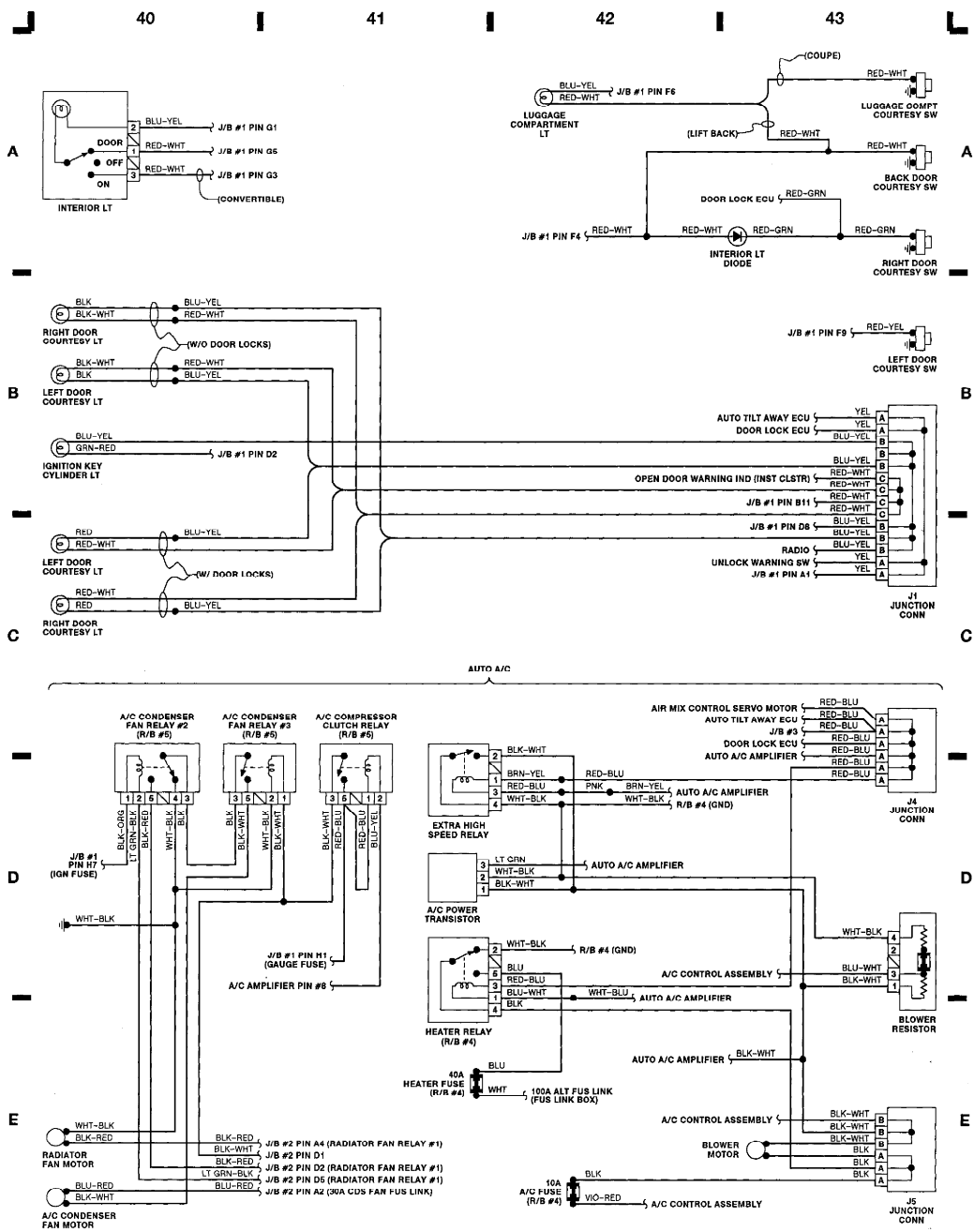


Fig. 11: Fan Relays, Courtesy Lights, Heater Relay (Grid 40-43)



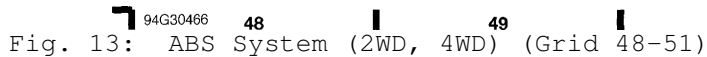


Fig. 13: ABS System (2WD, 4WD) (Grid 48-51)



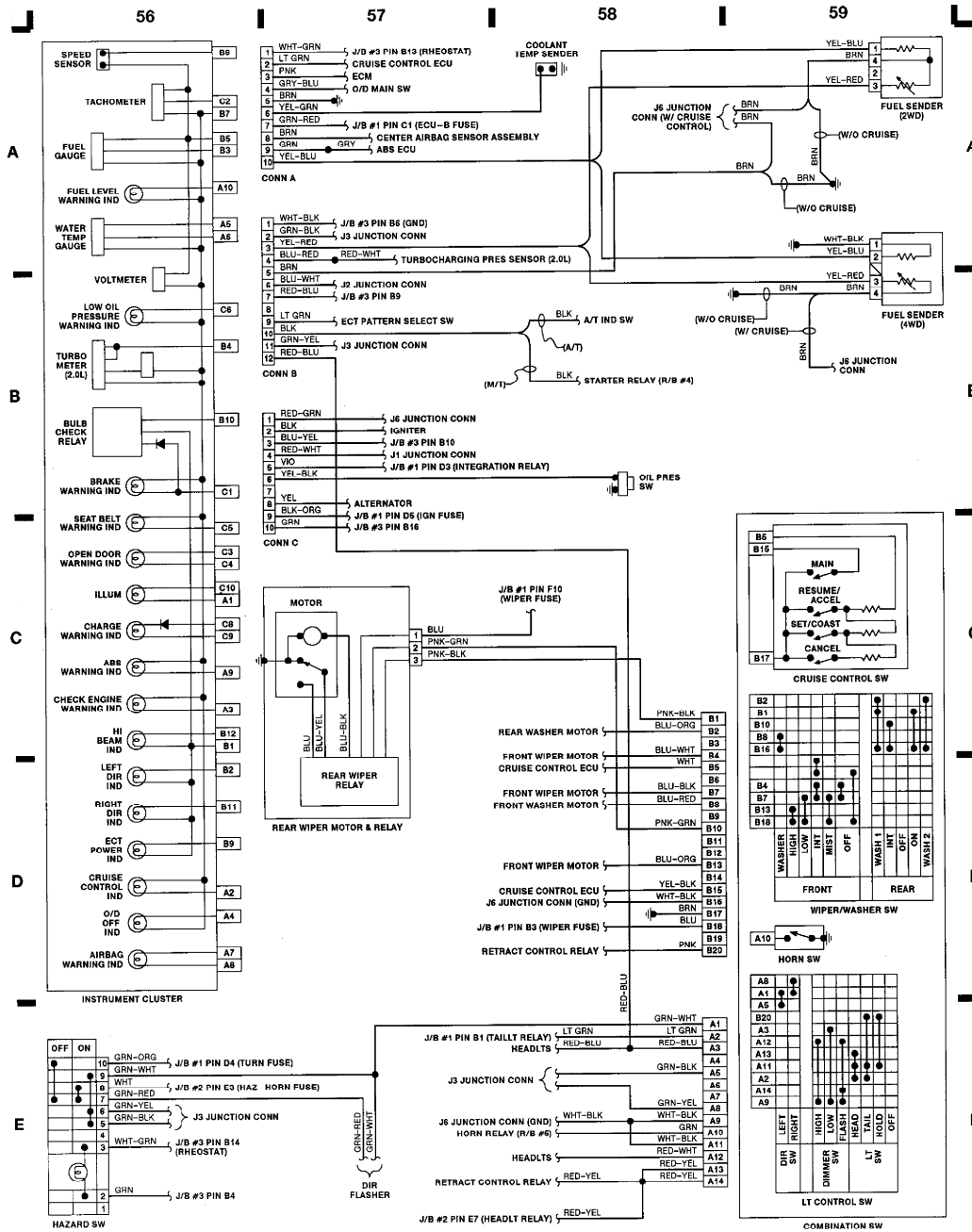


Fig. 15: Instrument Cluster, Combination Switch (Grid 56-59)



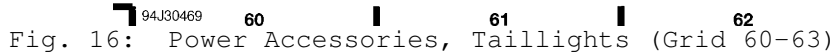


Fig. 16: Power Accessories, Taillights (Grid 60-63)

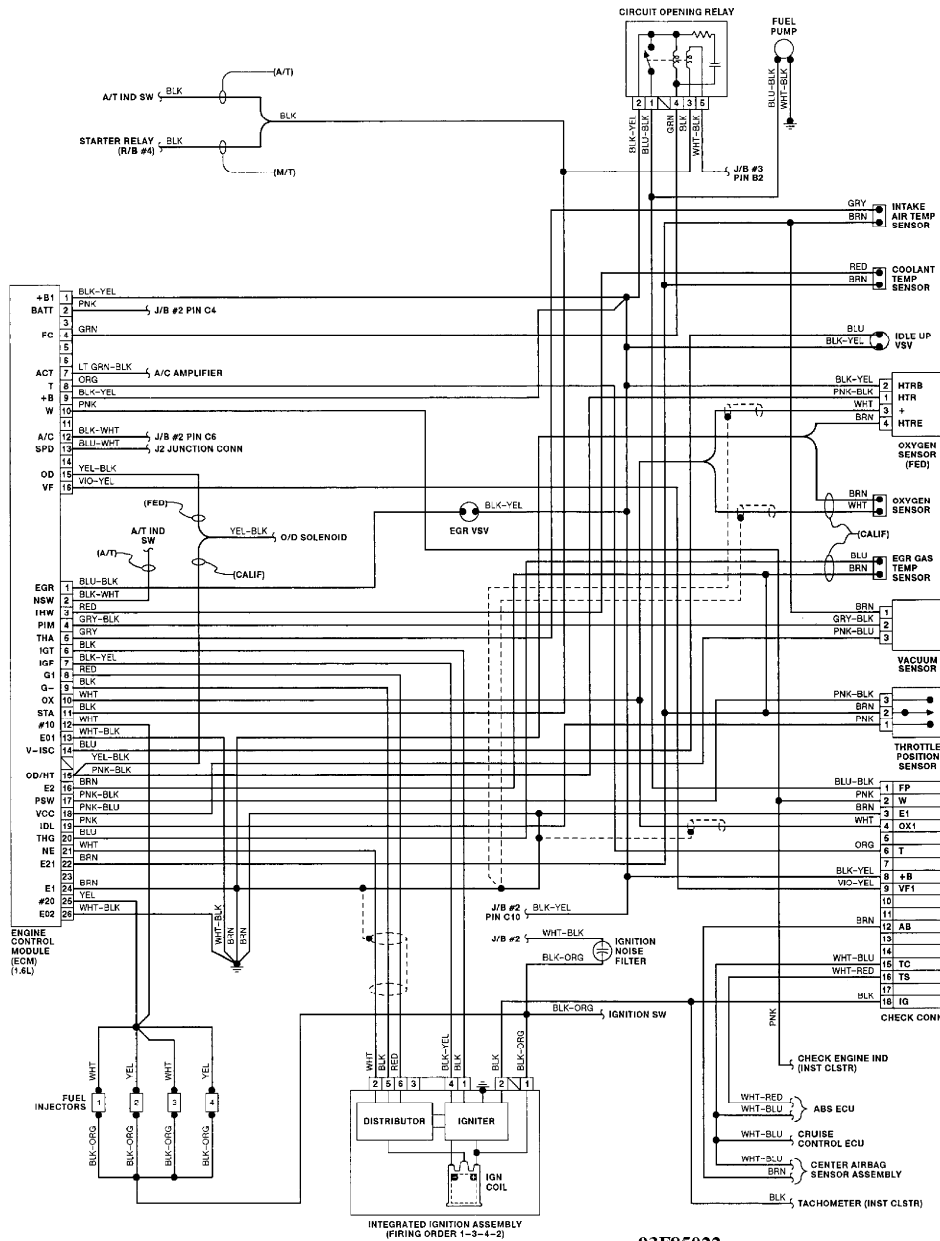
# L - WIRING DIAGRAMS

1993 Toyota Celica

1993 ENGINE PERFORMANCE  
Toyota Wiring Diagrams

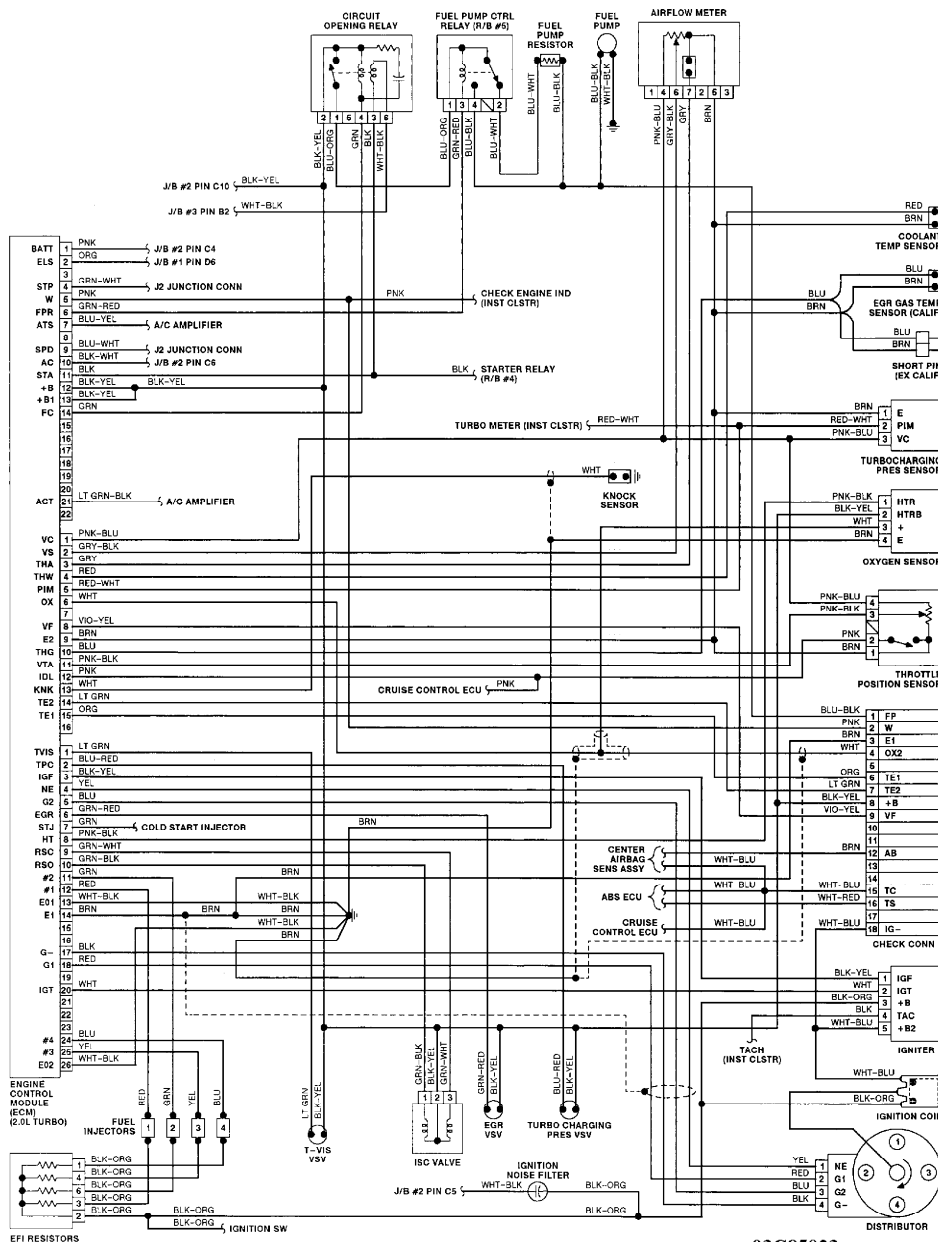
Celica

## WIRING DIAGRAMS



93F85022

Fig. 1: Wiring Diagram (Celica 1.6L)



93G85023

Fig. 2: Wiring Diagram (Celica 2.0L Turbo)

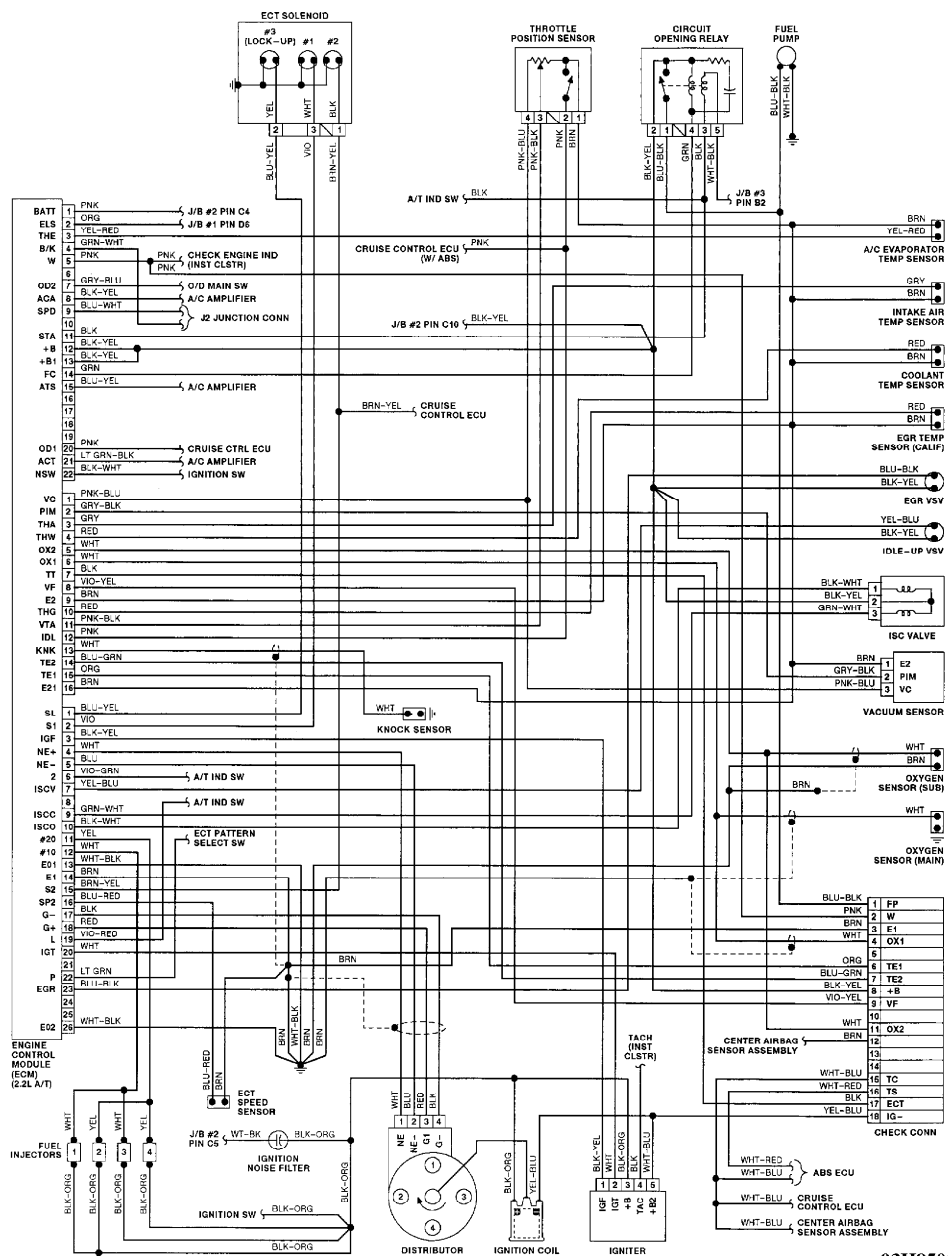
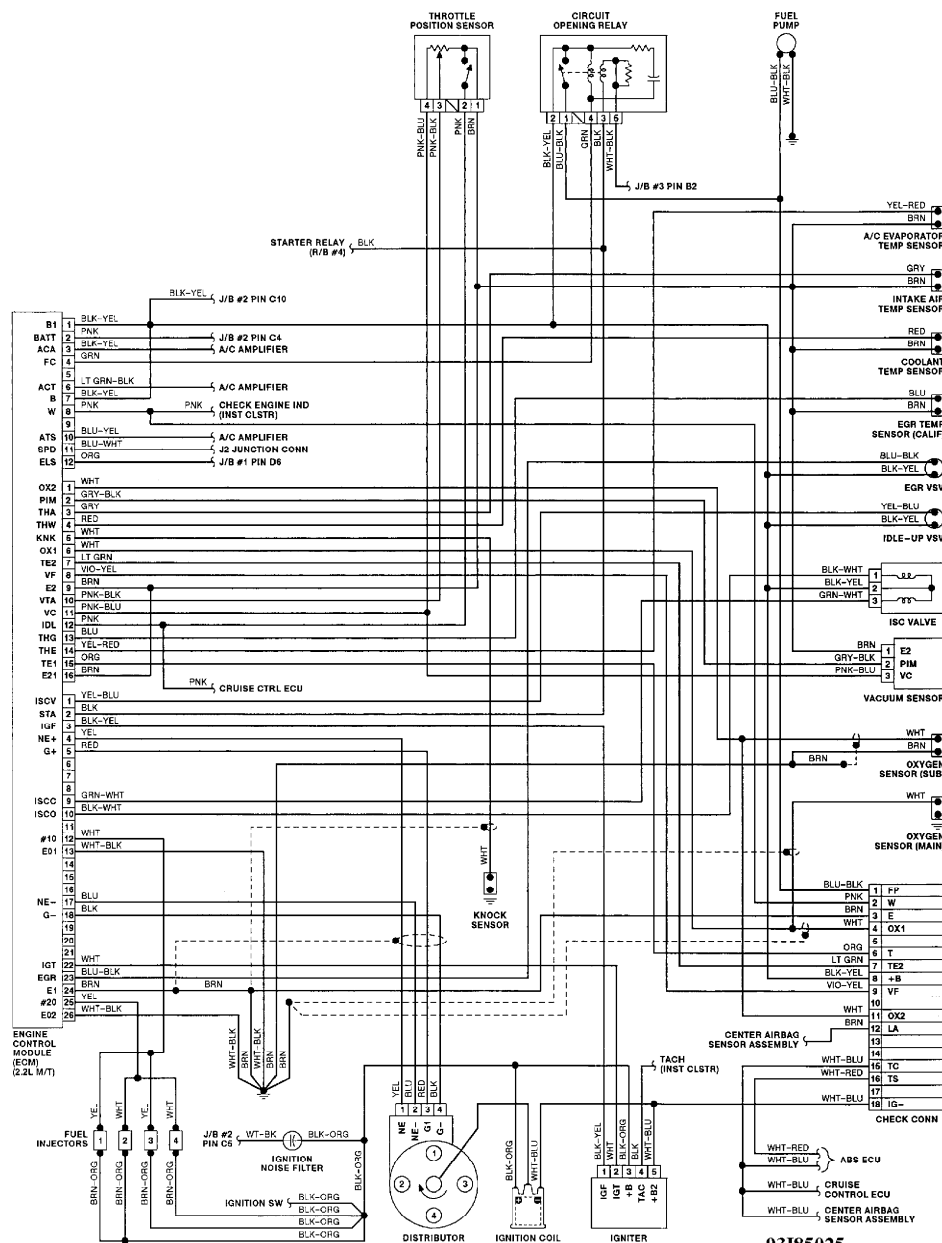


Fig. 3: Wiring Diagram (Celica 2.2L A/T)

93H8502.



93185025

Fig. 4: Wiring Diagram (Celica 2.2L M/T)

# WIRING DIAGRAM SYMBOLS

1993 Toyota Celica

WIRING DIAGRAMS  
How To Use The Wiring Diagrams

## WIRING DIAGRAMS

### INTRODUCTION

The wiring diagrams and technical service bulletins, containing wiring diagram changes, are obtained from the domestic and import manufacturers. These are checked for accuracy and are all redrawn into a consistent format for easy use.

All diagrams are arranged with the front of the vehicle at the left side of the first page and the rear of the vehicle at the right side of the last page. Accessories are shown near the end of the diagram.

Components are shown in their approximate location on the vehicle. Due to the constantly increasing number of components on vehicles today, it is impossible to show exact locations.

In the past, when cars were simpler, diagrams were simpler. All components were connected by wires, and diagrams seldom exceeded 4 pages in length. Today some wiring diagrams require more than 16 pages. It would be impractical to expect a service technician to trace a wire from page 1 across every page to page 16.

Removing some of the wiring maze reduces eyestrain and time wasted searching across several pages. Today, the majority of diagrams now follow a much improved format, which permits space for internal switch details and connector shapes.

Any wires that don't connect directly to their components are identified on the diagram to indicate where they go. There is a legend on the first page of each diagram, detailing component location. It refers you to sub-systems, using grid NUMBERS at the top and bottom of the page and grid LETTERS on each side. This grid system works in a manner similar to that of a road map.

### HOW TO USE THE WIRING DIAGRAMS

1) On the first page of the diagram, you will find a listing of major electrical components or systems. Locate the specific component or system you wish to trace. A grid number and letter will follow the component's name.

2) Use the grid NUMBERS (arranged horizontally across the top and bottom of each page) to find the page of the wiring diagram that contains the component you're seeking. When you reach this page, use the grid LETTERS on the side of the page to determine the component's vertical location.

3) Locate the circuit you need to service. The internals are shown for switches and relays to assist you in understanding how the circuit operates.

NOTE: In some of the newer wiring diagram articles in this product, there is a Legend for the wiring diagrams that has been created to make locating components easier. For these articles, there will be a COMPONENT LOCATION MENU title in the article main menu. These articles will also have the original legend available on the first graphic.

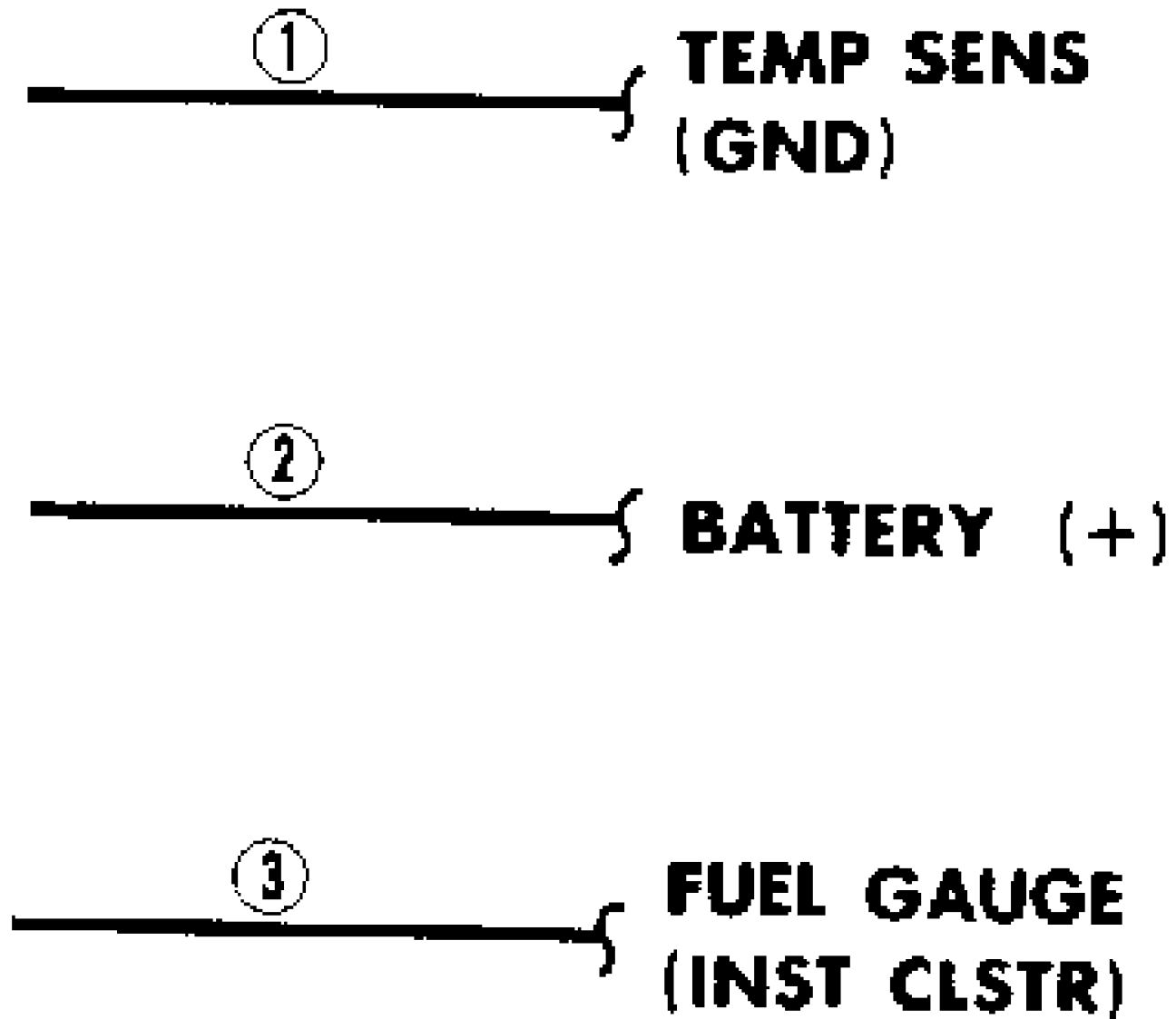


Fig. 1: Identifying Tie-Off Symbols

4) If the wires are not drawn all the way to another component (across several pages), a reference will tell you their final destination.

5) Again, use the legend on the first page of the wiring diagram to determine the grid number and letter of the referenced component. You can then turn directly to it without tracing wires across several pages.

6) The symbols shown in Fig. 1 are called tie-offs. The first tie-off shown indicates that the circuit goes to the temperature sensor, and is also a ground circuit.

7) The second symbol indicates that the circuit goes to a battery positive parallel circuit. The third symbol leads to a particular component and the location is also given.

8) The lines shown in Fig. 2 are called options. Which path or option to take depends on what engine or systems the vehicle has.

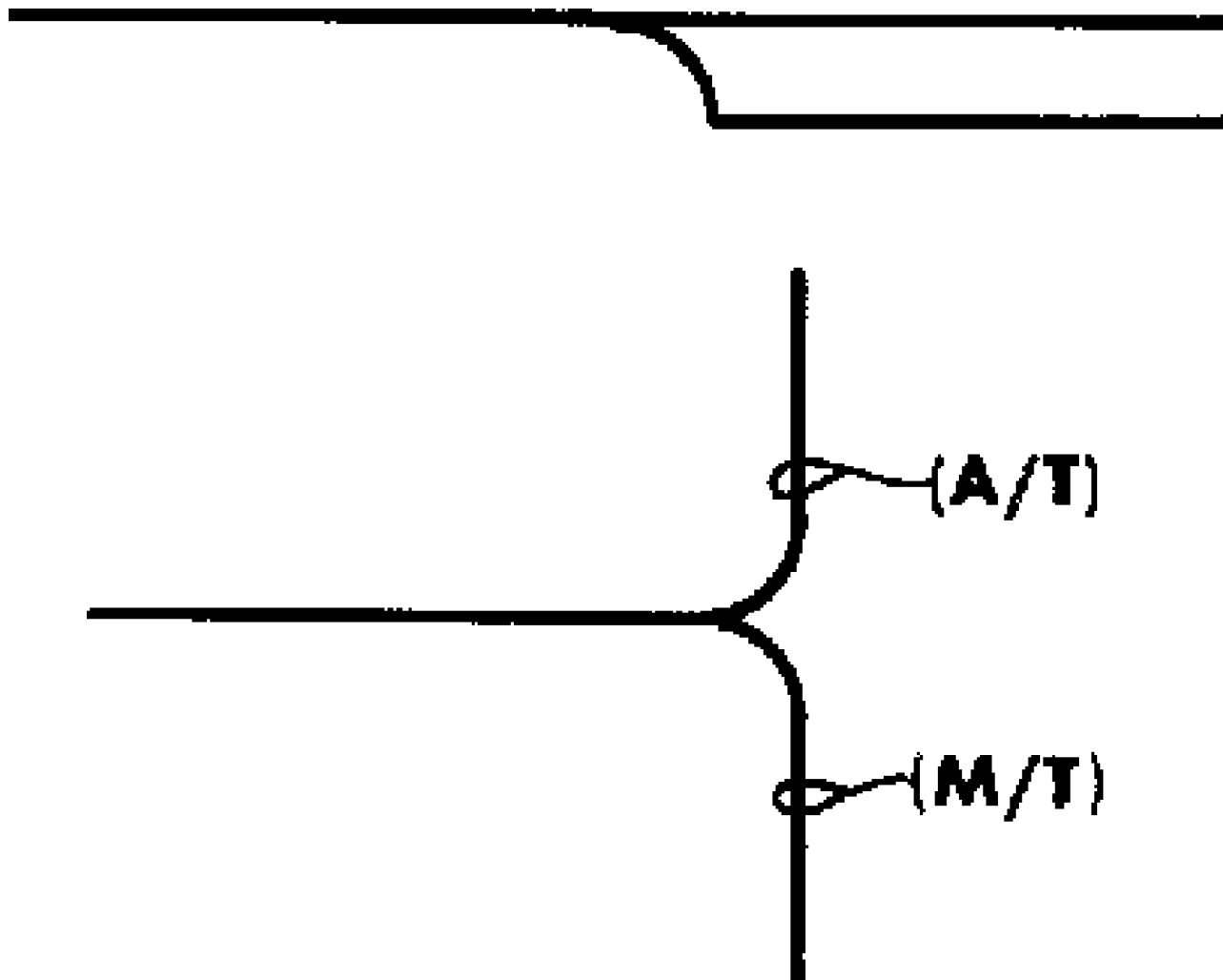


Fig. 2: Identifying Option Symbols

## COLOR ABBREVIATIONS IDENTIFICATION

### COLOR ABBREVIATIONS

| Color             | Normal       | Optional |
|-------------------|--------------|----------|
| Black .....       | BLK .....    | BK       |
| Blue .....        | BLU .....    | BU       |
| Brown .....       | BRN .....    | BN       |
| Clear .....       | CLR .....    | CR       |
| Dark Blue .....   | DK BLU ..... | DK BU    |
| Dark Green .....  | DK GRN ..... | DK GN    |
| Green .....       | GRN .....    | GN       |
| Gray .....        | GRY .....    | GY       |
| Light Blue .....  | LT BLU ..... | LT BU    |
| Light Green ..... | LT GRN ..... | LT GN    |
| Orange .....      | ORG .....    | OG       |
| Pink .....        | PNK .....    | PK       |
| Purple .....      | PPL .....    | PL       |
| Red .....         | RED .....    | RD       |
| Tan .....         | TAN .....    | TN       |
| Voilet .....      | VIO .....    | VI       |



|        |       |     |       |    |
|--------|-------|-----|-------|----|
| White  | ..... | WHT | ..... | WT |
| Yellow | ..... | YEL | ..... | YL |

WIRING DIAGRAM SYMBOL IDENTIFICATION

NOTE: Standard wiring symbols are used on diagrams. The list below will help clarify any symbols that are not easily understood at a glance. Most components are labeled "Motor", "Switch" or "Relay" in addition to being drawn with the standard symbol.

WIRING DIAGRAM SYMBOLS

Views of the symbols used in the WIRING DIAGRAM articles are in the following graphics. See Figs. 3 through 25.



CIRCUIT BREAKER

Fig. 3: Circuit Breaker



COIL (Internal)

Fig. 4: Coil (Internal)



CONNECTOR

Fig. 5: Connector



DIODE (In-Line)

Fig. 6: Diode (In-Line)



Fig. 7: Diode (Internal)



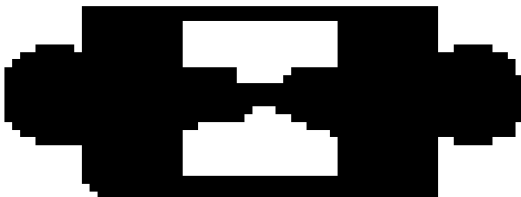
**DIODE (Light Emitting)**

Fig. 8: Diode (Light Emitting)



**DEFOGGER GRID**

Fig. 9: Defogger Grid



**FUSE**

Fig. 10: Fuse



**FUSIBLE LINK**

Fig. 11: Fusible Link



**GROUND**

Fig. 12: Ground



**GLOW PLUG, RESISTOR (In-Line)  
MIRROR HEATER**

Fig. 13: Glow Plug Resistor (In-Line) or Mirror Heater



**INJECTOR, PHOTOCELL**

Fig. 14: Injector (Diesel) or Photocell (Gasoline)



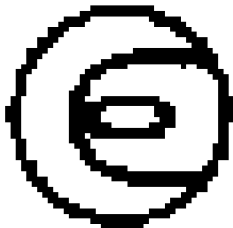
**INTERNAL FUSE,  
THERMAL LIMITER**

Fig. 15: Internal Fuse, Thermal Limiter



**LAMP (Dual Element)**

Fig. 16: Lamp (Dual Element)

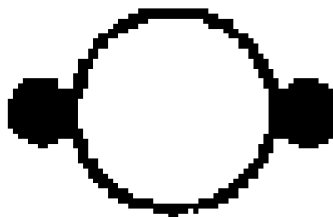


**LAMP (Single Element)**

Fig. 17: Lamp (Single Element)



**OR**



**MOTOR**

Fig. 18: Motor



**RESISTOR (Internal)**

Fig. 19: Resistor (Internal)



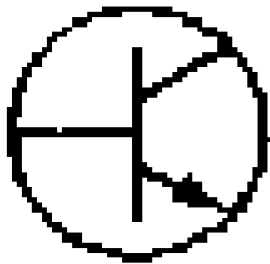
**SENSOR, THERMISTOR**

Fig. 20: Sensor, Thermistor



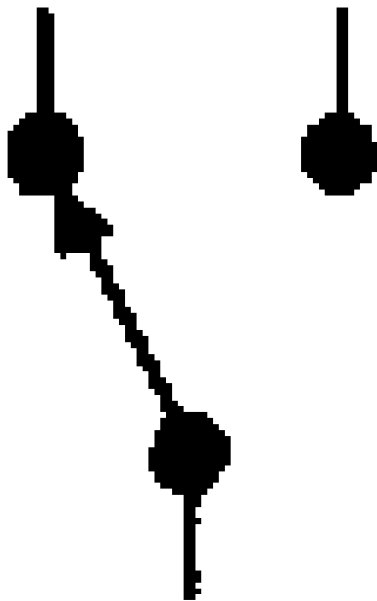
**SOLENOID**

Fig. 21: Solenoid



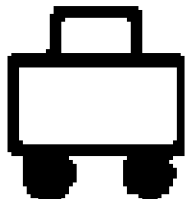
**SOLID STATE DEVICE.  
TRANSISTOR**

Fig. 22: Solid State Device, Transistor



**SWITCH (Internal)**

Fig. 23: Switch (Internal)



**TWO PIN SWITCH**

Fig. 24: Two Pin Switch



**VARIABLE RESISTOR  
OR POTENTIOMETER**

Fig. 25: Variable Resistor or Potentiometer

